Frankenstein Transformer: Unified Encoder-Decoder Library, CLI, and Research-Grounded Design Notes

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Abstract

Transformer architectures dominate sequence modeling across natural language processing, computer vision, speech, and computational biology, but the design space around them has fragmented. Attention has diversified beyond dense softmax into recurrent and retentive blocks, sparse patterns, gated mechanisms, latent projections, memory-augmented layers, and fast-weight rules; optimizers have expanded beyond AdamW with variance-reduction, memory-efficient, schedule-free, second-order, and low-rank methods. Comparing these ideas under consistent conditions remains costly, because each variant must be implemented, wired into a training pipeline, and maintained separately—a duplication of effort that hinders reproducibility and makes published comparisons difficult to align. We present Frankenstein Transformer, a configuration-driven CLI toolkit (`frankenstein-transformer`) that unifies model definition, training, deployment, and inference behind a single strict YAML schema. The toolkit covers forty-nine sequence mixers across nine families and thirty-eight optimizer families with per-parameter-group hyperparameter control, and supports encoder, decoder, Vision Transformer and audio model classes for masked language modeling, causal language modeling, sentence embeddings, image tasks, speech recognition and audio classification. Its contribution is a validated experimentation contract: invalid configurations fail before training starts, every architecture and optimizer shares one operational interface, and the same document drives training, quantized deployment and export, and a schema-generated web builder. The contract is exercised end-to-end in a measurement campaign of 868 configurations (905 CPU-hours) that trains, exports, quantizes, and evaluates every mixer pair, optimizer, and deployment path through the same interface, published as an interactive report with downloadable datasets. We describe the architecture taxonomy, optimizer routing, deployment paths, and integration surface, and defer mathematical and implementation detail to the appendices.

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1 Introduction

1.1 Motivation and Problem Statement

Transformer architectures are the dominant paradigm for sequence modeling in natural language processing [159], computer vision, and computational biology, with models such as BERT [33] and GPT establishing the standard for representation learning. The field has since diversified rapidly: recurrent and retentive blocks, sparse attention, gated mechanisms, latent projections, memory-augmented layers, fast-weight rules, and continuous-depth formulations each address a different limitation of dense attention, while the optimizer landscape has expanded beyond AdamW with variance-reduction, memory-efficient, schedule-free, second-order, and low-rank methods [22, 152, 54, 11, 134, 167, 120, 174, 144, 182, 31, 57, 160].

The practical consequence is a fragmented ecosystem. Comparing sequence mixers or optimizers under identical training conditions requires re-implementing each variant, wiring it into a training pipeline, and maintaining the result across framework changes. This effort hinders reproducibility, slows scientific progress, and raises the barrier to entry for new researchers.

1.2 Proposal and Contributions

We propose **Frankenstein Transformer** (`https://github.com/erickfmm/frankenstein-transformer`), a configuration-driven toolkit that treats architecture, optimizer, and training choices as a single validated document. The project provides:

1. **Schema-first design**: a strict YAML contract supporting forty-nine sequence mixers across nine families and thirty-eight optimizer families, enforced before training begins.
2. **Architecture-agnostic training**: one pipeline for dense baselines, recurrent and retentive blocks, sparse patterns, gated mechanisms, latent compression, memory layers, fast-weight attention, and geometric-field attention, with encoder and decoder modes [186, 21, 168, 181, 125].
3. **Optimizer routing**: prefixed hyperparameter groups that control embeddings, normalization layers, attention blocks, and remaining parameters independently for each optimizer family [184].
4. **End-to-end workflows**: training (including sentence-embedding objectives), deployment and export (quantized, standard, HuggingFace safetensors at 8/4/2 bits, and BitNet GGUF formats), inference, and a schema-driven web builder inspired by SBERT-style workflows [135].
5. **Vision support**: a Vision Transformer model class covering masked patch prediction, classification, and segmentation under the same mixer and optimizer infrastructure.
6. **Validation tooling**: unit tests, YAML preset validation, and continuous integration across supported Python versions.

The command surface is `frankenstein-transformer`, with four subcommands: `train`, `deploy`, `infer`, and `web-server`. The last launches a Streamlit configuration builder that renders schema metadata as forms, validates input live, previews the resulting YAML, and generates the equivalent CLI command, serving both as a configuration aid and as documentation of the available design space; a hosted instance runs at `https://erickfmm.github.io/frankenstein-transformer/index.html`.

1.3 Reading Guide

This document is organized as a technical reference:

1. **Prerequisites**: Appendix 1 introduces transformers and attention for readers new to the field.

2. **Configuration contract**: Section 3.1 describes the schema and Section 3.2 its validation rules.

3. **Architecture selection**: Section 4.1 compares sequence-mixer families; Appendices 2, 3, 4, and 5 collect the supporting literature, with normalization details in Appendix 10, BitNet quantization in Appendix 11, and conditional-memory details in Appendix 13, and manifold-constrained residuals in Appendix 16.

4. **Optimization**: Section 5 summarizes optimizer routing; Appendix 9 provides the full family analysis.

5. **Deployment and inference**: Sections 6 and 7 cover quantized deployment and sentence embeddings.

2 Related Work

This work sits at the intersection of deep learning frameworks, training tooling, and configuration-driven experimentation. While Sections 4.1 and 5 survey the architectural and optimization literature in depth, this section focuses on the software ecosystem for building, training, and configuring transformer models.

2.1 Base Deep Learning Frameworks

The modern transformer research ecosystem rests on several foundational frameworks that provide automatic differentiation, GPU acceleration, and high-level model-building APIs.

PyTorch [123] has become the dominant framework for transformer research. Its imperative programming model and dynamic computation graphs enable flexible debugging and rapid prototyping of novel architectures. The `torch.nn.Module` abstraction, automatic mixed-precision training via `torch.cuda.amp`, and the `torch.compile` JIT compiler form the backbone of most contemporary transformer implementations. Frankenstein Transformer is built entirely on PyTorch, leveraging its module system for composable architecture definitions.

TensorFlow [1] pioneered production-oriented deep learning with its static computation graph paradigm and the Keras high-level API. While its influence in transformer research has waned relative to PyTorch, TensorFlow's serving infrastructure (`TF Serving`, `TF Lite`) and the `tensorflow-text` library remain relevant for deployment scenarios.

JAX [16] offers a functional programming model built around composable transformations (`jit`, `vmap`, `grad`) and XLA compilation. Its design enables clean implementations of complex algorithms and has been adopted by several recent transformer libraries (e.g., Flax, Haiku). JAX's stateless design and automatic vectorization are particularly well-suited for research on novel attention mechanisms and optimization algorithms.

scikit-learn [124] provides classical machine learning algorithms and utilities (preprocessing, metrics, model selection) that serve as baselines and complementary tools in transformer workflows. While not designed for deep learning, its consistent API and extensive documentation make it a standard reference point for comparing transformer-based approaches against traditional methods.

2.2 Out-of-the-Box Training Frameworks

Several libraries provide complete training pipelines that abstract away the complexity of distributed training, data loading, and hyperparameter management.

Hugging Face Transformers [165] is the de facto standard for accessing, fine-tuning, and sharing pretrained transformer models. Its `Trainer` API, model hub with over 500,000 models, and tight integration with the `datasets` and `tokenizers` libraries have made it the primary entry point for transformer-based NLP. However, its architecture is optimized for fine-tuning existing pretrained models rather than experimenting with novel sequence mixer designs or training from scratch with custom optimization strategies.

Oumi (https://github.com/oumi-ai/oumi, 2025) is an open-source end-to-end platform for training, fine-tuning, evaluating, and deploying foundation models. It provides a unified interface across multiple cloud providers and supports both supervised fine-tuning and preference optimization (DPO, RLHF). Oumi emphasizes production readiness with built-in monitoring, checkpointing, and deployment tooling.

LLaMA Factory (https://github.com/hiyouga/LLaMA-Factory, 2024) offers a unified framework for efficient fine-tuning of over 100 large language models. It provides both a web-based UI and a CLI, supporting LoRA, QLoRA, full-parameter fine-tuning, and various alignment techniques. Its focus is on adapting existing pretrained LLMs to downstream tasks rather than architectural experimentation.

Axolotl (https://github.com/axolotl-ai-cloud/axolotl, 2024) is a streamlined fine-tuning tool supporting LoRA, QLoRA, DeepSpeed, and FSDP. It emphasizes configuration-driven workflows via YAML files and has become popular in the open-source LLM fine-tuning community. Like LLaMA Factory, it targets fine-tuning of pretrained models rather than training novel architectures from scratch.

2.3 Low-Code and Efficiency-Oriented Tools

A growing category of tools prioritizes accessibility and computational efficiency, lowering the barrier to transformer training.

Unsloth (https://github.com/unslothai/unsloth, 2024) provides low-code fine-tuning with 2–5× speedup and significantly reduced VRAM requirements through hand-written CUDA kernels and optimized Triton operations. It supports GGUF export for quantized deployment and integrates with the Hugging Face ecosystem. Unsloth's focus is on making fine-tuning faster and more memory-efficient, not on architectural exploration.

dashAI (https://github.com/DashAISoftware/dashAI, v0.9.7.post2, 2026) is an open-source, local-first desktop workbench for machine learning developed by a Chilean academic consortium (Universidad de Chile/FCFM, CENIA, IMFD, and ANID funding). Its interface is fully schema-driven: configuration forms are generated directly from Pydantic component schemas with no hand-written frontend code, and its catalog spans roughly 76 multitask models—tabular classifiers, NLP models, translation, open-weight LLMs, and text-to-image generation—unified behind a dozen typed base abstractions. Extensibility is central to its design: plugins are ordinary PyPI packages discovered through the `dashai.plugins` entry-points group and installable from the UI, a mechanism comprehensive enough that even dashAI's own generative capabilities (e.g., Stable Diffusion, Flux) are distributed as plugins. Where earlier releases targeted classical ML workflows, the current platform spans predictive and generative modeling across domains, making it a natural host for transformer components via its plugin system; Section 8 presents the `dashai-frankenstein` adapter built on this mechanism.

2.4 Configuration-Driven Experimentation

A common thread across the tools surveyed above is their focus on fine-tuning existing pretrained models. Hugging Face Transformers, LLaMA Factory, Axolotl, and Unsloth all assume a pretrained checkpoint as the starting point. Oumi supports training from scratch but targets standard architectures. None of these tools provide a systematic framework for experimenting with novel sequence mixer architectures, comparing diverse optimization algorithms under controlled conditions, or exploring the interaction between architectural choices and training dynamics.

Frankenstein Transformer fills this gap by providing a schema-first configuration system purpose-built for architecture experimentation. Its key differentiators include:

- **49 sequence mixers** spanning dense attention, recurrent/retentive blocks, sparse attention patterns, gated mechanisms, latent compression, memory layers, fast-weight attention, and geometric-field attention, all selectable via a single YAML field.
- **23 optimizer families** with a prefixed hyperparameter routing system that enables per-parameter-group control across embeddings, normalization layers, attention blocks, and feed-forward networks.
- **Dual training modes** supporting both encoder-style masked language modeling (MLM) and decoder-style autoregressive (AR) next-token prediction, with specialized fine-tuning workflows for each.
- **Strict schema validation** via `additionalProperties: false` constraints that prevent configuration errors before training begins, ensuring reproducibility across experiments.
- **End-to-end workflows** spanning quantized deployment (ternary weight packing, INT8 activations) and sentence-embedding training inspired by SBERT [135].
- **Web-based configuration interface** providing schema-driven form rendering with inline documentation, real-time validation, and CLI command generation.

By decoupling architecture definition from training infrastructure through a validated configuration contract, Frankenstein Transformer enables systematic comparison of sequence mixer families and optimization strategies under identical training conditions—a capability not provided by any existing tool in the ecosystem.

3 System Design and Architecture

3.1 Configuration-Centric Architecture

Experimentation in this project is *schema first*. Rather than exposing loosely checked flags, model topology, optimizer family, training limits, and telemetry pass through a single validated configuration document, which reduces ambiguity when reproducing results and lets many architectures share one operational interface.

The authoritative contract is `src/schema.yaml`, a modular JSON Schema whose modules under `src/schema/` cover the model, model class, tokenizer, optimizer, SBERT, training, and dataset blocks. The top-level objects are `base_model`, `tokenizer`, `model_class`, `model`, `training`, `image`, `vision_dataset`, `audio`, `audio_dataset`, `text_dataset`, `inference`, and `plugins`.

Model class. The `model_class` field selects among three variants:

- `frankenstein_text_encoder`: mixed-architecture encoder with the full range of sequence mixers, MoE, and advanced normalization, trained with masked language modeling.
- `frankenstein_text_decoder`: autoregressive causal decoder for next-token generation; the runtime forces decoder masking.
- `frankenstein_vit`: Vision Transformer over image patches for masked patch prediction, classification, and segmentation; the runtime forces encoder masking and the configuration requires the `image` and `vision_dataset` blocks.

Training mode. `model.dims.mode` controls masking: `encoder` uses bidirectional attention for masked language modeling, and `decoder` uses causal masking for autoregressive prediction.

Declarative component registry. The runtime configuration mirrors the validated document one-to-one: the nested `model:` block is materialised as a typed configuration tree and per-component knobs are bound through the component's declarative record, with no nested-to-flat rewriting stage. Every pluggable component—sequence mixer, optimizer, normalization, activation, residual strategy, positional encoding and token-embedding variant—is described by one registry entry holding its name, lazily imported implementation, parameter schema fragment, runtime defaults, capability flags (KV-cache support, training-time sparsity, QK-normalization support, gated-FFN status) and bilingual documentation. The schema enumerations, per-mixer parameter objects, optimizer prefix reference and conditional prefix rules are *assembled from that registry* before validation, so a component is declared once and cannot drift across parallel lists; the closed-world validation guarantee (`additionalProperties: false`) holds for every registered component. The same mechanism is the extension surface: a plugin module registers new components through a public authoring API, either listed under the configuration's `plugins:` key or discovered through the `frankenstein.plugins` entry-point group. Plugin-declared components therefore validate strictly, appear in the schema-generated web builder, and are dispatched by the same core code as the built-ins, while a plugin that does not declare KV-cache support simply falls back to the full-recompute path (decision D5).

Mixer selection. `model.dims.layer_pattern` is an ordered list of mixer identifiers drawn from the eight families described in Section 4; the schema fixes the accepted identifiers, and the complete enumeration appears in Figure 2 and Appendix 27.

Optimizer selection. `training.optimizer.optimizer_class` selects one of thirty-eight optimizer families, summarized in Section 5 and detailed in Appendix 9.

3.2 Schema Scope and Validation Rules

Top-level and nested objects set `additionalProperties: false`, so unknown keys fail fast instead of being silently ignored. Optimizer parameters are further constrained by per-family prefix rules. Cross-component validation rejects incompatible combinations before training: hidden size must be divisible by the number of heads, the number of key–value heads must divide the number of heads, vocabulary size must match the tokenizer, a base model requires a task and (for MLM) a tokenizer, each task requires its matching optimizer block, BitNet router flags require BitNet to be enabled, and the decoder and vision classes force their respective masking modes.

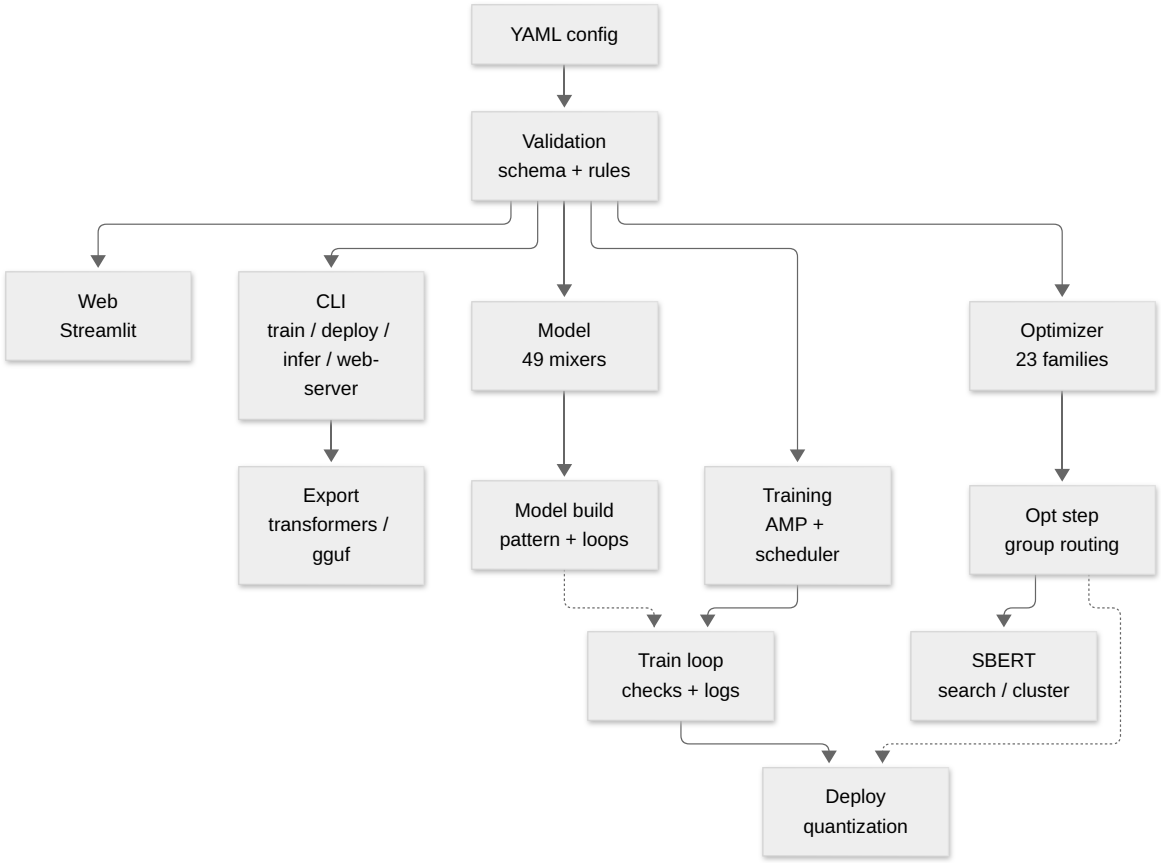


Figure 1: System architecture: configuration flows through validation, partitions into model, training, and optimizer components, executes at runtime, and produces deployment and SBERT artifacts.

3.3 Configuration Schema

The complete schema—fields, types, ranges, and validation rules—is documented in Appendix 26. Looped blocks are configured through `num_layers` and `num_loops`, giving `num_layers`×`num_loops` logical layers at fixed parameter count.

3.4 Training Tasks

`training.task` selects the objective: `mlm` (masked language modeling), `causal_lm` (autoregressive next-token prediction, decoder class only), `sbert` (sentence embeddings), `text_classification` (encoder class only), and the vision tasks `patch_prediction`, `classification`, and `segmentation` (Vision Transformer only). NLP tasks consume `text_dataset`; vision tasks consume `vision_dataset`. Sentence-embedding workflows are described in Section 7.

3.5 Training Safety and Runtime Semantics

The training step accumulates gradients over micro-batches, clips the global norm, checks for post-clip overflow, retries a bounded number of times on non-finite values, dispatches the configured scheduler, and rotates rolling and best checkpoints. When `training.test_percentage` is positive, a deterministic fraction of the dataset is held out before training and the final weights are evaluated on it; the loss and task accuracy (character error rate for ASR) are written to `test_metrics.json`. A single `training.seed` (default 42) controls the run's randomness: it seeds the Python, NumPy, PyTorch and CUDA generators, model weight initialization, DataLoader shuffle

generators and workers, and the held-out permutation, and it enables deterministic cuDNN, so repeated runs on the same hardware and software stack reproduce initialization, batch order and split. The full pseudocode and the split formulation are deferred to Appendix 13.

3.6 Normalization Variants

Normalization is both a modeling choice and a schema compatibility question. The accepted `norm.type` values are `layer_norm`, `dynamic_tanh`, `derf`, `rms_norm`, `prms_norm` (partial RMSNorm), and `flash_norm` (weightless RMSNorm). Their formulations and comparison are detailed in Appendix 10.

3.7 Residual-Stream Architectures

Residual connections are exposed as a macro-design choice through the `model.residuals` block:

- **Standard**: the classical identity residual [62].
- **None**: no skip connection, retained only as an ablation probe.
- **Attention Residuals** (`full_attn` or `block_attn`): learned attention over depth replaces the fixed residual sum, either over all previous layers or over block-wise summaries [153].
- **Manifold-constrained hyper-connections** (`mhc.enabled`): an n-stream residual whose mixing matrix is constrained to the Birkhoff polytope [168].

Attention Residuals compose with mHC and Mixture-of-Depths; the formulations, wiring, and reported results are collected in Appendices 17 and 16.

4 Architecture Taxonomy and Implementation

4.1 Attention and Sequence-Mixer Families

The system implements forty-nine sequence mixers organized into nine functional categories that reflect how each method balances expressivity, computational cost, and memory.

1. **Dense baselines** (4): standard softmax attention, sigmoid attention, Continuum Attention, and Grouped-Query Attention (GQA) provide full global routing at quadratic cost.
2. **Recurrent and retentive** (3): RetNet, Mamba, and ODE-style blocks maintain state for constant-time inference through recurrence, selective state spaces, or continuous depth.
3. **Sparse patterns** (11): Sparse Transformer, Longformer, SWA with attention sinks, BigBird, SparseK, NSA, SpargeAttn, FASA, MSA, SparDA, and LSA reduce quadratic cost through structured sparsity, token selection, training-free pruning, or semantic key clustering.
4. **Gated mechanisms** (9): GLA, DeltaNet, Gated DeltaNet, Gated DeltaNet-2, HGRN2, FoX, Gated Softmax, KDA, and First Passage Attention (FPA) introduce data-dependent control over retention, forgetting, and update strength.
5. **Latent compression** (10): MLA, GQLA, MLRA, Tucker, IHA, GTA, MTLA, CCA, CCGQA, and GMA compress key–value state into low-rank bottlenecks or route through a Gaussian-mixture latent memory.
6. **Memory** (3): Titans pairs a local attention window with a surprise-driven associative memory updated at test time, Engram provides test-time read/write memory through learned N-gram hash lookups, and Memory Attention replaces the dedicated value projection with a layer-specific token-ID table.

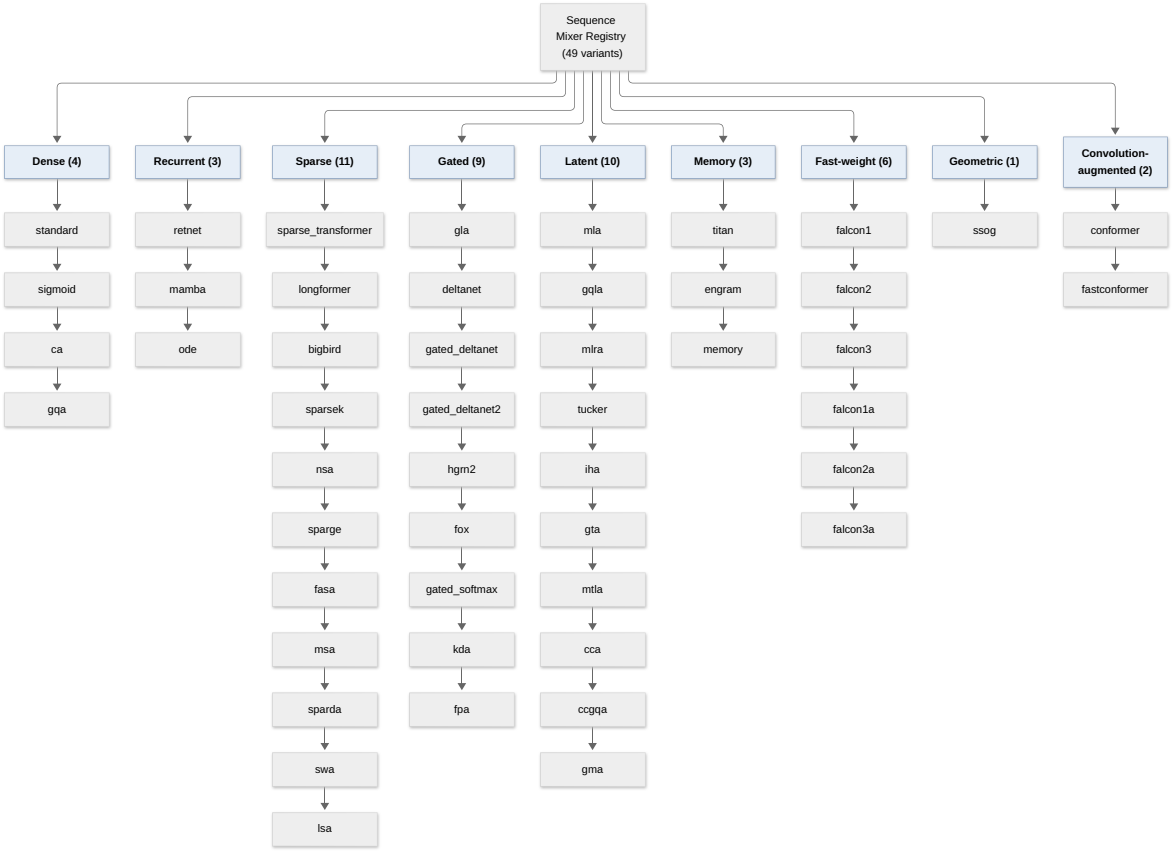


Figure 2: Taxonomy of the forty-nine sequence mixers across nine functional categories. Dense baselines provide full global routing at quadratic cost; recurrent methods compress state for constant-time inference; sparse variants restrict the attention pattern; gated mechanisms modulate retention and forgetting; latent methods compress key–value representations; the memory family pairs a test-time associative memory (Titans), hash-based N-gram lookups (Engram), and token-indexed value capacity (Memory Attention); fast-weight methods cast the state transition as an online learning rule; geometric methods encode structured priors; convolution-augmented methods add a local convolution branch beside attention.

7. **Fast-weight attention** (6): Falcon-1/2/3 and Falcon-1A/2A/3A treat the recurrent state transition as an online continual-learning rule [181].

8. **Geometric field** (1): SSOG applies a structured geometric prior over the attention matrix [125].

9. **Convolution-augmented** (2): the Conformer block [56] fuses multi-head self-attention with a depthwise-convolution module between two Macaron feed-forward layers, giving every layer a local branch alongside the global receptive field; Fast Conformer [136] keeps that design for long-form speech and replaces global self-attention with a limited local window plus a head-pinned global token (Appendix 8).

4.2 Dense and Recurrent Families

The dense baselines—standard softmax [159], sigmoid attention [134], and Continuum Attention [79], which replaces the softmax with exact integration of an exponential relevance density over piecewise-linear key/value trajectories—and the recurrent family—RetNet [152], Mamba [54],

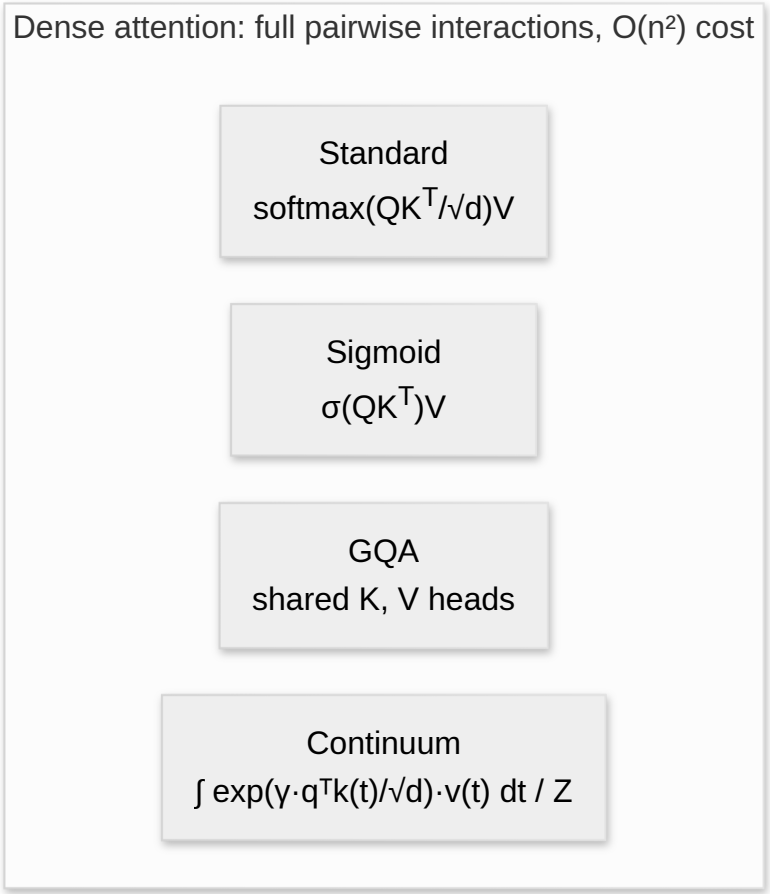


Figure 3: Dense attention baselines. Standard and sigmoid attention compute full pairwise interactions; Grouped-Query Attention shares key–value heads across query groups, reducing memory traffic while keeping full attention expressivity.

and ODE-style blocks [178]—define the expressiveness-versus-efficiency trade-off. Formulations, parallel and recurrent forms, pseudocode, and a comparison table are collected in Appendix 2.

4.3 Memory Family

The memory family groups three mechanisms that add storage beside the attention operator. Titans [11] runs dense attention over a local window while a surprise-driven associative memory is updated at test time; this repository implements the attention window only, and the neural-memory module is not part of the implementation. Engram [21] retrieves N-gram embeddings through per-order hash tables and gates them into the residual stream. Memory Attention [76] keeps the standard softmax attention operator but removes the dedicated value projection: values are the contextual keys plus a layer-specific, token-ID-indexed memory row, $V = K + \mathrm{Norm}(E[s])$, with RoPE applied only to the scoring pair. It increases parameter capacity without proportional dense computation and retains a conventional KV cache at inference.

4.4 Latent and Low-Rank Attention Family

This family compresses per-token key–value state into a low-rank latent: MLA, GQLA, MLRA, Tucker, IHA, GTA, and MTLA unify grouped-query, latent, factorized, and temporal designs. CCA and CCGQA formulate attention in a compressed latent with decoupled grouping, while

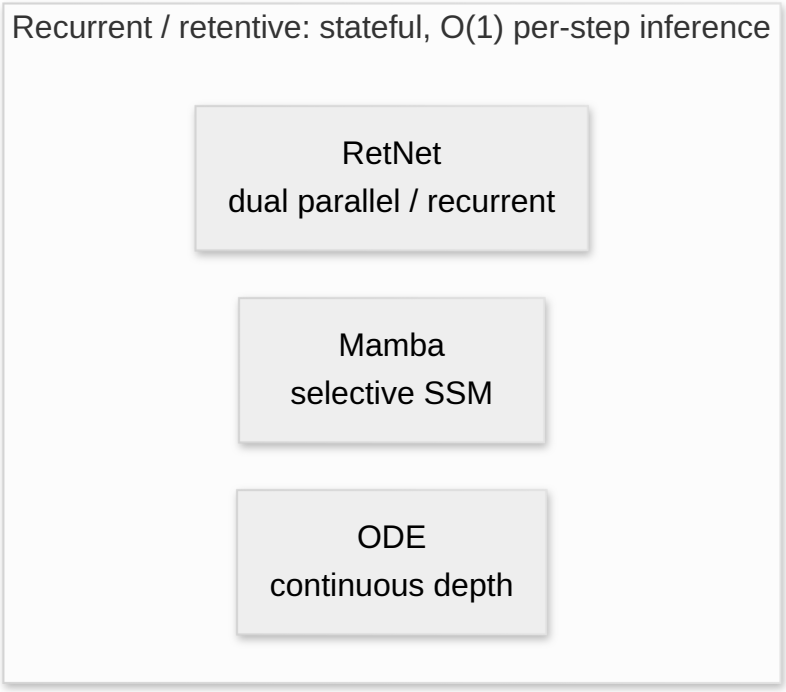


Figure 4: Recurrent and retentive family. RetNet offers parallel and recurrent forms, Mamba uses selective state-space parameters, and ODE blocks model continuous depth; all achieve constant-time per-step inference.

GMA replaces pairwise interaction with probabilistic routing through learned Gaussian-mixture components. Formulations, pseudocode, and a comparison table are collected in Appendix 3.

4.5 Sparse Attention Extensions

The registry includes eleven sparse blocks [22, 12, 75, 175, 108, 173, 180, 164, 91, 45, 89]. Two inference-time methods, `fasa` and `sparge`, apply their sparse heuristic only when the model is in evaluation mode: during training the enclosing hybrid layer substitutes standard dense attention over the same **q**/**k**/**v**/**o** projections, so the trained weights are exactly the ones the sparse path reads at inference. `swa` is also training-free but is a normal trainable mixer, so it runs its windowed pattern in both training and inference. LSA is the family's semantic selector: keys are pooled and Ward-clustered per 512-token block, and each query scores the cached bucket centroids instead of every key, selecting top-p buckets that expand into its k long-range keys on top of sinks and a one-block window. Formulations, pseudocode, per-block characteristics, and a comparison table are collected in Appendix 4.

4.6 Gated and Fast-Weight Attention

The gated family contains nine blocks [169, 170, 171, 60, 128, 99, 129, 83, 104]. Gating controls what information survives, either in recurrent state updates (GLA, DeltaNet variants, HGRN2, KDA) or in the full attention path (FoX, Gated Softmax, FPA). FPA replaces the path sum of a forget gate with a range minimum of a learned field, so a later token can erase newer keys while preserving older ones: the alive keys form a stack, and FoX is its non-rising-walk special case. Formulations, pseudocode, and a comparison table are collected in Appendix 5.

Fast-weight attention recasts the recurrent update as an online continual-learning rule; the six Falcon variants [181] unify DeltaNet-style regression and linear-attention writes under read-

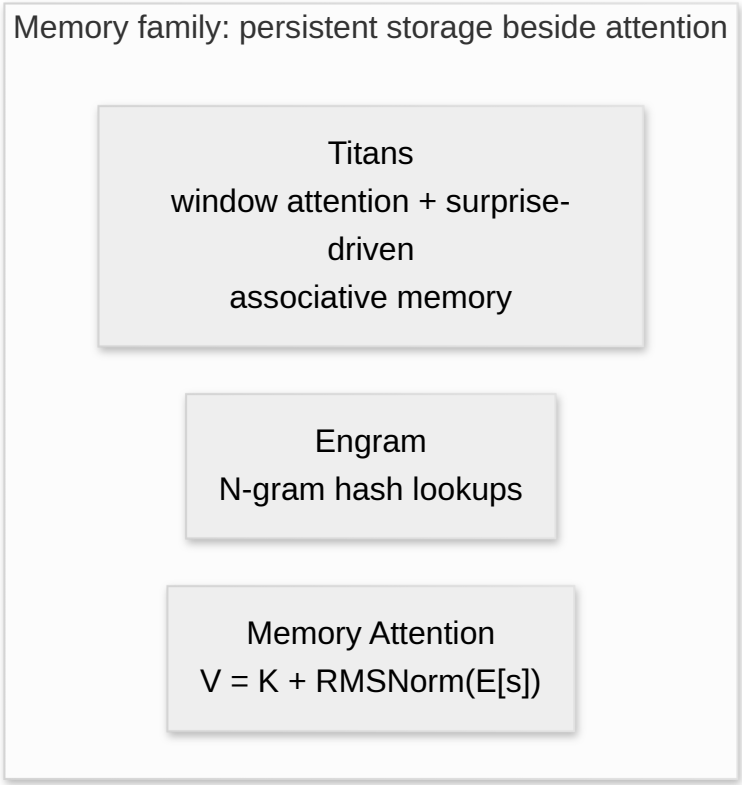


Figure 5: Memory family. Titans carries a surprise-driven associative memory across steps beside a local attention window; Engram retrieves N-gram embeddings through hash lookups; Memory Attention forms values from the contextual keys plus a normalized row of a per-layer token-ID memory table, replacing the value projection while keeping full quadratic attention weighting.

after-write semantics with normalized step sizes and explicit forgetting. Each variant ships a recurrent reference scan and a chunk-parallel kernel. Formulations, configuration knobs, and published results are collected in Appendix 7.

5 Optimizer Families and Training Dynamics

Optimizing large transformer architectures is complicated by non-convex landscapes, saddle points, and heterogeneous parameter groups. The system supports thirty-eight optimizer families in eight categories: classical baselines (SGD, SGD with momentum, RMSProp, AdamW, RAdam); momentum and variance reduction (Adan, ADOPT, AdEMAMix, MARS, Cautious AdamW); memory-efficient and low-rank methods (Adafactor, GaLore, Lion, APOLLO, APOLLO-Mini, Q-APOLLO, AdamPM); schedule-free and parameter-free methods (Schedule-Free AdamW, Prodigy) with large-batch scaling (LAMB); curvature-aware and second-order methods (Sophia, Shampoo, SOAP); geometry-oriented methods (Muon, Turbo-Muon, AF-Muon, AdaMuon, Dion3, ROOT, Muon^2, NorMuon, Conda, TUON, NAMO, NAMO-D); adaptivity-tunable optimizers (Anon); and evolutionary population-based hybrids (EvE, EA4LLM). Algorithms, pseudocode, and memory and complexity comparisons are collected in Appendix 9.

6 Quantization and Deployment

The deploy stage converts a trained checkpoint into a deployment artifact through post-training quantization. BitNet-style ternary packing [161] is complemented by two block-affine formats and a ternary safetensors format, all emitted inside a self-contained HuggingFace package whose only weight container is `model.safetensors`. In the block-affine formats, weights are grouped into blocks of 32: `Q8_0` stores one FP16 scale and 32 signed 8-bit values per block (8.5 bits per weight), whereas `Q4_0` stores one FP16 scale and 16 packed nibbles (4.5 bits per weight). The ternary `i2_s` layout packs four symbols per byte and adds one FP32 scale per tensor (2 bits per weight before the scale). These formats complement, rather than replace, the native standard and quantized artifacts and the BitNet GGUF export used by external runtimes.

The safetensors artifact supports two load-time runtimes. In the *dequantize-on-load* runtime (v1), packed blocks are expanded to FP32 parameters while the model is loaded, so inference executes exactly like the unquantized backbone. In the *packed-linear* runtime (v2), each quantized projection keeps its packed buffers and dequantizes inside `forward`; when a projection replaces a BitLinear layer, the input LayerNorm and INT8 activation quantization are preserved. Only two-dimensional linear weights whose last dimension is a multiple of 32 are quantized; embeddings, normalization parameters, biases and convolutions remain in FP32, which bounds the achievable compression ratio. The experiment harness exports `Q8_0`, `Q4_0` and `i2_s` in both runtimes and reports their task accuracy alongside the unquantized export. The formulations, block layouts and size estimates are deferred to Appendix 11.

7 SBERT Downstream Tasks

Sentence embedding is built on Siamese-style training [135]. For sentence pair (s_1, s_2) with embeddings (e_1, e_2):

$$\cos(e_1, e_2) = \frac{e_1^\top e_2}{\|e_1\|\|e_2\|}$$

and regression-style cosine loss:

$$\mathcal{L}_{\cos} = (\cos(e_1, e_2) - y)^2$$

with $y \in [-1, 1]$ in this pipeline.

Supported downstream modes:

- **Similarity**: pairwise score between two sentences.
- **Search**: top-k nearest neighbors over a corpus.
- **Cluster**: grouping embeddings (e.g., k-means).
- **Encode**: persistent embedding export for later retrieval.

The pseudocode for the SBERT inference mode router—dispatching among similarity, search, cluster, and encode modes over a shared encoder—is deferred to Appendix 13.

8 No-Code Integration: The `dashai-frankenstein` Plugin

The CLI and web builder described in Section 3 serve researchers comfortable with YAML configuration files. A complementary audience—domain specialists, educators, and analysts who need a trained model but not a training script—is better served by a graphical environment. Frankenstein Transformer therefore ships `dashai-frankenstein`, an adapter that registers its model classes as components of **dashAI** (https://github.com/DashAISoftware/dashAI), an

open-source, local-first machine-learning workbench developed by a Chilean academic consortium (Universidad de Chile/FCFM, CENIA, IMFD, and ANID funding). Both systems are schema-driven—dashAI renders forms from Pydantic schemas, Frankenstein validates YAML against a strict JSON Schema—so the integration leaves dashAI's backend untouched; the plugin is discovered through the standard `dashai.plugins` entry-points group documented in dashAI's plugin guide (https://docs.dash-ai.com/learn/guides/plugins/).

8.1 Registered Components

A single entry-points group registers five components:

- `FrankensteinMLMModel` (`BaseModel`, `TextClassificationTask`): encoder with a classification head over pooled output; `predict` returns per-class probabilities.
- `FrankensteinDecoderModel` (`BaseGenerativeModel`, `TextToTextGenerationTask`): forces the decoder class and exposes autoregressive top-k sampling.
- `FrankensteinViTClassifier` (`BaseModel`, `ImageClassificationTask`): uses the ViT classification head (Appendix 21).
- `FrankensteinViTSegmenter` (`BaseModel`, plugin `SegmentationTask`): uses the ViT segmentation head and returns per-pixel class maps.
- `SegmentationTask` (`BaseTask`): declares the image-in/mask-out contract that dashAI does not ship.

All share one thin facade over the engine API, so the plugin has a single heavy dependency.

8.2 Passthrough Configuration Schema

The Frankenstein JSON Schema remains the single source of truth instead of translating its fields into native dashAI forms. Each component exposes one `frankenstein_json` field holding a complete training configuration as single-line JSON. Before any training runs, the adapter parses the JSON, validates it against the resolved schema, and applies the config loader's cross-component checks; failures surface as a readable error tagged with the failing stage. Dataset adapters map dashAI datasets onto Frankenstein's expectations, inject the vocabulary size from the resolved tokenizer, and stream per-epoch metrics back to dashAI.

8.3 Status

The three implementation phases—classification MVP, generative decoder and ViT classifier, and ViT segmenter—are complete, verified inside a live dashAI instance (0.10.x and 0.11.0), and published on PyPI. Known limitations include a direct classification loop rather than MLM pretraining, deferred native form fields for high-leverage knobs, and unconfirmed rendering of segmentation masks. The adapter deliberately leaves dashAI's core untouched, so future releases cannot break it through coupling.

9 Empirical Evaluation

The preceding sections describe the toolkit in terms of what it can express: which mixers, optimizers, normalizations, and deployment paths the schema admits. This section turns from design to evidence and reports what happened when that design was exercised at scale. Every mixer pair, every optimizer, and every deployment path in the study was trained, exported, quantized, and evaluated through the same YAML interface, so the measurements that follow are

comparable by construction: a difference between two configurations reflects the configurations themselves, not differences in the plumbing around them.

The campaign is organised in *rounds*. A round is one complete pass of the experiment harness over a generated set of configurations, from training through reporting. Round 1 was a rehearsal at smoke scale—5,129 stage records, with training runs of roughly half a minute each—whose purpose was to debug the pipeline itself before committing serious computation. Round 2 is the measurement campaign analysed here: 868 configurations trained end-to-end, 2,656,372 logged optimisation steps, and 905 CPU-hours of training across those runs, published as an interactive report with downloadable datasets so that every number below can be rederived without re-running anything.[1]

Two appendices accompany this section and serve complementary purposes. Appendix 24 documents the harness—the five-stage pipeline, the three independent recorders, the consolidated datasets, and the reporting toolchain—in enough detail that launching a new round reduces to pointing the same scripts at a new folder. Appendix 25 reports Round 2 in full, with the parameter census, the training curves, the complete A×B atlas of mixer pairs, the ablation sweeps, and the deployment measurements. The present section states the findings and interprets them; the appendices carry the supporting evidence.

9.1 What One Round Measures

Every configuration passes through the same five stages, shown in Figure 12: it is trained for 100 epochs, its checkpoint is exported, the checkpoint is deployed twice (once in a standard format, once in a post-training-quantized format), the two deployed models are evaluated on four held-out masked prompts each, and a Hugging Face package is exported. A single YAML file drives all five stages: it selects the model (one mixer, a pair of mixers declared as `layer_pattern: [A, B]`, a vision transformer, or an ablation of the backbone), the task, and the training budget. Because the file is the only input, any measurement can be traced back to the configuration that produced it, and any configuration can be re-run to reproduce its measurement. The current harness extends the export stage to `Q8_0`, `Q4_0` and `i2_s` safetensors variants in both load runtimes and records their task accuracy; the Round 2 tables below were produced before that extension.

Three recorders observe the process independently and are joined only at reporting time: a per-step training log, a parameter census by module group, and a ledger with one row per stage execution. Independence is a deliberate safeguard—no single recorder can silently corrupt the picture, and a disagreement between recorders is itself a diagnostic. The report joins all three by configuration slug, a short identifier derived mechanically from the YAML file.

[1] Round 2 is published at https://erickfmm.github.io/frankenstein-transformer/experiments/2/index.html, and all rounds are indexed at https://erickfmm.github.io/frankenstein-transformer/experiments/index.html.

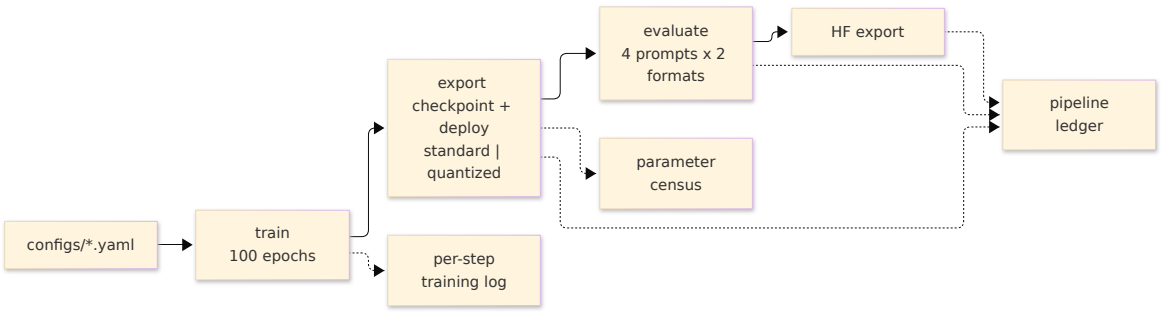


Figure 12: One configuration of a round. The five stages are driven by a single YAML file and logged by three independent recorders; the offline reporting toolchain that turns the logs into the published report is documented in Appendix 24.

Round 2 was generated programmatically rather than curated by hand. The hybrid pairs are all combinations of the 42-entry mixer list taken two at a time, materialised as a two-layer model whose layer 0 runs mixer A and layer 1 runs mixer B; single-mixer baselines use one layer. Because the combinations follow the fixed order of the list, no reversed pair (B, A) exists: the campaign measures `A__B` but never `B__A`. The "first versus second mixer" asymmetry reported below is therefore confounded with list order—position in the stack coincides with position in the list—and must be read descriptively rather than causally. Table 1 summarises the population by family.

Table 1: Round 2 population. Medians are computed per family; the held-out metric is top-1 for MLM pairs and accuracy for the vision, audio and real-data families.

Family	Configs	Median params	Median loss	Median score	Median tok/s	Median min
Hybrid pairs	703	220,928	9.591	4.08%	356	75.3
Singles	38	157,584	9.762	3.57%	366	24.8
ViT	48	315,694	1.205	75.97%	23	9.4
Optimizers	23	210,752	10.404	7.76%	391	21.2
Structural	20	210,752	10.397	5.71%	387	21.2
Positional enc.	12	210,752	10.381	7.76%	402	21.9
Normalization	6	210,512	10.348	1.61%	407	20.7
Real data	6	247,788	1.060	39.04%	392	7.6
Task	4	210,752	8.603	5.90%	457	19.4
Tokenizer	4	210,752	5.978	4.37%	4,202	1.1
Batch size	2	210,752	10.368	5.40%	596	13.0
BitNet/GGUF	1	209,856	10.369	—	562	—
Audio	1	147,171	1.099	0.00%	31,082	1.2

The families differ in what they hold fixed, and the distinction guides every comparison that follows. The hybrid pairs and singles vary the mixer stack itself. The optimizer, structural, normalization, and positional-encoding families share the same 210k-parameter backbone and vary exactly one component—which is precisely what makes them ablations. The task, tokenizer, batch-size, real-data, and audio families change the objective or the data, and therefore the loss scale, which is why they are never compared directly against the mixer sweep.

9.2 Findings

Six findings summarise the round. Each is stated as a claim, supported by a measurement, and followed by its interpretation.

The second mixer is the stronger lever. For each candidate mixer B, consider the mean final loss over all pairs in which B occupies the second position. This marginal mean ranges from 9.37 nats (`sparsek`) to 10.18 (`mamba`)—a spread of 0.81 nats. The corresponding range for the first position is only 0.55 nats (9.26–9.81, restricted to mixers with enough pairs for a stable mean). The same asymmetry appears when the atlas is read geometrically: column by column (fixed B, varying A), a good second mixer keeps almost every first mixer inside the 8.9–9.6 nat band, while row by row (fixed A, varying B), no first mixer rescues a bad second one. In other words, the pair inherits its ceiling from its second layer, and at this scale the choice of B is worth roughly 1.5× the choice of A. Losses are reported in nats—the natural-logarithm unit of cross-entropy—so a gap of 0.81 nats corresponds to a factor of $e^{0.81} \approx 2.2\times$ in the geometric-mean likelihood assigned to the correct token.

Pairing is not additive. If mixers contributed independently, the loss of a pair would be predictable from its parts. It is not. Define the marginal mean μ_M as the mean loss of all pairs containing mixer M, and the synergy of a pair as the average of its two marginals minus its measured loss, $S_{A,B} = (\mu_A + \mu_B)/2 - L_{A,B}$: positive synergy means the pair is better than its parts suggest, negative means worse. The best interaction is `nsa+engram` at +1.304 nats and the worst is `gqa+tucker` at −1.018. The positive tail consistently pairs a cheap, stable first mixer with a memory-heavy second one; the negative tail is dominated by second mixers that look average in isolation (`gta`, `gqa`, `tucker`) but clash when stacked after a strong first mixer. The practical consequence is that the atlas cannot be summarised by two ranked lists: interaction noise of ±0.3 nats is larger than the difference between most optimizers in the ablation.

Scale is a weak lever at this size. Within the hybrid pairs, the parameter count correlates with final loss at $r = -0.20$ and with held-out accuracy at $r = -0.07$: at this scale, adding parameters barely buys quality. What parameters do predict reliably is cost ($r = 0.68$ with step time, $r = 0.65$ with wall-clock). The largest model of the round (`engram__conformer`, 415,904 parameters) is not the best. Across all 868 rows the correlation appears much stronger ($r = -0.43$ for loss), but this is a family confound in the manner of Simpson's paradox: vision transformers are simultaneously larger and far easier to fit than MLM pairs, so the pooled coefficient measures the gap between families rather than the effect of size. The comparison of interest is the within-pairs one, and there size is close to irrelevant.

Post-training quantization is essentially free. Quantization here means converting trained weights to a lower-precision format after training, with no retraining. Of the 810 configurations deployed in both formats, 800 keep an identical top-5 hit rate; the mean top-1 moves from 5.62% to 5.43% and the mean top-5 from 13.36% to 13.24%—deltas of −0.19 and −0.12 percentage points, within the resolution of a four-sentence evaluation. Quantized deployment is *not* faster end-to-end—it wins in only 32.3% of runs, with a median ratio of 0.64×—because the deployment stage is dominated by serialisation and format conversion rather than by arithmetic. Evaluation itself, which does measure inference, is faster in 51.2% of runs with a median 1.03× speed-up. The quality claim is firm (quantization does not move the metrics); the speed claim is about the pipeline, not the format.

The sweep is honest about its failures and its blind spots. The ledger holds 5,271 stage records, of which 275 are failures (171 in training, 104 in deployment); all of them fall outside the 868 analysed configurations, which were selected precisely because they trained. Four mixers (`cca`, `ccgqa`, `msa`, `ssog`) never train in any configuration, which accounts exactly for the 158 missing cells of the 861-cell atlas. Four others (`standard`, `bigbird`, `longformer`, `titan`) produce bit-identical losses because their attention windows (at least 64 tokens) cover the whole 32-token sequence—at this context length they collapse to the same dense model, and the atlas measures one model four times. Separately, the ledger records 37 pairs under both `retnet` and the now-removed `retnet_attn` alias with identical metrics—one implementation under two names—which the analysis collapses. Reporting these defects explicitly is a design choice: a silent zero or a hidden duplicate would contaminate every downstream statistic.

The vision block works. Segmentation reaches 90.7% mean held-out accuracy, classification 75.8%, and patch prediction 69.9%—an order of magnitude above the MLM pairs. The gap reflects the difficulty of the tasks, not the quality of the mixers: guessing a masked token in a 512-token vocabulary is harder than the vision objectives at this scale. This is why the families are kept separate throughout the report, and why no cross-family ranking is stated.

9.3 Threats to Validity

The findings above are conditioned on the scale of the round, and four limitations should temper their reading.

- **Scale and convergence.** The models are small (127k–416k parameters), the vocabulary is 512 Spanish tokens, the sequences are 32 tokens long, and each configuration trains for 100 epochs with a single seed. Every learning curve is still descending at epoch 100, so the ranking rewards fast learners rather than converged quality.
- **Resolution of the evaluation signal.** Inference uses four sentences per configuration. One additional correct token moves a top-5 rate by 25 percentage points, so the held-out signal is a smoke detector for catastrophic failures, not a ranking instrument.
- **Optimizer ties.** Several optimizers converge to the same loss because the run is too short to separate them; the optimizer ranking is indicative at best.
- **Known failure modes.** One text-classification run collapsed to a constant prediction—a failure mode recorded as such, not a usable model—and the four never-training mixers and the `retnet` alias are carried forward as explicit caveats.

The rankings in Appendix 25 are therefore directional: they indicate where to look next, not final verdicts. The next round should re-run the atlas at 256–512 tokens, with a 200-sentence evaluation set, at least three seeds, and 300 epochs for the top pairs.

10 Discussion

The design choices in Frankenstein Transformer reflect several engineering and research tensions in modern deep learning tooling.

The campaign reported in Section 9 turns several of those tensions into measurements rather than opinions. The strict schema paid off operationally: 868 configurations trained, deployed and were evaluated through the same contract, and the failures that remained were concentrated in four mixers and in checkpoint serialisation, not in configuration drift. The prefixed optimizer system proved expressive but, at 100 epochs, undecisive—a dozen optimizers converged to the same loss—which is itself an argument for the schema: only a controlled sweep can show that a design dimension does not matter at a given scale.

10.1 Schema-Driven Design Trade-offs

The schema-first approach provides significant reproducibility benefits by enforcing explicit contracts and failing fast on invalid configurations. However, this approach also introduces rigidity: adding new architectures or optimizers requires schema extensions rather than loose command-line arguments. The prefixed hyperparameter system enables fine-grained control but increases configuration complexity for users accustomed to simpler interfaces.

The decision to enforce `additionalProperties: false` at all schema levels eliminates silent parameter swallowing that has plagued earlier configuration systems, but this strictness requires careful schema maintenance when extending system capabilities. Each new attention mechanism

or optimizer variant must be properly integrated into the validation framework, including schema field definitions with appropriate types and constraints, prefixed hyperparameter mapping for optimizer-specific groups, default values aligned with research best practices, and documentation strings for web interface rendering.

10.2 Architectural Coverage and Gaps

The forty-nine implemented mixers span the major research directions in sequence modeling, but certain gaps remain. The system lacks recent hybrid architectures such as Griffin [148] and Jamba [55], which combine gating with state-space models. MoE routing is implemented for FFN layers but not for attention computation, where recent work has shown benefits [94].

The sparse attention coverage is broad. The two inference-time methods (FASA, SpargeAttn) keep their training-free character in the sense that their sparsification rule is never learned; instead, the implementation trains the layer as standard dense attention over the same projections and activates the sparse selection/pruning path only at evaluation, so both mixers can be trained from scratch and their sparse behaviour is exactly the one described in the original papers.

The gated and fast-weight coverage spans the major categories, including the six Falcon variants [181]. Residual-stream macro-design is covered by Attention Residuals [153] and by the manifold-constrained n-stream residual (mHC) [168], whose Sinkhorn–Knopp projection onto the Birkhoff polytope restores the identity-mapping property lost by unconstrained hyper-connections; mHC is validated only at the level of the published 3B–27B MoE experiments and would benefit from independent reproductions at smaller scales (Appendix 16).

10.3 Optimizer Landscape Fragmentation

The support for thirty-eight optimizer families across eight categories demonstrates comprehensiveness but also highlights the fragmented state of optimization research. Users face significant decision complexity when choosing among variance-reduction methods (Adan, MARS), memory-efficient variants (GaLore, Adafactor, APOLLO), and curvature-aware approaches (Sophia, Shampoo). The prefixed hyperparameter system, while powerful, requires understanding of which parameters are relevant for each optimizer class.

The implementation quality varies across optimizers: classical methods (AdamW, SGD with momentum) are highly optimized in PyTorch, while newer methods (Muon, Turbo-Muon, AF-Muon, SOAP) may require custom implementations that affect numerical stability and performance characteristics.

10.4 Deployment and Production Considerations

The quantization pipeline demonstrates practical deployment concerns but makes specific engineering trade-offs. Ternary weight packing reduces storage to approximately 1.58 bits per parameter, but this aggressive compression may degrade performance, especially for smaller models where quantization error is more significant. The safetensors exporter instead offers block-affine `Q8_0` and `Q4_0` formats (8.5 and 4.5 bits per weight) and applies all three only to two-dimensional linear weights whose input dimension is divisible by 32; embeddings, normalization parameters and biases remain in FP32, so the effective compression ratio depends on the linear fraction of the model. The packed-linear runtime keeps the artifact compact at the cost of a dequantization step in every forward pass.

The SBERT workflows provide practical utility for semantic similarity and retrieval tasks, but implementation assumes standard pooling strategies (CLS token, mean pooling). Recent advances such as Matryoshka embeddings [117] and contrastive learning refinements [80] are not yet incorporated.

10.5 Integration and Extensibility Challenges

Extensibility is handled by a declarative component registry rather than ad-hoc dispatch. Every pluggable component—sequence mixer, optimizer, normalization, activation, residual strategy, positional encoding and token embedding—is described by a frozen `ComponentSpec` registered through `src/registry/`, and the schema enumerations, per-component parameter fragments, optimizer prefix reference and conditional prefix rules are assembled from that registry before validation (Section 3). A third-party plugin registers the same kind of spec through the public authoring API, either from the configuration's `plugins:` list or through the `frankenstein.plugins` entry-point group, so it validates strictly, appears in the schema-generated builder and is dispatched by the same core code as the built-ins. The maintenance pressure is therefore the size of the component catalog itself—forty-nine mixers and thirty-eight optimizers—rather than the dispatch logic, which contains no per-optimizer branches and no per-mixer selection ladder.

The web configuration interface provides significant usability improvements but introduces deployment complexity: running Streamlit alongside training jobs requires additional resources and infrastructure considerations that may not be appropriate for all environments, particularly HPC clusters without web access.

11 Conclusion

Frankenstein Transformer presents a unified, configuration-driven experimentation platform addressing three challenges in modern deep learning research: architectural fragmentation across dense attention, recurrent models, sparse patterns, gated mechanisms, latent compression, memory layers, fast-weight memories, and geometric priors; optimizer landscape complexity spanning classical baselines, variance-reduction methods, memory-efficient variants, schedule-free approaches, curvature-aware algorithms, and geometry-oriented methods; and end-to-end deployment workflows spanning quantization and sentence embedding.

The system's primary contributions are:

1. **Schema-First Design**: A strict YAML-based configuration contract with validation and prefixed hyperparameter routing enabling reproducible experiments across forty-nine sequence mixers and thirty-eight optimizer families.

2. **Comprehensive Architecture Support**: Implementation spanning dense baselines, recurrent and retentive blocks, sparse attention, gated mechanisms, latent compression, memory layers, fast-weight attention, and geometric-field attention, with encoder, decoder, and Vision Transformer model classes.

3. **Unified Optimizer Framework**: Prefixed hyperparameter groups enabling fine-grained control over embeddings, normalization layers, attention weights, and the remaining parameters across classical baselines (SGD+Momentum, AdamW, RAdam), variance reduction (Adan, ADOPT, AdEMAMix, MARS, Cautious), memory-efficient and low-rank methods (Adafactor, GaLore, Lion, APOLLO, APOLLO-Mini, Q-APOLLO), large-batch and schedule simplification (LAMB, Schedule-Free AdamW, Prodigy), curvature-aware (Sophia), second-order (Shampoo, SOAP), geometry-oriented (Muon, Turbo-Muon, AF-Muon), and adaptivity-tunable (Anon) optimizers.

4. **End-to-End Workflows**: Integrated deployment pipeline supporting ternary weight packing, block-affine 8/4-bit safetensors export at both load-time runtimes, and INT8 activation quantization; SBERT-inspired training and inference for semantic similarity, retrieval, and clustering tasks.

5. **Interactive Configuration**: Streamlit-based web interface providing schema-driven form generation, real-time validation, inline documentation, and CLI command synthesis.

This system enables rapid experimental iteration while maintaining reproducibility through strict configuration contracts. By consolidating diverse research contributions into a unified toolkit, it lowers barriers to exploring novel architectures and optimization strategies, particularly for researchers who may lack resources to implement and validate each variant independently.

11.1 Limitations and Future Directions

Several limitations and promising directions for future work emerge from this system's design and implementation:

1. **Architecture Integration**: Recent hybrid architectures (Griffin, Jamba, Mamba-X) demonstrate benefits of combining multiple mechanisms into unified blocks. Future versions should integrate these architectures and explore systematic composition patterns.

2. **Advanced Quantization**: Current safetensors quantization is weight-only and packs two-dimensional linear weights, dequantizing without fused integer kernels; layer-wise, channel-wise and importance-aware policies remain future work.

3. **Automated Hyperparameter Optimization**: The schema supports extensive hyperparameter spaces, but users must manually explore these spaces. Integration with Bayesian optimization, multi-armed bandit strategies, or gradient-based hyperparameter tuning could automate effective configuration discovery.

4. **Production Deployment**: The web interface improves usability but may not be appropriate for all deployment environments. Headless configuration modes, API-based configuration management, or improved CLI ergonomics could serve HPC and production workflows.

5. **Evaluation Benchmarking**: Round 2 (Section 9) provides the first comparative atlas across the mixer list, but it is a 32-token, single-seed, 100-epoch measurement. Future rounds should extend it to longer context, more held-out sentences and multiple seeds before the rankings are read as architectural conclusions.

6. **Training Stability Guarantees**: Current implementation includes NaN/Inf guards and gradient clipping, but formal analysis of stability conditions for different mixer-optimizer combinations, particularly with looped blocks and aggressive quantization, remains open.

7. **Multimodal and Task-Specific Extensions**: The current design covers text and single-image tasks through the Vision Transformer class. Extensions to vision-language models, multimodal architectures, and task-specific fine-tuning workflows (e.g., instruction tuning, RLHF) would broaden applicability.

The research trajectory of sequence modeling continues toward hybrid approaches that combine the strengths of multiple paradigms—compression from recurrence, selectivity from attention, gating for memory management, and sparsity for efficiency. A unified experimentation platform like Frankenstein Transformer becomes increasingly valuable as this convergence accelerates, enabling researchers to explore this expanding design space with reproducible, well-engineered infrastructure.

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1 Conceptual Introduction—Transformers and Attention for Beginners

This appendix is the on-ramp to the paper: it explains what a transformer is and how attention works, for readers with no deep-learning background. Every mechanism is introduced through one real-world analogy, and — this is the point of the appendix — the *same* analogy returns every time the mechanism returns, so a single mental image carries you from the first intuition

to the mathematics. Table 2 lists the cast; the running example is always the same sentence, "The cat eats fish".

Mechanism	Analogy reused throughout	Introduced in
Attention	listening to one voice in a noisy room	§1.2
Queries, keys, values	visitor, catalog card, and book in a library	§1.4
Attention heads	a team of analysts, one question each	§1.5
Stacked layers	editing an essay in successive passes	§1.6

Table 2: The cast of analogies of this tutorial. Each row is introduced once and then reused verbatim in every later section, including the takeaways.

1.1 What Is a Transformer? Words in Context

A transformer is a neural architecture that processes text (or any sequence) by interpreting the meaning of each word *in the light of all the other words around it*. The problem it solves is easiest to see with an ambiguous word. Read this:

"The bank was full of people waiting. Mary went to the bank."

What does "bank" mean in each case? Taken alone, the word has three unrelated meanings: a financial institution, a bench to sit on, the ground beside a river. Inside the first sentence, the neighbors ("full of people waiting") vote for the branch office. Inside the second, "Mary went to the…" leaves two readings alive — she could be heading to the counter or to the river. Figure 13 contrasts the two situations.

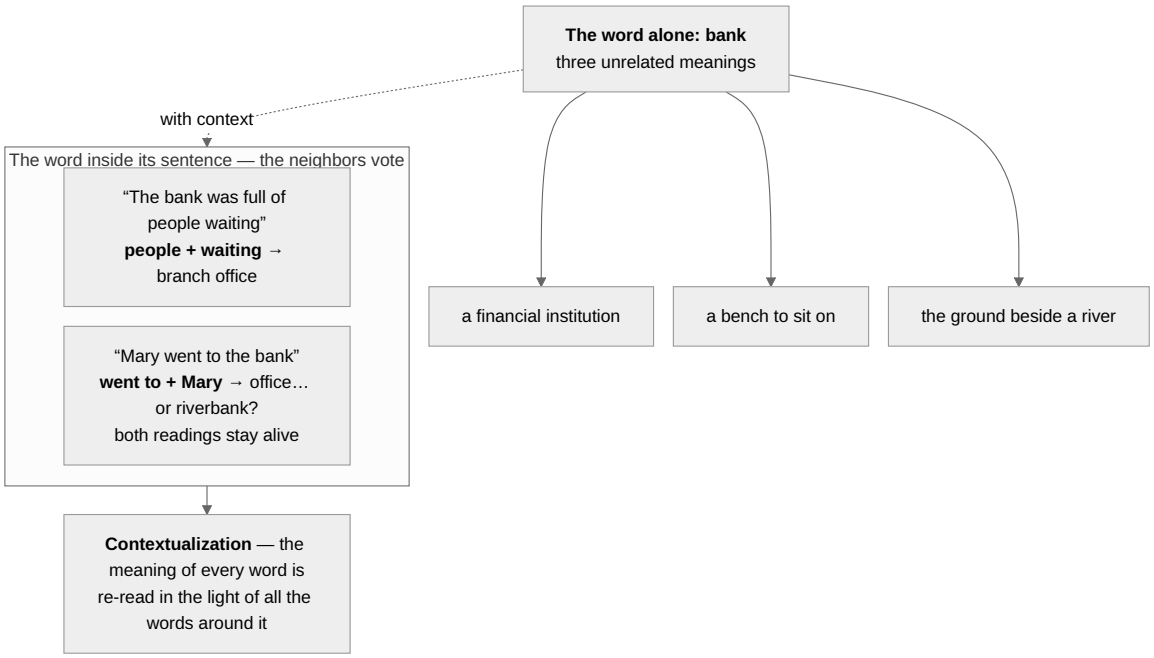


Figure 13: Contextualization in one picture: the word "bank" alone has three unrelated meanings (top); inside its sentence, the neighboring words vote and select the reading (middle). A transformer does this for every word simultaneously.

This ability to re-read every word in the light of the whole sentence is called *contextualization*, and it is the heart of every modern transformer. The question this appendix answers is *how* the model decides which neighbors to listen to. The answer is attention.

1.2 The Attention Mechanism: One Voice in a Noisy Room

Attention is the "magic trick" behind contextualization. Think of a conversation in a noisy room:

1. **Lots of noise**: many people talk at once, but one of them matters to you.
2. **You ignore the rest**: your brain automatically focuses on that one voice and filters the rest.
3. **You understand better**: by concentrating, you catch every word clearly.

The neural mechanism works the same way, with one twist: *every word in the sentence is such a listener*. Each word "asks" how important every other word is for understanding it, then focuses its attention on the most relevant ones. The room is the sentence, the voices are the words, and the focusing ability is attention. Keep this image: every formula below is just the arithmetic version of listening selectively.

1.3 A Worked Example: The Two Lives of "eats"

Let us watch the mechanism work on one word in two different sentences. The numbers that follow are illustrative, but they are exactly the kind of table a trained transformer computes — Figure 14 shows both cases.

1.3.1 Context 1: "The cat eats fish"

Processing "eats", the word-as-listener asks about every other voice in the room:

- How relevant is "The"? Little — it is just an article. **Low attention: 5%**.
- How relevant is "cat"? Very — it is the subject, who eats. **High attention: 55%**.
- How relevant is "eats"? A little — itself. **Low attention: 5%**.
- How relevant is "fish"? Very — it is what gets eaten. **High attention: 35%**.

The new representation of "eats" is dominated by "cat" and "fish": an action, with its subject and its object.

1.3.2 Context 2: "The restaurant eats into profit margins"

Same word, different room. Now the listeners that matter are different:

- "restaurant" (the subject): **30%**.
- "profit" (what is affected): **25%**.
- "margins" (what gets consumed): **35%**.
- "The", "into", and "eats" itself: the remaining **10%**.

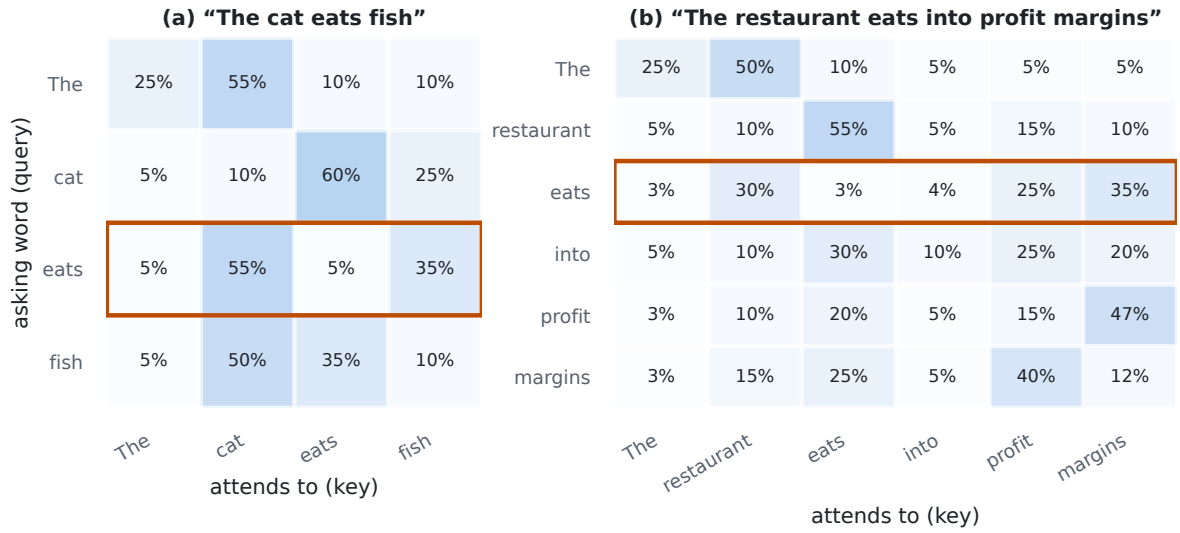


Figure 14: Two attention matrices (rows: asking word; columns: word attended to; the highlighted row is “eats”). In (a) the word “eats” listens mostly to “cat” (55%) and “fish” (35%); in (b) the same word listens to “restaurant”, “profit” and “margins”. Same word, different room, different focus — contextualization in action. Every row sums to 100% because attention is a limited budget.

The word “eats” ends up with a completely different representation in each sentence because it *attends* to different words. That is the whole trick: one word, many possible meanings, and attention picking the right neighbors to define it.

1.4 How Attention Works Mathematically: A Library Called Attention

Behind the listening there is arithmetic. The standard analogy — and the one this paper sticks to — is a library, shown in Figure 15.

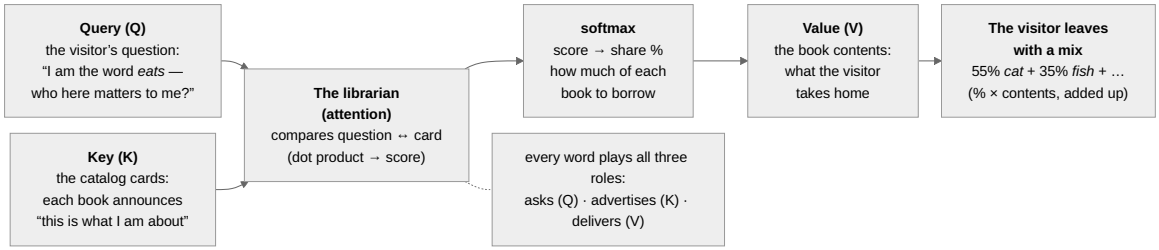


Figure 15: Queries, keys and values as a library visit. The visitor carries a question (Q), every book advertises itself with a catalog card (K), the librarian compares question and cards (the attention), and the visitor leaves with a weighted mix of the book contents (V).

1.4.1 Step 1: Queries, Keys, and Values

Each word is transformed into three versions of itself:

- **Query (Q)**: the visitor’s question — “what information do I need?”
- **Key (K)**: the catalog card — “what information do I offer?”
- **Value (V)**: the book’s content — “here is my actual information.”

Crucially, every word plays all three roles at once: it asks (Q), it advertises (K), it delivers (V). One visit to the library, with every book simultaneously a visitor and a shelf.

1.4.2 Step 2: Compatibility

The librarian compares the question with every card. Questions similar to a card get a high compatibility score; the score is computed by multiplying the query by the key (a *dot product*). For our "eats" query: s(eats → cat) = 3.40, s(eats → fish) = 2.95, s(eats → The) = 1.00. Figure 16 places the four steps in one pipeline.

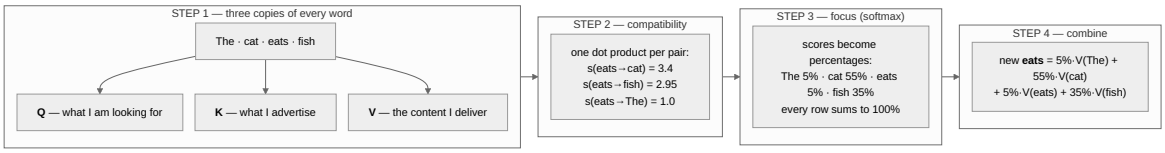


Figure 16: The full attention step for the query "eats" in "The cat eats fish": three copies per word (Step 1), one dot product per query–key pair (Step 2), softmax turning scores into percentages (Step 3), and the weighted combination of values (Step 4). The numbers are the same as in Figures 14 and 17.

1.4.3 Step 3: Focus

Raw scores are just numbers; attention is a *budget*. A function called `softmax` converts the scores into percentages that always sum to 100%: the "eats" query turns the scores above into **5%** / **55%** / **5%** / **35%** over "The / cat / eats / fish". Figure 17 shows the conversion. High compatibility means "focus attention here"; low compatibility means "ignore this" — back in the noisy room, the voice you decided to filter out.

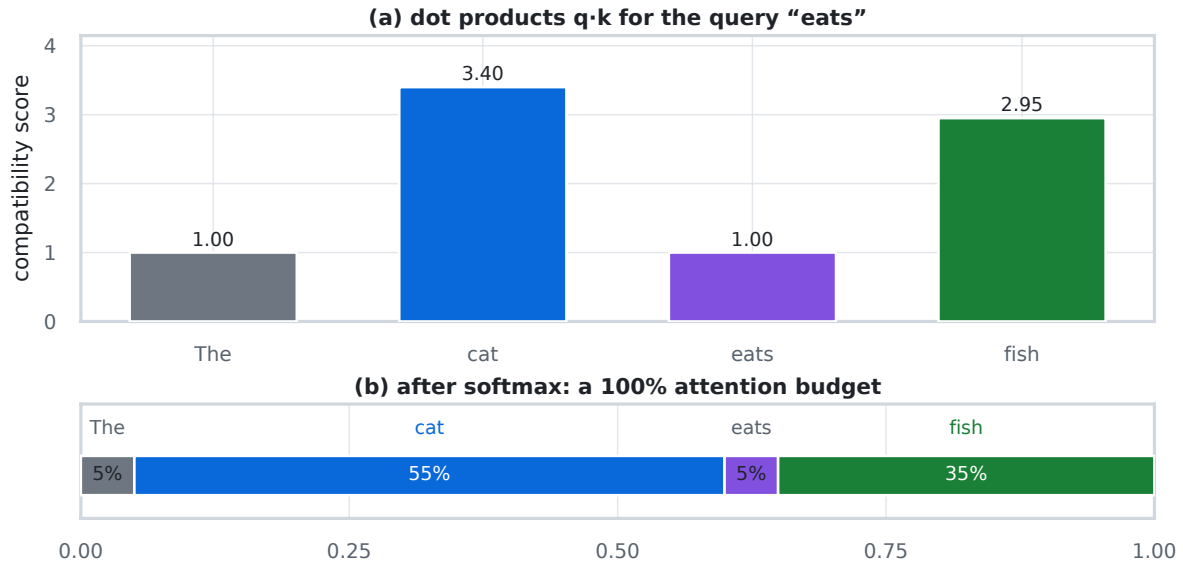


Figure 17: Step 2 → Step 3 in numbers. Top: the four raw compatibility scores (dot products) for the query "eats". Bottom: after `softmax`, the same four numbers become an attention budget that sums to exactly 100%.

1.4.4 Step 4: Combine

Finally the visitor leaves the library with a weighted mix of contents. If "eats" places 55% of its budget on "cat" and 35% on "fish", then 55% of cat's value and 35% of fish's value are blended into the new representation of "eats":

$$\text{new "eats"} \;=\; 5\% \cdot v_{\text{The}} + 55\% \cdot v_{\text{cat}} + 5\% \cdot v_{\text{eats}} + 35\% \cdot v_{\text{fish}}. \tag{1}$$

That blended vector is what the rest of the network sees: a representation of "eats" that already contains who eats and what gets eaten.

1.5 Multiple Attention Heads: A Team of Analysts

Transformers never run one attention but *many heads in parallel* — as if the sentence were analyzed by a team of specialists, each asking a different question about it (Figure 18):

- **Analyst 1 (grammar)**: "who does it?" — links *cat* → *eats*.
- **Analyst 2 (object)**: "what receives it?" — links *eats* → *fish*.
- **Analyst 3 (time)**: "when / how often?" — tracks verb tense and time words.
- **Analyst 4 (style)**: "which articles and modifiers?" — tracks "the", "a", adjectives.

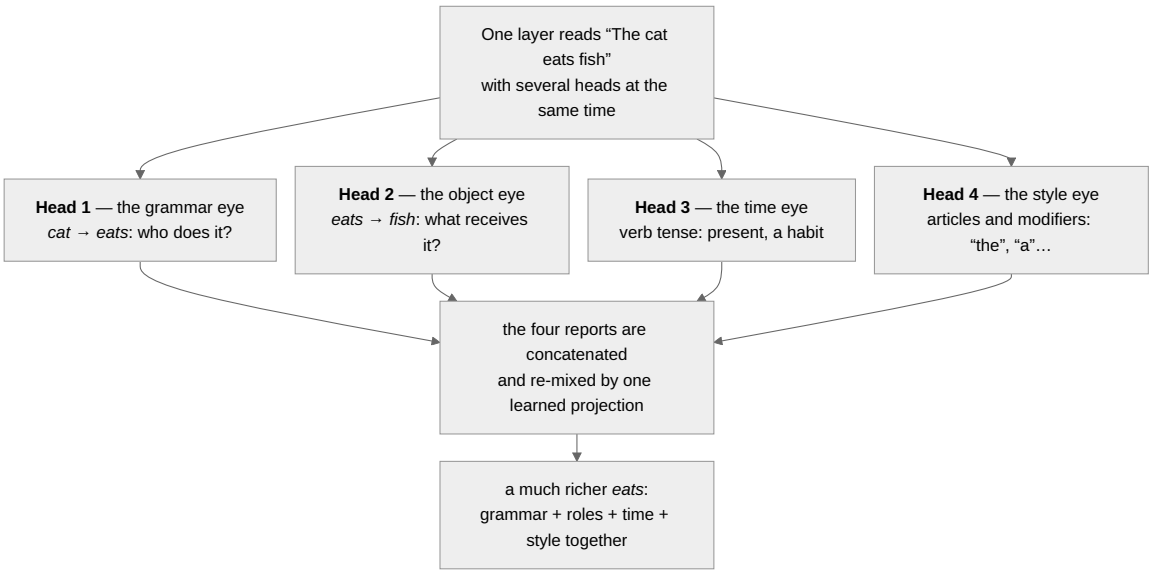


Figure 18: Several heads read the same sentence simultaneously, like a team of analysts with one question each. Their reports are concatenated and re-mixed into a single, much richer representation.

Each head learns its own focus pattern from data, and their reports are concatenated and re-mixed by one learned projection. Together they capture far more than any single head could — grammar, roles, time and style at once.

1.6 Stacking Layers: Editing the Essay in Passes

One attention step still sees shallow patterns, so transformers stack the same block many times (12 to 48 in practical models). The analogy: editing an essay in successive passes (Figure 19).

1. **First pass** — fix spelling and basic grammar: recognize words, forms, gender.
2. **Second pass** — improve clarity of each sentence: who does what to whom.
3. **Third pass** — ensure consistency and narrative flow: idioms, relations, abstract meaning.

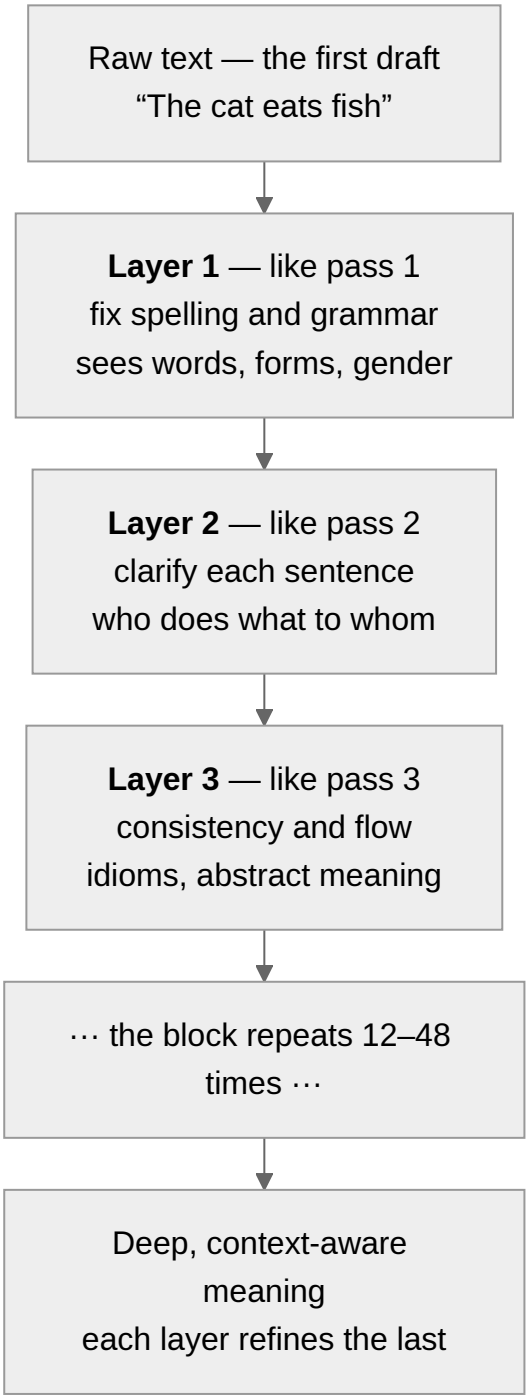


Figure 19: Layers as editing passes: the raw text enters at the top and each layer refines the previous layer's reading — from words and forms, through sentence structure, to abstract meaning. The block repeats 12–48 times in practice.

Each layer re-reads what the previous layer produced and refines it: early layers capture basic

features, later layers understand complex relations and abstract concepts. No layer is told what to look for — the passes organize themselves during training.

1.7 Why Does It Matter?

The attention mechanism was revolutionary because:

1. **Parallelism**: any word can be compared with any other word *at once* (no sequential processing, unlike earlier recurrent systems) — fast on modern hardware.
2. **Flexibility**: the patterns are learned from data; nobody programs the rules.
3. **Scalability**: the same mechanism works from a sentence to millions of words of context.
4. **Generality**: translation, summarization, question answering, generation, computer vision, audio — one mechanism, many tasks.

1.8 Practical Challenges: The Price of Listening to Everyone

Dense attention buys its power by comparing *every* word with *every* other word, and that price grows fast.

1.8.1 Computational Complexity

A 1,000-word document forces $1{,}000 \times 1{,}000 = 1{,}000{,}000$ word-to-word comparisons, and the cost squares as the document grows. Figure 20 shows the growth against the friendlier alternatives discussed below.

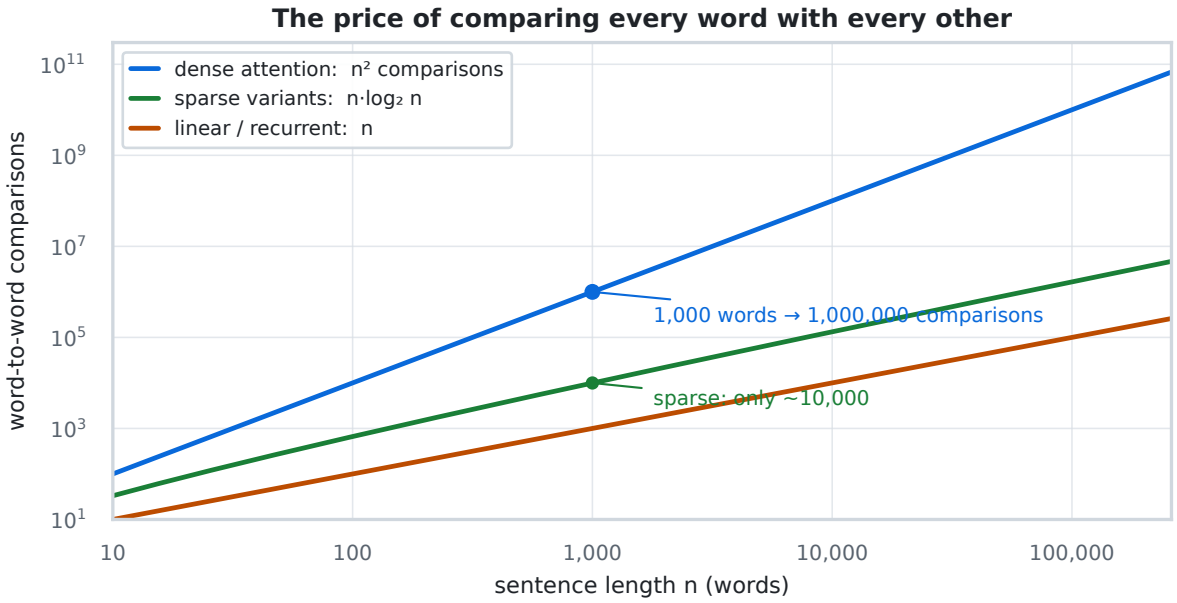


Figure 20: Word-to-word comparisons versus sentence length. Dense attention compares every word with every other (n^2): at $n = 1{,}000$ that is already one million comparisons. Sparse variants ($n \log_2 n$) and linear-time / recurrent models (n) grow far more gently.

1.8.2 Memory Requirements

During inference (using a trained model), the keys and values of every past word must be stored — the library keeps every book on the table just in case a later query needs it. For long contexts this cache dominates GPU memory.

1.8.3 Device Efficiency

Transformers were designed for powerful TPUs and GPUs; running them on phones or embedded devices is challenging.

1.9 Modern Solutions: Smarter Ways to Listen

The rest of this paper is a catalog of ways to keep the benefits of attention while paying less. Each keeps our analogies:

- **Sparse attention**: stop browsing *every* card in the library — visit only nearby shelves and a few strategic ones. Complexity drops from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$ or $\mathcal{O}(n)$.
- **Recurrent models** (e.g. Mamba): take notes as you read, instead of re-listening to the whole room for every new word — an old idea (recurrence) rebuilt with modern efficiency.
- **Gated attention**: a gatekeeper decides which of your notes are worth keeping and which to fade out, so less needs to be stored.
- **Quantization**: write each number with fewer digits (small integers instead of wide decimals), shrinking memory with minimal loss of precision.

These innovations are exactly what this toolkit (Frankenstein) lets you experiment with, and the annexes that follow are guided tours of each family.

1.10 Key Takeaways

- A **transformer** is a network that contextualizes: it re-reads every word in the light of all its neighbors (the ambiguous "bank").
- **Attention** is selective listening: each word focuses its limited 100% budget on the neighbors that define it (the noisy room).
- The arithmetic is a library visit: **query** asks, **key** advertises, **value** delivers; dot products score compatibility, `softmax` turns scores into a 100% budget, and the values are mixed by that budget.
- **Multiple heads** are a team of analysts with one question each; **stacked layers** are successive editing passes, each refining the last.
- The catch: dense attention compares everyone with everyone — $\mathcal{O}(n^2)$ — which is why sparse, recurrent, gated and quantized variants exist.
- Every later annex of this paper is a deeper tour of one of those variants; the analogies of Table 2 will keep coming back.

2 Dense, Recurrent, and Memory-Augmented Transformers

This annex synthesizes the dense, recurrent, and memory-augmented transformer families. The field has evolved toward several primary paradigms: dense global attention with variants, key-value head compression via grouped-query attention, recurrent state compression with decay, selective state-space models, continuous-depth numerical integration, token-indexed value memory, and memory-augmented architectures. Understanding this taxonomy illuminates fundamental tradeoffs between expressiveness, computational cost, memory footprint, and deployment simplicity.

2.1 Dense Attention Baselines: Standard, Sigmoid, and Continuum

2.1.1 Standard Softmax Attention

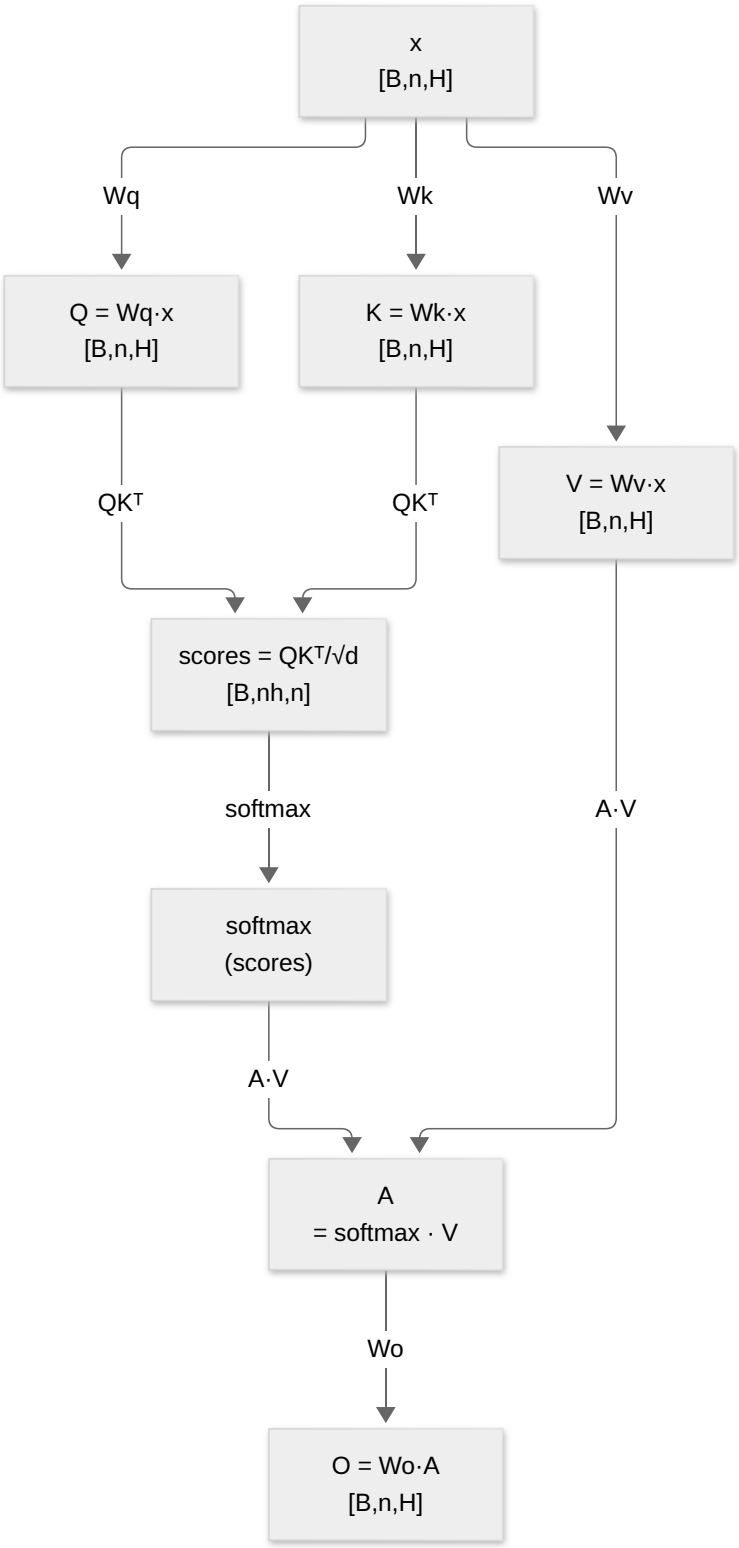


Figure 21: Dataflow of standard softmax attention: parallel Q/K/V projections, scaled dot-product scores, row-wise softmax, and output projection.

Standard multi-head self-attention [159] computes scaled dot-product similarity between token embeddings:

1: **Input:** Query matrix $\mathbf{Q} \in \mathbb{R}^{n \times d}$, Key matrix $\mathbf{K} \in \mathbb{R}^{n \times d}$, Value matrix $\mathbf{V} \in \mathbb{R}^{n \times d}$
2: scores $\leftarrow \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} \in \mathbb{R}^{n \times n}$ ▷ compute pairwise similarities
3: attention_weights $\leftarrow \texttt{softmax}(\text{scores}, \text{axis} = 1) \in \mathbb{R}^{n \times n}$ ▷ normalize across keys
4: **Output:** $\mathbf{Y} = \text{attention_weights} \cdot \mathbf{V} \in \mathbb{R}^{n \times d}$ ▷ weighted value combination

For autoregressive generation, KV caching stores past keys and values to avoid $\mathcal{O}(n^2)$ recomputation. However, this linearly growing cache occupies $\sim n \cdot d_{\text{hidden}}$ bytes, which can consume hundreds of gigabytes for billion-parameter models. Standard attention achieves **perfect expressiveness** within the context window—any token can attend to any other token with learned weights—but pays the price of dense computation.

- **Training complexity**: $\mathcal{O}(n^2 \cdot d)$ time, $\mathcal{O}(n^2)$ space (attention matrix materialization).
- **Inference complexity**: $\mathcal{O}(n)$ time per token, $\mathcal{O}(n)$ space (KV cache).
- **Strengths**: Unparalleled expressiveness; perfect history recall; highly parallelizable.
- **Weaknesses**: Quadratic bottleneck prohibits very long contexts; KV cache dominates memory during generation.

2.1.2 Sigmoid Attention

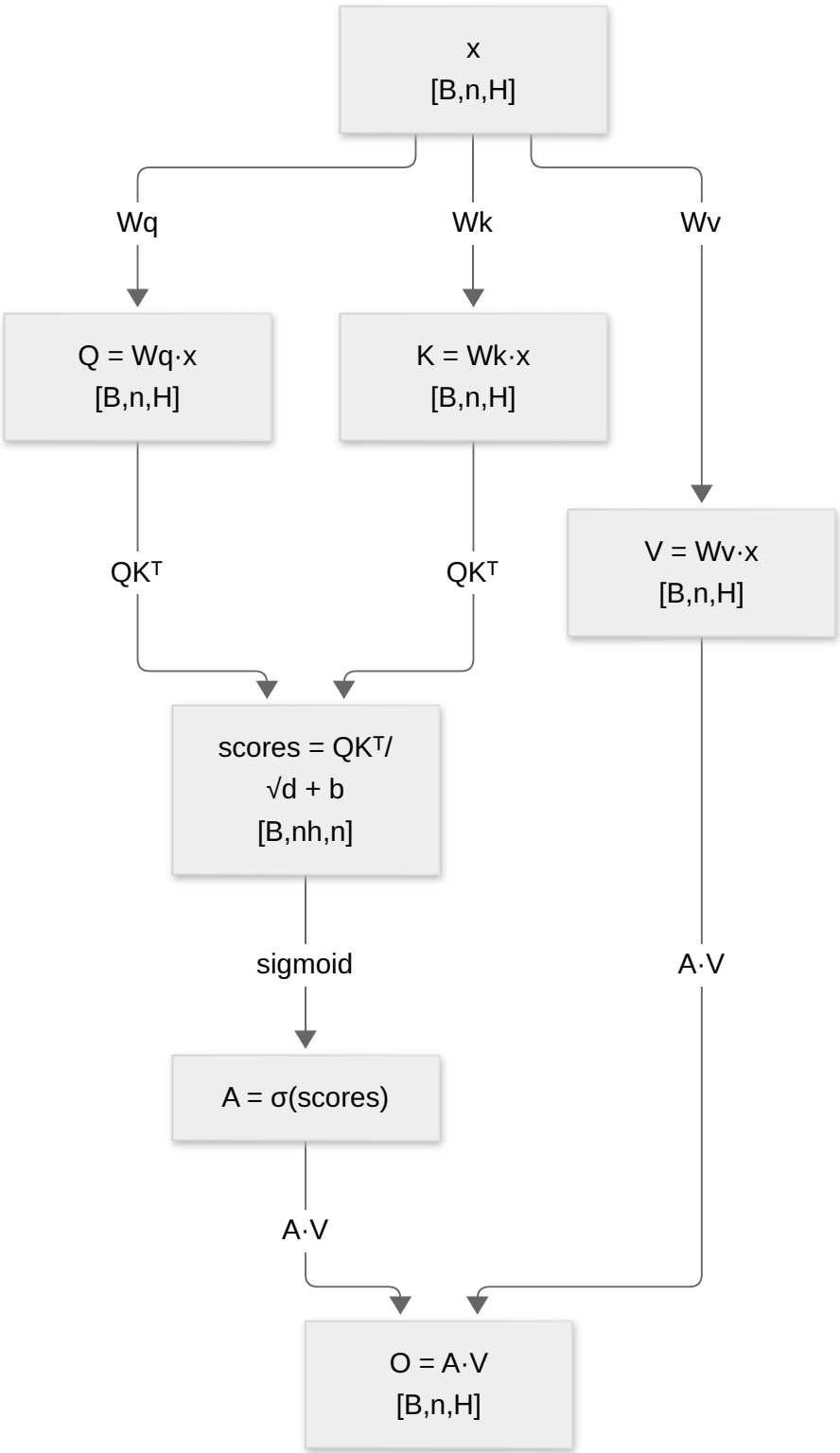


Figure 22: Dataflow of sigmoid attention: an element-wise sigmoid replaces the row-wise softmax, so attention weights need not sum to one.

Sigmoid attention replaces row-wise softmax with element-wise sigmoid activation [134]:

1: **Input:** $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ as above, learnable bias $\mathbf{b} \in \mathbb{R}^{n \times n}$
2: logits $\leftarrow \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{b}$
3: attention_weights $\leftarrow \sigma(\text{logits})$ ▷ element-wise sigmoid, not softmax
4: **Output:** $\mathbf{Y} = \text{attention_weights} \odot \mathbf{V}$

Unlike softmax, sigmoid does not enforce a probability distribution (weights need not sum to 1), enabling stronger token independence. Theoretical analysis via mixture-of-experts shows sigmoid achieves superior sample complexity: $\mathcal{O}(n^{-0.51})$ convergence for ReLU experts versus softmax's $\mathcal{O}(n^{-0.24})$. However, empirical training revealed gradient instabilities at scale. The remedy is **hybrid-norm**—adding normalization after the attention output—which stabilizes gradients without sacrificing the theoretical benefits of element-wise gating.

- **Training complexity**: $\mathcal{O}(n^2 \cdot d)$ (identical to standard), but element-wise operations enable 17% inference speedup via FlashSigmoid.
- **Inference complexity**: $\mathcal{O}(n)$ per token with KV cache (asymptotically same, but lower constant factors).
- **Strengths**: Overcomes zero-sum competition; avoids row-wise synchronization; hardware-friendly implementation.
- **Weaknesses**: Requires careful stabilization (hybrid-norm); training instability at large scales without auxiliary loss.

2.1.3 Grouped-Query Attention (GQA)

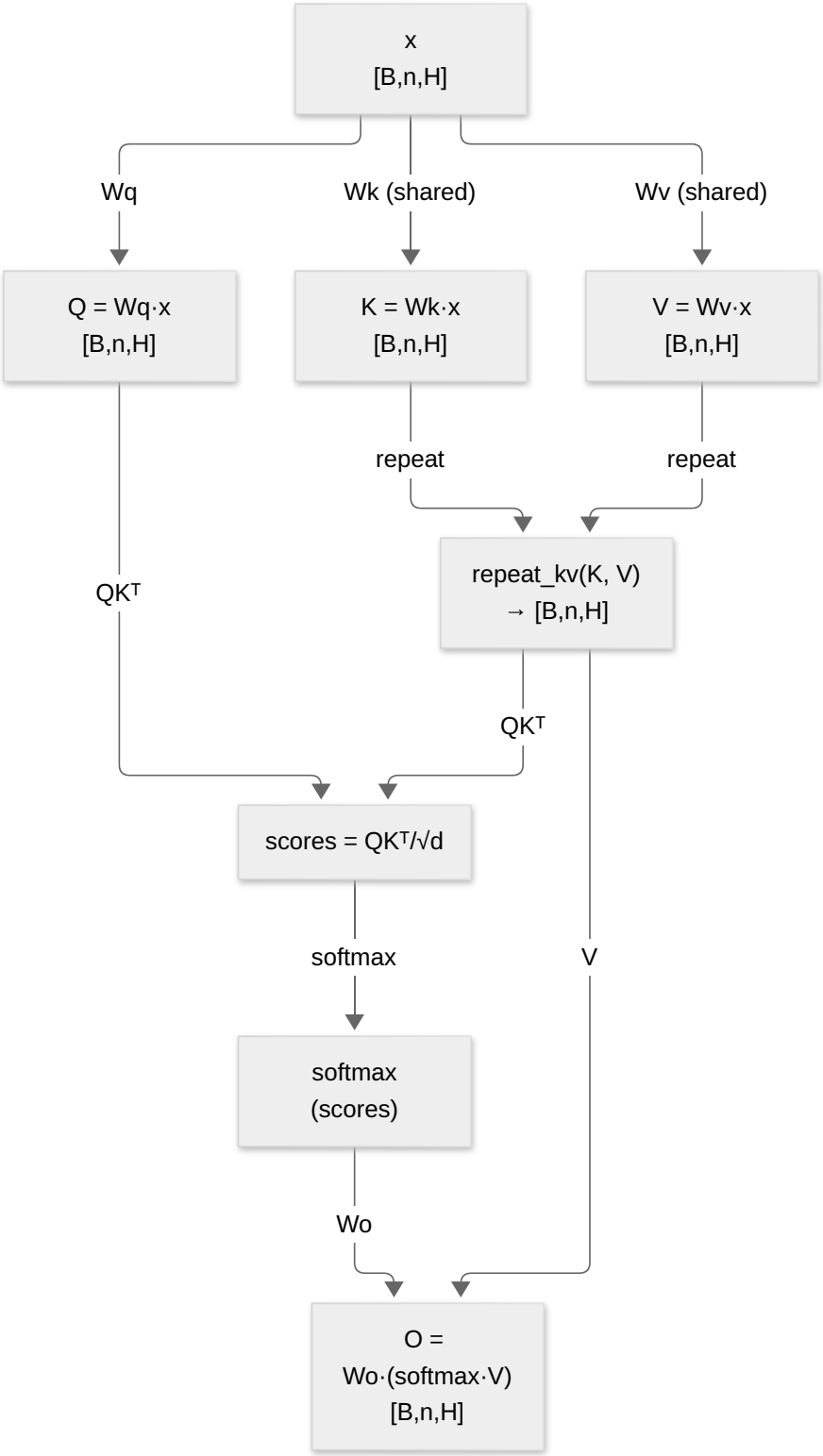


Figure 23: Dataflow of grouped-query attention: keys and values are projected to fewer heads, expanded by repeat-interleave, and shared across each group of query heads.

Grouped-query attention [2] bridges multi-head attention (MHA) and multi-query attention (MQA) by partitioning query heads into groups, each sharing a single key-value head. If h is the total number of attention heads, h_k is the number of key-value heads, and $G = h/h_k$ is the group size, then every group of G query heads shares one KV head:

Input: $\mathbf{Q} \in \mathbb{R}^{n \times h \cdot d_h}$, $\mathbf{K} \in \mathbb{R}^{n \times h_k \cdot d_h}$, $\mathbf{V} \in \mathbb{R}^{n \times h_k \cdot d_h}$
$\mathbf{K}_{\text{full}} \leftarrow \texttt{repeat_interleave}(\mathbf{K}, G)$ ▷ expand from h_k to h heads
$\mathbf{V}_{\text{full}} \leftarrow \texttt{repeat_interleave}(\mathbf{V}, G)$
$\text{scores} \leftarrow \frac{\mathbf{Q}\mathbf{K}_{\text{full}}^\top}{\sqrt{d_h}} \in \mathbb{R}^{n \times n}$ ▷ standard dot product after expansion
$\text{attention_weights} \leftarrow \texttt{softmax}(\text{scores}, \text{axis} = 1)$
Output: $\mathbf{Y} = \text{attention_weights} \cdot \mathbf{V}_{\text{full}} \in \mathbb{R}^{n \times h \cdot d_h}$

GQA interpolates smoothly between MHA ($h_k = h$, $G = 1$) and MQA ($h_k = 1$, $G = h$). The key advantage is a tunable tradeoff between KV-cache footprint and representational capacity. Each additional KV head consumes $2 \cdot d_h \cdot n$ extra bytes in the cache, so reducing h_k yields proportional memory savings.

- **Training complexity**: $\mathcal{O}(n^2 \cdot h \cdot d_h)$ (identical to MHA after head expansion).
- **Inference complexity**: $\mathcal{O}(n)$ per token, $\mathcal{O}(h_k \cdot n \cdot d_h)$ KV-cache space.
- **Strengths**: Flexible head-count tradeoff; proven at scale (Llama 2 & 3 use $h_k = 8$ KV heads with $h = 32$ query heads); reduces KV-cache by $G\times$; supports fast inference with FlashAttention.
- **Weaknesses**: Fewer KV heads can reduce representational capacity in small models; requiring h_k to divide h constrains architectural choices.

2.1.4 Continuum Attention

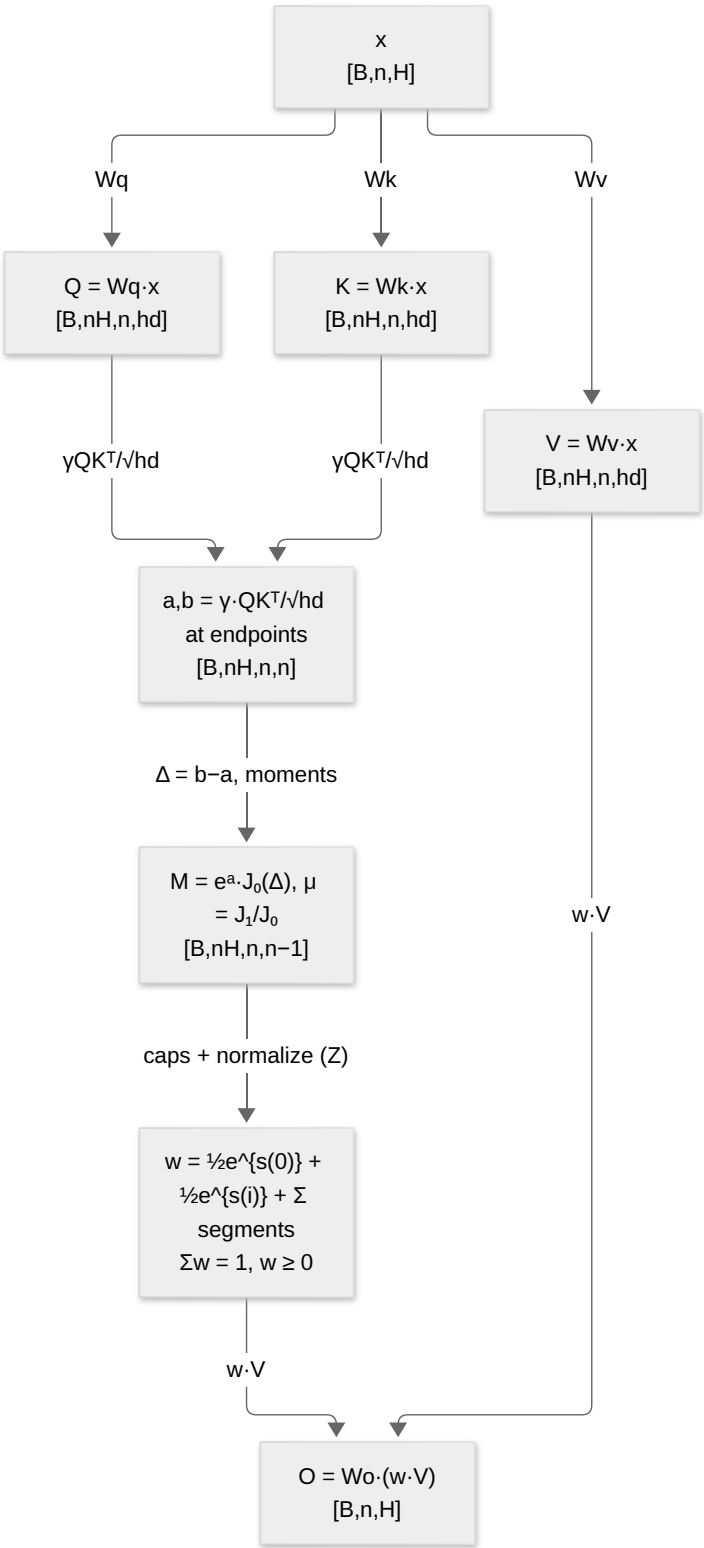


Figure 24: Dataflow of Continuum Attention: endpoint scores from the piecewise-linear key trajectory, exact segment integration through exponential moments, half-token boundary caps, and normalized effective weights on the original values.

Continuum Attention [79] replaces discrete softmax retrieval with exact integration over a continuous, piecewise-linear memory trajectory. Adjacent keys and values are interpolated so that $\mathbf{k}(t)$ and $\mathbf{v}(t)$ are defined at every position of the visible domain:

$$\mathbf{k}_j(u) = (1-u)\,\mathbf{k}_{j-1} + u\,\mathbf{k}_j, \qquad \mathbf{v}_j(u) = (1-u)\,\mathbf{v}_{j-1} + u\,\mathbf{v}_j, \tag{2}$$

with $u = t-(j-1) \in [0,1]$ on the interval $[j-1, j]$; constant half-token extensions cover $[-1/2, 0]$ and $[i, i+1/2]$. Each query induces an exponential relevance density over the visible prefix,

$$s_i(t) = \frac{\gamma}{\sqrt{d_h}}\,\mathbf{q}_i^\top \mathbf{k}(t), \qquad p_i(t) = \frac{\exp(s_i(t))}{Z_i}, \qquad \mathbf{o}_i = \int_{-1/2}^{i+1/2} p_i(t)\,\mathbf{v}(t)\,dt, \tag{3}$$

where $\gamma > 0$ is the score gain ($\gamma = 1/\tau$). Since the key trajectory is linear within a segment, the score is affine, $s_{ij}(u) = a_{ij} + u\,\Delta_{ij}$ with $a_{ij} = s_i(j-1)$, $b_{ij} = s_i(j)$ and $\Delta_{ij} = b_{ij} - a_{ij}$, so the integrals have closed forms in the exponential moments $J_n(\Delta) = \int_0^1 u^n e^{\Delta u}\,du$:

$$J_0(\Delta) = \frac{e^\Delta - 1}{\Delta}, \qquad J_1(\Delta) = \frac{(\Delta-1)e^\Delta + 1}{\Delta^2}, \tag{4}$$

$$M_{ij} = e^{a_{ij}} J_0(\Delta_{ij}), \qquad \mathbf{U}_{ij} = e^{a_{ij}}\big[(J_0 - J_1)\,\mathbf{v}_{j-1} + J_1\,\mathbf{v}_j\big], \tag{5}$$

$$Z_i = \tfrac{1}{2}e^{s_i(0)} + \tfrac{1}{2}e^{s_i(i)} + \sum_{j=1}^{i} M_{ij}, \qquad \mathbf{o}_i = \frac{\mathbf{U}_i^{\text{cap}} + \sum_{j=1}^{i} \mathbf{U}_{ij}}{Z_i}, \tag{6}$$

with $\mu(\Delta) = J_1/J_0$ the normalized retrieval position inside a segment. Because the contributions are nonnegative, CA is equivalently a normalized nonnegative weighted sum of the original values, $\mathbf{o}_i = \sum_{j \le i} w_{ij}\mathbf{v}_j$ with $w_{ij} \ge 0$ and $\sum_j w_{ij} = 1$; equal visible scores reduce exactly to uniform causal attention and $o_0 = v_0$. The learned-gain variant adds one scalar per head ($\gamma = \exp(\rho)$, initialized at $\gamma = 1$ in the 101M model; the fixed CA-P variant uses $\gamma = 1/0.30$ and adds no parameters). The encoder (bidirectional) form integrates over the full trajectory $[-1/2, n-1/2]$ – the natural symmetric extension of the causal-only paper.

Input: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{n \times d_h}$, gain $\gamma > 0$
$\mathbf{S} \leftarrow \gamma\,\mathbf{Q}\mathbf{K}^\top/\sqrt{d_h}$ ▷ endpoint scores, Eq. (4)
mask $\mathbf{S}$ to the visible prefix; subtract the row maximum m (stability)
$\Delta_{ij} \leftarrow \mathbf{S}_j - \mathbf{S}_{j-1}$; $J_0, J_1 \leftarrow$ Eqs. (11)–(12) (series near $\Delta = 0$)
$M_{ij} \leftarrow e^{a_{ij}-m} J_0$, $w_{ij}^{(L)} \leftarrow M_{ij}(1-\mu)$, $w_{ij}^{(R)} \leftarrow M_{ij}\mu$
add the half-token caps $\frac{1}{2}e^{s_i(0)-m}$ and $\frac{1}{2}e^{s_i(i)-m}$; normalize by Z_i
Output: $\mathbf{o}_i = \sum_{j \le i} w_{ij}\mathbf{v}_j$ ▷ Eq. (24)

- **Training complexity**: $\mathcal{O}(n^2 d)$ – identical to dense softmax; the closed-form moments replace the softmax (no quadrature).
- **Inference complexity**: $\mathcal{O}(n)$ per token with a KV cache; the effective weights are recomputed row-locally over the cached prefix each step (exact).
- **Strengths**: Exact continuous retrieval with normalized nonnegative effective weights; zero parameters beyond the projections in the fixed-gain form; strong multilingual and zero-shot results (FLORES macro 1.8915 BPB, a 5.8% reduction versus the strongest baseline and the best score in every language; HellaSwag and ARC-Challenge best among the baselines); 42% fewer training bytes to a matched 1.10 BPB threshold than a matched Transformer.
- **Weaknesses**: The quadratic attention arithmetic persists; source-code performance is the weakest of the five architectures (Stack-smol 3.3451 BPB); reduced robustness to byte deletion and "antspeak" perturbations.

2.2 Memory Attention

The memory family groups Titans (the attention window implemented here), Engram (Appendix 13) and Memory Attention; this annex covers the attention-side members.

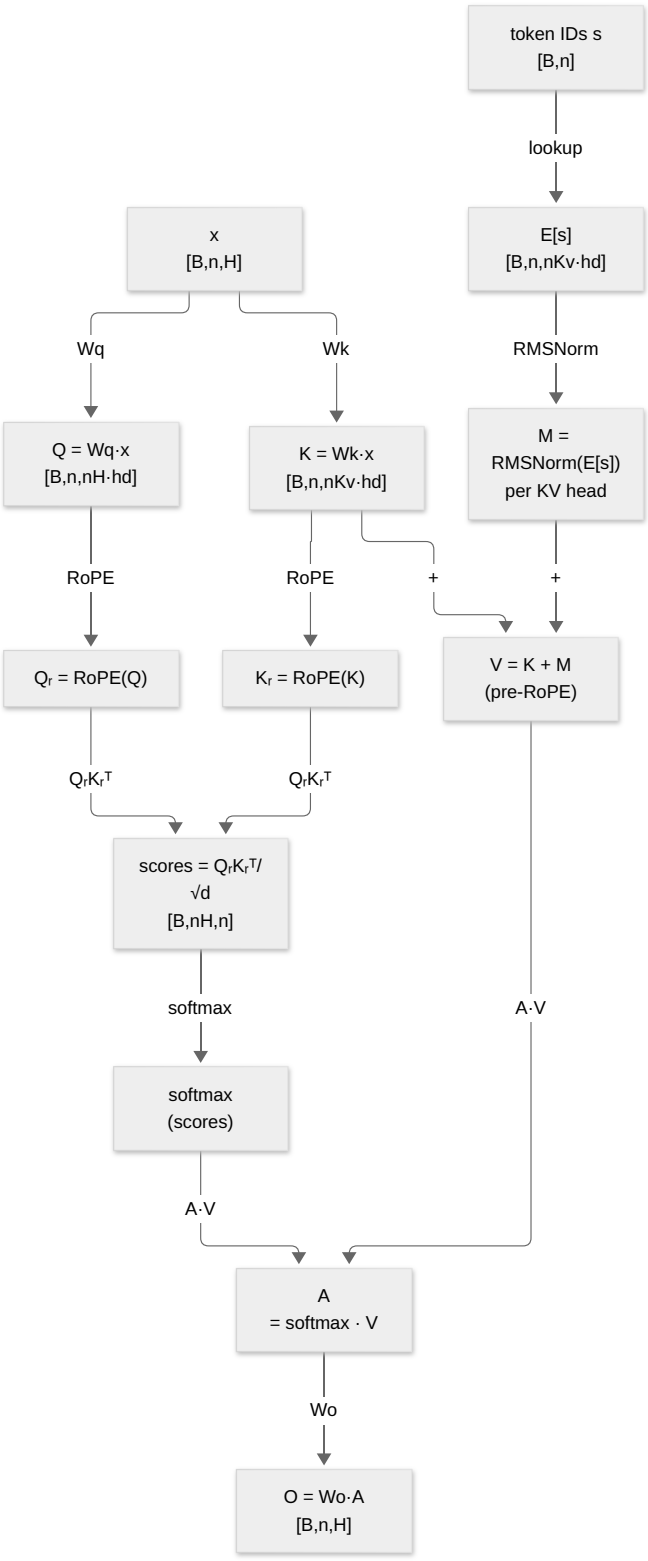


Figure 25: Dataflow of Memory Attention: the value projection is removed; values are the unrotated keys plus a per-KV-head RMSNorm'd row of a layer-specific token-ID memory table. RoPE applies only to the scoring pair.

Memory Attention [76] removes the dedicated value projection and constructs values from contextual keys plus layer-specific token memory. With token IDs s, vocabulary size N and $d_v = H_{KV} d_h$:

Input: hidden states $X \in \mathbb{R}^{T\times d}$, token IDs s, table $E \in \mathbb{R}^{N\times d_v}$
$M \leftarrow \mathrm{Norm}(E[s]) \in \mathbb{R}^{T\times d_v}$ ▷ RMSNorm per token vector, per key/value head
$Q \leftarrow XW_Q, \quad K \leftarrow XW_K, \quad V \leftarrow K + M$ ▷ no W_V; values from unrotated keys
$Q_R \leftarrow \mathrm{RoPE}(Q), \quad K_R \leftarrow \mathrm{RoPE}(K)$ ▷ rotation for scoring only
Output: $Y = \mathrm{softmax}\Big(\frac{Q_R K_R^\top}{\sqrt{d_k}} + C\Big) V$

The output decomposes as $O_{\mathrm{MA}} = A(K+M) = AK + AM$: a contextual retrieval term over keys and a memory retrieval term aggregated under the same attention distribution. Across L replaced layers, MA swaps the standard $L\,d\,d_v$ value-projection parameters for $L(Nd_v + p_{\mathrm{norm}})$ memory and normalization parameters — added capacity accessed by lookup instead of dense computation. At inference the per-row normalization is folded into the table ($E[i] \leftarrow \mathrm{Norm}(E[i])$), reducing online value construction to lookup and addition while retaining a conventional KV cache. The paper reports improved LAMBADA/WikiText perplexity and average downstream accuracy across MHA/GQA/MQA backbones at matched token budgets, with 1.42×/1.16× token efficiency; the comparison includes the additional memory parameters, so it does not isolate structure from capacity. MA-Offload (CPU-resident tables with prefetching) and MA-Recall (reconstructing historical values as $V_{1:T} = K_{1:T} + E[s_{1:T}]$ for a ∼50% smaller value cache) are analyzed but not implemented here.

- **Training complexity**: $\mathcal{O}(n^2 d)$ attention with $\mathcal{O}(nd_v)$ lookup+norm+add value construction (replacing the $\mathcal{O}(n\,d\,d_v)$ projection).
- **Inference complexity**: $\mathcal{O}(n)$ per token with a conventional KV cache; Nd_v extra parameters per layer.
- **Strengths**: Extra capacity without dense-compute growth; layer-specific token representations; exact standard-attention aggregation; KV-cache-compatible by construction.
- **Weaknesses**: Parameter count grows with the vocabulary ($N \gg d$); needs token IDs (text-only); the reported gains are not parameter-matched.

2.3 Recurrent and Retentive Architectures

2.3.1 Retentive Networks (RetNet)

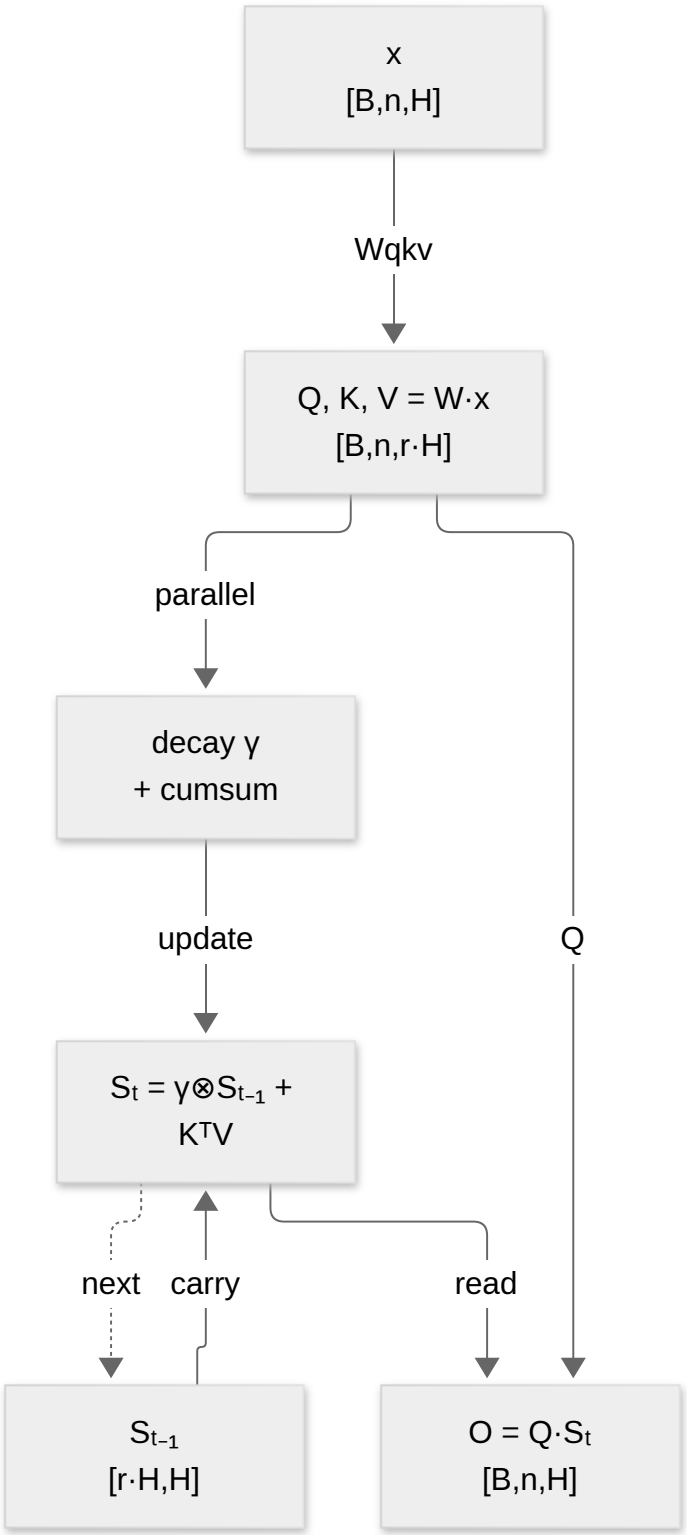


Figure 26: Dataflow of RetNet: parallel Q/K/V projections feed a decayed state recurrence; the state carries across steps and the output reads it with the query.

RetNet [152] unifies three computation paradigms: parallel training, recurrent inference, and chunkwise deployment. Its core innovation is the **retention mechanism**, which uses a fixed exponential decay matrix to model temporal importance:

Parallel (training) form:
decay_matrix$[i, j] \leftarrow \gamma^{i-j}$ for $i \geq j$, else 0 ▷ causal exponential decay
decay_matrix$[i, j] \leftarrow 0$ for $i < j$ ▷ causal masking
$\mathbf{Y}_{\text{parallel}} \leftarrow (\mathbf{Q}\mathbf{K}^\top \odot \text{decay_matrix})\mathbf{V}$ ▷ element-wise multiplication with decay
Recurrent (inference) form:
for $t = 1, \ldots, n$ **do**
    $\mathbf{s}_t \leftarrow \gamma \mathbf{s}_{t-1} + \mathbf{k}_t \mathbf{v}_t^\top$ ▷ state update with decay
    $\mathbf{y}_t \leftarrow \mathbf{q}_t \mathbf{s}_t$ ▷ output via query-state interaction
end for

The decay scalar $\gamma \in (0, 1)$ controls the temporal window. RetNet uses **multi-scale retention** with different γ values per head (e.g., $\gamma = 1 - 2^{-5}, 1 - 2^{-6}, \ldots$), allowing short-term and long-term dependencies simultaneously. Chunkwise recurrent mode divides sequences into chunks, processes each chunk in parallel, and threads a recurrent state between chunks.

- **Training complexity**: $\mathcal{O}(n^2 \cdot d)$ (parallel), or $\mathcal{O}(n \cdot c \cdot d)$ (chunkwise recurrent with chunk size c).
- **Inference complexity**: $\mathcal{O}(1)$ per token, $\mathcal{O}(d^2)$ state space (fixed matrix).
- **Strengths**: Constant-time inference; triple computation paradigm; multi-scale consolidation.
- **Weaknesses**: Fixed decay imposes rigid inductive bias; may truncate learned long-range patterns.

2.3.2 Mamba: Selective State-Space Models

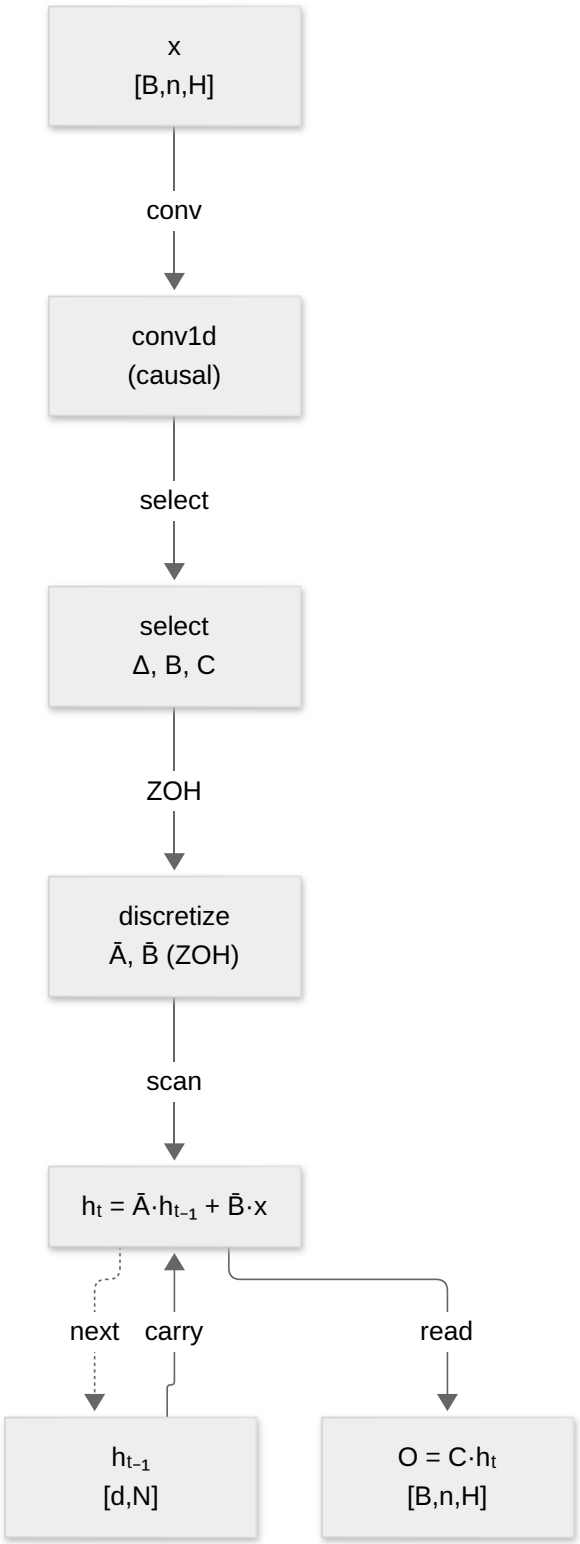


Figure 27: Dataflow of Mamba: a causal convolution feeds input-dependent Δ, B, C selections, ZOH discretization, and a selective state-space scan whose state carries across steps.

Mamba [54] frames recurrence as a continuous-time dynamical system with input-dependent parameters, achieving both linear training and constant-time inference:

Continuous dynamics: $h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t)$
Discretization with step size Δ_t:
$\bar{\mathbf{A}}_t \leftarrow \exp(\Delta_t \mathbf{A})$
$\bar{\mathbf{B}}_t \leftarrow (\Delta_t \mathbf{A})^{-1}(\exp(\Delta_t \mathbf{A}) - \mathbf{I})\Delta_t \mathbf{B}_t$
Recurrent update:
for $t = 1, \ldots, n$ **do**
    $\Delta_t \leftarrow \text{softplus}(\text{Linear}(x_t))$ ▷ input-dependent step size
    $\mathbf{B}_t \leftarrow \text{Linear}(x_t), \quad \mathbf{C}_t \leftarrow \text{Linear}(x_t)$ ▷ input-dependent projection matrices
    $h_t \leftarrow \bar{\mathbf{A}}_t h_{t-1} + \bar{\mathbf{B}}_t x_t$ ▷ state transition
    $y_t \leftarrow \mathbf{C}_t h_t$ ▷ output projection
end for

The critical innovation is that $\mathbf{A}$ (the system matrix) is *not* input-dependent, but Δ_t, $\mathbf{B}_t$, and $\mathbf{C}_t$ are, making the system **time-varying**. This selectivity allows the model to *ignore* irrelevant information by setting Δ_t near zero, effectively creating a gate. A hardware-aware parallel scan algorithm implements the recurrence efficiently on GPUs by fusing computation within SRAM, avoiding expensive HBM bandwidth.

- **Training complexity**: $\mathcal{O}(n \cdot d)$ via hardware-aware scan (linear in sequence length).
- **Inference complexity**: $\mathcal{O}(1)$ per token, $\mathcal{O}(d)$ state (hidden vector).
- **Strengths**: Achieves linear training and constant-inference simultaneously; practical on long sequences (millions of tokens).
- **Weaknesses**: State vector compression can weaken exact copying and dense associative recall versus full attention.

2.4 Continuous-Depth Transformers: ODE Integration

2.4.1 ODE Transformer

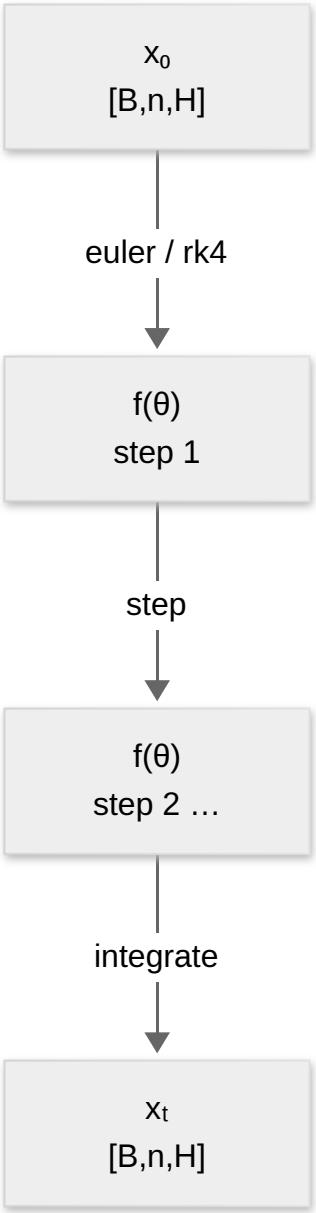


Figure 28: Dataflow of the ODE transformer block: the hidden state is integrated through one or more Euler/RK4 steps before the output is read.

The ODE Transformer [178] interprets network depth as numerical integration of a continuous dynamical system, using higher-order Runge-Kutta solvers to reduce truncation error:

1: **Continuous formulation:** $\frac{dh(t)}{dt} = f_\theta(h(t), t)$ where f_θ is the transformer sub-network
2: **Runge-Kutta-4 discrete approximation:**
3: $k_1 \leftarrow f_\theta(h_t, t)$
4: $k_2 \leftarrow f_\theta(h_t + \frac{1}{2}k_1, t + \frac{1}{2}\Delta t)$
5: $k_3 \leftarrow f_\theta(h_t + \frac{1}{2}k_2, t + \frac{1}{2}\Delta t)$
6: $k_4 \leftarrow f_\theta(h_t + k_3, t + \Delta t)$
7: $h_{t+1} \leftarrow h_t + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ ▷ weighted combination

Instead of simple Euler residual connections $h_{t+1} = h_t + f_\theta(h_t)$, the RK4 block computes four intermediate evaluations and combines them with classical RK4 weights. To avoid vanishing gradients, the architecture introduces **learned gating** that interpolates between intermediate approximations:

1: $g \leftarrow \sigma(\text{Linear}([k_1, k_2, k_3, k_4]))$ ▷ learnable gate
2: $h_{t+1} \leftarrow h_t + g \cdot k_1 + (1 - g) \cdot k_2$ ▷ interpolated refinement

This formulation reduces the effective number of parameters through weight sharing—the same f_θ is evaluated multiple times—while providing richer trajectory refinement.

- **Training complexity**: $\mathcal{O}(k \cdot n^2 \cdot d)$ where k is the RK order (4 for RK4).
- **Inference complexity**: $\mathcal{O}(k \cdot n)$ per token (higher constant overhead).
- **Strengths**: Significantly higher accuracy on generation tasks (state-of-the-art BLEU); parameter-efficient via weight sharing.
- **Weaknesses**: High per-step compute; inference latency increases by constant factor k; complex gating required for stability.

2.5 Test-Time Memory: Titans

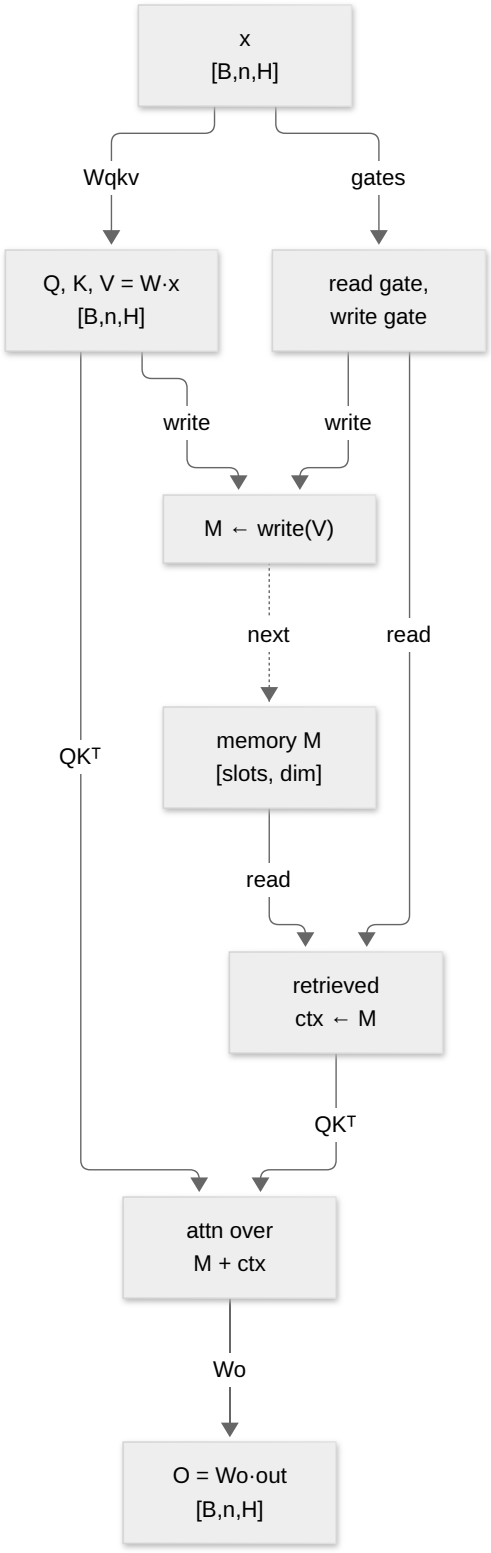


Figure 29: Dataflow of Titans test-time memory: dense Q/K/V attention runs over a local window while read and write gates update an associative memory that carries across steps.

The Titans architecture [11] introduces an orthogonal dimension: instead of static weights, the model maintains a learnable memory that is *updated during inference* based on a surprise-driven signal:

1: **Local short-term attention:** Apply standard or sparse attention within a fixed context window c
2: $y_t^{\text{local}} \leftarrow \text{Attention}(q_t, k_{[t-c:t]}, v_{[t-c:t]})$
3: **Memory update signal (surprise):**
4: $S_t \leftarrow \eta_t S_{t-1} - \theta_t \nabla_\ell(\mathcal{M}_{t-1}; x_t)$ ▷ surprise as gradient of loss w.r.t. memory
5: $\mathcal{M}_t \leftarrow (1-\alpha_t)\mathcal{M}_{t-1} + S_t$ ▷ memory updated via EMA
6: **Long-term memory retrieval:**
7: $y_t^{\text{memory}} \leftarrow \mathcal{M}_t^*(q_t)$ ▷ query memory module for retrieved information
8: **Output combination:**
9: $y_t \leftarrow y_t^{\text{local}} + \text{gate}(y_t^{\text{memory}})$ ▷ combine local and retrieved memory

The memory module $\mathcal{M}$ is literally updated during the forward pass by computing gradients of an associative loss and applying SGD steps with momentum. The decay rates $\eta_t, \theta_t, \alpha_t$ are themselves input-dependent, allowing the model to switch memory paradigms when context shifts.

- **Training complexity**: Roughly $\mathcal{O}(c^2 + n \cdot f)$ where c is local window and f is memory update overhead.
- **Inference complexity**: $\mathcal{O}(c^2)$ local attention plus $\mathcal{O}(1)$ memory retrieval per token.
- **Strengths**: Handles extreme context lengths; enables true associative recall; memory adapts to input.
- **Weaknesses**: Significantly more complex; inference includes gradient computation; higher coordination overhead.

2.6 Pattern-Driven Mixer Dispatch

The codebase selects a mixer per block from the configured `layer_pattern`. The dispatch logic routes inference-time sparse blocks (`fasa`, `sparge`) to standard dense attention while the model is in training mode, and to their sparse implementation during evaluation; all other blocks route to the corresponding family implementation.

The field exhibits a clear progression along two axes: (i) **computational efficiency**, moving from $O(n^2)$ to $O(n)$ or $O(1)$, and (ii) **memory adaptivity**, shifting from static weights to dynamic, test-time updated models. Standard attention remains the expressiveness baseline; Mamba and RetNet represent the practical efficiency frontier; Titans introduces an orthogonal innovation axis (test-time learning). The choice of architecture reflects the fundamental engineering constraint: expressiveness versus deployment cost. The latent-attention family, which compresses the KV cache into a low-rank latent, is covered in Appendix 3.

3 Latent and Low-Rank Attention Families

This annex covers the latent / low-rank KV compression family of attention mechanisms. These architectures compress the per-token key–value state into a low-rank latent vector that is the only quantity materialised in the KV cache during decoding, then reconstruct full multi-head keys and values on the fly. The family unifies grouped-query, latent, factorised, and temporal-latent designs under a common low-rank bottleneck.

Algorithm 1 Pattern-Driven Mixer Forward (Conceptual)

Require: Hidden states H, pattern P, layer index ℓ

$m \leftarrow P[\ell \bmod |P|]$
if $m \in \{\texttt{fasa}, \texttt{sparge}\}$ and model is in training mode **then**
    $H \leftarrow$ softmax-attention(H) ▷ dense surrogate, same projections
else if $m =$ `standard` **then**
    $H \leftarrow$ softmax-attention(H)
else if $m =$ `sigmoid` **then**
    $H \leftarrow$ sigmoid-attention(H)
else if $m =$ `ca` **then**
    $H \leftarrow$ continuum-attention(H) ▷ exact segment integration
else if $m =$ `memory` **then**
    $H \leftarrow$ memory-attention(H)
else if $m =$ `retnet` **then**
    $H \leftarrow$ retention(H)
else if $m =$ `mamba` **then**
    $H \leftarrow$ selective-ssm(H)
else if $m =$ `ode` **then**
    $H \leftarrow$ rk-step(H)
else if m is a sparse attention key **then**
    $H \leftarrow$ sparse-attention-family(H)
else if m is a gated attention key **then**
    $H \leftarrow$ gated-attention-family(H)
else
    $H \leftarrow$ memory-augmented-attn(H)
end if
return H

3.1 Multi-Head Latent Attention (MLA)

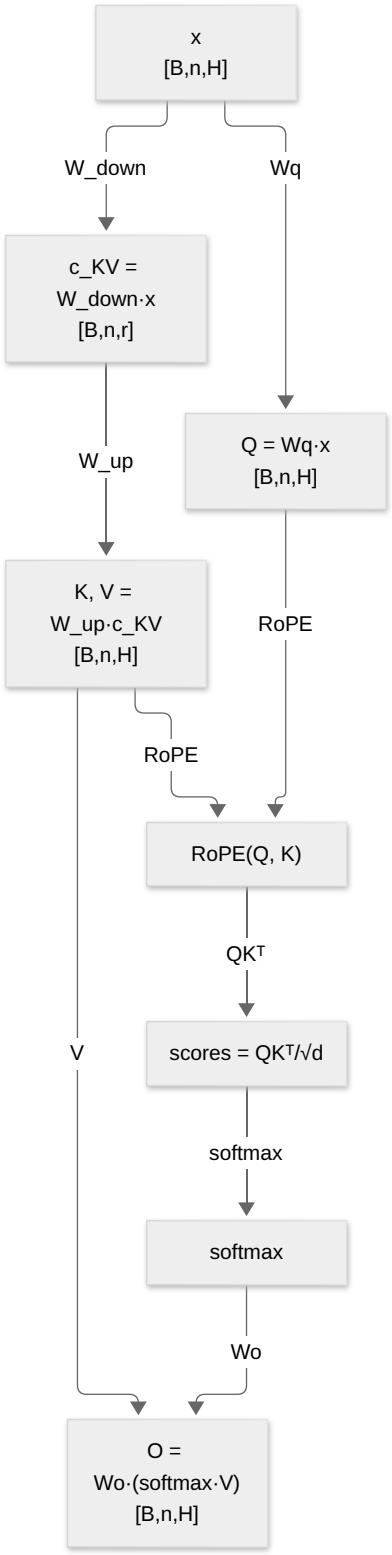


Figure 30: Dataflow of multi-head latent attention: keys and values are down-projected to a low-rank latent, up-projected back, and attended with the query after RoPE.

Multi-Head Latent Attention (MLA) [111] compresses the per-token key–value state into a single low-rank latent vector c_{KV} of rank r_{kv}, which is the only quantity materialised in the KV cache during decoding. A pair of up-projection matrices reconstructs the full multi-head keys and values on the fly, while RoPE is applied to the decompressed query and key tensors so that positional structure is preserved. The authors show that the Pareto-optimal latent width sits near $r_{kv} = d_{\text{hidden}}/2$, at which point MLA halves the KV cache yet matches multi-head attention (MHA) quality on small models.

3.1.1 Mathematical Formulation

$$c_{KV} = W_{DKV}\, x \in \mathbb{R}^{r_{kv}}, \qquad k = W_{UK}\, c_{KV}, \qquad v = W_{UV}\, c_{KV}, \qquad q = W_Q\, x.$$

$$\tilde{q} = \mathrm{RoPE}(q), \quad \tilde{k} = \mathrm{RoPE}(k), \qquad \text{attn} = \texttt{softmax}\left(\frac{\tilde{q}\,\tilde{k}^\top}{\sqrt{d_h}}\right) v.$$

3.1.2 Algorithmic Pseudocode

Input: hidden state $x \in \mathbb{R}^{n\times d}$, rank r_{kv}, projections $W_{DKV}, W_{UK}, W_{UV}, W_Q$
$c_{KV} \leftarrow W_{DKV}\, x$ ▷ down-project to latent of rank r_{kv} (cached at inference)
$k \leftarrow W_{UK}\, c_{KV}, \quad v \leftarrow W_{UV}\, c_{KV}$ ▷ up-project to full heads
$q \leftarrow W_Q\, x$
$\tilde{q} \leftarrow \mathrm{RoPE}(q), \quad \tilde{k} \leftarrow \mathrm{RoPE}(k)$ ▷ apply rotary embeddings
weights $\leftarrow \texttt{softmax}(\tilde{q}\,\tilde{k}^\top/\sqrt{d_h})$
Output: $\mathbf{Y} = \text{weights} \cdot v$

3.1.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot d)$ (full attention on decompressed heads).
- **Inference complexity**: $\mathcal{O}(n)$ per token, $\mathcal{O}(r_{kv} \cdot n)$ KV-cache space (latent only).
- **Trainable**: fully trainable; r_{kv} is a hyperparameter (Pareto-optimal $\approx d/2$).
- **Strengths**: Halves KV cache versus MHA at matched quality; RoPE-compatible; clean latent bottleneck.
- **Limitations**: Up-projection adds decode FLOPs; latent rank is a fixed global budget shared across heads.

3.2 Group-Query Latent Attention (GQLA)

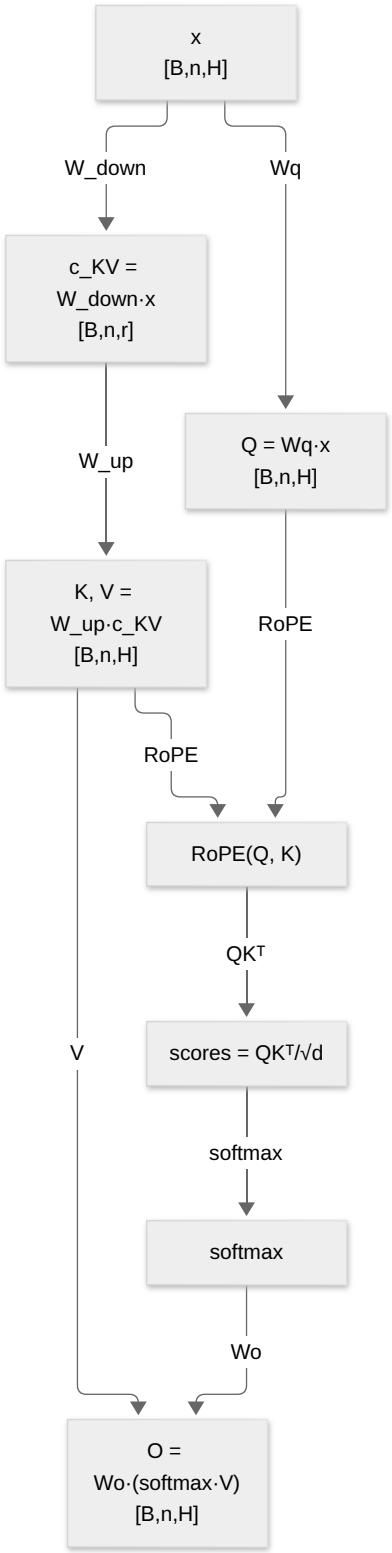


Figure 31: Dataflow of group-query latent attention: the shared latent cache is up-projected to key/value heads and combined with grouped queries.

Group-Query Latent Attention (GQLA) [113] is a minimal modification of MLA that exposes two algebraically equivalent decoding paths over the same trained weights: an MQA-absorb path (fusing the up-projection into the query, optimal on H100-class hardware) and a GQA-expanded path (materialising full keys/values, optimal on H20 / multi-token-prediction hardware). The two paths are mathematically identical because the up-projection is linear, so $\texttt{softmax}(q_{\text{absorbed}} \cdot c_{KV}^{\top}) = \texttt{softmax}((q \cdot W_{UK}^{\top}) \cdot c_{KV}^{\top})$, allowing hardware-adaptive dispatch with no retraining.

3.2.1 Mathematical Formulation

Same latent as MLA:

$$c_{KV} = W_{DKV}\, x, \qquad k = W_{UK}\, c_{KV}, \qquad v = W_{UV}\, c_{KV}, \qquad q = W_Q\, x.$$

$$\text{Path A (MQA-absorb): } \text{attn} = \texttt{softmax}\left(\frac{(q\, W_{UK}^{\top})\, c_{KV}^{\top}}{\sqrt{d_h}}\right)(W_{UV}\, c_{KV}).$$

$$\text{Path B (GQA-expanded): } \text{attn} = \texttt{softmax}\left(\frac{q\, k^{\top}}{\sqrt{d_h}}\right) v, \quad k = W_{UK} c_{KV},\ v = W_{UV} c_{KV}.$$

3.2.2 Algorithmic Pseudocode

Input: x, weights $W_{DKV}, W_{UK}, W_{UV}, W_Q$, hardware flag hw $\in \{\text{H100}, \text{H20}\}$
$c_{KV} \leftarrow W_{DKV}\, x$ ▷ shared latent, cached
$q \leftarrow W_Q\, x$
if hw == H100 **then** ▷ MQA-absorb path
  $q_{\text{abs}} \leftarrow q\, W_{UK}^{\top}$ ▷ absorb up-projection into query
  weights $\leftarrow \texttt{softmax}(q_{\text{abs}}\, c_{KV}^{\top} / \sqrt{d_h})$
  $\mathbf{Y} \leftarrow \text{weights} \cdot (W_{UV}\, c_{KV})$
else ▷ GQA-expanded path
  $k \leftarrow W_{UK}\, c_{KV}, \quad v \leftarrow W_{UV}\, c_{KV}$
  weights $\leftarrow \texttt{softmax}(q\, k^{\top} / \sqrt{d_h})$
  $\mathbf{Y} \leftarrow \text{weights} \cdot v$
end if
Output: $\mathbf{Y}$

3.2.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot d)$, identical to MLA.
- **Inference complexity**: $\mathcal{O}(n)$ per token, $\mathcal{O}(r_{kv} \cdot n)$ latent state; path chosen by hardware.
- **Trainable**: trained once; the two decoding paths reuse the same weights.
- **Strengths**: Hardware-adaptive decode with no quality loss; algebraic equivalence guarantees consistency.
- **Limitations**: Same latent-rank tradeoffs as MLA; path selection is a deployment-time heuristic.

3.3 Multi-Head Low-Rank Attention (MLRA)

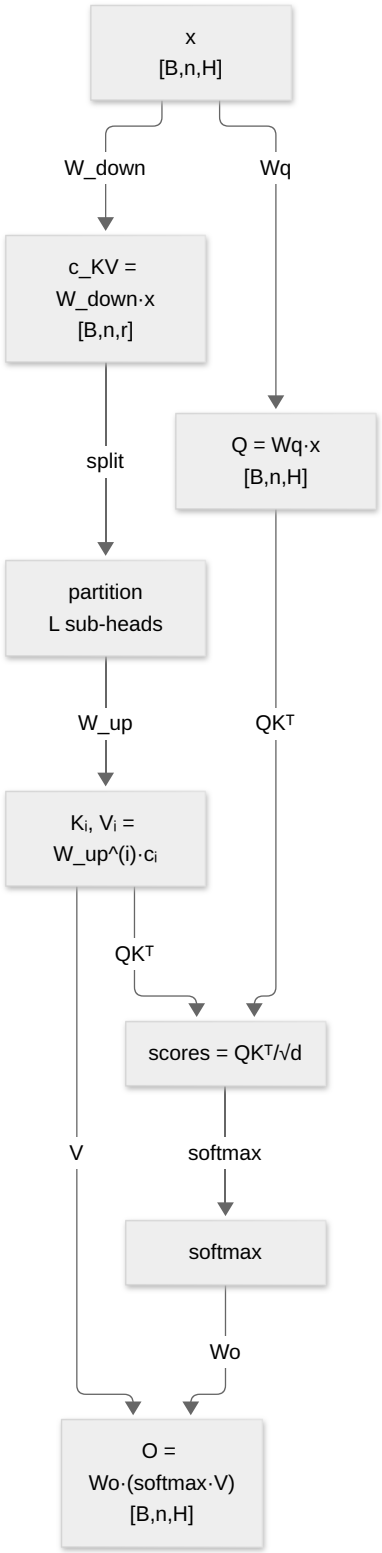


Figure 32: Dataflow of multi-head low-rank attention: the latent is partitioned across sub-heads, each with its own up-projection, before attention.

Multi-Head Low-Rank Attention (MLRA) [102] refactors the single MLA latent into L disjoint sub-heads each of rank r/L, so that the KV cache can be partitioned across L devices for 4-way tensor-parallel decoding (each device loads only $1/L$ of the cache). Per-sub-head up-projections reconstruct the corresponding slice of the full heads and are concatenated, yielding a 2.8× decode speedup over MLA while preserving the global latent budget.

3.3.1 Mathematical Formulation

$$c_{KV} = [c_1, c_2, \ldots, c_L], \quad c_i \in \mathbb{R}^{r/L}, \qquad c_{KV} = W_{DKV}\, x.$$

$$k_i = W_{UK}^{(i)}\, c_i, \quad v_i = W_{UV}^{(i)}\, c_i, \qquad k = [k_1; \ldots; k_L], \quad v = [v_1; \ldots; v_L].$$

$$\text{attn} = \texttt{softmax}\left(\frac{q\, k^\top}{\sqrt{d_h}}\right) v.$$

3.3.2 Algorithmic Pseudocode

Input: x, sub-head count L, per-sub-head projections $W_{UK}^{(i)}, W_{UV}^{(i)}$
$c_{KV} \leftarrow W_{DKV}\, x = [c_1, \ldots, c_L]$ ▷ partition latent into L blocks
for $i = 1, \ldots, L$ (parallel across L devices) **do**
    $k_i \leftarrow W_{UK}^{(i)}\, c_i, \quad v_i \leftarrow W_{UV}^{(i)}\, c_i$
end for
$k \leftarrow \texttt{concat}(k_1, \ldots, k_L), \quad v \leftarrow \texttt{concat}(v_1, \ldots, v_L)$
weights $\leftarrow \texttt{softmax}(q\, k^\top / \sqrt{d_h})$
Output: $\mathbf{Y} =$ weights $\cdot\, v$

3.3.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot d)$ (full attention on concatenated heads).
- **Inference complexity**: $\mathcal{O}(n)$ per token, $\mathcal{O}(r_{kv} \cdot n)$ state partitioned across L devices.
- **Trainable**: fully trainable; L controls partition granularity.
- **Strengths**: 2.8× decode speedup; clean 4-way tensor-parallel cache sharding; preserves total latent budget.
- **Limitations**: Requires L to divide r_{kv}; more projection matrices than MLA; inter-device all-reduce on output.

3.4 Tucker Attention

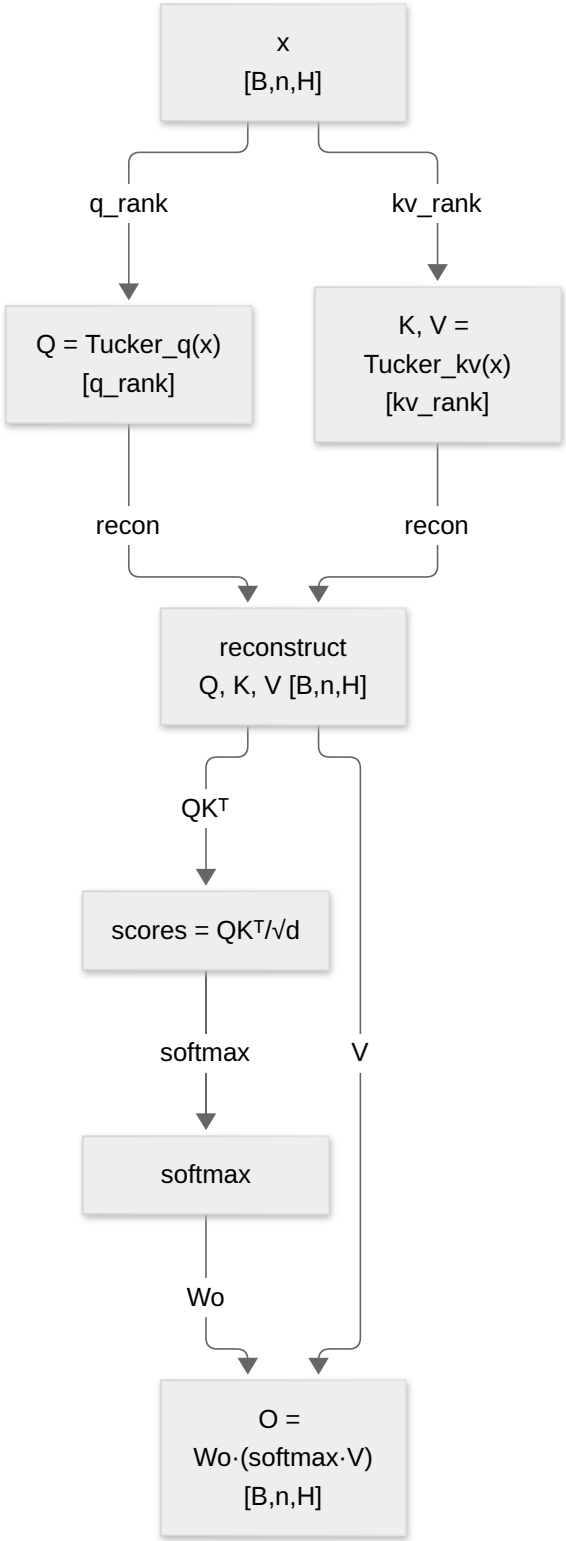


Figure 33: Dataflow of Tucker attention: queries and key/values are projected through Tucker factors and reconstructed before the scaled dot product.

Tucker Attention [86] unifies MHA, GQA and MLA under a single Tucker-style factorisation of the query/key/value weight tensors with independent ranks $q_{\text{rank}}, k_{\text{rank}}, v_{\text{rank}}$. MHA is recovered when all ranks equal the hidden size, GQA when $k_{\text{rank}} = v_{\text{rank}} < d$, and MLA when $k_{\text{rank}} = v_{\text{rank}}$ equals the latent rank. Because the factorisation is a pure linear reparameterisation, the method is fully compatible with FlashAttention and RoPE, and yields an order of magnitude fewer parameters for comparable metrics.

3.4.1 Mathematical Formulation

$$\mathbf{Q} = W_{Q,\text{core}}\, W_{Q,\text{factor}}\, x, \qquad \mathbf{K} = W_{K,\text{core}}\, W_{K,\text{factor}}\, x, \qquad \mathbf{V} = W_{V,\text{core}}\, W_{V,\text{factor}}\, x.$$

$$\text{attn} = \texttt{softmax}\left(\frac{\mathbf{Q}\,\mathbf{K}^\top}{\sqrt{d_h}}\right)\mathbf{V}.$$

Special cases: MHA $\Leftrightarrow q_{\text{rank}} = k_{\text{rank}} = v_{\text{rank}} = d$; GQA $\Leftrightarrow k_{\text{rank}} = v_{\text{rank}} < d$; MLA $\Leftrightarrow k_{\text{rank}} = v_{\text{rank}} = r_{kv}$.

3.4.2 Algorithmic Pseudocode

Input: x, ranks $q_{\text{rank}}, k_{\text{rank}}, v_{\text{rank}}$, core/factor matrices
$\mathbf{Q} \leftarrow W_{Q,\text{core}}\, W_{Q,\text{factor}}\, x$ ▷ rank-q_{rank} factorised query
$\mathbf{K} \leftarrow W_{K,\text{core}}\, W_{K,\text{factor}}\, x$ ▷ rank-k_{rank} factorised key
$\mathbf{V} \leftarrow W_{V,\text{core}}\, W_{V,\text{factor}}\, x$ ▷ rank-v_{rank} factorised value
Apply RoPE to $\mathbf{Q}, \mathbf{K}$ if positional encoding is used
weights $\leftarrow \texttt{softmax}(\mathbf{Q}\,\mathbf{K}^\top/\sqrt{d_h})$ ▷ FlashAttention-compatible
Output: $\mathbf{Y} = \text{weights} \cdot \mathbf{V}$

3.4.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot d)$ on the reconstructed tensors.
- **Inference complexity**: $\mathcal{O}(n)$ per token; cache size governed by k_{rank}.
- **Trainable**: ranks are hyperparameters; the factorisation is a drop-in weight reparameterisation.
- **Strengths**: Unifies MHA/GQA/MLA; order-of-magnitude parameter reduction at matched metrics; FlashAttention/RoPE compatible.
- **Limitations**: Two matmuls per projection add latency; rank selection is a per-task tuning problem.

3.5 Interleaved Head Attention (IHA)

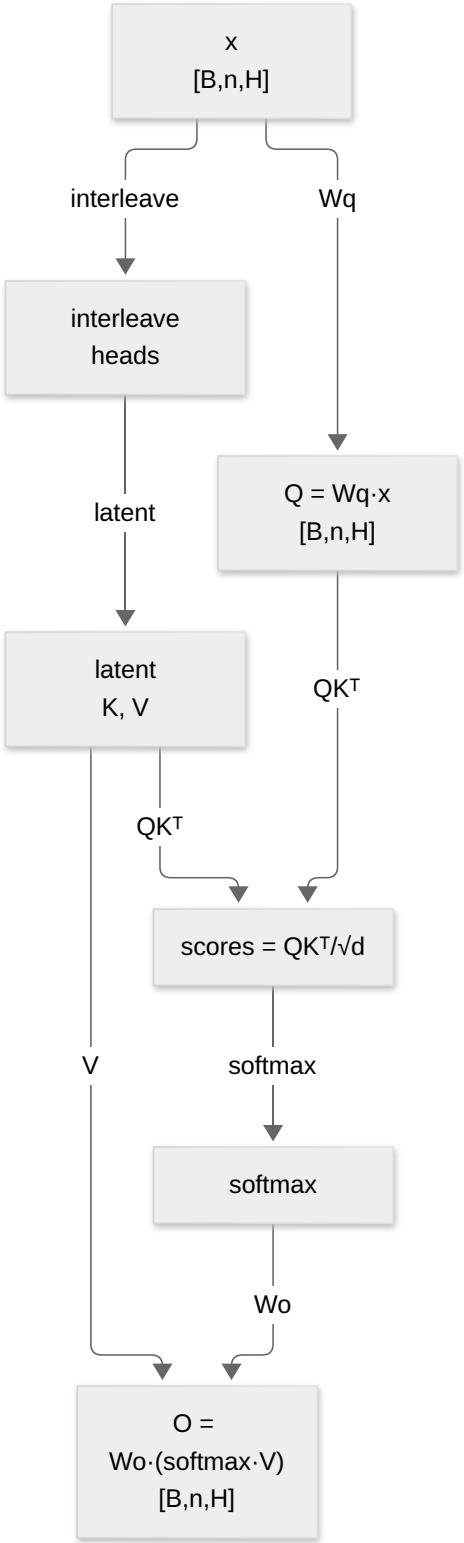


Figure 34: Dataflow of interleaved-head attention: heads are interleaved into a latent key/value representation that is attended by the query.

Interleaved Head Attention (IHA) [38] constructs P pseudo-heads per original head (typically $P = H$) where each pseudo query/key/value is a learned linear combination of all H original heads. This induces up to P^2 distinct attention patterns per head at an $\mathcal{O}(H^2P)$ overhead. Theoretically, IHA needs only $\Theta(\sqrt{k}\,n^2)$ parameters versus $\Theta(k\,n^2)$ for MHA on the Polynomial task; empirically it delivers +10–20% on RULER multi-key retrieval, +5.8% on GSM8K and +2.8% on MATH-500.

3.5.1 Mathematical Formulation

$$q_{\text{pseudo}}[p] = \sum_{h=1}^{H} \text{mix}_q[p,h]\, q[h], \quad k_{\text{pseudo}}[p] = \sum_{h} \text{mix}_k[p,h]\, k[h], \quad v_{\text{pseudo}}[p] = \sum_{h} \text{mix}_v[p,h]\, v[h].$$

$$\text{attn}_p = \texttt{softmax}\left(\frac{q_{\text{pseudo}}[p]\, k_{\text{pseudo}}[p]^\top}{\sqrt{d_h}}\right) v_{\text{pseudo}}[p], \qquad \mathbf{Y}[h] = \frac{1}{P}\sum_{p} \text{attn}_p[h].$$

3.5.2 Algorithmic Pseudocode

Input: per-head $q[h], k[h], v[h]$, mixing matrices $\text{mix}_q, \text{mix}_k, \text{mix}_v \in \mathbb{R}^{P\times H}$
for $p = 1, \ldots, P$ **do**
  $q_p \leftarrow \sum_h \text{mix}_q[p,h]\, q[h]$ ▷ learned mix across heads
  $k_p \leftarrow \sum_h \text{mix}_k[p,h]\, k[h], \quad v_p \leftarrow \sum_h \text{mix}_v[p,h]\, v[h]$
  $\text{attn}_p \leftarrow \texttt{softmax}(q_p\, k_p^\top/\sqrt{d_h})\, v_p$
end for
Output: $\mathbf{Y}[h] \leftarrow \frac{1}{P}\sum_p \text{attn}_p[h]$ ▷ average pseudo-head outputs per head

3.5.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot H \cdot P)$ with $P \approx H$, i.e. $\mathcal{O}(n^2 \cdot H^2)$.
- **Inference complexity**: $\mathcal{O}(n)$ per token with KV cache; extra $\mathcal{O}(H^2P)$ mixing overhead.
- **Trainable**: mixing matrices are learned; P is a hyperparameter.
- **Strengths**: $\Theta(\sqrt{k})$ parameter efficiency on Polynomial; large gains on retrieval and reasoning benchmarks.
- **Limitations**: Quadratic-in-heads mixing cost; more attention patterns to compute; added latency at large H.

3.6 Grouped-head laTenT Attention (GTA)

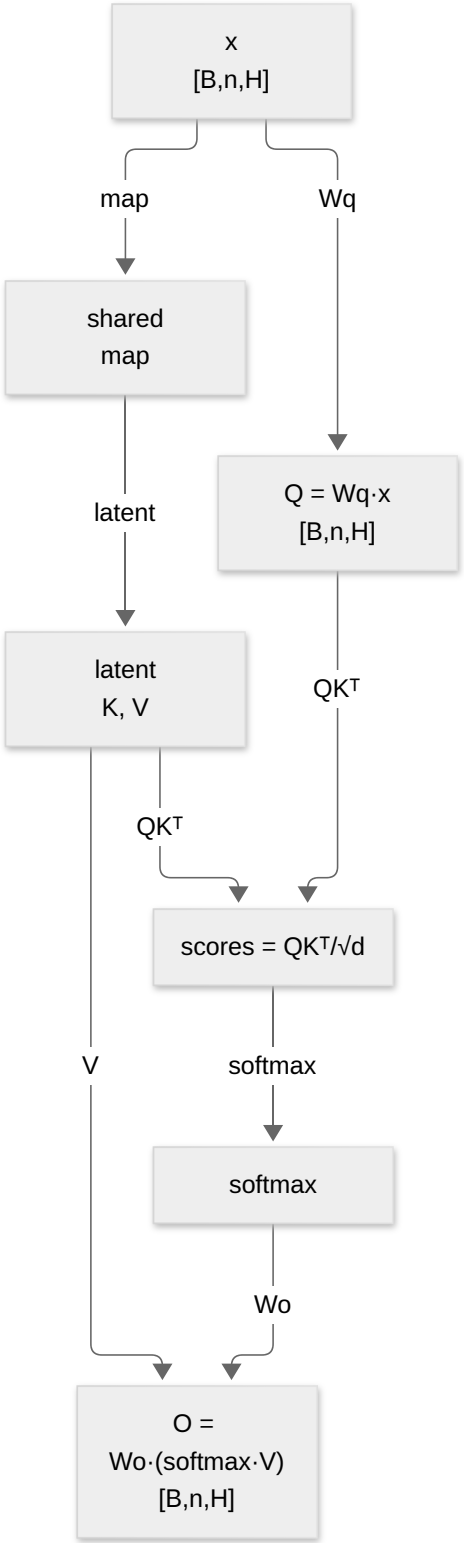


Figure 35: Dataflow of grouped-head latent attention: a shared map produces the latent key/value state consumed by the grouped heads.

Grouped-head laTenT Attention (GTA) [151] combines two compressions: (1) a *shared attention map* — one attention-score tensor per group of heads, reused across all heads in the group, shrinking the key cache; and (2) a *nonlinear value decoder* — the value cache is compressed to a latent by a down-projection and reconstructed by a SiLU-gated up-projection before the output projection. GTA cuts attention FLOPs by up to 62.5% versus GQA, shrinks the KV cache by up to 70%, and yields a 2× end-to-end inference speedup.

3.6.1 Mathematical Formulation

$$q_g = \texttt{mean}(q[\text{group } g]), \quad k_g = \texttt{mean}(k[\text{group } g]), \quad \text{attn}_g = \texttt{softmax}\left(\frac{q_g\, k_g^\top}{\sqrt{d_h}}\right) \text{ (reused across the group).}$$

$$v_{\text{latent}} = W_{DV}\, x, \qquad v = \text{SiLU}(W_{UV}\, v_{\text{latent}}), \qquad \mathbf{Y} = \text{attn}_g \cdot v.$$

3.6.2 Algorithmic Pseudocode

Input: q, k per head, hidden x, group partition $\{g\}$
$v_{\text{latent}} \leftarrow W_{DV}\, x$ ▷ compress values to latent (cached)
$v \leftarrow \text{SiLU}(W_{UV}\, v_{\text{latent}})$ ▷ nonlinear value decode
for each group g **do**
    $q_g \leftarrow \texttt{mean}(q[g]), \quad k_g \leftarrow \texttt{mean}(k[g])$ ▷ shared per-group queries/keys
    $\text{attn}_g \leftarrow \texttt{softmax}(q_g\, k_g^\top / \sqrt{d_h})$
    $\mathbf{Y}[g] \leftarrow \text{attn}_g \cdot v[g]$ ▷ one map reused for all heads in g
end for
Output: Y

3.6.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot G \cdot d_h)$ where G is the number of groups (vs H heads for GQA).
- **Inference complexity**: $\mathcal{O}(n)$ per token; value cache $\mathcal{O}(r_v \cdot n)$, key cache reduced by group sharing.
- **Trainable**: group assignment and latent rank are trainable/hyperparameters.
- **Strengths**: up to 62.5% attention-FLOP reduction, 70% KV-cache shrink, 2× end-to-end speedup.
- **Limitations**: Shared map reduces per-head diversity; nonlinear decoder adds latency; grouping must be chosen carefully.

3.7 Multi-head Temporal Latent Attention (MTLA)

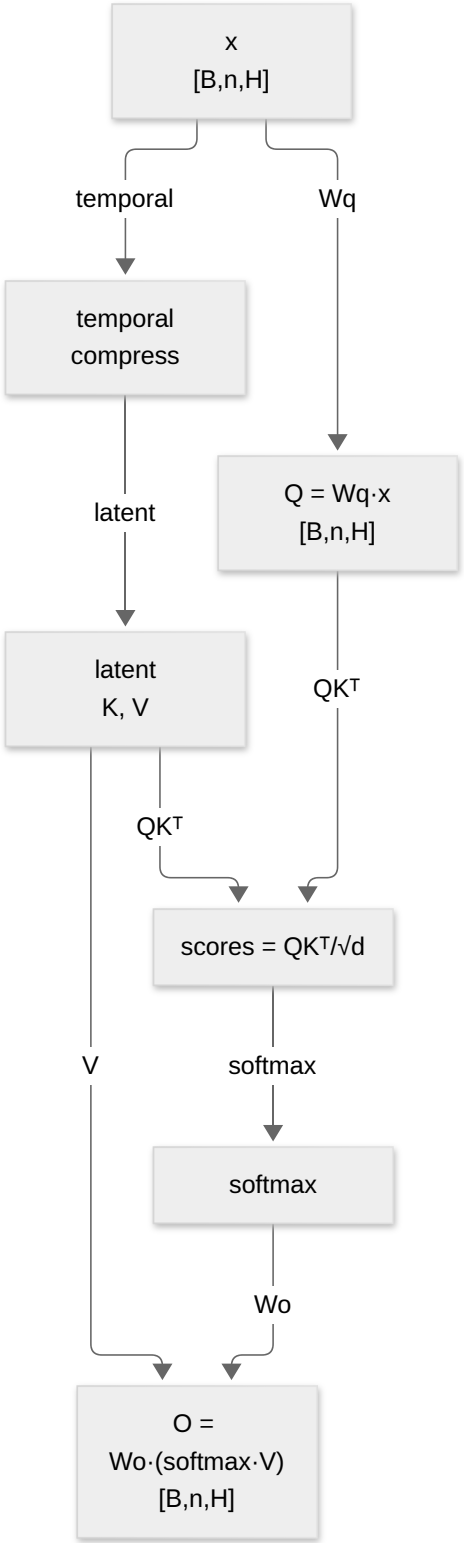


Figure 36: Dataflow of multi-head temporal latent attention: a temporal compression stage produces the latent key/value state attended by the query.

Multi-head Temporal Latent Attention (MTLA) [32] extends MLA along the temporal axis: a hyper-network dynamically merges m temporally adjacent KV-cache vectors into a single slot, shrinking the temporal length by a factor of m. A stride-aware causal mask keeps parallel training consistent with autoregressive inference, yielding a 5.3× decode speedup and 8.3× GPU-memory reduction versus MHA on En–De speech translation.

3.7.1 Mathematical Formulation

$$c_{KV}^{(t)} = W_{DKV}\, x^{(t)} \in \mathbb{R}^{r_{kv}} \text{ (per-token latent).}$$

$$\bar{c}_{KV}^{(j)} = \sum_{i=0}^{m-1} \alpha_i\, c_{KV}^{(sj+i)}, \qquad \alpha_i = \texttt{sigmoid}(\text{HyperNet}(c_{KV}^{(sj+i)})),$$

where s is the stride and m the merge window. Queries attend over merged slots $\bar{c}_{KV}$ with a stride-aware causal mask M_{stride}:

$$\text{attn} = \texttt{softmax}\left(\frac{q\,\bar{k}^{\top}}{\sqrt{d_h}} + M_{\text{stride}}\right)\bar{v}, \quad \bar{k} = W_{UK}\,\bar{c}_{KV},\ \bar{v} = W_{UV}\,\bar{c}_{KV}.$$

3.7.2 Algorithmic Pseudocode

Input: per-token latents $c_{KV}^{(t)}$, merge window m, stride s
Merge phase (hyper-network):
for each merged slot $j = 0, 1, \ldots$ **do**
    $\alpha_i \leftarrow \texttt{sigmoid}(\text{HyperNet}(c_{KV}^{(sj+i)}))$ for $i = 0, \ldots, m-1$
    $\bar{c}_{KV}^{(j)} \leftarrow \sum_{i=0}^{m-1} \alpha_i\, c_{KV}^{(sj+i)}$ ▷ gated average over window
end for
$\bar{k} \leftarrow W_{UK}\,\bar{c}_{KV}, \quad \bar{v} \leftarrow W_{UV}\,\bar{c}_{KV}$
weights $\leftarrow \texttt{softmax}(q\,\bar{k}^{\top}/\sqrt{d_h} + M_{\text{stride}})$ ▷ stride-aware causal mask
Output: $\mathbf{Y} = \text{weights} \cdot \bar{v}$

3.7.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot d/m)$ on the merged sequence (length $\lceil n/m \rceil$) plus hyper-network cost.
- **Inference complexity**: $\mathcal{O}(n/m)$ per token over merged slots; $\mathcal{O}(r_{kv} \cdot n/m)$ state.
- **Trainable**: hyper-network and merge window m are learned/chosen; stride-aware mask enforces consistency.
- **Strengths**: 5.3× decode speedup, 8.3× GPU-memory reduction vs MHA; temporal compression orthogonal to rank compression.
- **Limitations**: Merge window can blur fine temporal detail; hyper-network adds compute; mask design is non-trivial.

3.8 Architectural Comparison and Synthesis

3.9 Compressed Convolutional Attention (CCA)

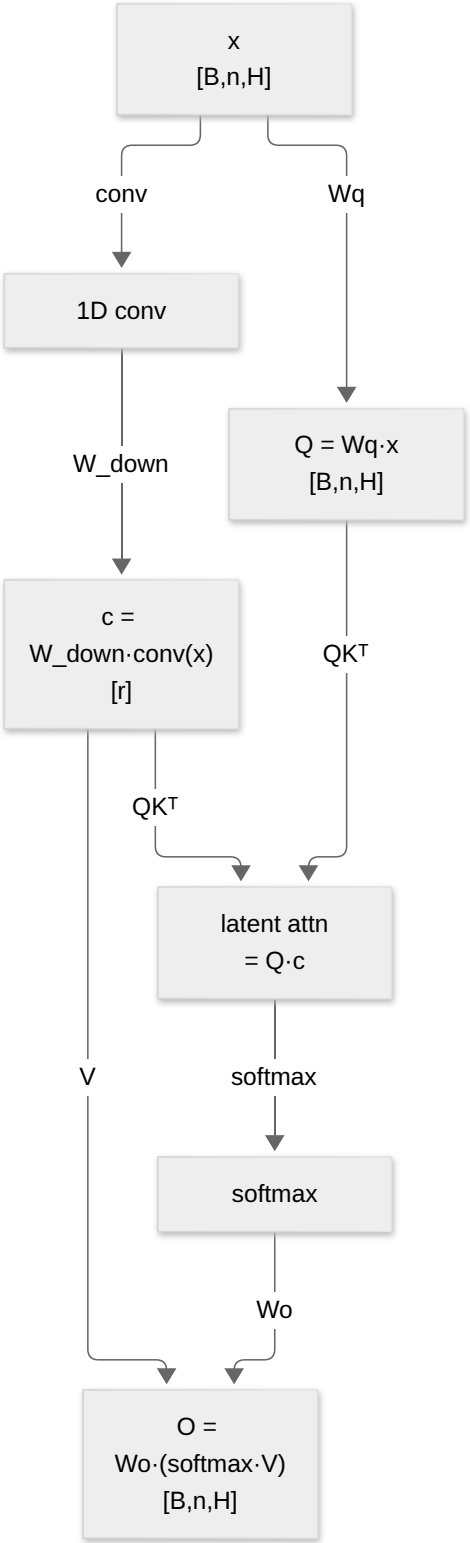


Figure 37: Dataflow of compressed convolutional attention: a 1D convolution feeds a latent compression c, which the query attends directly.

Compressed Convolutional Attention (CCA) [44] takes a fundamentally different approach to latent attention: rather than compressing the KV-cache and then up-projecting to full width before attending (as MLA/GQLA/MLRA do), CCA down-projects queries, keys, and values into a shared latent and performs the *entire* attention operation inside it. There are no Q/K/V up-projection matrices—only a single output up-projection $\tilde{W}_O$ maps the latent output back to the residual stream. This simultaneously reduces parameters, KV-cache size, **and** attention FLOPs by the compression factor $C = E/\tilde{e}$, whereas MLA only shrinks the cache.

To make attention in the fully compressed latent viable, CCA introduces three innovations: (1) two causal convolutions on the packed q/k tensor (a depth-wise sequence convolution and a head-wise grouped channel convolution), (2) q-k-mean (adds the pre-convolution mean of q and k to the post-convolution values), and (3) value-shift (each head receives half its values from the current token and half from the previous token via two independent projections). After these steps, QK L2-normalisation and a learnable key temperature β are applied, RoPE is applied *directly in the latent* (no separate RoPE head needed), and standard softmax attention is computed.

3.9.1 Mathematical Formulation

$$\tilde{q}_{\text{pre}} = \tilde{W}_Q\, x, \qquad \tilde{k}_{\text{pre}} = \tilde{W}_K\, x, \qquad \tilde{e} = E/C.$$

$$\tilde{q} = \text{Conv}(\tilde{q}_{\text{pre}}) + \tfrac{1}{2}(\tilde{q}_{\text{pre}} + \tilde{k}_{\text{pre}}), \qquad \tilde{k} = \text{Conv}(\tilde{k}_{\text{pre}}) + \tfrac{1}{2}(\tilde{q}_{\text{pre}} + \tilde{k}_{\text{pre}}).$$

$$\tilde{v} = [\tilde{W}_{V_1}\, x_t \parallel \tilde{W}_{V_2}\, x_{t-1}], \qquad \hat{q} = \frac{\tilde{q}\sqrt{d_h}}{\|\tilde{q}\|}, \quad \hat{k} = \frac{\tilde{k}\sqrt{d_h}}{\|\tilde{k}\|} e^{\beta}.$$

$$\text{attn} = \texttt{softmax}\left(\frac{\text{RoPE}(\hat{q})\,\text{RoPE}(\hat{k})^\top}{\sqrt{d_h}}\right)\tilde{v}, \qquad \mathbf{Y} = \tilde{W}_O\,\text{attn}.$$

3.9.2 Algorithmic Pseudocode

Input: hidden state $x \in \mathbb{R}^{n\times E}$, compression factor C, projections $\tilde{W}_Q, \tilde{W}_K, \tilde{W}_{V_1}, \tilde{W}_{V_2}, \tilde{W}_O$
$\tilde{e} \leftarrow E/C$ ▷ latent width
$\tilde{q}_{\text{pre}} \leftarrow \tilde{W}_Q\, x, \quad \tilde{k}_{\text{pre}} \leftarrow \tilde{W}_K\, x$ ▷ down-project to latent
$\tilde{q}_{\text{conv}} \leftarrow \text{ConvSeq}(\tilde{q}_{\text{pre}}), \quad \tilde{k}_{\text{conv}} \leftarrow \text{ConvSeq}(\tilde{k}_{\text{pre}})$ ▷ causal depth-wise + grouped convs
$\text{qk_mean} \leftarrow (\tilde{q}_{\text{pre}} + \tilde{k}_{\text{pre}})/2$ ▷ q-k-mean bias
$\tilde{q} \leftarrow \tilde{q}_{\text{conv}} + \text{qk_mean}, \quad \tilde{k} \leftarrow \tilde{k}_{\text{conv}} + \text{qk_mean}$
$\tilde{v}_1 \leftarrow \tilde{W}_{V_1}\, x_t, \quad \tilde{v}_2 \leftarrow \tilde{W}_{V_2}\, x_{t-1}$ ▷ value-shift
$\tilde{v} \leftarrow [\tilde{v}_1 \parallel \tilde{v}_2]$ ▷ concat half-heads
$\hat{q} \leftarrow \tilde{q}\,\sqrt{d_h}/\|\tilde{q}\|, \quad \hat{k} \leftarrow \tilde{k}\,\sqrt{d_h}/\|\tilde{k}\| \cdot e^{\beta}$ ▷ QK-norm + temperature
$\hat{q} \leftarrow \text{RoPE}(\hat{q}), \quad \hat{k} \leftarrow \text{RoPE}(\hat{k})$ ▷ RoPE in latent
$\text{weights} \leftarrow \texttt{softmax}(\hat{q}\,\hat{k}^\top/\sqrt{d_h})$
Output: $\mathbf{Y} = \tilde{W}_O\,(\text{weights} \cdot \tilde{v})$

3.9.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot d/C)$ (attention in compressed latent, $C\times$ fewer FLOPs than MHA).
- **Inference complexity**: $\mathcal{O}(n)$ per token, $\mathcal{O}(\tilde{e} \cdot n)$ KV-cache space.
- **Trainable**: C, conv kernel sizes, and all three tricks (convs, qk-mean, value-shift) are configurable.
- **Strengths**: Reduces params, cache, AND FLOPs simultaneously; RoPE integrates natively (no RoPE head); $>2\times$ fewer params than MLA at same compression.

- **Limitations**: Does not remove quadratic S^2 (only divides by C); conv/qk-mean/v-shift add inductive bias; fused kernel needed for theoretical speedup.

3.10 Compressed Convolutional Grouped Query Attention (CCGQA)

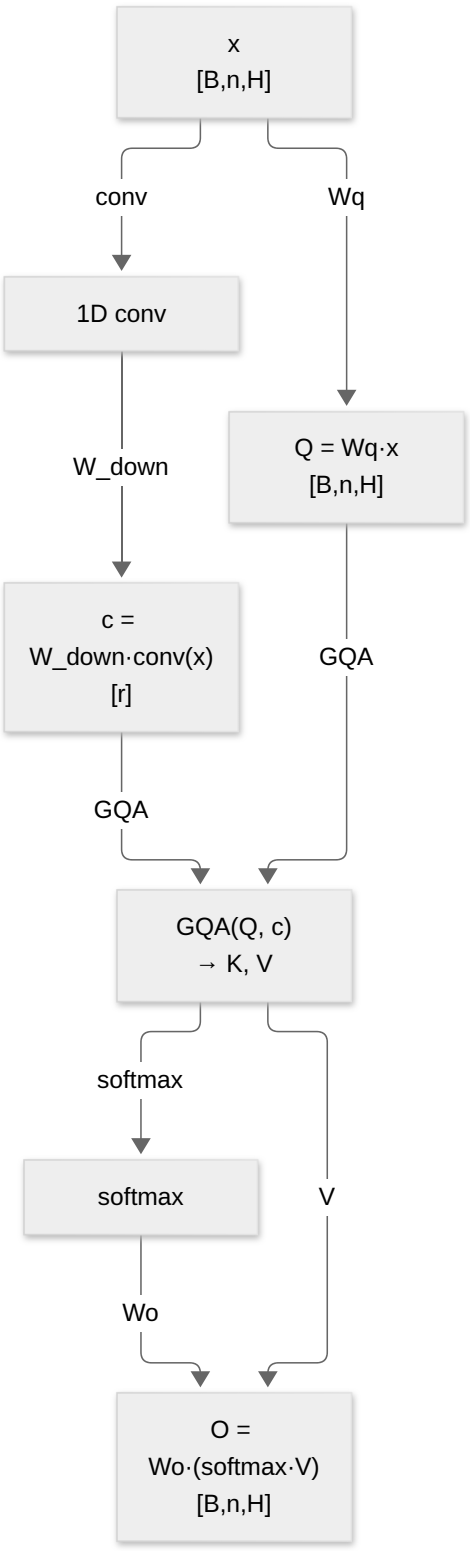


Figure 38: Dataflow of compressed convolutional grouped-query attention: the convolution latent supplies a grouped-query key/value expansion.

Compressed Convolutional Grouped Query Attention (CCGQA) [44] extends CCA with GQA-style key/value head sharing applied *inside* the compressed latent, and **decouples** the query and KV compression rates: queries are compressed by C_1 and keys/values by $C_2 \geq C_1$. The per-head latent dimension d_h must match between query and key heads, enforcing the constraint $C_2/C_1 = n_h/n_{kv}$. KV heads are shared per group (replicated to match query heads), and the qk-mean bias uses broadcast/reduce operators B_{group} (replicate KV heads to query heads) and E_{group} (average query heads within each group). CCGQA achieves the best reported loss at 8× KV-cache compression with no quality drop versus MHA.

3.10.1 Mathematical Formulation

Same pipeline as CCA, with decoupled latents:

$$\tilde{q}_{\text{pre}} = \tilde{W}_Q\, x \in \mathbb{R}^{E/C_1}, \qquad \tilde{k}_{\text{pre}} = \tilde{W}_K\, x \in \mathbb{R}^{E/C_2}, \qquad C_2 \geq C_1.$$

$$\text{qk_mean}_q = \tfrac{1}{2}(\tilde{q}_{\text{pre}} + B_{\text{group}}(\tilde{k}_{\text{pre}})), \qquad \text{qk_mean}_k = E_{\text{group}}(\text{qk_mean}_q).$$

$$\text{attn} = \texttt{softmax}\left(\frac{\text{RoPE}(\hat{q})\,\text{RoPE}(\hat{k})^\top}{\sqrt{d_h}}\right)\tilde{v}, \qquad \mathbf{Y} = \tilde{W}_O\,\text{attn}.$$

3.10.2 Key Characteristics

- **Training complexity**: $\mathcal{O}(n^2 \cdot d/C_1)$ (S^2 term scales by $1/C_1$; projection terms by $1/C_1 + 1/C_2$).
- **Inference complexity**: $\mathcal{O}(n)$ per token, $\mathcal{O}(\tilde{e}_{kv} \cdot n)$ KV-cache space (C_2-compressed, group-shared).
- **Trainable**: C_1, C_2, n_{kv}, and all CCA tricks are configurable. Constraint: $C_2/C_1 = n_h/n_{kv}$.
- **Strengths**: Decoupled compression enables smooth Pareto frontier; same arithmetic intensity as GQA; best loss at 8× cache reduction.
- **Limitations**: Same quadratic S^2 (only reduced by C_1); fused kernel needed; two value projections add complexity.

The latent family shows a clear progression along two axes: (i) **rank compression**, moving from full heads (MHA) to a shared latent (MLA) to partitioned sub-head latents (MLRA), and (ii) **deployment adaptivity**, moving from a single decode path to hardware-adaptive dispatch (GQLA) and temporal cache merging (MTLA). Tucker Attention provides the unifying algebraic frame: MHA, GQA, and MLA are all special cases of a Tucker factorisation with appropriate rank choices. CCA and CCGQA take a fundamentally different approach: rather than compressing the KV-cache and then up-projecting for full-width attention (MLA/GQLA/MLRA), they perform attention *entirely inside* the compressed latent space, reducing not just the cache but also the attention FLOPs by the compression factor. CCGQA further decouples query and KV compression rates, achieving the best reported loss at 8× KV-cache compression. Gaussian Mixture Attention (GMA) changes the routing philosophy entirely: instead of a low-rank linear bottleneck, it routes through K learned probabilistic Gaussian components, achieving linear-time $\mathcal{O}(NK)$ sequence mixing for fixed K while retaining a normalised attention-style interpretation.

3.11 Gaussian Mixture Attention (GMA)

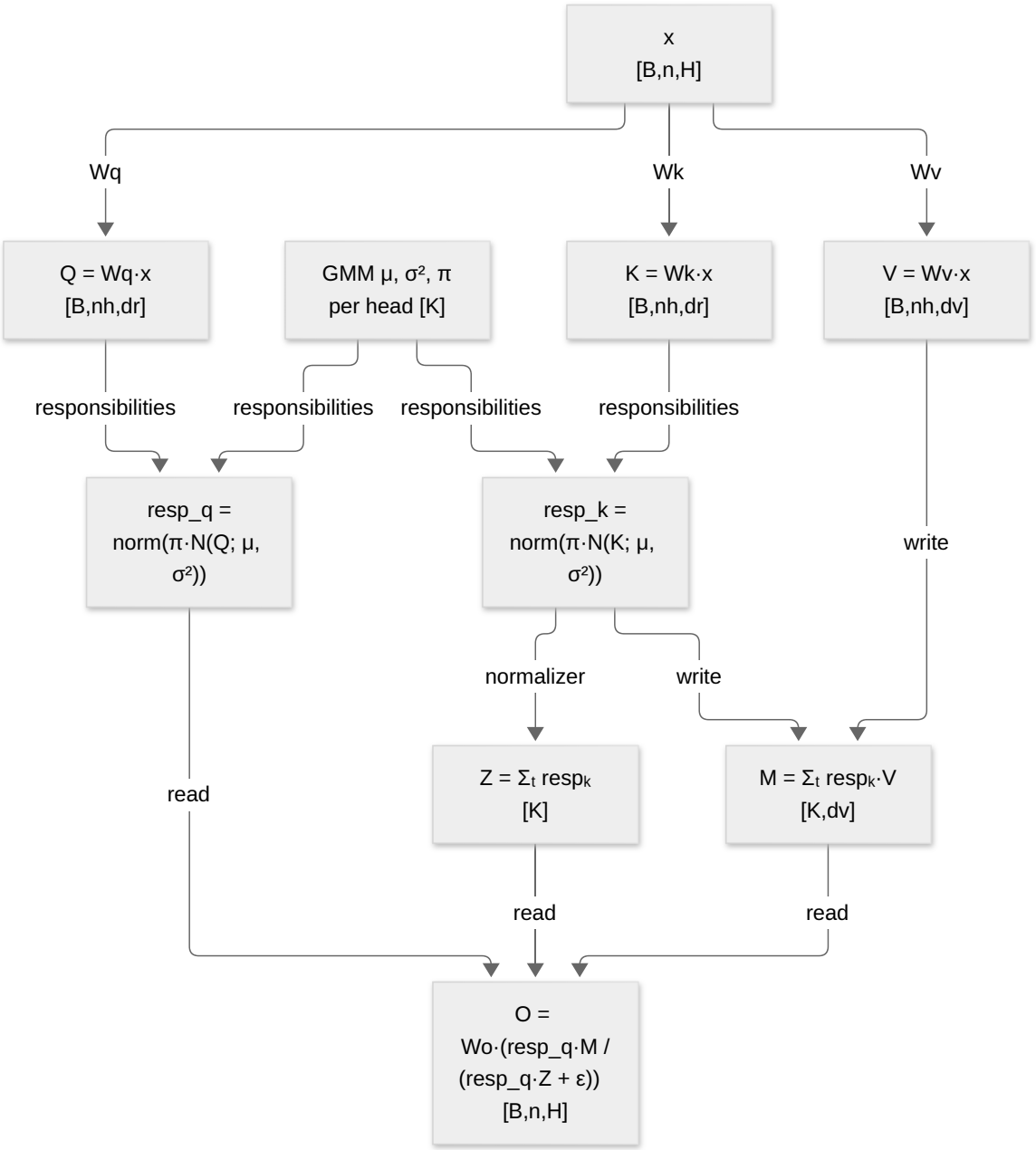


Figure 39: Dataflow of Gaussian-mixture attention: per-head Gaussian responsibilities route keys and values into a K-slot memory that the query reads back, normalized by Z.

Gaussian Mixture Attention (GMA) [72] replaces the explicit pairwise $QK^\top$ interaction of standard dot-product attention with probabilistic routing through K learned Gaussian-mixture components per head. Queries and keys are mapped to posterior *responsibility* vectors over a shared latent routing space; their overlap defines an implicit responsibility-space affinity, while values are written into and read from a K-slot latent memory. By exploiting the associativity of matrix multiplication, GMA avoids materialising the induced $N \times N$ affinity matrix and instead uses two $N \times K$ responsibility matrices whose dominant activation storage scales as $\mathcal{O}(NK)$ rather than $\mathcal{O}(N^2)$ for fixed K. The Gaussian mixture parameters—component means μ, diagonal covariances (reparameterised via $\sigma^2 = \text{softplus}(\omega) + \epsilon_\sigma$) and mixture-prior logits

α (with $\pi = \mathrm{softmax}(\alpha)$)—are fully differentiable learnable parameters optimised by standard backpropagation, with no EM loop or auxiliary clustering loss.

3.11.1 Mathematical Formulation

Per head, with routing dimension d_r and value dimension d_v:

$$Q_X = XW_Q \in \mathbb{R}^{N\times d_r}, \quad K_X = XW_K \in \mathbb{R}^{N\times d_r}, \quad V_X = XW_V \in \mathbb{R}^{N\times d_v}.$$

The posterior responsibility of component k for a routing vector x_i is

$$\gamma_{i,k} = p(z{=}k \mid x_i) = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_{\ell=1}^{K} \pi_\ell \mathcal{N}(x_i \mid \mu_\ell, \Sigma_\ell)}, \qquad \Sigma_k = \mathrm{diag}(\sigma_{k,1}^2, \ldots, \sigma_{k,d_r}^2),$$

yielding responsibility matrices $\Gamma^Q, \Gamma^K \in \mathbb{R}^{N\times K}$. The bidirectional output is computed by the associative write–read pair

$$\tilde{V} = (\Gamma^K)^\top V_X \in \mathbb{R}^{K\times d_v}, \qquad Z = (\Gamma^K)^\top \mathbf{1}_N \in \mathbb{R}^K, \qquad O = \frac{\Gamma^Q \tilde{V}}{\Gamma^Q Z + \epsilon},$$

which never materialises the implicit normalised affinity $A_{ij}^{\mathrm{GMA}} = \frac{\sum_k \gamma_{i,k}^Q \gamma_{j,k}^K}{\sum_\ell \sum_k \gamma_{i,k}^Q \gamma_{\ell,k}^K + \epsilon}$. For autoregressive modelling the global write statistics are replaced by prefix-restricted cumulative sums $\tilde{V}_k^{(i)} = \sum_{j\le i} \gamma_{j,k}^K V_{X,j}$ and $Z_k^{(i)} = \sum_{j\le i} \gamma_{j,k}^K$, enforcing causality while preserving the fixed-K linear scaling.

3.11.2 Algorithmic Pseudocode

Input: hidden state $X \in \mathbb{R}^{N\times d}$, K components, routing dim d_r, value dim d_v, params $\{\mu_k, \omega_k, \alpha_k\}_{k=1}^K$
$Q_X \leftarrow XW_Q, \quad K_X \leftarrow XW_K, \quad V_X \leftarrow XW_V$ ▷ head projections
$\sigma_k^2 \leftarrow \mathrm{softplus}(\omega_k) + \epsilon_\sigma, \quad \pi_k \leftarrow \mathrm{softmax}(\alpha)_k$ ▷ reparameterise
$\Gamma^Q \leftarrow \mathtt{softmax}_k\left(\log \pi_k - \frac{d_r}{2}\log 2\pi - \frac{1}{2}\sum_m \log \sigma_{k,m}^2 - \frac{1}{2}\sum_m \frac{(q_{i,m}-\mu_{k,m})^2}{\sigma_{k,m}^2}\right)$
$\Gamma^K \leftarrow$ same as Γ^Q applied to K_X
if causal **then**
    $\tilde{V} \leftarrow \mathtt{cumsum}_n(\Gamma^K \otimes V_X), \quad Z \leftarrow \mathtt{cumsum}_n(\Gamma^K)$ ▷ prefix-restricted
else
    $\tilde{V} \leftarrow (\Gamma^K)^\top V_X, \quad Z \leftarrow (\Gamma^K)^\top \mathbf{1}_N$ ▷ global write
end if
$O \leftarrow \Gamma^Q \tilde{V} \,/\, (\Gamma^Q Z + \epsilon)$ ▷ normalised read
Output: $\mathbf{Y} = OW_O$

3.11.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(NKd_r + NKd_v)$ for fixed K, linear in the sequence length N.
- **Inference complexity**: $\mathcal{O}(K)$ per token (responsibilities + latent read), $\mathcal{O}(NK)$ activation storage.
- **Trainable**: fully differentiable GMM parameters $\{\mu, \omega, \alpha\}$ per head, optimised end-to-end without EM.
- **Strengths**: Fixed-K linear-time sequence mixing; probabilistic interpretable routing; bidirectional and causal variants; non-negative low-rank affinity interpretation $A^{\mathrm{GMA}} = \mathrm{diag}(d)^{-1}\Gamma^Q(\Gamma^K)^\top$.

- **Limitations**: Quality depends on K and the routing dimension d_r; causal GMA trails optimised SDPA and state-space models on WikiText-103 in the paper's current implementation; not a universal replacement for softmax attention.

4 Comprehensive Sparse Attention Mechanisms

4.1 Executive Summary

Sparse attention mechanisms address the prohibitive $\mathcal{O}(n^2)$ computational and memory complexity of standard scaled dot-product attention by restricting the set of attended key-value pairs. Modern sparse attention spans a rich design space distinguished by four orthogonal dimensions: (i) **sparsity pattern** (fixed geometric, data-dependent, or frequency-based), (ii) **trainability** (end-to-end trained or inference-only), (iii) **sparsity unit** (individual tokens, local windows, or blocks), and (iv) **mechanism** (masking, selection, filtering, or compression). This annex synthesizes eleven major sparse attention methods—Sparse Transformer, Longformer, SWA with attention sinks, BigBird, SparseK, NSA, FASA, SpargeAttn, MSA, SparDA, and LSA—providing mathematical formulations, algorithmic pseudocode, and comprehensive architectural comparisons. Fast Conformer, the limited-context variant of the convolution-augmented Conformer block, belongs to the convolution-augmented family and is documented alongside the Conformer in Appendix 8.

4.2 Sparse Transformer: Factorized Strided and Fixed Patterns

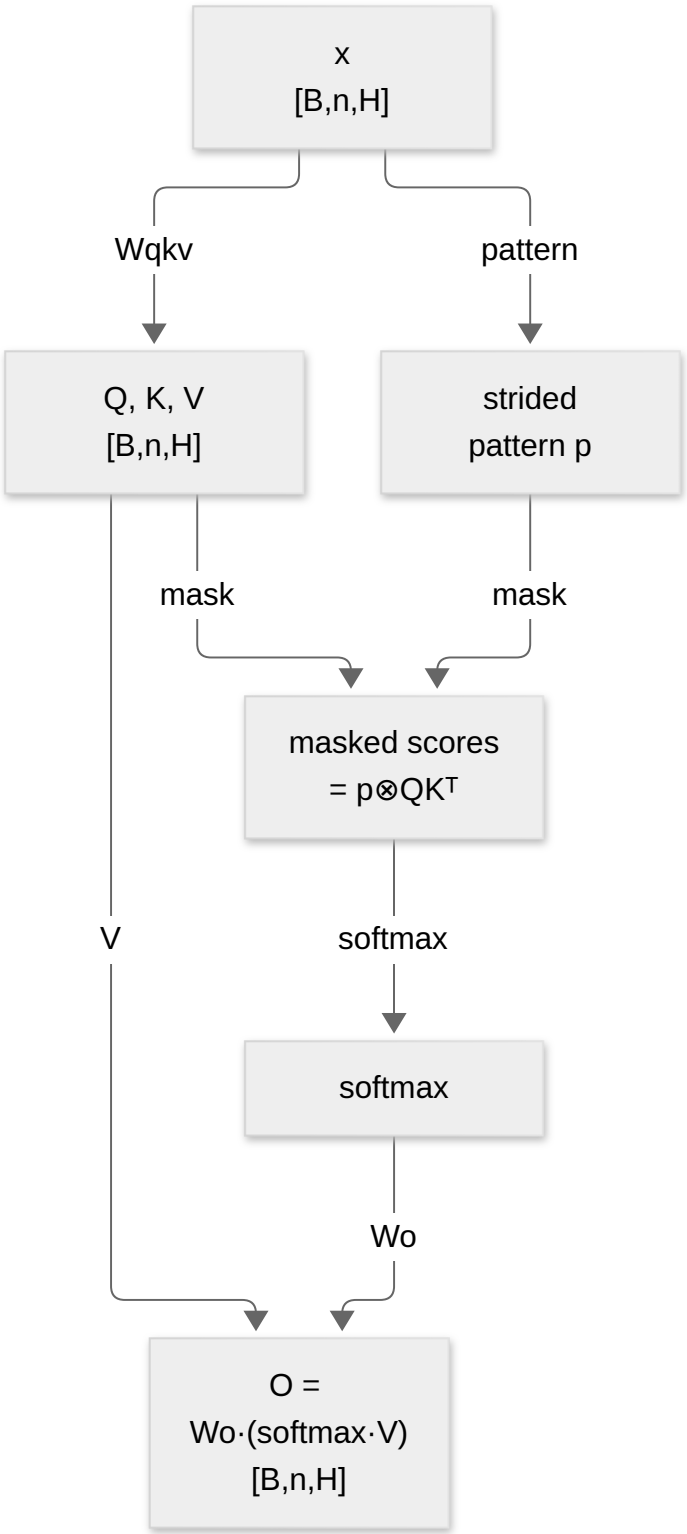


Figure 40: Dataflow of the Sparse Transformer: a strided factorized pattern masks the score matrix before softmax.

4.2.1 Mathematical Formulation

The Sparse Transformer [22] reduces quadratic complexity to $\mathcal{O}(n\sqrt{n})$ by factorizing sparse attention into two complementary sparse heads. Let n be the sequence length and $l = \lfloor \sqrt{n} \rfloor$ be the stride.

Strided attention head : Each position i attends to every l-th previous position:

$$\mathcal{A}_i^{(\text{strided})} = \{j : j \leq i,\ (i-j) \bmod l = 0\}$$

Fixed attention head : Each position i attends to positions within its local block plus fixed summary columns:

$$\mathcal{A}_i^{(\text{fixed})} = \{j : \lfloor j/l \rfloor = \lfloor i/l \rfloor\} \cup \{j : j \bmod l \in \{l-c, \ldots, l-1\}\}$$

where c is a hyperparameter controlling the number of summary columns (typically $c = 1$ or $c = 2$).

The attention computation for each head h follows standard scaled dot-product rules restricted to the sparse set:

$$\text{Attn}_h(Q, K, V)_i = \text{softmax}\left(\frac{q_i^h {K_{\mathcal{A}_i}^h}^\top}{\sqrt{d_k}}\right) V_{\mathcal{A}_i}^h$$

The key insight is that with the two factorized heads operating in parallel across a depth of L transformer layers, every position can reach every other position through a path of length at most $L + 1$, preserving long-range reachability with a fraction of dense attention cost.

4.2.2 Algorithmic Pseudocode

Algorithm 2 Sparse Transformer Attention: Strided + Fixed Dual Heads

Require: Query $\mathbf{Q} \in \mathbb{R}^{n \times d}$, Key $\mathbf{K} \in \mathbb{R}^{n \times d}$, Value $\mathbf{V} \in \mathbb{R}^{n \times d}$
Require: Stride $l = \lfloor \sqrt{n} \rfloor$, summary columns $c \in \{1, 2\}$
Ensure: Output $\mathbf{Y} \in \mathbb{R}^{n \times d}$

Head 1: Strided Attention
for $i = 1, \ldots, n$ **do**
  $\mathcal{A}_i \leftarrow \{j : j \leq i \text{ and } (i-j) \bmod l = 0\}$ ▷ Every l-th position
  $\text{scores}_i \leftarrow \mathbf{q}_i \mathbf{K}_{\mathcal{A}_i}^\top / \sqrt{d_k}$
  $\text{attn}_i \leftarrow \text{softmax}(\text{scores}_i)$
  $\mathbf{y}_i^{(1)} \leftarrow \text{attn}_i \mathbf{V}_{\mathcal{A}_i}$
end for
Head 2: Fixed Block + Summary Attention
for $i = 1, \ldots, n$ **do**
  block_start $\leftarrow \lfloor i/l \rfloor \cdot l$, block_end $\leftarrow \min(i+1, \text{block_start} + l)$
  $\mathcal{A}_i \leftarrow [\text{block_start}, \text{block_end}) \cup \{\text{cols from last } c \text{ columns}\}$
  $\text{scores}_i \leftarrow \mathbf{q}_i \mathbf{K}_{\mathcal{A}_i}^\top / \sqrt{d_k}$
  $\text{attn}_i \leftarrow \text{softmax}(\text{scores}_i)$
  $\mathbf{y}_i^{(2)} \leftarrow \text{attn}_i \mathbf{V}_{\mathcal{A}_i}$
end for
Concatenate: $\mathbf{Y} = [\mathbf{y}^{(1)} \| \mathbf{y}^{(2)}]$ and project via output linear layer **return** $\mathbf{Y}$

4.2.3 Key Characteristics

- **Complexity**: $\mathcal{O}(n\sqrt{n} \cdot d)$ during training; $\mathcal{O}(n)$ per token during generation.

- **Trainable**: Yes; full backpropagation through selected positions.
- **Sparsity pattern**: Fixed and data-agnostic; patterns do not adapt to content.
- **Strength**: Structured reachability guarantees long-range dependencies; proven effective on long sequences (Enwik8, text images).
- **Limitation**: Fixed patterns may miss important but non-contiguous relationships; requires custom CUDA kernels for practical efficiency.

4.3 Longformer: Sliding Window, Dilation, and Global Tokens

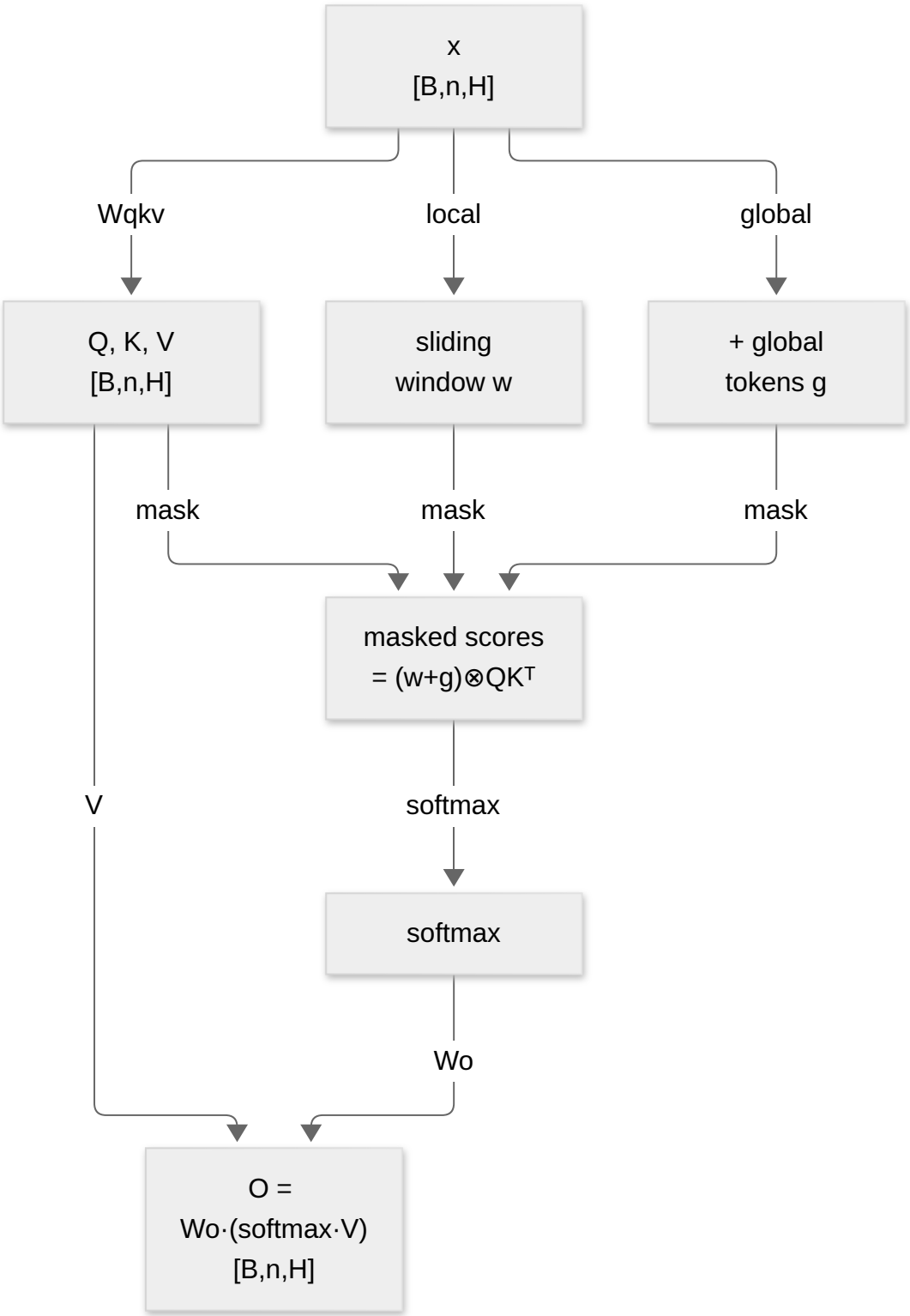


Figure 41: Dataflow of Longformer: sliding-window and global-token masks are applied to the score matrix before softmax.

4.3.1 Mathematical Formulation

Longformer [12] achieves linear $\mathcal{O}(n \cdot w)$ complexity by combining three complementary attention patterns.

Sliding window attention : Local neighborhood of size w:

$$\mathcal{A}_i^{(\text{window})} = \{j : |i - j| \leq w/2\}$$

Dilated sliding window : Windowed attention with dilation d, skipping stride $d + 1$:

$$\mathcal{A}_i^{(\text{dilated})} = \{j : |i - j| \leq (w/2)(d + 1) \text{ and } (i - j) \bmod (d + 1) = 0\}$$

Global attention : Designated global tokens (e.g., [CLS]) attend to all positions and are attended by all:

$$\mathcal{A}_i^{(\text{global})} = \begin{cases} \{1, \ldots, n\} & \text{if } i \in \mathcal{G} \\ \mathcal{A}_i^{(\text{window})} \cup \mathcal{G} & \text{otherwise} \end{cases}$$

The three patterns are applied via separate projections. The local and global attention can use either the same projections or task-specific separate ones.

4.3.2 Algorithmic Pseudocode

Algorithm 3 Longformer: Sliding Window + Dilated + Global

Require: Query $\mathbf{Q} \in \mathbb{R}^{n \times d}$, Key $\mathbf{K}$, Value $\mathbf{V}$, window size w, dilation d, global token indices $\mathcal{G}$
Ensure: Output $\mathbf{Y}$

for $i = 1, \ldots, n$ **do**
  if $i \in \mathcal{G}$ **then**
    $\mathcal{A}_i \leftarrow \{1, \ldots, n\}$ ▷ Global token attends to all
  else
    Local window: $\mathcal{A}^{(\text{local})} \leftarrow [i - w/2, i + w/2] \cap [1, n]$
    Dilated offsets: $\mathcal{A}^{(\text{dilated})} \leftarrow \{j : (i - j) \bmod (d + 1) = 0\} \cap \mathcal{A}^{(\text{dilated range})}$
    Combine: $\mathcal{A}_i \leftarrow \mathcal{A}^{(\text{local})} \cup \mathcal{A}^{(\text{dilated})} \cup \mathcal{G}$
  end if
  $\text{scores}_i \leftarrow \mathbf{q}_i \mathbf{K}_{\mathcal{A}_i}^\top / \sqrt{d_k}$
  $\text{attn}_i \leftarrow \text{softmax}(\text{scores}_i)$
  $\mathbf{y}_i \leftarrow \text{attn}_i \mathbf{V}_{\mathcal{A}_i}$
end forreturn $\mathbf{Y}$

4.3.3 Key Characteristics

- **Complexity**: $\mathcal{O}(n \cdot w)$ where w is the window size (linear when $w \ll n$).
- **Trainable**: Yes; drop-in replacement for standard attention.
- **Coverage**: With L layers and window size w, receptive field grows to $L \times w$, covering entire sequence with shallow networks.
- **Strength**: Practical for document-level tasks; dilation expands local receptive fields with minimal overhead; global tokens provide query-document correspondence.
- **Limitation**: Window size remains a design hyperparameter; suboptimal for tasks requiring dense attention patterns.

4.4 SWA with Attention Sinks: Sliding Window Plus a Sink Prefix

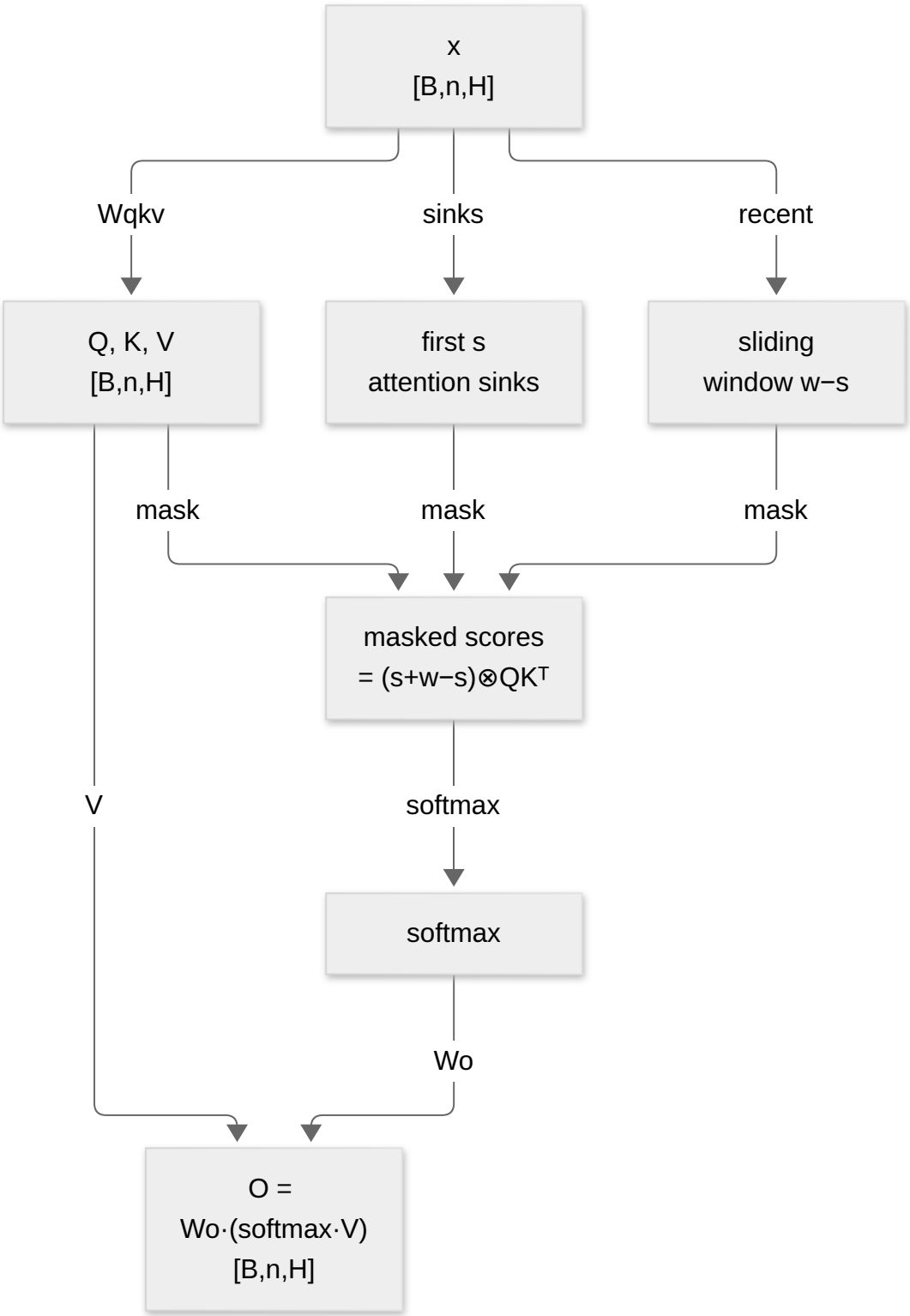


Figure 42: Dataflow of SWA with attention sinks: the first s sink tokens and a causal $w - s$ sliding window are unioned into a mask applied to the score matrix before softmax.

4.4.1 Mathematical Formulation

SWA with attention sinks [75] is the simplest KV-cache eviction policy: instead of compressing the cache or replacing attention with a recurrent state, it restricts every query to a small, fixed set of keys. Write the window size as w and the number of sinks as s, i.e. SWA(w, s). In decoder mode, query i attends to

$$\mathcal{A}_i = \underbrace{\{j : j < s,\ j \le i\}}_{\text{attention sinks}} \cup \underbrace{\{j : i - (w - s) + 1 \le j \le i\}}_{\text{sliding window of } w-s},$$

so each query sees at most w keys. The causal clip $j \le i$ on the sink term is essential: tokens before position s must not see future sinks.

Why sinks are indispensable Transformers learn to assign disproportionately large attention mass to the first few tokens even when they are not semantically relevant. These *attention sinks* [166] act as repositories for surplus attention. Once the prefix leaves a sink-free sliding window, the model loses its dump target and quality collapses catastrophically; re-including the first s tokens restores most of full-attention performance. The paper fixes $s = 4$ and evaluates $w \in \{64, 128, 256, 512\}$.

Encoder extension The paper defines SWA(w, s) for causal language models. For bidirectional encoders the local window is made symmetric, $\{j : |i - j| \le \lfloor (w - s)/2 \rfloor\}$, and unioned with the sink prefix—the same convention used by Longformer.

Because the intervention is a pure attention-mask change, it requires no additional parameters and no new QKV math: the standard projections W_q, W_k, W_v, W_o are reused. It can be applied to a pretrained checkpoint at inference time with zero training tokens, or trained from scratch as a normal mixer.

4.4.2 Algorithmic Pseudocode

Algorithm 4 SWA with Attention Sinks: Mask Construction and Attention

Require: Query $\mathbf{Q} \in \mathbb{R}^{n \times d}$, Key $\mathbf{K}$, Value $\mathbf{V}$, window w, sinks s
Ensure: Output $\mathbf{Y} \in \mathbb{R}^{n \times d}$

Row indices i, column indices j
causal $\leftarrow [j \le i]$
sinks $\leftarrow [j < s] \wedge$ causal ▷ Causal-clipped sink prefix
window $\leftarrow [i - (w - s) + 1 \le j \le i]$
$M \leftarrow$ sinks $\vee$ window
$S \leftarrow \mathbf{Q}\mathbf{K}^\top / \sqrt{d_k}$
$S \leftarrow \text{mask}(S, M)$ ▷ $-\infty$ wherever $M = 0$
$\mathbf{Y} \leftarrow \text{softmax}(S)\,\mathbf{V}$ **return** $\mathbf{Y}$

4.4.3 Key Characteristics

- **Complexity**: $\mathcal{O}(n \cdot w \cdot d)$ time and $\mathcal{O}(w)$ KV storage per layer at inference (fixed once the sequence exceeds w).
- **Trainable**: Both; training-free at inference on pretrained checkpoints, or trained from scratch.
- **Receptive field**: After L layers, $L(w - s)$ positions plus the always-visible sink prefix.

- **Strength**: Recovers $\approx 99\%$ of the average full-attention score across models from 1.3B to 72B with zero training tokens, and beats retrofitted linear attention by 2–10$\times$ on long-context reasoning (S-NIAH, BABILong) while decoding faster and using less memory.
- **Limitation**: The horizon is fixed and task-dependent—$w = 64$ suffices for log-likelihood benchmarks but collapses on generative math, where $w = 512$ is needed; there is no content-adaptive selection.

4.5 BigBird: Random, Local, and Global Sparse Graph

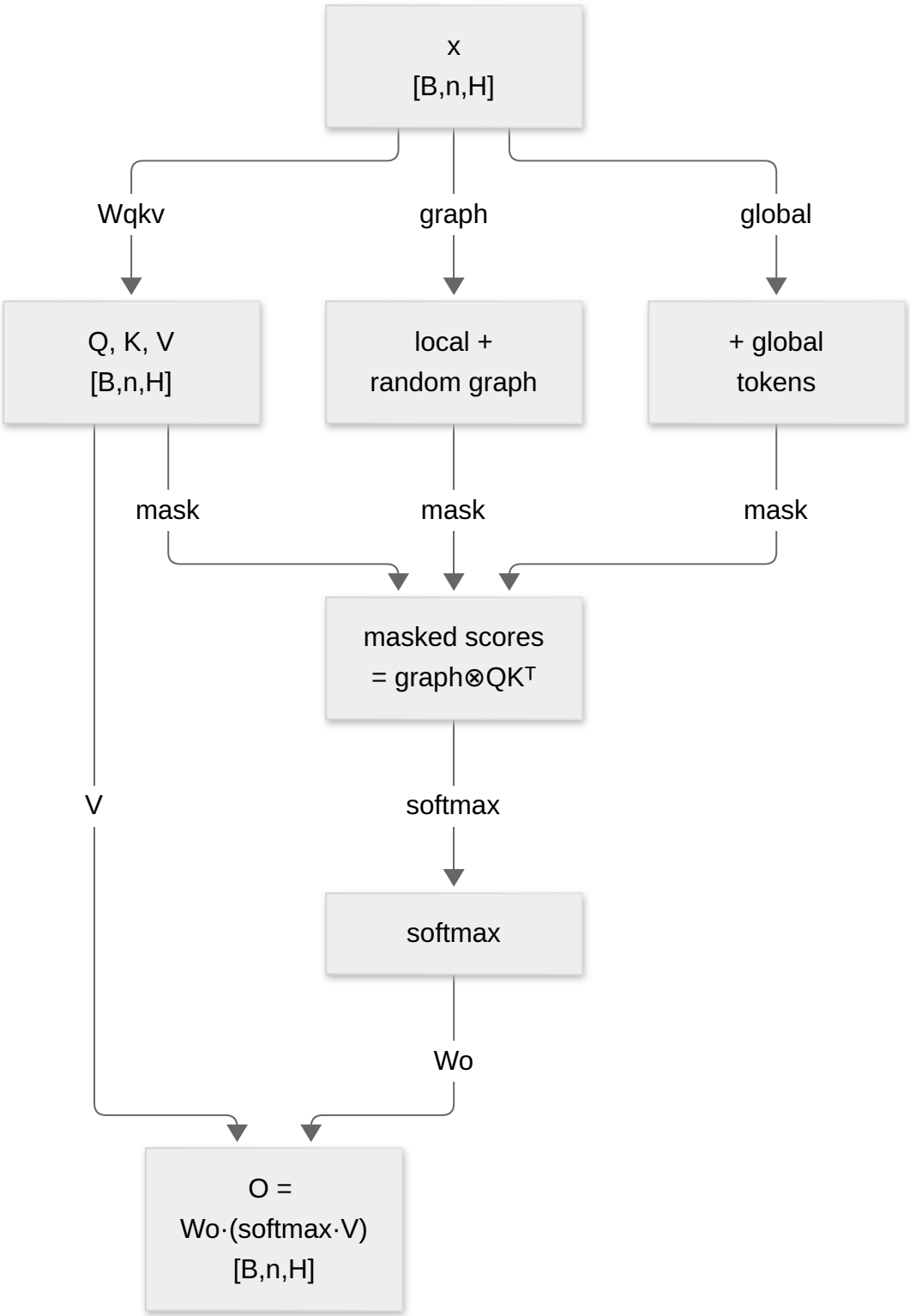


Figure 43: Dataflow of BigBird: local, random, and global adjacency masks the score matrix before softmax.

4.5.1 Mathematical Formulation

BigBird [175] preserves universal approximation and Turing completeness using a principled mix of three sparsity patterns.

Random connections : Each position connects to r randomly sampled positions:

$$\mathcal{A}_i^{(\text{random})} = \text{RandomSample}(\{1, \ldots, n\}, r)$$

Local window : Sliding window neighborhood:

$$\mathcal{A}_i^{(\text{local})} = \{j : |i - j| \leq w/2\}$$

Global tokens : Task-specific global anchors:

$$\mathcal{A}_i^{(\text{global})} = \begin{cases} \{1, \ldots, n\} & \text{if } i \in \mathcal{G} \\ \mathcal{A}_i^{(\text{random})} \cup \mathcal{A}_i^{(\text{local})} \cup \mathcal{G} & \text{otherwise} \end{cases}$$

The composite sparse mask M is:

$$M_{ij} = \mathbf{1}[j \in \mathcal{A}_i^{(\text{random})} \cup \mathcal{A}_i^{(\text{local})} \cup \mathcal{A}_i^{(\text{global})}]$$

In practice, BigBird groups tokens into blocks of size b and applies the sparsity decision at the block level, enabling efficient block-sparse implementations.

4.5.2 Theoretical Guarantee

The authors prove that with $O(1)$ random connections and global tokens, the sparse graph maintains the same universal approximation and Turing completeness properties as dense attention. Formally, any computable function can be represented and any sequence of operations can be simulated within the BigBird connectivity graph.

4.5.3 Algorithmic Pseudocode

Algorithm 5 BigBird: Random + Local + Global Block-Sparse Attention

Require: Query $\mathbf{Q}$, Key $\mathbf{K}$, Value $\mathbf{V}$, block size b, random links per block r, global indices $\mathcal{G}$
Ensure: Output $\mathbf{Y}$

Reshape into blocks: $n_b = \lceil n/b \rceil$ blocks
for block $i = 0, \ldots, n_b - 1$ **do**
    for position j in block i **do**
        Local window: $\mathcal{A}_j^{(\text{local})}$ = nearby positions in block $\cup$ boundary positions
        Random block sampling: Sample r blocks uniformly at random, add all positions from those blocks
        Global: Add all global token positions
        $\mathcal{A}_j \leftarrow \mathcal{A}_j^{(\text{local})} \cup \mathcal{A}_j^{(\text{random})} \cup \mathcal{G}$
    end for
end for
for $i = 1, \ldots, n$ **do**
    $\text{scores}_i \leftarrow \mathbf{q}_i \mathbf{K}_{\mathcal{A}_i}^\top / \sqrt{d_k}$
    $\text{attn}_i \leftarrow \text{softmax}(\text{scores}_i)$
    $\mathbf{y}_i \leftarrow \text{attn}_i \mathbf{V}_{\mathcal{A}_i}$
end forreturn $\mathbf{Y}$

4.5.4 Key Characteristics

- **Complexity**: $\mathcal{O}(n)$ or near-linear in optimal block-sparse implementation.
- **Trainable**: Yes.
- **Theoretical grounding**: Provably maintains universal approximation and Turing-completeness with sparse connectivity.
- **Strength**: Strong long-document performance on question answering, summarization, and genomics tasks.
- **Limitation**: Random sampling introduces variance and non-determinism; tuning of r, w, g hyperparameters remains task-dependent.

4.6 SparseK Attention: Differentiable Top-k Selection

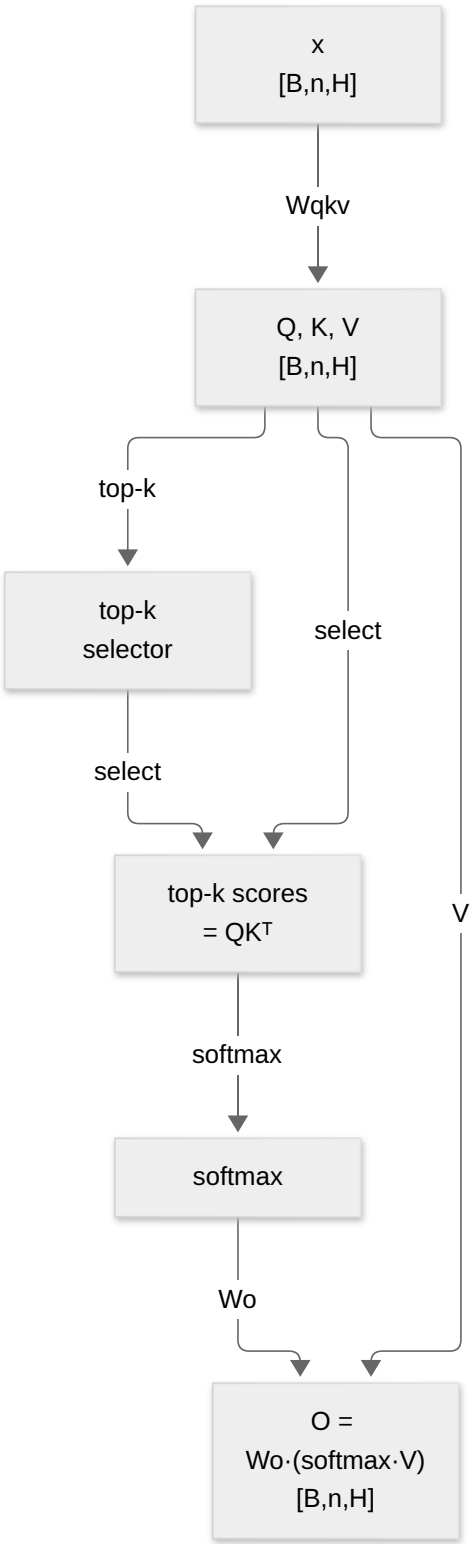


Figure 44: Dataflow of SparseK: a differentiable top-k selector keeps only the highest-scoring keys before softmax.

4.6.1 Mathematical Formulation

SparseK [108] enables learnable sparsity via a **differentiable top-k operator**. A scoring network ϕ_θ evaluates key-value pair importance:

$$u_j = \phi_\theta(\mathbf{k}_j, \mathbf{q}_i) \in \mathbb{R}$$

The **SparseK operator** selects the top-k scores while remaining differentiable. It computes a threshold $\tau(u)$ such that the sum of active scores equals k:

$$\text{SparseK}(u,k)_j = \max(u_j - \tau(u), 0), \quad \text{where} \quad \sum_j \max(u_j - \tau, 0) = k$$

The threshold τ is found via bisection. The attention becomes:

$$m_j = \text{SparseK}(u,k)_j, \quad \text{Attn}_i = \text{softmax}\left(\frac{\mathbf{q}_i K_{\text{sel}}^\top}{\sqrt{d_k}}\right) V_{\text{sel}}$$

where $K_{\text{sel}}, V_{\text{sel}}$ contain only the top-k entries (those with $m_j > 0$).

4.6.2 Algorithmic Pseudocode

Algorithm 6 SparseK: Differentiable Top-k Attention

Require: Query $\mathbf{q} \in \mathbb{R}^d$, Key matrix $\mathbf{K} \in \mathbb{R}^{n \times d}$, Value matrix $\mathbf{V} \in \mathbb{R}^{n \times d}$
Require: Scoring network ϕ_θ, target sparsity k
Ensure: Attention output $\mathbf{y}$

Compute importance scores:
for $j = 1, \ldots, n$ **do**
    $u_j \leftarrow \phi_\theta(\mathbf{k}_j, \mathbf{q})$
end for
Compute differentiable top-k via threshold:
$u_{\text{sorted}} \leftarrow \text{sort}(u, \text{descending} = \text{True})$
$\text{cumsum} \leftarrow \text{cumsum}(u_{\text{sorted}})$
Find $\rho = \max\{i : u_{\text{sorted}}[i] > 0 \text{ and } \text{cumsum}[i] \leq k\}$
$\tau \leftarrow (u_{\text{sorted}}[\rho] - k)/(\rho + 1)$ ▷ Threshold for k active elements
$m \leftarrow \max(u - \tau, 0)$ ▷ Differentiable selection mask
Apply mask and compute attention:
$K_{\text{sel}} \leftarrow K[m > 0]$, $V_{\text{sel}} \leftarrow V[m > 0]$
$\text{scores} \leftarrow \mathbf{q} K_{\text{sel}}^\top / \sqrt{d_k}$
$\text{attn} \leftarrow \text{softmax}(\text{scores})$
$\mathbf{y} \leftarrow \text{attn} \cdot V_{\text{sel}}$ **return** $\mathbf{y}$

4.6.3 Key Characteristics

- **Complexity**: $\mathcal{O}(n \cdot d)$ during training (linear in sequence length); $\mathcal{O}(k)$ per token during generation.
- **Trainable**: Yes; end-to-end gradient flow through the SparseK operator.
- **Incremental generation**: Supports efficient constant-memory autoregressive generation.
- **Strength**: Seamlessly integrates into existing LLM architectures; minimal fine-tuning needed.
- **Limitation**: Scattered memory access from top-k selection may limit cache efficiency on some hardware; scoring network adds overhead.

4.7 NSA (Native Sparse Attention): Hardware-Aligned Hierarchical Branches

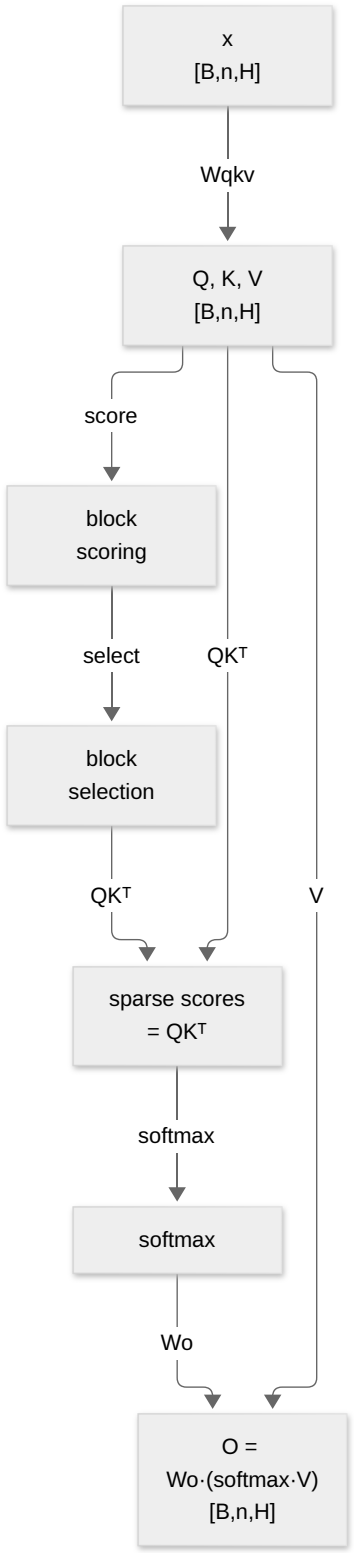


Figure 45: Dataflow of NSA: hierarchical block scoring and selection restrict the score matrix to selected blocks.

4.7.1 Mathematical Formulation

NSA [173] decomposes sparse attention into three parallel branches that are combined through learned gates.

Compression branch : A learnable MLP φ compresses key-value blocks:

$$\tilde{K}_t^{\text{cmp}} = [\varphi(k_{id+1:id+l})]_{1 \le i \le \lfloor (t-l)/d \rfloor} \quad , \quad \tilde{V}_t^{\text{cmp}} = [\varphi(v_{id+1:id+l})]$$

Selection branch : High-importance blocks are selected based on compressed scores:

$$p_t^{\text{cmp}} = \text{softmax}(q_t^\top \tilde{K}_t^{\text{cmp}} / \sqrt{d}), \quad I_t = \text{TopK}(p_t^{\text{cmp}}, n), \quad \tilde{K}_t^{\text{sel}} = \text{Gather}(K, I_t)$$

Sliding window branch : Fixed local context:

$$\tilde{K}_t^{\text{win}} = K_{t-w:t}, \quad \tilde{V}_t^{\text{win}} = V_{t-w:t}$$

Gated combination : The three attention outputs are combined via learned gates:

$$\alpha_t^{(\text{cmp})} = \sigma(\text{Linear}_{\text{cmp}}(q_t)), \quad \alpha_t^{(\text{sel})} = \sigma(\text{Linear}_{\text{sel}}(q_t)), \quad \alpha_t^{(\text{win})} = \sigma(\text{Linear}_{\text{win}}(q_t))$$

$$y_t = \alpha_t^{(\text{cmp})} \cdot \text{Attn}(q_t, \tilde{K}^{\text{cmp}}, \tilde{V}^{\text{cmp}}) + \alpha_t^{(\text{sel})} \cdot \text{Attn}(q_t, \tilde{K}^{\text{sel}}, \tilde{V}^{\text{sel}}) + \alpha_t^{(\text{win})} \cdot \text{Attn}(q_t, \tilde{K}^{\text{win}}, \tilde{V}^{\text{win}})$$

4.7.2 Algorithmic Pseudocode

Algorithm 7 NSA: Compressed + Selected + Window Branches with Learned Gating

Require: Query $\mathbf{q}_t$, cached Key/Value sequences, block size l, stride d, selection count n, window w
Require: Compression MLP φ, gate networks $\text{Linear}_{\text{cmp}}, \text{Linear}_{\text{sel}}, \text{Linear}_{\text{win}}$
Ensure: Output $\mathbf{y}_t$

Compression branch:
for $i = 1$ to $\lfloor t/d \rfloor$ **do**
  $k_i^{\text{cmp}} \leftarrow \varphi(\mathbf{K}_{id+1:id+l})$ ▷ Compress block via MLP
  $v_i^{\text{cmp}} \leftarrow \varphi(\mathbf{V}_{id+1:id+l})$
end for
$\tilde{\mathbf{K}}^{\text{cmp}} \leftarrow [k_1^{\text{cmp}}, \ldots, k_{\lfloor t/d \rfloor}^{\text{cmp}}]$, $\tilde{\mathbf{V}}^{\text{cmp}} \leftarrow [v_1^{\text{cmp}}, \ldots, v_{\lfloor t/d \rfloor}^{\text{cmp}}]$
$\mathbf{y}_t^{(\text{cmp})} \leftarrow \text{SDPA}(\mathbf{q}_t, \tilde{\mathbf{K}}^{\text{cmp}}, \tilde{\mathbf{V}}^{\text{cmp}})$
Selection branch:
Compute scores on compressed: $\mathbf{s} \leftarrow \text{softmax}(\mathbf{q}_t \tilde{\mathbf{K}}^{\text{cmp}\top} / \sqrt{d})$
Select top-n important blocks: $I \leftarrow \text{TopK}(\mathbf{s}, n)$
Gather full blocks: $\tilde{\mathbf{K}}^{\text{sel}} \leftarrow [\mathbf{K}_{I_i d+1:I_i d+l}]_i$, $\tilde{\mathbf{V}}^{\text{sel}} \leftarrow [\mathbf{V}_{I_i d+1:I_i d+l}]_i$
$\mathbf{y}_t^{(\text{sel})} \leftarrow \text{SDPA}(\mathbf{q}_t, \tilde{\mathbf{K}}^{\text{sel}}, \tilde{\mathbf{V}}^{\text{sel}})$
Window branch:
$\mathbf{y}_t^{(\text{win})} \leftarrow \text{SDPA}(\mathbf{q}_t, \mathbf{K}_{t-w:t}, \mathbf{V}_{t-w:t})$
Gated fusion:
$\alpha^{(\text{cmp})} \leftarrow \sigma(\text{Linear}_{\text{cmp}}(\mathbf{q}_t))$, $\alpha^{(\text{sel})} \leftarrow \sigma(\text{Linear}_{\text{sel}}(\mathbf{q}_t))$, $\alpha^{(\text{win})} \leftarrow \sigma(\text{Linear}_{\text{win}}(\mathbf{q}_t))$
$\mathbf{y}_t \leftarrow \alpha^{(\text{cmp})} \mathbf{y}_t^{(\text{cmp})} + \alpha^{(\text{sel})} \mathbf{y}_t^{(\text{sel})} + \alpha^{(\text{win})} \mathbf{y}_t^{(\text{win})}$ **return** $\mathbf{y}_t$

4.7.3 Key Characteristics

- **Complexity**: $\mathcal{O}(t/d + n \cdot l + w)$ tokens attended per step (significantly reduced versus $\mathcal{O}(t)$ for dense).
- **Trainable**: Yes; direct training with all branches differentiable.
- **Hardware alignment**: Designed for efficient tensor core utilization; compression and selection reduce memory bandwidth pressure.
- **Strength**: 11.6× decoding speedup and 9× forward speedup on 64k sequences; outperforms full attention on many tasks.
- **Limitation**: Requires custom Triton/CUDA kernels; complex multi-branch architecture increases engineering overhead.

4.8 FASA: Frequency-Aware Sparse Attention

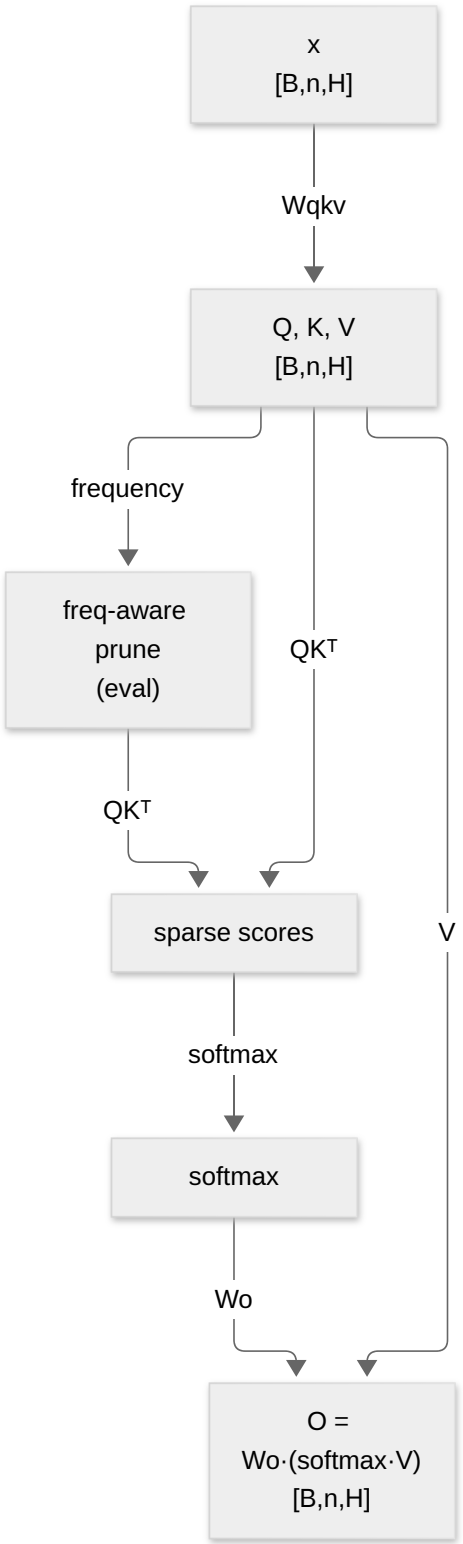


Figure 46: Dataflow of FASA: a frequency-aware prune stage (evaluation only) drops low-value key blocks before softmax.

4.8.1 Mathematical Formulation

FASA [164] is a **training-free inference-time method** that exploits rotary position embedding (RoPE) structure. Under RoPE, the token embedding is rotated by $\theta_i = B^{-2(i-1)/d}$, decomposing the d-dimensional space into $d/2$ frequency chunks (FCs).

Key insight : Only a small subset ($< 1\%$) of FCs matter for contextual awareness; most encode positional patterns.

Dominant frequency chunk identification : For each layer l and head h, identify dominant FCs via contextual agreement (CA):

$$\mathrm{CA}^{l,h,i} = \frac{|\mathrm{TopK}(\alpha^{l,h}) \cap \mathrm{TopK}(\alpha^{l,h,i})|}{K}$$

where $\alpha^{l,h}$ is the full attention mask and $\alpha^{l,h,i}$ is the mask using only FC i. Dominant FCs have high CA—they contribute meaningfully to the final attention pattern.

Token importance prediction (TIP) stage : Using only dominant FCs, compute lightweight importance per token:

$$S_t^{l,h} = \sum_{i \in \mathcal{I}_{\mathrm{dom}}^{l,h}} \alpha^{l,h,i}(\mathbf{q}_t, \mathbf{K}_{1:t}), \quad \mathcal{T}_t = \text{TopK-Indices}(S_t, N_{\mathrm{fac}})$$

Focused attention computation (FAC) stage : Compute full-precision attention on selected tokens:

$$\hat{\alpha}_{\mathrm{FAC}} = \mathrm{softmax}\left(\frac{\mathbf{q}_t \mathbf{K}_{\mathcal{T}_t}^\top}{\sqrt{d}}\right), \quad \mathbf{o}_t = \hat{\alpha}_{\mathrm{FAC}} \mathbf{V}_{\mathcal{T}_t}$$

4.8.2 Algorithmic Pseudocode

Algorithm 8 FASA: Frequency-Aware Sparse Attention (Inference-Time)

Require: Current query $\mathbf{q}_t$, cached Key/Value $\mathbf{K}_{1:t}, \mathbf{V}_{1:t}$, RoPE base B, dominant FCs $\mathcal{I}_{\text{dom}}$
Require: TIP token budget N_{tip}, FAC token budget N_{fac}
Ensure: Output $\mathbf{o}_t$

Stage 1: Token Importance Prediction (TIP)
for layer l and head h **do**
  for position $i = 1$ to t **do**
    Compute importance from dominant FCs: $s_i \leftarrow 0$
    for $fc \in \mathcal{I}_{\text{dom}}^{l,h}$ **do**
      Rotate query and key into FC fc: $\mathbf{q}^{(fc)}, \mathbf{k}_i^{(fc)}$
      $s_i^{(fc)} \leftarrow \text{softmax}(\mathbf{q}^{(fc)} \mathbf{k}_i^{(fc)\top} / \sqrt{d})$
      $s_i \leftarrow s_i + s_i^{(fc)}$
    end for
  end for
  $\mathcal{T}_t \leftarrow \text{TopK} - \text{Indices}([s_1, \ldots, s_t], N_{\text{fac}})$ ▷ Select top tokens
end for
Stage 2: Focused Attention Computation (FAC)
$\text{scores}_{\text{fac}} \leftarrow \mathbf{q}_t \mathbf{K}_{\mathcal{T}_t}^\top / \sqrt{d}$ ▷ Full-precision dot product
$\alpha_{\text{fac}} \leftarrow \text{softmax}(\text{scores}_{\text{fac}})$ ▷ Full softmax
$\mathbf{o}_t \leftarrow \alpha_{\text{fac}} \mathbf{V}_{\mathcal{T}_t}$ ▷ Weighted value sum **return** $\mathbf{o}_t$

4.8.3 Key Characteristics

- **Complexity**: TIP is $\mathcal{O}(t \cdot N_{\text{tip}})$; FAC is $\mathcal{O}(N_{\text{fac}} \cdot d)$.
- **Trainable**: No; training-free inference optimization.
- **Applicability**: Requires RoPE-based models; extended to ALiBi and MLA with modifications.
- **Strength**: Near-oracle accuracy with ≤ 256 tokens out of millions; up to 2.56× decoding speedup; orthogonal to quantization.
- **Limitation**: Offline offline calibration required; dominant FC identification adds preprocessing overhead.

Implementation note (training vs. inference). In this repository the sparse TIP/FAC path is inference-only. While the model is in training mode, the enclosing hybrid layer substitutes standard dense attention over the same **q/k/v/o** projections, so the mixer can be trained from scratch and the weights optimized during training are exactly the ones the sparse path reads at evaluation.

4.9 SpargeAttn: Two-Stage Block-Level Filtering

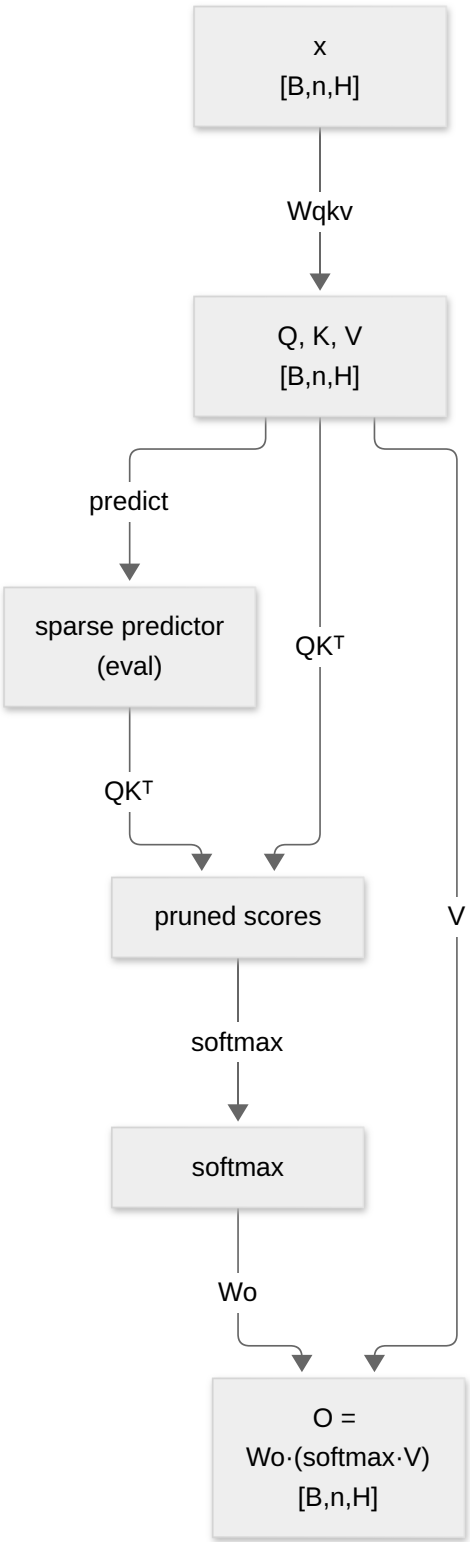


Figure 47: Dataflow of SpargeAttn: a sparse predictor (evaluation only) prunes key blocks before softmax.

4.9.1 Mathematical Formulation

SpargeAttn [180] is a **universal training-free method** that predicts which blocks of the attention matrix will contain negligible values and skips computation for those blocks.

Stage 1 — Sparse prediction : Compute low-cost proxy importance $\hat{s}_{ij}$ for block (i, j):

$$\hat{s}_{ij} = f_{\text{pred}}(\mathbf{Q}_i, \mathbf{K}_j; \text{hyperparams}), \quad \text{skip if } \hat{s}_{ij} < \epsilon_1$$

Common predictors include block-mean similarity or self-similarity statistics.

Stage 2 — Softmax-aware filtering : After computing $\tilde{P}_{ij} = \text{softmax}(Q_i K_j^\top / \sqrt{d})$, check if the block's maximum probability is negligible relative to the running softmax maximum:

$$\text{skip } P_{ij} V_j \quad \text{if} \quad \max(\tilde{P}_{ij}) < e^{m_{\text{old}} - m_{\text{new}}} \cdot \epsilon_2$$

where $m_{\text{old}}, m_{\text{new}}$ are online softmax running maxima (computed via FlashAttention-style numerics).

4.9.2 Algorithmic Pseudocode

Algorithm 9 SpargeAttn: Two-Stage Block-Sparse Filtering

Require: Query blocks $\mathbf{Q}_i$ for $i = 1, \ldots, n_b$, Key blocks $\mathbf{K}_j$ for $j = 1, \ldots, n_b$, Value blocks $\mathbf{V}_j$
Require: Prediction threshold ϵ_1, softmax threshold ϵ_2, block size b_s
Ensure: Output $\mathbf{Y}$

Initialize: online softmax max $m_{\text{global}} \leftarrow -\infty$
for block row $i = 1$ to n_b **do**
  Initialize: block row output $\mathbf{Y}_i \leftarrow 0$, local max $m_i \leftarrow -\infty$
  for block column $j = 1$ to n_b **do**
    Stage 1: Sparse Prediction
    Compute proxy importance: $\hat{s}_{ij} \leftarrow f_{\text{pred}}(\mathbf{Q}_i, \mathbf{K}_j)$
    if $\hat{s}_{ij} < \epsilon_1$ **then**
      skip this block $[i, j]$ ▷ Early termination
      continue ▷ Move to next block
    end if
    Stage 2: Compute and Filter
    $\tilde{P}_{ij} \leftarrow \text{softmax}(\mathbf{Q}_i \mathbf{K}_j^\top / \sqrt{d})$
    $m_{\text{block}} \leftarrow \max(\tilde{P}_{ij})$
    if $m_{\text{block}} < e^{m_i - m_{\text{global}}} \cdot \epsilon_2$ **then**
      Block contributes negligibly; skip ▷ Softmax-aware pruning
      continue
    end if
    Update global max: $m_i \leftarrow \max(m_i, m_{\text{block}})$
    Accumulate: $\mathbf{Y}_i \leftarrow \mathbf{Y}_i + \tilde{P}_{ij} \mathbf{V}_j$
  end for
  $m_{\text{global}} \leftarrow \max(m_{\text{global}}, m_i)$
end forreturn Y

4.9.3 Key Characteristics

- **Complexity**: Empirical $\mathcal{O}(n^2 \cdot s)$ where s is the fraction of non-skipped blocks (typically 0.2–0.5).

- **Trainable**: No; plug-and-play acceleration for existing models.
- **Universality**: Works on language models, image diffusion models, and video generation.
- **Strength**: 2.5–5× speedup compared to dense or previous sparse methods; compatible with quantization (SageAttention integration).
- **Limitation**: Speedup depends on inherent sparsity; block-level granularity may miss finer patterns; threshold tuning required per model family.

Implementation note (training vs. inference). In this repository the two-stage block filter is inference-only. While the model is in training mode, the enclosing hybrid layer substitutes standard dense attention over the same **q**/**k**/**v**/**o** projections, so the mixer can be trained from scratch and the weights optimized during training are exactly the ones the sparse path reads at evaluation.

4.10 MSA (MiniMax Sparse Attention): Blockwise Sparse Attention with Lightweight Index Branch

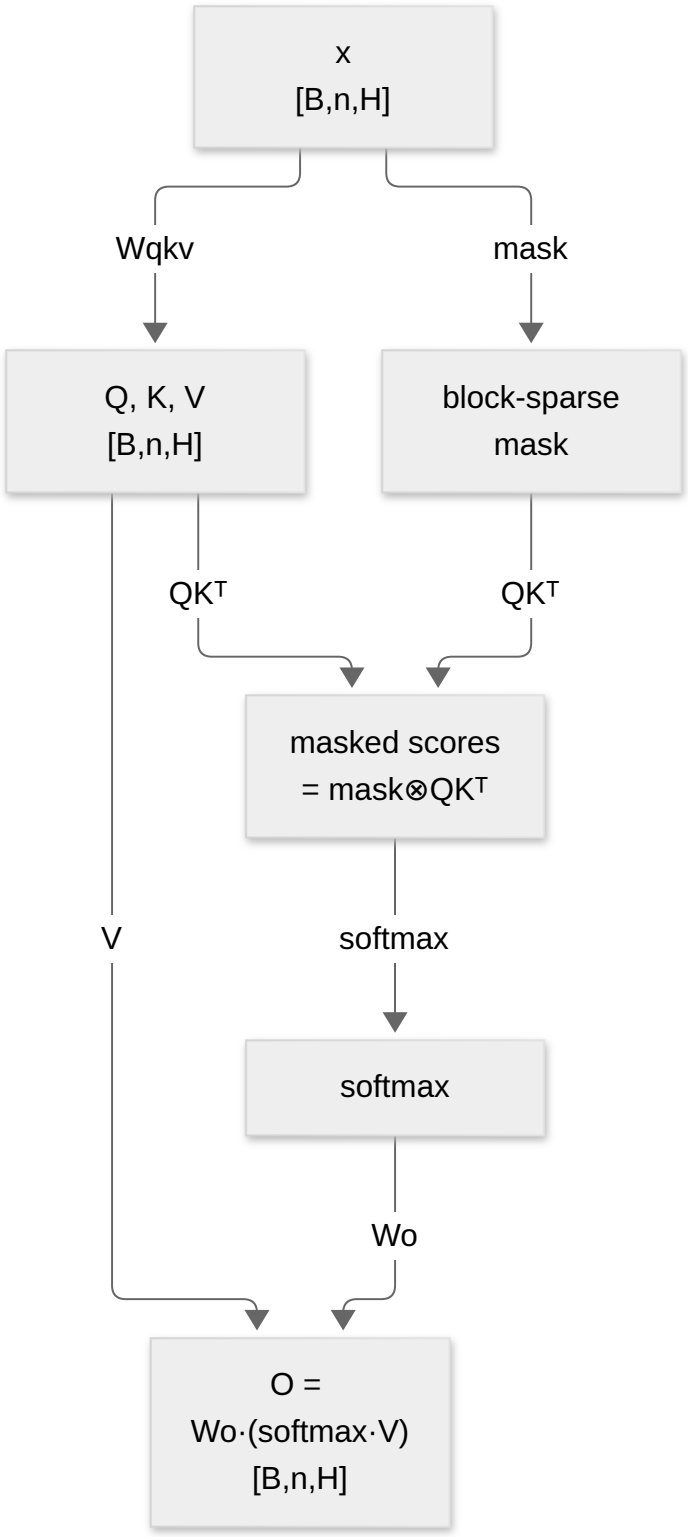


Figure 48: Dataflow of MSA: a block-sparse mask restricts the score matrix before softmax.

4.10.1 Mathematical Formulation

MSA [91] is a blockwise sparse attention mechanism built on grouped-query attention (GQA) that combines a lightweight **Index Branch** for block selection with a **Main Branch** for exact block-sparse softmax attention over the selected blocks. Let N be the sequence length, $B_k = 128$ the block size, and $k = 16$ the number of blocks selected per query (yielding a fixed per-query budget of $k \cdot B_k = 2048$ tokens regardless of N).

Index Branch : A low-dimensional projection produces index queries $\mathbf{q}_t^{\text{idx}} \in \mathbb{R}^{d_{\text{idx}}}$ and block representatives $\mathbf{r}_b \in \mathbb{R}^{d_{\text{idx}}}$ for each KV block b. The index scores per GQA group are:

$$s_{t,b}^{(g)} = \mathbf{q}_t^{\text{idx},(g)} \cdot \mathbf{r}_b, \qquad \mathcal{I}_t^{(g)} = \text{TopK}\left(\{s_{t,b}^{(g)}\}_{b=1}^{N/B_k},\ k\right)$$

The index input is **detached** (stop-gradient), so gradients do not flow back through the indexer's input; the Index Branch is trained instead by a KL alignment loss between its attention distribution and the Main Branch's distribution. Because softmax is order-preserving, the raw index scores feed directly into the top-k operator *without* an explicit softmax (**exp-free top-k**), reducing numerical cost.

Forced local block : The block containing the query position t is **always** included in $\mathcal{I}_t^{(g)}$, guaranteeing local context coverage.

Main Branch : Standard scaled dot-product attention is computed exactly over the union of selected blocks:

$$\mathbf{o}_t^{(h)} = \text{softmax}\left(\frac{\mathbf{q}_t^{(h)} \mathbf{K}_{\mathcal{I}_t}^{(h)\top}}{\sqrt{d_h}}\right) \mathbf{V}_{\mathcal{I}_t}^{(h)}$$

where $\mathbf{K}_{\mathcal{I}_t}, \mathbf{V}_{\mathcal{I}_t}$ gather the full-precision key/value tensors of the selected blocks. Selection is performed independently per GQA group, so different query groups within the same layer may attend to different block subsets.

4.10.2 Algorithmic Pseudocode

Algorithm 10 MSA: MiniMax Sparse Attention with Index + Main Branches

Require: Query $\mathbf{Q} \in \mathbb{R}^{N \times d}$, Key $\mathbf{K}$, Value $\mathbf{V}$, block size $B_k = 128$, top-k blocks $k = 16$
Require: Index projection W_{idx} (detached input), block representatives $\{\mathbf{r}_b\}$
Ensure: Output $\mathbf{Y} \in \mathbb{R}^{N \times d}$

Compute block reps: $\mathbf{r}_b \leftarrow \text{mean}(\mathbf{K}_{bB_k:(b+1)B_k})$ for $b = 1, \ldots, N/B_k$
for query position $t = 1, \ldots, N$ **do**
  for each GQA group g **do**
    $\mathbf{q}_t^{\text{idx}} \leftarrow \text{sg}(W_{\text{idx}} \mathbf{x}_t)$ ▷ detached index query
    Compute block scores: $s_b \leftarrow \mathbf{q}_t^{\text{idx}} \cdot \mathbf{r}_b$
    Identify local block: $b_{\text{local}} \leftarrow \lfloor t / B_k \rfloor$
    $\mathcal{I}_t^{(g)} \leftarrow \text{TopK}(\{s_b\}, k) \cup \{b_{\text{local}}\}$ ▷ exp-free top-k + forced local
  end for
  Gather selected KV: $\tilde{\mathbf{K}}_t \leftarrow \text{Gather}(\mathbf{K}, \mathcal{I}_t)$, $\tilde{\mathbf{V}}_t \leftarrow \text{Gather}(\mathbf{V}, \mathcal{I}_t)$
  Main branch SDPA: $\mathbf{y}_t \leftarrow \text{softmax}(\mathbf{q}_t \tilde{\mathbf{K}}_t^\top / \sqrt{d_h}) \tilde{\mathbf{V}}_t$
end for
Add KL alignment loss $\mathcal{L}_{\text{KL}} = \text{KL}(\text{index dist} \,\|\, \text{main dist})$ during training **return** $\mathbf{Y}$

4.10.3 Key Characteristics

- **Complexity**: Index branch $\mathcal{O}(H_{\text{kv}} \cdot d_{\text{idx}} \cdot N^2)$; Main branch $\mathcal{O}(H_q \cdot d_h \cdot N \cdot k \cdot B_k)$. Per-query budget fixed at $k \cdot B_k = 2048$ tokens regardless of N.
- **Trainable**: Yes; end-to-end (the Index Branch is trained via a KL alignment loss, not eval-only).
- **State**: $\mathcal{O}(N)$ KV cache (GQA-sized); no expansion of the recurrent/state dimension.
- **Strength**: Deployed on the 109B MiniMax-M3, MSA achieves a 28.4× per-token attention compute reduction at 1M context, with 14.2× prefill and 7.6× decode speedup on H800 GPUs.
- **Limitation**: Selection granularity is fixed at the block level ($B_k = 128$); the forced local block and per-group independent selection increase engineering complexity and require custom block-sparse kernels.

4.11 SparDA (Sparse Decoupled Attention): Forecast-Guided Lookahead Block Selection

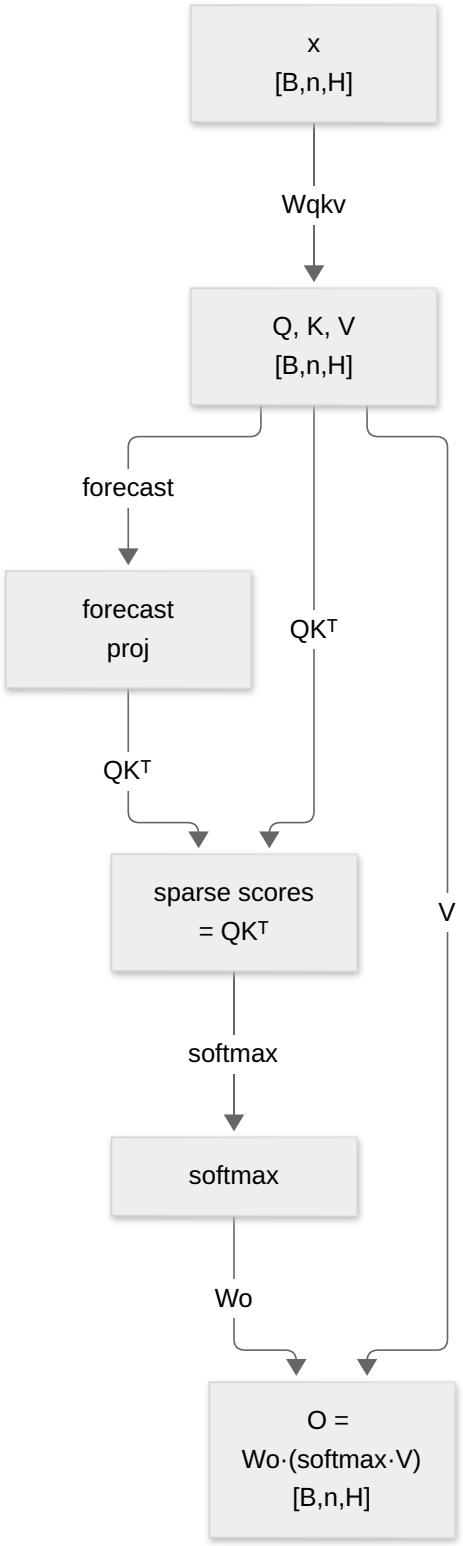


Figure 49: Dataflow of SparDA: a forecast projection guides lookahead block selection before softmax.

4.11.1 Mathematical Formulation

SparDA [45] introduces a **decoupled** sparse attention scheme that augments the standard per-layer projections Q, K, V with a fourth projection, the **Forecast**, which predicts the KV blocks that will be needed by the *next* layer. This lookahead enables overlapping CPU-to-GPU prefetch of the next layer's selected blocks with the current layer's execution, hiding the offload latency that plagues sparse-attention deployments on memory-constrained hardware.

Forecast projection : For input $\mathbf{x} \in \mathbb{R}^{N \times d}$,

$$\mathbf{F} = \mathbf{x}\mathbf{W}_F \in \mathbb{R}^{N \times H_{\text{kv}} \times d_f}$$

with **one Forecast head per GQA group**, which reduces selection overhead relative to multi-head selectors used by prior work.

Block scoring and selection : For each query i and KV block b with representative $\mathbf{r}_b$:

$$S_{i,b} = \mathbf{F}_i \cdot \mathbf{r}_b, \qquad \mathcal{I}_i = \text{TopK}(\{S_{i,b}\}_b, k)$$

Block-sparse attention : Standard scaled dot-product attention is then computed over the selected blocks:

$$\mathbf{o}_i = \text{softmax}\left(\frac{\mathbf{q}_i \mathbf{K}_{\mathcal{I}_i}^\top}{\sqrt{d_h}}\right)\mathbf{V}_{\mathcal{I}_i}$$

Training : SparDA adds fewer than 0.5% additional parameters. Only the Forecast projections are trained, by matching the original (frozen) sparse selector's attention distribution—a lightweight distillation that preserves the base model's behaviour while enabling lookahead prefetch.

4.11.2 Algorithmic Pseudocode

Algorithm 11 SparDA: Sparse Decoupled Attention with Forecast Prefetch

Require: Input $\mathbf{x} \in \mathbb{R}^{N \times d}$, cached Key/Value blocks, top-k blocks k
Require: Projections W_q, W_k, W_v, W_F; frozen base selector π_{orig}
Ensure: Output $\mathbf{Y}$; prefetched blocks for next layer

```
Compute Forecast: F ← xW_F                                   ▷ one head per GQA group
for query i = 1, ..., N do
    Compute block scores: S_{i,b} ← F_i · r_b for each block b
    Select: I_i ← TopK({S_{i,b}}, k)
    Prefetch K_{I_i}, V_{I_i} from CPU to GPU for the next layer   ▷ overlaps with current layer
end for
Compute block-sparse SDPA over current-layer selected blocks: Y ← BlockSparseSDPA(Q, K, V, {I_i})
Training loss: L ← KL(Forecast dist ‖ π_orig)                ▷ only W_F is updated
return Y
```

4.11.3 Key Characteristics

- **Complexity**: Forecast $\mathcal{O}(H_{\text{kv}} \cdot d_f \cdot N \cdot N_{\text{blocks}})$; Main attention $\mathcal{O}(H_q \cdot d_h \cdot N \cdot k \cdot B_k)$.
- **Trainable**: Yes; only the Forecast projections are trained ($< 0.5\%$ extra parameters).
- **Strength**: On two 8B sparse-pretrained models, SparDA achieves up to 1.25× prefill and 1.7× decode speedup over the sparse-attention offload baseline, and up to 5.3× decode throughput versus a non-offload sparse baseline.

- **Limitation**: Requires a CPU–GPU offload pipeline and a frozen base selector; the lookahead benefit is contingent on next-layer block demand being predictable from the current Forecast.

4.12 LSA (Latent Sparse Attention): Semantic Key Clustering with Top-k Selection

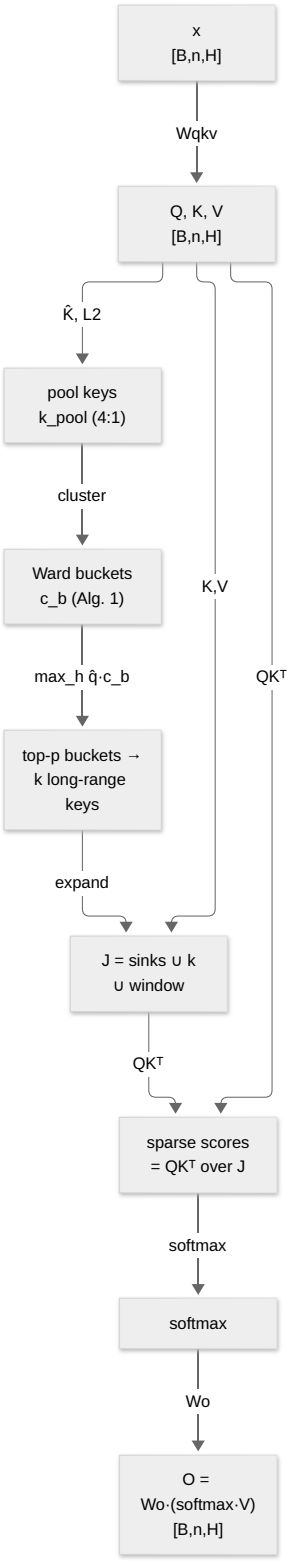


Figure 50: Dataflow of N-LSA: pooled keys are Ward-clustered into buckets; each query scores the centroids, expands the best buckets into a top-k long-range key set, and attends to sinks + selection + window.

4.12.1 Mathematical Formulation

Latent Sparse Attention [89] replaces the per-key (or per-block) scan of a sparse indexer with **semantic clustering**: keys are clustered by content in a learned latent space, and each query scores the bucket centroids instead of every key. The paper defines two trainable variants: Distilled-LSA (D-LSA), which adds small index projections to a frozen pre-trained model, and **Native-LSA (N-LSA)**, trained from scratch on the next-token loss alone. The implementation in this codebase follows N-LSA.

Semantic clustering : Each index key is first L2-normalised,

$$\hat{\mathbf{k}}_j = \frac{\mathbf{k}_j}{\|\mathbf{k}_j\|_2},$$

and every n_{pool} consecutive keys are summed into one pool key, so clustering runs over T/n_{pool} pool keys rather than T keys. Keys are processed in *cluster blocks* of n_{cluster} consecutive tokens; once a block is complete, its pool keys are merged bottom-up with Ward's criterion into $m = n_{\text{cluster}}/\bar{n}_b$ buckets whose unit-norm centroids are

$$\mathbf{c}_b = \frac{\mathbf{k}_b^{\text{sum}}}{\|\mathbf{k}_b^{\text{sum}}\|_2},$$

where $\mathbf{k}_b^{\text{sum}}$ is the sum of the keys in bucket b and $\bar{n}_b$ is the average number of keys per bucket. In every round all pairs that are each other's cheapest merge are merged in parallel; with the Gram matrix $G_{bb'} = \mathbf{k}_b^{\text{sum}} \cdot \mathbf{k}_{b'}^{\text{sum}}$ the merge cost is

$$a_b = \sqrt{G_{bb}}, \qquad a_{b\cup b'} = \sqrt{G_{bb} + G_{b'b'} + 2G_{bb'}}, \qquad \Delta(b, b') = a_b + a_{b'} - a_{b\cup b'},$$

and a merge is skipped when the combined bucket would exceed n_{max}^b keys. The bucket cache stores every centroid together with the positions of its keys. **Once a cluster block is clustered, its buckets never change**, which is what makes incremental decoding exact in this implementation.

Semantic search : Each query is L2-normalised, $\hat{\mathbf{q}}_i = \mathbf{q}_i/\|\mathbf{q}_i\|_2$, and scores the centroids of every bucket in the cluster blocks before its own. For N-LSA the score is the best similarity any query head gives the centroid,

$$s_{i,b} = \max_h \hat{\mathbf{q}}_{i,h} \cdot \mathbf{c}_b,$$

so a single index list $\mathcal{J}_i$ is shared by every query head. The top-p buckets are expanded best-first until k *long-range* keys are selected; the expansion always fills k because p is chosen such that $p\, n_{\text{pool}} \geq k$, and keys already covered by the sink prefix or the window are skipped. Every query additionally attends to n_{sink} sink keys and the $n_w = n_{\text{cluster}}$ most recent keys (its own, not-yet-clustered block), giving a set

$$\mathcal{J}_i \subseteq \{\text{sinks}\} \cup \{\text{top-}k \text{ long-range keys}\} \cup \{\text{own-block window}\}, \qquad |\mathcal{J}_i| \leq n_{\text{sink}} + k + n_w.$$

Attention : Standard scaled dot-product attention is computed only over $\mathcal{J}_i$,

$$z_{i,j} = \frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d_h}}, \qquad \mathbf{o}_i = \text{softmax}_{j\in\mathcal{J}_i}(z_{i,j})\, \mathbf{V}_{\mathcal{J}_i},$$

where the softmax is renormalised per head over the shared selection. N-LSA **adds no parameters**, uses **no positional encoding** (RoPE could be supported by clustering only the RoPE-free part of the keys), and **stops the gradient through the selection**: the clustering, centroid scoring and expansion run without gradients, so the next-token loss trains the attention projections only through the keys each query attends to.

4.12.2 Algorithmic Pseudocode

Algorithm 12 N-LSA: Ward clustering of one cluster block and flat semantic search

Require: Index keys $\mathbf{k}_j$ of a complete cluster block; sizes $n_{\text{cluster}}, n_{\text{pool}}, m, n^b_{\max}$
Require: Queries $\mathbf{q}_i$; sink budget n_{sink}; long-range budget k; window $n_w = n_{\text{cluster}}$
Ensure: Bucket cache with m centroids and their key positions; output $\mathbf{Y}$

$\hat{\mathbf{k}}_j \leftarrow \mathbf{k}_j / \|\mathbf{k}_j\|_2$ for all keys in the block
for all pool keys $\mathbf{k}^{\text{pool}}$, in parallel **do**
    $\mathbf{k}^{\text{pool}} \leftarrow$ sum of its n_{pool} consecutive $\hat{\mathbf{k}}_j$
end for
start one bucket b per pool key: $\mathbf{k}^{\text{sum}}_b \leftarrow \mathbf{k}^{\text{pool}}$, $|b| \leftarrow n_{\text{pool}}$
while more than m buckets remain **do**
    $G \leftarrow [\mathbf{k}^{\text{sum}}_b \cdot \mathbf{k}^{\text{sum}}_{b'}]$ ▷ Gram matrix
    $\Delta(b, b') \leftarrow \sqrt{G_{bb}} + \sqrt{G_{b'b'}} - \sqrt{G_{bb} + G_{b'b'} + 2G_{bb'}}$
    merge the cheapest mutual pairs with $|b| + |b'| \le n^b_{\max}$, at most as many as buckets above m
end while
$\mathbf{c}_b \leftarrow \mathbf{k}^{\text{sum}}_b / \|\mathbf{k}^{\text{sum}}_b\|_2$; store $\mathbf{c}_b$ and the positions of bucket b
$\hat{\mathbf{q}}_i \leftarrow \mathbf{q}_i / \|\mathbf{q}_i\|_2$ for all queries
$s_{i,b} \leftarrow \max_h \hat{\mathbf{q}}_{i,h} \cdot \mathbf{c}_b$ over the buckets before the query's own block
select the top-p buckets; expand them best-first into k long-range keys (skip sinks and the window)
$\mathcal{J}_i \leftarrow \{\text{sinks}\} \cup \{\text{long-range keys}\} \cup \{\text{own-block window}\}$
$\mathbf{Y} \leftarrow$ attention over $\mathcal{J}_i$ with a per-head softmax renormalised over the shared selection
return $\mathbf{Y}$

4.12.3 Key Characteristics

- **Complexity**: Flat search scores $\mathcal{O}(T/\bar{n}_b)$ centroids per query, i.e. $\mathcal{O}(T^2/\bar{n}_b + T(k + n_w + n_{\text{sink}})d_h)$ for prefill and $\mathcal{O}(T/\bar{n}_b + k + n_w + n_{\text{sink}})$ per decoded token; the tree variant of the paper makes the indexer $\mathcal{O}(T \log T)$.

- **Trainable**: Yes, natively from scratch on the next-token loss (N-LSA); the selection is non-differentiable and the gradient is stopped through it. No teacher, no index projections, no auxiliary loss—and **no extra parameters**.

- **Strength**: In the paper's 0.5B hybrid, N-LSA matches the full-attention validation loss at 128k tokens (1.085 against 1.085; PISA 1.100, NSA 1.117, SWA 1.154) while training up to 1.6× faster per attention layer at 128k. Because every cluster block is immutable once complete, cached decoding in this codebase is exact rather than an approximation.

- **Limitation**: The flat indexer remains quadratic in context length (with a $1/\bar{n}_b$ constant); tree search is left to future work. The paper's single index list assumes a single shared KV head ($n_{kv} = 1$), and N-LSA deliberately uses no positional encoding.

4.13 Design Space and Selection Criteria

The eleven sparse attention methods occupy complementary positions in a multidimensional design space:

- **Geometric/fixed patterns** (Sparse Transformer, Longformer): Simple to implement and analyze, but patterns do not adapt to content.

- **Fixed-memory KV eviction** (SWA with sinks): The cheapest possible policy—keep a sink prefix and a recent window, discard everything else. No extra parameters, no training required, bounded memory.
- **Learned selection** (SparseK, NSA): Enable adaptation through trainable selection networks, at the cost of higher implementation complexity.
- **Training-free acceleration** (FASA, SpargeAttn): Enable drop-in speedup of existing models without retraining, suited for deployed systems.
- **Semantic selection** (LSA): Clusters keys by content and caches one centroid per bucket, so the indexer scans a small multiple of $T/\bar{n}_b$ instead of T; the only member whose per-block statistics are immutable, which makes exact cached decoding possible.
- **Theoretical guarantees** (BigBird): Provide formal proofs of expressive completeness, important for understanding safety margins.

Selection guidance:

1. **New model training** where resources permit: NSA or SparseK for maximal inference speedup; BigBird for theoretical assurance; LSA when semantic clustering and exact cached decoding at ultra-long context matter; SWA with sinks as the strongest low-cost baseline.
2. **Long-context document understanding**: SWA with sinks (window 256–512) for the best accuracy per byte of KV cache; Longformer or FASA for practical simplicity; NSA for extreme scale; LSA when the indexer scan itself dominates.
3. **Accelerating existing models**: SWA with sinks as the default training-free baseline (beat most linear-attention retrofits in the original study); FASA for RoPE-based LLMs; SpargeAttn for any architecture.
4. **Constrained environments**: SWA with sinks for a fixed memory budget; Sparse Transformer for simplicity and proven efficiency at moderate scale.

5 Gated Attention Families—Complete Literature Analysis

5.1 Executive Summary: Gating for Memory Control

The emerging field of *gated attention mechanisms* addresses a fundamental challenge in sequence modeling: how should information be retained, forgotten, and updated as the model processes new tokens? Traditional additive recurrent states accumulate all information without erasure, leading to memory saturation and inability to adapt to changing context. Softmax attention provides full expressiveness but uses $\mathcal{O}(Ld)$ KV cache during inference, limiting context window size. Gated mechanisms provide a middle ground: selective control over information survival.

The unifying principle across gated architectures is that gating controls *what information survives*. Gates operate at three distinct levels:

1. **Recurrent state decay** (GLA, HGRN2): Multiplicative gates on the memory matrix S_t control how much previous state persists into the next step.
2. **Write strength in recurrent updates** (DeltaNet, Gated DeltaNet, Gated DeltaNet-2): Gates control the magnitude and channel-wise direction of new information written into memory.
3. **Softmax logit biasing** (FoX): Gates inject token-level recency bias into attention logit computation.

4. **Post-attention sparsity** (Gated Softmax): Gates selectively scale SDPA output channels.

This appendix synthesizes eight major gated-attention architectures published 2023–2026, analyzing their mathematical foundations, hardware efficiency characteristics, and empirical trade-offs.

5.2 1. Gated Linear Attention (GLA)

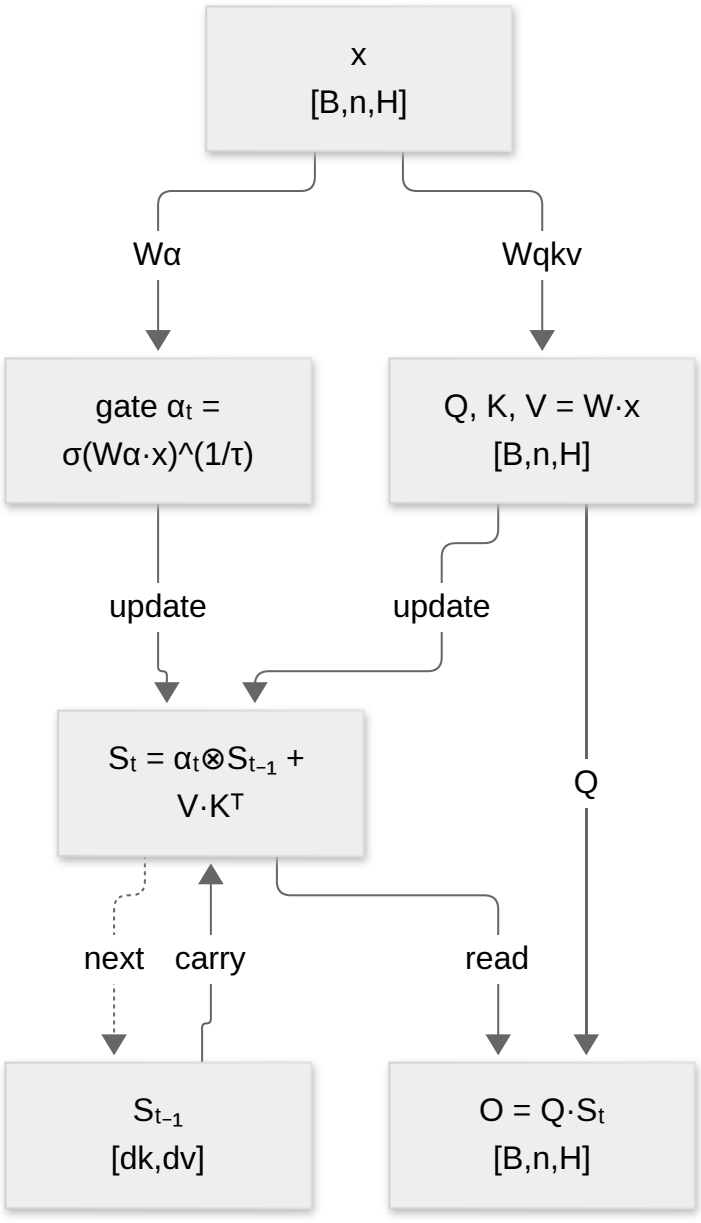


Figure 51: Dataflow of GLA: a sigmoid gate decays the recurrent state before the key/value outer product is added.

5.2.1 Mathematical Foundation

Gated Linear Attention [169] augments vanilla linear attention (which uses a matrix-valued recurrent state) with data-dependent multiplicative decay. Standard linear attention reformulates

the traditional softmax mechanism as a recurrence over hidden states:

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \mathbf{v}_t\mathbf{k}_t^\top, \quad \mathbf{o}_t = \mathbf{S}_t\mathbf{q}_t$$

where state $\mathbf{S}_t \in \mathbb{R}^{d_v \times d_k}$ accumulates outer products. This purely additive formulation suffers from "memory overload": information accumulates monotonically and cannot be erased, degrading performance on tasks requiring context selection.

GLA introduces a data-dependent **diagonal gating matrix** $\mathbf{G}_t \in [0,1]^{d_k \times d_k}$:

$$\mathbf{S}_t = \mathbf{G}_t \odot \mathbf{S}_{t-1} + \mathbf{v}_t\mathbf{k}_t^\top, \quad \mathbf{o}_t = \mathbf{S}_t\mathbf{q}_t$$

The gate $\mathbf{G}_t$ is parameterized with an outer-product structure:

$$\mathbf{G}_t = \mathbf{1}\boldsymbol{\alpha}_t^\top, \quad \boldsymbol{\alpha}_t = \text{logsigmoid}\left(\mathbf{W}_{gk}\mathbf{x}_t\right)/c$$

where $\mathbf{W}_{gk}$ is a low-rank projection ($\mathbb{R}^{d \to d_k}$) and $c \approx 16$ is a normalizer. The logsigmoid activation maps $\alpha_t \in \mathbb{R}$ to approximately $[-1/2, 0]$, and division by c further contracts this to near-identity for initialization. The resulting gate $G_t = 1 + \alpha_t \approx 1$ near initialization, then learns to decay ($\alpha_t \to -\infty$) historical information.

Algorithm 13 Gated Linear Attention (Recurrent Form)

Require: Input $\mathbf{x}_t$; prior state $\mathbf{S}_{t-1}$
Require: Projections: W_q, W_k, W_v (standard); W_{gk} (gate projection)
Ensure: Output $\mathbf{o}_t$ and updated state $\mathbf{S}_t$

$\mathbf{q}_t \leftarrow W_q\mathbf{x}_t$
$\mathbf{k}_t \leftarrow W_k\mathbf{x}_t$
$\mathbf{v}_t \leftarrow W_v\mathbf{x}_t$
Compute gate: $\boldsymbol{\alpha}_t \leftarrow \text{logsigmoid}(W_{gk}\mathbf{x}_t)/16$ ▷ Per-key-dim decay
Apply outer-product gating: $\mathbf{G}_t \leftarrow \mathbf{1}\boldsymbol{\alpha}_t^\top$ ▷ diagonal outer product
Decay previous state: $\mathbf{S}_t^{\text{decay}} \leftarrow \mathbf{G}_t \odot \mathbf{S}_{t-1}$ ▷ element-wise product
Add new association: $\mathbf{S}_t \leftarrow \mathbf{S}_t^{\text{decay}} + \mathbf{v}_t\mathbf{k}_t^\top$
Retrieve via query: $\mathbf{o}_t^{\text{raw}} \leftarrow \mathbf{S}_t\mathbf{q}_t$
Apply output gate: $\mathbf{o}_t \leftarrow \text{GroupNorm}(\mathbf{o}_t^{\text{raw}}) \otimes \text{SiLU}(W_g\mathbf{x}_t)$ **return** $\mathbf{o}_t$, $\mathbf{S}_t$

5.2.2 Hardware-Efficient Training

For parallel training, GLA uses a chunkwise block-wise algorithm that groups tokens into chunks C and computes recurrent transitions in parallel:

$$\mathbf{O}_{[t]} = \overleftarrow{\mathbf{Q}}_{[t]}\mathbf{S}_{[t]}^\top + \left(\mathbf{Q}_{[t]}\mathbf{K}_{[t]}^\top \odot \boldsymbol{\Gamma}_{[t]}\right)\mathbf{V}_{[t]}$$

where $\overleftarrow{\mathbf{Q}}_{[t]}$ are attending to the state at the chunk boundary (lookback), and $\boldsymbol{\Gamma}_{[t]}$ is the causal mask with decay-aware scaling:

$$\boldsymbol{\Gamma}_{[t],ij} = \begin{cases} \prod_{l=j}^{i-1} \alpha_l & \text{if } i > j \text{ (lookback)} \\ \prod_{l=1}^{i-j} \alpha_l & \text{if } i \le j \text{ (in-chunk)} \end{cases}$$

This design maximizes matmul operations susceptible to tensor-core acceleration, achieving sub-quadratic total complexity $\mathcal{O}(nLd^2/C)$ where L is the number of attention heads and C is chunk size.

5.2.3 Strengths and Limitations

Aspect	Strengths	Limitations
Computational efficiency	Sub-quadratic training via chunkwise parallelism. Constant-memory inference $O(d^2)$.	Requires custom CUDA/Triton kernels; not available in all frameworks.
Context generalization	Extends 2K-token training to 20K+ tokens with negligible perplexity degradation.	Still underperforms softmax on retrieval-heavy benchmarks (needle-in-haystack).
Selectivity	Per-key-dimension data-dependent decay.	Gate structure limits per-key-value selectivity; no direct control over which specific associations to erase.

5.3 2. DeltaNet: Error-Correcting Linear Attention

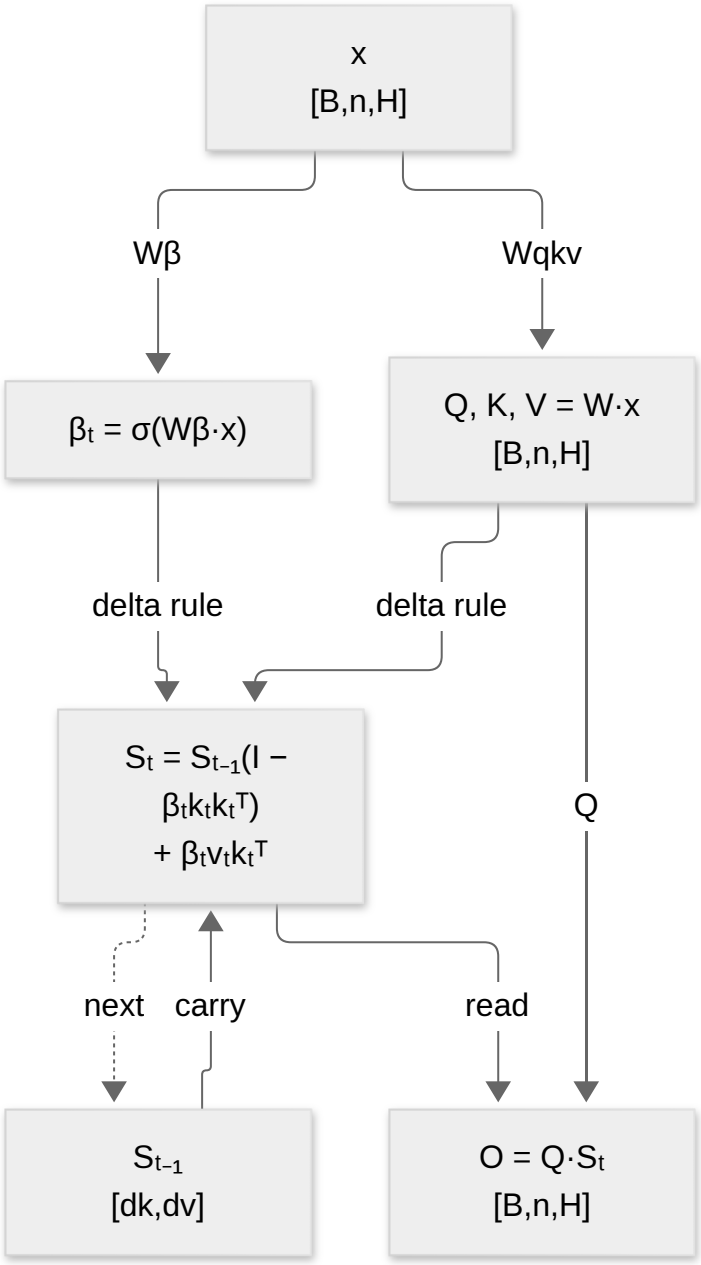


Figure 52: Dataflow of DeltaNet: the beta gate applies an error-correcting delta rule to the recurrent state.

5.3.1 Mathematical Foundation

DeltaNet [170] applies a classical **delta learning rule** to linear attention state updates. Instead of additive accumulation, DeltaNet computes the difference between the predicted value (retrieved from memory) and the target value, then performs an error-correcting update:

$$\mathbf{S}_t = \mathbf{S}_{t-1}(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top) + \beta_t \mathbf{v}_t \mathbf{k}_t^\top$$

where $\beta_t \in (0, 1)$ is a data-dependent **writing strength** scalar. This update can be decomposed as an erase-write operation:

$$\mathbf{v}_t^{\text{old}} = \mathbf{S}_{t-1}\mathbf{k}_t \quad \text{(current prediction)}, \quad \mathbf{v}_t^{\text{updated}} = \beta_t \mathbf{v}_t + (1 - \beta_t)\mathbf{v}_t^{\text{old}} \quad \text{(blend old/new)}$$

so that:

$$\mathbf{S}_t = \mathbf{S}_{t-1} - \mathbf{v}_t^{\text{old}}\mathbf{k}_t^\top + \mathbf{v}_t^{\text{updated}}\mathbf{k}_t^\top$$

The key insight is that this update rule minimizes an online MSE loss at each timestep:

$$\mathcal{L}_t(\mathbf{S}) = \tfrac{1}{2}\|\mathbf{S}\mathbf{k}_t - \mathbf{v}_t\|^2 \Rightarrow \nabla_{\mathbf{S}}\mathcal{L}_t = (\mathbf{S}\mathbf{k}_t - \mathbf{v}_t)\mathbf{k}_t^\top$$

One step of SGD with learning rate β_t yields the delta rule update. This connection to test-time training (TTT) provides theoretical grounding: DeltaNet is directly optimizing for value prediction at inference time.

5.3.2 Efficient Parallel Training

DeltaNet's transition matrix $\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top$ is a generalized Householder matrix (rank-1 update to identity). For chunkwise training, DeltaNet uses the **WY representation**, which represents the product of m such rank-1 updates compactly:

$$\prod_{i=1}^{m}(\mathbf{I} - \beta_i \mathbf{k}_i \mathbf{k}_i^\top) = \mathbf{I} - \mathbf{W}\mathbf{Y}^\top$$

where $\mathbf{W}, \mathbf{Y} \in \mathbb{R}^{d \times m}$ are constructed recursively. This factorization enables matmul-rich parallelism without materializing the full product, keeping chunkwise training efficient.

Algorithm 14 DeltaNet State Update with WY Representation

Require: Input chunk $\mathbf{X}_{[t]}$; prior state $\mathbf{S}_{t-1}$
Require: L2-normalized queries $\hat{\mathbf{Q}}_t$, keys $\hat{\mathbf{K}}_t$, values $\mathbf{V}_t$
Ensure: Output $\mathbf{O}_{[t]}$ and updated state $\mathbf{S}_t$

Chunkwise phase:
for position $i = 1$ to chunk_size **do**
  Normalize: $\hat{\mathbf{k}}_i \leftarrow \hat{\mathbf{K}}_t[i]/\|\hat{\mathbf{K}}_t[i]\|$, $\hat{\mathbf{q}}_i \leftarrow \hat{\mathbf{Q}}_t[i]/\|\hat{\mathbf{Q}}_t[i]\|$
  Compute write strength: $\beta_i \leftarrow \text{sigmoid}(W_\beta \mathbf{x}_i)$
  Prediction: $\hat{\mathbf{v}}^{\text{pred}} \leftarrow \mathbf{S}_{i-1}^{\text{in-chunk}}\hat{\mathbf{k}}_i$
  Error-aware blend: $\mathbf{v}_i^{\text{update}} \leftarrow \beta_i(\mathbf{v}_i - \hat{\mathbf{v}}^{\text{pred}})$
  Erase-write: $\mathbf{S}_i^{\text{in-chunk}} \leftarrow \mathbf{S}_{i-1}^{\text{in-chunk}} - \hat{\mathbf{v}}^{\text{pred}}\hat{\mathbf{k}}_i^\top + (\beta_i \mathbf{v}_i + (1 - \beta_i)\hat{\mathbf{v}}^{\text{pred}})\hat{\mathbf{k}}_i^\top$
  Output: $\mathbf{o}_i \leftarrow \mathbf{S}_i^{\text{in-chunk}}\hat{\mathbf{q}}_i$
end for
Inter-chunk phase: Use WY representation of transition products **return** $\mathbf{O}_{[t]}$, $\mathbf{S}_t$

5.3.3 Strengths and Limitations

Aspect	Strengths	Limitations
Associative recall	Perfect recall on Multi-Query Associative Recall (MQAR) benchmark; error-correcting semantics naturally fit retrieval patterns.	Without global forgetting, memory still crowds over extreme sequence lengths; requires periodic reset mechanisms for unbounded context.
Theory	Grounded in online MSE optimization; clear connection to test-time training paradigm.	Requires L2-normalized keys for numerical stability; double-width normalization adds overhead.
Throughput	WY representation enables efficient chunkwise training.	Slightly slower than Mamba2 per-token due to richer transition matrices (rank-1 instead of diagonal).

5.4 3. Gated DeltaNet: Synthesis of Gating and Error Correction

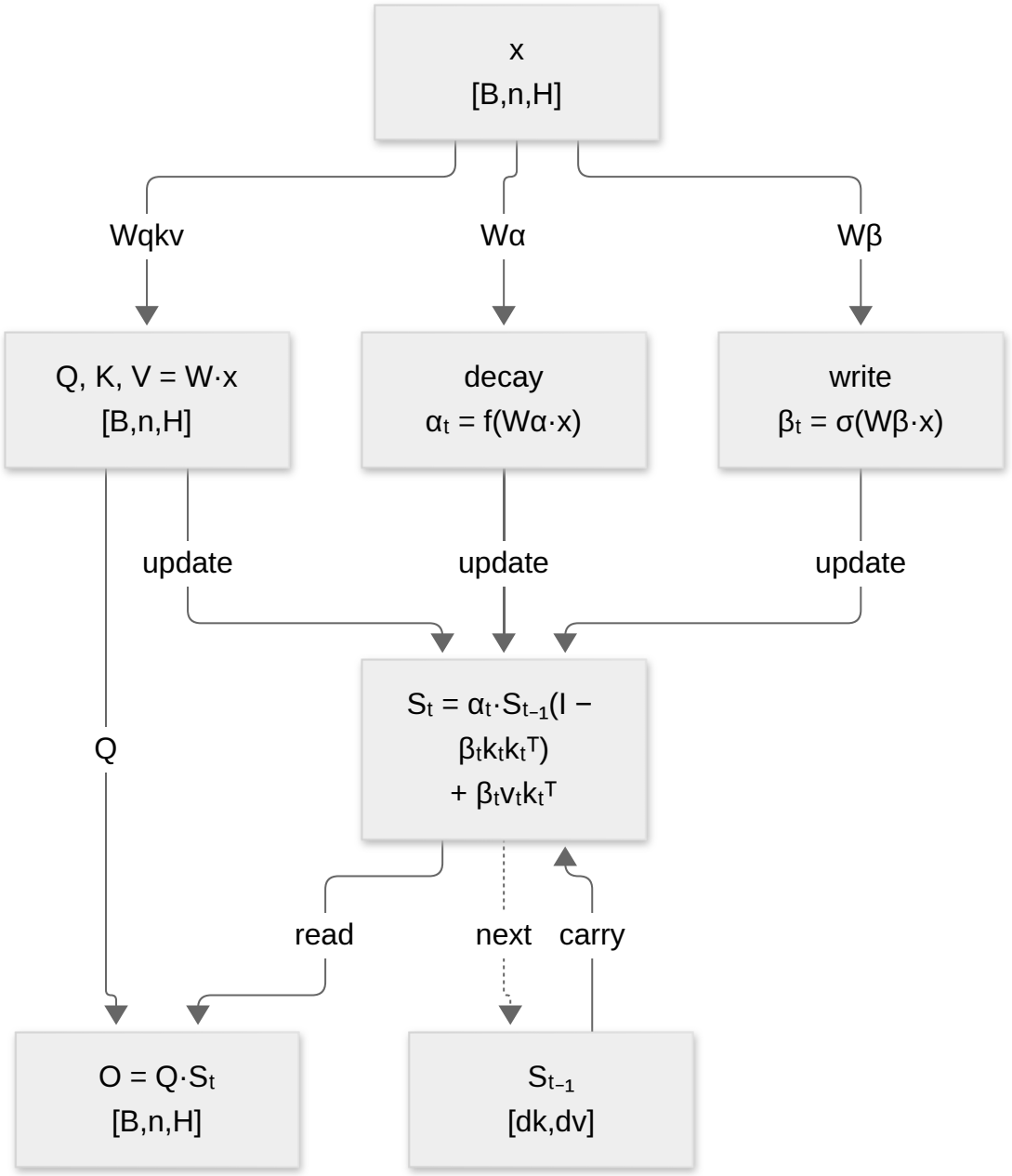


Figure 53: Dataflow of Gated DeltaNet: decay and write gates modulate the delta-rule state update.

5.4.1 Mathematical Foundation

Gated DeltaNet [171] synthesizes the strengths of GLA and DeltaNet by combining a **decay gate** α_t (from GLA) with the **writing strength gate** β_t (from DeltaNet):

$$\mathbf{S}_t = \mathbf{S}_{t-1}\left(\alpha_t(\mathbf{I} - \beta_t \mathbf{k}_t\mathbf{k}_t^\top)\right) + \beta_t \mathbf{v}_t\mathbf{k}_t^\top$$

Rewriting for clarity:

$$\mathbf{S}_t = \alpha_t \mathbf{S}_{t-1}(\mathbf{I} - \beta_t \mathbf{k}_t\mathbf{k}_t^\top) + \beta_t \mathbf{v}_t\mathbf{k}_t^\top$$

The two gates are complementary:

- $\alpha_t \to 0$ rapidly erases historical state: $\mathbf{S}_t \approx \beta_t \mathbf{v}_t \mathbf{k}_t^\top$ (context reset).
- $\alpha_t \to 1$ recovers pure delta-rule behavior: $\mathbf{S}_t \approx \mathbf{S}_{t-1}(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top) + \beta_t \mathbf{v}_t \mathbf{k}_t^\top$ (targeted updates).
- $\beta_t \to 0$ skips writing but decays: $\mathbf{S}_t \approx \alpha_t \mathbf{S}_{t-1}$ (pure forgetting).
- $\beta_t \to 1$ replaces memory aggressively: $\mathbf{S}_t \approx \alpha_t \mathbf{S}_{t-1} + \mathbf{v}_t \mathbf{k}_t^\top$ (replacement).

From an online learning perspective, Gated DeltaNet corresponds to:

$$\min_{\mathbf{S}_t} \|\mathbf{S}_t - \alpha_t \mathbf{S}_{t-1}\|_F^2 - 2\langle \mathbf{S}_t \mathbf{k}_t, \beta_t(\mathbf{v}_t - \alpha_t \mathbf{S}_{t-1}\mathbf{k}_t)\rangle$$

which introduces an adaptive weight decay α_t into an SGD-like update—analogous to decoupled weight decay in deep learning optimization.

Algorithm 15 Gated DeltaNet Parallel Chunkwise Forward

Require: Input chunk $\mathbf{X}_{[i]}$; prior state $\mathbf{S}_{i-1}$; chunk size C
Require: Projections: $W_q, W_k, W_v, W_\alpha, W_\beta$
Ensure: Output $\mathbf{O}_{[i]}$ and state $\mathbf{S}_i$
In-chunk computation:
for $j = 1$ to C **do**
  $\mathbf{q}_j \leftarrow W_q \mathbf{x}_j$, $\mathbf{k}_j \leftarrow W_k \mathbf{x}_j$, $\mathbf{v}_j \leftarrow W_v \mathbf{x}_j$
  $\alpha_j \leftarrow \text{sigmoid}(W_\alpha \mathbf{x}_j)$, $\beta_j \leftarrow \text{sigmoid}(W_\beta \mathbf{x}_j)$
  Apply combined gate: $\mathbf{S}_j \leftarrow \alpha_j \mathbf{S}_{j-1}(\mathbf{I} - \beta_j \mathbf{k}_j \mathbf{k}_j^\top) + \beta_j \mathbf{v}_j \mathbf{k}_j^\top$
  $\mathbf{o}_j \leftarrow \mathbf{S}_j \mathbf{q}_j$
end for
Inter-chunk recurrence:
$\mathbf{S}_i \leftarrow \mathbf{S}_C$ from in-chunk; propagate across chunks via cumulative $\prod \alpha$ masks
Output gating: $\mathbf{O}_{[i]} \leftarrow \text{GroupNorm}(\mathbf{O}^{\text{raw}}) \otimes \text{SiLU}(W_g \mathbf{X}_{[i]})$ **return** $\mathbf{O}_{[i]}$, $\mathbf{S}_i$

5.4.2 Empirical Performance and Trade-offs

Gated DeltaNet has been integrated into Alibaba's Qwen3-Next and Qwen3.5 production models. Empirically:

Task	Gated DeltaNet	Mamba2	DeltaNet	GLA
Language Modeling	**1.0**×	1.08×	1.05×	1.12×
Associative Recall	**1.0**×	$--$ (*noperfect*)	**1.0**×	0.75
Length Extrapolation	**1.0**×	0.98×	0.96×	0.94×
Throughput (tokens/sec)	0.85×	**1.0**×	0.82×	0.88×

Gated DeltaNet achieves the best balance across diverse tasks at the cost of slightly reduced throughput due to richer transition matrices.

5.5 4. HGRN2: Hierarchical Gating with Outer-Product Expansion

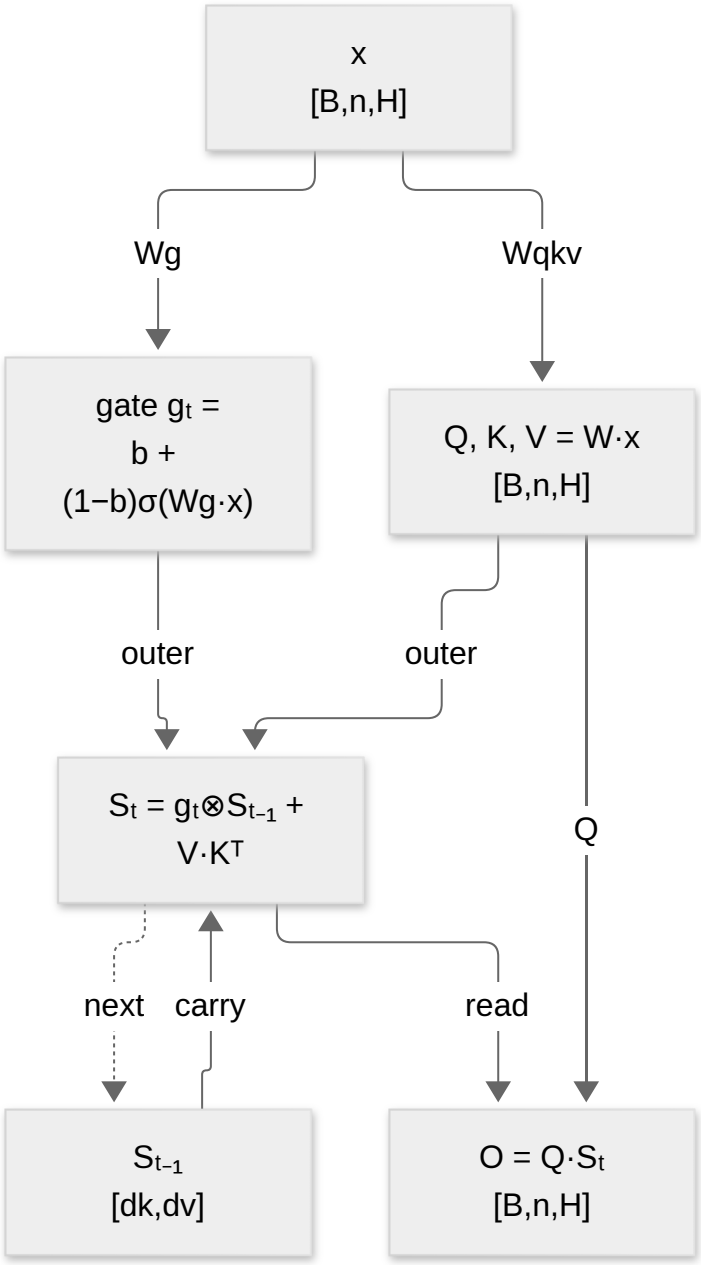


Figure 54: Dataflow of HGRN2: a hierarchical gate mixes the carried state with the key/value outer product.

5.5.1 Mathematical Foundation

HGRN2 [128] uses an outer-product-based state expansion with **hierarchically lower-bounded forget gates**. The state update is:

$$\mathbf{S}_t = \mathrm{diag}(\mathbf{g}_t) \cdot \mathbf{S}_{t-1} + \mathbf{v}_t \mathbf{k}_t^\top, \quad \mathbf{o}_t = \mathbf{S}_t \mathbf{q}_t$$

where $\mathbf{g}_t \in [b_\ell, 1]^d$ is a lower-bounded forget gate for layer ℓ. The bounds satisfy:

$$0 \leq b_{\ell_{\text{shallow}}} < b_{\ell_{\text{middle}}} < b_{\ell_{\text{deep}}} \leq 1$$

i.e., bounds increase (become less restrictive) in the deeper layers. The gate itself is computed as:

$$\mathbf{g}_t = b_\ell + (1 - b_\ell) \cdot \sigma(W_g \mathbf{x}_t)$$

where σ is sigmoid. When W_g outputs weakly negative logits, $\mathbf{g}_t \approx b_\ell$ (forced retention at shallow layers). When outputs are strongly positive, $\mathbf{g}_t \approx 1$ (memory refresh at any layer).

The hierarchical structure encourages different layers to specialize to different timescales: shallow layers model local dependencies (high retention due to binding b_ℓ), while deeper layers capture long-range structure (flexible gating with high upper bound).

Algorithm 16 HGRN2 Forward with Hierarchical Bounds

Require: Input $\mathbf{x}_t$; prior state $\mathbf{S}_{t-1}$; layer index $\ell \in [0, L-1]$
Require: Non-decreasing bounds: $0 \leq b_0 < b_1 < \cdots < b_{L-1} \leq 1$
Ensure: Output $\mathbf{o}_t$ and state $\mathbf{S}_t$
1: Layer-specific retain bound: $b \leftarrow b_\ell$
2: Project: $\mathbf{q}_t \leftarrow W_q \mathbf{x}_t$, $\mathbf{k}_t \leftarrow W_k \mathbf{x}_t$, $\mathbf{v}_t \leftarrow W_v \mathbf{x}_t$
3: Compute forget gate with binding: $\mathbf{g}_t \leftarrow b + (1 - b) \cdot \sigma(W_g \mathbf{x}_t)$ ▷ Constrained to $[b, 1]$
4: Diagonal multiplication (vectorized): $\mathbf{S}_t \leftarrow \mathbf{S}_{t-1} \circ \text{diag}(\mathbf{g}_t) + \mathbf{v}_t \mathbf{k}_t^\top$
5: Retrieve: $\mathbf{o}_t \leftarrow \mathbf{S}_t \mathbf{q}_t$
6: Optional normalization: $\mathbf{o}_t \leftarrow \text{RMSNorm}(\mathbf{o}_t)$ **return** $\mathbf{o}_t$, $\mathbf{S}_t$

5.5.2 Scaling and Empirical Results

At 3B scale on 100B tokens, HGRN2 slightly outperforms Mamba2 and LLaMA-architecture Transformers on language modeling. The hierarchical gating provides multi-scale temporal modeling without explicit layer-wise architectural variants. However, vector-valued diagonal gating (as opposed to element-wise selection) provides less per-item flexibility than delta-rule variants.

5.6 5. Forgetting Transformer (FoX): Gating in Softmax Logit Space

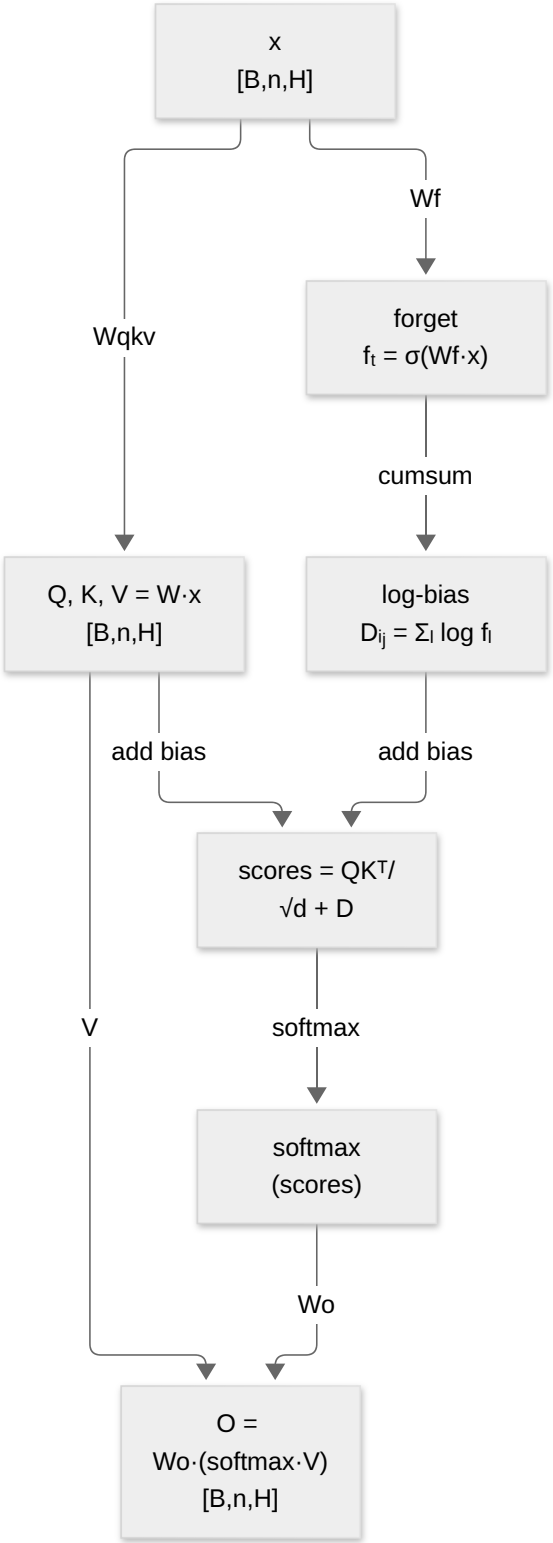


Figure 55: Dataflow of FoX: a learned forget gate becomes an additive log-bias on the score matrix.

5.6.1 Mathematical Foundation

Forgetting Transformer (FoX), proposed by Lin et al. [99], embeds a forget gate directly into softmax attention logits. Rather than replacing softmax with linear recurrence, FoX preserves full softmax expressiveness while adding recency control through a data-dependent logit bias.

For each token position t and all keys $j \leq t$, compute a scalar forget gate:

$$f_t = \sigma(w_f^\top \mathbf{x}_t + b_f)$$

The logit bias at position (i, j) is the cumulative log-forget product:

$$d_{ij} = \sum_{l=j+1}^{i} \log f_l = \log \prod_{l=j+1}^{i} f_l$$

The full attention output becomes:

$$\mathbf{O} = \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}} + \mathbf{D}\right)\mathbf{V}$$

where $\mathbf{D}_{ij} = d_{ij}$. Equivalently:

$$\text{Attention}(i, j) = \frac{\exp(\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k} + d_{ij})}{\sum_{j'=1}^{i} \exp(\mathbf{q}_i^\top \mathbf{k}_{j'} / \sqrt{d_k} + d_{ij'})}$$

This weighting down-weights older tokens multiplicatively: a token from 10 steps ago is scaled by $\prod_{l=j+1}^{i} f_l$, which is typically much less than 1 if $f_t < 1$ on average.

The mechanism is mathematically equivalent to a **data-dependent learnable variant of ALiBi (Attention with Linear Biases)**, where earlier work used fixed $d_{ij} = -\infty \cdot |i - j|$ or $d_{ij} = -(i - j)$.

Algorithm 17 FoX: Forgetting Attention with Recency Bias

Require: Queries $\mathbf{Q}$, keys $\mathbf{K}$, values $\mathbf{V}$; input $\mathbf{X}$
Require: Forget gate parameters: $w_f \in \mathbb{R}^d$, $b_f \in \mathbb{R}$
Ensure: Output attention $\mathbf{O}$

Compute forget gates:
for $t = 1$ to T **do**
    $f_t \leftarrow \sigma(w_f^\top \mathbf{x}_t + b_f)$ ▷ Per-token forget probability
end for
Compute cumulative log-forget biases:
for $i = 1$ to T **do**
    for $j = 1$ to i **do**
        $d_{ij} \leftarrow \sum_{l=j+1}^{i} \log f_l$ ▷ Cumulative product in log space
    end for
end for
Standard SDPA with bias:
Compute scores: $\mathbf{S} \leftarrow \mathbf{QK}^\top / \sqrt{d_k}$
Add bias: $\mathbf{S} \leftarrow \mathbf{S} + \mathbf{D}$
Apply softmax: $\mathbf{A} \leftarrow \text{softmax}(\mathbf{S}, \text{dim} = -1)$
Weight values: $\mathbf{O} \leftarrow \mathbf{AV}$ **return** $\mathbf{O}$

5.6.2 Integration with FlashAttention

The key advantage of FoX is compatibility with FlashAttention and existing optimized implementations. The forget biases are added in the softmax logit computation, which is already a core operation in FlashAttention's block-wise algorithm. The overhead is:

$$\text{Overhead} \approx 0.5 - -2\% \text{ wall-clock time}$$

since the bias addition is amortized across matmul operations.

5.6.3 Strengths and Limitations

FoX achieves state-of-the-art results on length extrapolation, near-perfect needle-in-haystack retrieval, and superior long-context understanding compared to Transformers. However, it remains $\mathcal{O}(L^2 d)$ at training and $\mathcal{O}(Ld)$ per step at inference with KV cache, limiting extreme sequence lengths.

5.7 6. Gated Attention (Post-SDPA Sigmoid Gating)

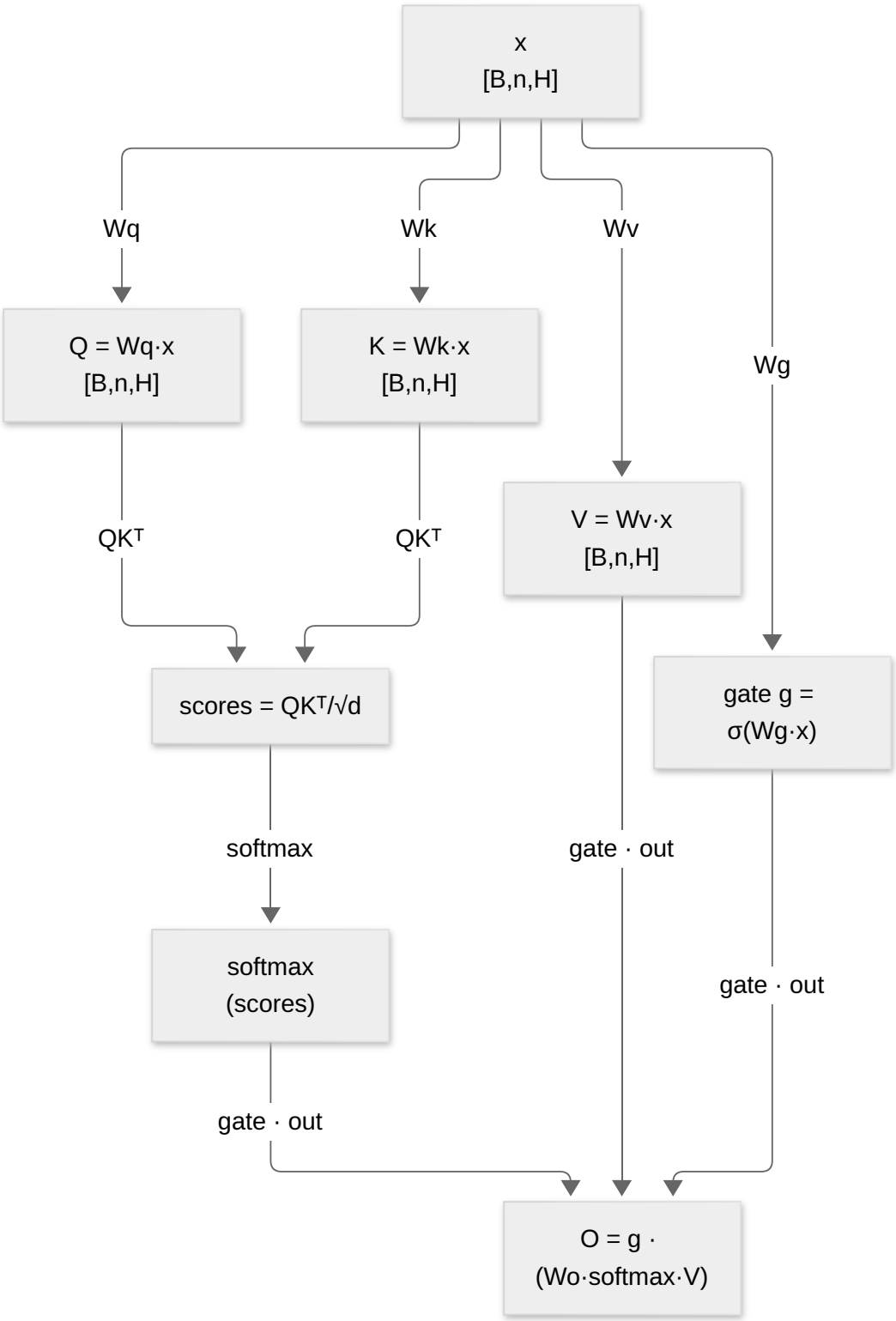


Figure 56: Dataflow of post-SDPA gated attention: a sigmoid gate scales the softmax output before the projection.

5.7.1 Mathematical Foundation

The NeurIPS 2025 Best Paper by the Qwen team proposes applying a sigmoid gate **after** scaled dot-product attention (SDPA):

$$\mathbf{Y} = \text{SDPA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

$$\mathbf{Y}' = \mathbf{Y} \odot \sigma\left(\mathbf{X}\mathbf{W}_\theta\right)$$

where the gate score $\sigma(\mathbf{X}\mathbf{W}_\theta)$ can have shape $\mathbb{R}^{n\times q\times d_k}$ (element-wise gating per query head) or $\mathbb{R}^{n\times q}$ (head-wise fixed gating). Across 30+ variants and 15B MoE + 1.7B dense models trained on 3.5T tokens, the team found:

- Headwise gating adds only $\sim 1.6M$ parameters to a 15B model (negligible overhead).
- Effective gates are **sparse**: mean activation ≈ 0.116 (i.e., 88% are zero or near-zero).
- Gates act as query-dependent filters, suppressing low-value channels while preserving high-value signal.
- Gates demonstrably eliminate the **attention sink** phenomenon, where the attention pattern becomes dominated by a single early token.

Algorithm 18 Gated Softmax Attention (Post-SDPA)

Require: Queries $\mathbf{Q}$, Keys $\mathbf{K}$, Values $\mathbf{V}$; input $\mathbf{X}$
Require: Gate projection matrix $W_g \in \mathbb{R}^{d\to d_{\text{gate}}}$ where $d_{\text{gate}} \in \{1, d_k\}$
Ensure: Gated output $\mathbf{Y}'$

Standard SDPA:
Compute attention: $\mathbf{A} \leftarrow \text{softmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d_k})$
Weight values: $\mathbf{Y} \leftarrow \mathbf{A}\mathbf{V}$
Compute gates:
Project input: $\mathbf{G} \leftarrow \mathbf{X}W_g$ ▷ shape: $[n, q, d_{\text{gate}}]$
Apply sigmoid: $\mathbf{G} \leftarrow \sigma(\mathbf{G})$ ▷ Element-wise gating
Apply gating:
if $d_{\text{gate}} = 1$ **then**
    Broadcast and scale: $\mathbf{Y}' \leftarrow \mathbf{Y} \cdot \mathbf{G}$ ▷ Head-wise scaling
else ▷ $d_{\text{gate}} = d_k$
    Element-wise modulation: $\mathbf{Y}' \leftarrow \mathbf{Y} \odot \mathbf{G}$ ▷ Per-dimension gating
end ifreturn $\mathbf{Y}'$

5.7.2 Key Findings

- **Attention-sink suppression**: Gating breaks the feedback loop where softmax concentrates on the first token, enabling more balanced attention.
- **Training stability**: Models with gating tolerate larger learning rates (+30% higher stable η_{max}).
- **Minimal overhead**: Latency impact is only 1.6% on H100 GPUs; throughput drops at most $2-3\%$.
- **Scaling law improvement**: Improves loss across model scales from 800M to 110B parameters consistently.

5.8 7. Gated DeltaNet-2: Decoupled Channel-wise Erase and Write

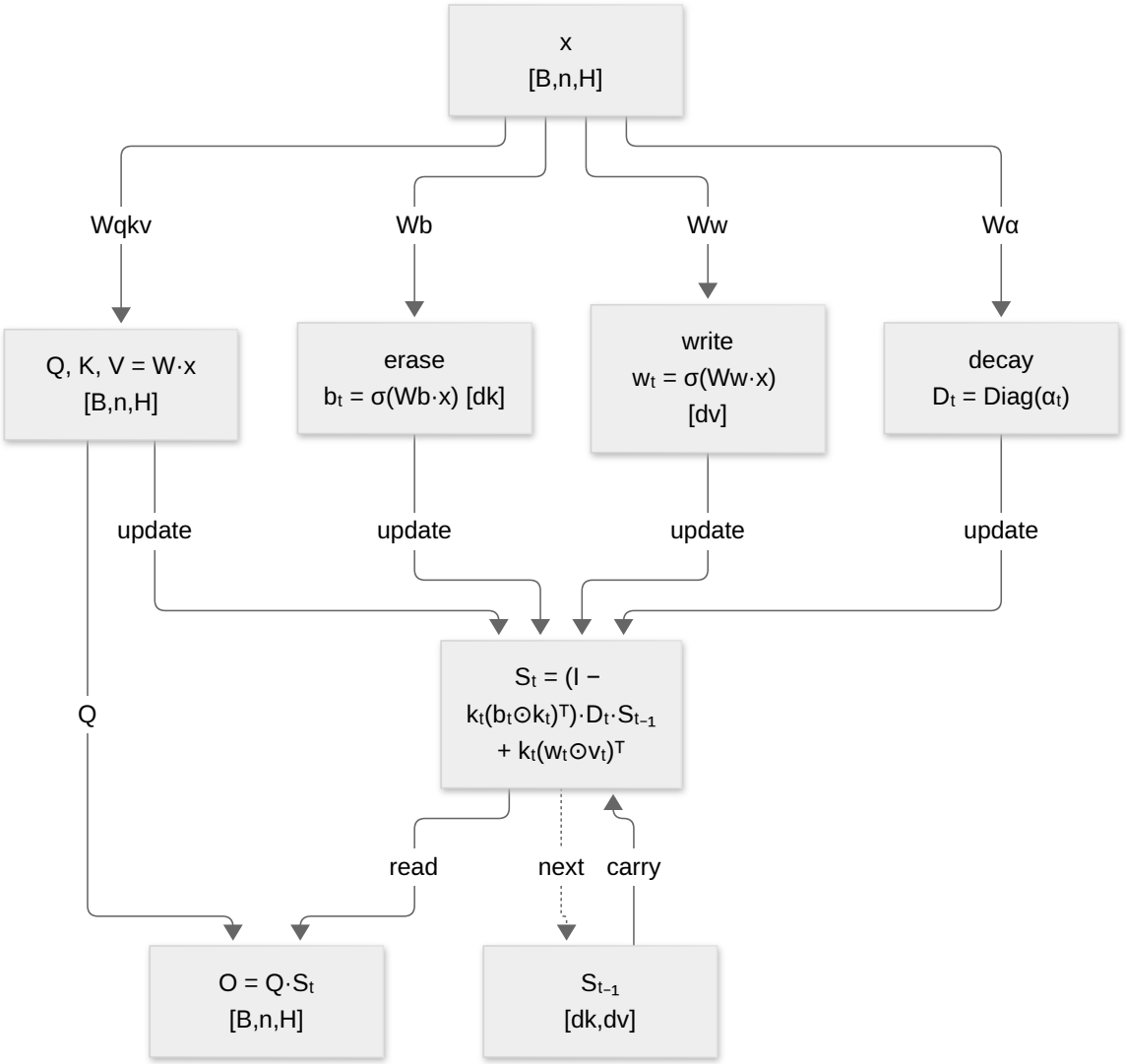


Figure 57: Dataflow of Gated DeltaNet-2: decoupled channel-wise erase and write gates act inside the delta rule.

5.8.1 Mathematical Foundation

Gated DeltaNet-2 [60] addresses a shared limitation of Gated DeltaNet and Kimi Delta Attention (KDA): both use a single *scalar* delta gate β_t to simultaneously control (i) how much old content is erased along the key-side read direction and (ii) how much new value is written along the value axis. Gated DeltaNet-2 **decouples** these two roles into a channel-wise erase gate $\mathbf{b}_t$ (key axis) and a channel-wise write gate $\mathbf{w}_t$ (value axis), while inheriting the channel-wise decay $\boldsymbol{\alpha}_t$ from KDA. With state $\mathbf{S}_t \in \mathbb{R}^{d_k \times d_v}$, decay $\mathbf{D}_t = \mathrm{Diag}(\boldsymbol{\alpha}_t)$, erase direction $\mathbf{e}_t = \mathbf{b}_t \odot \mathbf{k}_t$, and write target $\mathbf{z}_t = \mathbf{w}_t \odot \mathbf{v}_t$, the state update is:

$$\mathbf{S}_t = \left(\mathbf{I} - \mathbf{k}_t \mathbf{e}_t^\top\right) \mathbf{D}_t \, \mathbf{S}_{t-1} + \mathbf{k}_t \mathbf{z}_t^\top, \qquad \mathbf{o}_t = \mathbf{S}_t^\top \mathbf{q}_t$$

or, expanding the gate products:

$$\mathbf{S}_t = \left(\mathbf{I} - \mathbf{k}_t (\mathbf{b}_t \odot \mathbf{k}_t)^\top\right) \mathbf{D}_t \, \mathbf{S}_{t-1} + \mathbf{k}_t (\mathbf{w}_t \odot \mathbf{v}_t)^\top$$

The two gates are produced by independent linear projections followed by sigmoid:

$$\mathbf{b}_t = \sigma(\mathbf{W}_b \mathbf{x}_t), \qquad \mathbf{w}_t = \sigma(\mathbf{W}_w \mathbf{x}_t)$$

and the channel-wise decay follows a log-decay parameterization (computed in fp32 to avoid precision loss in long cumulative products):

$$\mathbf{g}_t = \mathbf{a} - \text{softplus}(\boldsymbol{\delta}), \qquad \boldsymbol{\alpha}_t = \exp(\mathbf{g}_t)$$

The structural roles of the two gates are complementary:

- $\mathbf{b}_t$ (key axis) makes the **read direction** channel-selective: it determines which key channels are removed before writing.
- $\mathbf{w}_t$ (value axis) makes the **write target** channel-selective: it determines which value channels are deposited into memory.
- $\boldsymbol{\alpha}_t$ (per key channel) provides global decay, as in KDA.

Reductions. Setting $\mathbf{b}_t = \beta_t \mathbf{1}_{d_k}$ and $\mathbf{w}_t = \beta_t \mathbf{1}_{d_v}$ recovers KDA; further collapsing $\boldsymbol{\alpha}_t$ to a scalar recovers Gated DeltaNet. From the fast-weight / online-learning perspective, the update is the minimizer of:

$$\mathcal{L}_t(\mathbf{S}) = \tfrac{1}{2}\|\mathbf{S} - \mathbf{D}_t \mathbf{S}_{t-1}\|_F^2 - 2\left\langle \mathbf{S}^\top \mathbf{k}_t,\ \mathbf{z}_t - (\mathbf{D}_t \mathbf{S}_{t-1})^\top \mathbf{e}_t \right\rangle$$

Algorithm 19 Gated DeltaNet-2 Recurrent Forward (Reference Form)

Require: Input $\mathbf{x}_t$; prior state $\mathbf{S}_{t-1} \in \mathbb{R}^{d_k \times d_v}$
Require: Projections: W_q, W_k, W_v, W_b, W_w; log-decay parameters $\mathbf{a}, \boldsymbol{\delta}$
Ensure: Output $\mathbf{o}_t$ and state $\mathbf{S}_t$

$\mathbf{q}_t \leftarrow \text{normalize}(W_q \mathbf{x}_t)$, $\mathbf{k}_t \leftarrow \text{normalize}(W_k \mathbf{x}_t)$, $\mathbf{v}_t \leftarrow \text{silu}(W_v \mathbf{x}_t)$
$\mathbf{b}_t \leftarrow \sigma(W_b \mathbf{x}_t)$ ▷ key-axis erase gate, $\mathbb{R}^{d_k}$
$\mathbf{w}_t \leftarrow \sigma(W_w \mathbf{x}_t)$ ▷ value-axis write gate, $\mathbb{R}^{d_v}$
$\boldsymbol{\alpha}_t \leftarrow \exp(\mathbf{a} - \text{softplus}(\boldsymbol{\delta}))$ ▷ channel-wise decay, fp32
Erase direction: $\mathbf{e}_t \leftarrow \mathbf{b}_t \odot \mathbf{k}_t$; write target: $\mathbf{z}_t \leftarrow \mathbf{w}_t \odot \mathbf{v}_t$
Decay state: $\mathbf{S}_t \leftarrow \text{Diag}(\boldsymbol{\alpha}_t)\, \mathbf{S}_{t-1}$
Read: $\mathbf{r}_t \leftarrow \mathbf{S}_t^\top \mathbf{e}_t$ ▷ sum over key axis
Delta-rule update: $\mathbf{S}_t \leftarrow \mathbf{S}_t - \mathbf{k}_t \mathbf{r}_t^\top + \mathbf{k}_t \mathbf{z}_t^\top$
Retrieve: $\mathbf{o}_t \leftarrow \mathbf{S}_t^\top \mathbf{q}_t$
Output gating: $\mathbf{o}_t \leftarrow \text{GroupNorm}(\mathbf{o}_t) \odot \text{silu}(W_g \mathbf{x}_t)$
return $\mathbf{o}_t$, $\mathbf{S}_t$

The chunkwise training algorithm extends KDA's WY representation by absorbing the channel-wise decay into an asymmetric delta recurrence on a decay-normalized state, enabling matmul-rich parallelism on tensor cores. Because $\mathbf{b}_t$ and $\mathbf{w}_t$ are per-channel diagonal operators varying per row, the scalar post-scaling shortcut of KDA breaks: the backward pass requires a **gate-aware WY** inversion that bakes the gates into each dot product accumulating the gradient.

5.8.2 Empirical Performance and Trade-offs

At 1.3B parameters trained on 100B FineWeb-Edu tokens, Gated DeltaNet-2 outperforms Mamba-2, Mamba-3, Gated DeltaNet, and KDA on language modeling, commonsense reasoning, and—most markedly—real-world and RULER long-context retrieval (especially multi-key needle-in-a-haystack). Its throughput on H100 stays near-flat (38.0 → 36.1 Kt/s) as sequence length grows from 2K to 16K, where a Transformer's drops sharply. Ablations show the erase gate $\mathbf{b}_t$

accounts for most of the gain; the write gate $\mathbf{w}_t$ contributes a smaller additional improvement. The hybrid variant (recurrent Gated DeltaNet-2 + sliding-window attention) further closes the gap with pure-attention models on tasks requiring local evidence aggregation.

Aspect	**Strength**	**Limitation**
Recall	Best-in-class among recurrent delta-rule models on multi-key NIAH.	Recurrent variant still gaps Transformer on NQ/DROP.
Throughput	Near-flat scaling 2K→16K; small overhead vs KDA.	Requires a new gate-aware WY Triton backward kernel.
Memory	Constant $\mathcal{O}(d^2)$ inference state.	Neg-eigenvalue erase range $[0, 2]$ gives no consistent gain at 1.3B scale.

Table 3: Gated DeltaNet-2 trade-offs at 1.3B / 100B-token scale [60].

5.9 8. Kimi Delta Attention (KDA): Channel-wise Decay Gating for the Delta Rule

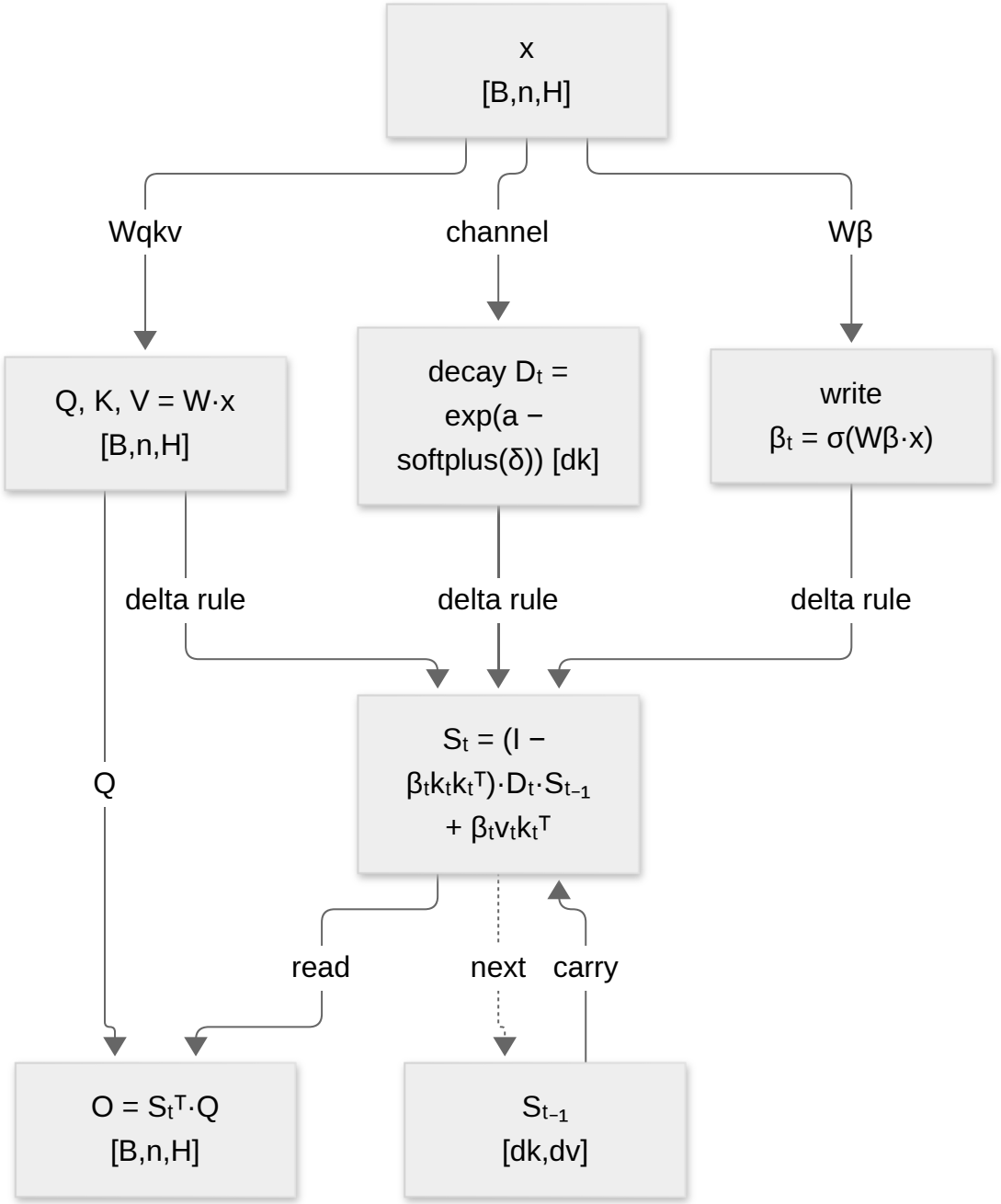


Figure 58: Dataflow of KDA: channel-wise decay and a scalar write gate modulate the delta-rule state update.

5.9.1 Mathematical Foundation

Kimi Delta Attention [83] is the core module of Kimi Linear and extends Gated DeltaNet with a finer-grained **channel-wise decay gate**. Where Gated DeltaNet uses a single scalar per-head decay α_t alongside the scalar delta write gate β_t, KDA replaces the scalar decay with a **per-key-channel** decay $\boldsymbol{\alpha}_t \in \mathbb{R}^{d_k}$, parameterised through a log-decay formulation and applied as a diagonal matrix $\mathbf{D}_t = \mathrm{diag}(\boldsymbol{\alpha}_t)$ before the delta-rule update. The delta rule itself retains a

scalar per-head write gate β_t:

$$\mathbf{S}_t = (\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top)\, \mathbf{D}_t\, \mathbf{S}_{t-1} + \beta_t \mathbf{v}_t \mathbf{k}_t^\top, \qquad \mathbf{o}_t = \mathbf{S}_t^\top \mathbf{q}_t$$

The channel-wise decay follows a log-decay parameterisation (computed in fp32 for numerical stability over long cumulative products):

$$\mathbf{g} = \mathbf{a} - \mathrm{softplus}(\boldsymbol{\delta}), \qquad \boldsymbol{\alpha}_t = \exp(\mathbf{g})$$

where $\mathbf{a}$ and $\boldsymbol{\delta}$ are learned per-channel parameters. This gives each key channel an independent forgetting timescale, allowing the model to retain some channels longer than others within the same head.

Reduction to Gated DeltaNet : When $\boldsymbol{\alpha}_t$ collapses to a scalar (i.e., all channels share the same decay), KDA reduces exactly to Gated DeltaNet. The channel-wise decay is thus a strict generalisation that adds per-channel selectivity on top of the scalar decay/write gating of Gated DeltaNet.

Chunkwise training : A bespoke chunkwise algorithm uses a specialised **Diagonal-Plus-Low-Rank (DPLR)** variant of the transition matrix to remain hardware-efficient. The product of per-step transitions $(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top)\mathbf{D}_t$ admits a DPLR decomposition that allows matmul-rich parallelism on tensor cores while preserving the channel-wise decay structure.

Algorithm 20 Kimi Delta Attention Recurrent Forward (Reference Form)

Require: Input $\mathbf{x}_t$; prior state $\mathbf{S}_{t-1} \in \mathbb{R}^{d_k \times d_v}$
Require: Projections: W_q, W_k, W_v, W_β; log-decay parameters $\mathbf{a}, \boldsymbol{\delta}$
Ensure: Output $\mathbf{o}_t$ and state $\mathbf{S}_t$

$\mathbf{q}_t \leftarrow W_q \mathbf{x}_t$, $\mathbf{k}_t \leftarrow W_k \mathbf{x}_t$, $\mathbf{v}_t \leftarrow W_v \mathbf{x}_t$
$\beta_t \leftarrow \mathrm{sigmoid}(W_\beta \mathbf{x}_t)$ ▷ scalar per-head write gate
$\boldsymbol{\alpha}_t \leftarrow \exp(\mathbf{a} - \mathrm{softplus}(\boldsymbol{\delta}))$ ▷ channel-wise decay, fp32
$\mathbf{D}_t \leftarrow \mathrm{diag}(\boldsymbol{\alpha}_t)$ ▷ diagonal decay matrix
Decay state: $\mathbf{S}_t \leftarrow \mathbf{D}_t \mathbf{S}_{t-1}$
Read: $\mathbf{r}_t \leftarrow \mathbf{S}_t^\top \mathbf{k}_t$ ▷ current prediction along key
Delta-rule update: $\mathbf{S}_t \leftarrow \mathbf{S}_t - \beta_t \mathbf{k}_t \mathbf{r}_t^\top + \beta_t \mathbf{v}_t \mathbf{k}_t^\top$
Retrieve: $\mathbf{o}_t \leftarrow \mathbf{S}_t^\top \mathbf{q}_t$
Output gating: $\mathbf{o}_t \leftarrow \mathrm{GroupNorm}(\mathbf{o}_t) \odot \mathrm{silu}(W_g \mathbf{x}_t)$
return $\mathbf{o}_t$, $\mathbf{S}_t$

5.9.2 Empirical Performance and Trade-offs

KDA is deployed in Kimi Linear, a hybrid model (3B activated / 48B total parameters) combining KDA layers with Multi-head Latent Attention (MLA). Kimi Linear outperforms a full-MLA baseline across all evaluated tasks, reduces the KV cache by up to 75%, and achieves up to 6× decoding throughput for 1M-token context. The channel-wise decay is the principal driver of the gain: it allows fine-grained control over which key channels persist, improving long-context recall relative to scalar-decay Gated DeltaNet.

Aspect	Strength	Limitation
Recall	Channel-wise decay improves long-context recall over scalar-decay baselines; deployed in Kimi Linear (3B/48B hybrid).	Recurrent matrix state $\mathcal{O}(d^2)$ is heavier than diagonal-gated linear attention.
Throughput	Up to 6× decoding throughput at 1M context; 75% KV cache reduction vs full MLA.	Requires DPLR-aware chunkwise kernels for hardware efficiency.
Memory	Constant $\mathcal{O}(d^2)$ inference state (matrix recurrent).	Log-decay fp32 computation adds overhead; reduction to Gated DeltaNet loses channel selectivity.

Table 4: KDA trade-offs in Kimi Linear (3B activated / 48B total) [83].

5.10 9. First Passage Attention (FPA): Forgetting by Path Extrema

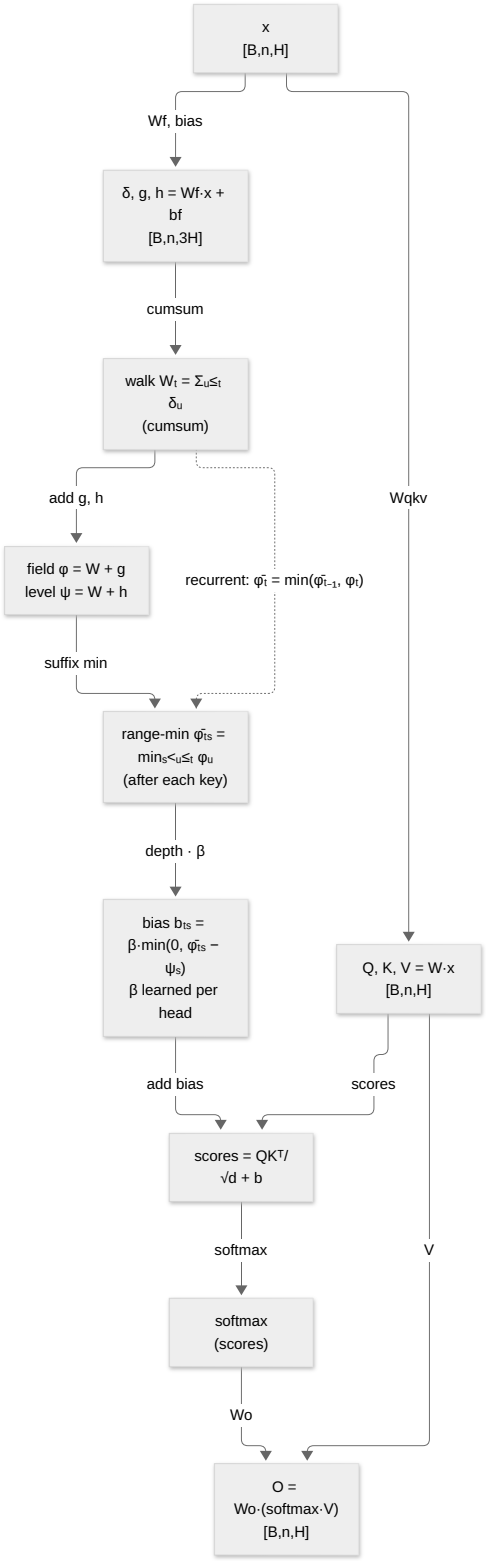


Figure 59: Dataflow of FPA: one linear map yields the per-head increment δ and the offsets g, h; the walk W is their prefix sum, the field is $\phi = W + g$, the key level is $\psi = W + h$, and the bias is β times the deepest dip of the field below the level over the span after each key.

5.10.1 Mathematical Foundation

First Passage Attention [104] keeps the softmax attention of Vaswani et al. [159] but replaces FoX's path *sum* of forget gates with a path *extremum*. A causal head of dimension d with queries $\mathbf{q}_t$, keys $\mathbf{k}_s$ and values $\mathbf{v}_s$ computes

$$\mathbf{o}_t = \sum_{s \le t} w_{ts} \mathbf{v}_s, \qquad w_{ts} \propto e^{\ell_{ts}}, \qquad \ell_{ts} = \frac{\mathbf{q}_t^\top \mathbf{k}_s}{\sqrt{d}} + b_{ts},$$

where the **first-passage bias** is

$$b_{ts} = \beta \min\Big(0, \min_{s < u \le t} \phi_u - \psi_s\Big), \qquad b_{tt} = 0,$$

with a learned positive strength β per head, a scalar field ϕ_u at every position and a scalar level ψ_s for every key. We write $\bar{\phi}_{ts} = \min_{s<u\le t} \phi_u$ for the range minimum. A key is *alive* at time t when $b_{ts} = 0$, and *dead* otherwise; a dead key's bias is $-\beta$ times the depth $\psi_s - \bar{\phi}_{ts}$ of its deepest dip. Death is permanent: $b_{t's} \le b_{ts}$ for all $t' < t$ (Proposition 2.3(a)), so the decoder can drop a dead key once its bias passes a threshold.

Each head derives the three scalars per position from the layer input $\mathbf{x}_t$ through one linear map, with a walk W_t and the field/level

$$W_t = \sum_{u \le t} \delta_u, \qquad \phi_u = W_u + g_u, \qquad \psi_s = W_s + h_s.$$

The increments δ move the reference of every later comparison, while g and h let a token dip or set a level without moving it. A layer of width D with H heads therefore gains $3H(D+1)+H$ parameters — 27,696 per layer at $D = 768$, $H = 12$, i.e. 332,352 over twelve layers, and +1.56% over keys and values in the cache.

Special cases (Proposition 2.2). The linear parameterisation contains several families:

- (a) If $g = h = 0$, then $b_{ts} = -\beta L_{ts}$ with $L_{ts} = \max\big(0, \max_{s<u\le t}(W_s - W_u)\big)$, the regulator of a queue that starts empty at time s (the *stack form*).
- (b) If in addition $\delta_u \le 0$ for all u, then $b_{ts} = \sum_{s<u\le t} \log f_u$ with $f_u = e^{\beta\delta_u} \in (0, 1]$ — **exactly FoX** with forget gates f_u. A rising walk is what lets FPA keep an old key while removing newer ones, which no forget gate can do.
- (c) If $\delta = 0$, $h = 0$, $g \le 0$ and $\beta = 1$, then $b_{ts} = \min_{s<u\le t} g_u$, a bottleneck bias.
- (d) If $\delta_u = -1$, $g = 0$ and $\psi_s = W_s - e_s$ with $e_s \ge 0$, then $b_{ts} = \beta \min(0, e_s - (t - s))$, the expiry rule of Expire-Span.

In the stack form the levels of the alive keys do not decrease with the time of writing, and a dip kills a suffix in time of the alive keys: the alive set evolves as a **stack** (Proposition 2.3(b)).

Depth, not duration. Selective Attention subtracts a sum of per-pair selections, so its suppression grows with the number of intervening steps, confounding the *depth* of an event with its *duration*. FPA's bias is $-\beta$ times the deepest dip and is unchanged when a dip no deeper than the deepest is held longer or inserted (Proposition 3.4); among aggregations of dips, functions of the maximum are the only ones with this duration invariance (Proposition 3.4(iii)).

Training (straight-through). The bias is piecewise linear in the field and the levels, and on the closed side of the outer minimum it passes no gradient to the field; in pilot runs that minimum could close for every key of a head early in training. FPA therefore trains with a straight-through estimator: the forward pass takes the hard range minimum and the backward pass takes, in its place, the gradient of the soft minimum $-\tau \log \sum_{s<u\le t} e^{-\phi_u/\tau}$ with $\tau = 1$. The outer minimum acts on the hard value, and evaluation uses the hard bias unchanged.

Algorithm 21 First Passage Attention (Parallel Reference Form)

Require: Normalised input $\mathbf{x} \in \mathbb{R}^{n\times D}$; heads H; strength β; temperature τ
Ensure: Output $\mathbf{O} \in \mathbb{R}^{n\times D}$

$\mathbf{Q}, \mathbf{K}, \mathbf{V} \leftarrow W_q\mathbf{x}, W_k\mathbf{x}, W_v\mathbf{x}$ ▷ per-head projections
$(\boldsymbol{\delta}, \mathbf{g}, \mathbf{h}) \leftarrow W_f\mathbf{x} + \mathbf{b}_f$ ▷ one map, three scalars per head
$W_t \leftarrow \sum_{u\le t} \delta_u$; $\phi_u \leftarrow W_u + g_u$; $\psi_s \leftarrow W_s + h_s$
$\bar{\phi}_{ts} \leftarrow \min_{s<u\le t} \phi_u$ ▷ range minimum after each key
$b_{ts} \leftarrow \beta \min(0, \bar{\phi}_{ts} - \psi_s)$; $b_{tt} \leftarrow 0$
$\mathbf{A} \leftarrow \mathrm{softmax}\big(\mathbf{Q}\mathbf{K}^\top/\sqrt{d} + \mathbf{B}\big)$; $\mathbf{O} \leftarrow W_o(\mathbf{A}\mathbf{V})$
return O ▷ backward uses ∂ of the soft minimum $-\tau \log \sum e^{-\phi_u/\tau}$

Cost and exact decoding. In a blocked computation with block size B, the bias between a query block and an earlier key block needs only the suffix minima of ϕ within the key block, the minimum over the blocks strictly between, and the prefix minima of ϕ within the query block: $O(B^2)$ operations per pair of blocks, as an element-wise bias such as FoX's, plus $O(T/B)$ block minima (Proposition 2.4(a)). In autoregressive decoding $\bar{\phi}_{ts} = \min(\bar{\phi}_{t-1,s}, \phi_t)$ for $s < t-1$ and $\bar{\phi}_{t,t-1} = \phi_t$, so each step updates the bias row in $O(t)$ and stores *two scalars per cached key and head* (Proposition 2.4(b)) — an exact plain-KV incremental branch, implemented in this repository as slot extras.

5.10.2 Empirical Performance and Trade-offs

The paper trains four 110M-parameter decoders (12 layers, width 768, 12 heads of dimension 64, SwiGLU-2048, RMSNorm, context 2048, 2.7B SlimPajama tokens) that differ only in attention. FPA (without positional encoding) matches or beats FoX in perplexity and extrapolates further:

Method	SlimPajama	WikiText-103	PG-19 2k–4k	Passkey 8k–16k	RULER 8k
FPA	**2.8640**	**3.1001**	**3.282**	**1.00**	**0.202**
FoX	2.8832	3.1425	3.316	0.20	0.117
Selective Attention	2.8686	3.1197	3.805	0.04	0.008
Softmax + RoPE	2.8908	3.1374	5.064	0.00	0.002

Table 5: 110M language models (losses in nats per token; passkey and RULER accuracies) [104]. FPA retrieves every passkey at lengths up to 8× its training context, even when its cache is compressed to about a quarter.

The cost is close to standard attention: one FPA layer takes 0.97× the forward-plus-backward time of PyTorch SDPA and 1.50× that of FoX, whose bias needs no range minimum; at batch size 1 and length 8192 decoding takes 22.0 ms per token against 20.2 ms for FoX and 23.9 ms for softmax. Because the bias of a key never rises again, a decoder can drop a key once its bias falls below a threshold: at $\theta = 10$ the FPA model loses 0.00044 nats at length 2048 and keeps 34% of its keys (26% at 8192), and it still retrieves every passkey at 8192 and 16,384 tokens with 24–27% of its cache.

Aspect	Strength	Limitation
Expressivity	One constant-dimension head realises every dip-induced alive set; softmax heads with positional biases or token-level forget gates need head dimension linear in the number of keys (Theorem 3.1).	In the hard limit Selective Attention and DCP realise the same alive sets; the separation is in the scalar-per-position bias and in depth versus duration.
Selectivity	A later token can erase a newer key while preserving older ones (the alive set is a stack); FoX is the special case of a non-rising walk.	Requires the straight-through soft-minimum gradient; the inner minimum can otherwise close early in training.
Cost	0.97× SDPA training time; two extra scalars per cached key/head (+1.56%); exact $O(t)$ decode; dead keys are evictable down to ∼26–34% of the cache at negligible loss.	Quadratic attention remains; the paper's fused Triton kernel is not reproduced here (the reference implementation materialises the $T \times T$ bias).
Evidence	110M LM matches FoX in perplexity and retrieves passkeys at 8×–16× the training length; preregistered one-layer probes agree with the theory.	One seed per 110M model; a first-layer bias hurt multi-query associative recall unless removed.

Table 6: FPA trade-offs [104]. The repository integrates the causal head exactly; encoder mode symmetrises the open-span range minimum (the paper defines only the causal head).

5.11 Unified Gating Principle

All nine architectures share a common principle: **gating is a mechanism to control information flow and selective memory retention**. The specific design choices diverge along several dimensions:

1. **Location of gating**: Recurrent state updates (GLA, DeltaNet, Gated DeltaNet, Gated DeltaNet-2, HGRN2) versus softmax logit/output space (FoX, Gated Softmax, FPA, RetNet).

2. **Granularity of control**: Dimension-wise (GLA, HGRN2 diagonal), key-specific (DeltaNet), token-wise (FoX), head-wise (Gated Softmax), channel-wise decoupled erase/write (Gated DeltaNet-2), or a per-key level compared against a head-specific field (FPA).

3. **Data dependence**: Fully data-dependent (GLA, DeltaNet, Gated DeltaNet, Gated DeltaNet-2, FoX, Gated Softmax, FPA) versus fixed schedules (RetNet with exponential decay).

4. **Aggregation along the key–query path**: No path term (Gated Softmax), a sum of log-forget gates (FoX), or a range minimum of a learned field (FPA) — sums accumulate with duration, extrema depend only on the deepest dip.

5. **Expressiveness trade-off**: Linear recurrent efficiency (GLA, DeltaNet, Gated DeltaNet, Gated DeltaNet-2, HGRN2, RetNet) versus full softmax expressiveness (FoX, Gated Softmax, FPA).

5.12 Implementation and Practical Guidance

5.12.1 When to Use Each Architecture

1. **GLA**: Fast inference with moderate-to-long contexts (2K–20K tokens), when custom CUDA kernels are acceptable, and retrieval performance is not critical.

2. **DeltaNet**: Strong associative recall and in-context learning tasks; good for synthetic tasks (MQAR) and key-value association testing.

3. **Gated DeltaNet**: Production models requiring the best balance of recall, forgetting, and throughput (proven in Qwen3-Next; ICLR 2025).

4. **Gated DeltaNet-2**: Long-context retrieval workloads (multi-key NIAH) where decoupled channel-wise erase/write yields the strongest recall; when a gate-aware WY kernel is available.

5. **RetNet**: Extremely efficient inference without KV caching; suitable for edge/mobile deployments or toy models.

6. **HGRN2**: Multi-scale temporal hierarchies; when layer-specific forget bounds are desirable.

7. **FoX**: Length extrapolation and long-context understanding; when quadratic training is acceptable and replacing softmax is not desired.

8. **Gated Softmax**: Quick improvement to existing softmax models with minimal code changes; best for practitioners wanting immediate gains without major refactoring.

9. **FPA**: Long-context and structured sequences where a later event invalidates recent tokens but not older ones (delimiters, backspaces, reassignment); when softmax quality and exact KV decoding are required and a small per-head scalar state is acceptable.

5.12.2 Hardware Considerations

Hardware Target	Recommended Architectures	Notes
GPU (H100/A100)	Gated Softmax, FoX, Gated DeltaNet, Gated DeltaNet-2	Mature softmax/linear implementations. GLA/DeltaNet require custom kernels.
TPU (high memory)	GLA, DeltaNet, Gated DeltaNet, Gated DeltaNet-2, HGRN2	Hardware supports efficient matmuls; linear recurrent methods shine.
Mobile/Edge	RetNet, lightweight Gated Softmax	Constant-memory inference critical.

6 SSOG: Separable Sum of Gaussians Attention

This annex covers SSOG (Separable Sum of Gaussians), a geometric-field attention mixer. SSOG [125] replaces the content-scored interaction of scaled dot-product attention with a learned mixture of Gaussians over *relative position*: each head owns a handful of Gaussian atoms—five numbers per atom, a center offset (μ_y, μ_x), a width per axis (σ_y, σ_x) and a mixture weight λ—forming a fixed field over the displacement between tokens. There are no query–key dot products anywhere: content never scores content, it only *steers* the field through small bounded residuals.

Because a 2D Gaussian factorises into a product of two 1D Gaussians, applying the field is two 1D filter passes per atom (rows, then columns of a $\sqrt{N} \times \sqrt{N}$ token raster) followed by a λ-mix contraction. The induced $N \times N$ attention matrix never exists, so the attention-specific cost is $\mathcal{O}(R \cdot N\sqrt{N} \cdot d)$ for R atoms instead of $\mathcal{O}(N^2 \cdot d)$. SSOG also skips the query and key projections entirely, paying $2d^2$ of projections per layer instead of the $4d^2$ of SDPA. In the reference recipe the full mechanism beats its matched SDPA twin on ImageNet-1k (65.28% vs 64.34% at ∼3M params; 72.02% vs 71.84% at ∼12M) while being ∼20% smaller and ∼30% cheaper per forward, and by +17 points on CIFAR-100 at matched recipe.

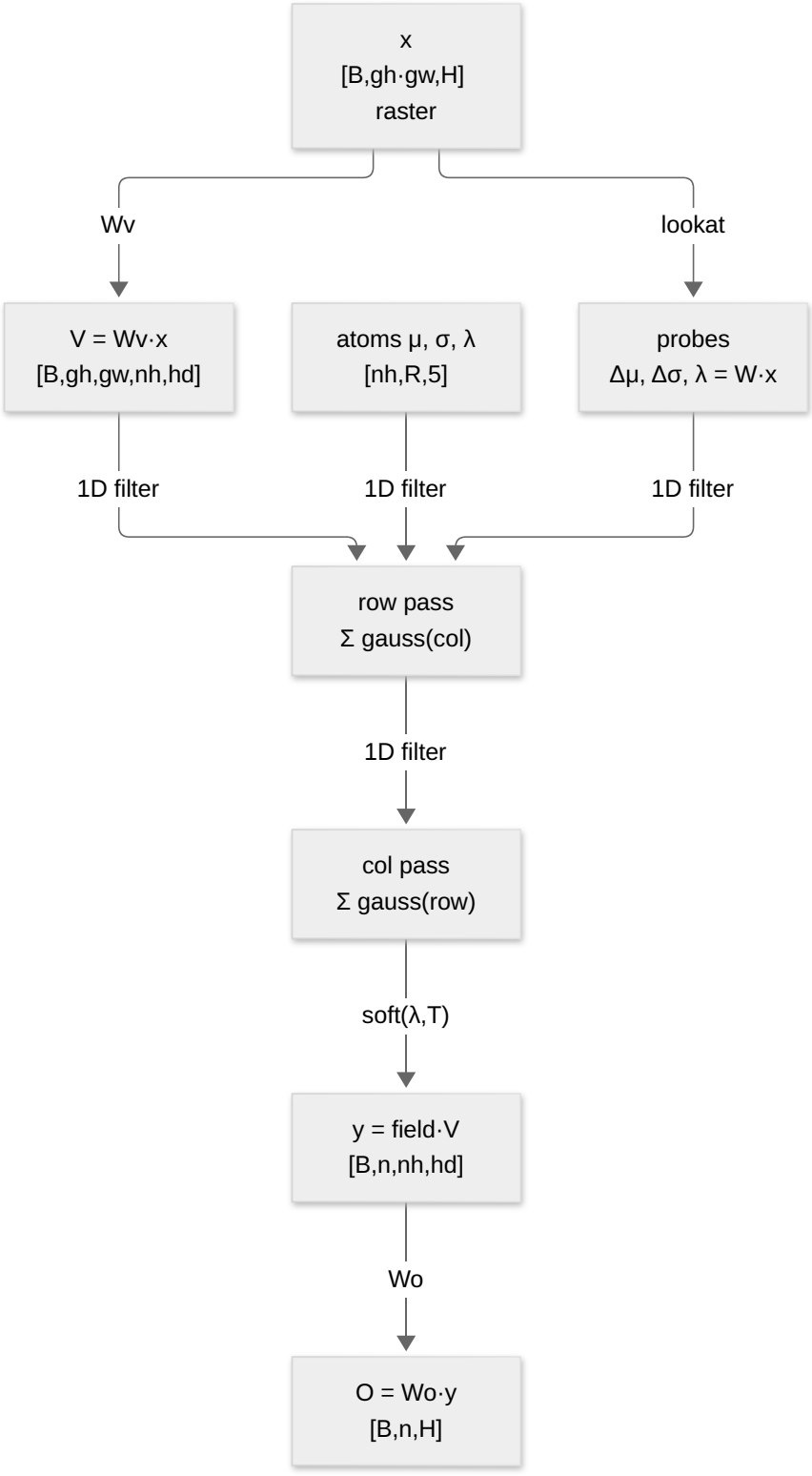


Figure 60: Dataflow of SSOG: values pass through separable row and column Gaussian filter passes per atom, with optional content steering of the atom parameters.

6.1 Mathematical Formulation

Let tokens be the row-major raster of a $G_h \times G_w$ grid. The attention weight from token p to token q is the softmax-tempered log-sum-exp of the atom log-weights evaluated at their displacement:

$$A(p,q) = \mathrm{softmax}_q\big(\tfrac{1}{\tau}\, s(p,q)\big)\,, \qquad s(p,q) = \mathrm{logsumexp}_r\,\big(\log \lambda_r + \log \mathcal{N}(p-q;\ \mu_r, \sigma_r)\big),$$

where τ is a learnable temperature and $\mathcal{N}(\cdot\,; \mu_r, \sigma_r)$ is a 2D Gaussian with diagonal covariance. Since the 2D Gaussian factorises, the softmaxed field applies separably: per axis, $a_y[i,j] = \mathrm{softmax}_j(\log \mathcal{N}(\Delta y_{ij}; \mu_y, \sigma_y)/\tau)$ and analogously a_x, followed by

$$Y = \sum_{r=1}^{R} \lambda_r \,\big(a_x^{(r)}\big(a_y^{(r)}\, V\big)\big), \qquad V = xW_V,$$

i.e. three einsum contractions per layer. With content steering (`lookat`) enabled, zero-initialised linear probes predict bounded per-token residuals on the field parameters behind cold-started softplus gates:

$$\mu \leftarrow \mu_0 + s_\mu \cdot m \cdot \tanh(W_\mu x), \qquad \sigma \leftarrow \sigma_0\, e^{s_\sigma \tanh(W_\sigma x)}, \qquad \lambda \leftarrow \mathrm{softmax}\,\big(\log \lambda_0 + s_\lambda \tanh(W_\lambda x)\big),$$

with m the maximum offset in grid cells and gate scales $s_\bullet = \mathrm{softplus}(\cdot)$ initialised at -8 ($\approx 3 \times 10^{-4}$), so the model begins as a frozen geometric field and learns how far to open each content tap. Widths are reparameterised as $\sigma = \mathrm{softplus}(\omega) + \epsilon + \sigma_{\text{floor}}$ to stay strictly positive.

6.2 Algorithmic Pseudocode

Input: hidden state $x \in \mathbb{R}^{B \times N \times d}$ as a $G_h \times G_w$ raster, atoms $\{\mu_r, \omega_r, \log \lambda_r\}_{r=1}^{R}$ per head, temperature τ
$V \leftarrow xW_V$ reshaped to (B, G_h, G_w, H, d_h) ▷ no Q/K projections
$\sigma_r \leftarrow \mathrm{softplus}(\omega_r) + \epsilon + \sigma_{\text{floor}}$ ▷ positive widths
if steering (lookat) **then**
    $\mu, \sigma, \lambda \leftarrow$ bounded per-token residuals via zero-init probes and gates ▷ content steers
end if
$a_y \leftarrow \mathrm{softmax}\big(\log \mathcal{N}(\Delta y; \mu_y, \sigma_y)/\tau\big)$, $a_x \leftarrow$ analogously ▷ 1D kernels, one per atom
$Y \leftarrow \sum_r \lambda_r\, a_x^{(r)}\big(a_y^{(r)} V\big)$ ▷ row pass, column pass, λ-mix: no $N \times N$
Output: $\mathbf{Y}W_O$ reshaped to (B, N, d)

6.3 Key Characteristics

- **Training complexity**: $\mathcal{O}(R \cdot N\sqrt{N} \cdot d)$ for fixed R atoms—near-linear by geometry alone, not by kernel approximation.
- **Inference complexity**: same per-forward cost; encoder-only (the separable field has no causal formulation), so there are no KV-cache semantics.
- **Trainable**: atoms $\{\mu, \omega, \log \lambda\}$, temperature, and the steering probes are fully differentiable; cold-started gates open during optimisation.
- **Strengths**: strong geometric prior (+17 pts over SDPA on small data at matched recipe; beats SDPA on ImageNet-1k at 3M and 12M params with ∼20% fewer parameters and ∼30% fewer FLOPs); inherently interpretable (each head is a few plottable blobs—early layers become convolutions in disguise, middle layers strip detectors, late layers go global); the field lives on coordinates, granting zero-shot resolution transfer.
- **Limitations**: encoder-only; requires a fixed $G_h \times G_w$ row-major raster (ViT needs `cls_token=false`; sequences need an explicit $1 \times L$ grid); the per-query steering kernels add a memory term; language results are unproven—content scoring is likely load-bearing there.

7 Fast-Weight Attention for Continual Learning

This annex covers the Falcon family [181], a fast-weight attention framework that treats the recurrent fast-weight state transition as an **online continual-learning rule** under *read-after-write* (RAW) autoregressive semantics. The framework's central observation is one of temporal alignment: for the prefix-prediction objective, the local fast-memory example revealed at step t is the **prefix-aligned pair** $(x_t, y_t) = (\phi(k_{t-1}),\, v_t)$ — one step shifted relative to the same-step association $(\phi(k_t), v_t)$ used by DeltaNet and standard linear attention. The same-step pairing remains causal, but it optimizes a different internal objective. The framework derives **normalized first-order updates** for two objectives (squared-error regression and negative inner product) and three step-size structures (scalar, per-column, sliding window), giving six variants: Falcon-1/2/3 (regression family) and Falcon-1A/2A/3A (inner-product family). All six are implemented in the codebase under `model.attention.falcon` with a shared configuration block, each providing both a recurrent reference scan and an SSD-style chunk-parallel kernel.

7.1 Preliminaries: Alignment, Normalization, and Stability

Read-after-write semantics. With feature maps $\phi(k_t)$ (QK-RMSNorm by default, ℓ_2 norm optional), the prefix-aligned stream sets $x_t = \phi(k_{t-1})$ for $t \geq 2$ with boundary sentinel $x_1 = 0$; the readout is $o_t = S_t^\top \phi(q_t)$. The write at step t therefore updates the memory *before* the query at step $t+1$ reads it, so no token's key ever associates with its own value.

Normalized first-order step sizes. A gradient step on the local objective $\ell_t(S)$ with the objective-matched denominator gives

$$\eta_t = \frac{\beta_t}{\text{statistic}_t + \lambda_t + \varepsilon}, \qquad \beta_t \in (0, 2),$$

where the statistic is the squared key energy $\|x_t\|^2$ (Falcon-1/1A/2/2A), the window spectral statistic $\mu_t^{(B)} = \lambda_{\max}(X^\top X)/B_t$ (Falcon-3), or the window mean energy $\bar{E}_t^{(B)}$ (Falcon-3A). Objective-matched normalization guarantees per-step descent and removes the scale sensitivity of fixed learning rates. The **plasticity** gain β_t, the **forgetting** ridge λ_t, the temporal **alignment**, and the rehearsal window B are thereby four previously entangled knobs, separated.

Positive-decay renormalization. When $\lambda_t > 0$, the per-step shrinkage $\alpha_t = \min(\eta_t \lambda_t,\, 1 - \varepsilon_\gamma)$ must be applied stably. The implementation clamps α_t strictly below 1, computes chunk-local log-prefixes in fp32 via $\log(1 - \alpha_t)$ and cumulative sums, rescales the drivers to the no-decay domain ($\tilde{\eta}_t = \eta_t/\gamma_t$, $\tilde{v}_t = v_t/c_{t-1}$ with $c_{t-1} = \prod_{r \leq t-1} \gamma_r$), and rescales outputs and the emitted state back by the accumulated δ. Boundary sentinels are $(\alpha_1, \gamma_1) = (0, 1)$ and $\eta_1 = 0$, so the first step is a pure read.

7.2 The Regression Family (Falcon-1/2/3)

The squared-error objective on the shifted pair is

$$\ell_t(S) = \tfrac{1}{2}\|v_t - S^\top x_t\|^2 + \tfrac{\lambda_t}{2}\|S\|_F^2, \qquad \nabla_S \ell_t = x_t (S^\top x_t - v_t)^\top + \lambda_t S,$$

which is L_t-smooth with $L_t = \|x_t\|^2 + \lambda_t$.

7.2.1 Falcon-1: Scalar NLMS Regression

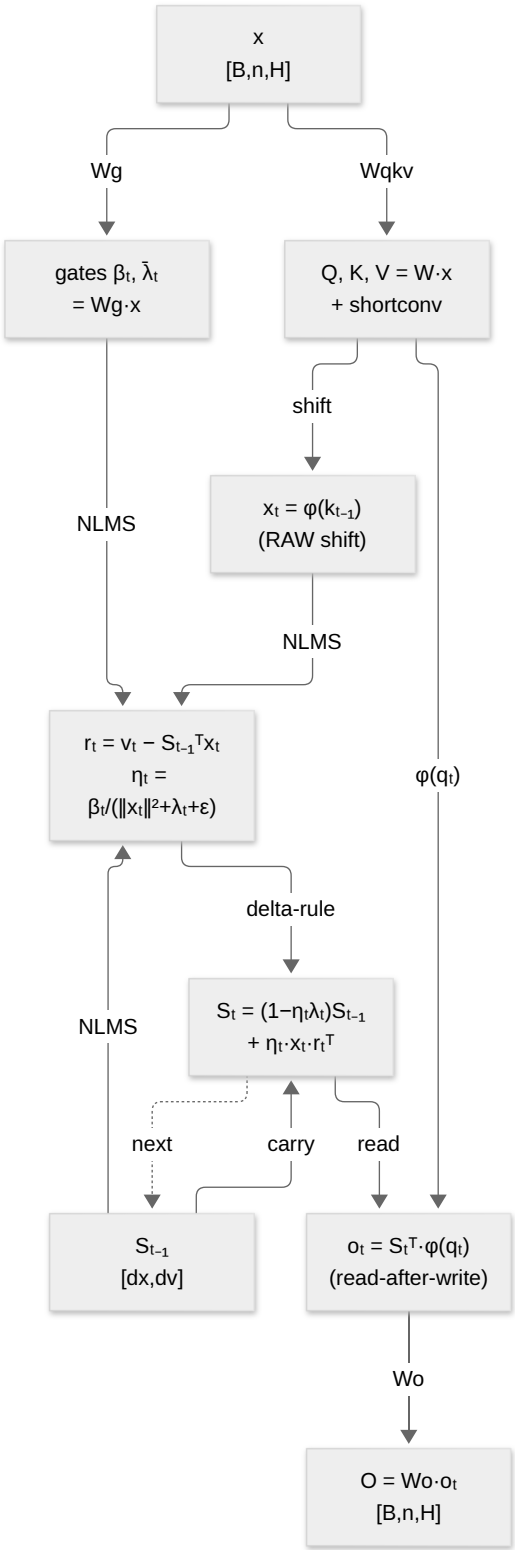


Figure 61: Dataflow of Falcon-1: the prefix-aligned key/value pair drives a scalar NLMS delta-rule update of the fast-weight state.

The scalar delta-rule variant — the classical NLMS recursion of adaptive filtering upgraded with the shifted write stream and an explicit ridge:

$$S_t = (1 - \eta_t \lambda_t)\, S_{t-1} + \eta_t\, x_t \big(v_t - S_{t-1}^\top x_t\big)^\top, \qquad o_t = S_t^\top \phi(q_t).$$

With $\varepsilon = 0$, $\lambda_t = 0$ and the unshifted assignment $x_t = \phi(k_t)$ this recovers DeltaNet exactly. The eigenstructure of the transition $A_t = \gamma_t I - \eta_t x_t x_t^\top$ (with $\gamma_t = 1 - \eta_t \lambda_t$) shows that the write direction contracts by $1 - \beta_t$ while the orthogonal subspace carries γ_t: $\beta_t > 1$ yields a stable sign-flip along the write direction, useful for state tracking. The chunk-parallel form uses the WY representation $\prod_s (I - \eta_s x_s x_s^\top) = I - UTU^\top$ with a single-residual unit-lower-triangular solve $L = \mathrm{tril}(G, -1) + I$ where $G = X^\top X$ (Algorithm 8 of the paper), after the positive-decay renormalization.

7.2.2 Falcon-2: Per-Column NLMS Regression

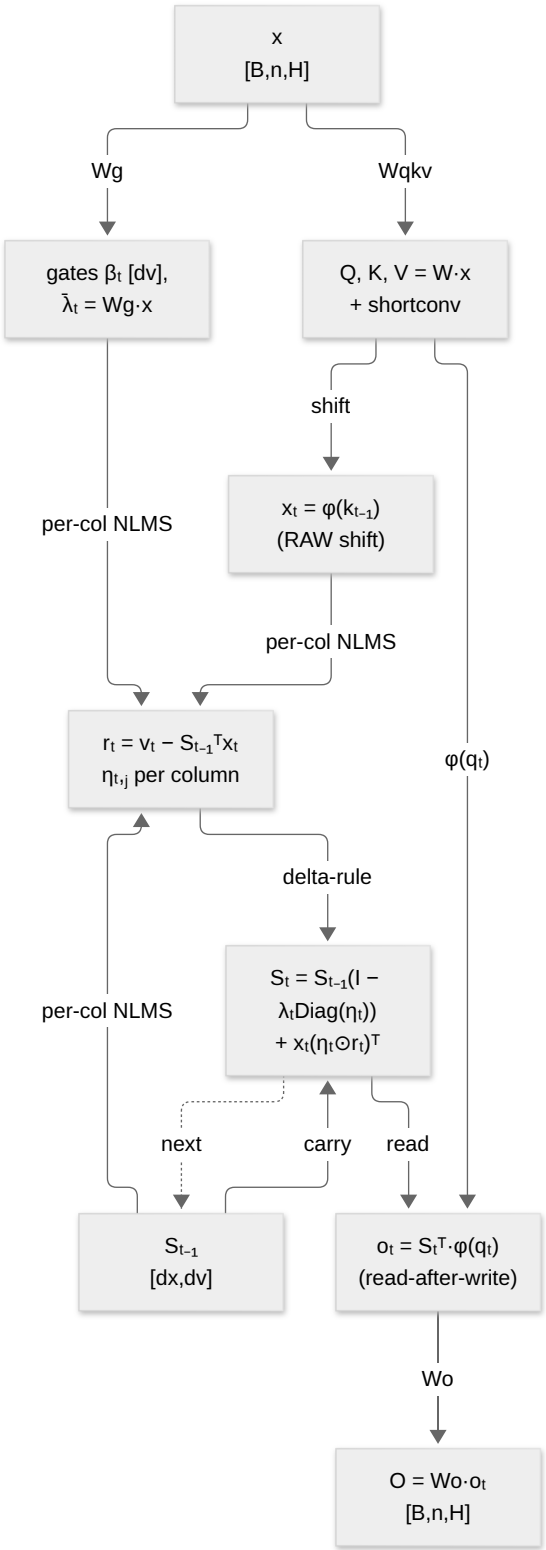


Figure 62: Dataflow of Falcon-2: per-column NLMS step sizes update the fast-weight state channel by channel.

The ridge loss decomposes across value coordinates, so each value channel j receives its own step size while sharing the NLMS normalizer:

$$\eta_{j,t} = \frac{\beta_{j,t}}{\|x_t\|^2 + \lambda_t + \varepsilon}, \qquad S_t = S_{t-1}\big(I - \lambda_t \operatorname{Diag}(\boldsymbol{\eta}_t)\big) + x_t\big(\boldsymbol{\eta}_t \odot \mathbf{r}_t\big)^\top.$$

The chunk-parallel kernel forms the shared key Gram $G = X^\top X$ and solves the batch of per-channel unit-lower-triangular systems $L_j = I + \operatorname{tril}\big(G \odot (\Sigma_j \Sigma_j^\top), -1\big)$ with $\Sigma = \sqrt{\boldsymbol{\eta}}$ (Algorithm 1). Defined but not separately benchmarked in the paper.

7.2.3 Falcon-3: Sliding-Window Minibatch Regression

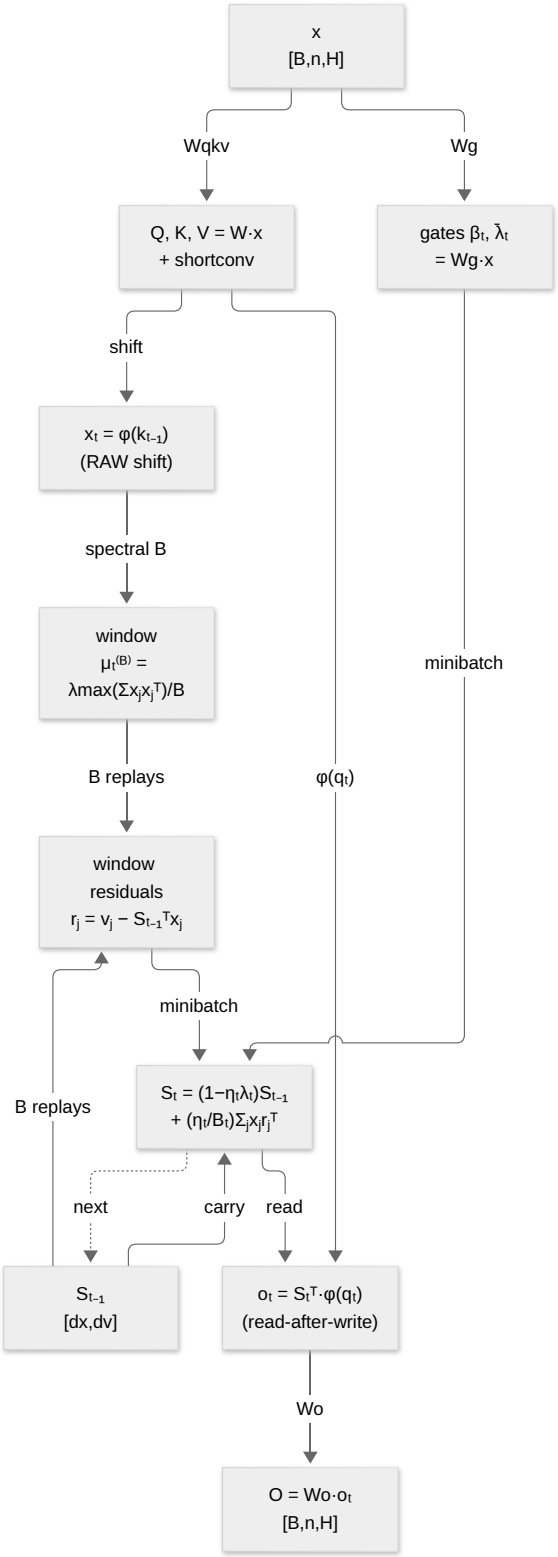


Figure 63: Dataflow of Falcon-3: a sliding window of prefix-aligned pairs provides the spectral statistic and the minibatch write.

Bounded rehearsal: each step regresses the fast memory on the last $B_t \le B$ prefix-aligned pairs, with *all window residuals evaluated at the pre-update state* S_{t-1}:

$$\mu_t = \frac{\lambda_{\max}\big(X_{\mathcal{I}_t}^\top X_{\mathcal{I}_t}\big)}{B_t}, \qquad \eta_t = \frac{\beta_t}{\mu_t + \lambda_t + \varepsilon}, \qquad S_t = (1 - \eta_t \lambda_t)\, S_{t-1} + \frac{\eta_t}{B_t} \sum_{j \in \mathcal{I}_t} x_j \big(v_j - S_{t-1}^\top x_j\big)^\top,$$

with $\mathcal{I}_t = \{\max(2, t - B + 1), \dots, t\}$. Window-averaging keeps the update magnitude invariant to B: each pair is replayed in exactly B consecutive updates at weight $1/B$, so its cumulative injection is η-independent of B. The state alone is *not* Markov: exact continuation requires a FIFO tail of the last $B - 1$ pairs, and the chunk kernels gather an extended slice with $|\mathcal{J}| \le C + B - 1$. The chunk-parallel form is a rank-B affine recurrence solved by block forward substitution over the (time $\times$ rank) extended index space, with same-time rank components uncoupled (Algorithm 3).

7.3 The Inner-Product Family (Falcon-1A/2A/3A)

The negative inner-product objective

$$\ell_t^{\mathrm{ip}}(S) = -\langle S^\top x_t,\, v_t \rangle + \tfrac{\lambda_t}{2} \|S\|_F^2 \qquad \Rightarrow \qquad \nabla_S \ell_t^{\mathrm{ip}} = -x_t v_t^\top + \lambda_t S$$

yields purely additive (Hebbian) writes with energy-normalized gains.

7.3.1 Falcon-1A and Falcon-2A

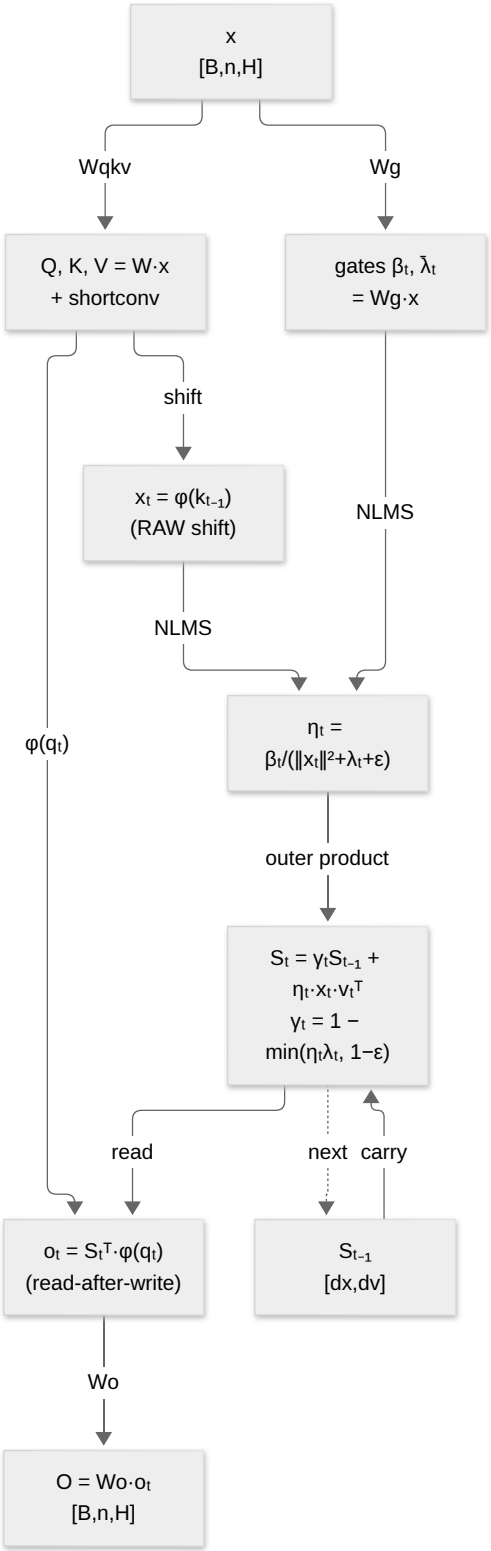


Figure 64: Dataflow of Falcon-1A: an energy-normalized additive (Hebbian) write with a derived decay.

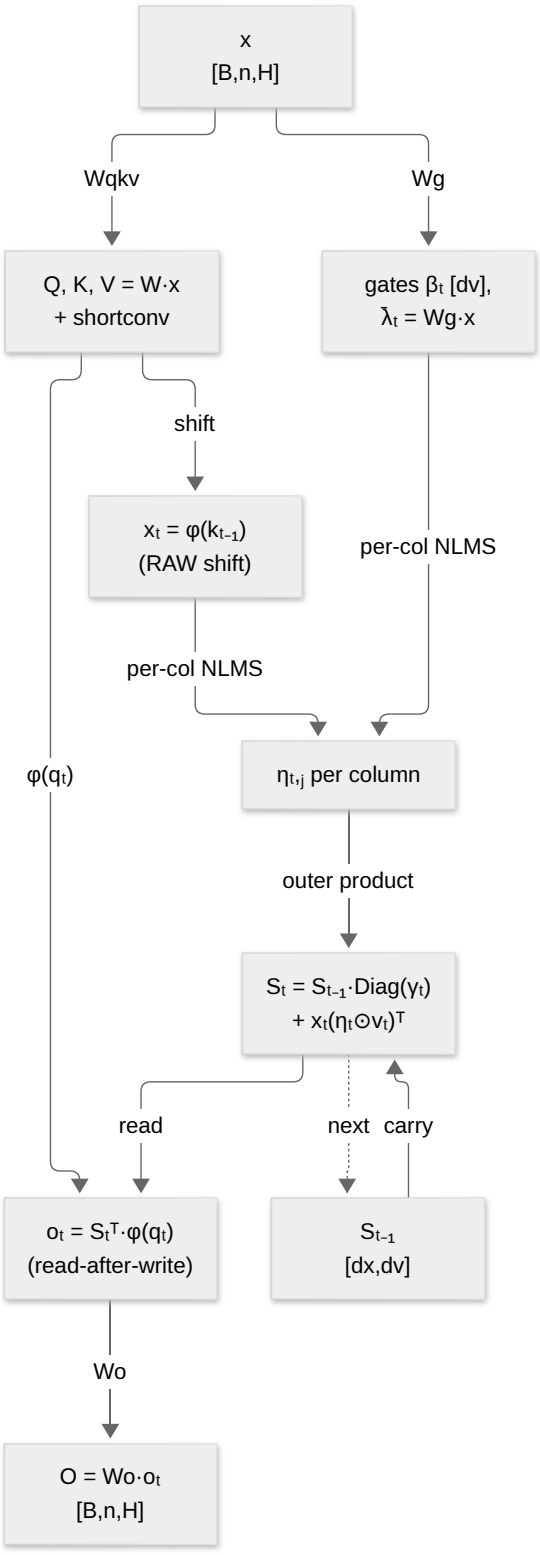


Figure 65: Dataflow of Falcon-2A: per-column additive writes with a per-channel decay.

$$\text{Falcon-1A:}\quad S_t = \gamma_t S_{t-1} + \eta_t\, x_t v_t^\top, \quad \gamma_t = 1 - \min(\eta_t \lambda_t,\, 1 - \varepsilon_\gamma);$$
$$\text{Falcon-2A:}\quad S_t = S_{t-1}\,\mathrm{Diag}(\boldsymbol{\gamma}_t) + x_t(\boldsymbol{\eta}_t \odot v_t)^\top.$$

With $\lambda_t = 0$ and the unshifted assignment this is exactly standard linear attention / Mamba-2 accumulation; the shifted stream makes it the one-step-shifted next-latent variant. The A-family's denominator is not required by curvature (the objective is linear) — it retains the

energy-normalized write as a magnitude stabilizer. The chunk-parallel form is decay-mask linear attention with $M_{t,i} = \eta_i \prod_{r=i+1}^{t} \gamma_r$ (per-channel masks for Falcon-2A).

7.3.2 Falcon-3A: Sliding-Window Inner-Product

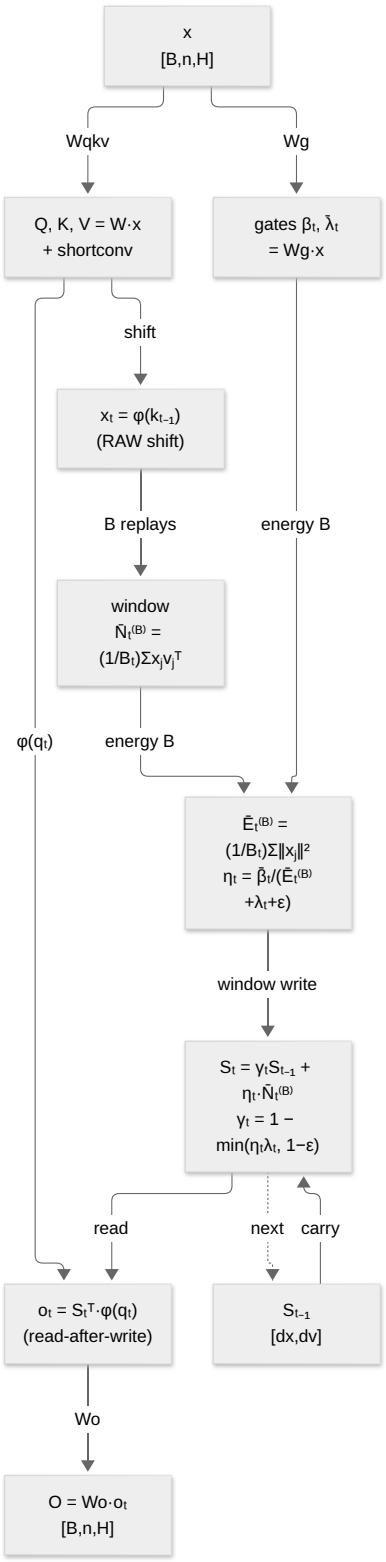


Figure 66: Dataflow of Falcon-3A: a sliding-window mean-energy statistic normalizes the additive window write.

The windowed additive variant — the family's best length-extrapolation configuration:

$$\bar{N}_t^{(B)} = \frac{1}{B_t}\sum_{j\in\mathcal{I}_t} x_j v_j^\top, \quad \bar{E}_t^{(B)} = \frac{1}{B_t}\sum_{j\in\mathcal{I}_t} \|x_j\|^2, \quad \eta_t = \frac{\beta_t}{\bar{E}_t^{(B)} + \lambda_t + \varepsilon}, \qquad S_t = \gamma_t S_{t-1} + \eta_t \bar{N}_t^{(B)}.$$

The window-banded decay mask factors as $M = D \cdot \mathrm{Diag}(\boldsymbol{\eta}) \cdot A$, where A is the B-banded window operator and D the causal decay kernel, evaluated over the extended slice (Algorithm 4).

7.4 Configuration Interface

All six variants share one nested block, `model.attention.falcon`, selected through the `layer_pattern` entries `falcon1`, `falcon2`, `falcon3`, `falcon1a`, `falcon2a`, `falcon3a`:

Key	Default	Description
`chunk_size`	64	Chunk size of the chunk-parallel kernels; `null` selects the recurrent reference scan
`qk_norm`	`rms_norm`	QK normalization (`rms_norm` or `l2_norm`); RMSNorm computed in fp32
`beta_mode`	`ctx_eta`	Plasticity gate parameterization: `static`, `ctx_beta` (content-conditioned gain), or `ctx_eta` (content-conditioned normalized step)
`lambda_mode`	`ctx`	Forgetting gate: `static` or `ctx` (scale-couples with the detached curvature statistic)
`beta` / `lambda`	1.0 / 0.0	Static gain and ridge (used when the corresponding mode is `static`)
`window`	4	Sliding window B (Falcon-3 / Falcon-3A only)
`short_conv` / `conv_kernel`	`true` / 4	Causal depth-wise short convolutions on the q/k/v projections
`eps` / `eps_gamma`	10^{-6} / 10^{-4}	Step-size stabilizer and positive-decay clamp floor

Table 7: Shared Falcon configuration knobs (`model.attention.falcon`). Each variant additionally exposes a per-mixer positional-encoding opt-out `falcon{1,2,3,1a,2a,3a}_use_pe`, default `False` like other recurrent/decay mixers.

Implementation notes: the recurrent and chunk-parallel kernels are verified numerically equivalent (chunk ≡ recurrent to $\leq 3.6 \times 10^{-7}$ across all six variants, including decay, multi-chunk, and windowed paths); gradients flow through the fp32 triangular-solve islands (feature drivers take only $\sqrt{\tilde{\eta}}$ while value targets take $\sqrt{\tilde{\eta}}\, e^{-u_{t-1}}$, with the square root clamped at 10^{-12} for NaN-free backward at the $\eta = 0$ boundary); positive exponents are clamped at 80.

7.5 Published Results

Model (124M, 49.2B FineWeb-Edu tokens)	WikiText ppl	LMB. ppl	FineEdu ppl	0-shot avg
Transformer (RoPE)	33.25	47.43	17.38	48.16
Mamba-2	34.53	48.74	17.70	48.80
DeltaNet	34.19	52.84	17.84	48.88
Gated DeltaNet	30.99	46.70	17.32	48.78
Falcon-1A.3	34.02	49.84	17.40	48.95
Falcon-1.3	33.00	48.70	**17.10**	**49.18**

Table 8: Language modeling results reported by Zhang et al. [181] at 124M parameters. On variable-digit addition length extrapolation (33–48 digits), Falcon-3A.3 reaches **87.2%** mean accuracy versus 65.8% for a RoPE Transformer and 85.9% for Falcon-1A.3: the inner-product family carries arithmetic, while the regression family carries perplexity.

7.6 Relation to Prior Work

- **DeltaNet**: the delta rule is the squared-error one-gradient-descent step with the same-step pairing; Falcon recovers it with the critical index shift plus objective-matched normalization and explicit λ_t shrinkage.
- **Gated DeltaNet**: adds a free learned carry gate; Falcon *derives* the carry $\gamma_t = 1 - \eta_t \lambda_t$ from the local objective and the normalized step-size parameterization instead of adding an independent gate, and defaults to QK-RMSNorm rather than ℓ_2 normalization.
- **Linear attention / Mamba-2**: the additive numerator is one gradient step on the inner-product objective with the unshifted assignment; the chunkwise kernels reuse the SSD scheme.
- **TTT / Titans / ATLAS**: fast-weight updates as test-time training; Falcon-3 is a sliding-window specialization of the internal-objective view with strict next-latent alignment.
- **Adaptive filtering (LMS/NLMS/RLS)**: Falcon imports the stability and normalization principles of NLMS into fast-weight attention under strict causality.

7.6.1 When to Use Each Variant

1. **Falcon-1**: best language-modeling perplexity of the family (FineEdu 17.10, beating Gated DeltaNet 17.32); principled forgetting tied to the local objective.
2. **Falcon-2**: finer-grained plasticity than Falcon-1 at the cost of heavier per-channel chunk kernels; unbenchmarked in the paper.
3. **Falcon-3**: controlled rehearsal horizon with exact window statistics; most complex kernel.
4. **Falcon-1A**: simple and stable additive variant; second-best length extrapolation (85.9%).
5. **Falcon-2A**: channel-wise forgetting granularity; unbenchmarked.
6. **Falcon-3A**: best length extrapolation of the paper (87.2% on 33–48-digit addition); carries arithmetic tasks.

8 Convolution-Augmented Attention: Conformer and Fast Conformer

8.1 Executive Summary

The Conformer block [56] is the toolkit's convolution-augmented sequence mixer, exposed as `conformer`. It keeps the global receptive field of scaled dot-product attention and adds a depthwise-convolution branch that models local, narrow-band structure—the inductive bias speech recognition needs and the one plain self-attention lacks. The block sandwiches multi-head self-attention (MHSA) with relative sinusoidal positional encodings between two Macaron-style half-step feed-forward layers [109], and inserts the convolution module between the attention and the second FFN. In the reference recipe it reaches 1.9%/3.9% WER on LibriSpeech *test-clean*/*test-other* without an external language model (Conformer-L, 118.8M parameters), and it is the mixer recommended for the audio model class of Appendix 22. Its long-form sibling, Fast Conformer [136], keeps the convolution module and the block layout but replaces global self-attention with limited-context attention—a fixed local window plus a global token pinned at the sequence head—so hours-long audio decodes with bounded memory; it is documented below. The toolkit runs both feed-forward layers at full step (the paper's "full-step" variant of its Table 5) and lets the outer layer own the second one; Table 9 lists the deviations.

The mixers are available to every model class: the text encoder, the causal decoder (where depthwise padding is shifted to stay streaming-safe), the Vision Transformer, and `frankenstein_audio`. Because the blocks live inside `HybridLayer`, they compose with all 48 other mixers, the 6 norms, the 4 residual strategies, MoE, Mixture-of-Depths, mHC, AttnRes, BitNet and the 38 optimizer families.

8.2 Conformer: Convolution-augmented Attention

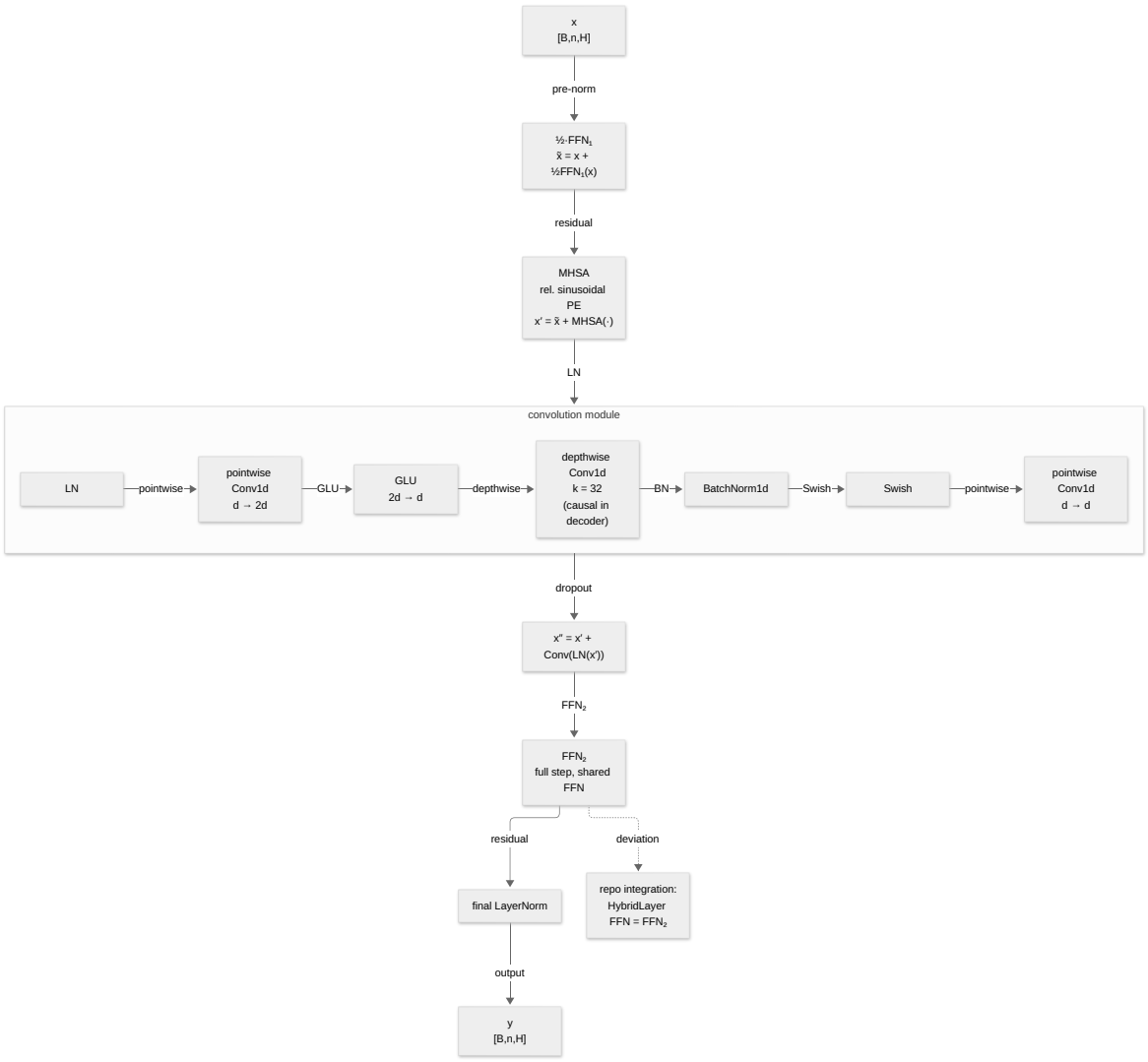


Figure 67: Dataflow of the Conformer block: a Macaron half-step feed-forward layer, multi-head self-attention with relative sinusoidal positional encodings, the convolution module (pointwise $\rightarrow$ GLU $\rightarrow$ depthwise $\rightarrow$ BatchNorm $\rightarrow$ Swish $\rightarrow$ pointwise), and the second feed-forward layer followed by the final LayerNorm. In this toolkit the shared `HybridLayer` feed-forward network plays the FFN_2 slot at a full residual step (dotted annotation).

8.2.1 Mathematical Formulation

For a block input $x \in \mathbb{R}^{B\times n\times d}$ the reference block of Fig. 1 of Gulati et al. [56] is

$$\tilde{x} = x + \tfrac{1}{2}\,\mathrm{FFN}_1\big(\mathrm{LN}(x)\big), \tag{7}$$
$$x' = \tilde{x} + \mathrm{MHSA}\big(\mathrm{LN}(\tilde{x})\big), \tag{8}$$
$$x'' = x' + \mathrm{Conv}\big(\mathrm{LN}(x')\big), \tag{9}$$
$$y = \mathrm{LN}\Big(x'' + \tfrac{1}{2}\,\mathrm{FFN}_2\big(x''\big)\Big). \tag{10}$$

The two feed-forward terms carry the Macaron *half-step* coefficient: each contributes half of a full residual update, one before and one after the attention–convolution pair.

Feed-forward module. With hidden width d_{ff},

$$\mathrm{FFN}(x) = W_2\,\mathrm{Dropout}\big(\mathrm{Swish}(W_1\,\mathrm{LN}(x) + b_1)\big) + b_2, \qquad \mathrm{Swish}(a) = a\,\sigma(a), \tag{11}$$

where $W_1 \in \mathbb{R}^{d_{ff}\times d}$, $W_2 \in \mathbb{R}^{d\times d_{ff}}$ and σ is the logistic sigmoid. The paper uses $d_{ff} = 4d$.

Multi-head self-attention with relative positional encoding. Queries, keys and values are linear projections of the normalized block input, $Q = \mathrm{LN}(\tilde{x})W_Q$, $K = \mathrm{LN}(\tilde{x})W_K$, $V = \mathrm{LN}(\tilde{x})W_V$, split into H heads of width $d_h = d/H$. Following the Transformer-XL relative scheme [27], the score of query i against key j adds a sinusoidal function of the *displacement* $i - j$ instead of an absolute position:

$$e_{ij} = q_i^\top k_j + (q_i + v)^\top R_{i-j}, \qquad A_{ij} = \mathrm{softmax}_j\Big(\frac{e_{ij}}{\sqrt{d_h}}\Big), \qquad O = AVW_O, \tag{12}$$

with v a learned bias vector and, for $m = 0, \ldots, d_h/2 - 1$,

$$R_{\delta,\,2m} = \sin\big(\delta \,/\, 10000^{2m/d_h}\big), \qquad R_{\delta,\,2m+1} = \cos\big(\delta \,/\, 10000^{2m/d_h}\big). \tag{13}$$

Because the encoding depends only on $\delta = i - j$, the block generalizes across utterance lengths. The toolkit routes positional information through the model-wide machinery that every other mixer uses: `model.positional_encoding` (RoPE by default) is applied to Q/K, and `conformer_use_pe` (default `true`) lets a layer opt out. The paper's relative sinusoidal scheme is closest to `sinusoidal_rotary` (sinusoidal frequencies applied as a rotation on Q/K, relative by construction); `alibi`/`bam` supply a distance bias instead of a rotation. A dedicated Transformer-XL-style relative-sinusoidal module is not implemented. In decoder mode the score matrix receives the usual causal mask before the softmax, and the fixed-softmax path of Appendix 15.3 is used when `numerics.fixed_softmax` is enabled.

Convolution module. The module applies a gated depthwise convolution over the time axis:

$$h = \mathrm{LN}(x'), \tag{14}$$
$$[u,\, g] = \mathrm{Conv1d}_{d\to 2d}(h), \qquad \tilde{h} = u \odot \sigma(g), \tag{15}$$
$$z_t = \sum_{k=0}^{K-1} w_k\, \tilde{h}_{t+k-\lfloor K/2\rfloor}, \tag{16}$$
$$c = \mathrm{Dropout}\Big(\mathrm{Conv1d}_{d\to d}\big(\mathrm{Swish}(\mathrm{BN}(z))\big)\Big), \tag{17}$$

where $\odot$ is the elementwise product (a GLU that halves the $2d$ channels back to d), $w \in \mathbb{R}^K$ is the depthwise kernel and BN normalizes each channel over the batch and time axes,

$$\mathrm{BN}(z)_{b,t,c} = \gamma_c\,\frac{z_{b,t,c} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}} + \beta_c. \tag{18}$$

In decoder mode the sum in Equation 16 is shifted to $z_t = \sum_{k=0}^{K-1} w_k \tilde{h}_{t-K+1+k}$, so that no future frame enters the convolution; encoder and ViT modes keep the symmetric "same" padding of the paper.

8.2.2 Algorithmic Pseudocode

Algorithm 22 Conformer block (`conformer`)

Require: block input $x \in \mathbb{R}^{B\times n\times d}$, kernel $K = 32$, heads H, Macaron flag, mode
Ensure: block-level delta Δ such that $x + \Delta$ enters the outer $\text{norm}_2 + \text{FFN}_2$

$\tilde{x} \leftarrow x + \text{FFN}_1(\text{LN}(x))$ ▷ Macaron FFN_1 ($\frac{1}{2}$ in the paper)
$Q, K_{proj}, V \leftarrow \text{LN}(\tilde{x})W_Q,\ \text{LN}(\tilde{x})W_K,\ \text{LN}(\tilde{x})W_V$
$e_{ij} \leftarrow q_i^\top k_j + (q_i + v)^\top R_{i-j}$ ▷ relative sinusoidal PE, Eq. 12
if mode = decoder **then**
    $e_{ij} \leftarrow -\infty$ for $j > i$ ▷ causal mask
end if
$x' \leftarrow \tilde{x} + \text{softmax}(e/\sqrt{d_h})\, V\, W_O$
$h \leftarrow \text{LN}(x')$
$[u, g] \leftarrow \text{Conv1d}_{d\to 2d}(h); \quad \tilde{h} \leftarrow u \odot \sigma(g)$ ▷ GLU
$z \leftarrow \text{DepthwiseConv1d}_K(\tilde{h})$ ▷ causal shift in decoder mode
$c \leftarrow \text{Dropout}(\text{Conv1d}_{d\to d}(\text{Swish}(\text{BN}(z))))$
$x'' \leftarrow x' + c$
return x'' ▷ `HybridLayer` adds FFN_2 and the final LN

8.2.3 Implementation Mapping and Deviations

Table 9 maps the paper block onto the toolkit. The substantive deviations are the two residual scalings and the second feed-forward layer: the paper keeps both $\frac{1}{2}$ Macaron coefficients and wraps FFN_2 in the block, whereas `conformer` uses full steps and returns the state after the convolution module, letting the outer `HybridLayer` apply its shared $\text{norm}_2 + \text{FFN}$ (also at full step). The decomposition is deliberate—it keeps MoE routing, Mixture-of-Depths, mHC/Hyperloop streams, AttnRes, FlashNorm fusion, BitNet projections and the per-parameter-group optimizer routing available for the second FFN, none of which would survive a private FFN inside the mixer.

Paper component	Toolkit implementation	Notes
$\frac{1}{2}$FFN$_1$ (pre-norm)	private FFN inside `ConformerAttention`; Swish; full-step residual	`conformer.macaron: true` (default); the $\frac{1}{2}$ coefficient of the paper is dropped; disabling leaves attention + convolution only
MHSA, relative sinusoidal PE	own $q/k/v/out$ projections; model-wide PE via `conformer_use_pe` (default `true`)	paper's relative sinusoid $\approx$ `sinusoidal_rotary`; no dedicated Transformer-XL module (RoPE is the model-wide default)
Convolution module	LN $\rightarrow$ pointwise $\rightarrow$ GLU $\rightarrow$ depthwise $\rightarrow$ BN $\rightarrow$ Swish $\rightarrow$ pointwise $\rightarrow$ dropout	`conformer.kernel_size` (32), `conformer.conv_norm` (`batch`/`group`/`layer`)
$\frac{1}{2}$FFN$_2$ at half step	`HybridLayer` norm$_2$ + FFN, full step	justified by Table 5 of Gulati et al. [56]: 1.9/4.4 vs. 1.9/4.5 dev WER (both coefficients dropped)
Final LayerNorm	`HybridLayer` final norm	unchanged from every other mixer
4$\times$ convolutional subsampling	`frankenstein_audio` frontend	Appendix 22; text/ViT use their own embeddings
Dropout ($p = 0.1$)	`model.dims.dropout`	global, per-block

Table 9: Implementation mapping of the Conformer block. The deviations are the dropped $\frac{1}{2}$ Macaron scalings, the second FFN (supplied by the outer layer at full step) and the positional-encoding plumbing.

Why the full-step FFN is safe. Dropping the Macaron $\frac{1}{2}$ changes the per-branch residual scale, not the expressive form of the block: the full-step variant is exactly what Gulati et al. [56] ablate in their Table 5, where it moves dev-clean/dev-other WER from 1.9/4.4 (half step) to 1.9/4.5 (full step), i.e. one tenth of a point. The toolkit therefore uses full steps for both feed-forward layers; the first stays private to the mixer, and the second is the shared one, which preserves the composability described above.

8.3 Published Ablations

Table 10 summarizes the ablations that motivate the default configuration. Values are the ones reported by Gulati et al. [56]; the qualitative ordering is what the defaults encode.

Ablation	Settings compared	Reported outcome
Convolution module (Table 3)	present vs. removed	the largest single degradation of the block when removed; the local branch is load-bearing, not decorative
Macaron FFN pair (Table 3)	$\frac{1}{2}$FFN$_1$ + $\frac{1}{2}$FFN$_2$ vs. one FFN	removing the pair costs roughly 0.2 dev-clean / 0.5 dev-other WER
Half vs. full step (Table 5)	$\frac{1}{2}$ residuals vs. full residuals	1.9/4.4 vs. 1.9/4.5 dev WER: at most 0.1 point
Kernel size (Table 7)	$K \in \{3, 7, 17, 32, 65\}$	$K = 32$ is best; small kernels under-fit the local context, $K = 65$ adds cost without gain
Conv placement	after MHSA, before FFN$_2$	reference order retained; moving the branch does not improve on it

Table 10: Conformer ablations and the defaults they justify (`kernel_size: 32`, `macaron: true`, full-step shared FFN$_2$).

8.4 Reuse of Existing Infrastructure

Because `ConformerAttention` returns a delta and inherits the `mixer(config, pos_encoder=...)` constructor, the entire stack is reused without special cases:

- **Model classes:** text encoder (`mlm`/`sbert`/`text_classification`), causal decoder (`causal_lm`), Vision Transformer (`patch_prediction`/`classification`/`segmentation`) and `frankenstein_audio` (`asr`, `audio_classification`).
- **Normalization:** all 6 variants, including partial RMSNorm and FlashNorm fusion on norm_1/norm_2; the convolution module keeps its own BatchNorm unless `conformer.conv_norm` selects group or layer normalization.
- **Routing and residuals:** MoE, Mixture-of-Depths, mHC, Hyperloop, AttnRes and the BitNet ternary projections (the private FFN and the convolution projections are BitLinear-aware).
- **Optimizers:** all 23 families with per-group prefixes; mixer parameters, including the depthwise kernel and the convolution norms, route to the `attention` group.
- **KV cache:** `conformer` is registered as *not* cacheable (`supported=False`), because the depthwise-convolution state and the BatchNorm running statistics would have to be carried across steps. `generate()` falls back to the full-recompute path, which is exact.

8.5 When to Use It

- **Speech:** the default mixer for `asr` and `audio_classification`; at 40 ms frames after 4× subsampling the kernel of 32 spans ∼1.28 s of context, roughly the width of a phone sequence.
- **Text and vision:** a strong drop-in for `standard` when the data has local structure (character-level models, code, OCR, patch sequences); it costs one depthwise convolution and two extra norms per layer over a plain attention block.
- **Streaming decoding:** usable in `mode: decoder` with causal convolution padding, but without the incremental cache of Appendix 23; the depthwise recurrence is the missing piece.

8.6 Fast Conformer: Limited-Context Attention with a Head-Pinned Global Token

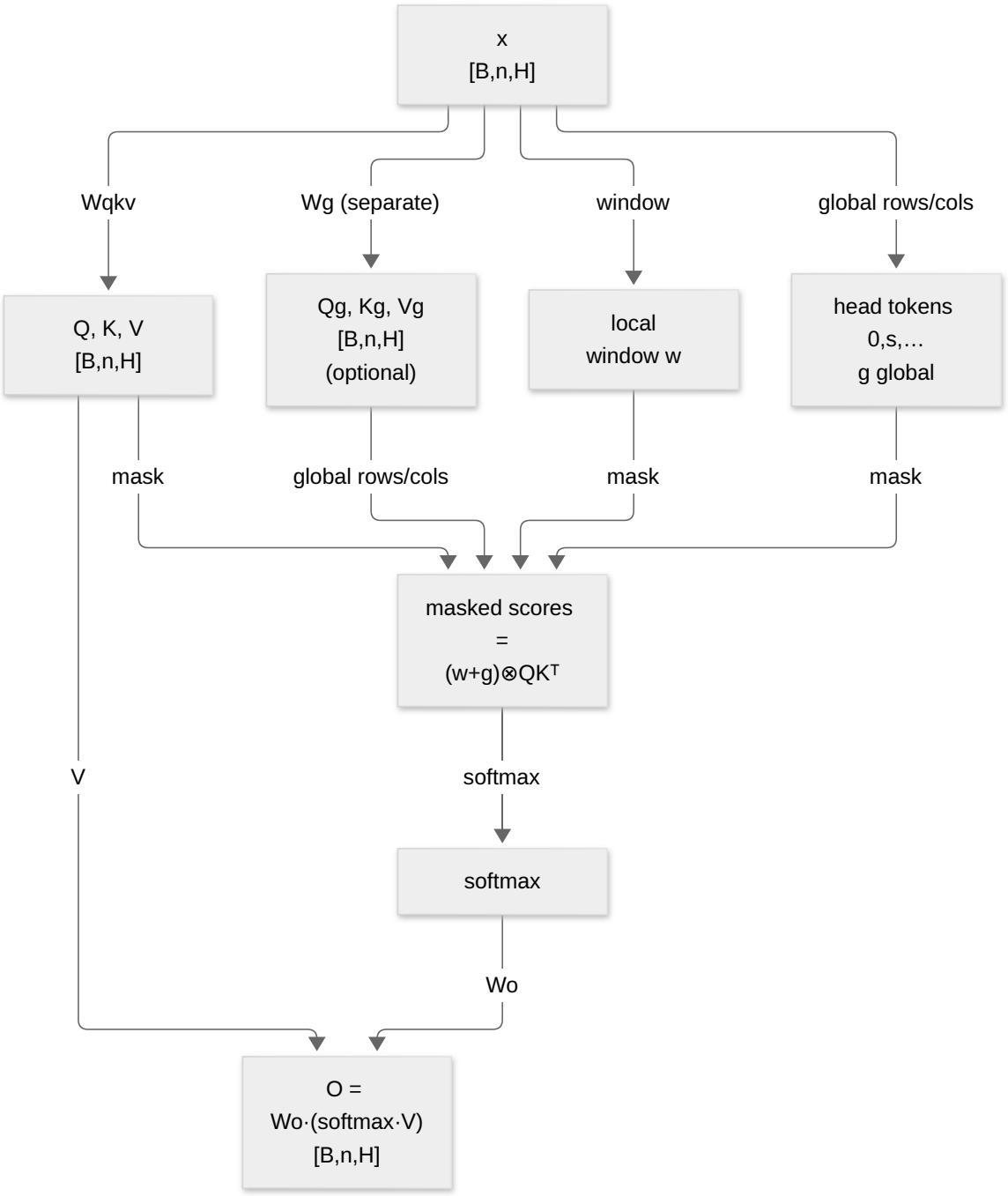


Figure 68: Dataflow of Fast Conformer limited-context attention: every query attends to a symmetric local window, while a small set of global attention tokens pinned at the head of the sequence attend to every position and are attended by every position; the global token may use its own query/key/value projections (paper §2.2).

8.6.1 Mathematical Formulation

Fast Conformer [136] attacks the quadratic cost of Conformer self-attention in two ways: it raises the front-end downsampling to 8×, and, for long-form inputs, it replaces global attention

with *limited-context* attention. The paper describes the mixer in prose rather than in display equations: "we use a single global attention token, which attends to all other tokens and has all other tokens attend to it, using a separate set of query, key and value linear projections, while others attend in a fixed-size window surrounding each token" (§2.2), while the evaluated long-form model "has shared query, key and value projection layers that are used for global and local attention" (§3.4) and sets "the size of the limited context ... to 128 steps on each side of a token". This implementation encodes exactly that description and inherits the global-attention dataflow from Longformer [12], which the paper explicitly cites for the mechanism and for its overlapping-chunk kernel.

Let $X \in \mathbb{R}^{n \times H}$ be the layer input, h the number of heads and $d = H/h$. The shared projections are

$$\mathbf{Q} = X\mathbf{W}_Q, \qquad \mathbf{K} = X\mathbf{W}_K, \qquad \mathbf{V} = X\mathbf{W}_V,$$

with $\mathbf{W}_* \in \mathbb{R}^{H \times H}$ and no bias. The optional global projections of the paper's §2.2 variant are a second set,

$$\mathbf{Q}^g = X\mathbf{W}_Q^g, \qquad \mathbf{K}^g = X\mathbf{W}_K^g, \qquad \mathbf{V}^g = X\mathbf{W}_V^g,$$

which adds $3H^2$ parameters; the §3.4 evaluated model sets $\mathbf{W}_*^g = \mathbf{W}_*$.

The global positions are the head of the sequence, spaced s apart,

$$\mathcal{G} = \{\, 0, s, 2s, \ldots, (g-1)s \,\},$$

so a global token is a designated input position rather than an extra learned embedding (the paper's "addition of a global token"). The attention pattern is

$$M_{ij} = \big[\, |i-j| \le w/2 \,\big] \;\vee\; \big[\, j \in \mathcal{G} \,\big] \;\vee\; \big[\, i \in \mathcal{G} \,\big],$$

where the first term is the local window, the second lets every query attend to the global tokens, and the third lets a global token attend to every position; M is intersected with the causal mask in decoder mode. With the effective keys and values

$$\tilde{\mathbf{k}}_j = \begin{cases} \mathbf{k}_j^g & j \in \mathcal{G} \text{ and separate,} \\ \mathbf{k}_j & \text{otherwise,} \end{cases} \qquad \tilde{\mathbf{v}}_j = \begin{cases} \mathbf{v}_j^g & j \in \mathcal{G} \text{ and separate,} \\ \mathbf{v}_j & \text{otherwise,} \end{cases}$$

the output is standard scaled dot-product attention over the allowed pairs,

$$z_{ij} = \frac{\mathbf{q}_i \cdot \tilde{\mathbf{k}}_j}{\sqrt{d}}, \qquad \mathbf{o}_i = \sum_{j:\, M_{ij}} \operatorname{softmax}_j(z_{ij})\, \tilde{\mathbf{v}}_j, \qquad \mathbf{Y} = \mathbf{O}\mathbf{W}_O.$$

In the separate variant a global query row is computed with the global projections over the whole causal prefix, $\mathbf{o}_i = \sum_{j \le i} \operatorname{softmax}_j(\mathbf{q}_i^g \cdot \mathbf{k}_j^g/\sqrt{d})\, \mathbf{v}_j^g$, matching Longformer's global-attention design. With shared projections all three cases collapse to one $\mathbf{Q}/\mathbf{K}/\mathbf{V}$ set: the global tokens are simply positions with unmasked rows and columns.

8.6.2 Algorithmic Pseudocode

Algorithm 23 Fast Conformer limited-context attention (masked-dense reference form)

Require: Input X; window w; global count g; spacing s; separate flag
Ensure: Output $\mathbf{Y}$

$\mathbf{Q}, \mathbf{K}, \mathbf{V} \leftarrow X\mathbf{W}_Q, X\mathbf{W}_K, X\mathbf{W}_V$
if separate **then**
  $\mathbf{Q}^g, \mathbf{K}^g, \mathbf{V}^g \leftarrow X\mathbf{W}_Q^g, X\mathbf{W}_K^g, X\mathbf{W}_V^g$
  $\tilde{\mathbf{K}}, \tilde{\mathbf{V}} \leftarrow \mathbf{K}, \mathbf{V}$; replace rows $\mathcal{G}$ with $\mathbf{K}^g, \mathbf{V}^g$
else
  $\tilde{\mathbf{K}}, \tilde{\mathbf{V}} \leftarrow \mathbf{K}, \mathbf{V}$
end if
$\mathcal{G} \leftarrow \{0, s, \ldots, (g-1)s\}$ clipped to n
$M \leftarrow [\,|i-j| \le w/2\,] \vee [\,j \in \mathcal{G}\,] \vee [\,i \in \mathcal{G}\,]$
if decoder **then** $M \leftarrow M \wedge [\,j \le i\,]$
end if
$Z \leftarrow \mathbf{Q}\tilde{\mathbf{K}}^\top/\sqrt{d}$ masked by M
if separate **and** $\mathcal{G} \neq \emptyset$ **then**
  $Z[\mathcal{G}, :] \leftarrow \mathbf{Q}^g[\mathcal{G}]\mathbf{K}^{g\top}/\sqrt{d}$ (causal-clipped)
end if
$\mathbf{O} \leftarrow \mathrm{softmax}(Z)\tilde{\mathbf{V}}$
if separate **and** $\mathcal{G} \neq \emptyset$ **then**
  $\mathbf{O}[\mathcal{G}, :] \leftarrow \mathrm{softmax}(Z[\mathcal{G}, :])\mathbf{V}^g$ ▷ global rows aggregate global values
end if
$\mathbf{Y} \leftarrow \mathbf{O}\mathbf{W}_O$

In incremental decoding the retained state is a bounded ring: the pinned head region covers $\mathcal{G}$ and the ring keeps the last $w/2 + 1$ keys, so each step costs $\mathcal{O}(w + g)$. The exact pattern is evaluated against the cached absolute positions (unlike a retention-equals-pattern shortcut, because a spaced global prefix may contain non-global positions).

8.6.3 Key Characteristics

- **Complexity**: $\mathcal{O}(n\,(w+g)\,d)$ training and $\mathcal{O}(w+g)$ per decoded token with the paper's overlapping-chunk kernel; the reference implementation here masks a dense score matrix, which is mathematically identical but $\mathcal{O}(n^2)$ in compute. Memory is bounded by $w + g$.
- **Trainable**: Yes, natively from scratch (the pattern runs in both training and evaluation). The paper fineTunes a full-context checkpoint for 10k steps at peak LR 10^{-6} with a 1k-step warmup and cosine decay, but the mixer's weights are ordinary attention projections.
- **Strength**: On long-form benchmarks the global token is the decisive ingredient: normalized greedy WER improves from 9.18 (Conformer) to 8.25 (limited context) to **7.5** (plus global token) on TED-LIUM v3, and from 18.26 to 16.08 to **11.85** on Earnings-21, while the single-pass maximum duration grows from 15 to 675 minutes on one A100. The encoder is 2.8× faster and uses 2.9× fewer multiply-accumulates than the original Conformer.
- **Limitation**: The paper gives no closed-form equations for the global-token path (the formulation above is a documented reconstruction from its prose, Longformer and the authors' NeMo recipe). The default is a single global token with shared projections; the separate-projection variant costs $3H^2$ extra parameters and is only exact in the cached path when the global queries are pre-filled from position zero. The efficient chunked kernel is not implemented.

9 Optimizer Families

9.1 The Evolution of Optimization in Neural Networks

The optimizer survey frames transformer optimization as a response to three structural pressures: non-convex loss landscapes, severe curvature heterogeneity across parameter blocks, and the memory cost of storing optimizer state for very large models. The report argues that the field has diverged into several trajectories: adaptive first-order baselines, variance-reduction methods, memory-efficient methods, structured second-order preconditioners, schedule-free methods, and orthogonality-oriented updates.

9.2 Tutorial: From Gradient Descent to Optimizer Families

The rest of this appendix is a reference: one entry per optimizer, with its update rule, pseudocode, and memory profile. This subsection is the on-ramp. It introduces the five ideas that every entry reuses—the loss surface, the gradient, the step size, the batch, and the two statistics that later methods add—and then compares the families as families, without deriving any single method. A reader who finishes it should be able to open any of the following sections and recognise which of the five ideas it is modifying.

9.2.1 One problem, many recipes

Training a neural network is an optimization problem wrapped in bookkeeping. Write θ for the vector of all trainable numbers (weights and biases, tens of millions for a small transformer), $L(\theta)$ for the loss, and $f(\theta)$ for the objective actually minimised, usually L plus a regulariser. The architecture only decides *which* surface L is: the number of layers, the attention pattern, the normalisation, the activation. The optimizer decides *how the walk over that surface proceeds.* Every optimizer in this appendix can be written as one line,

$$\theta_{t+1} \;=\; \theta_t \;-\; \eta_t \, u_t, \qquad u_t = u(g_t, g_{t-1}, \ldots, \theta_t)\,, \tag{19}$$

where $g_t = \nabla f(\theta_t)$ is the gradient and u_t is a direction built from the current gradient and, possibly, a small amount of remembered state. Gradient descent takes $u_t = g_t$. Momentum takes u_t to be a decaying average of past gradients. Adaptive methods divide u_t coordinate by coordinate. Second-order methods rotate it with a curvature estimate. Orthogonality methods reshape it as a matrix. Nothing else changes: the sign in front of η_t is always negative, because the gradient points uphill.

The usual picture is a hiker descending a mountain in fog [58]. The hiker feels the slope underfoot but cannot see the valley, so the strategy reduces to three questions: how confident is the slope reading (one example or a thousand?), how long is the stride, and how much memory of the terrain is worth carrying. Figure 69 turns those questions into the map of the families that follow.

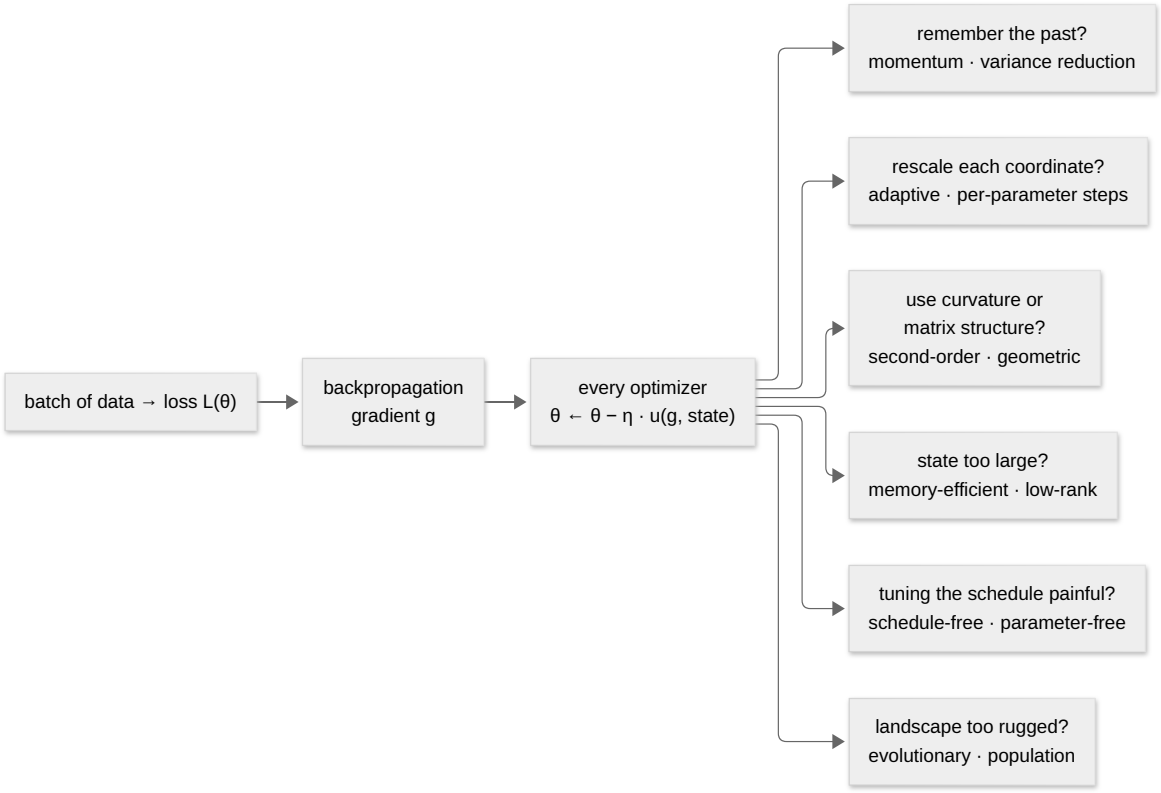


Figure 69: The whole appendix on one page. Data produces a loss, backpropagation turns the loss into a gradient, and every optimizer applies the same update rule with a different answer to one of six questions. The questions are independent, which is why methods can be combined; they are also the only dimension on which the families in Section 9 differ.

9.2.2 Gradient descent: walk downhill

For a single weight w, the derivative $f'(w)$ is the slope: positive means the loss grows to the right, so the useful move is to the left. In several dimensions the gradient ∇f collects all partial derivatives and points in the direction of steepest increase; its negative is the direction of steepest decrease. Taking a step of length η along it gives

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t), \tag{20}$$

the oldest algorithm in the appendix, usually credited to Cauchy in the nineteenth century and analysed by Curry in 1944 [47]. The scalar $\eta > 0$ is the *learning rate* or step size, and it is the only tunable number.

Two facts explain almost everything that follows. First, the step size has a hard ceiling. If the gradient changes at most L times faster than the parameters change (the gradient is L-*smooth*), then (20) decreases the loss for $\eta < 2/L$ and diverges beyond it; the safe monotone choice is $\eta \le 1/L$ [47]. Second, contour lines of a quadratic loss are ellipses whose axes have very different lengths, and the negative gradient is perpendicular to the contour but generally does *not* point at the minimum—except when the ellipses are circles [67]. The walk therefore zig-zags across the steep direction while creeping along the flat one. Figure 70 shows the three regimes on the same bowl, and Figure 71 shows what they look like as a loss curve.

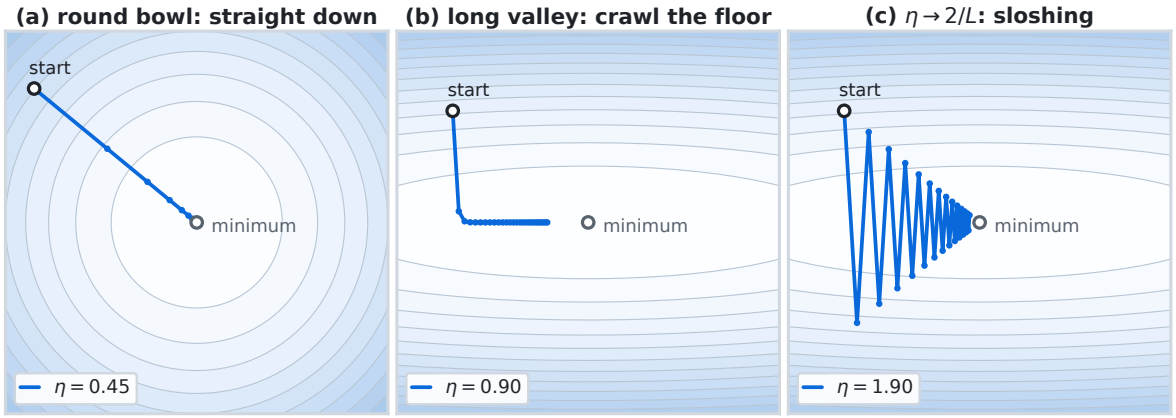


Figure 70: Gradient descent on a quadratic bowl, same start, 26 steps. (a) An isotropic bowl: the gradient aims at the minimum and the walk is a straight line. (b) A valley whose curvature differs by a factor of twenty between its axes: the steep direction is solved in a few steps, then the walk crawls along the floor because the same η is far too small there. (c) The same valley with η pushed towards the stability limit $2/L$: the steep direction now alternates sign and the path sloshes. One global step size cannot serve both directions—the observation that motivates every per-coordinate method in this appendix [67].

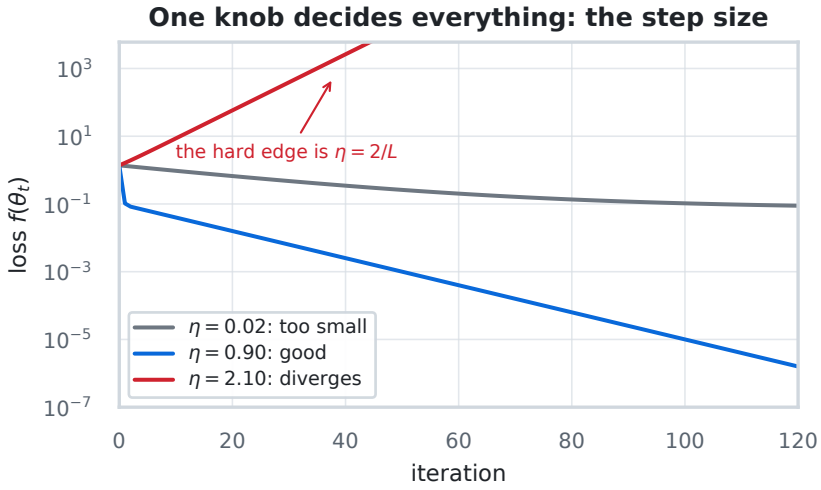


Figure 71: The same valley, one knob. With η far below $1/L$ the loss barely moves; at $\eta \approx 1/L$ it falls geometrically; past $\eta = 2/L$ every iteration multiplies the error by a factor larger than one and the run diverges. The two rates that matter—$1/L$ for monotone decrease and $2/L$ for stability—depend on the curvature, which is exactly what a practitioner does not know in advance.

When the curvature is unknown, the classical remedy is a *line search*: try $\eta = 1$, halve it until the loss actually decreases, and repeat [47]. Deep learning rarely does this, because every trial costs a full pass over the data; it schedules η by hand instead. Theory supplies the pace: for a convex L-smooth objective, gradient descent reaches $f(\theta_t) - f^\star \le 2L\|\theta_0 - \theta^\star\|^2/(t+1)$, that is, $O(1/t)$ [47]. Halving the distance to the optimum costs a doubling of iterations—a fair summary of why first-order methods need so many steps.

9.2.3 Where the gradient comes from: backpropagation

Nothing above said how to obtain g_t. For a network with millions of weights there is no formula to differentiate by hand, and finite differences would cost one forward pass per parameter. Backpropagation computes all partial derivatives in a single backward sweep, using only the chain rule.

In a feed-forward network each layer alternates an affine map and an activation, $a^{(\ell)} = W^{(\ell)} z^{(\ell-1)} + b^{(\ell)}$ and $z^{(\ell)} = \sigma(a^{(\ell)})$. The forward pass stores every a and z. The backward pass then propagates one number per neuron, $\delta^{(\ell)} = \partial e / \partial a^{(\ell)}$, from the loss towards the input,

$$\delta^{(L)} = \frac{\partial e}{\partial z^{(L)}} \odot \sigma'(a^{(L)}), \qquad \delta^{(\ell)} = \sigma'(a^{(\ell)}) \odot \left(W^{(\ell+1)\top} \delta^{(\ell+1)} \right), \qquad \frac{\partial e}{\partial W^{(\ell)}} = \delta^{(\ell)} \, z^{(\ell-1)\top}, \tag{21}$$

where $\odot$ is the element-wise product. The gradient of a weight is therefore the product of a backward number and a forward number—the two quantities drawn in Figure 72. The recursion is linear in the layer index, so the whole sweep costs about the same as one extra forward pass, roughly twice the forward cost per step, and it must keep the activations of every layer in memory until they are used [47, 141].

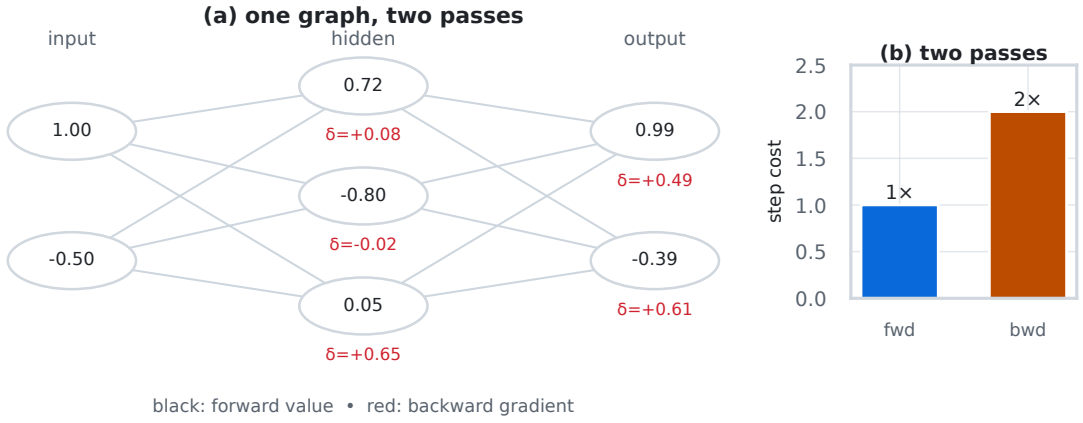


Figure 72: One training step on a two-hidden-layer network with exact values. Black numbers are the forward activations z; red numbers are the backward quantities $\delta = \partial e / \partial a$. Every weight gradient is the product of one red and one black number (equation 21), so one backward sweep yields all of them at once. Panel (b): the backward pass costs about twice the forward pass and keeps the activations alive, which is why optimizer memory is not the only memory bill in training.

Algorithm 24 Backpropagation: the gradient factory

Require: Input x, target y, weights $\{W^{(\ell)}, b^{(\ell)}\}_{\ell=1}^{L}$
Ensure: Gradients $\{\partial e / \partial W^{(\ell)}\}_{\ell=1}^{L}$

Forward: $z^{(0)} \leftarrow x$; **for** $\ell = 1, \dots, L$ **do** $a^{(\ell)} \leftarrow W^{(\ell)} z^{(\ell-1)} + b^{(\ell)}$, $z^{(\ell)} \leftarrow \sigma(a^{(\ell)})$
Loss: $e \leftarrow \ell(z^{(L)}, y)$
$\delta^{(L)} \leftarrow \partial e / \partial z^{(L)} \odot \sigma'(a^{(L)})$
for $\ell = L, L-1, \dots, 1$ **do**
    $\partial e / \partial W^{(\ell)} \leftarrow \delta^{(\ell)} z^{(\ell-1)\top}$
    $\partial e / \partial b^{(\ell)} \leftarrow \delta^{(\ell)}$
    $\delta^{(\ell-1)} \leftarrow \sigma'(a^{(\ell-1)}) \odot \left(W^{(\ell)\top} \delta^{(\ell)} \right)$
end for
return all weight gradients

The important consequence for this appendix is that backpropagation is not an optimizer. It is a gradient factory, and it always returns the exact gradient of whatever batch it was given. Every difference between the families that follow lives after the factory, in how u_t in (19) is formed.

9.2.4 Stochastic and mini-batch gradient descent

With n examples the loss is an average, $f(\theta) = \frac{1}{n}\sum_i f_i(\theta)$, so the exact gradient is an average of n per-example gradients and one step costs n backward passes. For a dataset of a billion tokens that is unaffordable, and it is also wasteful: neighbouring examples usually say almost the same thing about the parameters.

Stochastic gradient descent, in the line of the Robbins–Monro stochastic approximation [137], replaces the average by one term, $\hat{g}_t = \nabla f_{i_t}(\theta_t)$ with i_t drawn uniformly, giving

$$\theta_{t+1} = \theta_t - \eta_t \, \nabla f_{i_t}(\theta_t). \tag{22}$$

The estimate is unbiased, $\mathbb{E}[\hat{g}_t] = g_t$, so on average it points the right way; its error falls as $\sigma/\sqrt{b}$ when a batch of b examples is used instead of one [47, 15]. Modern libraries call this *mini-batch* method "SGD" and almost never use $b = 1$: one pass over the data (an *epoch*) then contains $\lfloor n/b \rfloor$ updates rather than one [58, 47]. Figure 73 shows both faces of the trade: with the same number of steps and the same step size, small batches wander, and the noise they inject is precisely the quantity that shrinks as $1/\sqrt{b}$.

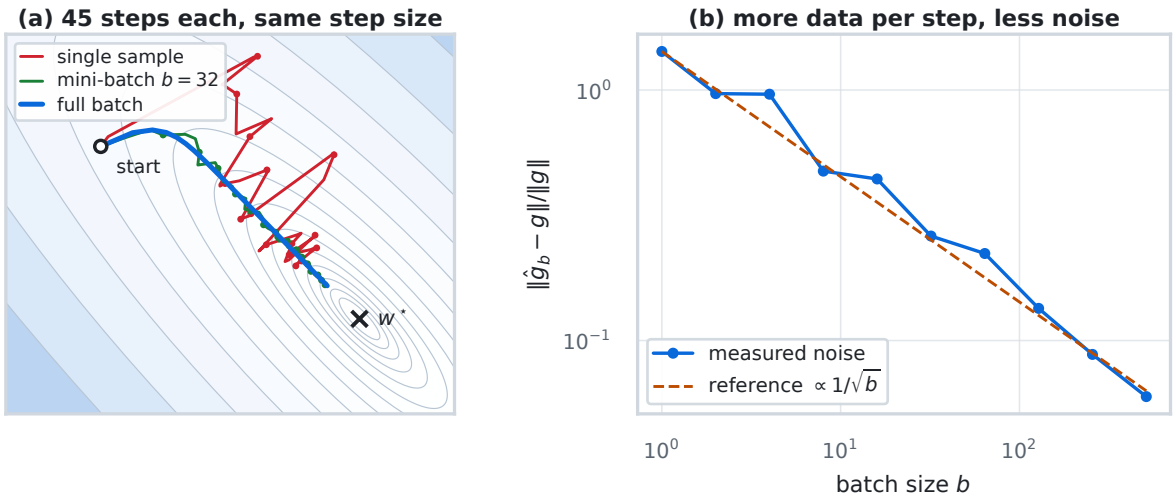


Figure 73: (a) A real least-squares surface in two weights (correlated features, so the valley is genuine), with 45 steps from the same start and the same step size. The full batch descends smoothly; a mini-batch of 32 tracks it with a visible wobble; single-sample gradients never settle. (b) Relative error of the batch gradient against the full gradient, measured on the same data: the measured points follow the $1/\sqrt{b}$ law, the noise is halved by quadrupling the batch.

Noise is not only a cost. Because each step is a random approximation, the iterate can leave a shallow basin or slide across a saddle that a full-batch method would treat as a stopping point [58]. The price appears in the rate. For a convex L-smooth objective, SGD with $\eta_t \propto 1/\sqrt{t}$ satisfies $f(\theta_t) - f^\star = O(\log t/\sqrt{t})$, and with a constant step it converges to a neighbourhood of size $O(\eta)$ and then stalls [47]. Full-batch gradient descent is therefore more accurate per iteration and much slower per iteration; SGD is the opposite. Figure 74 puts the two statements on the same axes, once per iteration and once per unit of wall-clock, which is the comparison that actually matters when a step costs n/b times less.

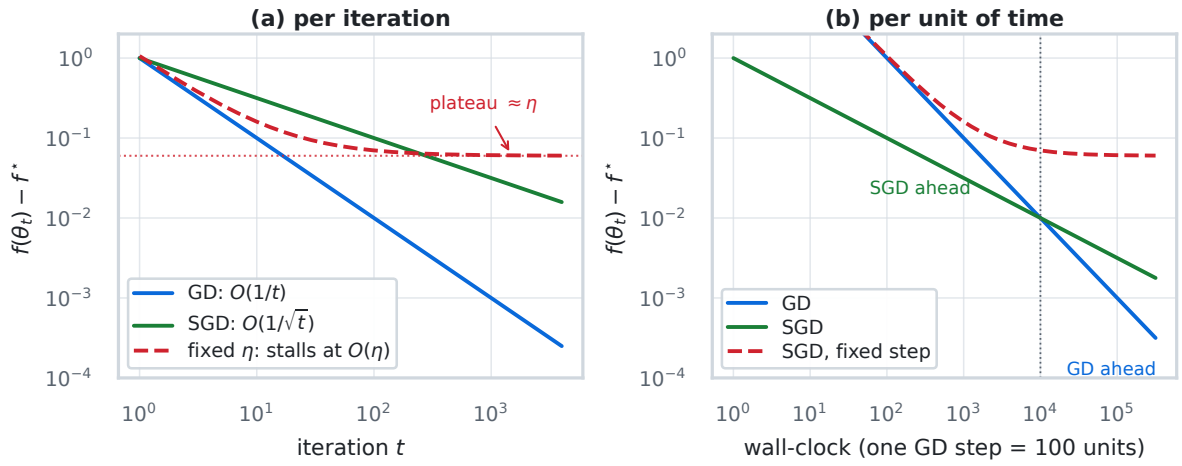


Figure 74: (a) Per iteration, gradient descent decays as $O(1/t)$ while SGD decays as $O(1/\sqrt{t})$ and, with a fixed step, flattens at a plateau of height $O(\eta)$ [47]. The plateau is the reason every practical recipe eventually reduces η. (b) The same curves against wall-clock, with one full-batch step costing a hundred mini-batch steps: SGD is ahead for most of the run and gradient descent only wins once the plateau binds. Curves are schematic, with constants chosen for legibility.

9.2.5 Two knobs that every later method turns

Everything after this point is a refinement of two observations about equation (20), and it is worth separating them before meeting the families.

Memory of the past. A ball rolling down the same valley keeps its velocity: it accumulates speed along a consistent slope and cancels the oscillation when the slope keeps changing sign. Momentum implements exactly that, replacing the gradient by a decaying sum of gradients,

$$v_t = \beta\, v_{t-1} - \eta\, g_t, \qquad \theta_{t+1} = \theta_t + v_t, \tag{23}$$

with $\beta \in [0, 1)$ the decay. The effective step is $\eta/(1-\beta)$ along a steady direction, which is how a small η can still move quickly. Nesterov momentum makes the same move but reads the gradient at the position the velocity is heading towards, so the correction arrives before the overshoot does [67, 154]. Figure 75 shows the composition of the two terms and its effect on the valley.

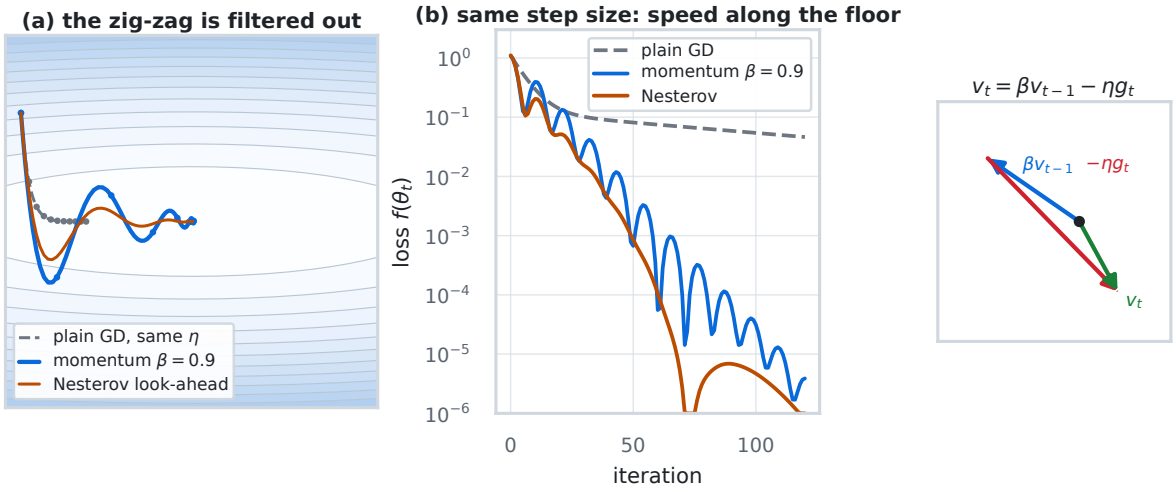


Figure 75: Momentum as memory. (a) On the valley of Figure 70(b), with the same step size, plain gradient descent still crawls along the floor after 120 steps while momentum and its Nesterov variant reach the minimum. (b) The same run as a loss curve: the remembered direction buys roughly an order of magnitude. (c) The update is the head-to-tail sum of the decayed velocity (blue) and the fresh gradient step (red); the result (green) is smoother than either term alone.

One step size per coordinate. The second observation is that the valley of Figure 70(b) has no single good η: what is stable along the steep axis is negligible along the flat one. Adaptive methods keep a running average of squared gradients and divide by its square root, so a coordinate that has been receiving large gradients gets a smaller step and a quiet coordinate gets a larger one,

$$s_t = \gamma\, s_{t-1} + (1-\gamma)\, g_t^2, \qquad \theta_{t+1} = \theta_t - \eta\, \frac{g_t}{\sqrt{s_t} + \epsilon}. \tag{24}$$

The scheme grew out of rprop, which used only the sign of the gradient, and its mini-batch version, rmsprop [67, 157]; AdaGrad is the cumulative rather than decaying version [37]. Figure 76 shows the effect: the two axes of the valley receive comparable displacements, at the cost of a per-coordinate state and a step that no longer shrinks on its own—the residual jitter of size $\approx \eta$ around the minimum is what later motivates decaying or schedule-free step sizes.

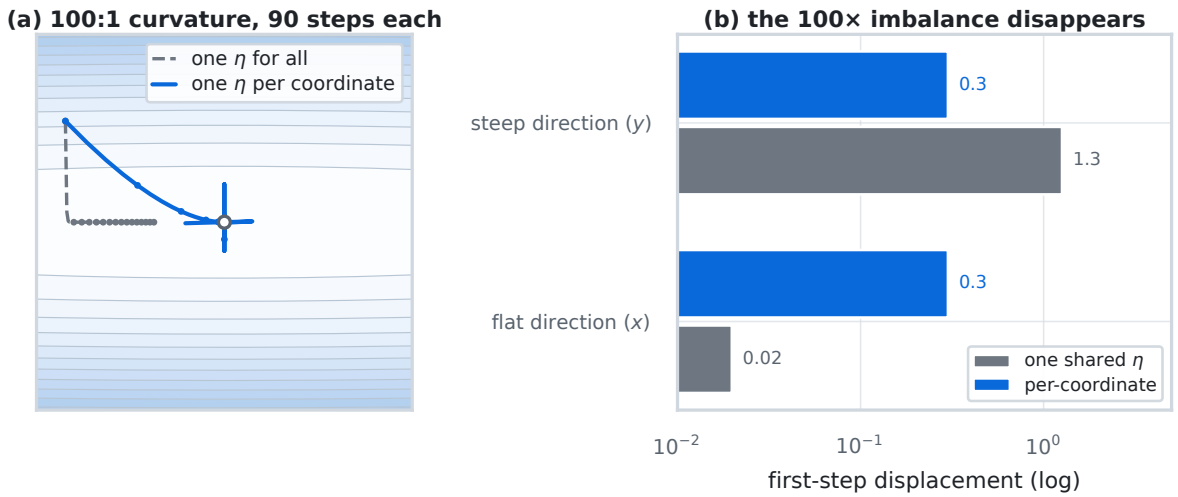


Figure 76: One step size per coordinate. (a) On a valley with a hundred-to-one curvature ratio, a shared step size either overshoots the steep axis or ignores the flat one; normalising each coordinate by the root-mean-square of its recent gradients reaches the basin and jitters around it. (b) First-step displacement per coordinate, on a logarithmic axis: the shared step size produces a factor of sixty between the axes, the normalised one produces none.

The two knobs compose, and their composition is the reason AdamW is the default in practice: a first moment for direction, a second moment for scale, a bias correction for the empty first steps, and a decoupled weight decay for regularisation [84]. Nothing in equations (23) and (24) is specific to transformers; what differs across the families is which statistics are kept, how they are combined, and how many bytes per parameter that costs. A compact survey of the resulting zoo is given by Ruder [140].

9.2.6 How the families differ

The appendix that follows contains thirty-eight methods, which is too many to hold in mind at once. They are easier to remember as eight families, each answering one of the questions in Figure 69 and each distinguished by what it remembers and what it pays.

- **First-order baselines** (SGD with momentum, AdamW, RAdam) keep one or two statistics per parameter and assume the landscape is smooth enough for a locally linear model. They are the reference against which everything else is measured.
- **Momentum and variance reduction** (Adan, ADOPT, AdEMAMix, MARS, Cautious AdamW) keep more than two buffers, or a gradient difference, to reduce the variance of the update itself. They buy stability and speed at the cost of memory and of a second gradient evaluation or a longer history.
- **Memory-efficient and low-rank** (Adafactor, GaLore, APOLLO, Lion, AdamPM) accept a worse estimate of the second moment—a factored one, a projected one, or none—precisely to shrink the state. Lion goes as far as keeping only the sign of the update.
- **Schedule-free and parameter-free** (Schedule-Free AdamW, Prodigy, LAMB for large batches) move the burden of choosing η_t from the practitioner to the algorithm, usually by averaging iterates or by estimating the step from the loss itself.
- **Curvature and second-order** (Sophia, Shampoo, SOAP) estimate how the gradient changes, not just its size, and precondition the step with that estimate. They converge in fewer iterations and cost more per iteration.

- **Geometric and orthogonality** (Muon, Turbo-Muon, AF-Muon, ROOT, NorMuon, Dion3) treat a weight matrix as a matrix: they orthogonalise or otherwise whiten the update so that all singular directions move at the same rate, which is a cheap surrogate for curvature on matrix-shaped parameters.
- **Adaptivity-tunable** (Anon) keeps the standard state but adapts its own coefficients, so the amount of adaptivity is a hyperparameter rather than a fixed design choice.
- **Evolutionary and population** (EvE, EA4LLM) do not follow a gradient at all: they sample perturbations or maintain a population of models, which is attractive when gradients are noisy, unavailable, or when many hyperparameters must be tuned at once, and expensive when the model is large.

Table 11 states the same comparison in the two currencies that decide a real training run: what is stored per parameter and what each step costs. Figure 77 turns it into a fingerprint: reading down a row tells you which ingredients a family uses, reading across a column tells you which families share one.

Table 11: The eight families by the question they answer and what they cost. State is approximate, in floats per parameter; k is a rank or block count, P a population size. No family is uniformly better: each line is a trade.

Family	Question answered	State per parameter	Main risk
First-order baseline	What is the slope?	0–2	needs a hand-tuned schedule
Momentum, variance reduction	Which way is it going?	2–3	more memory and bookkeeping
Memory-efficient, low-rank	Can the state be smaller?	≤ 1	biased second moment
Schedule-free, parameter-free	Can η_t be chosen for me?	1	theory assumes convexity
Curvature, second-order	How does the slope change?	$2 + 2k$	expensive factorisations
Geometric, orthogonality	Is the update well shaped?	1	matrices only
Adaptivity-tunable	How adaptive is it?	2	extra hyperparameters
Evolutionary, population	Can we search without gradients?	$P\times$	prohibitive at scale

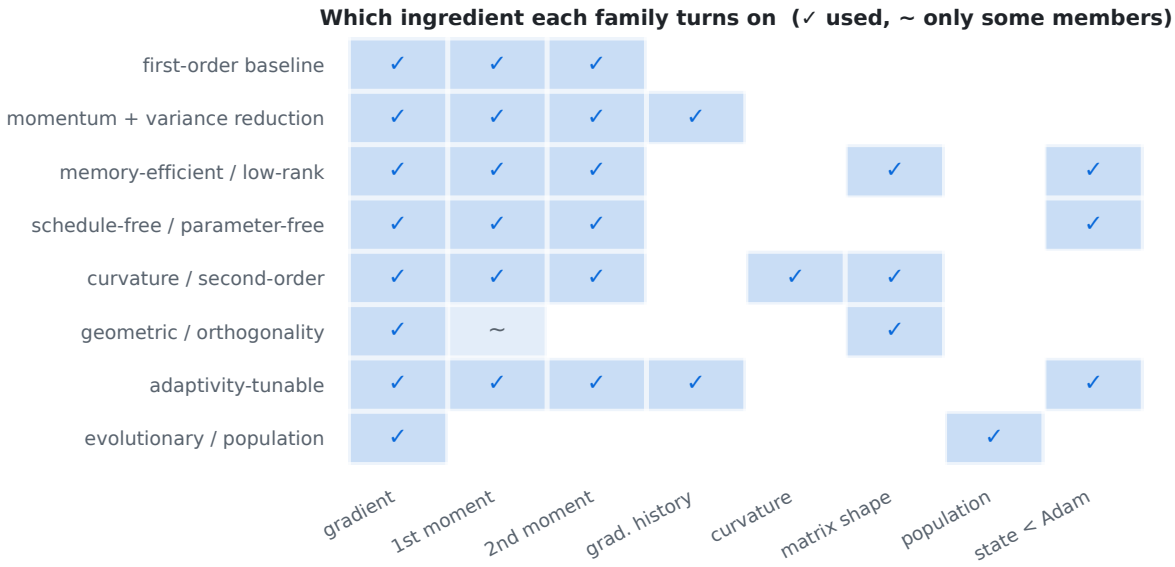


Figure 77: A fingerprint of the eight families. A tick marks an ingredient the family uses, a tilde marks one that only some members use. The first column is ticked everywhere—every family in this appendix is a gradient method except the last—and the remaining columns are the actual axes of variation: direction memory, per-coordinate scale, curvature, matrix structure, population, and a state smaller than Adam's. The map is deliberately coarse; the entries that follow give the exact rules.

Algorithm 25 makes the comparison operational: the families differ only in which slots they fill. Reading it side by side with Table 12 is enough to place any method from the rest of the appendix without memorising its algebra.

Algorithm 25 Any optimizer in this appendix, with the family-specific slots left open

Require: Parameters θ_0, learning rate η, slots $\mathcal{S}$ (state), $\mathcal{D}$ (direction), $\mathcal{P}$ (preconditioner), $\mathcal{T}$ (step rule)

Ensure: Updated parameters θ_T

Initialize the state required by $\mathcal{S}$
for $t = 1, 2, \ldots, T$ **do**
  Sample a batch $\mathcal{B}_t$ and compute $g_t \leftarrow \nabla f_{\mathcal{B}_t}(\theta_{t-1})$ ▷ backpropagation
  Update the state: $\mathcal{S} \leftarrow \textsc{Update}(\mathcal{S}, g_t, \theta_{t-1})$ ▷ 1st/2nd moment, curvature, population
  Form the direction $u_t \leftarrow \mathcal{D}(g_t, \mathcal{S})$ ▷ raw gradient, momentum, sign, orthogonalised
  Apply the preconditioner $u_t \leftarrow \mathcal{P}(u_t, \mathcal{S})$ ▷ per-coordinate scaling, factorised or full
  Choose the step $\eta_t \leftarrow \mathcal{T}(\eta, t, \mathcal{S})$ ▷ constant, schedule, implicit, self-tuned
  $\theta_t \leftarrow \theta_{t-1} - \eta_t\, u_t$ ▷ optionally after decoupled weight decay
end for
return θ_T

Table 12: The slots of Algorithm 25 and the knob each one turns. A method is identified by the slots it fills, not by its name.

Slot	What it decides	Knob turned	Seen in
$\mathcal{B}_t$	how many examples per step	variance	SGD, mini-batch, full batch
$\mathcal{S}$	what is remembered	state and memory	momentum, Adam, Sophia, Shampoo
$\mathcal{D}$	which direction is followed	smoothing	momentum, Nesterov, sign methods
$\mathcal{P}$	how coordinates are rescaled	conditioning	AdaGrad, RMSProp, Adam, Muon
$\mathcal{T}$	how long the step is	schedule	cosine, warmup, schedule-free, Prodigy
decay	how parameters are regularised	generalisation	L2, decoupled weight decay

9.2.7 Reading a loss curve

The practical value of the five ideas is diagnostic. A single loss curve already says which of them is wrong, and the answer is usually visible within a few hundred steps (Figure 78).

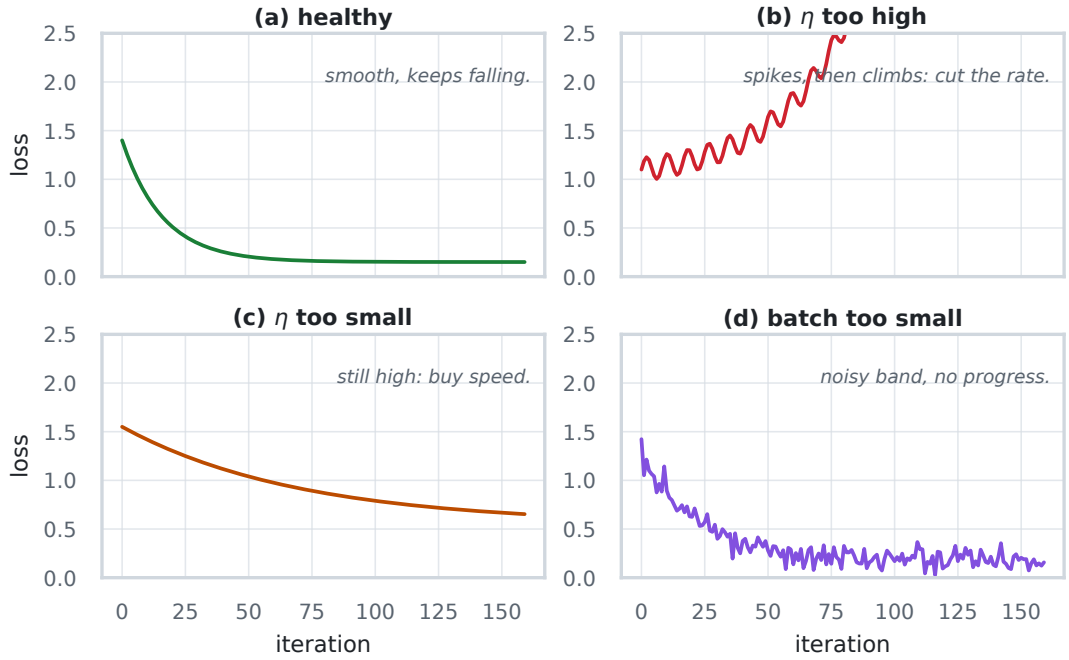


Figure 78: Four loss curves and the sentence each one is saying. A smooth curve that keeps falling is healthy; spikes followed by a rise mean the step size crossed $2/L$; a curve that is smooth but far too high means the step size is far below $1/L$ or the schedule decayed too early; a curve that reaches a noisy band and stops improving usually means the batch is too small or the step size has hit its $O(\eta)$ plateau.

Five rules of thumb close the tutorial, all of them consequences of the figures above rather than of any particular method.

1. Start from a first-order baseline with both moments (AdamW). It is the cheapest configuration that survives heterogeneous curvature.

2. If the optimizer state does not fit in memory, change family before changing hyper parameters: the memory-efficient and low-rank families exist for exactly that constraint.

3. If tuning the schedule consumes the experiment budget, the schedule-free and parameter-free families remove the knob instead of tuning it.

4. If most parameters live in matrices (attention and feed-forward projections), geometric methods exploit a structure that element-wise methods ignore; if the model is small and the search is over many configurations, population methods become affordable.

5. Change one slot at a time. Because the families are orthogonal compositions, a run that changes two of them cannot attribute its result.

The rest of this appendix is the reference behind these statements: one entry per method, each stating its update rule, its pseudocode, its memory profile, and the trade it makes.

9.3 Standard Baseline and Adaptive Optimizers

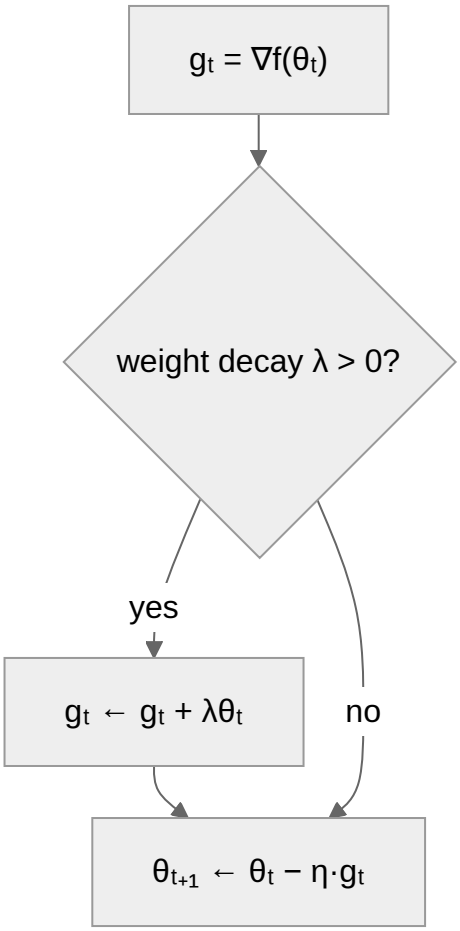


Figure 79: Step flow of mini-batch SGD: optional L2 weight decay and the plain scaled-gradient update, with no optimizer state carried between steps.

Vanilla SGD (Mini-Batch Gradient Descent). The plainest first-order method: update the weights with the gradient of the current mini-batch instead of the full batch [67]. On redundant datasets the mini-batch gradient is nearly as informative as the full one, and computing it with

matrix–matrix products makes it markedly cheaper than online updates, especially on GPUs. At iteration t with mini-batch B:

$$g_t = \frac{1}{|B|} \sum_{i \in B} \nabla_\theta E_i(\theta_t) \tag{25}$$

$$\theta_{t+1} = \theta_t - \eta\, g_t \tag{26}$$

where η is the only hyperparameter of the method. The deck's own tuning recipe is heuristic: reduce the rate when the error oscillates or diverges, increase it when the error falls consistently but slowly, and turn it down towards the end of training [67]. Momentum is a separate extension (next paragraph); this entry is deliberately the zero-state baseline of the catalog.

Algorithm 26 Mini-Batch Stochastic Gradient Descent

Require: Initial parameters θ_0, learning rate η, weight decay λ
Ensure: Updated parameters θ_T

for $t = 0, 1, 2, \ldots, T-1$ **do**
  Sample a mini-batch B_t; compute $g_t \leftarrow \frac{1}{|B_t|} \sum_{i \in B_t} \nabla f_i(\theta_t)$
  if weight_decay > 0 **then**
    $g_t \leftarrow g_t + \lambda \theta_t$ ▷ L2 regularization
  end if
  Update parameters: $\theta_{t+1} \leftarrow \theta_t - \eta\, g_t$
end forreturn θ_T

Cons.

- A single global rate cannot suit heterogeneous curvature: the method zig-zags across narrow ravines and stalls along flat directions.
- Strong dependence on hand-tuned learning-rate schedules; no per-parameter adaptivity.

Pros.

- Zero optimizer state: the memory floor of the catalog and the cheapest step.
- Fully deterministic given the batch order; the reference point against which every other entry is measured.

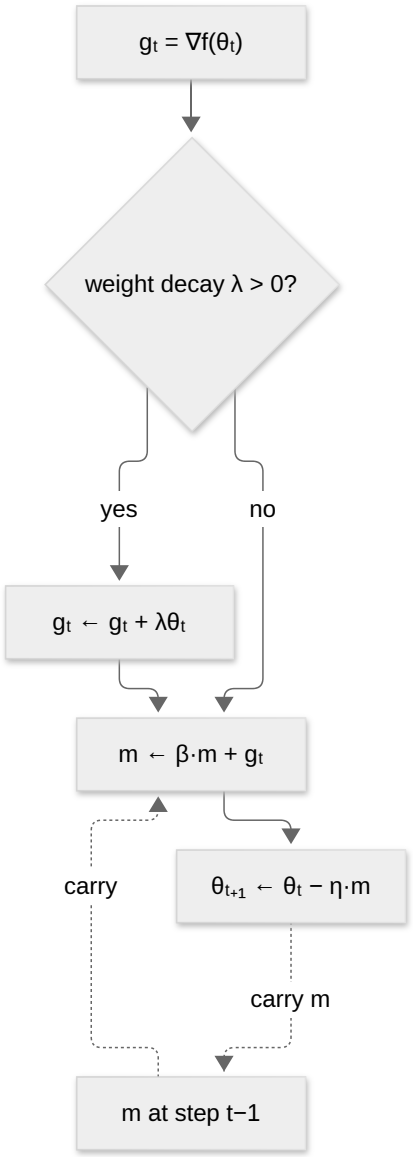


Figure 80: Step flow of SGD with momentum: optional L2 weight decay, exponential momentum accumulation, and the fixed-step parameter update.

SGD with Momentum. The classical update accumulates a momentum buffer and then applies a fixed learning rate. At each iteration t, the algorithm maintains an exponential moving average m_t of past gradients:

$$m_t = \beta m_{t-1} + g_t \tag{27}$$

$$\theta_{t+1} = \theta_t - \eta m_t \tag{28}$$

where $g_t = \nabla f(\theta_t)$, $\beta \in [0.8, 0.99]$ controls the momentum decay, and η is the fixed learning rate. The momentum term acts as velocity: it accelerates movement in consistent directions while dampening oscillations in variable directions.

Algorithm 27 SGD with Momentum

Require: Initial parameters θ_0, learning rate η, momentum coefficient β, weight decay λ
Ensure: Updated parameters θ_T

Initialize: momentum buffer $m \leftarrow 0$
for $t = 0, 1, 2, \ldots, T-1$ **do**
    Compute gradient: $g_t \leftarrow \nabla f(\theta_t)$
    if weight_decay > 0 **then**
        $g_t \leftarrow g_t + \lambda \theta_t$ ▷ L2 regularization
    end if
    Update momentum: $m \leftarrow \beta \cdot m + g_t$
    Update parameters: $\theta_{t+1} \leftarrow \theta_t - \eta \cdot m$
end forreturn θ_T

Cons.

- Poor robustness to heterogeneous curvature: a single global rate cannot suit every parameter group.
- Strong dependence on carefully engineered learning-rate schedules.

Pros.

- Low memory overhead: a single momentum buffer per parameter.
- Strong generalization when tuned carefully, and a simple update to implement.

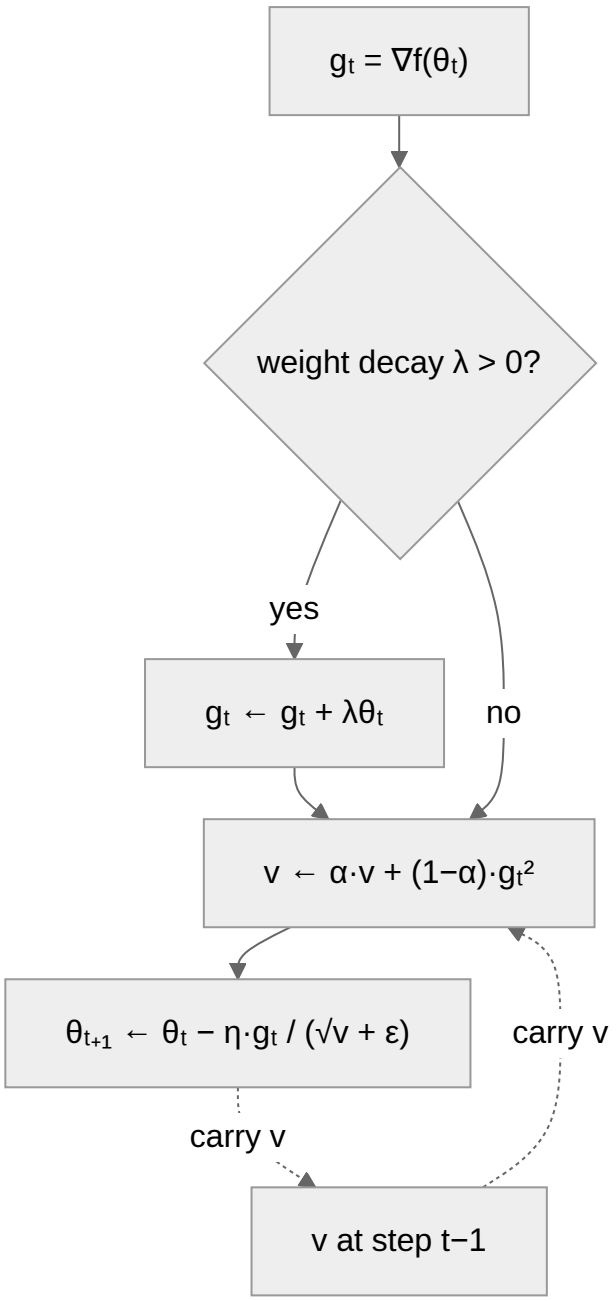


Figure 81: Step flow of RMSProp: optional L2 weight decay, the running average v_t of squared gradients, and the per-coordinate division by $\sqrt{v_t}$.

RMSProp. RMSProp is the mini-batch-safe version of rprop: instead of dividing by the size of the current gradient, force the divisor to be nearly identical for adjacent mini-batches by keeping a moving average of the squared gradient for each weight [157]:

$$v_t = 0.9\, v_{t-1} + 0.1\, g_t^2 \tag{29}$$

$$\theta_{t+1} = \theta_t - \eta\, \frac{g_t}{\sqrt{v_t} + \epsilon} \tag{30}$$

where $g_t = \partial E / \partial w(t)$, the 0.9/0.1 split is the exponential decay of the slides (exposed as the `alpha` hyperparameter), and ϵ is a numerical guard added by the implementation (the slides define none). No bias correction and no first moment appear in the original method [157]; combining

it with momentum is explicitly left open by the deck ("momentum does not help as much as it normally does"). The running average is the only state buffer and the direct predecessor of the adaptive family: AdaGrad's diagonal idea [37] made mini-batch-safe, later extended into Adam.

Algorithm 28 RMSProp

Require: Initial parameters θ_0, learning rate η, decay $\alpha = 0.9$, epsilon ϵ, weight decay λ
Ensure: Updated parameters θ_T

Initialize: running average $v \leftarrow 0$
for $t = 0, 1, 2, \ldots, T-1$ **do**
    Compute gradient: $g_t \leftarrow \nabla f(\theta_t)$
    if weight_decay > 0 **then**
        $g_t \leftarrow g_t + \lambda\theta_t$ ▷ L2 regularization
    end if
    Update running average: $v \leftarrow \alpha v + (1-\alpha) g_t^2$
    Update parameters: $\theta_{t+1} \leftarrow \theta_t - \eta\, g_t / (\sqrt{v} + \epsilon)$
end forreturn θ_T

Cons.

- No bias correction: the divisor is inflated early in training, when v_t is still a poor estimate.
- Still tuning-sensitive (rate, decay, epsilon) and largely superseded by Adam for transformers.

Pros.

- One buffer per parameter and an $\mathcal{O}(n)$ step — the cheapest adaptive method in the catalog.
- Per-coordinate scaling keeps the step robust to heterogeneous gradient magnitudes across layers.

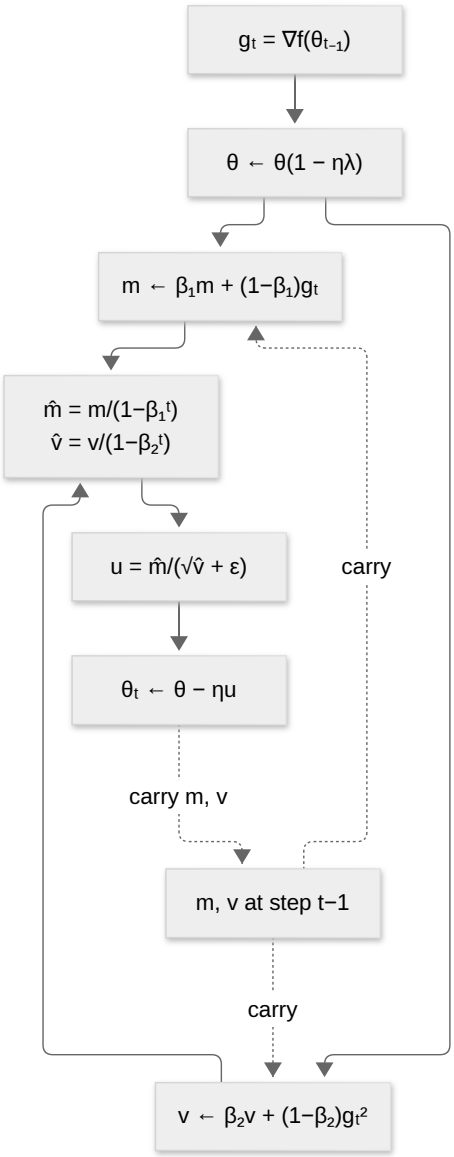


Figure 82: Step flow of AdamW: decoupled weight decay, first- and second-moment EMAs, bias correction, and the element-wise adaptive update.

Adam and AdamW. Adam tracks exponential moving averages of both first moment (mean) and second moment (uncentered variance) to achieve element-wise adaptive learning rates. The update rule is:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{31}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \tag{32}$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad \text{(bias correction)} \tag{33}$$

$$\theta_{t+1} = \theta_t - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \right) \tag{34}$$

AdamW introduces a crucial modification: weight decay is applied directly to parameters (decoupled from gradients): $\theta_t \leftarrow \theta_t(1 - \eta\lambda)$ rather than adding $\lambda\theta_t$ to the gradient. This decoupling prevents the adaptive scaling from interfering with regularization strength.

Algorithm 29 AdamW (Adam with Decoupled Weight Decay)

Require: Initial parameters θ_0, learning rate η, exponential decay rates $\beta_1, \beta_2 \in [0, 1)$
Require: Momentum constant ϵ, weight decay λ
Ensure: Updated parameters θ_T

Initialize: first moment $m \leftarrow 0$, second moment $v \leftarrow 0$, step counter $t \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
    Compute gradient: $g_t \leftarrow \nabla f(\theta_{t-1})$
    Weight decay (decoupled): $\theta_{t-1} \leftarrow \theta_{t-1}(1 - \eta\lambda)$
    Update moments: $m \leftarrow \beta_1 m + (1 - \beta_1) g_t$
    $v \leftarrow \beta_2 v + (1 - \beta_2) g_t^2$
    Bias correction: $\hat{m} \leftarrow m/(1 - \beta_1^t)$, $\hat{v} \leftarrow v/(1 - \beta_2^t)$
    Update parameters: $\theta_t \leftarrow \theta_{t-1} - \eta(\hat{m}/(\sqrt{\hat{v}} + \epsilon))$
end forreturn θ_T

Cons.

- Stores first and second moment tensors for every parameter, doubling optimizer-state memory relative to SGD.
- Element-wise adaptive scaling can interfere with regularization unless weight decay is decoupled.

Pros.

- Practical baseline for transformer fine-tuning: fast convergence and forgiving to hyperparameter variation.
- Decoupled weight decay keeps the regularization strength independent of the adaptive scaling.

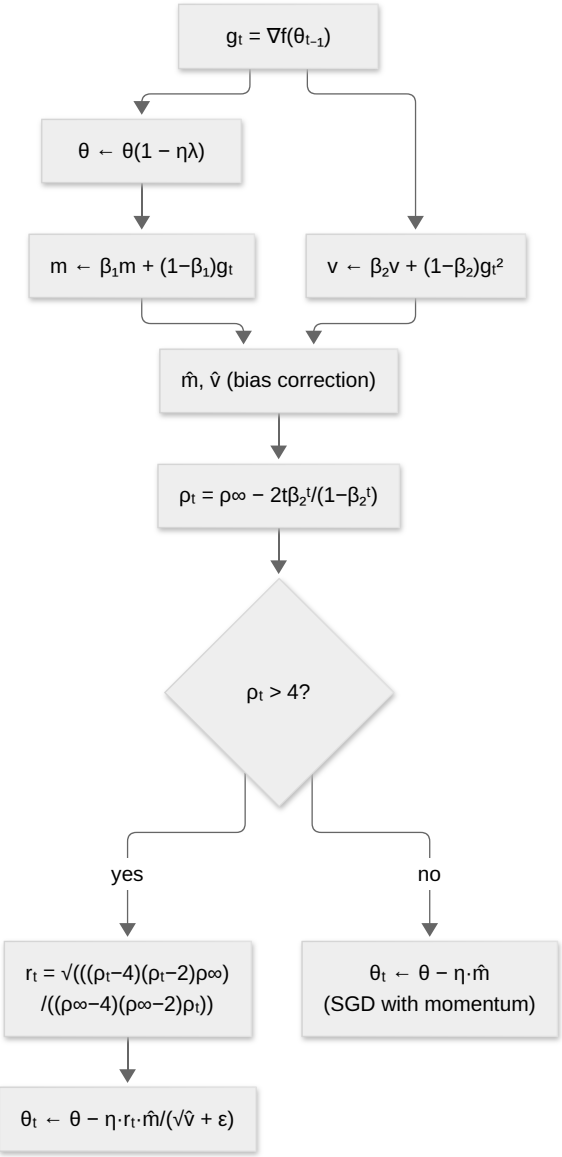


Figure 83: Step flow of RAdam: SMA-window rectification with an early-step fallback to momentum SGD.

RAdam (Rectified Adam). RAdam addresses Adam's instability in early training by dynamically rectifying the adaptive learning rate. The key observation is that the second moment v_t has very high variance in the first few steps, causing unreliable adaptive scaling. RAdam computes the effective simple moving average (SMA) window length:

$$\rho_t = \rho_\infty - \frac{2t\beta_2^t}{1-\beta_2^t}, \quad \text{where} \quad \rho_\infty = \frac{2}{1-\beta_2} - 1$$

When $\rho_t > 4$ (sufficient samples for variance estimation), RAdam applies adaptive scaling with a rectification term:

$$r_t = \sqrt{\frac{(\rho_t - 4)(\rho_t - 2)\rho_\infty}{(\rho_\infty - 4)(\rho_\infty - 2)\rho_t}}, \quad \theta_{t+1} = \theta_t - \eta r_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

When $\rho_t \leq 4$, RAdam falls back to SGD with momentum: $\theta_{t+1} = \theta_t - \eta \hat{m}_t$.

Algorithm 30 RAdam (Rectified Adam)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, weight decay λ
Ensure: Updated parameters θ_T

Initialize: $m \leftarrow 0$, $v \leftarrow 0$, $t \leftarrow 0$
Compute: $\rho_\infty \leftarrow 2/(1-\beta_2) - 1$
for $t = 1, 2, \ldots, T$ **do**
    $g_t \leftarrow \nabla f(\theta_{t-1})$
    if weight_decay > 0 **then**
        $\theta_{t-1} \leftarrow \theta_{t-1}(1 - \eta\lambda)$
    end if
    $m \leftarrow \beta_1 m + (1-\beta_1) g_t$
    $v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
    $\hat{m} \leftarrow m/(1-\beta_1^t)$, $\hat{v} \leftarrow v/(1-\beta_2^t)$
    $\rho_t \leftarrow \rho_\infty - 2t\beta_2^t/(1-\beta_2^t)$
    if $\rho_t > 4$ **then**
        $r_t \leftarrow \sqrt{((\rho_t - 4)(\rho_t - 2)\rho_\infty)/((\rho_\infty - 4)(\rho_\infty - 2)\rho_t)}$
        $\theta_t \leftarrow \theta_{t-1} - \eta r_t \hat{m}/(\sqrt{\hat{v}} + \epsilon)$
    else
        $\theta_t \leftarrow \theta_{t-1} - \eta\hat{m}$ ▷ SGD with momentum
    end if
end forreturn θ_T

Cons.

- Adds rectification bookkeeping on top of Adam's moment state.
- The benefit shrinks when a well-tuned warmup schedule is already in use.

Pros.

- Eliminates the need for manual learning-rate warmup by rectifying the adaptive rate.
- Falls back gracefully to momentum SGD while variance estimates are unreliable, improving robustness.

9.4 Advanced Momentum and Variance Reduction (2024–2025)

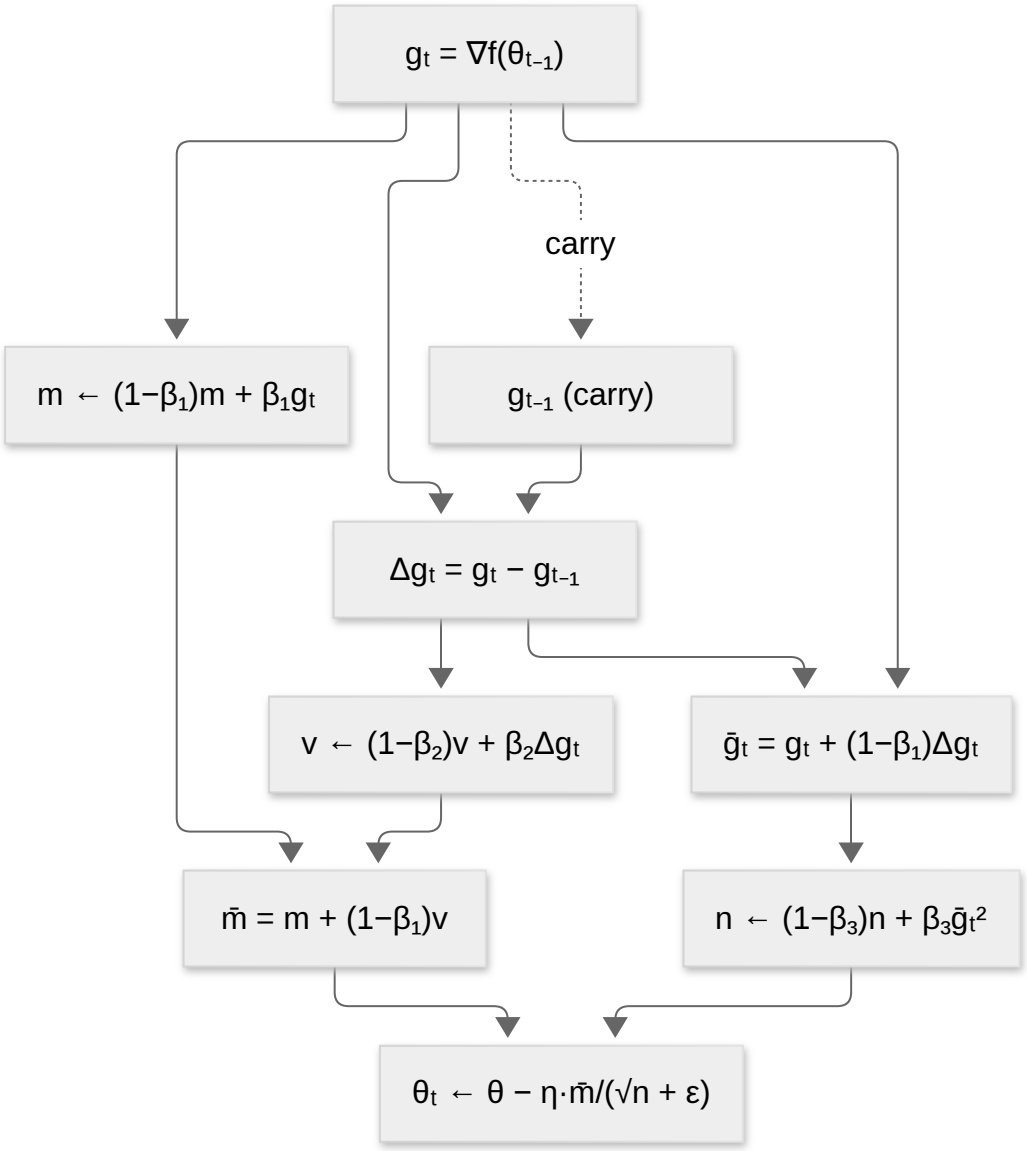


Figure 84: Step flow of Adan: Nesterov momentum estimation from the gradient difference, with three state buffers and a combined momentum step.

Adan (Adaptive Nesterov Momentum). Adan reformulates Nesterov acceleration without the extra gradient computation required by classical Nesterov SGD. The algorithm maintains three momentum buffers:

$$m_t = (1-\beta_1)m_{t-1} + \beta_1 g_t \quad \text{(first moment)} \tag{35}$$

$$v_t = (1-\beta_2)v_{t-1} + \beta_2(g_t - g_{t-1}) \quad \text{(velocity/gradient difference)} \tag{36}$$

$$n_t = (1-\beta_3)n_{t-1} + \beta_3[g_t + (1-\beta_1)(g_t - g_{t-1})]^2 \quad \text{(Nesterov second moment)} \tag{37}$$

The **Nesterov Momentum Estimation** (NME) term $\bar{g}_t = g_t + (1-\beta_1)(g_t - g_{t-1})$ estimates the gradient at a future position without evaluating it. The update combines momentum with velocity:

$$\bar{m}_t = m_t + (1-\beta_1)v_t, \quad \theta_{t+1} = \theta_t - \eta \frac{\bar{m}_t}{\sqrt{n_t} + \epsilon}$$

Algorithm 31 Adan (Adaptive Nesterov Momentum)

Require: Initial parameters θ_0, learning rate η, $\beta_1, \beta_2, \beta_3 \in (0, 1)$
Ensure: Updated parameters θ_T
Initialize: $m \leftarrow 0$, $v \leftarrow 0$, $n \leftarrow 0$, $g_{-1} \leftarrow 0$ (previous gradient)
for $t = 1, 2, \ldots, T$ **do**
    $g_t \leftarrow \nabla f(\theta_{t-1})$
    $m \leftarrow (1 - \beta_1)m + \beta_1 g_t$
    $\Delta g_t \leftarrow g_t - g_{t-1}$
    $v \leftarrow (1 - \beta_2)v + \beta_2 \Delta g_t$
    Nesterov estimation: $\bar{g}_t \leftarrow g_t + (1 - \beta_1)\Delta g_t$
    $n \leftarrow (1 - \beta_3)n + \beta_3 \bar{g}_t^2$
    Combined momentum: $\bar{m} \leftarrow m + (1 - \beta_1)v$
    $\theta_t \leftarrow \theta_{t-1} - \eta(\bar{m}/(\sqrt{n} + \epsilon))$
    $g_{t-1} \leftarrow g_t$
end forreturn θ_T

Cons.

- Three momentum-like buffers increase memory overhead relative to Adam.
- Additional gradient-difference bookkeeping per step.

Pros.

- Nesterov-style acceleration without an extra gradient evaluation.
- Fast convergence across diverse architectures (CNNs, GANs, transformers).

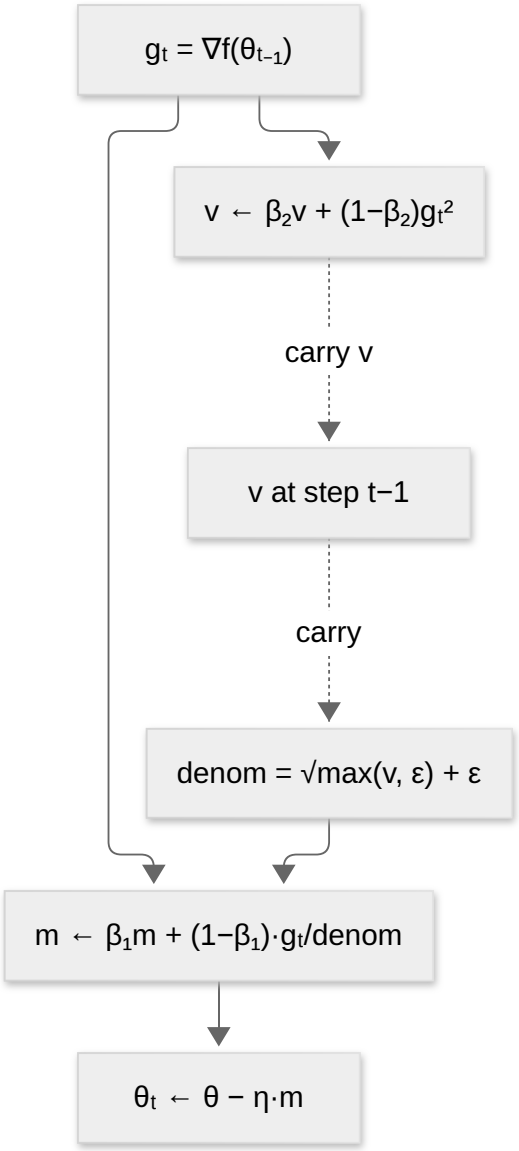


Figure 85: Step flow of ADOPT: the second moment from the previous step normalizes the current gradient before the first-moment update.

ADOPT (Adam with Optimal Pruned Tuning). ADOPT fixes a fundamental theoretical issue in Adam: the gradient appears in both the first moment and second moment estimates, creating circularity. ADOPT **decouples** by using the previous step's second moment for the denominator:

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \tag{38}$$

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) \frac{g_t}{\sqrt{v_{t-1}} + \epsilon} \quad (\text{uses } v_{t-1}, \text{ not } v_t) \tag{39}$$

$$\theta_{t+1} = \theta_t - \eta m_t \tag{40}$$

Algorithm 32 ADOPT (Adam with Optimal Pruned Tuning)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, tolerance ϵ
Ensure: Updated parameters θ_T

Initialize: $m \leftarrow 0$, $v \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
    $g_t \leftarrow \nabla f(\theta_{t-1})$
    if weight_decay > 0 **then**
        $\theta_{t-1} \leftarrow \theta_{t-1}(1 - \eta\lambda)$
    end if
    Use **previous** second moment: denom $\leftarrow \sqrt{\max(v, \epsilon)} + \epsilon$
    Update first moment using previous variance: $m \leftarrow \beta_1 m + (1 - \beta_1)(g_t/\text{denom})$
    Update second moment with **current** gradient: $v \leftarrow \beta_2 v + (1 - \beta_2) g_t^2$
    $\theta_t \leftarrow \theta_{t-1} - \eta m$
end forreturn θ_T

Cons.

- Requires keeping the previous second moment as an explicit denominator.
- Empirical gains are architecture- and tuning-dependent.

Pros.

- Removes Adam's gradient circularity and attains the optimal convergence rate $O(1/\sqrt{T})$ with **any** $\beta_2 \in (0, 1)$, without bounded-noise assumptions.
- Drop-in Adam replacement with comparable or superior empirical performance across vision, NLP, RL, and generative modeling.

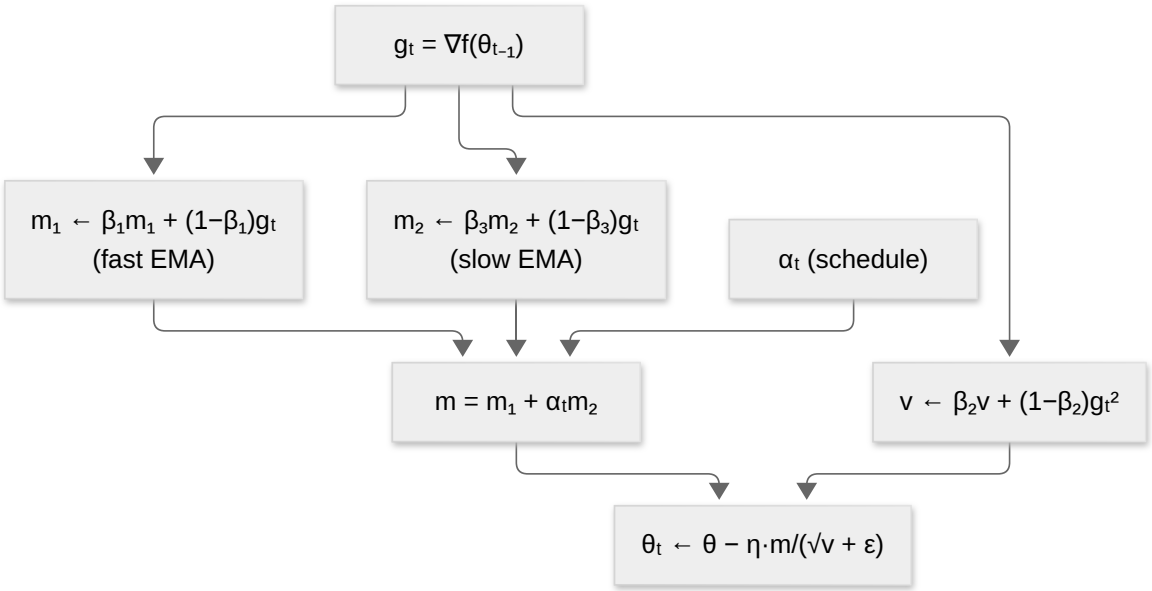


Figure 86: Step flow of AdEMAMix: parallel fast and slow gradient EMAs, mixed with a scheduled weight α_t before the adaptive update.

AdEMAMix (Exponential Moving Average Mixture). AdEMAMix replaces Adam's single EMA of gradients with a **mixture of two EMAs**: one fast-decaying and one slow-decaying. This addresses the observation that gradients remain informative over tens of thousands of steps, not just hundreds:

$$m_{1,t} = \beta_1 m_{1,t-1} + (1-\beta_1) g_t \quad \text{(fast EMA, } \beta_1 \approx 0.9\text{)} \tag{41}$$

$$m_{2,t} = \beta_3 m_{2,t-1} + (1-\beta_3) g_t \quad \text{(slow EMA, } \beta_3 \approx 0.9999\text{)} \tag{42}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \tag{43}$$

$$m_t = m_{1,t} + \alpha_t m_{2,t} \quad \text{(mixture with scheduled weight } \alpha_t\text{)} \tag{44}$$

$$\theta_{t+1} = \theta_t - \eta \frac{m_t}{\sqrt{v_t} + \epsilon} \tag{45}$$

The mixture weight α_t typically increases during training, allowing the fast EMA to provide immediate adaptation while the slow EMA accumulates long-range gradient correlations.

Algorithm 33 AdEMAMix (Exponential Moving Average Mixture)

Require: Initial parameters θ_0, learning rate η, β_1 (fast), β_3 (slow), β_2 (second moment)
Require: Mixture weight α_t (typically increasing), tolerance ϵ
Ensure: Updated parameters θ_T

Initialize: $m_1 \leftarrow 0$ (fast EMA), $m_2 \leftarrow 0$ (slow EMA), $v \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
  $g_t \leftarrow \nabla f(\theta_{t-1})$
  if weight_decay > 0 **then**
    $\theta_{t-1} \leftarrow \theta_{t-1}(1 - \eta\lambda)$
  end if
  $m_1 \leftarrow \beta_1 m_1 + (1-\beta_1) g_t$ ▷ Fast EMA
  $m_2 \leftarrow \beta_3 m_2 + (1-\beta_3) g_t$ ▷ Slow EMA
  $v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
  $m_t \leftarrow m_1 + \alpha_t m_2$ ▷ Mixture
  $\theta_t \leftarrow \theta_{t-1} - \eta (m_t / (\sqrt{v} + \epsilon))$
end forreturn θ_T

Cons.

- Two gradient EMAs plus the second moment roughly triple Adam's optimizer state.
- The mixture schedule α_t adds a hyperparameter to tune.

Pros.

- Exploits gradient information that remains informative over tens of thousands of steps.
- A 1.3B model on 101B tokens matches AdamW on 197B tokens (95% data-efficiency gain), suggesting reduced forgetting.

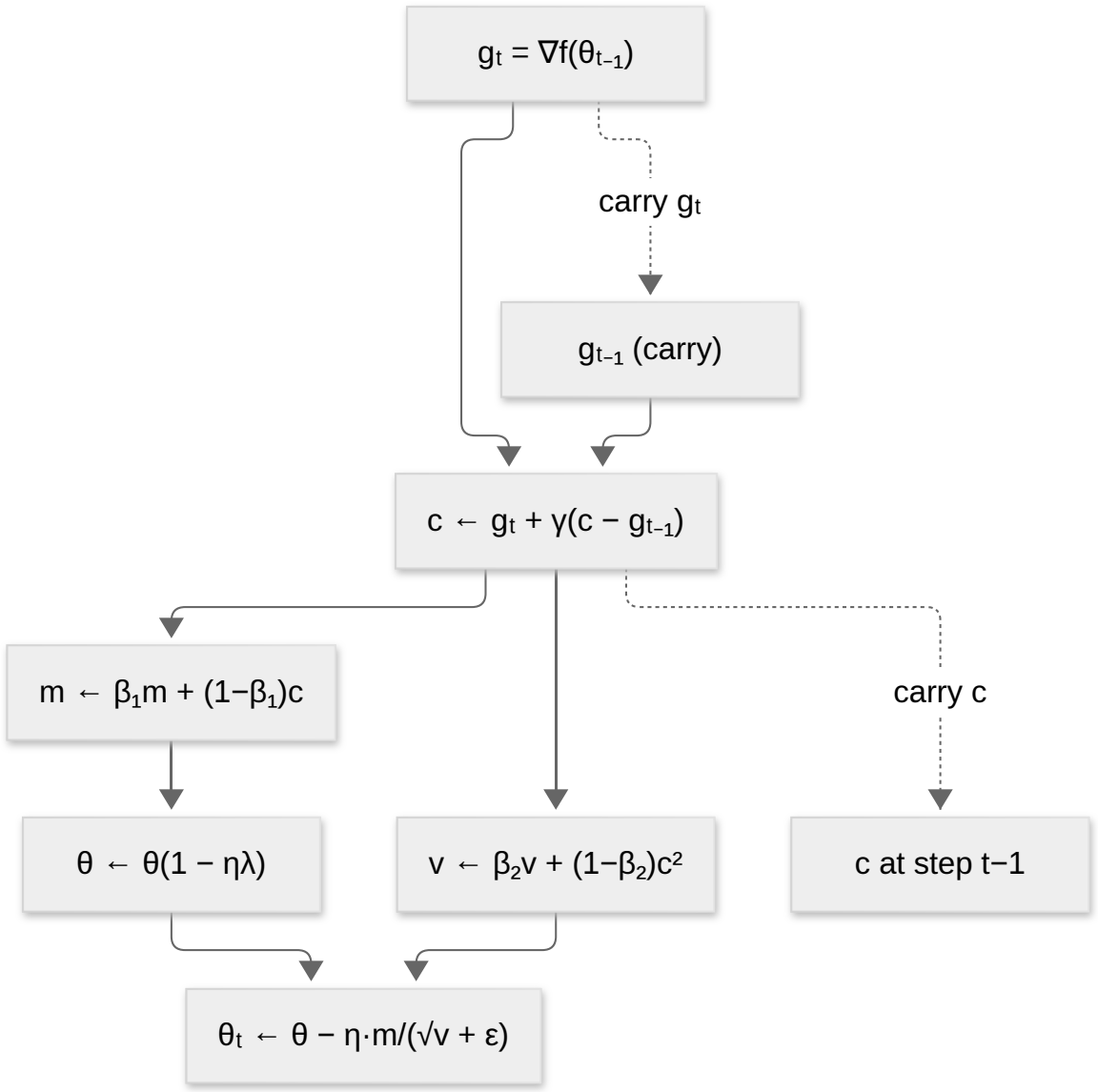


Figure 87: Step flow of MARS: a recursive variance-reduced gradient estimate feeding Adam-style first and second moments.

MARS (Make Adaptive learning Rates Shine). MARS combines preconditioned gradient methods (e.g., AdamW) with **variance reduction** via scaled stochastic recursive momentum. The core innovation is a variance-reduced gradient estimate:

$$c_t = g_t + \gamma(c_{t-1} - g_{t-1}) \quad \text{(SVRG-style recursive estimate)} \tag{46}$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)c_t \tag{47}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)c_t^2 \tag{48}$$

$$\theta_{t+1} = \theta_t - \eta \frac{m_t}{\sqrt{v_t} + \epsilon} \tag{49}$$

where $\gamma \in [0.01, 0.1]$ controls how much historical gradient information is retained. The variance-reduced gradient c_t acts as an implicit noise filter, amplifying consistent signal directions and dampening contradictory noise.

Algorithm 34 MARS (Make Adaptive learning Rates Shine)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, weight decay λ
Require: Variance reduction coefficient $\gamma \in [0.01, 0.1]$
Ensure: Updated parameters θ_T
Initialize: $m \leftarrow 0$, $v \leftarrow 0$, $c \leftarrow 0$ (variance-reduced gradient), $g_{-1} \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
    $g_t \leftarrow \nabla f(\theta_{t-1})$
    Variance reduction: $c \leftarrow g_t + \gamma(c - g_{t-1})$
    $m \leftarrow \beta_1 m + (1 - \beta_1)c$
    $v \leftarrow \beta_2 v + (1 - \beta_2)c^2$
    if weight_decay > 0 **then**
        $\theta_{t-1} \leftarrow \theta_{t-1}(1 - \eta\lambda)$
    end if
    $\theta_t \leftarrow \theta_{t-1} - \eta(m/(\sqrt{v} + \epsilon))$
    $g_{t-1} \leftarrow g_t$
end forreturn θ_T

Cons.

- Stores the variance-reduced gradient and the previous gradient in addition to Adam's moments.
- The variance-reduction coefficient $\gamma \in [0.01, 0.1]$ must be tuned.

Pros.

- Acts as an implicit noise filter, amplifying consistent directions and dampening contradictory noise.
- MARS-AdamW consistently outperforms bare AdamW by significant margins on GPT-2 pretraining.

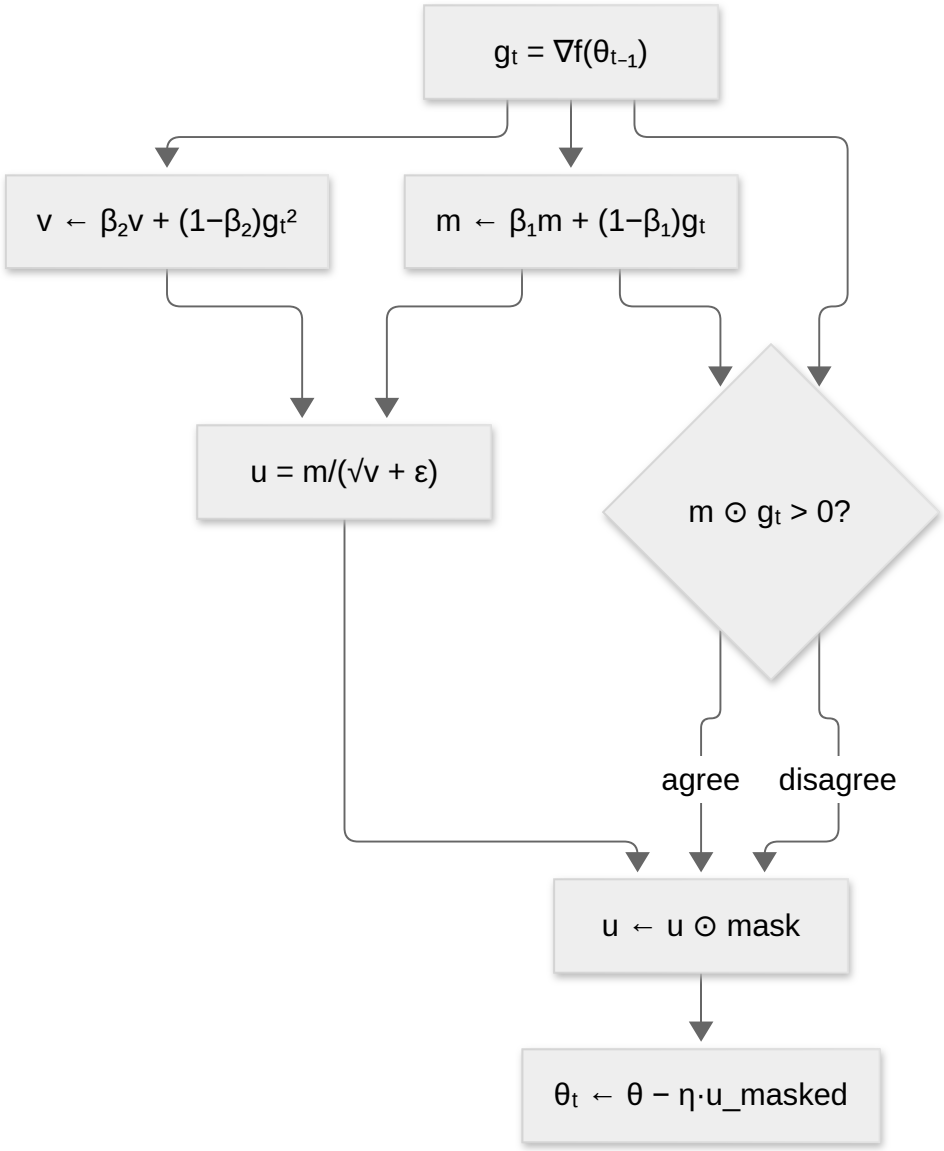


Figure 88: Step flow of Cautious AdamW: a consensus mask keeps only update coordinates where momentum and gradient agree.

Cautious Optimizers (Cautious AdamW, Cautious Lion). The Cautious framework applies a **one-line modification** to any momentum-based optimizer: mask the update so that only dimensions where momentum and gradient directions agree are applied:

$$\text{mask}_i = \begin{cases} 1 & \text{if } m_t[i] \cdot g_t[i] > 0 \quad \text{(agreement)} \\ 0 & \text{otherwise} \end{cases} \tag{50}$$

$$u_t = \left(\frac{m_t}{\sqrt{v_t} + \epsilon} \right) \odot \text{mask} \quad \text{(element-wise masking)} \tag{51}$$

$$\theta_{t+1} = \theta_t - \eta u_t \tag{52}$$

The intuition: momentum m_t estimates the gradient direction from history; the current gradient g_t is instantaneous signal. When both agree, the optimizer is confident and should update aggressively. When they disagree, the optimizer is conflicted—the historical trend points toward a region that may have been good before, but current evidence contradicts it. By masking conflicted dimensions, Cautious becomes more conservative and avoids corrupted steps.

Algorithm 35 Cautious AdamW (Consensus-based Update Masking)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, weight decay λ
Ensure: Updated parameters θ_T

Initialize: $m \leftarrow 0$, $v \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
  $g_t \leftarrow \nabla f(\theta_{t-1})$
  $m \leftarrow \beta_1 m + (1 - \beta_1) g_t$
  $v \leftarrow \beta_2 v + (1 - \beta_2) g_t^2$
  Consensus mask: $\text{mask}_i \leftarrow 1$ if $m[i] \cdot g_t[i] > 0$, else 0 ▷ Agreement
  Base Adam update: $u \leftarrow m/(\sqrt{v} + \epsilon)$
  Apply mask: $u_{\text{masked}} \leftarrow u \odot \text{mask}$ ▷ Element-wise
  if weight_decay > 0 **then**
    $\theta_{t-1} \leftarrow \theta_{t-1}(1 - \eta\lambda)$
  end if
  $\theta_t \leftarrow \theta_{t-1} - \eta u_{\text{masked}}$
end forreturn θ_T

Cons.

- Masking can discard useful updates when momentum and gradient temporarily disagree.
- Requires a base momentum optimizer plus the element-wise mask.

Pros.

- A one-line modification applicable to any momentum-based optimizer.
- Up to **1.47× speedup** on Llama and MAE pretraining while preserving convergence guarantees.

9.5 Large-Batch, Memory-Efficient, and Parameter-Free Optimizers

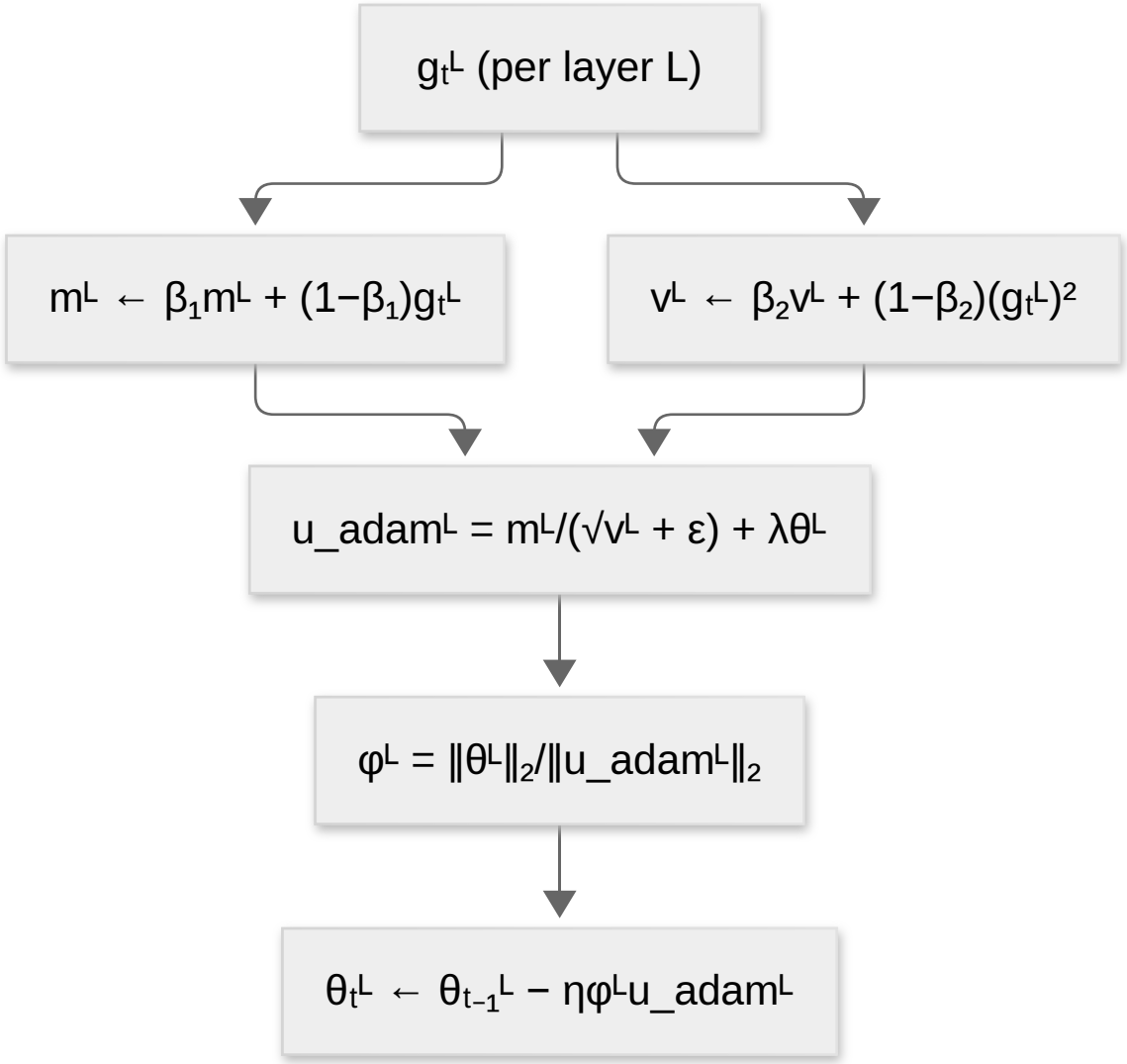


Figure 89: Step flow of LAMB: per-layer Adam step rescaled by the trust ratio between weight and update norms.

LAMB (Layer-wise Adaptive Moments optimizer for Batch training). LAMB extends Adam with **layer-wise adaptive rate scaling** (inspired by LARS), enabling stable training with extreme batch sizes (e.g., 64K on BERT):

$$m_t^L = \beta_1 m_{t-1}^L + (1-\beta_1) g_t^L \quad \text{(layer-wise first moment)} \tag{53}$$

$$v_t^L = \beta_2 v_{t-1}^L + (1-\beta_2)(g_t^L)^2 \tag{54}$$

$$u_{\text{adam}}^L = \frac{m_t^L}{\sqrt{v_t^L} + \epsilon} + \lambda \theta_{t-1}^L \quad \text{(base adaptive step with decay)} \tag{55}$$

$$\phi^L = \frac{\|\theta_{t-1}^L\|_2}{\|u_{\text{adam}}^L\|_2} \quad \text{(trust ratio: layer normalization)} \tag{56}$$

$$\theta_t^L = \theta_{t-1}^L - \eta \cdot \phi^L \cdot u_{\text{adam}}^L \tag{57}$$

The **trust ratio** $\phi^L = \|\theta^L\|_2 / \|u_{\text{adam}}^L\|_2$ normalizes the effective update step relative to weight magnitude. In very large batches, stochastic gradient noise can exceed signal, destabilizing

layer-wise learning rates. By scaling updates proportionally to weight norms, LAMB ensures relative changes rather than absolute ones, preventing small confident updates in large vectors from dominating.

Algorithm 36 LAMB (Layer-wise Adaptive Moments optimizer for Batch training)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, weight decay λ
Ensure: Updated parameters θ_T
Initialize: For each layer L: $m^L \leftarrow 0$, $v^L \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
  for each layer L **do**
    $g_t^L \leftarrow \nabla f(\theta_t^L)$
    $m^L \leftarrow \beta_1 m^L + (1 - \beta_1) g_t^L$
    $v^L \leftarrow \beta_2 v^L + (1 - \beta_2)(g_t^L)^2$
    Base Adam: $u_{\text{adam}}^L \leftarrow m^L / (\sqrt{v^L} + \epsilon)$
    if weight_decay > 0 **then**
      $u_{\text{adam}}^L \leftarrow u_{\text{adam}}^L + \lambda \theta_{t-1}^L$
    end if
    Trust ratio: $\phi^L \leftarrow \|\theta_{t-1}^L\|_2 / \|u_{\text{adam}}^L\|_2$ if both nonzero, else 1
    $\theta_t^L \leftarrow \theta_{t-1}^L - \eta \phi^L u_{\text{adam}}^L$
  end for
end forreturn θ_T

Cons.

- Layer-wise norm bookkeeping and trust-ratio computation add per-step overhead.
- The trust ratio is unstable when layer or update norms are close to zero.

Pros.

- Enables stable training with extreme batch sizes (e.g., 64K on BERT).
- Preserves small-batch generalization benefits while scaling the batch.

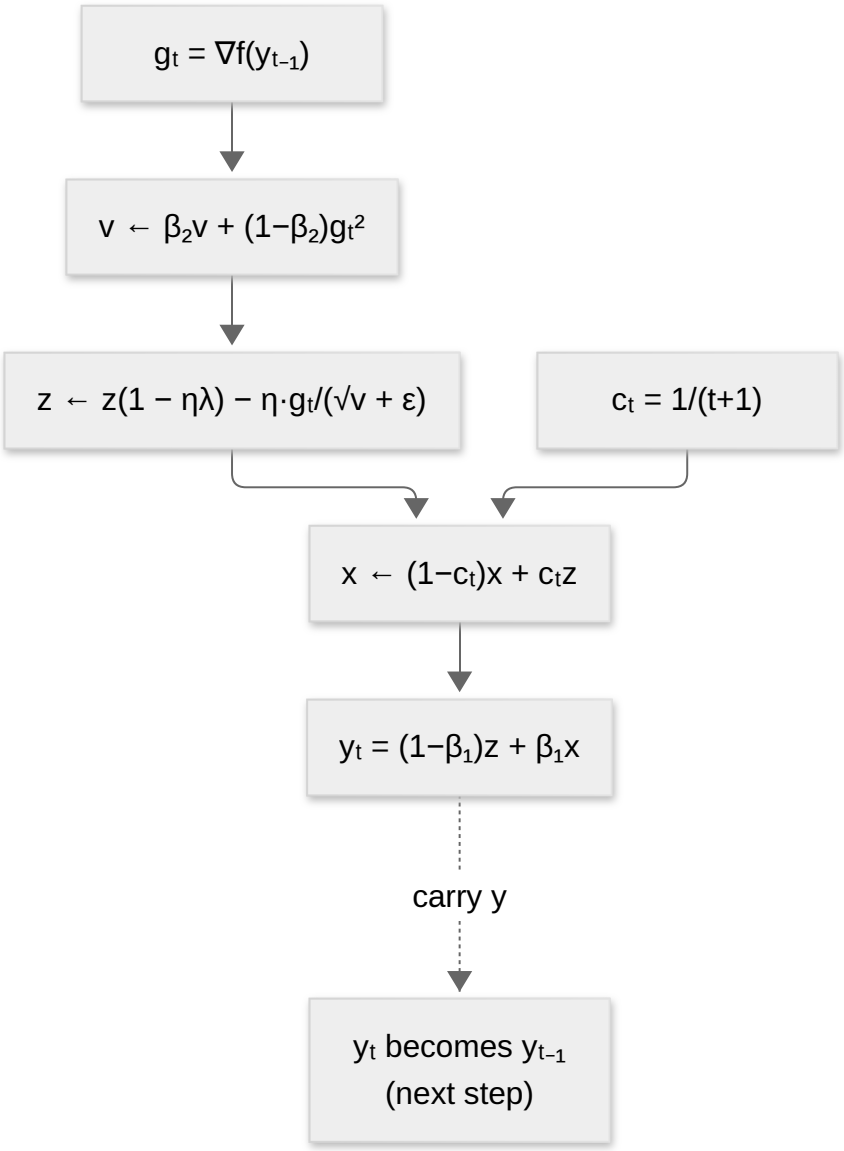


Figure 90: Step flow of Schedule-Free AdamW: exploration stream z, iterate average x, and interpolation point y, with implicit $1/(t+1)$ averaging.

Schedule-Free AdamW. Schedule-free methods remove explicit scheduler design from the optimization recipe. The algorithm maintains two parameter streams: z_t (exploration) and x_t (smooth average):

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \quad \text{(variance)} \tag{58}$$

$$z_t = z_{t-1}(1-\eta\lambda) - \eta \frac{g_t}{\sqrt{v_t} + \epsilon} \quad \text{(adaptive step with decay)} \tag{59}$$

$$c_t = \frac{1}{t+1} \quad \text{(averaging coefficient, decreases over time)} \tag{60}$$

$$x_t = (1-c_t)x_{t-1} + c_t z_t \quad \text{(iterate averaging)} \tag{61}$$

$$y_t = (1-\beta) z_t + \beta x_t \quad \text{(interpolation for evaluation point)} \tag{62}$$

The averaging coefficient $c_t = 1/(t+1)$ implements an implicit learning-rate schedule without explicitly specifying total steps T. This framework unifies scheduling and iterate averaging: the algorithm explores via z_t while accumulating stable direction via x_t. The evaluation point y_t

(used for gradient computation) interpolates between exploration and stability.

Algorithm 37 Schedule-Free AdamW

Require: Initial parameters θ_0, learning rate η (fixed), β_1, β_2, weight decay λ
Ensure: Updated parameters θ_T or averaging x_T
Initialize: $z \leftarrow \theta_0$ (exploration), $x \leftarrow \theta_0$ (average), $v \leftarrow 0$, $t \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
  $g_t \leftarrow \nabla f(y_{t-1})$ (gradient at interpolation point)
  $v \leftarrow \beta_2 v + (1 - \beta_2) g_t^2$ ▷ Variance
  Weight decay on z: $z \leftarrow z(1 - \eta\lambda)$
  $z \leftarrow z - \eta g_t / (\sqrt{v} + \epsilon)$ ▷ Adaptive step on raw
  $c_t \leftarrow 1/(t + 1)$ ▷ Averaging coefficient
  $x \leftarrow (1 - c_t)x + c_t z$ ▷ Iterate averaging
  $y_t \leftarrow (1 - \beta_1)z + \beta_1 x$ ▷ Interpolation for next eval
end forreturn x_T or y_T

Cons.

- Maintains two parameter streams (z and x) and requires explicit handling of the evaluation point.
- Behavior departs from scheduled baselines, which complicates comparisons.

Pros.

- Removes the explicit learning-rate schedule and its total-step hyperparameter.
- State-of-the-art convergence across convex optimization, large-scale deep learning, and reinforcement learning.

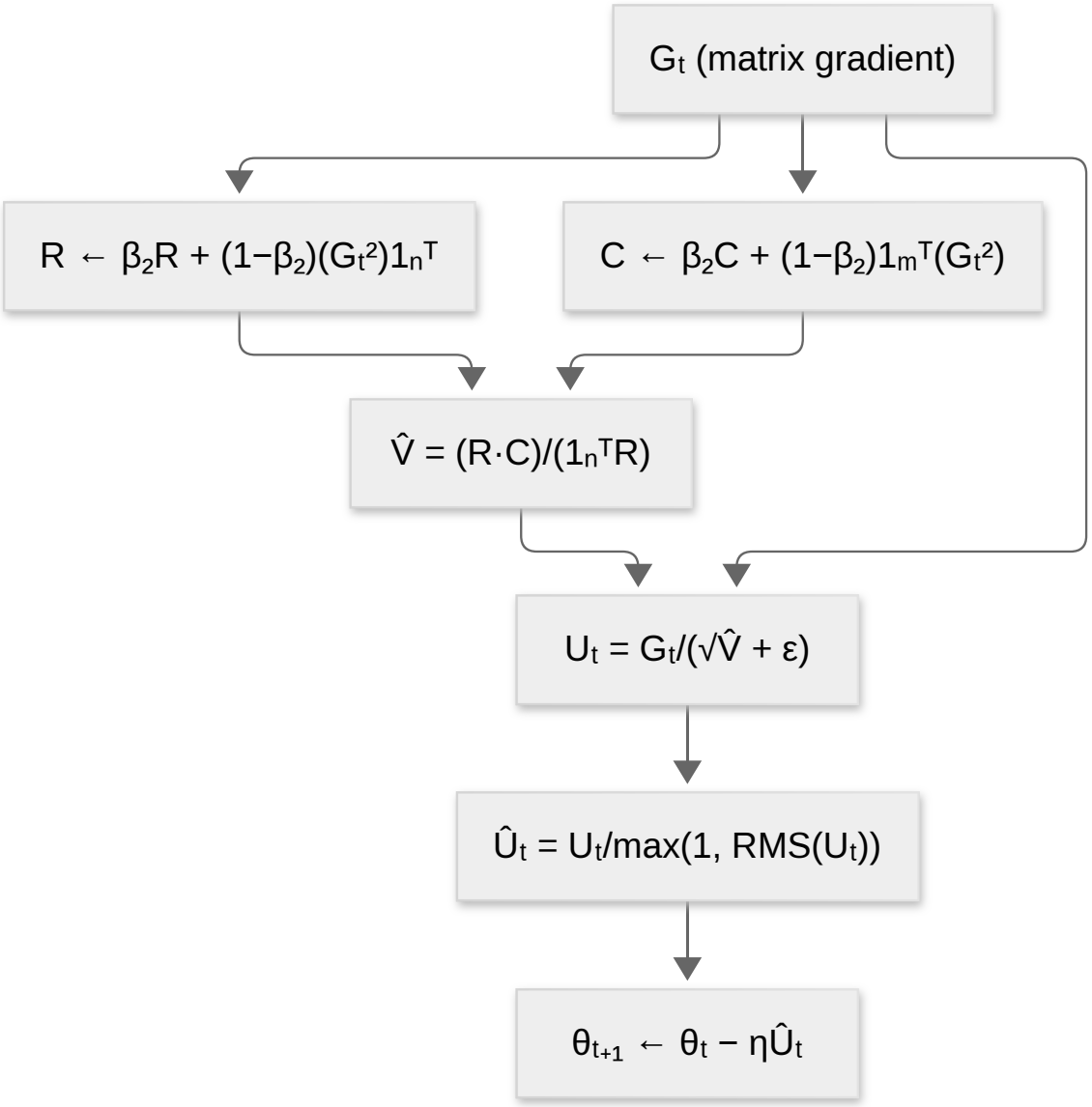


Figure 91: Step flow of Adafactor: factorized row/column second-moment accumulators, outer-product variance reconstruction, and RMS clipping.

Adafactor. Adafactor reduces optimizer memory by **factorizing** second-moment statistics for matrix-shaped parameters, storing only row and column accumulators rather than dense variance states. For a gradient matrix $G_t \in \mathbb{R}^{m \times n}$:

$$R_t = \beta_2 R_{t-1} + (1-\beta_2)(G_t^2)\mathbf{1}_n^T \quad \text{(row variance)} \tag{63}$$

$$C_t = \beta_2 C_{t-1} + (1-\beta_2)\mathbf{1}_m^\top (G_t^2) \quad \text{(column variance)} \tag{64}$$

$$\hat{V}_t = \frac{R_t C_t}{\mathbf{1}_n^\top R_t} \quad \text{(reconstructed variance via outer product)} \tag{65}$$

$$U_t = \frac{G_t}{\sqrt{\hat{V}_t} + \epsilon} \quad \text{(normalized adaptive step)} \tag{66}$$

$$\hat{U}_t = \frac{U_t}{\max(1, \mathrm{RMS}(U_t))} \quad \text{(stability clipping)} \tag{67}$$

Algorithm 38 Adafactor (Factorized Second-Moment Statistics)

Require: Gradient matrix $G_t \in \mathbb{R}^{m \times n}$, learning rate η, $\beta_2 \approx 0.999$
Ensure: Updated parameters θ_T

Initialize: Row accumulators $R \leftarrow \epsilon \mathbf{1}_m^T$, Column $C \leftarrow \epsilon \mathbf{1}_n$
for $t = 1, 2, \ldots, T$ **do**
  $G_t \leftarrow \nabla f(\theta_t)$ (matrix gradient)
  $R \leftarrow \beta_2 R + (1 - \beta_2)(G_t^2)\mathbf{1}_n^T$ ▷ Row inner products
  $C \leftarrow \beta_2 C + (1 - \beta_2)\mathbf{1}_m^\top (G_t^2)$ ▷ Column inner products
  Reconstructed variance: $\hat{V} \leftarrow (R \cdot C)/(\mathbf{1}_n^\top R)$ ▷ Via outer product
  Normalized adaptive step: $U_t \leftarrow G_t/(\sqrt{\hat{V}} + \epsilon)$
  Per-row RMS clipping: $\hat{U}_t \leftarrow U_t / \max(1, \mathrm{RMS}(U_t))$
  $\theta_{t+1} \leftarrow \theta_t - \eta \hat{U}_t$
end forreturn θ_T

Cons.

- Factorized variance is an approximation and can be unstable on some tasks.
- Reduced optimization expressiveness relative to full Adam.

Pros.

- Memory drops from $O(n_{\text{params}})$ to $O(\sqrt{n_{\text{params}}})$ for matrix parameters.
- Most attractive when VRAM is dominated by optimizer state rather than activations.

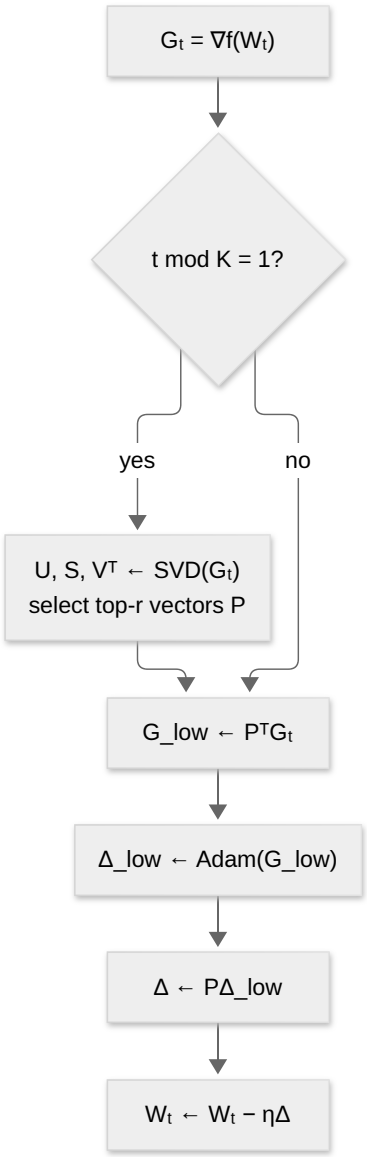


Figure 92: Step flow of GaLore: periodic SVD projection of the gradient into a low-rank subspace, Adam in that subspace, and reconstruction.

GaLore (Gradient Low-Rank Projection). GaLore projects 2D gradients into a low-rank subspace before optimization, reducing optimizer-state memory while maintaining adaptive step scaling. For a 2D parameter matrix $W \in \mathbb{R}^{m\times n}$:

$$U, S, V = \text{SVD}(G_t) \quad \text{(compute singular decomposition)} \tag{68}$$

$$P \in \mathbb{R}^{n\times r} \quad \text{or} \quad P^\top \in \mathbb{R}^{r\times m} \quad \text{(select top-}r\text{ singular vectors)} \tag{69}$$

$$G_{\text{low}} = P^\top G_t \quad \text{(project to low-rank space)} \tag{70}$$

$$\Delta_{\text{low}} = \text{Adam}(G_{\text{low}}) \quad \text{(optimize in compressed space)} \tag{71}$$

$$\Delta = P\Delta_{\text{low}} \quad \text{(reconstruct in original space)} \tag{72}$$

By projecting into a rank-r subspace (typically $r \ll \min(m, n)$), optimizer state is reduced from $O(mn)$ to $O(r(m + n))$ for 2D parameters. This complementary memory-saving approach is especially relevant when the model is too large for full-rank optimizer state.

Algorithm 39 GaLore (Gradient Low-Rank Projection)

Require: 2D parameter matrix $W \in \mathbb{R}^{m \times n}$, learning rate η, rank $r \ll \min(m, n)$
Ensure: Updated parameters θ_T

Initialize: Adam state in low-rank space (if rank-2 variant)
for $t = 1, 2, \ldots, T$ **do**
  $G_t \leftarrow \nabla f(W_t)$
  if $t \mod K = 1$ **then** ▷ Periodic SVD
    $U_t, S_t, V_t^\top \leftarrow \text{SVD}(G_t)$ (full or thin)
    Select projection: $P \leftarrow U_t[:, :r]$ or $P \leftarrow V_t[:, :r]$
  end if
  Project gradient: $G_{\text{low}} \leftarrow P^\top G_t$ (or $G_t P^\top$ for right projection)
  Run Adam on low-rank: $\Delta_{\text{low}} \leftarrow \text{Adam}(G_{\text{low}})$
  Reconstruct in original space: $\Delta \leftarrow P\Delta_{\text{low}}$ (or $\Delta_{\text{low}} P^\top$)
  $W_t \leftarrow W_t - \eta\Delta$
end forreturn θ_T

Cons.

- Periodic SVD adds compute and introduces a projection-refresh interval to tune.
- Low-rank projection can discard gradient information.

Pros.

- Optimizer state reduced from $O(mn)$ to $O(r(m+n))$ for 2D parameters.
- Works synergistically with other techniques and has shown strong results on billion-parameter models.

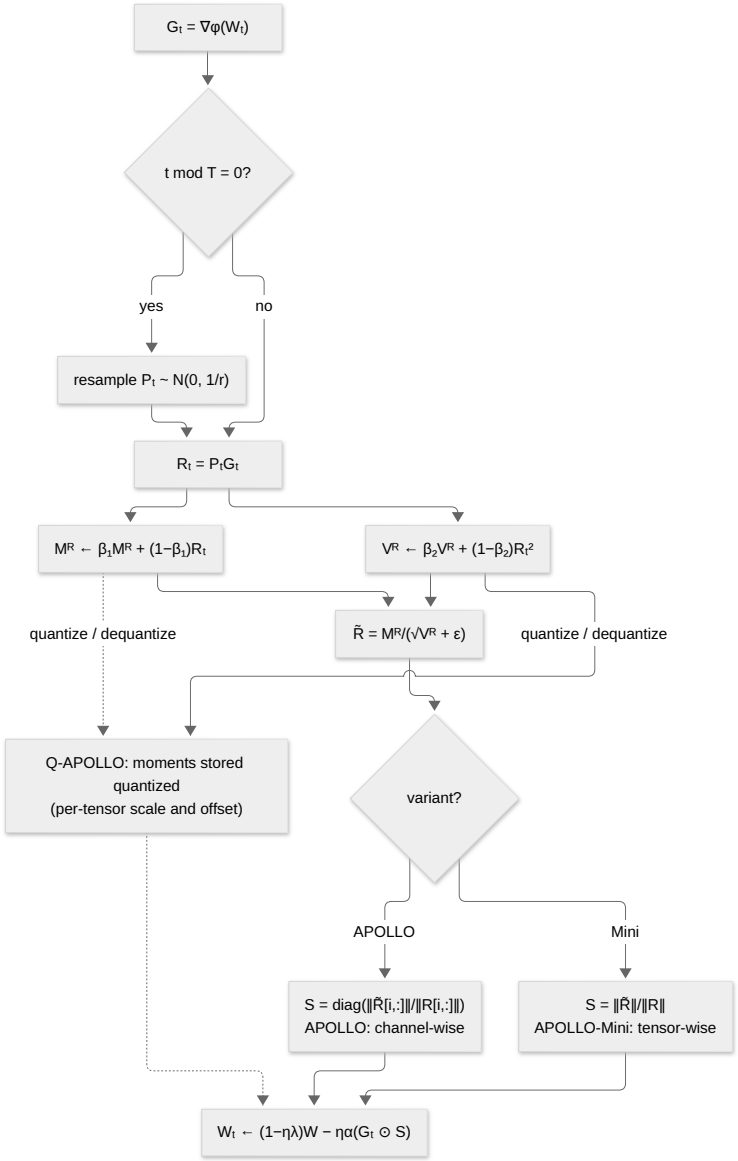


Figure 93: Step flow of the APOLLO family: Gaussian-projected moments, structured gradient scaling (channel-wise for APOLLO, scalar for APOLLO-Mini), and quantized state for Q-APOLLO.

APOLLO, APOLLO-Mini, and Q-APOLLO. The APOLLO family [184] begins from the observation that AdamW's element-wise denominator can be **coarsened into a structured learning-rate update**. Rather than storing dense moments for every parameter entry, APOLLO projects a matrix gradient $G_t \in \mathbb{R}^{m\times n}$ into a compact random subspace and tracks Adam-style moments there:

$$R_t = P_t G_t \quad \text{or} \quad R_t = G_t P_t^\top \tag{73}$$

$$M_t^R = \beta_1 M_{t-1}^R + (1-\beta_1) R_t \tag{74}$$

$$V_t^R = \beta_2 V_{t-1}^R + (1-\beta_2) R_t^2 \tag{75}$$

$$\widetilde{R}_t = \frac{M_t^R}{\sqrt{V_t^R} + \epsilon} \tag{76}$$

The projected state is *not* expanded back into a dense low-rank update as in SVD-based methods. Instead, APOLLO estimates a structured scaling tensor S_t in the original space. In the standard

APOLLO variant, that scaling is **channel-wise**: each row or channel receives its own norm ratio. In APOLLO-Mini, the scaling is reduced to a single **tensor-wise** scalar, corresponding to the rank-1 extreme described in the paper. The resulting parameter update is therefore Adam-like in adaptation but much closer to SGD in state cost:

$$W_t \leftarrow (1 - \eta\lambda)W_{t-1} - \eta\,\alpha\,(G_t \odot S_t)$$

where α is the extra scale factor used to stabilize highly compressed variants.

Algorithm 40 APOLLO / APOLLO-Mini Structured Gradient Scaling

Require: Weight matrix $W \in \mathbb{R}^{m\times n}$ with $m \le n$, learning rate η, scale factor α, decay rates (β_1, β_2), weight decay λ, rank r, projection refresh interval T
Ensure: Updated parameters W_T
Initialize projected moments: $M^R \leftarrow 0$, $V^R \leftarrow 0$, step $t \leftarrow 0$
repeat
  Compute gradient: $G_t \leftarrow \nabla_W \phi(W_t)$
  if $t \bmod T = 0$ **then**
    Sample Gaussian projector $P_t \sim \mathcal{N}(0, 1/r)$ with a fresh seed
  end if
  Project gradient: $R_t \leftarrow P_t G_t$ ▷ or $G_t P_t^\top$ depending on layout
  Update projected AdamW moments: $M_t^R, V_t^R \leftarrow \mathrm{AdamWState}(R_t; \beta_1, \beta_2)$
  Normalize projected state: $\widetilde{R}_t \leftarrow M_t^R/(\sqrt{V_t^R} + \epsilon)$
  if APOLLO **then**
    $S_t \leftarrow \mathrm{diag}(s_1^R, \ldots, s_m^R)$, where $s_i^R = \|\widetilde{R}_t[i,:]\|_2/\|R_t[i,:]\|_2$
  else
    $S_t \leftarrow s_t^R$, where $s_t^R = \|\widetilde{R}_t\|_2/\|R_t\|_2$
  end if
  Update weights: $W_t \leftarrow (1 - \eta\lambda)W_{t-1} - \eta\,\alpha\,(G_t \odot S_t)$
  $t \leftarrow t + 1$
until convergence

Cons.

- Structured scaling is a coarser approximation of AdamW's element-wise denominator.
- Projectors must be refreshed periodically, and very small ranks make the scaling noisy.

Pros.

- Replaces repeated SVD with periodically refreshed Gaussian random projection, so the systems cost is ordinary matrix multiplication rather than spectral decomposition.
- Unusually tolerant to extreme compression: even APOLLO-Mini, which keeps only rank-1 auxiliary state, remains competitive with or better than AdamW while approaching SGD-level memory.

APOLLO-Mini. APOLLO-Mini is the family member optimized for **extreme memory efficiency**. The paper motivates it by arguing that, in a rank-1 compact space, channel-wise scaling becomes too noisy, so the update is coarsened further to a single tensor-wise scale. The loss of granularity is partially offset by the explicit scaling factor α (the paper discusses values such as 128 in this regime), giving a useful point on the optimization Pareto frontier: very small state, no SVD overhead, and still strong pre-training behavior.

Cons.

- A single tensor-wise scale loses channel granularity, and rank-1 state can be noisy.
- Depends on the extra scale factor α (e.g., 128) to remain stable.

Pros.

- Smallest state in the APOLLO family, with no SVD overhead.
- Strong pre-training behavior at near-SGD memory cost.

Q-APOLLO. The APOLLO paper also stresses that the family combines naturally with **quantization** for ultra-low-memory training. This repository turns that systems observation into a concrete optimizer variant, `q_apollo`. In the implementation, APOLLO's low-rank first and second moments are stored in quantized form together with per-tensor scales and offsets, and are dequantized only when needed for the next step. Q-APOLLO therefore preserves APOLLO's projected-gradient logic while reducing the precision of the remaining optimizer state.

Cons.

- Quantized moments introduce approximation error and dequantization overhead.
- Requires managing per-tensor scales and offsets.

Pros.

- Reduces the remaining optimizer state to ultra-low precision.
- Most aggressive memory-saving member of the local optimizer stack while preserving APOLLO's projected-gradient logic.

AdamPM (Pooled-Moment Adam). AdamPM [73] is a **minimal, model-agnostic modification of Adam**: it keeps the element-wise first moment untouched and compresses only the second moment by **group-mean pooling of adjacent entries** along each row (the default, rows being output features) or each column (the paper's *AdamPM-col*). For a weight matrix $W \in \mathbb{R}^{m\times n}$ and group size s, every row is partitioned into $q = \lceil n/s \rceil$ groups, the squared gradient is averaged within each group *before* the exponential moving average, and only the compressed statistic $\bar{V}_t \in \mathbb{R}^{m\times q}$ is persisted:

$$M_t = \beta_1 M_{t-1} + (1-\beta_1)\, G_t \qquad \text{(first moment, element-wise, Eq. 1)} \tag{77}$$

$$\bar{S}_t = \frac{1}{s}\sum_{k=1}^{s} \mathrm{reshape}(G_t \odot G_t,\, m, q, s)[:,:,k] \qquad \text{(within-group means, Eq. 2)} \tag{78}$$

$$\bar{V}_t = \beta_2 \bar{V}_{t-1} + (1-\beta_2)\, \bar{S}_t \qquad \text{(compressed second-moment EMA, Eq. 3)} \tag{79}$$

$$W_t = W_{t-1} - \eta_t \frac{M_t/(1-\beta_1^t)}{\sqrt{\hat{\bar{V}}_t \otimes \mathbf{1}_{1\times s} + \epsilon}}, \qquad \hat{\bar{V}}_t = \frac{\bar{V}_t}{1-\beta_2^t} \qquad \text{(update, Eq. 4)} \tag{80}$$

When n is not divisible by s the final group is padded with its own mean, so its statistic averages only the entries it actually contains; s=1 recovers Adam exactly, and $s \geq n$ collapses each row (or column) to one scalar, the paper's *AdamPM-inf* limit. AdamPM pools the second moments of **all** weight matrices—embeddings, the LM head, attention and MLP weights—and treats

one-dimensional parameters such as RMSNorm scales as single-row or single-column matrices, so no parameter group is left on a fallback path. The persistent state is $mn(1 + 1/s)$ elements against Adam's $2mn$.

Algorithm 41 AdamPM (Pooled-Moment Adam)

Require: Initial parameters $W_0 \in \mathbb{R}^{m\times n}$, steps T, group size s, learning rate η_t, decay rates β_1, β_2, stability constant ϵ
Ensure: Updated parameters W_T

$t \leftarrow 0$; $q \leftarrow \lceil n/s \rceil$ ▷ Number of groups per row
$M_0 \leftarrow \mathbf{0}_{m\times n}$; $\bar{V}_0 \leftarrow \mathbf{0}_{m\times q}$ ▷ First moment and compressed second moment
repeat
  $t \leftarrow t + 1$
  $G_t \leftarrow \nabla_W \mathcal{L}(W_{t-1})$ ▷ Gradient
  $M_t \leftarrow \beta_1 M_{t-1} + (1 - \beta_1) G_t$ ▷ First-moment EMA (Eq. 1)
  $\bar{S}_t \leftarrow \mathrm{mean}_k \left(\mathrm{reshape}(G_t \odot G_t, m, q, s)[:, :, k]\right)$ ▷ Group means (Eq. 2)
  $\bar{V}_t \leftarrow \beta_2 \bar{V}_{t-1} + (1 - \beta_2)\bar{S}_t$ ▷ Compressed second-moment EMA (Eq. 3)
  $\hat{M}_t \leftarrow M_t/(1 - \beta_1^t)$; $\hat{\bar{V}}_t \leftarrow \bar{V}_t/(1 - \beta_2^t)$ ▷ Bias corrections
  broadcast $\hat{\bar{V}}_t$ over each group ▷ Kronecker product with $\mathbf{1}_{1\times s}$
  $W_t \leftarrow W_{t-1} - \eta_t \hat{M}_t/(\sqrt{\hat{\bar{V}}_t} + \epsilon)$ ▷ Update (Eq. 4)
until $t = T$ or convergence **return** W_t

Empirically, AdamPM matches a carefully tuned Adam on LLaMA pre-training from 60M to 1.3B parameters while cutting second-moment memory by 87.5% (s=8) to 93.75% (s=16), with the same pattern on GLUE fine-tuning of RoBERTa-base and on TinyLLaVA-1B supervised fine-tuning; group sizes between 2 and 8 are reported as robust, and three or five—not only powers of two—work because the incomplete final group is averaged over its own entries. Theorem 3.1 gives a stationarity bound independent of the group size s, and under a relative-noise condition AdamPM attains the same $\mathcal{O}(\log T/\sqrt{T})$ rate as Adam. Algorithm 1 of the paper defines no weight decay; this implementation applies the repository's decoupled `wd_*` decay before the adaptive update, and the shared per-group `betas`/`eps` suffixes parameterize the bias corrections and the denominator.

Cons.

- The diagonal preconditioner becomes coarser as s grows; the paper's AdamPM-inf limit shows a measurable perplexity loss at 350M and 1.3B.
- The saving covers optimizer state only: the first moment stays full-size, so optimizer memory is $1 + 1/s$ of Adam's, not $1/s$.
- Pooling is one-dimensional (rows or columns); two-dimensional groups and selective per-layer pooling remain unexplored (paper limitations).

Pros.

- Roughly halves optimizer-state memory at the default s=8 (second moment at 1/8) with Adam-level perplexity through 1.3B parameters and negligible per-step overhead—no SVD, no projection, no extra state.
- Inherits Adam's learning rate and decay rates directly, so existing Adam presets transfer without retuning and β_1=0.9, β_2=0.95 match the repository defaults.

- Model-agnostic and applies uniformly to every parameter group; the preconditioned updates grow more similar to Adam's as model scale grows.

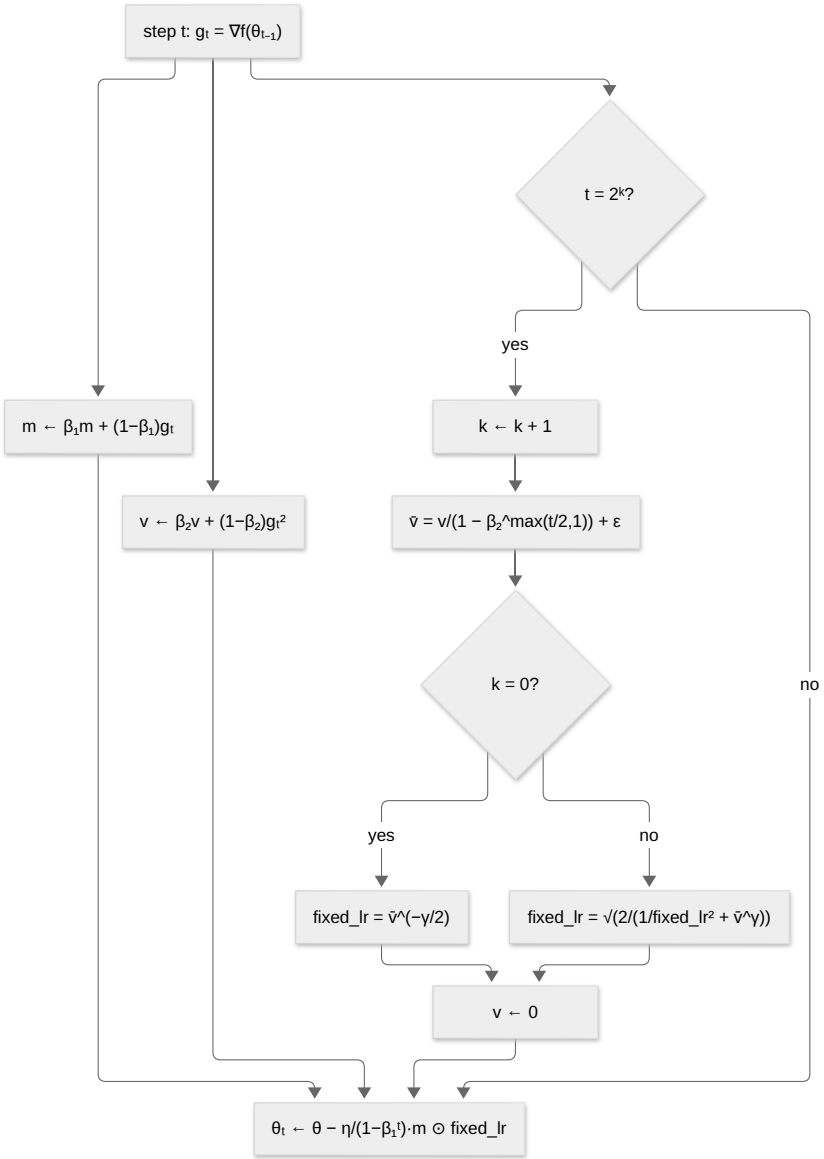


Figure 94: Step flow of Anon: the IDU preconditioner refreshed at powers of two, with tunable adaptivity γ and a bias-corrected momentum update.

Anon (Adaptivity Non-restricted Optimizer with Novel convergence technique). Anon [6] introduces continuously tunable adaptivity via a single parameter $\gamma \in \mathbb{R}$, bridging the gap between SGD-like ($\gamma = 0$) and Adam-like ($\gamma = 1$) behaviors, and extrapolating beyond both. The core innovation is the Incremental Delay Update (IDU) mechanism, which enables stable convergence across the entire real-valued adaptivity spectrum:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \quad \text{(first moment)} \tag{81}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \quad \text{(second moment)} \tag{82}$$

$$\bar{v}_t = \frac{v_t}{1-\beta_2^{\max(t,1)}} + \epsilon \quad \text{(bias-corrected second moment)} \tag{83}$$

$$\text{fixed_lr}_k = \begin{cases} \bar{v}_k^{-\gamma/2} & \text{if } k = 0 \\ \sqrt{2 / \left(\frac{1}{\text{fixed_lr}_{k-1}^2} + \bar{v}_k^{\gamma} \right)} & \text{if } k > 0 \end{cases} \tag{84}$$

$$\theta_{t+1} = \theta_t - \frac{\eta}{1-\beta_1^t} \cdot m_t \odot \text{fixed_lr}_{\lfloor \log_2 t \rfloor} \tag{85}$$

Unlike standard adaptive optimizers, IDU updates the preconditioner only at powers of two ($t = 2^k$), accumulating gradient statistics between updates and aggregating them recursively. This soft, multi-scale accumulator avoids the hard max-tracking of AMSGrad while provably stabilizing convergence for any $\gamma \in \mathbb{R}$. The parameter γ directly controls adaptivity: $\gamma > 1$ accelerates escape from saddle points (favorable for complex architectures), $\gamma = 1$ recovers Adam-like behavior, $\gamma = 0$ yields SGD-like scaling, and $\gamma < 0$ biases toward flatter minima (beneficial for classical architectures like CNNs).

Algorithm 42 Anon (Adaptivity Non-restricted Optimizer with IDU)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, tolerance ϵ, adaptivity $\gamma \in \mathbb{R}$
Ensure: Updated parameters θ_T

Initialize: $m \leftarrow 0$, $v \leftarrow 0$, fixed_lr $\leftarrow 0$, step $t \leftarrow 0$, IDU counter $k \leftarrow -1$
for $t = 1, 2, \ldots, T$ **do**
    $g_t \leftarrow \nabla f(\theta_{t-1})$
    $m \leftarrow \beta_1 m + (1-\beta_1) g_t$
    $v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
    if $t = 2^k$ **then**
        $k \leftarrow k + 1$
        $\bar{v} \leftarrow v / (1 - \beta_2^{\max(t/2,1)}) + \epsilon$
        if $k = 0$ **then**
            fixed_lr $\leftarrow \bar{v}^{-\gamma/2}$
        else
            fixed_lr $\leftarrow \sqrt{2/(1/\text{fixed_lr}^2 + \bar{v}^{\gamma})}$
        end if
        $v \leftarrow 0$
    end if
    $\theta_t \leftarrow \theta_{t-1} - \frac{\eta}{1-\beta_1^t} \cdot m \odot$ fixed_lr
end forreturn θ_T

Cons.

- Maintains three state buffers (m, v, fixed_lr) and refreshes the preconditioner on a logarithmic schedule.
- The adaptivity parameter γ must be selected per architecture.

Pros.

- Same $\mathcal{O}(n)$ per-step complexity class as AdamW, with continuously tunable adaptivity.

- IDU provably attains $\mathcal{O}(\sqrt{T})$ regret for convex problems and $\mathcal{O}(\ln T/\sqrt{T})$ for non-convex settings.

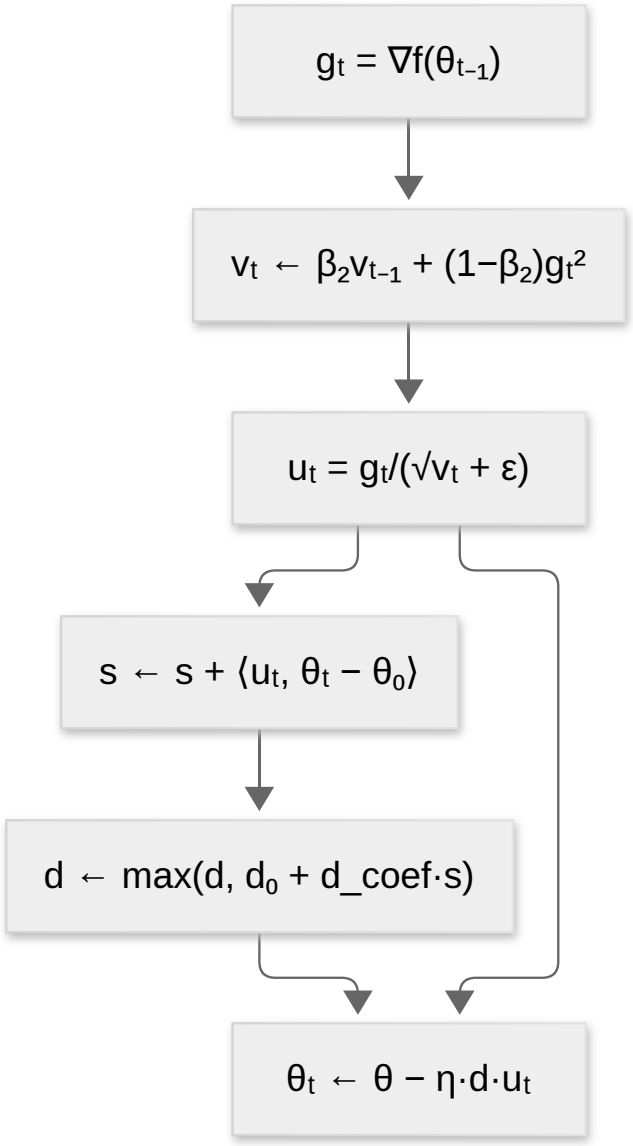


Figure 95: Step flow of Prodigy: a cumulative signed-distance estimate that scales the adaptive step without a learning-rate schedule.

Prodigy (Approximating the Distance Estimate). Prodigy adapts the effective step scale through a running distance-like statistic, eliminating the need for explicit learning-rate tuning. The algorithm maintains a cumulative distance estimate:

$$u_t = \frac{g_t}{\sqrt{v_t} + \epsilon} \quad \text{(unnormalized adaptive step)} \tag{86}$$

$$s_t = s_{t-1} + \langle u_t, \theta_t - \theta_0 \rangle \quad \text{(cumulative signed distance)} \tag{87}$$

$$d_t = \max(d_{t-1}, d_0 + d_{\text{coef}} \cdot s_t) \quad \text{(distance estimate with lower bound)} \tag{88}$$

$$\theta_{t+1} = \theta_t - \eta \cdot d_t \cdot u_t \tag{89}$$

where d_0 is an initial bound and d_{coef} controls how aggressively the estimate adapts. The distance estimate d_t captures the combined magnitude of past gradients weighted by actual parameter

displacement, implementing an implicit effective learning rate.

Algorithm 43 Prodigy (Approximating the Distance Estimate)

Require: Initial parameters θ_0, learning rate η, coeff d_{coef}, initial dist $d_0 > 0$
Ensure: Updated parameters θ_T

Initialize: $s \leftarrow 0$ (cumulative signed distance), $d \leftarrow d_0$
for $t = 1, 2, \ldots, T$ **do**
  $g_t \leftarrow \nabla f(\theta_{t-1})$
  $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ ▷ Second moment
  $u_t \leftarrow g_t / (\sqrt{v_t} + \epsilon)$ ▷ Unnormalized adaptive
  Update distance: $s \leftarrow s + \langle u_t, \theta_t - \theta_0 \rangle$ ▷ Signed distance
  $d \leftarrow \max(d, d_0 + d_{\text{coef}} \cdot s)$ ▷ Lower-bounded distance
  $\theta_t \leftarrow \theta_{t-1} - \eta \cdot d \cdot u_t$ ▷ Distance-scaled update
end forreturn θ_T

Cons.

- Introduces d_0 and d_{coef} hyperparameters for the distance estimate.
- Adds an inner-product bookkeeping step per iteration.

Pros.

- Removes explicit learning-rate tuning and scheduling.
- Robust convergence across problem scales and batch sizes.

9.6 Second-Order, Geometric, and Orthogonality Optimizers

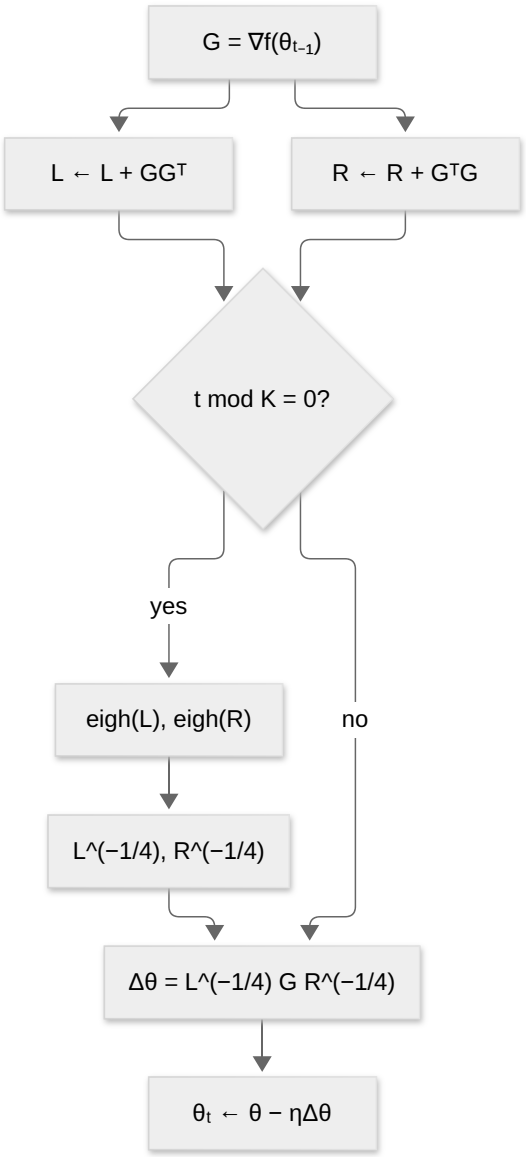


Figure 96: Step flow of Shampoo: Kronecker-factored Gram matrices, periodic eigendecomposition, and fourth-root preconditioning of the gradient.

Shampoo (Matrix Preconditioning via Kronecker Products). Shampoo is a **structured second-order method** that computes matrix preconditioners from Kronecker-structured outer-product statistics. For a 2D parameter matrix $W \in \mathbb{R}^{m \times n}$:

$$L_t = L_{t-1} + G_t G_t^\top \quad \text{(left/row Gram matrix, } m \times m\text{)} \tag{90}$$

$$R_t = R_{t-1} + G_t^\top G_t \quad \text{(right/column Gram matrix, } n \times n\text{)} \tag{91}$$

$$L_t^{-1/4} = Q_L (\Lambda_L)^{-1/4} Q_L^\top \quad \text{(via eigendecomposition)} \tag{92}$$

$$R_t^{-1/4} = Q_R (\Lambda_R)^{-1/4} Q_R^\top \tag{93}$$

$$\Delta W_t = L_t^{-1/4} G_t R_t^{-1/4} \quad \text{(preconditioned gradient)} \tag{94}$$

Shampoo approximates the full-matrix Adagrad preconditioner $H^{-1/2}$ (where H is the Hessian) using Kronecker-factored structure. Instead of storing a $(mn) \times (mn)$ preconditioner, Shampoo

maintains two smaller matrices ($m \times m$ and $n \times n$), capturing cross-parameter correlations within and across groups. Eigendecompositions are performed periodically (e.g., every $K = 10$ steps) to amortize cost.

Algorithm 44 Shampoo (Matrix Preconditioning via Kronecker Products)

Require: Initial parameters θ_0, learning rate η, eigendecomposition interval K
Ensure: Updated parameters θ_T

Initialize: $L \leftarrow \epsilon I_m$, $R \leftarrow \epsilon I_n$ (left and right Gram matrices)
for $t = 1, 2, \ldots, T$ **do**
  $G \leftarrow \nabla f(\theta_{t-1})$ ▷ Gradient
  Update Gram matrices: $L \leftarrow L + GG^\top$, $R \leftarrow R + G^\top G$
  if $t \mod K == 0$ **then**
    $Q_L, \Lambda_L \leftarrow \text{eigh}(L)$ ▷ Eigendecomposition of L
    $Q_R, \Lambda_R \leftarrow \text{eigh}(R)$ ▷ Eigendecomposition of R
    $L^{-1/4} \leftarrow Q_L(\Lambda_L + \epsilon)^{-1/4} Q_L^\top$
    $R^{-1/4} \leftarrow Q_R(\Lambda_R + \epsilon)^{-1/4} Q_R^\top$
  end if
  $\Delta\theta \leftarrow L^{-1/4} G R^{-1/4}$ ▷ Preconditioned gradient
  $\theta_t \leftarrow \theta_{t-1} - \eta \cdot \Delta\theta$
end forreturn θ_T

Cons.

- Periodic $O(m^3 + n^3)$ eigendecompositions and higher per-step compute.
- Assumes 2D parameter structure; embeddings and biases need separate handling.

Pros.

- Kronecker-factored second-order preconditioning without a dense $(mn) \times (mn)$ matrix.
- Proven effective at production scale and well suited to high-rank transformer matrices, with improved conditioning and accelerated convergence.

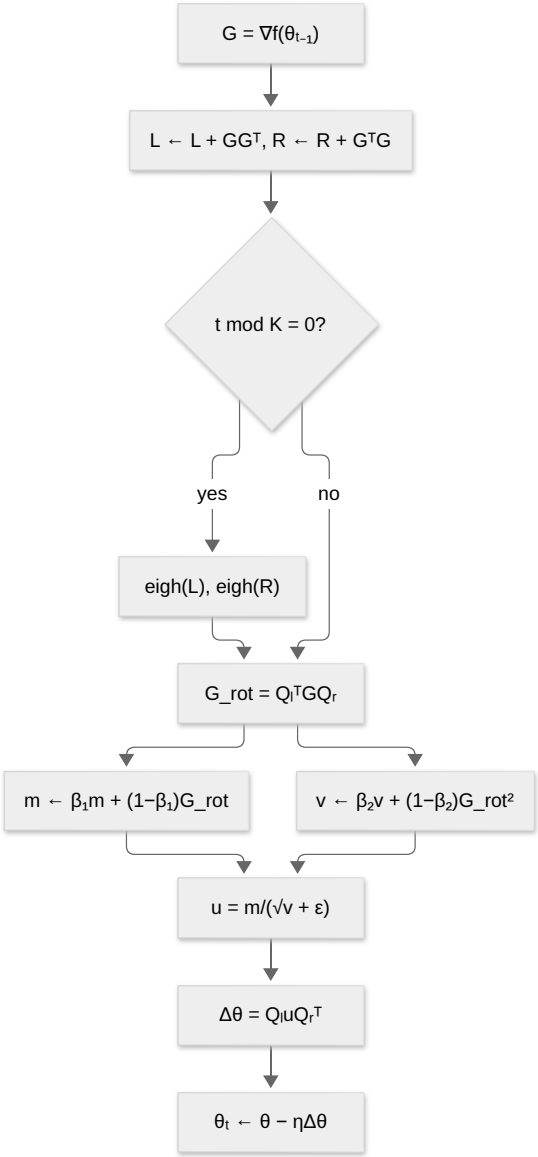


Figure 97: Step flow of SOAP: Adam moments tracked in the Shampoo eigenbasis and rotated back to parameter space.

SOAP (Shampoo with Adam in eigenbasis). SOAP is a simplified variant of Shampoo that decouples preconditioning from momentum tracking. Instead of complex matrix algebra on preconditioned gradients, SOAP runs standard Adam in the eigenbasis of Shampoo's preconditioners:

$$L_t = L_{t-1} + G_t G_t^\top, \quad R_t = R_{t-1} + G_t^\top G_t \quad \text{(Gram accumulation)} \tag{95}$$

$$Q_L, \Lambda_L = \text{eigh}(L_t), \quad Q_R, \Lambda_R = \text{eigh}(R_t) \quad \text{(periodic eigendecomposition)} \tag{96}$$

$$G_t^{\text{rot}} = Q_L^\top G_t Q_R \quad \text{(rotate gradient into eigenbasis)} \tag{97}$$

$$m_t^{\text{rot}} = \beta_1 m_{t-1}^{\text{rot}} + (1-\beta_1) G_t^{\text{rot}} \tag{98}$$

$$v_t^{\text{rot}} = \beta_2 v_{t-1}^{\text{rot}} + (1-\beta_2)(G_t^{\text{rot}})^2 \quad \text{(Adam in rotated space)} \tag{99}$$

$$U_t^{\text{rot}} = \frac{m_t^{\text{rot}}}{\sqrt{v_t^{\text{rot}}} + \epsilon}, \quad U_t = Q_L U_t^{\text{rot}} Q_R^\top \quad \text{(rotate back)} \tag{100}$$

The key insight: all momentum tracking occurs in the well-conditioned eigenbasis, simplifying numerical stability and theoretical analysis.

Algorithm 45 SOAP (Shampoo with Adam in Eigenbasis)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, eigendecomposition interval K
Ensure: Updated parameters θ_T

Initialize: $L \leftarrow \epsilon I_m$, $R \leftarrow \epsilon I_n$, $m \leftarrow 0$, $v \leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
  $G \leftarrow \nabla f(\theta_{t-1})$ ▷ Gradient
  Update Gram: $L \leftarrow L + GG^\top$, $R \leftarrow R + G^\top G$
  if $t \mod K == 0$ **then**
    $Q_L, \Lambda_L \leftarrow \text{eigh}(L)$, $Q_R, \Lambda_R \leftarrow \text{eigh}(R)$
  end if
  $G^{\text{rot}} \leftarrow Q_L^\top G Q_R$ ▷ Rotate gradient into eigenbasis
  $m \leftarrow \beta_1 m + (1 - \beta_1) G^{\text{rot}}$ ▷ Exponential moving average (bias-correct outside)
  $v \leftarrow \beta_2 v + (1 - \beta_2)(G^{\text{rot}})^2$ ▷ Second moment (bias-correct outside)
  $u \leftarrow m/(\sqrt{v} + \epsilon)$ ▷ Adaptive step in eigenbasis
  $\Delta\theta \leftarrow Q_L u Q_R^\top$ ▷ Rotate back to parameter space
  $\theta_t \leftarrow \theta_{t-1} - \eta \cdot \Delta\theta$
end forreturn θ_T

Cons.

- Inherits Shampoo's periodic eigendecomposition cost.
- Rotation and inverse rotation add matrix multiplications per step.

Pros.

- Combines Shampoo's explicit curvature structure with Adam's proven momentum mechanics.
- Often cleaner and more stable than full Shampoo, with eigendecompositions amortized every K steps.

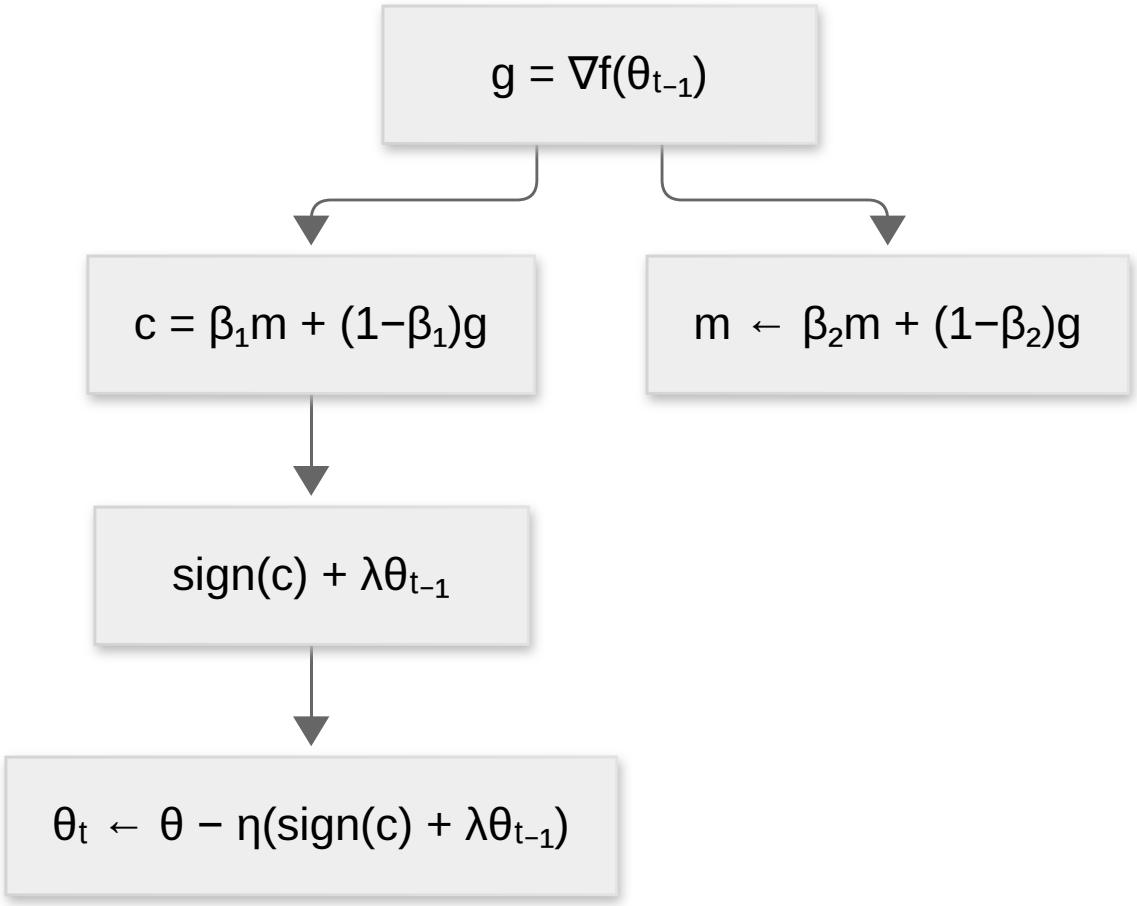


Figure 98: Step flow of Lion: sign-of-momentum update with decoupled momentum accumulation and weight decay.

Lion (EvoLved Sign Momentum). Lion achieves **minimal memory overhead** by using sign-based updates instead of full adaptive scaling:

$$c_t = \beta_1 m_t + (1-\beta_1) g_t \quad \text{(momentum input)} \tag{101}$$

$$\theta_{t+1} = \theta_t - \eta \left(\text{sign}(c_t) + \lambda \theta_t\right) \quad \text{(sign operator update)} \tag{102}$$

$$m_{t+1} = \beta_2 m_t + (1-\beta_2) g_t \quad \text{(momentum accumulation)} \tag{103}$$

All element-wise scaling operations are replaced with `sign()`, which outputs $\{-1, 0, +1\}$. This dramatically reduces memory compared to Adam-like methods: Lion stores only the momentum buffer, no second moment variance.

Algorithm 46 Lion (EvoLved Sign Momentum)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, weight decay λ
Ensure: Updated parameters θ_T

Initialize: $m \leftarrow 0$ (momentum buffer)
for $t = 1, 2, \ldots, T$ **do**
  $g \leftarrow \nabla f(\theta_{t-1})$ ▷ Gradient
  $c \leftarrow \beta_1 m + (1-\beta_1) g$ ▷ Momentum input
  $\theta_t \leftarrow \theta_{t-1} - \eta(\text{sign}(c) + \lambda \theta_{t-1})$ ▷ Sign-based update with weight decay
  $m \leftarrow \beta_2 m + (1-\beta_2) g$ ▷ Momentum accumulation (decoupled)
end forreturn θ_T

Cons.

- Sign-based updates sacrifice the element-wise learning-rate adaptation that makes Adam effective.
- Can be less robust than Adam-family methods on tasks that need adaptive scaling.

Pros.

- Stores only the momentum buffer, with no second moment variance.
- High-throughput, low-memory option for large-batch regimes where activation memory dominates.

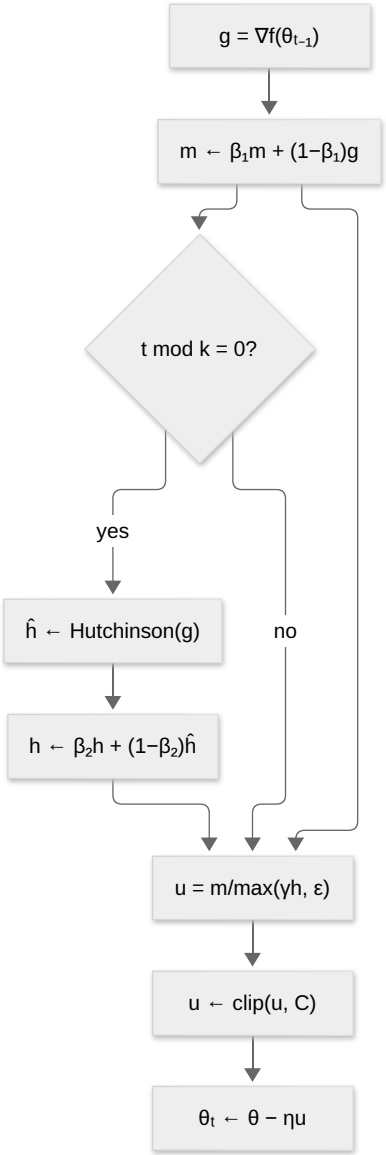


Figure 99: Step flow of Sophia: diagonal Hutchinson Hessian estimate with clipped, curvature-aware element-wise steps.

Sophia (Second-Order Hessian Information with Optimized Approximation). Sophia uses **diagonal Hessian estimates** for curvature-aware scaling without the memory and compute

cost of dense second-order methods.

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \quad \text{(first moment)} \tag{104}$$

$$h_t = \beta_2 h_{t-(k-1)} + (1-\beta_2)\hat{h}_t \quad \text{(diagonal Hessian, updated every } k \text{ steps)} \tag{105}$$

$$\text{clip}(x, C) = \min(\max(x, -C), C) \quad \text{(element-wise clipping)} \tag{106}$$

$$\theta_{t+1} = \theta_t - \eta \cdot \text{clip}\left(\frac{m_t}{\max(\gamma h_t, \epsilon)}, 1\right) \tag{107}$$

where $\hat{h}_t$ is a diagonal estimate (e.g., Hutchinson-trace estimator) and γ is a scaling factor. The diagonal Hessian h_t captures local curvature, allowing the optimizer to take smaller steps in sharp directions and larger steps in flat directions. The clipping operation $\text{clip}(\cdot, 1)$ prevents adaptive steps from exploding.

Algorithm 47 Sophia (Diagonal Hessian with Clipped Updates)

Require: Initial parameters θ_0, learning rate η, β_1, β_2, Hessian update freq k, clip threshold C
Ensure: Updated parameters θ_T

Initialize: $m \leftarrow 0$, $h \leftarrow \epsilon$ (diagonal Hessian estimate)
for $t = 1, 2, \ldots, T$ **do**
  $g \leftarrow \nabla f(\theta_{t-1})$
  $m \leftarrow \beta_1 m + (1-\beta_1) g$ ▷ First moment (bias-correct outside)
  if $t \mod k == 0$ **then**
    $\hat{h} \leftarrow \text{HutchinsonEstimate}(g)$ ▷ Diagonal Hessian via Hutchinson
    $h \leftarrow \beta_2 h + (1-\beta_2)\hat{h}$ ▷ Update Hessian estimate
  end if
  $u \leftarrow \frac{m}{\max(\gamma h, \epsilon)}$ ▷ Adaptive step (element-wise)
  $u \leftarrow \text{clip}(u, C)$ ▷ Element-wise clipping to $[-C, C]$
  $\theta_t \leftarrow \theta_{t-1} - \eta \cdot u$
end forreturn θ_T

Cons.

- Hutchinson estimates add periodic compute and noise to the curvature signal.
- The Hessian estimate can become stale between refreshes.

Pros.

- Curvature-aware scaling without the $O(n^2)$ or $O(n^3)$ cost of full second-order methods.
- Clipped steps prevent explosions and work particularly well in second-pass fine-tuning.

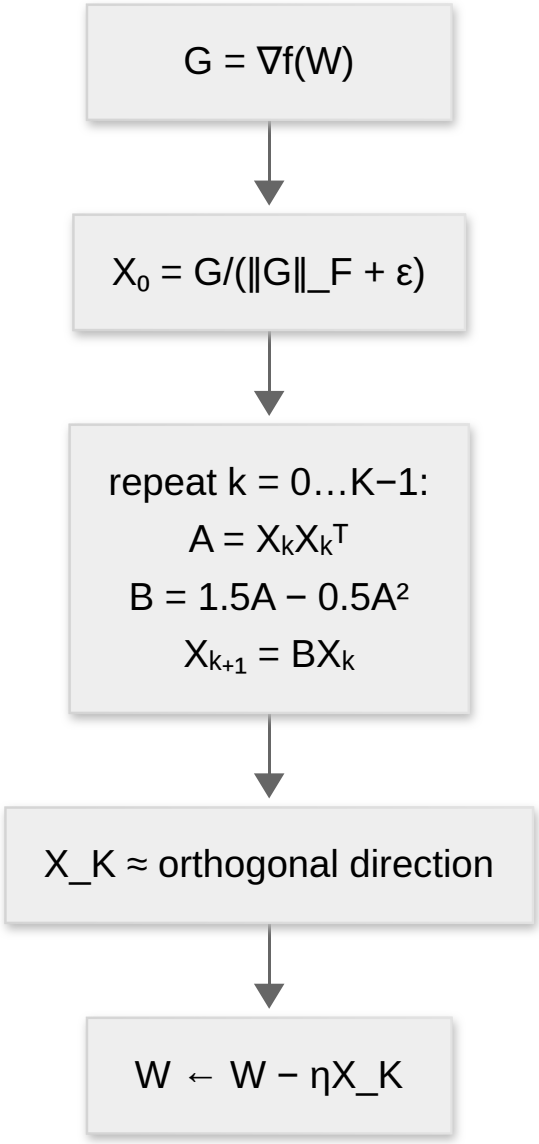


Figure 100: Step flow of Muon: Frobenius normalization followed by Newton-Schulz iterations that orthogonalize the update direction.

Muon (Newton-Schulz Orthogonalization). Muon is an **orthogonality-oriented optimizer** that reshapes update geometry using Newton-Schulz polynomial iterations for orthogonalization. For a 2D parameter matrix W with gradient G_t:

$$X_0 = \frac{G_t}{\|G_t\|_F + \epsilon} \quad \text{(normalized gradient)} \tag{108}$$

$$A_k = X_k X_k^\top \quad \text{(Gramian)} \tag{109}$$

$$B_k = bA_k + cA_k^2 \quad \text{(polynomial step, coefficients } b, c \text{ from Newton-Schulz)} \tag{110}$$

$$X_{k+1} = aX_k + B_k X_k \quad \text{(iteration } k = 0, 1, \ldots, 4) \tag{111}$$

$$W_{t+1} = W_t - \eta X_K \quad \text{(update with orthogonalized direction)} \tag{112}$$

Muon performs 5 Newton-Schulz iterations to produce an approximately orthogonal direction, ensuring updates respect geometric constraints. The rationale: orthogonal updates preserve norms during training, avoiding the norm creep observed in momentum-based methods.

Following the hybrid recipe of Shen et al. [146], the orthogonalized update is reserved for hidden-layer weight matrices (`attention`, `ffn` and other hidden-block groups). Embedding tables, learned positional encodings, output heads and every non-matrix parameter (biases, normalization scales, convolution kernels) are trained with decoupled AdamW instead of being skipped, using the group-level learning rate, weight decay, β_1, β_2 and ϵ settings exposed by the per-group configuration.

Algorithm 48 Muon (Newton-Schulz Orthogonalization)

Require: Initial parameters θ, learning rate η, Newton-Schulz iters K
Ensure: Updated parameters θ_T

for each parameter P **do**
  if P is a hidden-layer 2D matrix W **then**
    $G \leftarrow \nabla f(W)$ ▷ Gradient
    $X_0 \leftarrow \frac{G}{\|G\|_F + \epsilon}$ ▷ Normalize gradient by Frobenius norm
    for $k = 0, 1, \ldots, K-1$ **do**
      $A_k \leftarrow X_k X_k^\top$ ▷ Gramian (cost: $O(mn^2)$ or $O(m^2 n)$ for tall/wide)
      $B_k \leftarrow (3/2)A_k - (1/2)A_k^2$ ▷ Newton-Schulz coefficients
      $X_{k+1} \leftarrow B_k X_k$ ▷ Polynomial step
    end for
    $W \leftarrow W - \eta X_K$ ▷ Update with orthogonalized direction
  else
    $P \leftarrow \mathrm{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Hybrid fallback: embeddings, pos, heads, non-matrices
  end if
end forreturn θ_T

Cons.

- Several matrix multiplications per Newton-Schulz iteration; efficient custom kernels are required.
- The orthogonalized construction is defined only for 2D matrices; the remaining parameter classes depend on the AdamW fallback and its per-group learning-rate calibration.

Pros.

- Orthogonalized updates preserve norms and avoid the norm creep of momentum-based methods.
- Promising results on large-scale training with fundamentally better-conditioned update geometry.

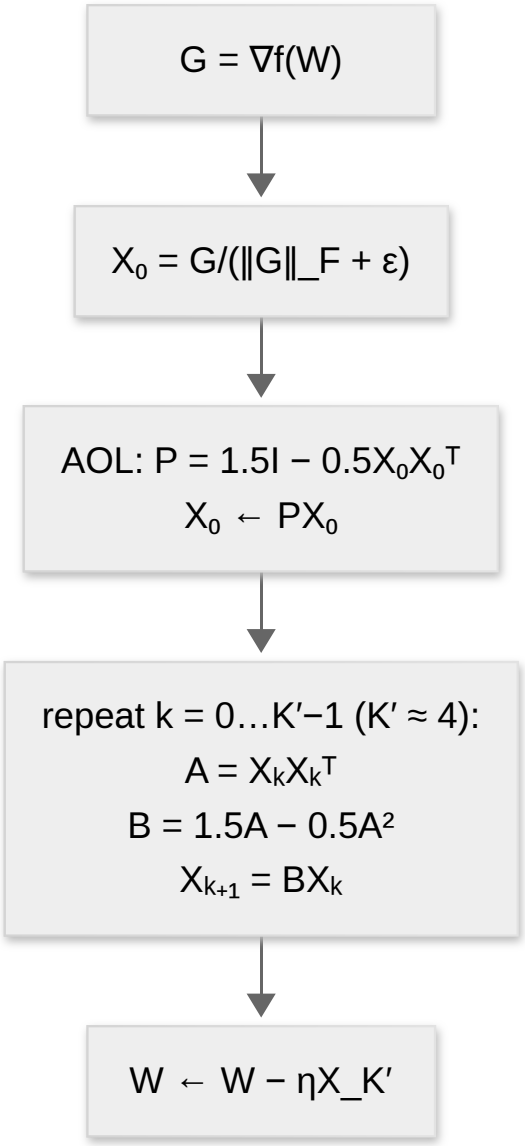


Figure 101: Step flow of Turbo-Muon: almost-orthogonal preconditioning (AOL) before the Newton-Schulz iterations, reducing their count.

Turbo-Muon (AOL-Preconditioned Orthogonalization). Turbo-Muon adds an **almost-orthogonal preconditioning** (AOL) step before the Newton-Schulz iterations. The preconditioner $P = 1.5I - 0.5X_0X_0^\top$ is applied to the normalized gradient, reducing the number of required orthogonalization steps to four while preserving the orthogonalized update geometry. It inherits Muon's hybrid routing: only hidden-layer 2D matrices are orthogonalized, while embedding tables, learned positional encodings, output heads and non-matrix parameters take decoupled AdamW updates.

Algorithm 49 Turbo-Muon (AOL-Preconditioned Orthogonalization)

Require: Initial parameters θ, learning rate η, Newton-Schulz iters K' (typically 4)
Ensure: Updated parameters θ_T

for each parameter P **do**
  if P is a hidden-layer 2D matrix W **then**
    $G \leftarrow \nabla f(W)$ ▷ Gradient
    $X_0 \leftarrow \frac{G}{\|G\|_F + \epsilon}$ ▷ Normalize gradient
    AOL (Almost-Orthogonal preconditioner): ▷ Preconditioning step
    $P \leftarrow (1.5I - 0.5 X_0 X_0^\top)$ ▷ Preconditioning matrix
    $X_0 \leftarrow P X_0$ ▷ Preconditioned start
    for $k = 0, 1, \ldots, K' - 1$ **do**
      $A_k \leftarrow X_k X_k^\top$
      $B_k \leftarrow (3/2) A_k - (1/2) A_k^2$
      $X_{k+1} \leftarrow B_k X_k$ ▷ Reduced iterations due to AOL preconditioning
    end for
    $W \leftarrow W - \eta X_{K'}$ ▷ Update with preconditioned orthogonalized direction
  else
    $P \leftarrow \mathrm{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Hybrid fallback: embeddings, pos, heads, non-matrices
  end if
end forreturn θ_T

Cons.

- Still compute-heavy; AOL adds a preconditioning matrix multiplication.
- Requires the same custom-kernel support as Muon, and shares its dependence on the AdamW fallback for non-hidden parameter classes.

Pros.

- AOL reduces the Newton-Schulz iteration count (typically five to four).
- Retains orthogonalized update geometry at lower iteration cost.

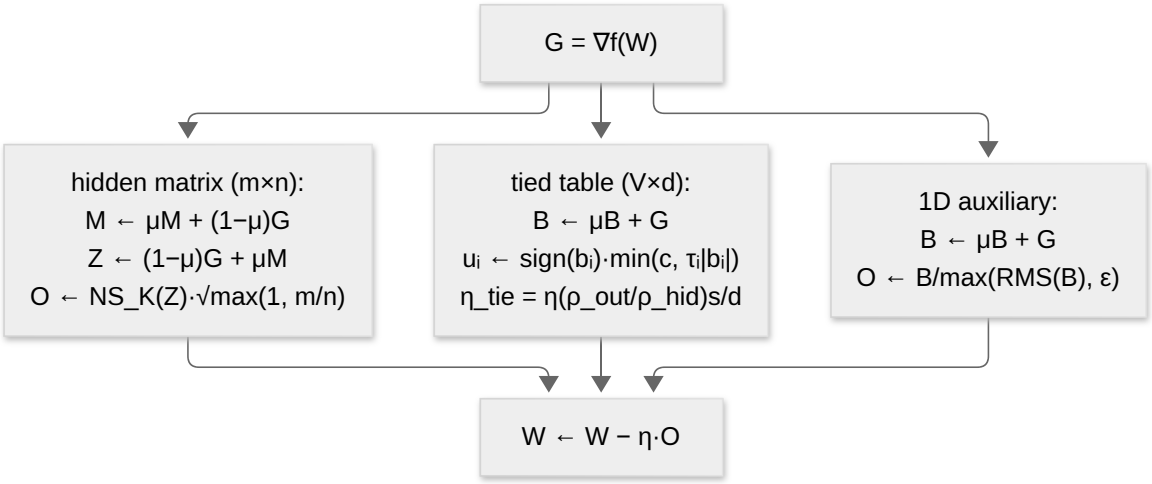


Figure 102: Step flow of AF-Muon: Muon's spectral update for hidden matrices, a support-aware finite-cap clipped-row LMO for the tied vocabulary table, and RMS-normalized updates for one-dimensional auxiliary parameters, all with a single first-moment buffer.

AF-Muon (AdamW-Free Tied-Embedding Muon). AF-Muon is an **AdamW-free extension of Muon** that removes the auxiliary AdamW fallback of the Hybrid Muon recipe by giving every parameter class its own norm-constrained geometry. Hidden matrices keep the spectral update with a Nesterov signal; the vocabulary table, which simultaneously receives sparse input-lookup and dense output-softmax gradients, is updated row by row with a *support-aware finite-cap linear minimization oracle*; one-dimensional auxiliary tensors receive an RMS-normalized update. No parameter class carries second-moment state, so a single first-moment buffer per parameter replaces the AdamW state of the hybrid recipe, which the authors report as roughly 20% less optimizer-state memory at matched training.

For the tied table $W \in \mathbb{R}^{V\times d}$ with momentum row b_i, the oracle solves the row problem

$$u_i = \operatorname{sign}(b_i) \odot \min\left(c\,\mathbf{1},\ \tau_i\,|b_i|\right), \tag{113}$$

$$\|u_i\|_2^2 = \min\left(d,\ c^2\left|\operatorname{supp}(b_i)\right|\right), \tag{114}$$

where the per-row scale τ_i is obtained by bisection in FP32 over complete-row chunks. Rows whose support cannot fill the row-ℓ_2 radius saturate every useful coordinate at the cap c, so sparse lookup rows are handled without inflating absent coordinates. The cap interpolates between two known endpoints: $c = 1$ recovers the SCION-style Sign update and $c \geq \sqrt{d}$ reduces to a row-RMS update. The tied step size is $\eta_{tie} = \eta_M(\rho_{out}/\rho_{hid})\, s/d$; the paper's main setting is $c = 3$ and $s = 0.5$ with $\rho_{out}/\rho_{hid} = 60$, and its diagnostics report the finite cap active on under 0.3% of coordinates while affecting most rows.

Algorithm 50 AF-Muon (AdamW-Free Tied-Embedding Muon)

Require: Initial parameters θ, matrix LR η_M, vector LR η_{vec}, momentum μ, matrix weight decay λ_M, tied scale s, cap c, radii ρ_{out}, ρ_{hid}, Newton-Schulz iters K

Ensure: Updated parameters θ_T

for each parameter tensor W with gradient G_t **do**
  if W is a hidden matrix in $\mathbb{R}^{m\times n}$ **then**
    $M_t \leftarrow \mu M_{t-1} + (1-\mu)G_t$ ▷ First moment
    $Z_t \leftarrow (1-\mu)G_t + \mu M_t$ ▷ Nesterov signal
    $O_t \leftarrow \mathrm{NS}_K(Z_t)\sqrt{\max(1, m/n)}$ ▷ Spectral update
    $W_t \leftarrow (1-\eta_M\lambda_M)W_{t-1} - \eta_M O_t$
  else if $W \in \mathbb{R}^{V\times d}$ is the tied vocabulary table **then**
    $B_t \leftarrow \mu B_{t-1} + G_t$ ▷ Unnormalized first moment
    $\eta_{tie,t} \leftarrow \eta_M(\rho_{out}/\rho_{hid})s/d$
    for each row $i = 1, \ldots, V$ **do**
      $u_i \leftarrow L_c\big(\mathrm{row}_i(B_t)\big)$ ▷ Capped-row LMO
    end for
    $W_t \leftarrow W_{t-1} - \eta_{tie,t}U$
  else
    $B_t \leftarrow \mu B_{t-1} + G_t$ ▷ 1D auxiliary tensor
    $O_t \leftarrow B_t/\max(\mathrm{RMS}(B_t), \epsilon)$
    $W_t \leftarrow W_{t-1} - \eta_{vec}O_t$
  end if
end forreturn θ_T

Cons.

- Every parameter must be assigned an explicit role (matrix, tied table, or vector); an untied output head or an extra embedding table is not a tied-table block.
- The tied-table oracle requires a per-row bisection with complete-row chunking; the paper reports roughly 1% median step-time overhead at matched training.
- The tied-table claim is established for tied-vocabulary models; frontier-scale pretraining and universal endpoint superiority are not claimed.

Pros.

- AdamW-free: one first-moment buffer per parameter and no second-moment state, reported as roughly 20% less optimizer-state memory than Hybrid Muon.
- The finite cap preserves within-row magnitudes while bounding coordinate concentration, and recovers Sign ($c = 1$) and row-RMS ($c \geq \sqrt{d}$) as limit cases.
- Improves mean validation loss and perplexity over both Hybrid Muon and a SCION-style Sign endpoint across nine tied-token settings spanning text, image-token, protein, MoE, and encoder-decoder models.

AdaMuon (Adaptive Muon). AdaMuon is an **adaptive extension of Muon** that keeps the spectral (polar) update while adding coordinate-wise variance normalization. Muon orthogonalizes the momentum matrix into a direction-only update, which removes anisotropic scaling but assigns the same magnitude to every coordinate; AdaMuon addresses the resulting mismatch between noisy and informative coordinates by tracking the element-wise variance of the *orthogonalized*

direction rather than of the raw gradient or of the momentum. Because the polar factor is globally scale-invariant, the sign transform applied before orthogonalization preserves the update direction while stabilizing its element-wise magnitudes; the paper proves that sign is the unique element-wise map that is scale-free, sign-consistent, odd and bounded (Theorem 1). The second-moment estimate therefore sees a stable signal, and its variance-normalized version is rescaled to an RMS of 0.2, matching Adam's empirical update scale so that existing learning-rate schedules are reused without tuning.

For a weight $W_t \in \mathbb{R}^{n\times m}$, momentum buffer M_t, momentum β, Newton–Schulz steps T and small constant ε:

$$M_t = \beta M_{t-1} + G_t, \tag{115}$$

$$O_t = \mathrm{NewtonSchulz}\big(\mathrm{sign}(M_t), T\big), \tag{116}$$

$$V_t = \beta V_{t-1} + (1-\beta)\, O_t \odot O_t, \tag{117}$$

$$\widehat{O}_t = O_t \oslash \big(\sqrt{V_t} + \varepsilon \mathbf{1}\big), \tag{118}$$

$$\gamma_t = \frac{0.2\sqrt{mn}}{\|\widehat{O}_t\|_F}, \tag{119}$$

$$W_{t+1} = W_t - \eta\big(\gamma_t \widehat{O}_t + \lambda W_t\big). \tag{120}$$

The Newton–Schulz map uses $X_0 = \mathrm{sign}(M_t)/\|\mathrm{sign}(M_t)\|_F$ and the cubic iteration $X_k = aX_{k-1} + b(X_{k-1}X_{k-1}^{\mathsf{T}})X_{k-1} + c(X_{k-1}X_{k-1}^{\mathsf{T}})^2 X_{k-1}$ with $(a,b,c) = (3.4445, -4.7750, 2.0315)$ and $T = 5$. No bias correction is applied to either moment: a constant factor in V_t is cancelled exactly by $\gamma_t \propto \sqrt{c}$, and a constant factor in M_t is removed by the polar map, $\mathrm{polar}(cM) = \mathrm{polar}(M)$.

Algorithm 51 AdaMuon (Adaptive Muon)

Require: Initial weights $W_0 \in \mathbb{R}^{n\times m}$, learning rate η, weight decay λ, momentum β, Newton–Schulz steps T, small constant ε

Ensure: Updated weights W_T

$M_0 \leftarrow \mathbf{0}$, $V_0 \leftarrow \mathbf{0}$
for each iteration $t = 1, 2, \ldots$ **do**
  $G_t \leftarrow \nabla_{W_t} \mathcal{L}(W_t)$
  $M_t \leftarrow \beta M_{t-1} + G_t$
  $O_t \leftarrow \mathrm{NewtonSchulz}(\mathrm{sign}(M_t), T)$
  $V_t \leftarrow \beta V_{t-1} + (1-\beta)\, O_t \odot O_t$
  $\widehat{O}_t \leftarrow O_t \oslash (\sqrt{V_t} + \varepsilon \mathbf{1})$
  $\gamma_t \leftarrow 0.2\sqrt{mn}/\|\widehat{O}_t\|_F$
  $W_{t+1} \leftarrow W_t - \eta(\gamma_t \widehat{O}_t + \lambda W_t)$
end forreturn W_T

Cons.

- One additional second-moment buffer per matrix parameter (two buffers versus Muon's single first-moment buffer), plus per-step element-wise EMA, square-root and normalization work on the update.
- The update is defined for two-dimensional parameters; one-dimensional parameters (biases, norm scales) require an auxiliary optimizer in the paper's recipe.
- The RMS target (0.2) is a fixed constant rather than a learned quantity, so the method assumes Adam-like update scales when reusing existing schedules.

Pros.

- No new hyperparameters: the second-moment EMA reuses Muon's momentum coefficient β, and the paper reports low sensitivity across $\beta \in [0.8, 0.99]$.
- The sign transform makes the variance estimate stable enough to normalize, and the RMS alignment cancels any constant bias in V_t, so no bias correction is needed for either moment.
- RMS alignment matches Adam's update scale, allowing direct reuse of learning-rate schedules; the authors report up to roughly 40% training-efficiency gains over AdamW and consistent wins over Muon on GPT-2 (125M–1.5B) and Qwen2.5 (1.5B, 7B).

Dion3 (Fractional Gram-Newton-Schulz Orthogonalization). Dion3 [4] is a **revision of Muon** that reduces the cost of the orthogonalization step at the algorithmic level. It replaces standard Newton–Schulz with *Gram Newton–Schulz*, a mathematically equivalent iteration that runs on the small symmetric Gram matrix XX^T, and it orthogonalizes only a fraction $f \in (0, 1]$ of the momentum rows at each step, updating only those rows of the weight matrix. *Error feedback* decays only the selected rows of the momentum buffer, so rows that were not selected keep accumulating and are eventually chosen. At $f = 1$ the update reduces to Muon in exact arithmetic (or to NorMuon with the optional row-wise normalization). For a weight $W \in \mathbb{R}^{n\times m}$ (operating on the smaller dimension, $n \le m$), fraction f, $k = \lceil fn \rceil$, momentum μ, learning rate η and weight decay λ:

$$M_t = M_{t-1} + G_t \quad \text{(momentum accumulation; damping deferred)} \tag{121}$$
$$S_t = \mathrm{topk}_i\big(\|M_{t,i,:}\|_1,\ k\big) \quad \text{(select the } k \text{ largest rows)} \tag{122}$$
$$O_t = \mathrm{GramNS}\big(M_t[S_t, :]\big) \quad \text{(orthogonalize the } k \times m \text{ submatrix)} \tag{123}$$
$$W_t = (1 - \eta\lambda)\, W_{t-1}, \qquad W_t[S_t, :] \leftarrow W_t[S_t, :] - \eta\, O_t \tag{124}$$
$$M_t[S_t, :] \leftarrow \mu\, M_t[S_t, :] \quad \text{(error feedback: decay only selected rows)} \tag{125}$$

Gram Newton–Schulz normalizes $X \leftarrow M_t[S_t, :]/(\|M_t[S_t, :]\|_F + \epsilon)$ with $\epsilon = 10^{-7}$, casts to `float16`, forms $R_0 = XX^\mathsf{T}$ and $Q_0 = I$, and runs T iterations

$$Z_t = b_t R_{t-1} + c_t R_{t-1}^2, \qquad Q_t = Q_{t-1} Z_t + a_t Q_{t-1}, \tag{126}$$
$$(RZ)_t = R_{t-1} Z_t + a_t R_{t-1}, \qquad R_t = Z_t (RZ)_t + a_t (RZ)_t, \tag{127}$$

returning $Q_T X$. A *restart* after iteration 2 ($X \leftarrow Q_2 X$, $R_2 \leftarrow XX^\mathsf{T}$, $Q_2 \leftarrow I$) removes the spurious negative eigenvalues that half precision introduces in the Gram matrix; the coefficients (a_t, b_t, c_t) are the Polar Express five-step triples [3]. The cost falls from $T(3\alpha + 1)n^3$ FLOPs for standard Newton–Schulz to $(4T + 3\alpha - 3)n^3$ for Gram Newton–Schulz ($\alpha = m/n$), and selecting $k = \lceil fn \rceil$ rows both shrinks the iteration and raises the aspect ratio by $1/f$. The optional NorMuon flavor adds a row-wise second moment $v_i \leftarrow \beta_2 v_i + (1 - \beta_2)\ \mathrm{mean}_j(O_{ij}^2)$ for $i \in S$, rescales $\tilde{O}_{ij} = O_{ij}/(\sqrt{v_i} + \epsilon)$, and renormalizes $\tilde{O} \leftarrow \tilde{O}\, \|O\|_F / \|\tilde{O}\|_F$ in `float32` before the update.

Algorithm 52 Dion3 (Fractional Orthogonalization with Error Feedback)

Require: Matrices W, learning rate η, momentum μ, fraction f, Gram Newton–Schulz steps T, ϵ, weight decay λ

Ensure: Updated matrices W_T

for each 2D parameter matrix W with gradient G **do**
  $M \leftarrow M + G$ ▷ Momentum accumulation
  $S \leftarrow$ indices of the $k = \lceil fn \rceil$ rows of M with largest ℓ_1 norm ▷ Selection
  $O \leftarrow \mathrm{GramNS}_T(M[S,:])$ ▷ Orthogonalize the selected submatrix
  if NorMuon **then**
    $v_i \leftarrow \beta_2 v_i + (1-\beta_2)\,\mathrm{mean}_j(O_{ij}^2)$ for $i \in S$
    $\tilde{O}_{ij} \leftarrow O_{ij}/(\sqrt{v_i} + \epsilon)$; $\tilde{O} \leftarrow \tilde{O}\,\|O\|_F/\|\tilde{O}\|_F$
  end if
  $W \leftarrow (1-\eta\lambda)W$ ▷ Weight decay on all rows
  $W[S,:] \leftarrow W[S,:] - \eta\,\tilde{O}$ ▷ Update selected rows only
  $M[S,:] \leftarrow \mu\,M[S,:]$ ▷ Error feedback
end forreturn W_T

Cons.

- Each step updates only a fraction of the rows, so the optimization trajectory differs from Muon; the paper reports the change is benign (Section 8.1 of the Dion3 report).
- Row selection and error feedback add top-k bookkeeping and a row-wise momentum decay per step.
- The largest wall-clock gains in the paper come from symmetric GEMM kernels and megabatched communication, which are hardware- and cluster-specific; only the algorithmic reduction and the single first-moment buffer apply here.

Pros.

- Gram Newton–Schulz is mathematically equivalent to Newton–Schulz and cuts orthogonalization FLOPs by about 55% at typical transformer aspect ratios with five iterations.
- Fractional row selection reduces the orthogonalized submatrix to $k \times m$ and raises its aspect ratio by $1/f$, compounding the Gram savings; the paper reports up to $6\times$ faster optimizer steps at matched loss.
- At $f = 1$ the update reduces to Muon (or NorMuon), and a single momentum buffer per matrix parameter suffices; error feedback lets ignored rows participate eventually.

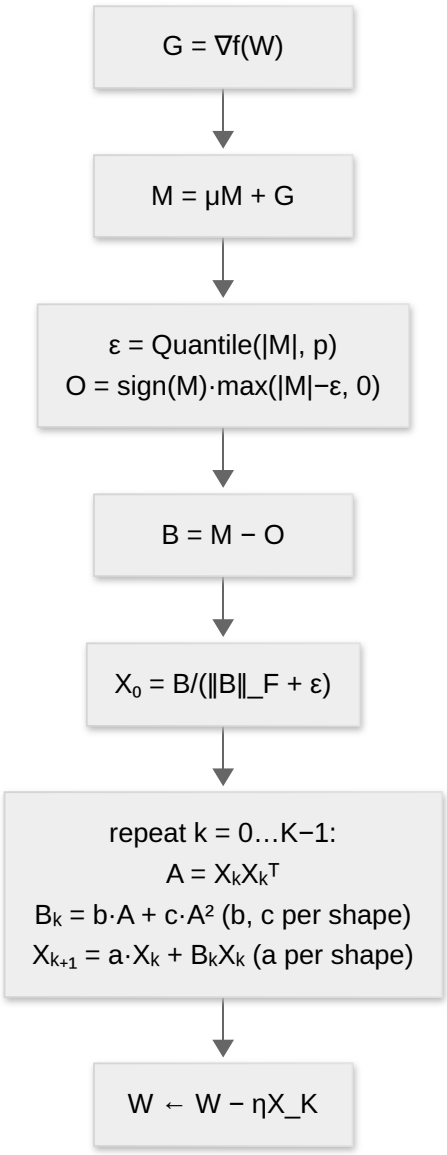


Figure 103: Step flow of ROOT: momentum accumulation, soft-threshold outlier separation, and shape-calibrated Newton–Schulz orthogonalization.

ROOT (Robust Orthogonalized Optimizer). ROOT [64] is a **robustness-oriented revision of Muon** that attacks two fragilities of the fixed-coefficient Newton–Schulz pipeline. First, the universal coefficients $(a, b, c) = (3.4445, -4.7750, 2.0315)$ are calibrated for an "average" matrix shape and lose precision exactly on square and near-square layers: the orthogonalization MSE ranges from 0.0499 at 2048×2048 to 0.0003 at 2048×16384 in Table 1 of the paper. ROOT replaces them with per-shape coefficients $\{a^{(m,n)}, b^{(m,n)}, c^{(m,n)}\}$ calibrated offline over momentum singular-value distributions (Mixed 1:3 real-to-random calibration), a scheme the authors call *AdaNewton*. Second, heavy-tailed gradient components distort the Frobenius normalization required by Newton–Schulz and are amplified by the polynomial iteration; ROOT models the momentum as $M_t = B_t + O_t$ and solves a proximal L1 problem whose closed-form solution is soft-thresholding. Only the robust base component B_t is orthogonalized; the sparse outlier component O_t is discarded. For a hidden-layer weight matrix $W \in \mathbb{R}^{m \times n}$, momentum μ, learning

rate η, percentile p and iteration count K:

$$M_t = \mu M_{t-1} + \nabla \mathcal{L}(\theta_{t-1}) \quad \text{(momentum, Eq. 1)} \tag{128}$$

$$\varepsilon_t = \text{Quantile}(|M_t|, p) \quad \text{(dynamic threshold, } p = 0.90 \text{ default)} \tag{129}$$

$$O_t = \mathcal{T}_{\varepsilon_t}[M_t], \quad \mathcal{T}_\varepsilon[x]_i = \text{sign}(x_i)\max(|x_i| - \varepsilon, 0) \quad \text{(soft-threshold, Eqs. 7–10)} \tag{130}$$

$$B_t = M_t - O_t \quad \text{(clipped robust component, Eq. 9)} \tag{131}$$

$$X_0 = \frac{B_t}{\|B_t\|_F}, \qquad X_k = a^{(m,n)} X_{k-1} + b^{(m,n)} X_{k-1}(X_{k-1}^\top X_{k-1}) + c^{(m,n)} X_{k-1}(X_{k-1}^\top X_{k-1})^2 \tag{132}$$

$$\theta_t = \theta_{t-1} - \eta X_K \quad \text{(update, } K = 5) \tag{133}$$

Table 13: AdaNewton coefficients from Table 1 of He et al. [64] (Mixed 1:3 calibration). Shapes outside the table fall back to Muon's fixed $(3.4445, -4.7750, 2.0315)$.

m	n	$a^{(m,n)}$	$b^{(m,n)}$	$c^{(m,n)}$	MSE
2048	2048	3.3334	−4.2591	1.7791	0.0352
4096	4096	3.3732	−4.6134	2.0576	0.0470
8192	8192	3.3886	−4.9026	2.3105	0.0587
3072	2048	2.9091	−3.8108	1.8600	0.0024
4096	2048	2.7739	−3.4911	1.6992	0.0010
8192	2048	2.5925	−3.0010	1.4138	0.0003
16384	2048	2.7045	−3.2335	1.5336	0.0003

Algorithm 53 ROOT (Robust Orthogonalized Optimizer)

Require: Initial parameters θ, learning rate η, momentum μ, percentile p, Newton–Schulz iters K

Ensure: Updated parameters θ_T

for each parameter P **do**
  if P is a hidden-layer 2D matrix W **then**
    $G \leftarrow \nabla f(W)$ ▷ Gradient
    $M \leftarrow \mu M + G$ ▷ Momentum accumulation
    $\varepsilon \leftarrow \text{Quantile}(|M|, p)$ ▷ Dynamic threshold
    $O \leftarrow \text{sign}(M)\max(|M| - \varepsilon, 0)$ ▷ Outlier separation (soft-threshold)
    $B \leftarrow M - O$ ▷ Clipped robust component
    $X_0 \leftarrow \frac{B}{\|B\|_F + \epsilon}$ ▷ Normalize by Frobenius norm
    for $k = 0, 1, \ldots, K-1$ **do**
      $A_k \leftarrow X_k X_k^\top$ ▷ Gramian
      $B_k \leftarrow b^{(m,n)} A_k + c^{(m,n)} A_k^2$ ▷ Shape-calibrated coefficients
      $X_{k+1} \leftarrow a^{(m,n)} X_k + B_k X_k$ ▷ Polynomial step
    end for
    $W \leftarrow W - \eta X_K$ ▷ Update with orthogonalized direction
  else
    $P \leftarrow \text{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Hybrid fallback: embeddings, pos, heads, non-matrices
  end if
end forreturn θ_T

Following the hybrid recipe of Muon, only hidden-layer 2-D matrices take the ROOT path; embedding tables, learned positional encodings, output heads and every non-matrix parameter

are trained with decoupled AdamW using the group-level learning rate, weight decay, β_1, β_2 and ϵ settings. The paper's LLM runs apply a 0.2 update-RMS scaling factor following Moonlight; this implementation lets the per-group learning rates absorb that calibration, so no scale constant is applied. Shapes absent from Table 13 fall back to Muon's fixed coefficients, and the dynamic quantile costs an additional sort of $|M_t|$ per matrix per step.

Cons.

- The percentile threshold requires an $O(N \log N)$ sort of the momentum matrix at every step, on top of the Newton–Schulz matmuls.
- The calibrated coefficient table covers seven shapes only; every other shape falls back to Muon's fixed coefficients.
- Outliers are discarded rather than shrunk, so an extremely noisy gradient can also lose informative signal; the paper's ablation shows the trade-off is governed by the percentile (0.85 truncates too much, 0.99 too little).

Pros.

- Dimension-invariant orthogonalization precision: shape-calibrated coefficients remove the precision cliff that fixed coefficients show on square and near-square layers.
- Outlier-robust without retuning the Muon recipe; the paper reports lower loss and a better zero-shot average than Muon and AdamW at 1B scale (60.12 vs 59.59 vs 59.05) and gains on ViT/CIFAR-10.
- Same single momentum buffer per matrix parameter as Muon, and the `softthresh`/`ada_newton` toggles expose the paper's ablations directly.

Muon² (Second-Moment Preconditioned Orthogonalization). Muon² [106] generalizes Muon by preconditioning the momentum with an **Adam-style second-moment EMA before the Newton–Schulz orthogonalization**. The paper shows that the practical bottleneck of Muon's polar approximation is the ill-conditioned momentum matrix (early-training spectra span 10^{-4}–1 with roughly half the singular values in the NS "dead zone"); dividing by $\sqrt{V_t}$ tightens the spectrum, so every NS step aligns better with the true orthogonalized update (cosine similarity 0.916 at K=3 vs Muon's 0.931 at K=5, rising to 0.975 at K=5). For a hidden-layer weight matrix $W \in \mathbb{R}^{n \times m}$ with learning rate η:

$$M_t = \beta_1 M_{t-1} + (1-\beta_1) G_t \qquad \text{(momentum EMA, Eq. 2)} \tag{134}$$

$$V_t = \beta_2 V_{t-1} + (1-\beta_2)\,(G_t \odot G_t) \qquad \text{(second-moment EMA, Eq. 3)} \tag{135}$$

$$\widetilde{M}_t = M_t \oslash \left(\sqrt{V_t} + \epsilon \mathbf{1}\right) \qquad \text{(preconditioning, Eq. 4)} \tag{136}$$

$$X_0 = \frac{\widetilde{M}_t}{\|\widetilde{M}_t\|_F + \epsilon}, \qquad X_{k+1} = a X_k + b\, X_k X_k^\top X_k + c\, X_k (X_k X_k^\top)^2 X_k \qquad \text{(NS, Eq. 5)} \tag{137}$$

$$W_{t+1} = W_t - \eta \sqrt{m/n}\, X_K \qquad \text{(update, Eq. 6)} \tag{138}$$

with the same quintic coefficients $(a, b, c) = (3.4445, -4.7750, 2.0315)$ as Muon and the $\sqrt{m/n}$ learning-rate factor proposed by Bernstein (2025). The memory-efficient variant **Muon²-F**

replaces V_t with Adafactor-style row/column statistics:

$$r_t = \beta_2 r_{t-1} + (1-\beta_2)\,\mathrm{sum}_{\mathrm{col}}(G_t \odot G_t), \qquad c_t = \beta_2 c_{t-1} + (1-\beta_2)\,\mathrm{sum}_{\mathrm{row}}(G_t \odot G_t), \tag{139}$$

$$\widehat{V}_t = \frac{r_t\, c_t^\top}{\mathrm{sum}(r_t)}, \tag{140}$$

keeping memory within ∼0.3% of Muon at 7B scale (47.53 vs 47.29 GB) while retaining most of Muon^2's benefit. Across GPT, LLaMA and Mixture-of-Experts pre-training up to 13B parameters, Muon^2 outperforms Muon at equal NS iterations and matches it with K=3 instead of 5 (a 40% reduction), saving up to a quarter of training time at equal loss.

Algorithm 54 Muon^2 (Second-Moment Preconditioned Orthogonalization)

Require: Initial parameters θ, learning rate η, β_1, β_2, NS iters K, preconditioner ϵ
Ensure: Updated parameters θ_T

for each parameter P **do**
  if P is a hidden-layer 2D matrix $W \in \mathbb{R}^{n\times m}$ **then**
    $G \leftarrow \nabla f(W)$ ▷ Gradient
    $M \leftarrow \beta_1 M + (1-\beta_1)G$ ▷ Momentum EMA (Eq. 2)
    $V \leftarrow \beta_2 V + (1-\beta_2)(G \odot G)$ ▷ Second-moment EMA (Eq. 3); factored variant keeps row/col sums only
    $\widetilde{M} \leftarrow M \oslash (\sqrt{V} + \epsilon \mathbf{1})$ ▷ Precondition (Eq. 4)
    $X_0 \leftarrow \widetilde{M}/(\|\widetilde{M}\|_F + \epsilon)$ ▷ Frobenius normalization
    for $k = 0, 1, \ldots, K-1$ **do**
      $A_k \leftarrow X_k X_k^\top$ ▷ Gramian
      $X_{k+1} \leftarrow 3.4445\, X_k - 4.7750\, A_k X_k + 2.0315\, A_k^2 X_k$ ▷ Quintic NS step
    end for
    $W \leftarrow W - \eta\sqrt{m/n}\, X_K$ ▷ Scaled update (Eq. 6)
  else
    $P \leftarrow \mathrm{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Hybrid fallback: embeddings, pos, heads, non-matrices
  end if
end forreturn θ_T

Following the hybrid recipe of Muon, only hidden-layer 2-D matrices take the Muon^2 path; embedding tables, learned positional encodings, output heads and every non-matrix parameter are trained with decoupled AdamW using the group-level learning rate, weight decay, β_1, β_2 and ϵ settings — the same $\beta_1, \beta_2, \epsilon$ also parameterize Eqs. 2–4, so one per-group knob set configures both paths.

Cons.

- One extra full-size state buffer over Muon (two moment EMAs per matrix), doubling optimizer memory on the orthogonalized path unless the factored variant is enabled.
- The $\sqrt{m/n}$ update scale differs from this repository's plain `muon` path, so learning rates do not transfer one-to-one between the two registry entries.
- β_1, β_2 defaults are not fixed by the paper; this implementation inherits the shared per-group defaults $(0.9, 0.95)$.

Pros.

- Better orthogonalization at equal NS iterations and equal quality with 40% fewer iterations (3 instead of 5), cutting NS compute and communication.
- Consistent gains over Muon across GPT, LLaMA and Mixture-of-Experts pre-training up to 13B parameters, with up to a quarter of training time saved at matched loss.
- The `factored` toggle exposes Muon^2-F, which keeps memory within $\sim$0.3% of Muon while preserving most of the benefit.

NorMuon (Neuron-wise Normalized Muon). NorMuon [96] keeps Muon's orthogonalized update and adds a **neuron-wise (row-wise) adaptive learning rate** on top of it. Muon's Newton–Schulz iteration equalizes the singular values of the update, but the L_2 norm of the individual neuron rows remains highly non-uniform: some neurons receive disproportionately large updates while others are barely trained. NorMuon tracks the mean squared magnitude of each row of the *orthogonalized* direction in a second-moment EMA, divides every neuron row by its RMS, and rescales the whole update to Adam's empirical update scale so existing learning-rate schedules transfer unchanged. For a hidden-layer weight matrix $W \in \mathbb{R}^{m\times n}$ with learning rate η:

$$M_t = \beta_1 M_{t-1} + (1-\beta_1) G_t \qquad \text{(momentum, line 5)} \tag{141}$$

$$O_t = \mathrm{NS}_5(M_t) \qquad \text{(orthogonalization, line 6)} \tag{142}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\,\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t) \qquad \text{(neuron EMA, line 7)} \tag{143}$$

$$V_t = \mathrm{ExpandRows}(v_t), \qquad \widehat{O}_t = O_t \oslash \left(\sqrt{V_t} + \epsilon\right) \qquad \text{(row normalization, lines 8–9)} \tag{144}$$

$$\hat{\eta} = \frac{0.2\,\eta\,\sqrt{mn}}{\|\widehat{O}_t\|_F}, \qquad W_{t+1} = W_t - \eta\lambda W_t - \hat{\eta}\,\widehat{O}_t \qquad \text{(update, lines 10–11)} \tag{145}$$

with $v_t \in \mathbb{R}^m$ (one scalar per neuron) and the same quintic Newton–Schulz coefficients $(a, b, c) = (3.4445, -4.7750, 2.0315)$ as Muon. The per-neuron statistic costs m scalars against the $m \times n$ first moment, so optimizer state is $m(n+1)$ — within a $1/n$ factor of Muon and roughly half of Adam's $2mn$. On 1.1B and 5.4B pretraining NorMuon needs 21.74% and 13.91% fewer steps than AdamW to reach the same validation loss; on Modded-NanoGPT (124M/350M) it reaches Muon's loss with 6% and 15% fewer iterations, at a 2.9% training-step-time overhead over AdamW and 21.19 GB of optimizer state versus 40.56 GB for AdamW at 5.4B.

Algorithm 55 NorMuon (Neuron-wise Normalized Muon)

Require: Initial parameters θ, learning rate η, β_1, β_2, NS iters K, row-normalization ϵ, weight decay λ

Ensure: Updated parameters θ_T

1: **for** each parameter P **do**
2: **if** P is a hidden-layer 2D matrix $W \in \mathbb{R}^{m \times n}$ **then**
3: $G \leftarrow \nabla f(W)$ ▷ Gradient
4: $M \leftarrow \beta_1 M + (1 - \beta_1) G$ ▷ Momentum (line 5)
5: $O \leftarrow \mathrm{NS}_K(M)$ ▷ Newton–Schulz (line 6)
6: $v \leftarrow \beta_2 v + (1 - \beta_2)\,\mathrm{mean}_{\mathrm{cols}}(O \odot O)$ ▷ Per-neuron second moment (line 7)
7: $\widehat{O} \leftarrow O \oslash (\sqrt{\mathrm{ExpandRows}(v)} + \epsilon \mathbf{1})$ ▷ Row normalization (lines 8–9)
8: $\hat{\eta} \leftarrow 0.2\,\eta\,\sqrt{mn}/(\|\widehat{O}\|_F + \epsilon)$ ▷ RMS alignment (line 10)
9: $W \leftarrow (1 - \eta\lambda)\,W - \hat{\eta}\,\widehat{O}$ ▷ Update (line 11)
10: **else**
11: $P \leftarrow \mathrm{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Hybrid fallback: embeddings, pos, heads, non-matrices
12: **end if**
13: **end forreturn** θ_T

Following the hybrid recipe of Muon, only hidden-layer 2-D matrices take the NorMuon path; embedding tables, learned positional encodings, output heads and every non-matrix parameter are trained with decoupled AdamW using the group-level learning rate, weight decay, β_1, β_2 and ϵ settings. This implementation maps the paper's parameters to the registry as follows: the `momentum` suffix is β_1, the shared per-group `betas[1]` is β_2, the shared per-group `eps` is ϵ, and `ns_steps`/`ns_eps` drive the Newton–Schulz iteration. Because NS5 starts with a Frobenius normalization, the paper's $(1 - \beta_1)$ momentum scaling and this repository's standard momentum accumulation produce identical updates. The paper's second contribution, Algorithm 2, distributes the orthogonalization across FSDP2 ranks (round-robin parameter assignment, gather/NS5/scatter, shard-local row normalization) to avoid replicated work; that deployment strategy is orthogonal to the update rule and is not implemented here.

Cons.

- One extra per-row state vector per matrix parameter (stored in float32); small next to the $m \times n$ momentum, but non-zero.
- The normalization must follow orthogonalization: the paper's "NorMuon (Front)" ablation, which applies it before NS5, is slightly worse.
- ϵ is placed outside the root ($\sqrt{V_t} + \epsilon$), following Algorithm 2; Algorithm 1's rendering of line 9 is typographically ambiguous.
- The neuron-wise statistics do not replace the AdamW fallback: embeddings, pos, heads and non-2-D tensors still carry full first/second moments.

Pros.

- Combines Muon's low condition number with AdamW-like uniform per-neuron update norms, consistently outperforming both on 124M–5.4B pretraining.
- The $0.2\eta\sqrt{mn}/\|\widehat{O}\|_F$ rescaling targets RMS 0.2η, so Adam learning-rate schedules apply without retuning.

- Memory stays close to Muon ($m(n+1)$ versus mn), roughly halving Adam's optimizer state, and the per-step overhead is a few percent.

Conda (Column-Normalized Adam). Conda [162] keeps Adam's coordinate-wise adaptivity and adds a **milder, column-specific spectral normalization**. The authors first reformulate Muon as a first/second-moment method whose second moment is the row-constant matrix $\mathrm{diag}(\mathbf{\Sigma}_t)\mathbf{1}^\top$: Muon normalizes every column of the update by the same $1/\sigma_i$ factors, which over-amplifies directions with tiny singular values. Conda instead constrains the second moment to the same subspace as the first moment by projecting the stochastic gradient, tracking an element-wise EMA of the *squared projected gradients*, and dividing the projected first moment by its square root. For a weight matrix $W \in \mathbb{R}^{m\times n}$ with $m \le n$, learning rate η, decay rates β_1, β_2, subspace refresh period T, scale factor α and stability constant ϵ:

$$M_t = \beta_1 M_{t-1} + (1-\beta_1) G_t \tag{146}$$

$$\bar{U}_t = \begin{cases} \mathrm{SVD}(M_t), & t \bmod T = 0, \\ \bar{U}_{t-1}, & \text{otherwise}, \end{cases} \tag{147}$$

$$M'_t = \bar{U}_t^\top M_t, \qquad N_t = \beta_2 N_{t-1} + (1-\beta_2)\big(\bar{U}_t^\top G_t\big)^2, \tag{148}$$

$$\widehat{M}_t = \frac{M'_t}{1-\beta_1^t}, \qquad \widehat{N}_t = \frac{N_t}{1-\beta_2^t}, \tag{149}$$

$$W_t = W_{t-1} - \eta\,\alpha\,\bar{U}_t \frac{\widehat{M}_t}{\sqrt{\widehat{N}_t} + \epsilon}. \tag{150}$$

For $m > n$ the right singular basis V_t replaces $\bar{U}_t$ ($M'_t = M_t V_t$, N_t tracks $(G_t V_t)^2$, and the update is $(\widehat{M}_t/\sqrt{\widehat{N}_t})\,V_t^\top$), exactly as in the released reference implementation. The SVD is the only super-linear step and runs once every T steps ($T = 2000$ in the paper's pretraining runs, $T = 200$ in its LoRA runs); the wall-clock overhead is negligible for $T \ge 100$.

Algorithm 56 Conda (Column-Normalized Adam)

Require: Parameters θ, learning rate η, β_1, β_2, refresh period T, scale α, ϵ, weight decay λ
Ensure: Updated parameters θ_T

for each parameter P **do**
  if P is a 2-D matrix $W \in \mathbb{R}^{m\times n}$ **then**
    $G \leftarrow \nabla f(W)$ ▷ Gradient
    $M \leftarrow \beta_1 M + (1-\beta_1) G$ ▷ First moment
    if $t \bmod T = 0$ or P has no basis **then**
      $U, \Sigma, V^\top \leftarrow \mathrm{SVD}(M)$; $\bar{U} \leftarrow U$ if $m \le n$ else V ▷ Lazy subspace
    end if
    $G_{\mathrm{proj}} \leftarrow \bar{U}^\top G$; $M_{\mathrm{proj}} \leftarrow \bar{U}^\top M$ ▷ Project
    $N \leftarrow \beta_2 N + (1-\beta_2)\, G_{\mathrm{proj}}^2$ ▷ Projected second moment
    $M_{\mathrm{proj}} \leftarrow M_{\mathrm{proj}}/(1-\beta_1^t)$; $N \leftarrow N/(1-\beta_2^t)$ ▷ Bias correction
    $W \leftarrow (1-\eta\lambda)W - \eta\,\alpha\,\bar{U}\, M_{\mathrm{proj}}/(\sqrt{N}+\epsilon)$ ▷ Update
  else
    $P \leftarrow \mathrm{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Non-matrix fallback
  end if
end forreturn θ_T

On LLaMA 60M–1B pretraining Conda reaches validation perplexities of 28.32, 21.38, 16.44 and 13.59, against 34.06/25.08/18.80/15.56 for AdamW and 29.89/22.15/16.51/14.17 for Muon,

i.e. 2–2.5× AdamW's convergence speed in steps and wall-clock time; it also leads on GPT-2 (125M/355M: 17.40/13.92) and on LoRA commonsense fine-tuning. Unlike AdamW and Muon it remains stable at learning rates up to 2×10^{-2}, and the ablations show robustness to $T \in [2, 20000]$, $\beta_1 \in \{0.9, 0.95\}$ and $\beta_2 \in \{0.95, \ldots, 0.999\}$. Optimizer memory stays within 0.03–1.3 GB of AdamW from 60M to 1B. In this implementation the `update_proj_gap` suffix is T (default 2000), `scale` is α (default 1.0; the paper's pretraining uses 0.25), `correct_bias` toggles the $1-\beta^t$ corrections, and the shared per-group `betas`/`eps` are (β_1, β_2) (the paper uses 0.9/0.99) and ϵ; the shared `wd` is decoupled. Every 2-D parameter takes the Conda path regardless of group — there is no embedding/pos/head exclusion as in Muon — while every non-2-D parameter is trained with decoupled AdamW.

Cons.

- The SVD is a per-matrix cost spike every T steps; at $T = 2$ it raises LLaMA-60M wall-clock training time from 2h49min to 4h37min.
- The second moment must live in the same subspace as the first: the unprojected Adam estimator is unstable at scale and fails to converge on LLaMA-1B.
- Optimizer state is Adam-sized (full $m \times n$ first moment) plus the projection basis; the memory advantage over AdamW is small, not the $O(nr)$ of GaLore.
- Evaluated only up to 1B parameters and without Mixture-of-Experts; the paper flags both as future work.

Pros.

- Combines Adam's coordinate-wise adaptivity with improved spectral conditioning in a single update rule, consistently beating AdamW, Muon, SOAP and Adafactor.
- The column-wise normalizer is milder than Muon's uniform $1/\sigma_i$ scaling and preserves the relative order of singular values, avoiding overshoot in small-singular directions.
- A single hyperparameter set transfers across 60M–1B, and training is stable at learning rates 10–20× larger than AdamW's.
- Memory and per-step cost stay close to AdamW for $T \gg 1$; the lazy SVD makes the subspace refresh nearly free in wall-clock terms.

TUON (Tensor-Aware Optimization via T-SVD and Polarization). TUON [155] lifts Muon's layer-wise orthogonalization to **cross-layer tensors** using a transform-based tensor algebra derived from the Tensor-SVD. Layer-wise polar maps discard cross-layer correlations, while unfolding-based tensor optimizers (TEON) suffer dimension imbalance and cross-layer misalignment and degrade beyond two stacked layers or QKV-only modules. TUON instead *mixes then separates*: for one compatible group of K same-shape matrices with stacked momentum

$\mathcal{M}_t \in \mathbb{R}^{m\times n\times K}$:

$$\mathcal{M}_t = \beta\mathcal{M}_{t-1} + (1-\beta)\,\mathcal{G}_t \qquad \text{(momentum tensor, Alg. 1 line 2)} \tag{151}$$

$$C_t = \mathsf{M}_t\mathsf{M}_t^\top \in \mathbb{R}^{K\times K}, \qquad C_t = U_t\Lambda_t U_t^\top, \qquad Q_t = U_t^\top \qquad \text{(layer Gram + eigendecomposition, lines 3–5)} \tag{152}$$

$$\widetilde{\mathcal{M}}_t = \mathcal{M}_t \times_3 Q_t, \qquad \widetilde{P}_{t,j} = \mathrm{polar}\big(\widetilde{\mathcal{M}}_{t,j}\big) \qquad \text{(transform + slice polarization, lines 6–9)} \tag{153}$$

$$\mathcal{P}_t = \widetilde{\mathcal{P}}_t \times_3 Q_t^H \qquad \text{(inverse transform, line 10)} \tag{154}$$

$$W_{t+1} = (1-\eta_t\lambda)\,W_t - \eta_t\gamma_t\mathcal{P}_t, \qquad \gamma_t = \frac{s}{\mathrm{RMS}(\mathcal{P}_t)+\epsilon} \qquad \text{(RMS-scaled update, line 11)} \tag{155}$$

where M_t is the mode-3 unfolding (k-th row $= \mathrm{vec}(\mathcal{M}_t^{(k)})^\top$), the mode-3 product mixes slices as $\widetilde{\mathcal{M}}_{t,j} = \sum_k (Q_t)_{jk}\mathcal{M}_t^{(k)}$, $\mathrm{RMS}(\mathcal{P}_t) = \sqrt{(mnK)^{-1}\sum_{i,j,k}|\mathcal{P}_t(i,j,k)|^2}$, and the target stack RMS is $s = 1/\sqrt{\max(m,n)}$ (the rank–energy identity: full-rank slices give RMS $= 1/\sqrt{\max(m,n)}$, so $\gamma_t \approx 1$ and per-matrix update magnitudes match Muon). The default transform **DecorQ** takes the eigenvectors of the layer Gram matrix ($Q_t = U_t^\top$) and is refreshed every T steps (default 200); it diagonalizes the transformed Gram, so the transformed slices are mutually Frobenius-orthogonal and each polar map acts in its own subspace — the coupled K-layer problem decouples into K independent single-channel problems. Fixed unitary transforms (DCT, Haar, DFT) are valid but impose a prior mismatched to layer statistics; in the paper's ablation DecorQ (2.7598) beats DFT (2.7631), Haar (2.7635), static DecorQ (2.7658) and DCT (2.8100) on Qwen3-4B. Slice polarization uses the same quintic Newton–Schulz iteration as Muon, $(a,b,c) = (3.4445, -4.7750, 2.0315)$ with $J = 5$ steps after a Frobenius normalization and a transpose-when-tall.

Algorithm 57 TUON (Tensor-Aware Optimization via T-SVD and Polarization)

Require: Compatible matrices $\{W^{(k)}\}_{k=1}^K$, learning rate η, momentum β, weight decay λ, NS iters J, refresh interval T, RMS ϵ
Ensure: Updated parameters

for each compatible group **do**
  $\mathcal{G} \leftarrow \mathrm{stack}_3(\nabla f(W^{(1)}), \ldots, \nabla f(W^{(K)}))$ ▷ Gradient tensor (line 1)
  $\mathcal{M} \leftarrow \beta\mathcal{M} + (1-\beta)\,\mathcal{G}$ ▷ Momentum tensor (line 2)
  $C \leftarrow \mathsf{M}\mathsf{M}^\top$; $C = U\Lambda U^\top$; $Q \leftarrow U^\top$ ▷ Gram + eigendecomposition, refreshed every T steps (lines 3–5)
  $\widetilde{\mathcal{M}} \leftarrow \mathcal{M} \times_3 Q$ ▷ Mix slices (line 6)
  for $j = 1, \ldots, K$ in parallel **do**
    $\widetilde{P}_{:,:,j} \leftarrow \mathrm{NS}_J(\widetilde{\mathcal{M}}_{:,:,j})$ ▷ Slice polarization (lines 7–9)
  end for
  $\mathcal{P} \leftarrow \widetilde{\mathcal{P}} \times_3 Q^H$ ▷ Inverse transform (line 10)
  $\gamma \leftarrow s/(\mathrm{RMS}(\mathcal{P}) + \epsilon)$ ▷ Shared stack scale (line 11)
  $W \leftarrow (1-\eta\lambda)W - \eta\gamma\mathcal{P}$ ▷ Decoupled update (line 11)
end for

This implementation maps the paper's "one compatible group" to each *(parameter group, matrix shape)* bucket: same-shape hidden 2-D matrices of a group are stacked along the layer mode in registration order (`stack: full`); `stack: pairs` restricts the stack to consecutive pairs ($K = 2$). The `transform` suffix selects Q (`decorq` default; `dct`/`haar`/`dft` fixed transforms), `momentum` is β, `refresh_interval` is T, `ns_steps`/`ns_eps` drive the Newton–Schulz iteration,

and `rms_eps` is the ϵ of γ_t. For $K = 1$ the transform cancels exactly and TUON reduces to Muon (paper Prop. 2); following the hybrid recipe of Muon, only hidden-layer 2-D matrices take the TUON path — embedding tables, learned positional encodings, output heads and every non-matrix parameter are trained with decoupled AdamW using the group-level learning rate, weight decay, β_1, β_2 and ϵ settings. Theory: TUON is steepest descent under the Q-transformed tensor spectral norm (dual norm pair $\|\cdot\|_{Q,Ts}$ / $\|\cdot\|_{Q,Tn}$), its update equals the tensor polar factor of a T-product T-SVD, no unitary layer transform distorts the Muon geometry by more than $\sqrt{K}$, and at equal operator radius the TUON oracle is never weaker than TEON's and up to $\sqrt{K}$ stronger, with an $O(T^{-1/2})$ convergence certificate. Empirically TUON beats Muon and TEON on all of GPT-124M/350M and Qwen-0.6B/4B (validation loss 2.9357/2.6925/2.8887/2.7598), is the only method that improves from K=2 to K=full stacking on all four models, transfers from QKV to QKVO+MLP, and costs 0.2–1.1% step-time over TEON and at most 12.5% over AdamW; per-iteration complexity is $\mathcal{O}(Kmn \min(m, n) + K^2mn)$, the same dominant term as Muon.

Cons.

- Cross-layer mixing assumes the stacked matrices are genuinely compatible; this implementation stacks by exact shape within a parameter group, which is the closest faithful realization available from the unnamed factory param lists.
- The Gram/eigendecomposition adds $\mathcal{O}(K^2mn)$ per step and a $K \times K$ cached basis per bucket; negligible for $K \ll \min(m, n)$, but a per-bucket cost that Muon does not pay.
- β is required by Algorithm 1 but its default value is not stated in the paper; this implementation inherits the repo Muon-family convention (0.95).
- The `dft` transform runs the Newton–Schulz iteration over complex slices (Hermitian Gram); the real part is taken after the inverse transform, justified by the conjugate symmetry of the DFT of a real stack.

Pros.

- Exploits cross-layer structure that layer-wise Muon discards, with a convergence guarantee and exact steepest-descent interpretation under the induced tensor norm.
- DecorQ is data-dependent and refreshed every T steps; stacking K=full *improves* with depth instead of degrading as unfolding methods do.
- Memory stays at Muon's level (one momentum buffer per matrix plus a $K \times K$ basis), and the RMS stack scaling keeps updates at Muon's magnitude so existing learning rates transfer.
- Applies uniformly beyond QKV (attention, MLP and output projections), where prior tensor-aware optimizers fail to transfer.

NAMO (Norm-Based Adaptive Moment Estimation with Orthogonalized Momentum). NAMO [179] keeps Muon's orthogonalized update direction and adds an Adam-type **norm-based noise adaptation** on top of it. Matrix orthogonalization is an unbounded operation that can amplify stochastic noise, so Muon's effective stepsize does not shrink when gradients become noisy. NAMO tracks a single scalar EMA of the squared Frobenius norm of the gradient and rescales the whole orthogonalized update by the bias-corrected norm ratio $\|\widehat{M}_t\|_F/(\sqrt{\widehat{v}_t} + \epsilon_t)$ — an aggregated signal-to-noise statistic analogous to Adam's coordinate-wise ratio. Because the scaling is a single scalar, the update direction stays exactly orthogonal, and the optimizer state

grows by only one scalar per matrix over Muon. For a hidden-layer weight matrix $W \in \mathbb{R}^{m\times n}$ with learning rate η, momenta $\mu_1, \mu_2 \in [0,1)$, stability constant ϵ and decoupled weight decay λ:

$$M_t = \mu_1 M_{t-1} + (1-\mu_1) G_t \qquad \text{(momentum)} \tag{156}$$

$$v_t = \mu_2 v_{t-1} + (1-\mu_2)\,\|G_t\|_F^2 \qquad \text{(scalar norm EMA)} \tag{157}$$

$$O_t = \mathrm{NS}_K(M_t) \qquad \text{(orthogonalization)} \tag{158}$$

$$\alpha_t = \frac{\sqrt{1-\mu_2^t}}{1-\mu_1^t}\cdot\frac{\|M_t\|_F}{\sqrt{v_t}+\epsilon} = \frac{\|\widehat{M}_t\|_F}{\sqrt{\widehat{v}_t}+\epsilon_t}, \qquad \epsilon_t := \frac{\epsilon}{\sqrt{1-\mu_2^t}} \qquad \text{(adaptive stepsize)} \tag{159}$$

$$W_t = W_{t-1} - \eta\,\alpha_t\left(O_t + \lambda W_{t-1}\right) \qquad \text{(update)} \tag{160}$$

with the same quintic Newton–Schulz coefficients $(a,b,c) = (3.4445, -4.7750, 2.0315)$ as Muon. Under nuclear-norm smoothness and bounded-variance noise, NAMO attains the optimal deterministic rate $\mathcal{O}(T^{-1/2})$, and its stochastic bound decomposes into an optimization term $\mathcal{O}(T^{-1/4})$ plus a variance term $\mathcal{O}(\sqrt{\sigma}\,b^{-1/4}T^{-1/8})$, recovering the optimal $\mathcal{O}(T^{-1/4})$ once the batch size satisfies $b = \Omega(\sigma^2 T^{1/2})$.

Algorithm 58 NAMO (Norm-Based Adaptive Moment Estimation with Orthogonalized Momentum)

Require: Parameters θ, learning rate η, momenta μ_1, μ_2, NS iters K, stability constant ϵ, weight decay λ
Ensure: Updated parameters θ_T

for each parameter P **do**
  if P is a hidden-layer 2D matrix $W \in \mathbb{R}^{m\times n}$ **then**
    $G \leftarrow \nabla f(W)$ ▷ Gradient
    $M \leftarrow \mu_1 M + (1-\mu_1) G$ ▷ Momentum
    $v \leftarrow \mu_2 v + (1-\mu_2)\,\|G\|_F^2$ ▷ Scalar second moment
    $O \leftarrow \mathrm{NS}_K(M)$ ▷ Newton–Schulz
    $\alpha \leftarrow \frac{\sqrt{1-\mu_2^t}}{1-\mu_1^t}\cdot\frac{\|M\|_F}{\sqrt{v}+\epsilon}$ ▷ Adaptive stepsize
    $W \leftarrow W - \eta\,\alpha\,(O + \lambda W)$ ▷ Update
  else
    $P \leftarrow \mathrm{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Non-matrix fallback
  end if
end forreturn θ_T

On GPT-2 pretraining over OpenWebText (124M/355M, 2000-step linear warmup then constant LR), NAMO reaches validation losses of 3.0351/2.9516 versus 3.0643/2.9914 for AdamW and 3.0435/2.9684 for Muon, with optimal learning rates $\eta = 0.012/0.007$ (roughly $9\times$ AdamW's) at $\mu_1 = 0.95$, $\mu_2 = 0.99$, $\lambda = 0.01$, and it remains stable over a wider learning-rate grid than both baselines. In this implementation the global `namo-betas` suffix is (μ_1, μ_2) (default $[0.95, 0.99]$), `ns_steps`/`ns_eps` configure Newton–Schulz, the shared per-group `eps` is α_t's ϵ, the shared `wd` is decoupled and scaled by α_t as in the paper's update, and the shared `betas`/`eps` configure the AdamW fallback for the embeddings/pos/head groups and every non-2-D parameter. The reference release additionally exposes Nesterov momentum, an effective-stepsize clamp and a $0.2\sqrt{\max(m,n)}$ shape scaling; Algorithm 1 of the paper defines none of these, so this module follows the paper.

Cons.

- A single scalar stepsize carries no per-neuron information; the paper itself addresses this with the diagonal extension NAMO-D.

- The update is not RMS-aligned, so Adam learning-rate schedules do not transfer unchanged: the paper's optima are roughly 9× AdamW's.
- The theory assumes exact orthogonalization while practice uses K=5 Newton–Schulz iterations, and the empirical evidence is limited to GPT-2-scale pretraining.

Pros.

- First principled integration of orthogonalized momentum with Adam-type noise adaptation, with optimal deterministic and noise-adaptive stochastic convergence rates.
- The scalar scaling preserves orthogonality exactly, and the effective stepsize shrinks automatically under noise or near stationarity.
- Zero additional memory over Muon and negligible per-step overhead; outperforms AdamW and Muon on GPT-2 (124M/355M) over a wider stable learning-rate range.

NAMO-D (Diagonal Extension of NAMO). NAMO-D [179] replaces NAMO's single adaptive scalar with a diagonal matrix that right-multiplies the orthogonalized momentum, so every neuron receives its own adaptive stepsize. Column-wise scaling aligns with the near block-diagonal Hessian of neural networks and yields finer-grained noise adaptation, but it no longer preserves strict orthogonality of the update direction. With the column-norm operator $[N^c(M)]_j := \|M_{:,j}\|$ for $j = 1, \dots, n$:

$$M_t = \mu_1 M_{t-1} + (1-\mu_1) G_t \qquad \text{(momentum)} \tag{161}$$

$$[v_t]_j = \mu_2 [v_{t-1}]_j + (1-\mu_2) \, \|[G_t]_{:,j}\|^2 \qquad \text{(column norm EMA)} \tag{162}$$

$$d_t = \frac{\sqrt{1-\mu_2^t}}{1-\mu_1^t} \, N^c(M_t) \oslash \left(\sqrt{v_t} + \epsilon\right) \qquad \text{(column stepsizes)} \tag{163}$$

$$\bar{d}_t = \|d_t\|_1 / n, \qquad \tilde{d}_t = \min\left\{\max\{d_t, c\,\bar{d}_t \mathbf{1}\}, (\bar{d}_t/c)\,\mathbf{1}\right\} \qquad \text{(clamp)} \tag{164}$$

$$O_t = \mathrm{NS}_K(M_t), \qquad W_t = W_{t-1} - \eta \left(O_t + \lambda W_{t-1}\right) \mathrm{diag}(\tilde{d}_t) \qquad \text{(update)} \tag{165}$$

By Lemma 7 of the paper the entries satisfy $\|d_t\|_\infty \le \sqrt{(1-\mu_1)/(1-\mu_2)}$, and the clamp bounds the diagonal's condition number by $1/c^2$. The additional state over NAMO is one vector of length n per matrix; NAMO-D keeps the same optimal deterministic $\mathcal{O}(T^{-1/2})$ and noise-adaptive stochastic rates as NAMO (Theorems 5–6). The extra clamping constant c balances the two competing goals: small c clamps harder toward the average and keeps the direction close to orthogonal, while larger c allows more neuron-wise variation.

Algorithm 59 NAMO-D (Diagonal Extension of NAMO)

Require: Parameters θ, learning rate η, momenta μ_1, μ_2, NS iters K, stability constant ϵ, clamp c, weight decay λ

Ensure: Updated parameters θ_T

for each parameter P **do**
  if P is a hidden-layer 2D matrix $W \in \mathbb{R}^{m\times n}$ **then**
    $G \leftarrow \nabla f(W)$ ▷ Gradient
    $M \leftarrow \mu_1 M + (1-\mu_1)G$ ▷ Momentum
    $v \leftarrow \mu_2 v + (1-\mu_2)\, N^c(G) \odot N^c(G)$ ▷ Column-wise second moment
    $d \leftarrow \frac{\sqrt{1-\mu_2^t}}{1-\mu_1^t}\, N^c(M) \oslash (\sqrt{v}+\epsilon)$ ▷ Column stepsizes
    $\bar{d} \leftarrow \|d\|_1/n; \quad \tilde{d} \leftarrow \min(\max(d, c\bar{d}\mathbf{1}), (\bar{d}/c)\mathbf{1})$ ▷ Clamp
    $O \leftarrow \mathrm{NS}_K(M)$ ▷ Newton–Schulz
    $W \leftarrow W - \eta\,(O + \lambda W)\,\mathrm{diag}(\tilde{d})$ ▷ Update
  else
    $P \leftarrow \mathrm{AdamW}(P, \nabla f(P); \eta, \beta_1, \beta_2, \epsilon, \lambda)$ ▷ Non-matrix fallback
  end if
end forreturn θ_T

The paper fixes $c = 0.1$ for GPT-2 (124M) and sweeps $c \in \{0.12, 0.40, 0.75, 0.90\}$ for GPT-2 (355M), selecting $c = 0.9$; the reference implementation defaults to 0.75. NAMO-D reaches validation losses of 3.0246 and 2.9507 on GPT-2 (124M/355M), improving on NAMO's 3.0351/2.9516, at optimal learning rates $\eta = 0.009$ with the same $\mu_1 = 0.95$, $\mu_2 = 0.99$ and $\lambda = 0.01$. In this implementation the global `namo_d-betas` suffix is (μ_1, μ_2) (default $[0.95, 0.99]$), `namo_d-clamp` is c (default 0.1), `ns_steps`/`ns_eps` configure Newton–Schulz, the shared per-group `eps` is the denominator ϵ, the shared `wd` is decoupled and right-multiplied by the diagonal, and the shared `betas`/`eps` configure the AdamW fallback. As with NAMO, the official release's Nesterov momentum, effective-stepsize clamp and $0.2\sqrt{\max(m,n)}$ shape scaling are not part of Algorithm 2 and are not implemented.

Cons.

- The update no longer preserves strict orthogonality: the diagonal scaling trades the structural guarantee for finer-grained noise adaptation.
- The clamp constant c is a new hyperparameter whose optimum varies with model scale (0.1 at 124M, 0.9 at 355M), requiring a sweep.
- One extra $\mathbb{R}^n$ state per matrix over NAMO, and the theory still assumes exact orthogonalization.

Pros.

- Neuron-wise adaptation matches the near block-diagonal Hessian of neural networks and improves on NAMO on both GPT-2 sizes.
- The clamp bounds the diagonal's condition number by $1/c^2$, keeping the scaled direction well-conditioned under heavy-tailed or noisy gradients.
- Optimal deterministic and noise-adaptive stochastic convergence rates carry over from NAMO, with the same Newton–Schulz cost and hybrid AdamW routing.

9.7 Evolutionary and Population-Based Hybrids

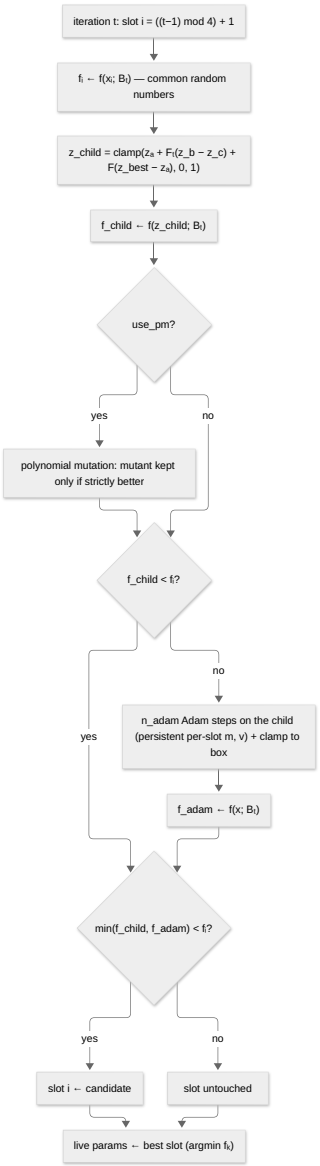


Figure 104: Step flow of EvE: a steady-state population of four challenges one round-robin slot per iteration with a DE child; a persistent-state Adam burst rescues the child only when it fails to improve, and greedy common-random-number selection keeps the best-so-far monotone.

EvE (Evolutionary Explorer). EvE [132] is a **population-based alternate to Adam** targeted at the search-heavy, budget-constrained regime one level above final-stage training: ranking or pruning candidate configurations inside hyperparameter and architecture search. Instead of refining a single point, EvE keeps a steady-state population of four candidates and challenges one round-robin slot per iteration with a *rand-to-pbest/1* differential-evolution child; only when the child fails to improve the slot's incumbent does EvE run a short burst of stock Adam on the same minibatch, keeping the refined point only if it beats the child. Selection is greedy, so the best-so-far value is monotone non-increasing on deterministic objectives, and because gradients are used only as a targeted rescue, per-iteration cost stays within a constant factor of a single Adam step regardless of dimension.

All internal search happens on the unit cube $z \in [0,1]^n$ via the affine map $\mathbf{x} = \mathbf{x}_l + \mathbf{z} \odot (\mathbf{x}_u - \mathbf{x}_l)$,

so a fixed-size step never overshoots narrow-range coordinates or under-moves wide ones. The population is initialized by proximity — one anchor with three Gaussian jitters, $\sigma = 0.1$ — anchored at the model's own initial weights (the paper's LoRA precedent), with the per-tensor search box centered on the anchor at the Kaiming half-width $\sqrt{6/\text{fan_in}}$. Each iteration draws three distinct donors $a, b, c \neq i$ and builds the child with a jittered difference weight (the jitter applies only to the difference term; the pbest pull uses the unjittered F):

$$z_{\text{child}} = \text{clamp}\left(z_a + F_t(z_b - z_c) + F(z_{\text{pbest}} - z_a),\ 0,\ 1\right), \quad F_t = F\big(1 + \gamma(u - \tfrac{1}{2})\big), \quad u \sim U(0,1), \tag{166}$$

with $F = 0.5$ and $\gamma = 10^{-4}$. An optional polynomial mutation (PM; off by default for neural-network training, since a single-coordinate nudge essentially never wins its accept/reject gate at network scale) perturbs exactly one gene at rate $p = 1/n$ with distribution index $\eta = 20$, the perturbation bounded and self-scaling to the distance to each boundary:

$$z_d \leftarrow z_d + \delta_q(z_u - z_l)_d, \quad \delta_q = \begin{cases} \left[2r + (1-2r)(1-\delta_1)^{\frac{1}{\eta+1}}\right]^{\frac{1}{\eta+1}} - 1, & r \leq 0.5, \\ 1 - \left[2(1-r) + 2(r-0.5)(1-\delta_2)^{\frac{1}{\eta+1}}\right]^{\frac{1}{\eta+1}}, & r > 0.5, \end{cases} \tag{167}$$

with $\delta_1 = (z_d - z_l)/(z_u - z_l)$, $\delta_2 = (z_u - z_d)/(z_u - z_l)$ and $r \sim U(0,1)$. When the (possibly mutated) child fails to beat the slot's value, EvE runs n_{adam} stock-Adam steps on it *in real coordinates*, with classic L2 weight decay folded into the gradient before the moment update (PyTorch's stock Adam, not AdamW's decoupled rule) and every iterate clamped back into the box:

$$g_t = \nabla_{\mathbf{x}} f(\mathbf{x}_{t-1}; B_t) + \lambda \mathbf{x}_{t-1}, \quad m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \tag{168}$$

$$\widehat{m}_t = \frac{m_t}{1-\beta_1^t}, \qquad \widehat{v}_t = \frac{v_t}{1-\beta_2^t}, \qquad \mathbf{x}_t = \text{clamp}\left(\mathbf{x}_{t-1} - \eta_{lr}\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon},\ \mathbf{x}_l,\ \mathbf{x}_u\right), \tag{169}$$

with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$. Each population slot keeps its own persistent (m, v, t) state across the whole run, so fallback bursts form one continuous Adam trajectory rather than repeated cold starts. Every evaluation of an iteration — the slot refresh, the child and the whole fallback burst — is scored on one freshly drawn minibatch B_t (common random numbers), and a candidate replaces the incumbent only if it is strictly better on that same batch. In this implementation the shared per-group `lr_*`, `wd_*`, `betas_*` and `eps_*` configure the Adam fallback per parameter group (`wd_*` is the L2 term λ inside g_t), the live model parameters always hold the best-so-far slot between steps, and the step is closure-based — the trainer re-evaluates the current batch on each candidate point inside `optimizer.step()`, which is why EvE requires `gradient_accumulation_steps: 1` and is not available for SBERT.

Algorithm 60 EvE (Evolutionary Explorer)

Require: Objective f with bounds $\mathbf{x}_l, \mathbf{x}_u$, iterations T, DE weight F, jitter γ, init jitter σ, PM index η, fallback steps n_{adam}, fallback LR η_{lr}, L2 decay λ, seed s, USE_PM

Ensure: Best point $\arg\min_k z_k$ and value $\min_k f_k$

$z_1 \sim U(0,1)^n$ (anchor: the initial weights); $z_k \leftarrow \text{clamp}(z_1 + \sigma\epsilon_k, 0, 1)$, $\epsilon_k \sim \mathcal{N}(0, I_n)$, $k = 2, 3, 4$ ▷ Proximity init

$f_k \leftarrow f(\mathbf{x}_l + z_k \odot (\mathbf{x}_u - \mathbf{x}_l))$ for $k = 1, \ldots, 4$; $\text{state}_k \leftarrow (\mathbf{0}, \mathbf{0}, 0)$ ▷ Persistent Adam moments

for $t = 1, \ldots, T$ **do**

  $i \leftarrow ((t-1) \bmod 4) + 1$ ▷ Round-robin target slot

  draw minibatch B_t; $f_i \leftarrow f(\mathbf{x}_i; B_t)$ ▷ Common random numbers

  draw distinct $a, b, c \in \{1, \ldots, 4\} \setminus \{i\}$; $F_t \leftarrow F(1 + \gamma(u - 0.5))$, $u \sim U(0,1)$

  $z_{\text{child}} \leftarrow \text{clamp}(z_a + F_t(z_b - z_c) + F(z_{\text{pbest}} - z_a), 0, 1)$

  $f_{\text{child}} \leftarrow f(z_{\text{child}}; B_t)$

  if USE_PM **then** ▷ PM's own accept/reject gate

    $z_{\text{mut}} \leftarrow \text{PM}(z_{\text{child}}, \eta)$; $f_{\text{mut}} \leftarrow f(z_{\text{mut}}; B_t)$

    if $f_{\text{mut}} < f_{\text{child}}$ **then** $z_{\text{child}}, f_{\text{child}} \leftarrow z_{\text{mut}}, f_{\text{mut}}$

    end if

  end if

  if $f_{\text{child}} < f_i$ **then**

    $z_{\text{cand}}, f_{\text{cand}} \leftarrow z_{\text{child}}, f_{\text{child}}$

  else

    $\mathbf{x} \leftarrow \mathbf{x}_l + z_{\text{child}} \odot (\mathbf{x}_u - \mathbf{x}_l)$

    for $n = 1, \ldots, n_{\text{adam}}$ **do** ▷ Targeted Adam fallback

      $\mathbf{x} \leftarrow \text{AdamStep}(\mathbf{x}, \nabla_{\mathbf{x}} f(\mathbf{x}; B_t), \eta_{lr}, \lambda, \text{state}_i)$; $\mathbf{x} \leftarrow \text{clamp}(\mathbf{x}, \mathbf{x}_l, \mathbf{x}_u)$

    end for

    $z_{\text{adam}} \leftarrow (\mathbf{x} - \mathbf{x}_l)/(\mathbf{x}_u - \mathbf{x}_l)$; $f_{\text{adam}} \leftarrow f(z_{\text{adam}}; B_t)$

    $(z_{\text{cand}}, f_{\text{cand}}) \leftarrow \arg\min_{(z,f) \in \{(z_{\text{child}}, f_{\text{child}}), (z_{\text{adam}}, f_{\text{adam}})\}} f$

  end if

  if $f_{\text{cand}} < f_i$ **then** $z_i \leftarrow z_{\text{cand}}$; $f_i \leftarrow f_{\text{cand}}$ ▷ Greedy selection

  end if

end for

return $\arg\min_k z_k$, $\min_k f_k$

Cons.

- A step behind Adam at matched evaluation budgets: roughly one accuracy point on MNIST and 9–11% higher relative test loss on the LoRA fine-tuning tasks; the paper does not claim final-stage superiority.
- 13× the live parameter count in total storage (four population copies plus eight persistent moment copies) against Adam's 4×; the measured peak GPU multiplier is lower (2.95×) but fixed and dimension-independent.
- The learning rate does not transfer from Adam's and needs its own search (3–100× higher depending on the task); the four-member population shows the widest seed-to-seed variance of the nine methods compared.
- Closure-based step: the trainer re-evaluates the batch inside `optimizer.step()`, so gradient accumulation is unsupported and the SBERT loop (sentence-transformers `fit`) cannot host it; polynomial mutation is pure cost at network scale and is off by default.

Pros.

- Monotone best-so-far by construction on deterministic objectives (greedy accept-only-if-better selection), with per-iteration cost within a constant factor of a single Adam step regardless of dimension.
- Wins or ties Adam on 76% of 70 (problem, dimension) benchmark cells across seven classical scalable problems up to 10^6 variables, most decisively on the multimodal ones.
- Finishes matched evaluation budgets 1.7–3.9× faster in wall-clock on real training (an MLP on MNIST and LoRA fine-tuning of a 1.5B-parameter model), trading a modest amount of final quality for speed.
- Inside successive halving, completes hyperparameter and architecture searches 3.1–3.5× faster while ranking candidate configurations about as consistently with Adam as Adam does with itself across seeds (Kendall's τ of 0.66–0.69).

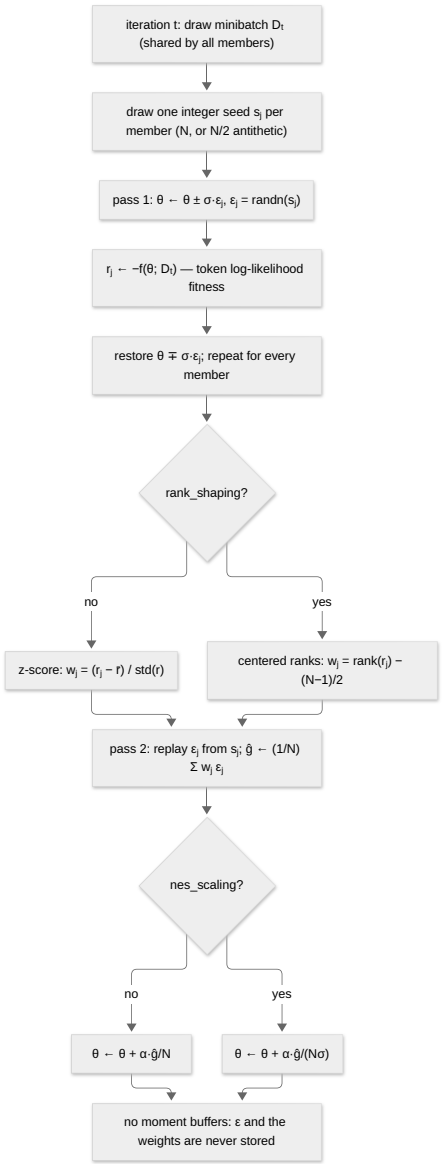


Figure 105: Step flow of EA4LLM: each iteration evaluates N Gaussian perturbations of the parameters on one minibatch shared by every member, scores them by the negated token-level log-likelihood, shapes the rewards into z-score or centered-rank weights, and replays the same seeded noise to accumulate the ES direction — no gradients and no per-parameter optimizer state.

EA4LLM (Evolution Strategies for Large Language Models). EA4LLM [103] is a **gradient-free alternative to Adam** that removes backpropagation and optimizer state entirely, and it is the first method the authors validate for full-parameter pretraining of LLMs from scratch, across Qwen3-style models from 0.5B to 32B parameters. Instead of differentiating a loss, EA4LLM treats the next-token likelihood as a black-box fitness and estimates a descent direction from a population of Gaussian perturbations. The fitness is the mean token-level log-probability of the true next token, masked for padding, which is exactly the standard causal-LM objective up to sign:

$$p_t = \mathrm{softmax}(z_t), \quad R(x;\theta) = \frac{1}{\sum_t m_t} \sum_{t=1}^{T} m_t \log p_\theta(x_{t+1} \mid x_{\le t}), \quad F(\theta) = \frac{1}{|D|} \sum_{x \in D} R(x;\theta), \quad (170)$$

with logits z_t, padding mask $m_t \in \{0,1\}$ and corpus D. Each iteration samples N independent noises $\epsilon_j \sim \mathcal{N}(0, I)$, evaluates the perturbed fitness $r_j = F(\theta + \sigma\epsilon_j)$ on one uniformly drawn subset $D_t \subset D$ of size M (the subset is shared by every member of the iteration, so the stochastic estimator stays unbiased, $\mathbb{E}[F_M(\theta)] = F(\theta)$), and shapes the rewards into weights by z-score normalization or centered ranks:

$$w_j = \frac{r_j - \bar{r}}{\mathrm{std}(r)} \qquad \text{or} \qquad w_j = \mathrm{rank}(r_j) - \frac{N-1}{2}, \tag{171}$$

where $\mathrm{rank}(\cdot)$ assigns $0, \ldots, N-1$ after sorting. The direction is the noise-weighted mean, optionally scaled NES-style by $1/\sigma$, and the update ascends the fitness:

$$\hat{g} = \frac{1}{N}\sum_{j=1}^{N} w_j \epsilon_j \qquad \text{or} \qquad \hat{g} = \frac{1}{N\sigma}\sum_{j=1}^{N} w_j \epsilon_j, \qquad \theta \leftarrow \theta + \alpha\hat{g}. \tag{172}$$

Antithetic sampling evaluates the $\pm\sigma\epsilon$ pair of $N/2$ seeds instead of one sign per seed. The paper's runner uses $N = 30$, $\sigma = 10^{-3}$, $\alpha = 5 \times 10^{-4}$ and $M = 5000$ evaluation samples per iteration ($T = 1000$ iterations, micro-batch 16, maximum sequence length 256); antithetic sampling, rank shaping, the NES $1/\sigma$ scaling, grouped/layered multipliers and step-size schedules are all *off* in those runs, and Appendix A.1 states that the deployed algorithm removes the grouped operators and multipliers outright. In this implementation the step is closure-based: the trainer's current minibatch supplies D_t and the loss is negated to form the fitness, the shared per-group `lr_*` is the step size α, and `wd_*` applies a decoupled multiplicative decay before the ES move (a repo-consistency extension — the paper defines no weight decay). Noise is regenerated from one integer seed per member in a second pass, so the full noise set is never stored: optimizer state is O(1) and no moment buffers exist. Like EvE, EA4LLM requires `gradient_accumulation_steps: 1` and is not available for SBERT (the closure cannot coexist with the SBERT fit loop), and AMP is not supported on the closure path.

Algorithm 61 EA4LLM (Zeroth-Order Evolution Strategies)

Require: Parameters θ, fitness F (negated loss), iterations T, population N, noise scale σ, step size α, subset size M, normalization (z-score or rank), seed s, USE_ANTITHETIC, USE_RANK, USE_NES

Ensure: Optimized parameters θ_T

seed the persistent generator with s
for $t = 1, \ldots, T$ **do**
  draw minibatch $D_t \subset D$, $|D_t| = M$ ▷ Shared by all members
  draw seeds $\{s_j\}_{j=1}^{N/2}$ (antithetic) or $\{s_j\}_{j=1}^{N}$
  rewards $\leftarrow [\,]$ ▷ Pass 1: perturb, evaluate, restore
  for each seed s_j and sign $\pm$ (or a single sign) **do**
    $\epsilon \leftarrow \mathrm{randn}(s_j)$; $\theta \leftarrow \theta \pm \sigma\epsilon$; $r \leftarrow -f(\theta; D_t)$; $\theta \leftarrow \theta \mp \sigma\epsilon$
    append r to rewards
  end for
  $w \leftarrow$ z-score(rewards) or centered ranks(rewards)
  $g \leftarrow 0$ ▷ Pass 2: replay the same noise
  for each seed s_j and sign $\pm$ (or a single sign) **do**
    $\epsilon \leftarrow \mathrm{randn}(s_j)$; $g \leftarrow g + w_j(\pm\epsilon)$
  end for
  $\theta \leftarrow \theta + \alpha g/(N\sigma)$ if USE_NES, else $\theta \leftarrow \theta + \alpha g/N$
end for
return θ_T

Cons.

- Requires N forward evaluations per step (30 in the paper), so sample efficiency trails first-order methods and the variance of the estimate is higher; the paper reports sensitivity to N, σ, α and the normalization choice.
- No gradient is available, so the gradient-clipping and AMP paths do not apply: the trainer's closure harness rejects `use_amp`, requires `gradient_accumulation_steps: 1`, and the SBERT loop cannot host it. Each evaluation also pays a backward pass through the shared closure even though ES discards it.
- At matched evaluation budgets the paper does not claim superior final quality to Adam; its case is memory ($>4\times$ reduction at $\geq$4B parameters) and gradient-free feasibility, with antithetic sampling, rank shaping and NES scaling disabled in its own large-scale runs.
- Because the z-score makes the update magnitude independent of σ and of the gradient scale, α must be tuned per task rather than transferred from Adam.

Pros.

- Zero optimizer state: no moments and no population copies, so the paper measures 60.01 vs 137.61 GB at 8B and 137.61 vs 431.76 GB at 32B against Adam.
- Provides the first stable full-parameter ES pretraining from scratch across 0.5B–32B, with steadily decreasing loss curves (Figure 1) and no gradient computation.
- Architecture-agnostic: the fitness only needs a forward pass and a probability, so non-differentiable operators become trainable.
- The subsampled fitness is an unbiased estimator of the full-corpus fitness, and the two-pass seeded replay makes the noise reproducible while keeping optimizer state at O(1).

Group	**Methods**	**Primary Goal**	**Interpretation from the Survey**
Classical baseline	SGD, SGD with momentum, RMSProp, AdamW, RAdam	stability and reference baselines	These define the comparison floor for newer optimizer claims.
Momentum redesign	Adan, AdEMAMix, MARS, Cautious AdamW	faster or safer first-order adaptation	Best when convergence speed or noisy-gradient stability is the main concern.
Large-batch and schedule simplification	LAMB, Schedule-Free AdamW	operational robustness at scale	Reduce brittleness from batch-size growth or schedule engineering.
Memory-efficient	Adafactor, GaLore, APOLLO, APOLLO-Mini, Q-APOLLO, Lion	optimizer-state reduction	Most useful when VRAM is dominated by optimizer state rather than activations; APOLLO-family methods replace dense AdamW moments with projected or quantized structured scaling.
Curvature-aware	Shampoo, SOAP, Sophia	better conditioning	Prefer when richer geometry is worth implementation and compute overhead.

Group	Methods	Primary Goal	Interpretation from the Survey
Geometry-oriented	Muon, Turbo-Muon, AF-Muon, AdaMuon, Dion3	orthogonalized update structure	Specialized options for matrix geometry and representation shaping; AdaMuon adds an element-wise second moment on the orthogonalized direction with RMS-aligned updates; AF-Muon assigns the tied vocabulary table a capped-row table geometry and one-dimensional parameters an RMS update, with no AdamW auxiliary state; Dion3 keeps the spectral update but orthogonalizes only a fraction of the momentum rows per step with Gram Newton–Schulz and error feedback.
Evolutionary hybrid	EvE, EA4LLM	fast, monotone candidate ranking for search loops; gradient-free full-parameter updates	A steady-state differential-evolution population with a targeted Adam rescue trades final quality for wall-clock speed; best used as the inner optimizer of hyperparameter or architecture search. EA4LLM removes gradients entirely — zeroth-order ES with token-level log-likelihood fitness and no optimizer state, the memory-first option for pretraining at scale.

10 Normalization Variants and Attention-Logit Control

This annex documents the normalization variants implemented by the toolkit and the query–key normalization mechanisms that control attention-logit growth. The main sections defer to this annex for the formulations and pseudocode; bibliography keys resolve against `docs/bibliography/`.

10.1 Normalization Variants: LayerNorm, RMSNorm, pRMSNorm, Dynamic Tanh, Dynamic Erf, and FlashNorm

Normalization determines how activation scale is controlled across depth. In this repository, normalization is not only a modeling choice but also a schema compatibility question, because only certain values are currently accepted by `norm_type`. The schema contract is:

$$\texttt{norm_type} \in \{\texttt{layer_norm}, \texttt{dynamic_tanh}, \texttt{derf}, \texttt{rms_norm}, \texttt{prms_norm}, \texttt{flash_norm}\}$$

The formulations most relevant to this codebase are:

10.1.1 LayerNorm (Baseline)

LayerNorm [7] is the default normalizer; it centers and rescales each token by its per-token mean and variance:

$$\mu = \frac{1}{d}\sum_{i=1}^{d} x_i, \quad \sigma^2 = \frac{1}{d}\sum_{i=1}^{d}(x_i-\mu)^2, \quad y_i = \gamma_i \frac{x_i-\mu}{\sqrt{\sigma^2+\epsilon}} + \beta_i.$$

10.1.2 RMSNorm

RMSNorm removes mean-centering and only rescales by root mean square magnitude [177]:

$$\mathrm{RMS}(x) = \sqrt{\frac{1}{d}\sum_{i=1}^{d} x_i^2 + \epsilon}, \qquad y_i = \gamma_i \frac{x_i}{\mathrm{RMS}(x)}.$$

Compared with LayerNorm, RMSNorm is computationally simpler (no subtraction of feature mean) and is often used when reducing normalization overhead is important (e.g., Llama, GPT-NeoX). It has a learnable per-dimension scale γ but no bias term.

10.1.3 Partial RMSNorm (pRMSNorm)

pRMSNorm [177] further reduces overhead by estimating the RMS from only the first $p\%$ of the hidden dimensions:

$$\mathrm{RMS}(x) = \sqrt{\frac{1}{k}\sum_{i=1}^{k} x_i^2 + \epsilon}, \qquad k = \lceil d \cdot p \rceil,$$

exploiting the assumption that neurons within a layer are approximately i.i.d. The same scale γ is then applied to all dimensions. The partial ratio p is controlled by the `prms_partial_ratio` config field (default 6.25%). The paper reports that competitive results are achievable with as little as 6.25% of the inputs, though gradient instability can arise with very small ratios.

10.1.4 Dynamic Tanh (DyT)

Dynamic Tanh proposes replacing explicit normalization with a bounded elementwise map [185]:

$$\mathrm{DyT}(x) = \tanh(\alpha x)$$

where α is learned. The core idea is that bounded nonlinear contraction can provide stable signal scaling without explicitly computing per-token normalization statistics.

10.1.5 Dynamic Erf (Derf)

Derf extends the same normalization-free direction by using an error-function based map [19]:

$$\mathrm{Derf}(x) = \mathrm{erf}(\alpha x + s)$$

with learnable scale/shift. Reported results in the cited work indicate stronger performance than DyT and common normalization baselines across multiple domains.

10.1.6 FlashNorm (Weightless RMSNorm)

FlashNorm [51] is not a new mathematical normalization; it is an *exact algebraic rewrite* of the ubiquitous "RMSNorm → Linear" pair that (i) eliminates the per-dimension learnable scale $\mathbf{g}$ by folding it into the subsequent linear layer's weights, and (ii) defers the scalar RMS reduction to the output of the matrix multiplication so that the matmul (matrix unit) and the RMS reduction (vector unit) execute in parallel. Three propositions formalize the rewrite.

Proposition 1 (Weight folding). Given activation $\mathbf{a} \in \mathbb{R}^n$, RMSNorm weight $\mathbf{g} \in \mathbb{R}^n$, and a subsequent linear $\mathbf{W} \in \mathbb{R}^{n \times k}$,

$$\mathbf{z} = \mathrm{RMSNorm}(\mathbf{a})\mathbf{W} = \frac{\mathbf{a}}{\mathrm{RMS}(\mathbf{a})}\mathbf{W}^*, \qquad W^*_{i,j} = g_i \cdot W_{i,j} \;\Leftrightarrow\; \mathbf{W}^* = \mathrm{diag}(\mathbf{g})\mathbf{W}.$$

The per-dimension scale $\mathbf{g}$ becomes redundant and is folded into $\mathbf{W}$ once at load time. The standalone `FlashNorm` module in this repository applies this directly: it has *no learnable parameters* (the scale is meant to be absorbed by the downstream projection).

Proposition 2 (Deferred normalization). For a bias-free linear $\mathbf{W}^*$, scalar–matrix commutativity yields

$$\frac{\mathbf{a}}{\mathrm{RMS}(\mathbf{a})}\mathbf{W}^* = (\mathbf{a}\mathbf{W}^*) \cdot \frac{1}{\mathrm{RMS}(\mathbf{a})}.$$

The matmul $\mathbf{a}\mathbf{W}^*$ and the RMS reduction are independent of each other; on hardware with distinct matrix and vector execution units they execute in parallel, and only a single vector–scalar multiply synchronizes them. The fused module `FlashNormLinear` implements this rewrite: on the bias-free path it computes the matmul on the *un-normalized* input and multiplies the result by the per-token reciprocal RMS. When the linear has a bias $\mathbf{c}$, the rewrite is no longer exact (Remark 1 of the paper): $(\mathbf{a}\mathbf{W}^* + \mathbf{c})/\mathrm{RMS}(\mathbf{a}) \neq \mathbf{a}\mathbf{W}^*/\mathrm{RMS}(\mathbf{a}) + \mathbf{c}$; the layer then applies the scalar to the input first and lets the linear add its bias as usual.

Proposition 3 (RMS cancellation). By RMS scale invariance, $\mathrm{RMS}(\alpha\mathbf{a}) = \alpha\,\mathrm{RMS}(\mathbf{a})$. Therefore an "RMSNorm → Linear → RMSNorm" pattern allows the *first* RMSNorm to be dropped entirely:

$$\mathrm{RMSNorm}\left(\frac{\mathbf{a}}{\mathrm{RMS}(\mathbf{a})}\mathbf{W}^*\right) = \mathrm{RMSNorm}(\mathbf{a}\mathbf{W}^*)\,.$$

This cancellation is relevant for QKV-normalization (Gemma 4) and MLA latent normalization (DeepSeek-V2, Mistral Small 4); it is not auto-applied in this repository but is documented for completeness.

Partial-RMS composition. FlashNorm composes naturally with the pRMSNorm trick (§10, pRMSNorm subsection): when `flashnorm_partial_ratio` $= p > 0$, the RMS is estimated from the first $k = \lceil d \cdot p \rceil$ dimensions of the input only, reducing the cost of the vector-unit RMS reduction further.

BitNet interaction. Folding a floating-point $\mathbf{g}$ into BitNet's ternary weight tensor $\{-1, 0, +1\}$ [161] would destroy the quantization, and BitLinear applies its own internal LayerNorm before activation quantization which needs the normalized input. The fused module `FlashNormBitLinear` therefore falls back to the sequential composition FlashNorm → BitLinear: the matmul and the RMS reduction still execute in parallel, but the post-matmul scalar multiply of Prop. 2 is sacrificed. Folding f.p. $\mathbf{g}$ into a 1.58-bit weight tensor is left to future kernel-level work.

PyTorch pseudocode.

Input: activation $\mathbf{a} \in \mathbb{R}^{T\times n}$, RMSNorm weight $\mathbf{g} \in \mathbb{R}^n$, linear $\mathbf{W} \in \mathbb{R}^{n\times k}$, bias $\mathbf{c} \in \mathbb{R}^k$ or `None`
`// Prop. 1: weight folding at load time`
$\mathbf{W}^* \leftarrow \mathbf{g} \odot \mathbf{W}$ ▷ column-wise; `g.unsqueeze(0) * W` in PyTorch
$\mathbf{g} \leftarrow \mathbf{1}$ ▷ the standalone norm is now weightless
`// Prop. 2: forward pass with deferred RMS (bias-free path)`
$\mathbf{b} \leftarrow \mathbf{a}\mathbf{W}^{*\top}$ ▷ matrix unit / tensor cores
rms_inv $\leftarrow \mathrm{rsqrt}(\mathrm{mean}(\mathbf{a}^2) + \varepsilon)$ ▷ vector unit, independent of $\mathbf{b}$
if $\mathbf{c} =$ `None` **then**
  return $\mathbf{b} \cdot$ rms_inv ▷ single vector–scalar multiply
else
  `// Remark 1: bias path is sequential; matmul and RMS still parallelize`
  return $(\mathbf{a} \cdot \mathrm{rms_inv})\mathbf{W}^{*\top} + \mathbf{c}$
end if

Reported latency (NVIDIA T4 GPU). The paper reports the following speedups on the norm-then-project operation, with an adaptive dispatch that falls back to the sequential path in the memory-bound (decode) regime:

Scale	Tokens	Sequential (ms)	FlashNorm (ms)	Speedup	Notes
SmolLM2-135M	4096	0.704	0.468	+33.6%	compute-bound (prefill)
SmolLM2-135M	8192	0.929	0.599	+35.5%	compute-bound (prefill)
Llama-7B	1024	1.882	1.654	+12.1%	compute-bound (prefill)
Llama-7B	4096	7.628	6.570	+13.9%	compute-bound (prefill)

Zero-loss weight folding validation. Weight folding (Prop. 1) is mathematically exact; the paper verified bit-identical outputs on three open-weight checkpoints:

Model	Precision	Result
SmolLM2-135M	fp16	max diff = 0.0, cosine sim = 1.0, identical text
Llama-3.2-1B	fp32	bit-identical greedy generation
Llama-3.1-8B	fp32	bit-identical greedy generation

Quality preservation on `lm-evaluation-harness` (wikitext word_ppl, MMLU 5-shot, HellaSwag 0-shot) stays within benchmark noise (worst $\Delta = +0.0017$ word_ppl).

10.1.7 Comparison

Method	Formula	Stats Needed	Notes
LayerNorm	$\gamma_i(x_i - \mu)/\sqrt{\sigma^2 + \epsilon} + \beta_i$	mean + variance	Default; full centering and rescaling [7].
RMSNorm	$\gamma_i x_i / \sqrt{\frac{1}{d}\sum_j x_j^2 + \epsilon}$	RMS only	Lower overhead; widely used baseline [177].
pRMSNorm	$\gamma_i x_i / \sqrt{\frac{1}{k}\sum_{j\le k} x_j^2 + \epsilon}$	Partial RMS	RMS from first p% of dims; `prms_partial_ratio` default 6.25% [177].
Dynamic Tanh	$\tanh(\alpha x)$	none	Normalization-free bounded transform; drop-in replacement [185].
Dynamic Erf	$\mathrm{erf}(\alpha x + s)$	none	Normalization-free alternative; improves over DyT [19].
FlashNorm	$x_i / \sqrt{\frac{1}{d}\sum_j x_j^2 + \epsilon}$ (weightless)	RMS only	Exact rewrite of RMSNorm→Linear: folds $\mathbf{g}$ into $\mathbf{W}$ (Prop. 1), defers scalar RMS to matmul output (Prop. 2); 12–35% latency reduction on T4 GPU [51].

10.2 Query–Key Normalization and Attention-Logit Control

Attention logits $\ell_{i,t,h} = \mathbf{q}_{i,h}^\top \mathbf{k}_{t,g(h)}/\sqrt{d_h}$ are the input to the softmax. Their unbounded growth is a canonical source of training instability: saturated attention, entropy collapse, loss spikes and, at scale, divergence [66, 81]. This repository exposes five mechanisms under `norm.qk_norm.method`; three act in the forward pass and two act on the optimizer state. They are mutually exclusive per configuration.

Forward-time QK normalization. QK normalization [66] normalizes queries and keys along the per-head feature dimension before the dot product. In RMSNorm form,

$$\hat{\mathbf{q}}_{i,h} = \frac{\mathbf{q}'_{i,h} \odot \boldsymbol{\gamma}_q}{S_q(\mathbf{q}'_{i,h})}, \qquad S_q(\mathbf{q}) = \sqrt{\frac{1}{d_h}\sum_{j=1}^{d_h} q_j^2 + \epsilon},$$

and identically for $\hat{\mathbf{k}}$ with $\boldsymbol{\gamma}_k$. The affine weights are shared across heads and broadcast over the head axis, so the parameter cost is $2d_h$ (`rms_norm`) or $2 \cdot 2d_h$ with bias (`layer_norm`); `l2_`

`norm` is weightless. Because the normalized norms satisfy $\|\hat{\mathbf{q}}\|_2 \approx \|\boldsymbol{\gamma}_q\|_2$ up to direction and ϵ, the residual logit growth is mediated by the learned affine product $\boldsymbol{\gamma}_q \odot \boldsymbol{\gamma}_k$ rather than by uncontrolled raw query/key norms.

QK normalization with Multi-Head Latent Attention. MLA caches a low-rank latent $\mathbf{c}_t$ and reconstructs keys by up-projection, so post-projection QK RMSNorm appears to require the full keys that MLA avoids caching. Han et al. [59] show the incompatibility is an implementation artifact: RMSNorm decomposes into a static affine weight and a dynamic scalar statistic,

$$\hat{\mathbf{k}}^c_{t,g} = \frac{(\mathbf{c}_t \mathbf{W}^K_g) \odot \boldsymbol{\gamma}^c_k}{S^c_k(\mathbf{c}_t \mathbf{W}^K_g)}, \qquad \ell^c_{i,t,h} = \left((\mathbf{q}'^c_{i,h} \odot \boldsymbol{\gamma}^c_q \odot \boldsymbol{\gamma}^c_k)(\mathbf{W}^K_{g(h)})^\top\right) \cdot \mathbf{c}_t \; \alpha^q_{i,h} \alpha^k_{t,g(h)},$$

where $\alpha^q = 1/S_q$ and $\alpha^k = 1/S_k$ are scalars. The static weight is absorbed into the query-side projection and only one inverse-RMS scalar per token and KV group is cached ($T \times G$ scalars versus $T \times G \times d_c$ full content keys), which keeps MLA's latent decode path intact; the paper reports lower loss than QK-Clip and $< 2\%$ decode overhead up to 256k tokens. In this repository the MLA variant of Mehta et al. [111] already re-expands keys from the cached latent on every step, so explicit per-head normalization of the decompressed $\mathbf{q}$ and $\mathbf{k}$ is *exactly* equivalent to the absorbed formulation in both training and cached decoding; no scalar cache is required. Content and RoPE blocks are normalized blockwise, matching the natural boundary between the latent content path and the small materialized rotary path.

QK-Clip. Kimi K2's MuonClip [81] targets the same failure mode after the dot product. For each head h it defines the per-head maximum pre-softmax logit over the batch,

$$S^h_{\max} = \frac{1}{\sqrt{d}} \max_{\mathbf{X} \in \mathcal{B}} \max_{i,j} \mathbf{Q}^h_i {\mathbf{K}^h_j}^\top,$$

and, after every optimizer step, rescales the projection weights with

$$\gamma_h = \min\left(1, \frac{\tau}{S^h_{\max}}\right), \qquad \mathbf{W}^h_q \leftarrow \gamma_h^\alpha \mathbf{W}^h_q, \qquad \mathbf{W}^h_k \leftarrow \gamma_h^{1-\alpha} \mathbf{W}^h_k.$$

The intervention never alters the current forward/backward pass; heads already below τ receive $\gamma_h = 1$, so the mechanism self-deactivates once training stabilizes (K2 used $\tau = 100$ and observed all heads below threshold after roughly 30% of steps). We apply it per head to the row blocks of `q_proj` and `k_proj`; for latent attention the key side is `uk_proj`, and for grouped-query attention each KV head receives the smallest γ_h of the query heads that read it, which bounds every logit in the group. `clip_alpha` defaults to 0.5 (i.e. $\sqrt{\gamma_h}$ on each side, the balanced K2 choice).

QuacK. Instead of a forward or weight intervention, QuacK [5] assigns parameter-dependent learning rates to the query and key weights. Storing the Frobenius norms at initialization, before each step it sets

$$\eta_{W_Q} = \tau\, \eta\, \frac{\|\mathbf{W}_K\|_{\text{init}}}{\|\mathbf{W}_K\|}, \qquad \eta_{W_K} = \tau\, \eta\, \frac{\|\mathbf{W}_Q\|_{\text{init}}}{\|\mathbf{W}_Q\|},$$

which follows from controlling the first-order change $\Delta\ell \approx (\mathbf{q}\,\Delta\mathbf{k}^\top + \Delta\mathbf{q}\,\mathbf{k}^\top)/\sqrt{d}$: when one side grows, the learning rate of the other decreases. Frobenius and spectral norms performed indistinguishably in the paper, so Frobenius is used. The relative rate τ defaults to 1.0; the paper reports stability comparable to QK-norm at high base learning rates and about a 10% speedup because two RMS reductions per block disappear. The mechanism is optimizer-agnostic here: the trainer splits the query/key weights into dedicated parameter groups, so any registered optimizer can carry it.

Schema contract and validation. `norm.qk_norm` is a sub-object with `enabled`, `method` $\in$ {`rms_norm`, `layer_norm`, `l2_norm`, `qk_clip`, `quack`}, `clip_tau`, `clip_alpha`, `quack_tau` and `eps`. The mechanism requires explicit per-head query/key projections: enabling it for a layer pattern containing a recurrent or state-space mixer is rejected at load time (`QK_NORM_MIXERS`); `qk_clip` additionally requires per-head dot-product logits and therefore excludes the Gaussian-mixture router (`QK_CLIP_MIXERS`).

PyTorch pseudocode (QK-Clip).

Input: model, threshold τ, exponent α, cached per-head $S^h_{\max}$
for each attention head h with $S^h_{\max} > \tau$ **do**
    $\gamma_h \leftarrow \tau / S^h_{\max}$
    $\mathbf{W}^h_q \leftarrow \gamma_h^{\alpha} \mathbf{W}^h_q$ ▷ row block $[hd_h : (h+1)d_h]$
    $\mathbf{W}^h_k \leftarrow \gamma_h^{1-\alpha} \mathbf{W}^h_k$ ▷ `k_proj` or `uk_proj`; min over GQA group
end for
clear $S^h_{\max}$ ▷ avoid reapplying a stale statistic

Comparison.

Method	Intervention	Extra state	Notes
`rms_norm`	per-head forward RMSNorm	$2d_h$	Standard QK-norm; exact with this repo's re-expanding MLA [66, 59].
`layer_norm`	per-head forward LayerNorm	$4d_h$	Mean-centering variant (SAN); slightly heavier.
`l2_norm`	per-head forward L2	none	Weightless ablation.
`qk_clip`	post-step weight rescale	per-head $S_{\max}$	Self-deactivating; needs per-head logits [81].
`quack`	per-step query/key learning rates	init norms	Optimizer-agnostic; ∼10% cheaper than QK-norm [5].

11 BitNet Quantization and Safetensors Block Formats

This annex documents the BitNet ternary weight path and the block-affine quantization formats written by the safetensors export. The deployment section defers to this annex for the layouts and size estimates; bibliography keys resolve against `docs/bibliography/`.

11.1 BitNet Ternary Quantization Path

When `use_bitnet=true`, BitLinear layers replace standard `nn.Linear` with ternary weight quantization $(-1, 0, +1)$ following BitNet [161]. Given weight tensor W, a practical scaling is:

$$s = \text{mean}(|W|), \qquad \tilde{W} = \text{clip}\left(\text{round}\left(\frac{W}{s}\right), -1, 1\right).$$

The packed mapping uses two bits per weight symbol for storage efficiency. This enables significant compression ($\sim 32\times$ versus FP32) but is experimental and may affect model quality; it is best suited for inference deployment.

Activation quantization for the INT8 path is:

$$q = \text{round}\left(x \cdot \frac{127}{\max(|x|) + \epsilon}\right), \qquad q \in [-128, 127]$$

with dequantization $x \approx q/\alpha$. For N parameters, size estimates before metadata and packing overhead are:

$$\text{FP32 size} \approx 4N, \quad \text{FP16 size} \approx 2N, \quad \text{1.58-bit size} \approx \frac{1.58}{8} N$$

which aligns with lightweight deployment goals [142, 17].

11.2 Block-Affine Quantization and Safetensors Export

The HuggingFace export applies block-affine quantization to every two-dimensional linear weight whose input dimension is a multiple of 32. Let $w_1, \ldots, w_{32}$ be one block. The `Q8_0` layout stores

$$d = \frac{\max_i |w_i|}{127}, \qquad q_i = \operatorname{round}\Big(\frac{w_i}{d}\Big) \in [-127, 127],$$

as one FP16 scale plus 32 signed bytes (34 bytes per block, 8.5 bits per weight). The `Q4_0` layout follows the signed-maximum convention: it selects the signed value m with the largest magnitude, sets $d = m/(-8)$, and stores

$$q_i = \min\big(15, \operatorname{trunc}(w_i/d + 8.5)\big) \in [0, 15],$$

as one FP16 scale plus 16 nibble bytes (18 bytes per block, 4.5 bits per weight). Element j occupies the low nibble and element $j + 16$ the high nibble; dequantization is $\hat{w}_i = d\,(q_i - 8)$. The ternary `i2_s` layout reuses the BitNet scale $s = \operatorname{mean}(|W|)$ (or $\max|W|$ when the weights are already ternary), packs four symbols per byte, and adds one FP32 scale per tensor (2 bits per weight). Nominal storage is therefore 8.5, 4.5 and 2.0 bits per quantized linear weight, before tensor metadata.

The safetensors artifact records the per-tensor format and original shape in `config.json`. Two runtimes consume it: *dequantize-on-load* (v1) reconstructs FP32 parameters while loading, while *packed-linear* (v2) keeps the packed buffers in a `PackedLinear` module and dequantizes in `forward`, preserving BitLinear's input normalization and INT8 activation quantization when applicable. The exporter does not use fused integer kernels, and the experiment harness evaluates `Q8_0`, `Q4_0` and `i2_s` in both runtimes.

12 Mixture-of-Depths Token Routing

This annex documents Mixture-of-Depths (MoD) token routing, the adaptive-depth mechanism that updates only a scored subset of tokens per block.

12.1 Mixture-of-Depths (MoD) Token Routing

When `use_mixture_of_depths=true`, each transformer block scores tokens with a lightweight router. Only the top `mixture_of_depths_capacity_ratio` subset is updated; skipped tokens pass through unchanged [186]. This allocates more depth to salient tokens, reducing per-layer compute.

Input: hidden states $H \in \mathbb{R}^{n \times d}$, router R, capacity ratio ρ
scores $\leftarrow R(H)$ ▷ per-token router score
$\mathcal{S} \leftarrow$ `top-k`$(\text{scores}, k = \lceil \rho \cdot n \rceil)$ ▷ select salient subset
$H_{\text{update}} \leftarrow$ `block`$(H[\mathcal{S}])$ ▷ apply transformer block to selected tokens
$H[\mathcal{S}] \leftarrow H_{\text{update}}$ ▷ write back; non-selected tokens bypass the block
Output: H ▷ auxiliary load-balancing loss added from `mixture_of_depths_router_aux_loss_weight`

12.1.1 Configuration

- `use_mixture_of_depths`: bool — enable MoD token routing.
- `mixture_of_depths_capacity_ratio`: float in $(0, 1]$ — fraction of tokens updated per layer.
- `mixture_of_depths_router_aux_loss_weight`: float ≥ 0 — auxiliary loss weight for router regularization.

13 Runtime Algorithms: Conditional Memory, Embeddings, Stability, and Inference Routing

This annex collects the runtime algorithms that support the configuration-driven pipeline: conditional memory (Engram), factorized embeddings and embedding convolution, training-step stability controls, the held-out evaluation split, and the SBERT inference mode router. Bibliography keys cited here resolve against `docs/bibliography/`.

13.1 Engram Conditional Memory

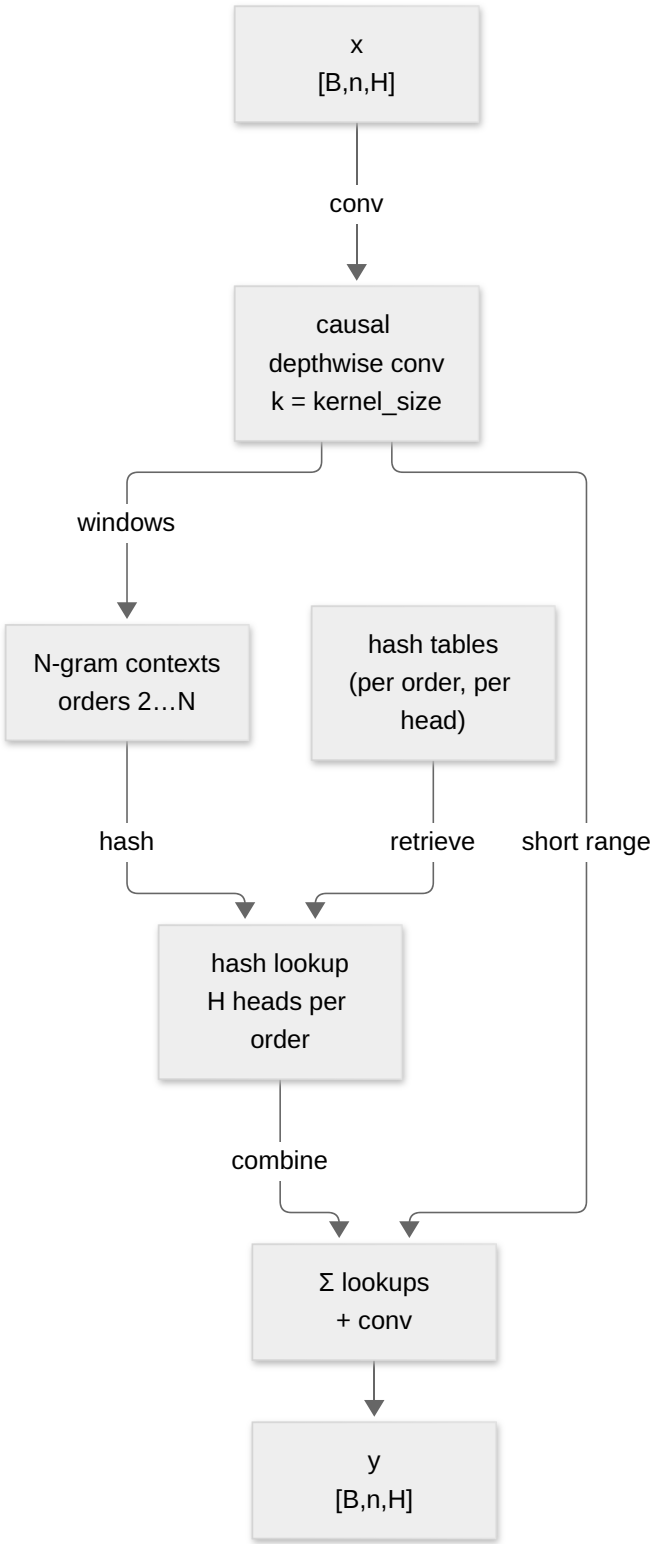


Figure 106: Dataflow of Engram conditional memory: a causal depthwise convolution supplies short-range context, N-gram hash lookups retrieve per-order embeddings, and the lookups are summed with the convolution output.

Engram (`engram`) implements conditional memory via scalable N-gram lookup [21]. It builds hash-based lookup tables for N-gram contexts from size 2 up to `engram_max_ngram_size`, with independent hash heads per N-gram order. A causal depthwise convolution (`engram_kernel_size`) provides short-range context before the N-gram lookup.

Input: token stream $x_{1:n}$, N-gram orders $\{2, \ldots, N\}$, hash heads per order H
$h_{\text{conv}} \leftarrow \text{DepthwiseConv1d}(x_{1:n}, k = \texttt{engram_kernel_size})$ ▷ short-range context
for each N-gram order $k \in \{2, \ldots, N\}$ **do**
    $\text{ctx}_t^{(k)} \leftarrow x_{t-k+1:t}$ ▷ N-gram context window
    $\text{lookup}_t^{(k)} \leftarrow \text{HashTable}^{(k)}[\text{hash}(\text{ctx}_t^{(k)})]$ ▷ per-head retrieval, H heads per order
end for
$y_t \leftarrow \sum_k \text{lookup}_t^{(k)} + h_{\text{conv},t}$ ▷ combine conditional memories
Output: $y_{1:n}$

13.1.1 Configuration

- `engram_max_ngram_size`: int ≥ 2 — highest N-gram order (default 3).
- `engram_n_heads_per_ngram`: int ≥ 1 — hash heads per N-gram order (default 4).
- `engram_embed_dim_per_head`: int ≥ 1 — embedding dim per hash head (default 32).
- `engram_kernel_size`: int ≥ 1 — causal depthwise conv kernel width (default 4).
- `engram_seed`: int — hash seed for reproducibility (default 42).

Total Engram hidden size = $(\texttt{max_ngram_size} - 1) \times \texttt{n_heads_per_ngram} \times \texttt{embed_dim_per_head}$.

13.2 Factorized Embeddings and Embedding Convolution

13.2.1 Factorized Embeddings

When `use_factorized_embedding=true`, the embedding matrix is factorized into two smaller matrices, reducing parameters from $O(V \times H)$ to $O(V \times R + R \times H)$ where $R = \texttt{factorized_embedding_dim}$.

- `use_factorized_embedding`: bool — enable factorization.
- `factorized_embedding_dim`: int ≥ 1 — intermediate dimension (typical: 64–256).

13.2.2 Embedding Convolution

When `use_embedding_conv=true`, a depthwise Conv1d is applied over the embedding stream before the first transformer block, with kernel size `embedding_conv_kernel`. This provides a lightweight local-context filter on token embeddings.

13.3 Training-Step Stability Controls

The schema-driven training step applies gradient accumulation, global-norm clipping, post-clip explosion guards, and bounded NaN/Inf retries. The accumulated gradient is:

$$g_{\text{acc}} = \frac{1}{K} \sum_{i=1}^{K} g_i, \quad K = \texttt{gradient_accumulation_steps}$$

$$g_{\text{clip}} = g_{\text{acc}} \cdot \min\left(1, \frac{\tau}{\|g_{\text{acc}}\|_2 + \epsilon}\right), \quad \tau = \texttt{grad_clip_max_norm}$$

then overflow guards use `inf_post_clip_threshold` and retry logic bounded by `max_nan_retries`.

Algorithm 62 Schema-Driven Training Step with Stability Controls

Require: Batch stream, config C

Initialize retry counter $r \leftarrow 0$
for each optimizer step **do**
  Accumulate gradients for $K = C$.`gradient_accumulation_steps` micro-batches
  Apply global norm clipping with $\tau = C$.`grad_clip_max_norm`
  if post-clip gradient exceeds C.`inf_post_clip_threshold` or NaN/Inf detected **then**
    if $r < C$.`max_nan_retries` **then**
      restore safe state / skip step; $r \leftarrow r + 1$
      continue
    else
      stop training with failure state
    end if
  end if
  run optimizer step selected by `optimizer_class`
  update scheduler (`cosine`, `constant`, or `linear_warmup_then_constant`)
  if step mod `checkpoint_every_n_steps`$= 0$ **then**
    save rolling checkpoint and prune to `max_rolling_checkpoints`
  end if
  update best checkpoints up to `num_best_checkpoints`
  emit CSV + telemetry following `gradient_log_interval` and `telemetry_log_interval`
end for

13.4 Held-Out Test Split and End-of-Training Evaluation

`training.test_percentage` reserves a deterministic fraction of the task dataset before the first optimizer step. With n dataset examples, held-out percentage $\rho =$ `test_percentage`$/100$, and a permutation π drawn once from the run seed `training.seed`, the partition is

$$n_{\text{test}} = \text{clip}(\text{round}(\rho n),\, 1,\, n-1)\,, \qquad \mathcal{D}_{\text{train}} = \{x_{\pi(i)}\}_{i=n_{\text{test}}+1}^{n}, \qquad \mathcal{D}_{\text{test}} = \{x_{\pi(i)}\}_{i=1}^{n_{\text{test}}},$$

so the same configuration always reserves the same examples regardless of DataLoader shuffling. After the last epoch the final weights are evaluated once with the task objective under `no_grad`, and the sample-weighted loss and accuracy

$$\hat{L} = \frac{1}{n_{\text{test}}} \sum_{x \in \mathcal{D}_{\text{test}}} \ell(f_\theta; x), \qquad \widehat{\text{acc}} = \frac{1}{n_{\text{test}}} \sum_{x \in \mathcal{D}_{\text{test}}} a(f_\theta; x)$$

are written to `test_metrics.json` next to the CSV log. The split covers masked and causal language modeling, text classification, the three vision tasks, ASR (where the accuracy slot carries the character error rate, labelled `metric: cer`), and audio classification. SBERT keeps its own validation partition through `training.sbert.max_eval_samples` and is unaffected. Setting the field to zero disables the split and restores the historical all-data behavior.

13.5 SBERT Inference Mode Router

The SBERT inference dispatcher selects among similarity scoring, corpus search, clustering, and persistent encoding modes over a single shared encoder [135]. The cosine similarity and regression loss are:

$$\cos(e_1, e_2) = \frac{e_1^\top e_2}{\|e_1\|\|e_2\|}, \qquad \mathcal{L}_{\cos} = (\cos(e_1, e_2) - y)^2$$

with $y \in [-1, 1]$ in this pipeline.

Algorithm 63 SBERT Inference Mode Router

Require: mode m, model E, inputs X

if $m =$ `similarity` **then**
    return $\cos(E(x_1), E(x_2))$
else if $m =$ `search` **then**
    return top-k by dot-product/cosine against corpus embeddings
else if $m =$ `cluster` **then**
    return clustering labels over $E(X)$
else
    return serialized embeddings $E(X)$
end if

14 Activation Functions

This annex documents the complete activation-function library available in the model. The library provides **43 feed-forward activation functions**: 40 elementwise activations and 3 gated-FFN variants. Activations are grouped into five families, following the taxonomy of Dubey et al. [36] and the analytic overview of Lederer [92], augmented with the learnable Rational Activation Function (RAF) of Fang et al. [42].

The selection is driven by the `ffn_activation` configuration field (a string enum) and dispatched at runtime by a factory function that mirrors the normalization-layer dispatch (Appendix 10).

$$\texttt{ffn_activation} \in \{\texttt{silu}, \texttt{gelu}, \ldots, \texttt{raf}, \texttt{swiglu}, \texttt{geglu}, \texttt{reglu}\}$$

14.1 Classical / Sigmoid–Tanh Family (12 activations)

The classical family comprises smooth, stateless activations rooted in logistic and algebraic functions [92, 36]. All share monotonicity or near-monotonicity and are continuously differentiable.

$$\text{Sigmoid}(x) = \frac{1}{1+e^{-x}} \qquad \text{Range: } (0,1),\ \text{monotonic, smooth} \quad [92] \tag{173}$$

$$\text{Tanh}(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}} = 2\,\sigma(2x) - 1 \qquad \text{Range: } (-1,1),\ \text{monotonic, smooth} \quad [92] \tag{174}$$

$$\text{Arctan}(x) = \arctan(x) \qquad \text{Range: } (-\tfrac{\pi}{2}, \tfrac{\pi}{2}),\ \text{monotonic, smooth} \quad [92] \tag{175}$$

$$\text{Softsign}(x) = \frac{x}{1+|x|} \qquad \text{Range: } (-1,1),\ \text{smooth, once-diff. at } 0 \quad [92] \tag{176}$$

$$\text{Elliott}(x) = \frac{x}{1+|x|} \qquad \text{Alias of Softsign (survey “elliottsig” notation)} \quad [92] \tag{177}$$

$$\text{Identity}(x) = x \qquad \text{Range: } (-\infty,\infty),\ \text{linear, } C^{\infty} \quad \text{(internal)} \tag{178}$$

$$\text{Softplus}(x) = \log(1 + e^x) \qquad \text{Range: } [0, \infty),\ \text{smooth, } C^\infty \quad [48] \tag{179}$$

$$\text{Mish}(x) = x\,\tanh\big(\text{softplus}(x)\big) = x\,\tanh\big(\log(1 + e^x)\big) \qquad \text{Range: } [\approx -0.31, \infty),\ \text{non-monotonic, smooth} \quad [116] \tag{180}$$

$$\text{GELU}(x) = x\,\Phi(x) = x \cdot \tfrac{1}{2}\Big(1 + \text{erf}\Big(\frac{x}{\sqrt{2}}\Big)\Big) \qquad \text{Range: } (\approx -0.17, \infty),\ \text{smooth, probabilistic} \quad [65] \tag{181}$$

$$\text{GELU-tanh}(x) = \tfrac{x}{2}\Big(1 + \tanh\Big(\sqrt{\tfrac{2}{\pi}}(x + 0.044715\,x^3)\Big)\Big) \qquad \text{Tanh approximation of GELU (faster)} \quad [65] \tag{182}$$

$$\text{ReLU}(x) = \max(0, x) \qquad \text{Range: } [0, \infty),\ \text{piecewise-linear, non-smooth at 0} \quad [118] \tag{183}$$

$$\text{SiLU}(x) = x\,\sigma(x) = \frac{x}{1 + e^{-x}} \qquad \text{Range: } (\approx -0.278, \infty),\ \text{non-monotonic, smooth} \quad [40] \tag{184}$$

SiLU (Sigmoid Linear Unit) is the default activation (`ffn_activation=silu`). When $\beta = 1$, the parametric Swish reduces to SiLU. GELU and its tanh approximation are the standard in BERT and GPT-2/3 [65]. Mish [116] is a self-regularized, non-monotonic composition of Softplus and Tanh, reported to outperform Swish on several benchmarks.

14.2 Rectified Family (12 activations)

The rectifier family extends ReLU with learnable slopes, bounded variants, and rectified-hyperbolic forms [118, 36].

$$\text{LeakyReLU}(x) = \begin{cases} x & x \geq 0 \\ a\,x & x < 0 \end{cases}, \quad a = 0.01 \qquad \text{Range: } (-\infty, \infty) \quad [110] \tag{185}$$

$$\text{ReLU6}(x) = \min\big(\max(0, x),\, 6\big) \qquad \text{Range: } [0, 6],\ \text{bounded, fixed-point friendly} \quad [70] \tag{186}$$

$$\text{HardSwish}(x) = x \cdot \frac{\text{ReLU6}(x + 3)}{6} \qquad \text{Range: } (\approx -1.67, \infty),\ \text{cheap Swish approx.} \quad [69] \tag{187}$$

$$\text{PReLU}(x) = \max(0, x) + \mathbf{p} \odot \min(0, x) \qquad \text{Learnable per-channel slope } \mathbf{p},\ \text{init 0.25} \quad [61] \tag{188}$$

$$\text{AbsReLU}(x) = \max(0, x) - \max(0, -x) = x\ \text{(sign-ramp)} \qquad \text{Range: } (-\infty, \infty) \quad [36] \tag{189}$$

$$\text{NLReLU}(x) = \beta\,\log\big(1 + \max(0, x)\big) \qquad \text{Range: } [0, \infty),\ \beta = 1.0,\ \text{logarithmic compression} \quad [36] \tag{190}$$

$$\text{BReLU}(x) = \min\big(\max(0, x),\, t\big), \quad t = 1.0 \qquad \text{Range: } [0, t],\ \text{bounded rectifier} \quad [36] \tag{191}$$

$$\text{VReLU}(x) = |x| \qquad \text{Range: } [0, \infty),\ \text{symmetric rectifier} \quad [36] \tag{192}$$

$$\text{Hexpo}(x) = a\,\max(0, x) - c\,\max(0, -x), \quad a{=}c{=}1.0 \qquad \text{Asymmetric two-sided rectifier} \quad [36] \tag{193}$$

$$\text{PenalizedTanh}(x) = \max\big(0,\, \tanh(x)\big) \qquad \text{Range: } [0, 1),\ \text{rectified tanh, smooth near 0} \quad [36] \tag{194}$$

$$\text{DisReLU}(x) = \max(0,\, x - \delta), \quad \delta = 0.0 \qquad \text{Right-shifted ReLU hinge} \quad [36] \tag{195}$$

$$\text{LiSHT}(x) = x\,\tanh(x) \qquad \text{Range: } [0, \infty)\ \text{(magnitude), non-monotonic, smooth} \quad [139] \tag{196}$$

PReLU [61] makes the negative slope a per-channel learnable parameter, enabling the network to learn the optimal leak per feature. **HardSwish** [69] is the cheap Swish approximation used in MobileNetV3. **ReLU6** [70] was introduced in MobileNetV1 for fixed-point quantization friendliness.

14.3 Exponential / ELU Family (12 activations)

The exponential-linear family provides smooth negative saturation and self-normalizing variants [23, 85, 36]. Several members introduce per-channel learnable parameters.

$$\mathrm{ELU}(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}, \quad \alpha = 1.0 \qquad \text{Range: } (-\alpha, \infty) \quad [23] \quad (197)$$

$$\mathrm{SELU}(x) = \lambda\, \mathrm{ELU}_{\alpha'}(x) \qquad \alpha' \approx 1.6733,\ \lambda \approx 1.0507,\ \text{self-normalizing (mean 0, var 1)} \quad [85] \quad (198)$$

$$\mathrm{CELU}(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^{x/\alpha} - 1) & x < 0 \end{cases}, \quad \alpha = 1.0 \qquad \text{Once-diff. at 0 for any } \alpha \quad [8] \quad (199)$$

$$\mathrm{PELU}(x) = \begin{cases} \frac{\alpha}{\beta}\, x & x \geq 0 \\ \alpha(e^{x/\beta} - 1) & x < 0 \end{cases} \qquad \alpha, \beta \text{ learnable per-channel} \quad [158] \quad (200)$$

$$\mathrm{MPELU}(x) = \mathrm{ReLU}(x) + \beta \cdot \mathrm{ELU}_\alpha(x) \cdot \mathbb{1}_{x<0} \qquad \alpha, \beta \text{ learnable per-channel} \quad [36] \quad (201)$$

$$\mathrm{FELU}(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases} \qquad \alpha \text{ learnable, clamped to } [0, 1] \quad [36] \quad (202)$$

$$\mathrm{EELU}(x) = \beta\, \mathrm{ReLU}(x) + \alpha(e^x - 1) \cdot \mathbb{1}_{x<0} \qquad \alpha, \beta \text{ learnable per-channel} \quad [36] \quad (203)$$

$$\mathrm{PDELU}(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^{x/\alpha} - 1) + (1 - \alpha)\, x & x < 0 \end{cases} \qquad \alpha \text{ learnable, interpolates CELU–identity} \quad [36] \quad (204)$$

$$\mathrm{PREU}(x) = \beta\, \mathrm{ReLU}(x) + \alpha(e^x - 1) \cdot \mathbb{1}_{x<0}^{\leq 0} \qquad \alpha, \beta \text{ learnable per-channel} \quad [36] \quad (205)$$

$$\mathrm{SoftExp}(x) = \begin{cases} \frac{1}{\alpha}\left(e^{\alpha x} - 1\right) + \alpha & \alpha > 0 \\ x & \alpha = 0 \\ -\frac{1}{\alpha}\ln\left(1 - \alpha(x + \alpha)\right) & \alpha < 0 \end{cases} \qquad \alpha \text{ learnable; exp/linear/log interpolation} \quad [49] \quad (206)$$

$$\mathrm{ELiSH}(x) = \begin{cases} x\, \sigma(x) & x \geq 0 \\ (e^x - 1)\, \sigma(x) & x < 0 \end{cases} \qquad \text{Swish pos. branch, ELU-gated neg. branch} \quad [10] \quad (207)$$

$$\mathrm{HardELiSH}(x) = \begin{cases} x \cdot \mathrm{hard_sigmoid}(x) & x \geq 0 \\ (e^x - 1) \cdot \mathrm{hard_sigmoid}(x) & x < 0 \end{cases} \qquad \text{Hard-sigmoid variant of ELiSH} \quad [10] \quad (208)$$

where $\mathrm{hard_sigmoid}(x) = \mathrm{ReLU6}(x + 3)/6$.
ELU [23] provides smooth negative saturation. **SELU** [85] induces self-normalization with fixed $\alpha' \approx 1.6733$, $\lambda \approx 1.0507$. **CELU** [8] is once-differentiable at 0 for any α (unlike ELU). **PELU** [158] makes both α, β learnable per-channel. The parametric variants (MPELU, FELU, EELU, PDELU, PREU) generalize ELU and PReLU by introducing learnable per-channel parameters for the negative and positive branches. **SoftExp** [49] continuously interpolates between logarithmic ($\alpha < 0$), linear ($\alpha = 0$), and exponential ($\alpha > 0$) regimes. **ELiSH** and **HardELiSH** [10] combine Swish and ELU behaviors via a piecewise definition.

14.4 Swish and Parametric Activations

Swish [133] is a family parameterized by a slope β:

$$\text{Swish}_\beta(x) = x\,\sigma(\beta x) = \frac{x}{1 + e^{-\beta x}}.$$

When $\beta = 1$ this is SiLU (the default). Large β approaches ReLU; $\beta \to 0$ approaches the linear function $x/2$.

- **Swish** (`swish`): fixed β (default 1.0, configured via `ffn_activation_config.swish_beta`). No learnable parameters.
- **SwishTrainable** (`swish_trainable`): β is a per-channel learnable parameter, initialized to 1.0. Equivalent to SiLU at initialization, then adapts during training.
- **Maxout** (`maxout`): computes k linear projections and takes the elementwise maximum: $\max_{i=0}^{k-1}(W_i x + b_i)$. A universal approximator for any continuous function given enough pieces. Parameter count increases by factor k (default $k = 2$, configured via `ffn_activation_config.maxout_pieces`).

References: Swish [133], Maxout [50].

14.5 Rational Activation Functions (RAF)

Following Fang et al. [42], a Rational Activation Function is a learnable ratio of two low-degree polynomials (a Padé approximant):

$$F(x) = \frac{P(x)}{Q(x)} = \frac{\displaystyle\sum_{j=0}^{m} a_j\, x^j}{1 + R(x)}$$

where $\mathbf{a} \in \mathbb{R}^{m+1}$ and $\mathbf{b} \in \mathbb{R}^n$ are *learnable*, and the denominator term $R(x)$ depends on the version. The default configuration matches the paper: degree $(m, n) = (5, 4)$, version "A" (safe), initialized by a least-squares fit to GELU on $[-3, 3]$.

The implementation supports five denominator variants:

- **Version A** (default, "safe"): $Q(x) = 1 + \sum_k |b_k\, x^{k+1}|$ (per-term absolute value). Guarantees $Q(x) \geq 1$, no division by zero.
- **Version B**: $Q(x) = 1 + \left|\sum_k b_k\, x^{k+1}\right|$ (absolute value of the whole sum). Also guarantees $Q(x) \geq 1$.
- **Version C**: $Q(x) = 0.1 + \left|\sum_k b_k\, x^k\right|$ with a 0.1 floor. Minimum denominator 0.1.
- **Version D**: like B, but injects uniform multiplicative noise $U(1-\varepsilon, 1+\varepsilon)$ on the denominator weights during *training only* ($\varepsilon = 0.1$). Regularization effect.
- **Version N** ("nunsafe"): $Q(x) = 1 + \sum_k b_k\, x^{k+1}$ without any absolute value. Can have poles (denominator zeros); use with care.

RAFT input scaling. Because a Padé approximant is a reliable universal approximator only on a bounded interval, the RAFT model preprocesses each token's pre-activations with a per-token min–max scaling into $[-3, 3]$:

$$\tilde{x} = \frac{x - \min(x)}{\max(x) - \min(x)} \cdot 6 - 3.$$

This is toggled by `ffn_activation_config.raf_input_scaling` (default `false`).

Initialization targets. The rational coefficients can be initialized by fitting to any of the following target functions: `gelu` (default), `relu`, `leaky_relu`, `leaky_relu_0.1`, `sigmoid`, `tanh`, `swish`/`silu`, or `identity`. The fit uses a least-squares solve on 2001 uniformly spaced samples over $[-3, 3]$.

Freezing. Setting `raf_trainable=false` freezes the rational parameters, useful for parameter-efficient fine-tuning. Frozen RAFs are typically paired with a separate (higher) learning rate for the remaining network parameters.

Note on naming. RAF stands for *Rational* Activation Function, not "Rectified"; it is the learnable polynomial ratio described above.

14.6 Gated FFN Variants: SwiGLU, GEGLU, ReGLU

SwiGLU, GEGLU, and ReGLU [143] are gated feed-forward units that replace the standard Linear → activation → Linear block. They own their own linear projections and are therefore implemented as FFN modules, *not* as elementwise activations:

$$\text{GatedFFN}(x) = \text{dropout}\Big(W_{\text{down}}\Big(\text{act}(x\,W_{\text{gate}}) \odot (x\,W_{\text{up}})\Big)\Big),$$

where $W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d \times d_{\text{ff}}}$ and $W_{\text{down}} \in \mathbb{R}^{d_{\text{ff}} \times d}$. The gate function is:

- **SwiGLU** (`swiglu`): act = SiLU (the default, used in Llama, PaLM).
- **GEGLU** (`geglu`): act = GELU (used in some T5 variants).
- **ReGLU** (`reglu`): act = ReLU (cheapest option).

When `ffn_activation` is one of `swiglu`/`geglu`/`reglu`, the feed-forward layer swaps the standard dense and MoE FFN blocks for a gated FFN built with the same projection class (BitLinear under BitNet). Bias is disabled by default.

14.7 Configuration Schema

The `ffn_activation_config` object groups all learnable-activation parameters. It is optional and ignored for stateless activations.

- `raf_degrees`: $[m, n]$ Padé numerator/denominator degrees (ints ≥ 1). Default: $[5, 4]$.
- `raf_version`: denominator form, one of `A` (default), `B`, `C`, `D`, `N`.
- `raf_approx_func`: initialization target. One of `gelu` (default), `relu`, `leaky_relu`, `leaky_relu_0.1`, `sigmoid`, `tanh`, `swish`, `silu`, `identity`.
- `raf_trainable`: freeze the rational parameters when `false`. Default: `true`.
- `raf_input_scaling`: per-token min–max scaling to $[-3, 3]$ before the rational (RAFT preprocessing). Default: `false`.
- `prelu_init`: initial negative slope for PReLU. Default: 0.25.
- `elu_alpha`: ELU saturation constant. Default: 1.0.
- `celu_alpha`: CELU saturation constant. Default: 1.0.
- `swish_beta`: fixed β for Swish / initial β for SwishTrainable. Default: 1.0.
- `leaky_relu_slope`: negative slope for LeakyReLU. Default: 0.01.

- `maxout_pieces`: number of linear pieces k for Maxout (int ≥ 1). Default: 2.
- Additional ELU-parametric keys: `pelu_alpha`, `mpelu_alpha`, `mpelu_beta`, `felu_alpha`, `eelu_alpha`, `eelu_beta`, `pdelu_alpha`, `preu_alpha`, `preu_beta`, `softexp_alpha`. All default to 1.0 (except `softexp_alpha` which defaults to 0.0).

14.8 Selection Guide

The following decision tree aids activation choice:

- **Safe default** → `silu` (SiLU/Swish$_1$, used in Llama, PaLM).
- **Classic stability** → `gelu` or `relu`.
- **Smooth non-monotonic** → `mish` or `swish`.
- **Mobile-friendly** → `hardswish` or `relu6`.
- **Learnable per-task shape** → `raf`, `prelu`, `swish_trainable`, or `maxout`.
- **Self-normalizing networks** → `selu` (with AlphaDropout).
- **Gated FFN (own projections)** → `swiglu`, `geglu`, or `reglu`.

With BitNet quantization, `silu` may be more robust than smooth activations that rely on precise gradient computation. Learnable activations (`raf`, `prelu`, `swish_trainable`) add parameters but can adapt to the task distribution.

15 Numerically Fixed Softmax and Cross-Entropy

Standard implementations of softmax, log-softmax and cross-entropy subtract the maximum logit before exponentiating, which prevents overflow but leaves two floating-point defects [9]. The first, *Softmax Collapse*, rounds the target (or argmax) gradient to zero once the logits become sufficiently confident, breaking the zero-sum property of the gradient and producing the "Slingshot" loss spikes observed with adaptive optimizers. The second, *Softmax Cancellation*, degrades soft-target (distillation) gradients when student and teacher logits nearly coincide. This annex documents the toolkit's opt-in implementation of the first fix — cancellation-free softmax, log-softmax and hard-target cross-entropy — its custom backward pass, its integration surface and its validation. Soft-target cross-entropy is out of scope because the toolkit exposes no distillation objective; the sentence-embedding path delegates its losses to `sentence-transformers`.

15.1 The Two Floating-Point Defects

For logits $z \in \mathbb{R}^K$ and target class t, the standard stabilised computations are

$$\tilde{z}_k = z_k - z_{\max}, \qquad \operatorname{logsoftmax}(z)_k = \tilde{z}_k - \log \sum_j e^{\tilde{z}_j}, \qquad q_k = \frac{e^{\tilde{z}_k}}{\sum_j e^{\tilde{z}_j}}, \tag{209}$$

and the cross-entropy is $L(z,t) = -\operatorname{logsoftmax}(z)_t$, whose exact gradient is $g = q - p$ for a target distribution p.

Softmax Collapse. Define the logit margin

$$m = z_t - \log \sum_{j \neq t} e^{z_j}. \tag{210}$$

When the target is also the maximum logit, $\tilde{z}_t = 0$ and the denominator becomes $1 + e^{-m}$. Floating-point numbers are much denser near zero than near one: in fp32, values near 0 resolve down to $\sim 10^{-38}$, while terms added to 1.0 are lost below the unit roundoff $u = 2^{-p}$. The denominator therefore rounds to exactly 1.0 once m exceeds the *precision cliff*

$$c_d = -\ln u = p \ln 2, \qquad c_{16} \approx 5.5, \quad c_{32} \approx 16.6, \quad c_{64} \approx 36.7, \tag{211}$$

for bf16 ($p = 8$), fp32 ($p = 24$) and fp64 ($p = 53$). Past the cliff, the forward loss becomes $\log(1) = 0$ and the target gradient $q_t - 1$ becomes exactly 0, while the non-target gradients $q_j = e^{z_j - z_t}/(1 + e^{-m})$ remain positive down to exponent underflow ($m \approx 88$ in fp32). The gradient no longer sums to zero,

$$\sum_k g_k \approx \sum_{j \neq t} q_j > 0, \tag{212}$$

which biases every parameter update and interacts badly with Adam's per-coordinate normalisation: when the missing target gradient later reappears, the second-moment estimate has decayed and the update spikes.

Softmax Cancellation. For soft targets $p = \mathrm{softmax}(z^T)$, the gradient $q - p$ is the difference of two nearly equal probability vectors. Computing them separately loses the small difference that carries the signal and leaves a non-zero residual even when $z^S = z^T$. The paper's fix computes the difference directly in log space with `expm1`/`log1p`; this toolkit has no distillation loss and therefore does not implement it.

15.2 Cancellation-Free Formulations

Both fixes exclude the argmax entry from the log-sum-exp and reintroduce it through a stable softplus. Let $a = \arg\max_k z_k$, $z_{\max} = z_a$, and

$$s = \mathrm{softplus}(-m) = \log\bigl(1 + e^{-m}\bigr) = \log \sum_{j \neq a} e^{z_j - z_{\max}}, \qquad m = z_{\max} - \log \sum_{j \neq a} e^{z_j}. \tag{213}$$

Then

$$\mathrm{logsoftmax}(z)_k = z_k - z_{\max} - s, \qquad q_k = \exp(z_k - z_{\max} - s)\,, \tag{214}$$

with the argmax entry evaluated as exactly $-s$ and $\exp(-s)$ respectively. Because s is computed as $\log(1 + e^{-m})$ (equivalently $\mathrm{log1p}(e^{-m})$), the tiny e^{-m} term survives far beyond the standard cliff, down to exponent underflow.

The complement of the maximum probability is the load-bearing quantity for the backward pass. Rather than forming $1 - q_{\max}$ (which cancels in the output dtype), the implementation uses the identity

$$1 - q_{\max} = -\,\mathrm{expm1}(-s), \tag{215}$$

which is accurate for arbitrarily small s.

Hard-target cross-entropy. Using the target logit as the reference, the loss is

$$L(z, t) = \mathrm{softplus}\bigl(\mathrm{logsumexp}_{j \neq t}\,(z_j - z_t)\bigr) = \log\left(1 + \sum_{j \neq t} e^{z_j - z_t}\right), \tag{216}$$

which is algebraically identical to $\log(1 + e^{-m})$ but never adds the small term to one. Writing z_t through the differences $z_j - z_t$ makes automatic differentiation return

$$g_t = -\frac{e^{-m}}{1 + e^{-m}}, \qquad g_j = \frac{e^{z_j - z_t}}{1 + e^{-m}} \quad (j \neq t), \tag{217}$$

so the target gradient is obtained as minus the sum of the non-target gradients and $\sum_k g_k = 0$ holds by construction, at any margin. The magnitude $e^{-m}/(1 + e^{-m})$ is preserved down to exponent underflow — precisely the component the standard implementation rounds to zero.

Softmax and log-softmax Jacobians. For a standalone softmax the forward values of Equation 214 may still round to one in the output dtype (that is unavoidable for the stored probability), but the Jacobian must not collapse. Given an upstream gradient g, the softmax vector–Jacobian product is

$$\frac{\partial L}{\partial z} = q \odot (g - \langle g, q \rangle), \tag{218}$$

whose argmax entry is rewritten as

$$\left.\frac{\partial L}{\partial z_a}\right|_{\text{fixed}} = q_{\max} \left(g_{\max} (1 - q_{\max}) - \sum_{j \neq a} g_j q_j \right), \tag{219}$$

with $1 - q_{\max}$ taken from Equation 215. The true Jacobian entry $q_{\max}(1 - q_{\max}) \approx e^{-m}$ thus survives the cliff. For log-softmax ($\partial \operatorname{logsoftmax}_k / \partial z_j = \delta_{kj} - q_j$) the argmax entry is likewise rewritten as

$$\left.\frac{\partial L}{\partial z_a}\right|_{\text{fixed}} = -\sum_{j \neq a} g_j + (1 - q_{\max}) \sum_k g_k. \tag{220}$$

Table 17 contrasts the two regimes. The values are the exact quantities of Equations 217–219; the standard column is what Equation 209 produces once $m > c_d$.

Quantity	**Standard**	**Fixed**
$q_{\max}$	1.0 (rounded)	e^{-s} (values; may round)
$1 - q_{\max}$	0.0 (rounded)	$-\operatorname{expm1}(-s)$
CE loss	$\log(1) = 0$	$\operatorname{softplus}(s) > 0$
g_t	0.0	$-e^{-m}/(1 + e^{-m})$
$\sum_k g_k$	$\sum_{j \neq t} q_j > 0$	0 (by construction)
$\partial q_{\max} / \partial z_{\max}$	0.0	$q_{\max}(1 - q_{\max})$

Table 17: Behaviour of the standard and fixed computations past the precision cliff $m > c_d$. The fixed column preserves every exponentially small quantity in Equations 217–219; the standard column rounds the target/argmax terms to zero.

15.3 Implementation in the Toolkit

The fixes live in `src/model/numerics.py` and are exposed as two boolean checkboxes under the `model.numerics` schema group, both defaulting to `false`:

```
model:
  numerics:
    fixed_softmax: true          # every explicit softmax / log-softmax
    fixed_cross_entropy: true    # hard-target task losses
```

Neither flag adds parameters, buffers or state-dict entries, so checkpoints, parameter counting and export (PyTorch, Transformers, GGUF) are unaffected; the flags travel with the model configuration. When disabled, the code paths reduce to the original `torch` calls with a single attribute lookup per softmax, so the default behaviour and performance are unchanged. The fixed losses accept the same `ignore_index` and reduction semantics as `torch.nn.functional.cross_entropy` and compute in the input dtype (no implicit upcast), which keeps the fix effective in bf16 as well as fp32/fp64.

15.3.1 Backward Pass

PyTorch's autograd graph for Equation 214 still contains the subtraction that produces $1 - q_{\max}$, and evaluating it there re-introduces the cancellation. The implementation therefore defines the two softmax functions as `torch.autograd.Function` objects whose `forward` computes Equation 214 and whose `backward` uses Equations 218–220. The cross-entropy needs no custom backward: the graph of Equation 216 already yields Equation 217 exactly, because z_t enters every shifted logit with coefficient -1 and the target row is excluded from the log-sum-exp.

Algorithm 64 Fixed softmax (forward + vector–Jacobian product)

Require: logits $z \in \mathbb{R}^K$ along a normalisation axis, upstream gradient g

$a \leftarrow \arg\max_k z_k, \quad z_{\max} \leftarrow z_a$
$\ell \leftarrow \operatorname{logsumexp}\{z_k - z_{\max} : k \neq a\}$ ▷ argmax excluded
$s \leftarrow \operatorname{softplus}(\ell)$ ▷ $= \log(1 + e^{-m})$
$c \leftarrow -\operatorname{expm1}(-s)$ ▷ $= 1 - q_{\max}$, no cancellation
$q \leftarrow \exp(z - z_{\max} - s)$ ▷ forward output
backward:
$\tau \leftarrow \sum_k g_k q_k, \quad \sigma \leftarrow \sum_{k \neq a} g_k q_k$
$\nabla z \leftarrow q \odot (g - \tau)$
$\nabla z_a \leftarrow q_a \, (g_a \, c - \sigma)$ ▷ accurate argmax entry
return q; **return** ∇z

15.3.2 Integration Surface

A single switch must reach every place where the toolkit materialises probabilities or logits. `HybridLayer` attaches the resolved flag to each mixer as `use_fixed_softmax`; the residual factory does the same for the depth-wise attention residuals; the KV-cache hooks take an explicit `use_fixed_softmax` argument. A dispatch helper then selects the implementation per call site. Table 18 lists the covered surfaces.

Surface	Covered sites
Attention mixers	`standard`, `titan`, `gqa`, `ode`; latent family (`mla`, `gqla`, `mlra`, `tucker`, `iha`, `gta`, `mtla`, `cca`, `ccgqa`, `gma`); gated (`gated_softmax`, `fox`, `fpa`); sparse (`fasa`, `sparge`, `msa`, `nsa`, `bigbird`, `longformer`, `sparse_transformer`, `sparsek`, `swa`, `sparda`); dense surrogate of training-free mixers
Routers	MoE and Mixture-of-Depths top-k softmax
Residuals	Full and Block AttnRes depth-wise softmax
KV cache	recompute/fallback normaliser in `src/model/cache/hooks.py`
Decoding	top-p nucleus filtering and final sampling distribution
Deployment	`inference.predict` probability readout
Training losses	`mlm`, `causal_lm`, `classification`, `segmentation` (CE and Dice softmax), `text_classification`, `patch_prediction`

Table 18: Coverage of the `fixed_softmax` and `fixed_cross_entropy` flags. SSOG's internal separable-Gaussian LSE kernels keep their own formulation: they are not categorical softmaxes over the same axis.

The loss side is dispatched by a single trainer helper, so every task shares the same semantics:

```
def _task_cross_entropy(self, logits, targets, **kwargs):
    if self._use_fixed_cross_entropy():
        return fixed_cross_entropy(
            logits, targets, ignore_index=kwargs.get("ignore_index", -100))
    return F.cross_entropy(logits, targets, **kwargs)
```

15.4 Cost, Scope and Validation

Cost. The fixed paths have the same asymptotic complexity as the standard ones and touch each row with a constant number of extra element-wise operations (one masked log-sum-exp and one softplus; one `expm1` in the backward). There are no matrix multiplications, no additional memory of order K beyond temporaries, and no parameters. On ordinary logits the fixed and standard values agree to within floating-point tolerance; only the sub-roundoff regime that the standard path collapses changes.

Scope. The flags target hard-target classification-style losses. Soft-target cross-entropy (knowledge distillation, KL regularisation) would need the paper's `expm1`/`log1p` difference construction, which this toolkit does not implement because it has no distillation path. The SSOG mixer's internal LSE kernels, the sigmoid attention normaliser (which is not a softmax), and the Continuum Attention integral normaliser are intentionally left unchanged.

Validation. The repository ships `tests/test_fixed_numerics.py`, which asserts (i) forward parity with `torch` for softmax, log-softmax and cross-entropy in fp32/fp64, including masked ($-\infty$) and single-key inputs; (ii) the cliff regressions of Table 17 at margins $25 > c_{32}$ and $40 > c_{64}$ — the standard target/argmax gradients are exactly zero and the fixed ones are not, with a zero-sum gradient in the fixed case; (iii) `ignore_index`, reduction and dispatch semantics; (iv) full-model forward parity (fixed versus standard is mathematically identical) and finite backward passes; (v) KV-cache prefill+decode parity with the flag enabled; and (vi) trainer dispatch. The combinatoric end-to-end harness additionally trains real MLM and causal-LM models with `fixed_softmax`, `fixed_cross_entropy`, both together, and the interplay with SSMax and BitNet, all on CPU. A ready-to-run preset is `configs/examples/numerics_fixed.yaml`.

When to enable. The fix matters when logit margins can grow large during training: small algorithmic datasets that exhibit grokking, long training with weight decay and adaptive optimizers, and any regime where the loss becomes extremely confident. In ordinary large-scale training the standard path is usually indistinguishable; the fixed path is a safe default candidate because it is mathematically identical in the forward pass and strictly better conditioned in the backward pass.

16 Manifold-Constrained Hyper-Connections (mHC)

This annex documents the integration of *Manifold-Constrained Hyper-Connections* (mHC) [168], a residual-connection framework that replaces the standard identity residual with an n-stream residual whose stream-mixing matrix is constrained to the Birkhoff polytope. The main body delegates to this annex for the mathematical formulation, the implementation details, and the reported results; the bibliography keys cited here resolve against `docs/bibliography/`.

16.1 Motivation: The Identity-Mapping Property

The standard residual connection [62] propagates signals across depth unmodified: with $x_l \in \mathbb{R}^C$ and a layer function F,

$$x_{l+1} = x_l + F(x_l, W_l),$$

so that, recursively, $x_L = x_l + \sum_{i=l}^{L-1} F(x_i, W_i)$. The direct term x_l (the *identity mapping*) lets information flow from shallow to deep layers unchanged, which He et al. [62] identified as key to stable training of very deep networks.

Hyper-Connections (HC) [168] widen the residual stream by an expansion factor n: the stream becomes $x_l \in \mathbb{R}^{n\times C}$ while each layer's internal function F still operates at dimension C. Three learnable linear mappings govern the stream each layer:

- $H_l^{\text{pre}} \in \mathbb{R}^{1\times n}$ aggregates the nC-dimensional stream into the C-dimensional layer input;
- $H_l^{\text{post}} \in \mathbb{R}^{1\times n}$ maps the layer output back onto the stream;
- $H_l^{\text{res}} \in \mathbb{R}^{n\times n}$ mixes features *within* the residual stream.

The single-layer propagation is

$$x_{l+1} = H_l^{\text{res}} x_l + \big(H_l^{\text{post}}\big)^\top F\big(H_l^{\text{pre}} x_l, W_l\big),$$

and, recursively, the deep-to-shallow signal is governed by the composite mapping $\prod_{i=L-1}^{l} H_{i+1}^{\text{res}}$. Because H_l^{res} is unconstrained in HC, this composite deviates from the identity, the stream norm is not conserved, and the forward/backward signal can be amplified or attenuated without bound. On a 27B-parameter MoE model the authors report composite gain magnitudes peaking near 3000 (versus 1 for the identity), correlating with gradient-norm spikes and an unexpected loss surge around step 12k. HC also multiplies memory-access cost by roughly n (the "memory wall").

16.2 Formulation

Let $x_l \in \mathbb{R}^{n\times C}$ be the stream at layer l. The coefficients are computed from the flattened stream $\tilde{x}_l = \mathrm{vec}(x_l) \in \mathbb{R}^{1\times nC}$:

$$\tilde{H}_l = \frac{1}{r_l}\Big(\alpha \odot \big(\tilde{x}_l\, \varphi_l\big)\Big) + b_l, \qquad r_l = \frac{\|\tilde{x}_l\|_2}{\sqrt{nC}},$$

where $\varphi_l \in \mathbb{R}^{nC\times(n^2+2n)}$ is a learned linear projection, $b_l \in \mathbb{R}^{1\times(n^2+2n)}$ a learned bias, and $\alpha^{\text{pre}}, \alpha^{\text{post}}, \alpha^{\text{res}} \in \mathbb{R}$ are learnable gating scalars initialized to a small value (0.01 in the paper

and in this codebase). The scaling by $1/r_l$ is mathematically equivalent to RMSNorm on the flattened stream, with the per-dimension scale absorbed into φ_l. The three coefficient blocks are then:

$$H_l^{\text{pre}} = \sigma\big(\tilde{H}_l^{\text{pre}}\big), \qquad H_l^{\text{post}} = 2\,\sigma\big(\tilde{H}_l^{\text{post}}\big), \qquad H_l^{\text{res}} = \text{SK}\big(\exp(\tilde{H}_l^{\text{res}})\big),$$

where σ is the logistic sigmoid (enforcing non-negativity and preventing signal cancellation from mixed-sign coefficients) and $\text{SK}(\cdot)$ is the Sinkhorn–Knopp projection onto the set of doubly stochastic matrices

$$\mathcal{M}^{\text{res}} = \big\{\, H \in \mathbb{R}^{n\times n} \;:\; H \geq 0,\;\; H\,\mathbf{1}_n = \mathbf{1}_n,\;\; \mathbf{1}_n^\top H = \mathbf{1}_n^\top \,\big\},$$

i.e., the Birkhoff polytope (the convex hull of permutation matrices). Starting from $M^{(0)} = \exp(\tilde{H}_l^{\text{res}})$, the projection alternates row and column normalizations,

$$M^{(t)} = T_c\big(T_r(M^{(t-1)})\big),$$

for $t = 1, \ldots, t_{\max}$ rounds ($t_{\max} = 20$ by default in both the paper and this codebase). Finally the layer input and the updated stream are

$$F_l^{\text{pre}} = H_l^{\text{pre}} x_l \in \mathbb{R}^{1\times C}, \qquad x_{l+1} = H_l^{\text{res}} x_l + \big(H_l^{\text{post}}\big)^\top \otimes F\big(F_l^{\text{pre}}, W_l\big).$$

16.3 Properties of the Manifold Constraint

Constraining H_l^{res} to the Birkhoff polytope restores the conservation behavior of the identity residual and guarantees three properties [168]:

1. **Norm preservation**: the spectral norm satisfies $\|H_l^{\text{res}}\|_2 \leq 1$ (non-expansive map), mitigating gradient explosion.
2. **Compositional closure**: doubly stochastic matrices are closed under multiplication, so the composite $\prod_i H_i^{\text{res}}$ remains doubly stochastic at arbitrary depth, preserving the identity-mapping property between *any* two depths.
3. **Geometric interpretation**: since the Birkhoff polytope is the convex hull of permutation matrices, $H_l^{\text{res}} x$ is a convex combination of feature permutations, i.e., a monotonic information-mixing mechanism across streams.

When $n = 1$ the doubly stochastic condition degenerates to the scalar 1, recovering the exact identity mapping. With finite $t_{\max}$ the projection is approximate, and the composite backward gain deviates from 1 but stays bounded (around 1.6 in the paper's 27B runs, a $\sim$ 3-orders-of-magnitude reduction versus HC's ≈ 3000).

16.4 Implementation in Frankenstein Transformer

mHC is an opt-in replacement for the standard residual connection, controlled by the `model.mhc` sub-object (hierarchical schema) or the flat `use_mhc` keys, with `additionalProperties: false` at the sub-object level:

- `enabled` (`use_mhc`, bool, default `false`): replace the standard residual with the n-stream residual.
- `expansion_rate` (`mhc_expansion_rate`, int ≥ 1, default 4): stream expansion factor n; the stream becomes $n \times$ `hidden_size` while layer internals stay at `hidden_size`. 1 recovers the identity mapping.
- `sinkhorn_iters` (`mhc_sinkhorn_iters`, int ≥ 1, default 20): Sinkhorn–Knopp normalization rounds $t_{\max}$.

- `gating_init` (`mhc_gating_init`, float > 0, default 0.01): initial value of the gating scalars α.
- `checkpoint` (`mhc_checkpoint`, bool, default `false`): gradient checkpointing on mHC layers, trading compute for memory to mitigate the $\sim n\times$ activation-memory increase of the n-stream residual.
- `full_prec_under_bitnet` (`mhc_full_prec_under_bitnet`, bool, default `true`): keep the coefficient projection φ_l a full-precision `nn.Linear` even when `use_bitnet` is `true`, avoiding ternary-quantization noise on the small mHC coefficients; when `false` (and BitNet enabled), φ_l uses `BitLinear`.

The module lives in `src/model/mhc.py`:

- `SinkhornKnoppFunction` (`torch.autograd.Function`): the forward pass applies the alternating row/column normalizations of $\exp(\cdot)$; the backward pass recomputes the full iteration under `torch.enable_grad` and differentiates through it, obtaining the exact Jacobian-vector product (the recompute-on-chip strategy described in the paper).
- `ManifoldHyperConnections` (`nn.Module`): holds φ_l (`proj`), b_l (`bias`) and the three gating scalars `alpha_pre`, `alpha_post`, `alpha_res`; exposes `fpre` (project the stream down to a layer input), `recombine` (update the stream with a layer output) and `mappings` (compute $H^{\text{pre}}, H^{\text{post}}, H^{\text{res}}$).

Wiring in `src/model/frankenstein_model.py`:

- `HybridLayer` gains optional `mhc_attn` and `mhc_ffn` modules (one mHC module per layer function); the `_forward_dense_mhc` path runs attention and FFN as layer functions over the shared (B, S, n, C) stream.
- `FrankensteinTransformer` expands the C-dimensional embedding to (B, S, n, C) on entry via `mhc_in_proj` and collapses back to C via `mhc_out_proj` before the head.
- mHC parameters route into the `other` optimizer parameter group.
- mHC is **incompatible with** `use_mixture_of_depths`: MoD token routing operates on a single C-dimensional stream, conflicting with the n-stream residual; a `ValueError` is raised if both are enabled.

Example configuration: `configs/examples/es_arch_mhc_adamw.yaml`.

16.5 Reported Results

The paper evaluates mHC (and HC) with $n = 4$, $t_{\max} = 20$ and α-init 0.01 on DeepSeek-V3-style MoE models (MLA attention, RoPE, RMSNorm, Loss-Free load balancing) from 3B to 27B parameters [168]:

- **Stability**: mHC removes HC's loss surges (e.g., near step 12k on the 27B model) and keeps the composite residual gain bounded (≈ 1.6 versus HC peaks ≈ 3000).
- **Quality**: on the 27B model mHC improves final training loss by 0.021 over the baseline, beats the baseline on all 8 downstream benchmarks (BBH, DROP, GSM8K, HellaSwag, MATH, MMLU, PIQA, TriviaQA) and HC on 7 of 8, with the largest gains on reasoning tasks (+2.1 BBH, +2.3 DROP over HC).
- **Scaling**: the relative loss improvement over baseline is maintained across compute scales (3B $\rightarrow$ 9B $\rightarrow$ 27B) and across 1T tokens of token scaling.
- **Overhead**: with fused kernels (TileLang), mixed-precision coefficient computation, selective recomputation, and DualPipe communication overlap, mHC adds only 6.7% additional training time at $n = 4$ on the 27B model.

16.6 Hyperloop: Loop-Level Hyper-Connections on Looped Transformers

The *Hyperloop Transformer* [176] combines looped Transformers — which reuse layer weights across depth — with *loop-level* hyper-connections, targeting parameter efficiency rather than raw depth. The layer stack is partitioned *middle-cycle* into a begin block (run once), a looped middle block, and an end block (run once); the looped middle block is wrapped in hyper-connections applied only after each loop iteration. With $Y_t^{(l)} \in \mathbb{R}^{n\times C}$ the matrix-valued stream at loop l, $z_t^{(l)} = \mathrm{RMSNorm}(\mathrm{flatten}(Y_t^{(l)}))$, and per-loop parameters $\{W_l^\tau, b_l^\tau, \alpha_l^\tau\}$ for $\tau \in \{\mathrm{pre}, \mathrm{post}, \mathrm{res}\}$ plus a loop position embedding $e_l \in \mathbb{R}^C$, the recurrence is

$$y_t^{(l+1)} = H_l^{\mathrm{res}}\, y_t^{(l)} + H_l^{\mathrm{post}} \otimes \big(F(H_l^{\mathrm{pre}}\, y_t^{(l)}) + e_l\big), \tag{221}$$

with the data-dependent coefficients

$$H_l^{\mathrm{pre}} = \sigma\big(\alpha_l^{\mathrm{pre}}(W_l^{\mathrm{pre}} z_t^{(l)}) + b_l^{\mathrm{pre}}\big), \qquad H_l^{\mathrm{post}} = 2\sigma\big(\alpha_l^{\mathrm{post}}(W_l^{\mathrm{post}} z_t^{(l)}) + b_l^{\mathrm{post}}\big),$$
$$H_l^{\mathrm{res}} = \mathrm{diag}\big(\sigma(\alpha_l^{\mathrm{res}}(W_l^{\mathrm{res}} z_t^{(l)}) + b_l^{\mathrm{res}})\big) \qquad W_l^{\mathrm{res}} \in \mathbb{R}^{n\times nC}$$

(the diagonal H^{res} substitutes mHC's sinkhorn$(\cdot)$ over a dense $n^2 \times nC$ projection with a sigmoid over a diagonal gate — the paper's Table 6 ablation finds this both cheaper *and* more accurate: 14.40 perplexity versus 14.59 Sinkhorn and 14.61 identity).

Four properties distinguish the Hyperloop wiring from per-sublayer mHC (§16): (1) hyper-connections fire once per *loop*, not after every sub-layer — also the most performant placement in the paper's Table 5 ablation; (2) the stream is created by *copying* the begin-block output n times and collapsed by *averaging* the streams before the end block, replacing mHC's learned in/out projections; (3) the parameters are *per loop*, relaxing the strict weight sharing of ordinary looped Transformers — the paper's representation-similarity analysis (Table 8) shows looped-layer representations deviate more across loops, which it identifies as the source of the quality gains; and (4) H^{res} defaults to the diagonal-sigmoid parameterization.

The Frankenstein integration is controlled by the `mhc.hyperloop` boolean (flat `mhc_hyperloop`; requires `mhc.enabled` and `dims.num_loops` ≥ 2), with three additional keys:

- `hyperloop_begin_layers` / `hyperloop_end_layers` (int ≥ 0, default 0): sizes of the begin/end blocks, which run exactly once. The paper allocates ∼25% of parameters to each and 50% to the middle block (unrolled depth = begin + middle × loops + end; e.g. $2L \to 4L(\times 3) \to 2L$ at 16 layers). 0/0 loops the whole stack.
- `hyperloop_res_parameterization` (enum, default `diagonal`): `diagonal`, `sinkhorn` (full mHC-style doubly stochastic mixing, reusing `SinkhornKnoppFunction`) or `identity`.

Implementation: `src/model/hyperloop.py` hosts `HyperloopConnections` with per-loop `projs` / `biases` / `alphas_` / `loop_pos_embs` (e_l, zero-initialized so that at initialisation the update reduces to the plain looped-Transformer recurrence). `FrankensteinEncoder` (and `FrankensteinViT`) build it *instead of* the `mhc_in_proj`/`mhc_out_proj` pair; the forward pass runs begin → copy → per-loop read/middle/update → average → end. `HybridLayer` disables its per-layer `mhc_attn`/`mhc_ffn` modules under Hyperloop, so middle-block layers run the standard C-dimensional path — which restores **Mixture-of-Depths compatibility** (per-layer mHC forbids MoD). Hyperloop is incompatible with the AttnRes residuals (`residuals.type full_attn` / `block_attn`), which assume a per-logical-layer n-stream residual. The per-loop parameter budget is $3n^2C + 3n + 3 + C$ (diagonal): for the paper's 136M model ($C = 1024$, $n = 4$, 3 loops) only ≈150K parameters beyond the plain looped Transformer.

Example configuration: `configs/examples/hyperloop_mhc_adamw.yaml`.

Reported results (arXiv:2604.21254)

Trained on FineWeb-Edu with $n = 4$ streams and 3 loops (25%/50%/25% parameter split), against depth-matched Transformer and mHC baselines:

- **Parameter efficiency**: at 240M/1B/2B depth-matched scale, Hyperloop uses ≈50% fewer parameters (136M/580M/991M) yet matches or beats the baselines in perplexity (14.40 vs 14.65 at 240M scale; 9.65 vs 10.19 at 1B; 8.49 vs 8.60 at 2B).
- **Quantization**: INT4 (GPTQ) perplexity degrades less than the vanilla looped model (14.68 vs 15.18 at 240M scale), so the parameter savings survive post-training compression.
- **Overhead**: loop-level placement plus the diagonal H^{res} keep the training throughput within ≈5% of the plain looped Transformer in unoptimized PyTorch (per-layer mHC loses ≈33% without fused kernels).
- **Ablations**: every-loop placement beats every-layer (Table 5); diagonal H^{res} beats Sinkhorn and identity (Table 6); stream count n shows diminishing returns beyond 4 (Table 4); LoRA-per-loop (relaxed recursive Transformers) helps far less per parameter than the hyper-connection relaxation (Table 7).

17 Attention Residuals (AttnRes)

This annex documents the integration of *Attention Residuals* (AttnRes) [153], a generalization of the residual connection that replaces the fixed unit-coefficient residual sum with *learned softmax attention over depth*. The main body delegates to this annex for the mathematical formulation, the four available strategies (standard, no-residual, Full AttnRes, Block AttnRes), the implementation wiring, the integration with mHC and Mixture-of-Depths, and the reported results. The bibliography keys cited here resolve against `docs/bibliography/`.

17.1 Motivation: From a Fixed Sum to Depth-Wise Attention

The standard residual connection [62] propagates signals across depth unmodified: with $x_l \in \mathbb{R}^C$ and a layer function F,

$$x_{l+1} = x_l + F(x_l, W_l),$$

so that, recursively, $x_L = x_l + \sum_{i=l}^{L-1} F(x_i, W_i)$. The identity term x_l (the *identity mapping*) lets information flow from shallow to deep layers unchanged, which He et al. [62] identified as key to stable training of very deep networks. Highway networks [149] generalize this with learned element-wise gates $\mathbf{g}_l \in [0,1]^C$,

$$x_{l+1} = (1 - \mathbf{g}_l) \odot x_l + \mathbf{g}_l \odot F(x_l, W_l),$$

which interpolates between the identity path and the transformation path with input-dependent weights.

Attention Residuals [153] take the next step: instead of weighting a single compressed state x_l, layer l *attends* over all earlier layer outputs. Let $v_0 = h_1$ (the token embedding) and $v_i = F(x_i, W_i)$ for $i \geq 1$. AttnRes computes, for each layer l,

$$q_l = w_l, \qquad k_i = \text{RMSNorm}(v_i), \qquad \alpha_{i \to l} = \text{softmax}_i(q_l^\top k_i), \qquad x_{l+1} = \sum_{i=0}^{l} \alpha_{i \to l}\, v_i,$$

where $w_l \in \mathbb{R}^C$ is a learned per-layer pseudo-query, *initialized to zero* per the paper. At zero-init the softmax is uniform, so AttnRes degenerates to an equal-weight average of all previous outputs — matching the standard residual at step zero and avoiding training volatility. The RMSNorm on the keys prevents layers with naturally large magnitudes from dominating the softmax (ablation in the paper: § 5.3).

17.2 Four Strategies

The schema field `model.residuals.type` selects one of four strategies, each implemented as a separate module under `src/model/residuals/`.

17.2.1 Standard Residual (`"standard"`)

The default backwards-compatible strategy:

$$x_{l+1} = x_l + F(x_l, W_l).$$

Stateless, no extra parameters, no extra compute. Implemented as a no-op pass-through in `StandardResidual` that delegates the merge to `HybridLayer`.

17.2.2 No Residual (`"none"`, experimental)

Drops the skip connection entirely:

$$x_{l+1} = F(x_l, W_l).$$

Stateless, no extra parameters. Useful only as an ablation probe; the identity term is critical for deep stacks (paper § 2.1) and training is typically unstable without it. Implemented as `NoResidual`.

17.2.3 Full Attention Residuals (`"full_attn"`, paper § 3.1)

Each layer attends over *all* previous layer outputs:

$$x_{l+1} = \sum_{i=0}^{l} \alpha_{i \to l}\, v_i, \qquad \alpha_{i \to l} = \mathrm{softmax}_i\big(w_l^\top \mathrm{RMSNorm}(v_i)\big).$$

Adds $L \cdot C$ parameters (L = logical depth = `num_layers` × `num_loops`, C = `hidden_size`), one query vector per layer. Memory and compute grow as $O(L\,d)$ per token; at typical depth ($L < 1000$) this is negligible. Implemented as `FullAttentionResidual`; supports `attnres_gradient_checkpoint` to wrap the attention in `torch.utils.checkpoint` for memory-constrained runs.

17.2.4 Block Attention Residuals (`"block_attn"`, paper § 3.2)

Partitions the L logical layers into N blocks of $S = \lceil L/N \rceil$ layers each. Within a block the layer outputs accumulate as a standard partial sum b_n^i; across blocks attention is applied over the N complete block representations and the running partial sum. For layer l in block n,

$$x_{l+1} = \sum_{m=0}^{n} \alpha_{m \to l}\, v_m, \qquad \alpha_{m \to l} = \mathrm{softmax}_m\big(w_l^\top \mathrm{RMSNorm}(v_m)\big),$$

with $v_m = b_m$ for $m < n$ and $v_n = b_n^{il}$ (the current intra-block partial sum). N interpolates between Full AttnRes ($N = L$, one block per layer) and standard residuals ($N = 1$, the embedding isolated as b_0). The paper's empirical sweet spot is $N \approx 8$; this codebase defaults to $N = 8$ and validates that $N \leq L$. Implemented as `BlockAttentionResidual`, which also follows Algorithm 1 of the paper (two-phase computation with online softmax merging).

17.3 Implementation in Frankenstein Transformer

The `src/model/residuals/` package hosts one class per strategy (`StandardResidual`, `NoResidual`, `FullAttentionResidual`, `BlockAttentionResidual`) plus an abstract `ResidualBase` that defines the lifecycle hooks `set_streams`, `register_state`, `reset_state`, `forward`, and `finalize`. The `build_residual(config)` factory in `src/model/residuals/factory.py` selects the right class from `FrankensteinModelConfig`.

Wiring in `src/model/frankenstein_encoder.py`:

- `FrankensteinEncoder` owns the `ResidualBase` module as `self.residual`; the factory is invoked once in `__init__`.
- Before the looped-depth loop the encoder calls `reset_state` and, for AttnRes variants, `set_embedding(x)` so the depth-wise attention knows the $v_0 = h_1$ source.
- After each `HybridLayer` call, if `self.residual.is_attn_res` is true, the encoder applies `x = self.residual(layer_idx, x)` to overwrite the stream with the attended aggregation. For stateless strategies (`standard`, `none`) this step is skipped and `HybridLayer` continues to apply its own residual merge internally.

Wiring in `src/model/hybrid_layer.py`:

- `HybridLayer` reads `residual_type` from the config and dispatches the residual merge accordingly. For `standard` the merge is the original `x = residual + mixer(norm(x))`; for `none` the skip is dropped (`x = mixer(norm(x))`); for the AttnRes variants the layer performs the standard merge internally because the depth-wise attention is applied externally by the encoder.
- `_forward_dense_mhc` (the mHC path) is unaffected because mHC has its own stream-expansion residual; combining AttnRes with mHC is handled by the `mhc_stream_mode` field of the residual module.

Schema (hierarchical, with the flat-key renames shown in parentheses) and defaults:

- `model.residuals.type` (`residual_type`, enum `[standard, none, full_attn, block_attn]`, default `standard`): selects the strategy.
- `model.residuals.full_attn.init_query_zero` (`full_attn_init_query_zero`, bool, default `true`): zero-init the per-layer query vectors $\mathbf{w}_l$.
- `model.residuals.full_attn.use_rmsnorm_keys` (`full_attn_use_rmsnorm_keys`, bool, default `true`): apply RMSNorm on keys before the dot product.
- `model.residuals.block_attn.num_blocks` (`block_attn_num_blocks`, int ≥ 1, default 8): number of block representations N; validated $N \leq L$.
- `model.residuals.block_attn.init_query_zero` (`block_attn_init_query_zero`, bool, default `true`): zero-init the per-layer query vectors.
- `model.residuals.block_attn.use_rmsnorm_keys` (`block_attn_use_rmsnorm_keys`, bool, default `true`): RMSNorm on keys.
- `model.residuals.mhc_stream_mode` (`attnres_mhc_stream_mode`, enum `[independent, joint]`, default `independent`): how AttnRes interacts with mHC's n-stream residual.
- `model.residuals.gradient_checkpoint` (`attnres_gradient_checkpoint`, bool, default `false`): wrap the AttnRes attention in gradient checkpointing.

17.4 Integration with mHC and Mixture-of-Depths

The residual strategy is *orthogonal* to mHC and Mixture-of-Depths (MoD): all combinations are supported. The interaction modes are:

mHC × AttnRes (`mhc_stream_mode`). When `use_mhc=true` and the residual is `full_attn` or `block_attn`, the depth-wise attention is applied either *per stream* (`"independent"`) or over the *flattened* nC-dim projection (`"joint"`). Independent mode runs n parallel attentions, one per stream — matching the paper's framing where each stream is a parallel "view" of the residual. Joint mode treats the n streams as a single fat residual, more expressive but more parameter-hungry. The mHC coefficient projections (φ_l, b_l, α gates) are unaffected; AttnRes adds its own parameters on top.

Mixture-of-Depths × AttnRes. MoD token routing operates *within* a layer (selects which tokens get the full layer update) and is orthogonal to the residual choice. When `use_mixture_of_depths=true` together with an AttnRes residual, only the tokens selected by MoD contribute to the layer-output buffer (Full AttnRes) or the intra-block partial sum (Block AttnRes); the rest pass through unchanged and are scattered back by MoD's existing logic. MoD is mutually exclusive with mHC (a `ValueError` is raised in `HybridLayer._init__` if both are enabled); the corresponding combinations are (no_mHC, MoD, AttnRes), (mHC, no_MoD, AttnRes), or (mHC, no_MoD, no_AttnRes).

17.5 Reported Results

The paper [153] validates AttnRes through scaling laws, ablations, and a 48B-parameter Kimi Linear extension:

- **Scaling laws**: both Full and Block AttnRes consistently outperform the PreNorm baseline across five model sizes. At 5.6 PFLOP/s-days, Block AttnRes reaches loss 1.692 versus the baseline's 1.714 — equivalent to a 1.25× compute advantage. The gap between Full and Block AttnRes narrows with scale (Full 1.737 vs. Block 1.746 on the 16-layer model).
- **Ablations**: input-dependent softmax attention (1.737) outperforms input-independent scalar mixing (1.749) and sigmoid attention (1.741). RMSNorm on keys is essential (Full 1.737 vs. 1.743 without; Block 1.746 vs. 1.750). Multi-head depth aggregation ($\texttt{H} = 16$) hurts Block AttnRes (1.752 vs. 1.746), indicating the optimal mixture is largely uniform across channels. Sliding-window attention over the last $W = 8$ layers falls short of Block AttnRes (1.764 vs. 1.746).
- **Kimi Linear extension**: a 48B-parameter (3B activated) Kimi Linear model with Block AttnRes ($N \approx 9$, $S = 6$) pre-trained on 1.4T tokens improves over the baseline across all 14 downstream benchmarks (MMLU-Pro, GPQA-Diamond, BBH, HellaSwag, TriviaQA, GSM8K, MGSM, Math, CMath, HumanEval, MBPP, CMMLU, C-Eval, ARC-C). The largest gains are on multi-step reasoning (GPQA-Diamond +7.5, Minerva Math +3.6) and code generation (HumanEval +3.1). Training-dynamics analysis shows that AttnRes mitigates PreNorm dilution: hidden-state magnitudes remain bounded across depth and gradient norms distribute more uniformly across layers.
- **Overhead**: training overhead is marginal when pipeline parallelism is not enabled; with pipeline parallelism Block AttnRes adds $< 4\%$ end-to-end training overhead (cross-stage caching eliminates redundant transfers) and $< 2\%$ inference latency overhead (two-phase computation with online softmax merging).

Example configurations: `configs/examples/full_attn_res_adamw.yaml`, `configs/examples/block_attn_res_adamw.yaml`, `configs/examples/mhc_full_attn_res_adamw.yaml`.

18 Mixture-of-Experts: Theory and Implementation

This appendix documents the Mixture-of-Experts (MoE) contract implemented by `src/model/moe/` and declared in `src/schema/_model/_moe.yaml`. The schema is the executable source of truth: every key discussed here exists in the schema, is materialised in the nested `FrankensteinModelConfig` `moe` block, and is exercised by the test suite. The first two sections are the on-ramp: they explain what an MoE layer does—one equation, one analogy, one worked example—and then the two failure modes (routing collapse and the hot-expert straggler) that force most of the design. The rest is the reference: router scoring, routing strategy, load balancing, experts, layer layout, capacity and monitoring, closing with the schema-contract and preset tables. A reader who finishes Section 18.2 should be able to open any later section and know which of the two failures, or which of the seven design questions, it is answering.

18.1 Overview: what a Mixture-of-Experts layer is

The feed-forward half of a transformer block is *dense*: one network, applied identically to every token, no matter what the token is. That is a clinic with a single physician who must treat every complaint. An MoE layer turns the clinic into a roster of E specialists with a receptionist: the *router* reads each token and sends it to the k most relevant specialists, the *experts* $\{E_e\}_{e=1}^{E}$. Formally,

$$y_t = g \sum_{j=1}^{k} \hat{w}_{t,j}\, E_{i_{t,j}}(x_t) + y_t^{\text{shared}}, \tag{222}$$

read from the inside out: expert $i_{t,j}$ processes the token and produces a vector, the mixing weight $\hat{w}_{t,j}$ scales it, the k contributions are summed, the routed scaling factor g restores the branch's magnitude (Section 18.3), and y_t^{shared} is the optional always-on shared-expert branch (Section 18.6). Only the selected experts run, which is the whole bargain: total parameters grow with E while per-token compute stays proportional to k [145, 43].

The bargain in numbers: with $E = 8$ experts of the dense width and $k = 2$, the layer stores $8\times$ the feed-forward parameters but each token is processed by only 2 experts—25% of the pool—plus a router of $d \times E$ weights, which is negligible next to any expert. Memory grows with E; FLOPs grow with k. Those are two independent budgets, and most of the schema is the machinery for spending them well.

Seven questions span the design space, and Table 19 maps each to its knobs and its section: (i) how logits become weights, (ii) who selects the experts, (iii) how load is kept balanced, (iv) what the experts look like, (v) which layers are MoE at all, (vi) what happens when an expert overflows, and (vii) how all of it is observed during training.

Table 19: The seven questions of an MoE layer, the knobs that answer them and the section that develops each answer. Every `model.moe.*` key of the schema contract (Table 20) belongs to exactly one row.

Axis	Question	Knobs	Section
Scoring	how much weight does each selected expert get?	`router.*`	18.3
Routing	who picks whom?	`routing.type`, `routing.hash_*`	18.4
Balancing	how do we keep every expert employed?	`balancing.*`	18.5
Experts	how many, how wide, shared or latent?	`num_experts`, `expert_hidden_size`, `shared.*`, `latent.*`, `experts.*`	18.6
Layout	which layers are MoE at all?	`layout.*`	18.7
Capacity	what happens when an expert overflows?	`routing.capacity_factor`	18.8
Monitoring	how is all of it observed?	logged scalars	18.9

18.2 The two failure modes: collapse and the straggler

Four of the seven questions exist because of two failure modes. Both follow from one design decision—the router is learned and free to concentrate—and every mechanism in the rest of this appendix (auxiliary loss, expert bias, expert choice, hash tables, capacity factor) is an answer to one of them.

Collapse. Routing decisions are uncoordinated: each token independently picks its own k favourites, and nothing stops every token from picking the same expert. The loop is self-reinforcing. An expert that starts slightly ahead receives more tokens; more tokens mean more gradient, so it improves faster than the others; improving faster, it wins even more tokens on the next batch. Run the loop forward and a layer built with E experts routes through two or three of them, and the rest never train at all. Switch Transformers observed whole layers collapsing to a single expert [43]. This is the default trajectory of an unbalanced MoE, not a rare accident, and it is why Section 18.5 exists: the auxiliary loss and the expert bias are two independent ways to break the loop, expert choice (Section 18.4) removes it by construction, and hash layers remove it by fixing the selection outright.

The straggler. Even a layer that balances well on average fluctuates per step: one batch may send a third of its tokens to one expert. Two costs follow. The step cannot finish until the busiest expert finishes, so a single hot expert sets the time of everything, and in distributed training—where different experts live on different devices—the imbalance becomes rows of idle devices waiting for one. And an expert that receives a burst needs buffer memory proportional to the burst. The capacity factor (Section 18.8) bounds both by capping how many assignments an expert must honour. The gauge for both failure modes is MaxVio (Section 18.5): collapse and stragglers both read as the busiest expert pulling away from the mean.

18.3 Router scoring: from logits to weights

The router is one linear layer, $z_t = W_r x_t \in \mathbb{R}^E$ with $W_r \in \mathbb{R}^{E \times d}$: one logit per expert. A gating function converts the E logits into E scores, and the top-k scores select the experts. It helps to separate the two jobs of the gating function: it *ranks* (which experts are selected) and it *weights* (how much each selected expert contributes). The implementation supports the three gating

functions used by current open models:

$$\text{softmax}: \quad s_{t,e} = \frac{e^{z_{t,e}}}{\sum_{e'} e^{z_{t,e'}}}, \tag{223}$$

$$\text{sigmoid}: \quad s_{t,e} = \sigma(z_{t,e}), \tag{224}$$

$$\text{sqrt-softplus}: \quad s_{t,e} = \sqrt{\text{softplus}(z_{t,e})}. \tag{225}$$

All three are monotone in $z_{t,e}$—a larger logit always yields a larger score—and monotone functions never reorder. So for a given z_t all three select *exactly the same experts*; the choice of gating function is a choice about the weights, not about the selection. In words:

- **softmax** makes experts compete for a fixed pie. The scores sum to one, so expert e's score depends on every other logit through the denominator: raising one logit lowers all the others. As logits grow, the distribution sharpens towards winner-takes-all.
- **sigmoid** judges each expert in isolation. $\sigma(z_{t,e})$ depends only on its own logit, so there is no competition: all experts can score low together (the selected weights sum to less than one and the routed branch shrinks) or high together. The renormalisation and scaling factor below exist largely to manage exactly this freedom [28].
- **sqrt-softplus** sits between the two: like sigmoid it is per-expert, but $\sqrt{\text{softplus}(z)}$ decays more gently on the negative side and never saturates at exactly zero [29].

`router.scoring` selects the function (default `softmax`).

Output scale. Two knobs control the magnitude of the routed branch, and they only make sense as a pair. With `router.renormalize` (default true), the selected scores are rescaled to sum to one,

$$\hat{w}_{t,j} = \frac{s_{t,i_{t,j}}}{\sum_{l=1}^{k} s_{t,i_{t,l}}}, \tag{226}$$

so the routed branch becomes a weighted *average* of k expert outputs rather than a sum. Averaging shrinks. Take $k = 4$ experts whose outputs each have root-mean-square 1 and are uncorrelated, and near-equal weights of $\approx 1/4$: the variance of the average is $\frac{1}{4^2}(1+1+1+1) = \frac{1}{4}$, so its RMS is $1/\sqrt{k} = 0.5$. The average is half the size of any single expert's answer, and the shrinkage worsens as k grows. Renormalisation still pays for itself—it removes the token-to-token variation of the top-k probability mass and lowered validation loss in the granularity study of Krajewski et al. [88]—so the routed scaling factor g (`router.routed_scaling_factor`, default 1) is there to undo the shrinkage: $g \approx \sqrt{k}$ restores the RMS of a single expert. The published values match the derivation. Kimi K2 uses $2.827 \approx \sqrt{8}$; DeepSeek-V3 uses $g = 2.5$ at $k = 8$, slightly below $\sqrt{8} \approx 2.83$ because real expert outputs are partly correlated and cancel less than independent ones [28]; and Kimi K3 uses $g = 1.0$ at $k = 16$ with two shared experts carrying part of the branch [82].

Pre-softmax. With `router.pre_softmax` the softmax runs over all E experts *before* the top-k cut, so the kept weights are raw probabilities from the full distribution and are not renormalised. To see why softmax routing at k=1 requires this, trace the default path step by step: select first, then softmax over the *selected* logits only. With one selected logit that softmax is $e^z/e^z = 1$—always, regardless of how well the expert matches the token. A weight that is constantly one has derivative zero, so the router would receive no gradient and could never learn which expert to prefer: selection would be frozen at its initialisation. Pre-softmax repairs it: the kept weight is the top probability of the full softmax, a genuine number in $(0, 1)$ that moves whenever any logit moves, so the gradient flows. The configuration layer detects that combination, enables `pre_softmax` automatically, and emits a warning instead of rejecting legacy configs.

Sharpness and precision. Three mechanisms control how peaked, and how numerically trustworthy, the router is. *Input jitter* (`router.jitter_eps`) multiplies the token by $x \cdot (1 + \epsilon(2u - 1))$ with u uniform before the projection: random multiplicative noise that penalises routing decisions built on fragile directions of x_t, a regulariser Switch was trained with [43]. *Router precision* (`router.dtype`) defaults to `float32`; the router is tiny ($d \times E$ parameters), so full precision costs nothing, and what it buys is stability—a bf16 router diverged in Switch Transformers, and in low precision tied top-k logits are broken arbitrarily [43]. Finally, the *z-loss* of ST-MoE penalises logit drift. The log-sum-exp is a smooth maximum of the logits (large whenever *any* logit is large), so

$$\mathcal{L}_z = \beta \frac{1}{T} \sum_{t=1}^{T} \Big(\log \sum_{e=1}^{E} e^{z_{t,e}} \Big)^2, \tag{227}$$

grows quadratically whenever any router logit grows in magnitude. Minimising it keeps the logits small, the softmax unsaturated and the router from becoming overconfident [187]. Here β is `router.z_loss_coeff` (default 0; ST-MoE uses 10^{-3}); the loss is returned as `moe_router_z_loss` and added to the task loss.

18.4 Routing strategy: who selects, and how

Section 18.3 fixed *how much* each chosen expert weighs; this one fixes *who chooses whom*. The literature has three answers, and each has a structural consequence.

Token choice. The default (`routing.type: token_choice`): each token keeps its own k best experts under $s_t + b$, where b is the balancing bias of Section 18.5. The consequence is that nobody guarantees any expert work: an expert can win every token or none, which is exactly the collapse loop of Section 18.2. Token choice therefore ships with a balancing mechanism. It is the strategy trained by Megatron and used by most open models [43, 28].

Expert choice. Flip the direction of the decision: with `routing.type: expert_choice`, each expert keeps its best Tk/E tokens [183]. Every expert receives exactly the same number of tokens, so loads are equal by construction and the balancing machinery is not merely unnecessary but contradictory—the schema forbids combining expert choice with an auxiliary loss or an expert bias. The price is causality: whether a token is served now depends on the *other* tokens in the batch, including later ones. That is harmless in an encoder, where the whole sequence is available at once, and fatal in autoregressive decoding, where token t must be routed before tokens $t{+}1, t{+}2, \ldots$ exist; training-time and inference-time routing would disagree by construction. Hence the recommendation: encoder and training experiments yes, autoregressive serving no.

Hash layers. Dispense with the learned decision altogether: the first `routing.hash_layers` MoE layers select experts from a fixed table $\text{tid} \to (e_1, \ldots, e_k)$ [138], as DeepSeek-V4 does in its first three MoE layers [29]. The observation behind it: in the first layers, tokens are still close to their embeddings and the routing decision is mostly surface identity—which token it is, not what it means—and a lookup table captures that. The table never changes, so it cannot collapse. The learned router still produces the mixing weights, so it keeps training through them; it just receives no gradient from the selection itself. The table is built from a seeded random assignment (`hash_table: random`) or a round-robin cycle (`round_robin`); Roller et al. found random and frequency-greedy tables reach similar perplexity, with the greedy table better balanced [138]. Because a fixed table can never be rebalanced, the schema forbids combining it with `balancing.bias_update`.

18.5 Load balancing

Section 18.2 described the collapse loop: an expert slightly ahead receives more tokens, trains faster, and pulls further ahead. Two independent remedies are implemented; either works alone, and DeepSeek-V3 combines both [28]. They differ in *how* they reach the routing: the auxiliary loss works through the gradient, the bias works around it.

Auxiliary loss. The obstacle to fixing imbalance directly is that the thing to fix—how many times each expert was chosen—is a count, and counts have no derivative: being chosen 500 times instead of 499 is not an operation backpropagation can influence. The Switch surrogate [43] multiplies each expert's share of the assignments by its mean router probability,

$$\mathcal{L}_{\text{aux}} = \alpha\, E \sum_{e=1}^{E} f_e P_e, \qquad f_e = \frac{1}{Tk} \sum_{t,j} \mathbb{1}[i_{t,j} = e], \qquad P_e = \frac{1}{T} \sum_{t} s_{t,e}, \tag{228}$$

where f_e (the fraction of the Tk assignments) carries the imbalance but no gradient, and P_e (the mean score) carries gradient. The product is the trick: lowering the loss means lowering P_e precisely where f_e is high, so the optimiser leans on the router's probabilities for exactly the overloaded experts, and future selections spread. Under softmax scoring the scores sum to one per token and the minimum sits at the uniform distribution, $f_e = k/E$ and $P_e = 1/E$; the prefactor E then makes the loss at the minimum independent of the expert count (exactly α at k=1), so one coefficient means the same thing at E=8 and E=256. The coefficient itself is a trade-off: too large and it fights the language-model loss, too small and MaxVio stays high [163]; the default is 0.01, close to Switch and DeepSeekMoE [43, 26]. The scope is selected by `balancing.aux_loss`: `batch` (one micro-batch, the default), `sequence` (one loss per sequence, DeepSeek-V3's guard against a single sequence piling onto a few experts—balance the batch and one long document can still monopolise its favourite experts), or `none`. Balancing per sequence or per micro-batch can prevent domain experts from forming, whereas global-batch balancing leaves each device's load uneven [130]; the single-device runtime implements the first two scopes and documents the global scope as future work.

Auxiliary-loss-free bias. The second remedy ignores gradients entirely and works like a thermostat: measure the temperature, nudge the knob, repeat. With `balancing.bias_update` set to `sign` or `centered_sign`, each expert carries a bias b_e added to the scores *before selection only—never to the mixing weights*—and moved once per optimizer step:

$$b_e \leftarrow b_e + \gamma\, \operatorname{sign}(\bar{c} - c_e)\,, \qquad \bar{c} = \frac{1}{E} \sum_{e'} c_{e'}, \tag{229}$$

where c_e counts the step's assignments and γ is `balancing.bias_update_rate` (default 10^{-3}). An expert above the mean load ($c_e > \bar{c}$) has its bias lowered by γ, so it wins fewer tokens on the next step; an expert below the mean has its bias raised. Because the bias never touches the mixing weights, it redirects traffic without changing what the selected experts contribute—which is why it costs no language-model loss, unlike the auxiliary loss [163].

Why the schema restricts the bias to sigmoid or sqrt-softplus scoring: under softmax every score depends on every logit through the denominator, so shifting one expert's logit rescales all E scores at once and the bias would steer the selection *and* distort every mixing weight at the same time. Under sigmoid or sqrt-softplus each score is a function of its own logit alone, so b_e moves expert e in the ranking and touches nothing else—steering stays pure.

The `centered_sign` variant subtracts the mean update ($\frac{1}{E}\sum_e \operatorname{sign}(\bar{c} - c_e) = 0$ in exact arithmetic, not necessarily in reduced precision) so the bias mean does not drift, which matters when the bias is stored in reduced precision [163]. The bias is a non-persistent fp32 buffer: it is

training state, never an inference parameter, and `eval()` freezes it. The update is applied by `update_moe_expert_bias()` after `optimizer.step()` in the trainer.

MaxVio. Balance is read from the maximal violation,

$$\text{MaxVio} = \frac{\max_e c_e - \bar{c}}{\bar{c}}, \tag{230}$$

a single number per layer, logged as `moe/max_vio`. How to read it: 0 is perfectly even; 0.2–0.5 is ordinary fluctuation; 1 means the busiest expert receives twice the mean load [163]. The absolute value matters less than the trend—a MaxVio that climbs over steps is the collapse loop of Section 18.2 running in real time.

18.6 Experts: count, width, latent space, sharing

Count and width. The two budgets of Section 18.1 are spent separately here: `num_experts` × `expert_hidden_size` is (roughly) the total routed parameters—what memory pays—while `top_k` × `expert_hidden_size` is the per-token compute—what the user pays. `expert_hidden_size` sets each expert's intermediate width and falls back to `ffn.hidden_size` when unset. Keeping the two widths separate is essential for real architectures: DeepSeek-V3 has a dense width of 18432 but experts of width 2048 [28]—its dense layers are nine times wider than its experts. Granularity is the refinement of the same trade: at a fixed parameter budget, "16 experts of width 4096" and "64 experts of width 1024" hold the same total. The finer split lets each expert specialise more narrowly, which improved loss up to 128 experts of width $d/4$ with top-8 in one study [98]; but narrower experts also see fewer tokens each per step, and once training throughput is priced in, the same total and active parameters are predicted to favour a *less* sparse configuration [88]. The honest summary: finer helps at fixed FLOPs, and the evidence at fixed wall-clock is mixed.

Latent MoE. Ship a compressed copy instead of the whole token. LatentMoE wraps the experts between a down- and an up-projection: each token is projected from d to a latent width ℓ before dispatch, and the combined result is projected back after [39],

$$\ell_t = W_{\downarrow} x_t, \qquad y_t = W_{\uparrow} \sum_{j=1}^{k} \hat{w}_{t,j} E_{i_{t,j}}(\ell_t). \tag{231}$$

Each of the k dispatched copies now carries ℓ numbers instead of d. With compression ratio $\alpha = d/\ell$, moving and processing one copy costs α times less, so the same expert parameters, compute and traffic afford α times more experts and a larger top-k. The bet is that experts can work on a lossy sketch of the token; NVIDIA reports it holds up to $\alpha \leqslant 4$ (+5.65 MMLU-Pro points at 95B total with $\alpha = 4$) [39]. Kimi K3 adds an RMSNorm before the up-projection to stop routed-branch activation growth at 2.8T parameters, exposed as `latent.norm_before_up_proj` [82].

Shared experts. A routed layer has no guarantee that common knowledge gets represented: each expert learns whatever its visitors need, so E experts may each re-learn the same syntax and punctuation. `shared.count` always-on experts run outside the router on *every* token—the general practitioners of the clinic in Section 18.1—absorbing the knowledge every token needs and freeing the routed specialists to spend their capacity on what differs across tokens [26]. Their width is `shared.hidden_size` (default: the routed width). They are added at weight one, or multiplied by a learned per-token gate $\sigma(w_s^\top x_t) \in (0, 1)$ when `shared.gate: sigmoid` (Qwen3-Next) [131]; the fixed-weight form matches DeepSeek-V3 [28]. They are not free: a

shared expert runs on every token, so per-token compute grows by its width, and measured quality gains are small and inconsistent [95]. Add one when you would otherwise raise `top_k`, not on top of it.

Activation clamp. A rare but real pathology: individual experts whose output norms grow while the loss stays flat—Step 3.5 Flash found such outliers, and their growth was contained by clipping at unchanged loss [71]. `experts.activation_clamp` bounds the gated intermediate,

$$\mathrm{GatedFFN}(x) = W_{\mathrm{down}}\big(\phi\big(\mathrm{clamp}(xW_{\mathrm{gate}}, -c, c)\big) \odot (xW_{\mathrm{up}})\big)\,, \tag{232}$$

with $c =$ `activation_clamp` (0 disables it). DeepSeek-V4.1 uses $c = 10$ and Step 5 Preview $c = 7$ [30]. Treat it as insurance against rogue experts, not as a default: it exists so that the monitoring signal of Section 18.9 (expert-norm ratio rising) has a cheap remedy. The exact SiTU-GLU smooth cap used by Kimi K3 is approximated here by a gate-input clamp and documented as a deviation [82].

18.7 Layer layout

Not every layer needs to be an MoE layer. The first layers of a transformer compute generic features that every token needs regardless of content; specialisation pays later. `layout.dense_layers` keeps the first N layers dense (DeepSeek-V3 and GLM keep three), and `layout.every_n_layers` interleaves dense and MoE layers (Llama 4 Maverick makes every second layer an MoE layer) [28, 114]. Dense layers use `ffn.hidden_size`; MoE layers use `expert_hidden_size`. The model config exposes `is_moe_layer(i)`, `moe_layer_mask()` and `moe_layer_count()`; the encoder, decoder, ViT and audio stacks build a routed or dense FFN per physical layer accordingly. Leading dense layers reduce the number of MoE layers at little cost at matched FLOPs [88], but they do not by themselves fix early-layer imbalance: the first MoE layers after the prefix are still the most collapse-prone and remain the ones to monitor.

18.8 Capacity and token dropping

By default routing is dropless: an expert accepts every token the router sends it (`routing.capacity_factor: 0`), and per-step loads fluctuate freely. A positive factor κ caps each expert at

$$C = \left\lceil \kappa \frac{Tk}{E} \right\rceil \tag{233}$$

assignments. The formula is the mean load with headroom: T tokens each make k requests, giving Tk assignments shared by E experts, so the mean is Tk/E and κ is the slack. Worked example: $T = 4096$ tokens, $E = 8$, $k = 2$ gives 8192 assignments and a mean of 1024 per expert; $\kappa = 1.5$ sets $C = 1536$. An expert that receives 2000 requests serves its 1536 highest-scoring ones and drops the other 464: those tokens leave the layer without that expert's contribution (its weight is zeroed), and `dropped_fraction` records the share of removed assignments (464 of 8192 $\approx 5.7\%$ in the example). What the cap buys is the straggler remedy of Section 18.2: a hard upper bound on each expert's buffer (memory) and, in distributed training, the guarantee that no hot expert sets the step time. The cost is silent quality loss: Pangu Ultra MoE dropped about 6–8% of tokens at $\kappa = 1.5$ and its dropless run kept a lower loss, a gap that grew with the expert count [121], and MegaBlocks reached its lowest loss when no token was dropped [46]. Dropless remains the default and the recommended setting for fine-tuning, where dropped tokens can fall inside instruction responses.

18.9 Monitoring

Each `MoELayer` exposes `last_auxiliary_losses` (coefficient-weighted `moe_load_balancing_loss` and `moe_router_z_loss`) and `last_moe_stats`. The model aggregates them over layers through `aggregate_moe_aux_losses()` and adds the losses to the task loss; the trainer logs the scalars to the metrics callback and to Weights & Biases. Read as five questions, one signal each:

- **Is the load even?** MaxVio (`moe/max_vio`) per layer: near 0 healthy, $\geqslant 1$ severe (Section 18.5); a value rising across steps is the collapse loop in progress.
- **Has the router committed?** Routing entropy and top-1 weight. A flat gate that merely averages its top-k experts shows up as entropy near $\log k$ and top-1 weight near $1/k$ (no specialisation); a winner-takes-all gate shows entropy near 0 and top-1 near 1. Healthy training drifts from the first state towards committed, structured routing without reaching the second extreme.
- **Are the experts healthy?** The expert-norm ratio: the largest per-expert output norm over the layer median. Far above one means a rogue expert dominating the branch—the pathology the activation clamp contains; pinned at zero means a dead expert nobody selects [71].
- **How hard is balancing pulling, and is the router drifting?** The load-balancing loss and the z-loss (coefficients already divided out in the logged statistics): the first should relax as balance improves, the second tracks logit drift.
- **Is capacity binding?** The dropped fraction: share of assignments removed by the capacity factor; it is 0 unless `routing.capacity_factor` is positive.

Five rules of thumb close the on-ramp, each a consequence of the sections above rather than new machinery:

1. Start with the defaults: dropless routing, batch-scope auxiliary loss at 0.01, no bias. They are the configuration most presets converge to after their own deviations.
2. If balance is the problem, reach for the bias before raising the auxiliary coefficient: the bias redirects traffic through the selection only and costs no language-model loss [163], while a larger α fights the task loss.
3. Watch MaxVio on the first MoE layers after any dense prefix; that is where collapse starts (Section 18.7).
4. Softmax at k=1 needs `pre_softmax` or the router never learns (Section 18.3); the loader repairs it, but the warning is worth understanding.
5. Budget compute as `top_k` $\times$ `expert_hidden_size` + `shared` before enabling: shared experts and latent width change the per-token bill, and the cheapest fix for slow experts is a smaller budget, not more machinery.

18.10 Schema contract

Table 20 lists every `model.moe.*` key, its default, what it controls, and the governing reference. The JSON schema is the executable contract; runtime validation adds the cross-key rules: $top_k \leqslant num_experts$, softmax at $k = 1$ implies pre-softmax, the bias requires sigmoid or sqrt-softplus scoring, expert choice is incompatible with the balancing terms, hash layers require token choice and forbid the bias, the latent width must lie in $(0, d)$, and the layer count must be positive.

Table 20: MoE schema contract (`model.moe.*`).

Key	Default	Controls	Reference
`enabled`	`true`	routed vs. dense FFN	Shazeer et al. [145]
`num_experts`	8	size of the routed pool	Fedus et al. [43]
`top_k`	2	experts used per token	Fedus et al. [43]
`expert_hidden_size`	`null`	expert width (falls back to `ffn.hidden_size`)	DeepSeek-AI [28]
`router.scoring`	`softmax`	logits→scores function	DeepSeek-AI [28, 29]
`router.pre_softmax`	`false`	softmax before the top-k cut	Fedus et al. [43]
`router.renormalize`	`true`	selected weights sum to one	Krajewski et al. [88]
`router.routed_scaling_factor`	1.0	routed-branch magnitude g	DeepSeek-AI [28], Kimi Team [82]
`router.dtype`	`float32`	router compute precision	Fedus et al. [43], Zoph et al. [187]
`router.jitter_eps`	0.0	input jitter strength	Fedus et al. [43]
`router.z_loss_coeff`	0.0	logit-drift penalty β	Zoph et al. [187]
`routing.type`	`token_choice`	who picks whom	Zhou et al. [183]
`routing.capacity_factor`	0.0	per-expert cap κ (0: dropless)	Pangu Team [121]
`routing.hash_layers`	0	first N layers routed by table	Roller et al. [138], DeepSeek-AI [29]
`routing.hash_table`	`random`	table construction	Roller et al. [138]
`routing.hash_seed`	42	table seed	Roller et al. [138]
`balancing.aux_loss`	`batch`	loss scope	Dai et al. [26], DeepSeek-AI [28]
`balancing.aux_loss_coeff`	0.01	loss weight α	DeepSeek-AI [28]
`balancing.bias_update`	`none`	thermostat off / `sign` / centred	Wang et al. [163]
`balancing.bias_update_rate`	0.001	thermostat step γ	Wang et al. [163]
`shared.count`	0	always-on experts	Dai et al. [26]
`shared.hidden_size`	`null`	shared-expert width	Dai et al. [26]
`shared.gate`	`none`	fixed weight or learned gate	Qwen Team [131]
`latent.enabled`	`false`	compress before dispatch	Elango et al. [39]
`latent.dim`	`null`	latent width ℓ	Elango et al. [39]
`latent.norm_before_up_proj`	`false`	RMSNorm before $W_\uparrow$	Kimi Team [82]
`layout.dense_layers`	0	dense prefix N	DeepSeek-AI [28]
`layout.every_n_layers`	1	MoE every n-th layer	Meta AI [114]
`experts.activation_clamp`	0.0	gate-input clamp c (0: off)	Huang et al. [71]

18.11 Named presets

The presets in `configs/` mirror the published configurations, with the documented single-device deviations. Table 21 summarises the MoE block of each MoE preset; the values were checked against the models' Hugging Face `config.json` where available. Reading the table against Section 18.2: every recent frontier preset answers collapse with the bias, the auxiliary loss, or both.

Table 21: MoE presets and their real-architecture MoE blocks.

Preset	Experts / top-k	Expert width	Router	Shared	Balancing / layout
DeepSeek-V2	160 / 6	1536	softmax, g=16	2	seq-aux 10^{-3}; 1 dense
DeepSeek-V3	256 / 8	2048	sigmoid, g=2.5	1	bias + seq-aux 10^{-4}; 3 dense
GLM-4.5/4.7	160 / 8	1536	sigmoid, g=2.5	1	bias; 3 dense
GLM-5	256 / 8	2048	sigmoid, g=2.5	1	bias; 3 dense
Kimi K2	384 / 8	2048	sigmoid, g=2.827	1	bias + seq-aux; 1 dense
Kimi K3	896 / 16	3072	sigmoid, g=1	2	centred bias; latent 3584; 1 dense
gpt-oss-120b	128 / 4	2880	softmax	0	aux, clamp 7
Llama 4 Scout	16 / 1	8192	softmax, pre	0	every layer
Llama 4 Maverick	128 / 1	8192	softmax, pre	0	every 2nd layer
MiniMax-Text-01	32 / 2	9216	sigmoid	0	aux 10^{-3}
MiniMax-M2	256 / 8	1536	sigmoid	0	bias + aux 10^{-3}
Qwen3-Next-80B	512 / 10	512	softmax	1, gated	aux 10^{-3}
Qwen3.8 2.4T-A95B	512 / 10	2048	softmax	1	aux 10^{-3}

Single-device deviations. The runtime is CPU-first and single-device, so it implements the model-level semantics and leaves distributed mechanisms to future work. Documented deviations: DeepSeek-V2's device-level loss is approximated by the batch auxiliary loss; Kimi K3's quantile/PID balancing is approximated by the centred sign rule; Kimi K3's SiTU-GLU smooth cap is approximated by `activation_clamp: 4`; Llama 4's input-side score scaling is not applied (the router weights are applied to the expert output); group-limited routing, expert parallelism, token dispatchers, expert replicas/EPLB, and FP8/FP4 expert execution are out of scope. DeepSeek-V4's hash layers are supported through `routing.hash_layers`.

18.12 Training interaction

The expert bias is updated after the optimizer step from counts accumulated across the step's forwards, so gradient accumulation and micro-batching do not change its semantics. The router receives no bias gradient, and the bias itself is a non-persistent buffer; exported checkpoints therefore contain no training state—they remain inference-clean. Auxiliary losses are coefficient-weighted tensors in `last_auxiliary_losses` and are summed into the task loss by the trainer; they are excluded from evaluation. The router and the experts are categorised as `router` and `ffn` parameter blocks respectively, so per-block gradient norms and parameter counts match the estimator taxonomy. Finally, the MoE layer is token-wise and therefore cache-safe: the KV-cache parity suites keep the FFN dense because a discrete top-k can flip under the tiny numerical differences between full and cached forwards—a property of discrete routing, not of the cache.

18.13 Related systems and future work

The appendix's design follows the MoE handbook's anatomy (router, routing, balancing, experts, layout, capacity) and the systems literature it surveys. Future work in this codebase: expert parallelism and all-to-all dispatchers [93, 43]; global-batch auxiliary losses [130]; quantile/PID balancing [74]; dense-to-MoE upcycling and expert growth [87]; frequency-balanced hash tables [138]; exact SiTU-GLU; and shortcut-connected or parallel MoE blocks that overlap dispatch with compute [18, 112].

19 Model-Wide Positional Encodings

This annex documents the unified positional-encoding (PE) subsystem that applies a single `model.positional_encoding` choice to every attention mixer in the network. The schema contract is:

$$\texttt{positional_encoding} \in \{\texttt{rope}, \texttt{hope}, \texttt{nope}, \texttt{alibi}, \texttt{bam}, \texttt{pape}, \texttt{pape_efficient}, \texttt{pape_ri}, \texttt{sinusoidal_absolute}, \texttt{sinusoidal_rotary}, \texttt{learned_absolute}, \texttt{none}\}$$

The implementation lives in `src/model/embeddings/` and the shared injection helpers in `src/model/attention/common.py`. Bibliography keys cited here resolve against `docs/bibliography/`.

19.1 Architecture: One Shared Module, Per-Mixer Opt-Out

A single PE module is built once in the encoder (`FrankensteinEncoder` or `FrankensteinViT`) via `build_pos_encoder(config)` and injected into every `HybridLayer`, which forwards it to each mixer's `__init__(config, pos_encoder=None)` and `forward(..., pos_encoder=None)`.

Each mixer carries a per-mixer `use_pe` flag (`<mixer>_use_pe`, default `True` for attention mixers, `False` for recurrent/decay mixers such as `retnet`, `mamba`, `ode`, `gla`, `deltanet`, `mtla`, and the six Falcon mixers `falcon{1,2,3,1a,2a,3a}`). When `use_pe` is `False` the mixer ignores the injected module entirely.

The PE is applied in two phases via the shared helpers `apply_pe_to_qk()` (called before the attention-score computation) and `apply_pe_to_scores()` (called after). The helper dispatches on the PE type and falls through to a no-op when the mixer's `use_pe` flag is `False`.

19.2 Injection Styles

The 11 enum values fall into four injection styles:

1. **Rotation** (`rope`, `hope`, `sinusoidal_rotary`): rotates the query and key tensors before the score computation. The inner product $\mathbf{q}_i^\top \mathbf{k}_j$ becomes a function of the relative offset $i - j$.
2. **Score bias** (`alibi`): adds a distance-proportional bias to the attention-score matrix *after* it is computed.
3. **Q/K augmentation** (`pape`, `pape_efficient`, `pape_ri`): concatenates extra positional dimensions onto the query and key vectors before the dot product, increasing the effective head dimension.
4. **Additive at embedding level** (`sinusoidal_absolute`, `learned_absolute`): added directly to the token embeddings before the first layer; no per-mixer action is taken in the attention block.

`nope` and `none` are both no-ops: `nope` explicitly disables positional encoding (the model must learn position implicitly), while `none` is the null module returned by the factory when no PE is desired.

19.3 Mathematical Formulations

19.3.1 RoPE — Rotary Position Embedding

RoPE [150] rotates each head's query and key by an angle proportional to the token index. For head dimension d_h and token position m, the rotation matrix $\mathbf{R}_m \in \mathbb{R}^{d_h \times d_h}$ acts on 2-D

sub-spaces (x_{2i}, x_{2i+1}):

$$\mathbf{R}_m = \mathrm{diag}\left(\begin{bmatrix} \cos m\theta_0 & -\sin m\theta_0 \\ \sin m\theta_0 & \cos m\theta_0 \end{bmatrix}, \ldots, \begin{bmatrix} \cos m\theta_{d_h/2-1} & -\sin m\theta_{d_h/2-1} \\ \sin m\theta_{d_h/2-1} & \cos m\theta_{d_h/2-1} \end{bmatrix} \right),$$

with frequencies $\theta_i = \texttt{rope_base}^{-2i/d_h}$. The attention score becomes $\mathbf{q}_m^\top \mathbf{R}_{m-n} \mathbf{k}_n$, a function of the relative offset $m-n$. A `rope_scaling` factor s rescales the effective position to m/s for length-extrapolation modes.

19.3.2 HoPE — Hyperbolic Rotary Positional Encoding

HoPE [25] generalises RoPE to the hyperbolic geometry of the Lorentz model. Instead of Euclidean rotations, HoPE applies Lorentz rotations built from hyperbolic functions, enforcing a *monotonic decay* of attention weights with increasing token distance — a property RoPE lacks because its oscillatory attention pattern can re-spike at long ranges. RoPE is recovered as a special case. The `hope_base` and `hope_damping` knobs control the hyperbolic frequency and the decay rate.

19.3.3 NoPE — No Positional Encoding

NoPE [77] is the explicit absence of any positional injection. The model must discover position implicitly from the causal mask (decoder) or from the data itself. The cited work shows that NoPE outperforms explicit PE methods (including RoPE and ALiBi) on length-generalisation benchmarks while requiring no additional computation, because a causal-attention decoder can implicitly encode relative position through its attention patterns.

19.3.4 ALiBi — Attention with Linear Biases

ALiBi [127] does not modify the embeddings or the query/key vectors. Instead it adds a static, head-specific linear bias to the attention-score matrix:

$$\mathrm{score}_{i,j}^{(h)} = \frac{\mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)}}{\sqrt{d_h}} - m_h \cdot (i-j), \qquad m_h = \frac{1}{2^{\lfloor h/\lceil H/8 \rceil \rfloor}},$$

where m_h is a geometric slope per head h and the bias penalises distant keys. The `alibi_num_heads` knob controls the slope schedule's head count. ALiBi enables length extrapolation: a model trained on 1024 tokens generalises to 2048 without retraining.

19.3.5 BAM — Bayesian Attention Mechanism

BAM [13] reframes positional encoding as a probabilistic prior over positions and unifies existing methods: NoPE corresponds to a Uniform prior, ALiBi to a Uniform×Laplace prior. The practical instantiation, GGD-BAM, adds a Generalized-Gaussian relative-position bias matrix $\mathbf{B}$ to the attention logits, exactly like ALiBi but with a non-linear, learnable shape:

$$B_{i,j}^{(h)} = -\left(|j - \alpha_h\, i - \mu_h| + \varepsilon\right)^{\beta_h}, \qquad \alpha_h = e^{-\theta_\beta^{(h)}}\, \theta_\alpha^{(h)} \geq 0,$$

with per-head learnable $\theta_\alpha, \theta_\beta$ (init `bam_theta_init`=0, the Uniform prior) and optional location θ_μ (fixed to 0 by default via `bam_learn_mu=false`). The floor ε (`bam_eps`, default 10^{-5}) stabilises the exponentiation when $\beta < 0$, a relaxed regime that suppresses local context and acts as a long-range "retrieval head". Special cases: $\beta = 1, \mu = 0$ recovers ALiBi; $\beta = 2$ gives a Normal prior; $0 < \beta < 1$ yields heavier tails than Laplace. The paper pairs BAM with Scalable Softmax (SSMax), a transversal logit rescale $s \cdot \ln(n)$ (learnable per-head s) that counteracts "attention fading" and is composable with any PE via the top-level `use_ssmax` flag. BAM SSMax achieves accurate information retrieval at 500× the training context length.

19.3.6 PaPE — Parabolic Position Encoding

PaPE [119] is a vision-centric encoding that augments the query and key with extra positional dimensions built from parabolic basis functions. Given P parabolas (`pape_num_parabolas`) and S positions (`pape_num_positions`), the positional feature for coordinate $\mathbf{p} \in \mathbb{R}^D$ is

$$\phi_k(\mathbf{p}) = \sum_{j=1}^{S} \alpha_{k,j}\, f_j(\mathbf{p}), \qquad f_j(\mathbf{p}) = -\|\mathbf{p} - \mathbf{c}_j\|^2 + b_j,$$

a sum of S paraboloids centred at $\mathbf{c}_j$ with offset b_j. The $P \cdot S$ extra dimensions are concatenated to the query and key before the dot product, so the effective head dimension grows from d_h to $d_h + P \cdot S$. Three variants are provided:

- `pape` — the reference implementation.
- `pape_efficient` — a pure-PyTorch reformulation that avoids custom Triton kernels while preserving the same math.
- `pape_ri` — the rotation-invariant variant (PaPE-RI in the paper), which satisfies rotation invariance by construction and is controlled by `pape_rotation_invariant`.

PaPE is designed for n-D vision tokens (images, video, event cameras, point clouds) and is the recommended PE for the ViT path.

19.3.7 Sinusoidal Absolute

The original Transformer absolute encoding [159] adds a fixed sinusoidal vector to the token embedding:

$$\mathrm{PE}_{(m,2i)} = \sin\left(\frac{m}{\texttt{sinusoidal_base}^{2i/d}}\right), \quad \mathrm{PE}_{(m,2i+1)} = \cos\left(\frac{m}{\texttt{sinusoidal_base}^{2i/d}}\right),$$

with `sinusoidal_max_len` capping the precomputed table and `sinusoidal_scale` multiplying the embedding. This is an additive PE: it is applied once at the embedding level and the mixers do not apply any per-layer PE.

19.3.8 Sinusoidal Rotary

A rotary variant of the sinusoidal encoding: the same sinusoidal frequencies are used to build a rotation matrix applied to the query and key (as in RoPE), but with the `sinusoidal_base` and `sinusoidal_scale` knobs. This is a rotation-style PE applied per-layer via `apply_pe_to_qk`.

19.3.9 Learned Absolute

A learnable absolute encoding: an `learned_max_len` $\times\, d$ parameter matrix is added to the token embeddings at the embedding level. Initialisation standard deviation is controlled by `learned_init_std`. As with sinusoidal absolute, the mixers take no per-layer action.

19.4 Per-Mixer `use_pe` Flags

Each mixer exposes a `<mixer>_use_pe` config field (e.g. `standard_use_pe`, `titan_use_pe`). The defaults are:

- **True** (attention mixers): `standard`, `ca`, `titan`, `sigmoid`, `gated_softmax`, `gqa`, all latent mixers (`mla`, `gqla`, `mlra`, `tucker`, `iha`, `gta`, `cca`, `ccgqa`, `gma`), all sparse mixers (`longformer`, `bigbird`, `sparse_transformer`, `sparsek`, `nsa`, `msa`, `sparda`, `fasa`, `sparge`), and `engram`, and `memory` (RoPE on the scoring pair only; values are built from the unrotated keys).

- **False** (recurrent/decay mixers): `retnet`, `mamba`, `ode`, `gla`, `deltanet`, `gated_deltanet`, `gated_deltanet2`, `hgrn2`, `fox`, `fpa`, `kda`, `mtla`.

The `False` default for recurrent/decay mixers reflects that these architectures bake their own temporal/positional inductive bias into their recurrence (e.g. RetNet's complex rotation, Mamba's selective scan). Overriding `use_pe=True` is allowed but is typically redundant.

19.5 Vision Transformer Unification

The ViT path (`FrankensteinViT`) previously used its own `pos_embedding_type` field with values `learned_1d`, `none`. This has been unified with the model-wide enum: `learned_1d` is migrated to `learned_absolute`, and `none` maps to `none`. The ViT builds the same `build_pos_encoder(config)` module and injects it into its HybridLayers, so a ViT can now use RoPE, HoPE, ALiBi, PaPE, or any other PE in the enum.

19.6 Legacy Compatibility

The legacy `use_hope` boolean flag is deprecated and mapped to the new enum: `use_hope=True` → `positional_encoding=hope`, `use_hope=False` → `positional_encoding=rope`. The per-mixer `positional_encoding` override that previously existed only on `titan` is now a no-op: the model-wide enum is the single source of truth, and the TitanAttention allow-list that restricted PE to `{hope, rope, none, nope}` has been removed — all 12 values now work with every mixer.

19.7 Architectural Profile: Positional Encodings

Table 22 summarises the 12 supported positional encodings.

Name	Style	Key Idea	Params	Reference
rope	Rotation	Frequency-based rotation of Q/K; relative position via inner-product geometry	`rope_base`, `rope_scaling`	[150]
hope	Rotation	Hyperbolic Lorentz rotation; monotonic attention decay with distance	`hope_base`, `hope_damping`	[25]
nope	None	Explicit no-PE; model learns position implicitly from causal mask	—	[77]
alibi	Score bias	Static linear distance penalty added to attention scores; length extrapolation	`alibi_num_heads`	[127]
bam	Score bias	Generalized-Gaussian relative-position bias; learnable per-head shape (β) and scale (α); recovers ALiBi at $\beta = 1$; SSMax rescale via `use_ssmax`	`bam_theta_init`, `bam_learn_mu`, `bam_eps`, `use_ssmax`, `ssmax_s_init`	[13]
pape	Q/K aug.	Parabolic basis functions concatenated to Q/K; vision-centric, n-D	`pape_num_parabolas`, `pape_num_positions`	[119]
pape_efficient	Q/K aug.	Pure-PyTorch PaPE; no custom Triton kernels	same as pape	[119]
pape_ri	Q/K aug.	Rotation-invariant PaPE (PaPE-RI)	`pape_rotation_invariant`	[119]

continued on next page

Name	Style	Key Idea	Params	Reference
sinusoidal_absolute	Additive	Fixed sinusoidal added to embeddings (original Transformer)	`sinusoidal_base`, `sinusoidal_max_len`, `sinusoidal_scale`	[159]
sinusoidal_rotary	Rotation	Sinusoidal frequencies as a rotation matrix on Q/K	same as sinusoidal	[159]
learned_absolute	Additive	Learnable parameter matrix added to embeddings	`learned_max_len`, `learned_init_std`	[159]
none	None	Null module; no positional information injected	—	—

Table 22: Positional encoding catalog. "Style" is the injection mechanism (rotation, score-bias, Q/K-augmentation, additive). "Params" lists the principal config knobs.

20 Dynamic Token Embeddings (EELBERT)

20.1 Executive Summary

`embedding.dynamic` replaces the trainable $V \times H$ input embedding table with an on-the-fly *n-gram pooling hash* of the token string, following EELBERT [24]. The lookup table disappears from the parameter count, from checkpoints and from native deployment artifacts. The effect is largest in the small models where the lookup layer dominates the budget—in BERT-tiny the input embedding accounts for roughly 90% of the parameters—and it is complementary to quantization, pruning and distillation.

Dynamic embeddings are *frozen*: they are never trained and never receive gradients. They only change how a token id becomes a vector; everything downstream (mixers, normalization, residuals, routing, optimizers) is unchanged. As in the paper, the output head remains an independent, untied matrix: during masked-LM pre-training it still stores a $V \times H$ matrix, so the size advantage materialises in the backbone-only or post-fine-tuning artifact (the training checkpoint is not automatically 15× smaller). The implementation lives in `src/model/embeddings/dynamic_embedding.py`, the configuration contract is the `model.embedding.dynamic.*` block, and the tokenizer vocabulary is injected by the training, deployment and inference pipelines through `src/utils/tokenizer_vocab.py`.

20.2 Algorithm

Let w be a token string of l bytes with byte values $b_1, \ldots, b_l$, and let $d = H$ be the embedding width. The algorithm has five steps, with N the maximum n-gram size and B the hash bucket.

1. **Seeds.** Draw d seeds from a single stored random state s,

$$\mathbf{h} = (h_1, \ldots, h_d) \in \{0, \ldots, B-1\}^d, \qquad \mathbf{h} \sim \text{Uniform}(\text{seed} = s).$$

   Only the integer s is persisted; the seed vector is a deterministic function of it.

2. **i-gram signatures.** For $i = 1, \ldots, N$ compute the polynomial rolling hash of every i-gram,

$$\sigma_i(j) = \Big(\sum_{t=0}^{i-1} b_{j+t}\, a^{\,i-1-t} \Big) \bmod B, \qquad j = 1, \ldots, k_i, \qquad k_i = \max(l - i + 1, 0),$$

   with $a = 257$ (prime, larger than any UTF-8 byte). The hashes are computed incrementally: the $(i{+}1)$-gram hash reuses the i-gram hash and appends one byte, so the whole token costs $\mathcal{O}(lN)$ integer operations.

3. **Partitioned projection.** Split $\mathbf{h}$ into N contiguous parts $\mathbf{h}_i$ of length d_i with

$$\sum_{i=1}^{N} d_i = d, \qquad d_i \propto i,$$

so longer n-grams receive more dimensions. The projection matrix of each gram size is the outer product

$$\mathbf{P}_i = \sigma_i \otimes \mathbf{h}_i \in \mathbb{Z}^{k_i \times d_i}.$$

4. **Folding.** Map each entry into $[-1, 1]$ through

$$\phi(x) = \frac{(x \bmod B) - \mathbf{1}[x \bmod B > B/2]\, B}{B/2} \in (-1, 1],$$

which keeps the embedding bounded without storing or normalising a table.

5. **Pooling and concatenation.** Average each projection across its k_i rows and concatenate the N pieces:

$$\mathbf{e}_i = \frac{1}{k_i} \sum_{j=1}^{k_i} \phi\big(\mathbf{P}_i[j, :]\big), \qquad \mathbf{e} = [\mathbf{e}_1; \ldots; \mathbf{e}_N] \in \mathbb{R}^d.$$

When a token is shorter than i, $k_i = 0$ and that partition contributes a zero vector.

Because the hash is polynomial over characters, morphologically related tokens (e.g. `runner` and `running`) share many 1/2/3-grams and land close in embedding space. The paper shows this matters most for small models: replacing the n-gram hash with a purely random (position-preserving) hash costs 5.0 GLUE points for BERT-mini but only 0.3 for BERT-base.

20.3 Configuration Contract

Key (`model.embedding.dynamic.*`)	Default	Effect
`enabled`	`false`	Replace the input lookup table with the frozen hash embedding. Mutually exclusive with `embedding.factorized.enabled`.
`ngram_max`	`3`	Maximum n-gram size N; must satisfy $1 \le N \le H$.
`bucket`	`1000000007`	Hash bucket B ($10^9 + 7$ in the paper). Changing it changes every embedding.
`seed`	`42`	Random state s for the seeds; identical seeds reproduce identical embeddings.
`precompute`	`true`	Memoise the $V \times H$ table in a *non-persistent* buffer (pure caching; never saved or exported).

Table 23: Dynamic-embedding configuration. The block composes with `embedding.conv.*`, which applies a Conv1d over the H-wide dynamic stream.

The schema enforces the mutual exclusion with factorized embeddings twice: a conditional `if/then` rule in `src/schema/_conditional_rules.yaml` and a `__post_init__` check in `FrankensteinModelConfig`. The block is only valid for the text classes (`frankenstein_text_encoder` and `frankenstein_text_decoder`); the ViT uses a convolutional patch embedding and the audio frontend is not a vocabulary lookup, so `config_loader` rejects the combination for those classes.

20.4 Vocabulary Injection and Lifecycle

Hash embeddings consume token *strings*, while models receive integer ids. The configuration therefore carries a runtime-only, non-schema field `token_vocab` (id-ordered token strings). `inject_token_vocab(config, tokenizer)` extracts it from a HuggingFace tokenizer (`get_vocab()`) or from the legacy SentencePiece wrapper (`vocab`, token → id), attaches it before `build_model`, and aligns `vocab_size` with the tokenizer ids. The training engine (`src/engine.py` and `src/training/main.py`), deployment checkpoint loader and inference engine all call the helper, so the vocabulary need not appear in the YAML. Constructing `DynamicEmbedding` without a vocabulary raises an actionable error.

The checkpoint stores no embedding weights: `state_dict()` contains nothing for the input layer, and the vocabulary travels with the tokenizer saved next to the checkpoint. With `precompute: true` the computed table is a non-persistent buffer—it follows `.to(device)` but is excluded from checkpoints and exports, so it is exactly equivalent to the on-the-fly computation. With `precompute: false` the hash is recomputed per batch (lower resident memory, slower steps); the two paths agree up to float32 reduction order.

The native deployment artifacts and the GGUF exporter therefore keep the size benefit. The HuggingFace Transformers export cannot run the custom hash, so it materialises the computed table as a frozen `nn.Embedding`, records the conversion in the compatibility report and emits a warning that the EELBERT size advantage is not preserved in that artifact.

20.5 Size Accounting and Paper Results

The removed block is $V \times H$ weights; the added block is only the optional Conv1d ($kH^2 + H$) when `embedding.conv.enabled` is true. In our preset `configs/eelbert_tiny.yaml` (BERT-tiny geometry: 2 layers, $H = 128$, 2 heads, FFN reduced to 128, $V = 30{,}522$), the input table removes 3.91M weights while the untied MLM head keeps 3.94M, leaving 4.15M total parameters—the paper's own accounting, where the head is dropped only for deployment.

Table 24 summarises the reference results reported by Cohn et al. [24].

Model	Params	Size	GLUE	ΔGLUE	Latency
BERT-base	109.5M	438 MB	0.775	—	162.0 ms
EELBERT-base	86.1M	344 MB	0.760	−1.5	165.0 ms
BERT-mini	11.17M	44.8 MB	0.753	—	7.0 ms
EELBERT-mini	3.36M	13.4 MB	0.746	−0.7	9.9 ms
BERT-tiny	4.39M	17.7 MB	0.671	—	1.7 ms
EELBERT-tiny	0.48M	2.04 MB	0.666	−0.5	3.9 ms
UNO-EELBERT	0.31M	1.24 MB	0.632	−3.9	—

Table 24: Reference EELBERT results (Table 2 of 24); latency on an M1 MacBook Pro with 32 GB of RAM. UNO-EELBERT is EELBERT-tiny with the intermediate size reduced to 128.

The complexity of a token embedding is $\mathcal{O}(l\,d)$ against the $\mathcal{O}(1)$ table lookup of BERT: for the 128–768 widths used by BERT the additional latency is negligible at the base scale but becomes visible on tiny models (about 2.3× for EELBERT-tiny in the table). The paper proposes three mitigations that this implementation leaves as engineering options: a sparse fixed-width mask on the outer product ($\mathcal{O}(l\,s)$ with $s \ll d$), parallel per-token hashing, and a bounded cache of the most frequent n-grams.

20.6 Implementation Map

- `src/model/embeddings/dynamic_embedding.py` — `DynamicEmbedding`: seeds, chunked rolling hash, partitioned outer product, folding, pooling, optional Conv1d, `embedding_`

`table()`.

- `src/model/frankenstein_encoder.py` — `emb` dispatch (`dynamic` → `factorized` → `nn.Embedding`); the decoder reuses it through its encoder backbone. Mutual exclusion and range checks live in `src/model/config.py`.
- `src/utils/tokenizer_vocab.py` — `token_strings()` and `inject_token_vocab()`, shared by the engine, the CLI training path, deployment and inference.
- `src/deploy/transformers\_export.py` — frozen-table materialisation and the export warning.
- `tests/test\_dynamic\_embedding.py` — determinism per seed, $[-1, 1]$ bounds, morphology similarity, precompute vs. on-the-fly parity, zero trainable parameters, vocabulary and configuration errors, encoder/decoder forwards, schema rule and export materialisation.
- `configs/eelbert_tiny.yaml` and `configs/examples/embedding_dynamic.yaml` — presets; both are smoke-tested, and the E2E harness covers the combination as `trans_dynamic_emb`.

20.7 Scope and Limitations

Dynamic embeddings are a compression mechanism, not a new representation-learning objective: they do not approximate a trained BERT embedding, and they cannot be trained or fine-tuned themselves. Because the mapping is a deterministic function of (string, seed, B, N), changing any of those invalidates previously trained checkpoints. The design also couples the model to the tokenizer that produced the training ids: the vocabulary (and therefore the token strings) is part of the artifact contract, and a tokenizer swap silently changes every input embedding. Finally, the HuggingFace export loses the size advantage by construction, so on-device deployments should prefer the native checkpoint, the deployment package or GGUF.

21 Vision Transformer: Patch Prediction, Classification, and Segmentation

This annex documents the `frankenstein_vit` model class, a Vision Transformer (ViT) implementation [34] that reuses the modality-agnostic `HybridLayer` stack from the encoder. It supports three tasks: autosupervised masked patch prediction, image classification, and image segmentation (per-pixel or EoMT-style [78]).

21.1 Patch Embedding and Positional Encoding

Following Eq. 1 of the ViT paper, an input image $x \in \mathbb{R}^{H\times W\times C}$ is split into $N = HW/P^2$ non-overlapping $P \times P$ patches, each linearly projected to D dimensions. A Conv2d with kernel and stride P performs both the splitting and the projection in a single operation. A learnable `[CLS]` token is optionally prepended, and a learnable positional embedding is added:

$$z_0 = \left[\, x_{\text{class}};\, x_p^1 E;\, x_p^2 E;\, \ldots;\, x_p^N E \,\right] + E_{\text{pos}}, \qquad E \in \mathbb{R}^{(P^2C)\times D}, \quad E_{\text{pos}} \in \mathbb{R}^{(N+1)\times D}.$$

Positional encoding. The default positional encoding is a learnable 1D `nn.Parameter` of shape $(1, N+1, D)$, faithful to the original ViT. The alternative `none` drops the explicit embedding and relies on RoPE/HoPE applied inside the attention mixers (Appendix 19), which is consistent with the encoder's modality-agnostic handling of rotary embeddings.

`[CLS]` **token.** A learnable `[CLS]` token is optionally prepended at position 0 for classification: the output of position 0 after the final normalization is fed to the task head. When `[CLS]` is disabled, classification relies on Global Average Pooling (GAP) over the patch tokens.

21.2 Masked Patch Prediction (Autosupervised)

The autosupervised task mirrors the BERT-style objective of the ViT paper (App. B.1.2). A corruption ratio of 50% is applied to the patch tokens: of the corrupted subset, 80% are replaced by a learnable mask token, 10% are replaced by a random patch, and 10% are kept unchanged.

Three reconstruction targets are supported:

1. **3-bit mean color** (default): each patch's mean color is quantized to 3 bits per channel ($2^{3\times 3} = 512$ bins), and a 512-way cross-entropy is computed. This target is the most stable and best-performing in the ViT paper.
2. 4×4 **downsampled 3-bit**: the patch is downsampled to a 4×4 grid of 3-bit colors, producing 16×512 cross-entropy targets.
3. **Full-patch L2**: mean squared error on the raw pixels of the masked patches, equivalent to the pixel reconstruction loss of Masked Autoencoders [63].

The model predicts only at masked positions; the loss is computed on corrupted patches only, so the network is never asked to reconstruct patches it was allowed to observe.

21.3 Image Classification

Classification fine-tuning follows ViT Sec. 3.2:

Pooling. The representation is pooled either by taking the `[CLS]` token output at position 0 or by Global Average Pooling over the patch tokens.

Head. A zero-initialized `Linear`(D, K) projects the pooled vector to K class logits. The zero initialization guarantees that the fine-tuning starts from the pretrained linear projection and that no class bias is introduced before the first optimizer step.

Loss. Standard cross-entropy over the K classes.

21.4 Image Segmentation

Two head types are selected by the `seg_head_type` config field.

21.4.1 Per-Pixel Head (`pixel`)

A `Linear`(D, `num_seg_classes`) is applied per patch token, and the resulting per-patch logits are reshaped to $(H/P, W/P)$. A ViTDet-style transposed-convolution upsampler then scales the logits to the full image resolution (H, W). The training loss is the unweighted sum of cross-entropy and Dice:

$$\mathcal{L}_{\text{pixel}} = \text{CE}(\hat{y}, y) + \text{Dice}(\hat{y}, y).$$

This head supports both 1D grayscale (2 classes) and multicolor non-overlapping ($N + 1$ classes) segmentation.

21.4.2 Encoder-only Mask Transformer (`eomt`)

The EoMT head [78] repurposes the ViT backbone as a mask decoder by splitting the layer stack into two contiguous groups:

- L_1 (patch-only): the first L_1 blocks process the patch tokens as in a plain ViT.
- L_2 (patches + queries): the remaining L_2 blocks receive a concatenation of K learnable object queries and the patch tokens. The multi-head self-attention inside each L_2 block therefore computes all four attention quadrants (patch↔patch, query↔query, query→patch, patch→query) in a single operation, halving the decoder compute relative to an encoder–decoder Mask2Former.

Mask module. The mask module predicts, for each query, class logits via a linear layer and a mask embedding via a 3-layer MLP. The mask embedding is dot-producted with the upscaled patch feature map $\tilde{F}_4$ to produce per-pixel mask logits.

Loss with Hungarian matching. Following Mask2Former, predictions are first matched to ground-truth masks via the Hungarian algorithm, then supervised with:

$$\mathcal{L}_{\text{eomt}} = 5\,\text{BCE} + 5\,\text{Dice} + 2\,\text{CE},$$

where the BCE and Dice terms act on mask logits and the CE term acts on class logits (factor 2 for matched queries, factor 0 for the no-object class).

Mask annealing. A fraction P_{mask} of patches is masked during training, with the fraction decaying polynomially (decay factor 0.9) to 0 at inference time. When $P_{\text{mask}} = 0$ the model reverts to a plain ViT and the L_2 blocks operate on patches only, yielding $\sim 4\times$ faster inference than at training time while preserving mask quality.

21.5 Reuse of Existing Infrastructure

Because the `HybridLayer` stack is modality-agnostic and operates purely on (B, S, D) tensors, the Vision Transformer reuses the entire encoder-side machinery without modification:

- **Attention mixers:** all 34 sequence mixers are available, except `engram`, which requires discrete token IDs and is therefore not meaningful for continuous patch embeddings.
- **Normalization:** all 6 norm variants (LayerNorm, RMSNorm, pRMSNorm, DyT, Derf, FlashNorm).
- **Residuals:** all 4 residual variants.
- **Activations:** all 42 activation functions (including GLU gated FFN units).
- **Optimizers:** all 23 optimizer families with prefixed per-parameter-group hyperparameter control.
- **Training configuration:** the three vision tasks read `batch_size`, `num_epochs`, `optimizer`, `scheduler_type`, `grad_clip_max_norm`, and checkpointing from the same top-level `training:` block as `mlm` and `causal_lm`. The optional `training.<task>:` sub-block carries only task-specific knobs: `label_smoothing` for classification, and `seg_loss_bce_weight` / `seg_loss_dice_weight` / `seg_loss_ce_weight` / `mask_annealing_factor` for EoMT segmentation.

The `FrankensteinViT` class mirrors the `FrankensteinDecoder` wrapper pattern: it owns a `PatchEmbed` module, a `HybridLayer` stack, and a task-specific head, and it forces `mode='encoder'` so that the bidirectional attention pattern required by vision tasks is preserved.

21.6 PyTorch Pseudocode

The forward pass for each task is sketched below.

Classification.

$x \leftarrow$ `patch_embed(pixel_values)` ▷ (B, N, D)
$x \leftarrow [\,$`cls_token` $;\, x\,]$ ▷ $(B, N+1, D)$
$x \leftarrow x +$ `pos_embed` ▷ learnable 1D
$x \leftarrow$ `dropout`(x)
for each `layer` in `layers` **do**
    $x \leftarrow$ `layer`(x)
end for
$x \leftarrow$ `final_norm`(x)
`logits` $\leftarrow$ `head`$(x[:, 0])$ ▷ `[CLS]` token

Masked patch prediction.

$x \leftarrow$ `patch_embed(pixel_values)` ▷ (B, N, D)
$(x,\, m) \leftarrow$ `mask`$(x,\, r = 0.5)$ ▷ m: binary mask of corrupted positions
$x \leftarrow [\,$`cls_token` $;\, x\,] +$ `pos_embed`
$x \leftarrow$ `dropout`(x)
for each `layer` in `layers` **do**
    $x \leftarrow$ `layer`(x)
end for
$x \leftarrow$ `final_norm`(x)
$\hat{y} \leftarrow$ `head`$(x[:, 1:])$ ▷ predict at patch positions only
$\mathcal{L} \leftarrow$ `loss`$(\hat{y}[m],\, y[m])$ ▷ loss on corrupted patches only

EoMT segmentation.

$x \leftarrow$ `patch_embed(pixel_values)` ▷ (B, N, D)
$x \leftarrow x +$ `pos_embed`
$x \leftarrow$ `mask_anneal`(x) ▷ mask P_{mask} patches when training
for each `layer` in L_1 **do**
    $x \leftarrow$ `layer`(x) ▷ patch-only blocks
end for
$q \leftarrow$ `learnable_queries` ▷ (B, K, D)
$x \leftarrow [\,x;\, q\,]$ ▷ concatenate patches and queries
for each `layer` in L_2 **do**
    $x \leftarrow$ `layer`(x) ▷ four-quadrant self-attention
end for
$\hat{c},\, \hat{m} \leftarrow$ `mask_module`$(x_{\text{patches}},\, x_{\text{queries}},\, \tilde{F}_4)$ ▷ class + mask logits
$(\sigma,\, \hat{y}_\sigma) \leftarrow$ `Hungarian`$(\hat{c},\, \hat{m},\, y)$ ▷ match to ground truth
$\mathcal{L} \leftarrow 5\,\text{BCE} + 5\,\text{Dice} + 2\,\text{CE}$ ▷ supervise matched predictions

22 Audio: Speech Recognition and Audio Classification

22.1 Executive Summary

`frankenstein_audio` is the fourth model class of the toolkit. It owns an audio frontend—log-mel filterbanks or a raw spectrogram, optional SpecAugment [122], and two stride-2 convolutions that reduce the frame rate by 4×—and then runs the same `HybridLayer` stack as the text encoder over the resulting frame sequence. Because the stack is modality-agnostic, every sequence

mixer (including the Conformer of Appendix 8), normalization, residual strategy, positional encoding, optimizer and routing block is available unchanged. The class supports two tasks: **automatic speech recognition** (`asr`, with a CTC [53] or an RNN-Transducer [52] loss) and **audio classification** (`audio_classification`, clip-level cross-entropy). The same frontend, driven with `audio.feature: raw`, accepts multichannel one-dimensional sensor signals such as EEG, EMG or accelerometer traces for classification.

The reference recipe is the Conformer frontend of Gulati et al. [56]: 80-channel log-mel filterbanks with a 25 ms window and a 10 ms stride, two 3×3 convolutions with stride 2, a linear projection to the model width, and SpecAugment with frequency width $F = 27$, ten time masks and a maximum time-mask ratio $p_S = 0.05$. The three reference sizes (S/M/L) are reproduced in Table 25.

22.2 The `frankenstein_audio` Model Class

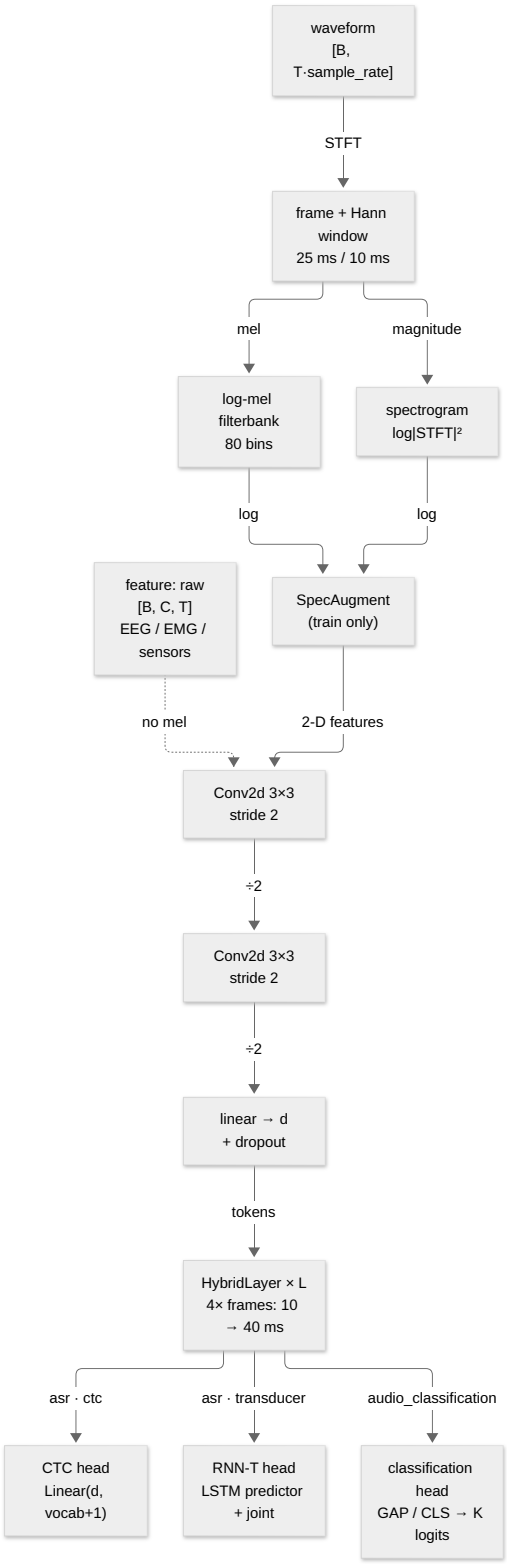


Figure 107: Dataflow of the audio frontend: waveform → 25 ms/10 ms framing → 80-bin log-mel (or spectrogram, or raw multichannel frames) → SpecAugment (training only) → two stride-2 Conv2d stages (4×) → linear projection → `HybridLayer` stack → CTC, RNN-T or classification head.

22.2.1 Audio Frontend

The frontend module turns a batch of waveforms $w \in \mathbb{R}^{B\times T}$ (or a batch of multichannel clips $w \in \mathbb{R}^{B\times C\times T}$ in `raw` mode) into a feature map:

1. **Framing:** the waveform is cut into overlapping windows of `window_ms` = 25 ms every `stride_ms` = 10 ms, with a Hann window, at `sample_rate` = 16000 Hz. At the reference rate this yields frames of 400 samples with a 160-sample hop, i.e. 100 frames per second.
2. **Spectral representation** (`audio.feature`):
   - `logmel` (default): an 80-bin mel filterbank (Equation 235) followed by a log compression;
   - `spectrogram`: $\log|\mathrm{STFT}(w)|^2$ with no mel projection;
   - `raw`: no spectral transform; the clip is fed as a $C \times T$ map (Section 22.8).
3. **SpecAugment** [122]: F frequency channels are masked per clip, and up to `time_masks` time bands of width at most $p_S \cdot T$ are masked, both with zero fill (training only).

The mel filterbank is

$$M_k(f) = \max\Big(0,\ \min\Big(\tfrac{f-f_{k-1}}{f_k-f_{k-1}},\ \tfrac{f_{k+1}-f}{f_{k+1}-f_k}\Big)\Big), \tag{234}$$

for $K = 80$ triangular filters spaced on the mel scale, $\mathrm{mel}(f) = 2595\log_{10}(1 + f/700)$, and the log-mel feature is

$$x_k = \log\Big(\sum_f M_k(f)\,|X(f)|^2 + \epsilon\Big). \tag{235}$$

When `torchaudio` is importable, the frontend uses `torchaudio.transforms.MelSpectrogram` for Equation 235; otherwise it falls back to a pure-PyTorch implementation built on `torch.stft` and a cached filterbank matrix. The two paths are numerically equivalent up to filterbank convention (HTK vs. Slaney mel spacing), and the pure-PyTorch path is what keeps CPU-only CI free of the optional `audio` extra.

22.2.2 Convolutional Subsampling

Two convolutional stages reduce the frame rate by 4× (10 ms → 40 ms), which is what makes a 30-second utterance a ∼750-token sequence instead of 3000:

$$F = \mathrm{Dropout}\Big(W_p\,\mathrm{Conv2d}_{3\times3,\,s=2}\big(\mathrm{Conv2d}_{3\times3,\,s=2}(X)\big)\Big), \qquad W_p \in \mathbb{R}^{d\times d_c}, \tag{236}$$

where X is the `n_mels` × T' feature map with `conv_channels` channels after the first stage, and W_p projects the flattened channels to the model width d. Both stages are fixed at 3×3 kernels with stride 2; `audio.conv_channels` (32) sets the channel width of the first stage and `audio.n_fft` (400) the STFT size.

22.2.3 Forced Encoder Mode and the HybridLayer Stack

`frankenstein_audio` forces `mode: encoder`, exactly as `frankenstein_vit` does: the audio tasks are bidirectional (a CTC or transducer encoder must see both past and future frames), and the Conformer's convolution padding is symmetric in encoder mode. The class mirrors the ViT wrapper pattern—frontend, `HybridLayer` stack, task head—and its layer loop keeps the same hooks, so AttnRes, mHC, Hyperloop, looped depth, MoE, Mixture-of-Depths and BitNet all work without audio-specific code. All task heads are built after the final normalization:

- **CTC:** Linear$(d, V + 1)$ over the frame sequence, where the extra class is the blank.

- **RNN-T:** a one-layer LSTM prediction network over the previous non-blank labels (`audio.asr_decoder_dim`, `asr_decoder_layers`, `asr_decoder_dropout`), and a joint network $z_{t,u} = \tanh(W_e h_t + W_p h_u + b)$, $\text{logits}_{t,u} = W_o z_{t,u}$.
- **Classification:** a pooled vector (global average pooling, or a learned `[CLS]` token prepended to the frame sequence, per `pooling_mode`) projected to `num_classes` logits.

22.3 Automatic Speech Recognition

22.3.1 CTC

Connectionist temporal classification [53] marginalizes over all frame-level alignments that collapse to the target label sequence y under the many-to-one map $\mathcal{B}$ (merge repeats, delete blanks):

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_{t=1}^{T} p(\pi_t \mid x), \tag{237}$$

computed with `torch.nn.functional.ctc_loss` in log space with `zero_infinity=False`, so that impossible alignments do not poison the gradient.

22.3.2 RNN-Transducer

The RNN-T loss [52] sums over *label-and-blank* alignments on the $T \times (U+1)$ lattice. With $\alpha_{t,u}$ the log-probability of the prefix $y_{1:u}$ after t encoder frames, the forward recursion is

$$\alpha_{t,u} = \text{logsumexp}\Big(\underbrace{\alpha_{t-1,u} + \log p(\varnothing \mid t-1, u)}_{\text{advance time}}, \underbrace{\alpha_{t,u-1} + \log p(y_u \mid t, u-1)}_{\text{emit label}} \Big), \tag{238}$$

with $\alpha_{0,0} = 0$ and $-\infty$ elsewhere, $p(\cdot \mid t, u) = \text{softmax}(\text{logits}_{t,u})$, and

$$\mathcal{L}_{\text{RNN-T}} = -\alpha_{T,U}. \tag{239}$$

The implementation is autograd-friendly in pure PyTorch (log-domain dynamic programming on the lattice, with the gradient obtained by the standard backward recursion), and takes the `torchaudio.functional.rnnt_loss` fast path when the optional dependency is installed. The lattice is $O(T \cdot U)$ in memory, which is why the predictor is kept shallow (`audio.asr_decoder_layers`, default 1) and narrow (`audio.asr_decoder_dim`, default 320). The loss is selected with `audio.asr_loss` (`ctc` by default).

22.3.3 Decoding and Error Rate

Greedy CTC decoding takes the per-frame argmax, collapses repeats and removes blanks. Greedy RNN-T decoding alternates a label emission with a frame advance until the blank is scored higher than the next label. Both are scored with the character error rate

$$\text{CER} = \frac{S + D + I}{N}, \tag{240}$$

where S, D, I are the substitutions, deletions and insertions of the Levenshtein alignment against the reference transcript and N is the reference length.

22.4 Audio Classification

Classification reuses the pooling and head conventions of the Vision Transformer (Appendix 21): `pooling_mode: cls` prepends a learned token and reads position 0, `pooling_mode: gap` (the default) averages the frame outputs. The head is a zero-initialized $\text{Linear}(d, K)$ trained with cross-entropy and the same label smoothing knob as the vision tasks. SpecAugment applies to the feature map before subsampling, so it augments the input, never the token sequence.

22.5 SpecAugment

Following Park et al. [122] in its "LibriSpeech basic" configuration, the two mask types are applied independently per clip:

$$\Delta f \sim \mathcal{U}[0, F], \quad f_0 \sim \mathcal{U}[0, \nu - \Delta f], \qquad \Delta t \sim \mathcal{U}[0, p_S \cdot T], \quad t_0 \sim \mathcal{U}[0, T - \Delta t], \tag{241}$$

with $F = 27$, `time_masks` $= 10$ and $p_S = 0.05$, each mask zeroing a $\Delta f \times \Delta t$ rectangle. Time *warping*, the third SpecAugment transform, is not implemented—it requires a per-sample warp of the time axis that interacts poorly with the fixed-length bucketing of the dataset. Frequency and time masking carry most of the regularization benefit in the paper's own ablations, and both are disabled at evaluation time.

22.6 Reference Sizes

Variant	Layers	d	Heads	Kernel	Params	RNN-T d_{dec}
Conformer-S	16	144	4	32	10.3M	320
Conformer-M	17	256	4	32	30.7M	640
Conformer-L	17	512	8	32	118.8M	640

Table 25: Reference Conformer sizes of Gulati et al. [56] (Table 1), which the example presets follow. The RNN-T decoder width is only used when `audio.asr_loss: transducer`.

22.7 Example Configurations

The ASR examples use `astroza/chilean-jokes-festival-de-vina` (Spanish audio at 16 kHz with transcripts, 50 rows), a dataset small enough to run end-to-end on CPU as a smoke test. The classification example falls back to the synthetic `DummyAudioDataset` when no dataset is reachable.

```
model_class: frankenstein_audio
model:
  dims: {hidden_size: 144, num_layers: 16, num_heads: 4}
  dims.layer_pattern: [conformer, ...]   # 16 entries
  attention.conformer: {kernel_size: 32, macaron: true, conv_norm: batch}
audio:
  feature: logmel
  sample_rate: 16000
  n_mels: 80
  window_ms: 25
  stride_ms: 10
  n_fft: 400
  conv_channels: 32
  asr_loss: ctc          # or transducer
  asr_decoder_dim: 320
  asr_decoder_layers: 1
  asr_decoder_dropout: 0.1
  spec_augment: {enabled: true, freq_mask_width: 27,
                 time_masks: 10, time_mask_ratio: 0.05}
audio_dataset:
  dataset_name: astroza/chilean-jokes-festival-de-vina
  split: test
  audio_column: audio
```

```
  text_column: text
training:
  task: asr
  asr: {loss: ctc}       # optional override of audio.asr_loss
```

SpecAugment is part of the acoustic frontend ("`audio.spec_augment`"), so it is applied to the feature map before subsampling and is active only in training mode. The optional `training.asr.loss` block overrides `audio.asr_loss` for presets that want to switch between CTC and RNN-T without editing the architecture block; the two must agree because the transducer head is built from the model-side knob.

22.8 Generic 1-D Signals: EEG and Beyond

The frontend is not tied to speech. With `audio.feature: raw`, the mel filterbank and SpecAugment are skipped and the clip is treated as a multichannel $C \times T$ map, with `audio.in_channels: C`. The two stride-2 convolutions then act as a learned filter bank over time (and, as a side effect, over the channel axis), the linear projection produces d-wide tokens, and the `HybridLayer` stack sees a sequence of frames exactly as it does for speech. This makes the class usable, at least for **classification**, on:

- **EEG:** C electrodes $\times$ T samples per epoch, labelled with a cognitive state, sleep stage or pathology; electrodes play the role of the mel channels.
- **EMG / ECG / EDA:** one or a few channels, labelled with gesture, activity or anomaly class.
- **Accelerometer / IMU:** $C \in \{3, 6, 9\}$ axes, labelled with activity or device state.
- **Industrial and environmental sensors:** vibration, pressure, acoustic emission traces, labelled per window.

What carries over. Everything downstream of the frontend: all 45 mixers (Conformer is a good default because its convolution branch matches the local morphology of biosignals), all 6 norms, AttnRes/mHC/MoE/MoD, the 23 optimizers, the frozen-encoder `text_classification` recipe, and the checkpoint/deployment paths. Length bucketing, padding masks and the CER/CER-style evaluation are dataset-side and are unaffected by the modality.

Limitations. Raw mode is a convenience, not a replacement for a domain frontend:

1. **Channel axis downsampling.** The two stride-2 `Conv2d` stages are fixed and halve both spatial axes, so a montage with fewer than four channels loses its channel axis entirely; keep $C \geq 4$ (or project the montage to a wider map first).
2. **No montage geometry.** Electrode positions, referential vs. bipolar montages and inter-channel correlations are not encoded; the convolution sees channels as an unordered stack, unlike EEG-specific models with topographic or graph priors.
3. **No spectral prior.** `logmel` and `spectrogram` assume audio band limits and mel spacing that do not apply below 20 Hz or above 8 kHz; raw mode avoids the mismatch but also loses the band-aligned inductive bias.
4. **Classification only.** CTC and RNN-T require a token-level target sequence. Sequence-to-sequence labelling of sensor windows (e.g. sleep-stage sequences) is out of scope; use per-window classification and a post-hoc decoder instead.

5. **Statistics and normalization.** The convolution module's BatchNorm uses per-batch statistics; for EEG the amplitude scale varies by subject and session, so per-channel normalization in the dataset is recommended over relying on the frontend.

6. **Sampling-rate assumptions.** `sample_rate` is used to convert `window_ms`/`stride_ms` into samples; for raw signals set it to the true acquisition rate and choose the window in milliseconds accordingly.

22.9 PyTorch Pseudocode

Input: waveforms w, lengths ℓ, transcripts y (ASR) or labels c (classification)
$F \leftarrow \texttt{frontend}(w)$ ▷ log-mel / spectrogram / raw, $(B, C, 80, T')$
$F \leftarrow \texttt{spec_augment}(F)$ ▷ training only
$F \leftarrow \texttt{conv2d}_{3\times3,s=2}(F)$; $F \leftarrow \texttt{conv2d}_{3\times3,s=2}(F)$
$x \leftarrow \texttt{dropout}(\texttt{linear}(F))$ ▷ $(B, T'/4, d)$ tokens
$x \leftarrow x + \texttt{pos_encoder}(x)$ ▷ frame positions, not mel bins
for each `layer` in `layers` **do**
    $x \leftarrow \texttt{layer}(x)$ ▷ `HybridLayer`, forced encoder mode
end for
$x \leftarrow \texttt{final_norm}(x)$
if task = asr and loss = ctc **then**
    logits $\leftarrow \texttt{ctc_head}(x)$; $\mathcal{L} \leftarrow \texttt{ctc_loss}(\log\text{softmax}(\text{logits}), y, \ell/4)$
else if task = asr and loss = transducer **then**
    $h^{pred} \leftarrow \texttt{predictor}(y)$; $\text{logits}_{t,u} \leftarrow \texttt{joint}(x_t, h_u^{pred})$
    $\mathcal{L} \leftarrow \texttt{rnnt_loss}(\text{logits}, y, \ell/4)$
else
    logits $\leftarrow \texttt{class_head}(\texttt{pool}(x))$; $\mathcal{L} \leftarrow \text{CE}(\text{logits}, c)$
end if

22.10 Deviations and Implementation Notes

- **Optional torchaudio.** The frontend prefers `torchaudio.transforms` but never requires it; the pure-PyTorch fallback keeps the CPU CI and the `cpu` extra torchaudio-free (decision D7 of the architecture guide).

- **Time warping.** Not implemented (Equation 241 covers frequency and time masking only); time warping is the transform the paper uses least.

- **Shallow predictor.** The RNN-T prediction network is a single LSTM layer by default (`audio.asr_decoder_layers`), trading the paper's deeper predictor for an $O(T \cdot U)$ lattice that fits in memory at the reference batch sizes.

- **Export.** The HuggingFace and GGUF exports reject audio artifacts explicitly, as they do for SBERT: there is no upstream `FrankensteinForAudioClassification` mapping. `deploy -format standard|quantized` works unchanged.

- **Cache.** There is no incremental encoder cache for audio; ASR consumes a whole utterance per forward.

23 KV Cache and Incremental Decoding

Autoregressive decoding is dominated by a structural asymmetry: the prefill stage processes the whole prompt in one quadratic pass, while each decode step appends a single token and needs

only one new query row against an already-computed history. Recomputing the full sequence at every step turns generation into an $\mathcal{O}(S^2)$-per-token procedure; maintaining the per-layer history turns it into $\mathcal{O}(S)$ per token with constant-size work per new position. This annex documents the toolkit's KV-cache subsystem (`src/model/cache/`), the correctness contracts that make cached and uncached execution numerically interchangeable, the per-family incremental recipes, and the write-time quantizers and read-time eviction policies layered on top of it.

Three design commitments frame everything below. First, the cache is *opt-in*: with `inference.kv_cache.enabled` unset (default), generation executes the original full-recompute loop bit-identically. Second, the cache is *central and declarative*: a single manager is threaded through the model, and a per-mixer capability registry decides which layers may cache — never the other way around. Third, the cache is *never fatal* (decision D5): a layer whose source formulation does not admit an exact incremental form silently runs uncached, so enabling the cache can only ever change speed, not correctness or availability.

23.1 Motivation and Scope

Let S be the prompt length and T the number of generated tokens. Without a cache, step t feeds $S + t - 1$ tokens through every layer and discards all intermediate keys and values, for total attention work

$$\mathcal{O}\left(\sum_{t=0}^{T-1}(S+t)^2\right) \;=\; \mathcal{O}\big(TS^2 + T^2S + T^3\big)\,.$$

With a cache, the prefill costs $\mathcal{O}(S^2)$ once and each decode step costs $\mathcal{O}(S + t)$:

$$\mathcal{O}\left(S^2 + \sum_{t=0}^{T-1}(S+t)\right) \;=\; \mathcal{O}\big(S^2 + TS + T^2\big)\,.$$

The difference is the familiar one between re-encoding a conversation and continuing it.

The subsystem covers every one of the forty-nine dispatchable layer types. Thirty-one have parity-tested incremental branches; the remainder are registered with an explicit, paper-based rationale for running uncached (§23.4.5). Quantization and eviction are orthogonal to the mixer mathematics: codecs transform tensors at *write* time and policies act at *read* time, so the same `int8` or `int4_kivi` codec serves a dense, latent or ring slot without per-mixer code [105, 68, 35].

23.2 Cache Architecture

23.2.1 Central Manager and Layer Identity

The unit of caching is a logical layer, not a module. The encoder may re-execute the same physical stack `num_loops` times (looped depth) and, under Hyperloop, partitions it into a begin block, a looped middle block and an end block. The same `HybridLayer` object therefore plays several distinct roles during one forward pass. The cache keys every slot by the global **logical layer index**

$$\ell \;=\; 0, \ldots, \texttt{num_layers} \times \texttt{num_loops} - 1,$$

which the encoder already threads into every mixer call. Two logical roles never share storage even though they share weights.

Algorithm 65 Cached generation loop (prefill + decode)

Require: prompt ids $\mathbf{x}_{1:S}$; generation config θ; model f
Ensure: generated ids

$\mathcal{C} \leftarrow \textsc{Cache.new}(f, \theta)$ ▷ lazy slots, token tail, write cursor
$\mathbf{L}, \mathcal{C} \leftarrow f(\mathbf{x}_{1:S}; \text{past} = \mathcal{C}, \text{use_cache} = \top, \text{return_cache} = \top)$ ▷ prefill
$\mathbf{z} \leftarrow \mathbf{L}_{:,S,:}$
for $t \leftarrow 0$ **to** $T - 1$ **do**
    $y \leftarrow \textsc{Sample}(\mathbf{z}; \theta)$ ▷ greedy, top-k, top-p
    append y to the output
    if $t < T - 1$ **then**
        $p \leftarrow \mathcal{C}.\text{next_position}()$
        $\mathbf{L}, \mathcal{C} \leftarrow f(y; \text{past} = \mathcal{C}, \text{use_cache} = \top, \text{cache_position} = p, \text{return_cache} = \top)$
        $\mathbf{z} \leftarrow \mathbf{L}_{:,-1,:}$
    end if
end for

The manager owns the per-layer slot dictionary, a FIFO of recent token ids (consumed by token-level mixers such as Engram), a write cursor advanced by each forward, and the resolved quantization codec and eviction policy. Slot creation is lazy and driven by the layer on first use, so a heterogeneous `layer_pattern` — say a dense layer, a latent layer and a recurrent layer in one stack — holds three different slot kinds without special-casing. Rollback is a first-class operation: `truncate(n)` keeps the first n positions across all slots (cheap list truncation with length bookkeeping; recurrent-state slots are no-ops because they are not position-indexed), which is the primitive required by speculative decoding and early-exit verification [41].

23.2.2 Typed Slots

Six slot kinds cover the taxonomy of §23.4. Position-indexed slots store an ordered list of *parts*; a part is either full precision or an encoded payload, and reading concatenates [sinks | history | protected window].

Slot kind	Stores	Used by
`kv`	post-RoPE keys and values (B, H_{kv}, T, d); absolute positions; optional per-token extras	quadratic softmax mixers
`latent_kv`	named low-rank streams: latent $\mathbf{c}_{KV}$, decoupled RoPE key, grouped keys, latent values	latent family
`state`	one recurrent matrix (+ gate accumulators), updated in place; constant memory	recurrent / linear family
`ring`	capacity-bounded K/V with a pinned sink region	sliding-window family
`conv_tail`	last $k - 1$ pre-convolution rows (FIFO)	causal short convolutions
`token_tail`	last n token ids	Engram n-gram lookup

Table 26: Typed cache slots. Position-indexed kinds (`kv`, `ring`, `latent_kv`) carry absolute position metadata, so eviction gaps keep correct relative distances; `state` and `conv_tail` are position-less and never evicted.

23.2.3 Capability Registry

One frozen record per `layer_pattern` enum value declares whether and how a layer caches:

$$\textsc{CacheSpec} \;=\; \big(\texttt{supported},\ \texttt{slot_type},\ \texttt{slot_types},\ \texttt{extras},\ \texttt{notes}\big).$$

The registry is the single source of truth consulted both at runtime (slot resolution) and statically (configuration warnings). A layer receives a slot only when its mixer's formulation admits a cache *and* an incremental branch is implemented and parity-tested; otherwise it receives `None` and executes the legacy path. The registry is therefore also the project's honest inventory of what caching means for each paper: `supported` records the source formulation, the implemented set records engineering status, and `notes` records the reason.

23.2.4 Forward Plumbing and Fallback Semantics

All signature changes are additive and default to the legacy behaviour:

$$f_{\text{mixer}}(\mathbf{x}; \ell, \text{pe}, \texttt{cache} = \varnothing, \texttt{cache_position} = \varnothing),$$
$$f_{\text{model}}(\mathbf{x}; \texttt{past_key_values} = \varnothing, \texttt{use_cache} = \bot, \texttt{return_cache} = \bot).$$

`HybridLayer` resolves the slot once per call and honours the four historical dispatch shapes: stateless linear (`mamba`) is called without arguments because it has nothing to cache; the two whole-stream mixers (`ode`, `retnet`) receive the slot only when their spec supports it; Engram additionally receives `input_ids`; every other mixer receives `cache` and `cache_position`. Unsupported or disabled layers are called exactly as before, which keeps the uncached path bit-identical.

23.3 Correctness Contracts

Cached and uncached execution must agree on the query row they share: the last row of a full forward and the row produced by a decode step. The contracts below establish that agreement.

23.3.1 Post-RoPE Storage and the Absolute Schedule

Keys are stored *after* the rotary transform, the industry convention: the cache holds the rotated vectors, and no re-rotation is needed when a ring buffer wraps. The prerequisite is that rotation use absolute positions on both sides of the cache boundary. RoPE, HoPE and sinusoidal-rotary accept a position offset π_0 and place the incoming tokens at

$$\pi_i = \big(\pi_0 + i\big) \cdot \texttt{scaling}, \qquad i = 0, \ldots, S_{\text{new}} - 1,$$

so a decode step at cursor p rotates exactly as the full forward rotated position p. The slot metadata versions the scaling factor; a runtime change of the RoPE schedule invalidates already-rotated keys and is reported as a configuration error at generation start, never mid-decode [150, 25].

23.3.2 Additive Score Biases

ALiBi and BAM inject position through the score matrix, so they carry no positional tensor in the cache — precisely their appeal for length extrapolation [127, 13]. The full-square builder generalizes to a windowed builder over absolute query and key positions:

$$b_{h,i,j} \;=\; -\,\text{slope}_h \cdot \big|q_i - \kappa_j\big| \quad \text{(ALiBi)}, \qquad b_{h,i,j} \;=\; -\Big(\big|\kappa_j - \alpha_h q_i - \mu_h\big| + \varepsilon\Big)^{\beta_h} \quad \text{(BAM)},$$

where $q_i = \pi_0 + i$ for the incoming query positions and κ_j are the cached absolute key positions. With $\pi_0 = 0$ and $\kappa_j = j$ this reproduces the legacy matrix exactly; after eviction, κ_j may be non-contiguous, and the distances remain correct by construction.

23.3.3 Length-Dependent Rescaling

Scalable Softmax multiplies pre-softmax logits by $s \cdot \ln n$ where n is the context length. In cached decoding the context of the newest query is the accumulated cache, so the rescale uses

$$\ln\left(n_{\text{cached}} + S_{\text{new}}\right)$$

rather than the size of the incoming chunk. For the final row of a full forward over $S_{\text{cached}} + S_{\text{new}}$ tokens both expressions coincide, which is exactly the row the parity tests compare.

23.3.4 Causal Masking from Positions

Masks are derived from metadata rather than from array indices:

$$M_{ij} \;=\; \big[\, q_i < \kappa_j \,\big] \;=\; \big[\, \pi_0 + i < \kappa_j \,\big],$$

which reduces to the classical upper-triangular mask when the cache starts empty and $\kappa_j = j$. For a single decode query every cached key satisfies $\kappa_j \le q_i$, so the mask is empty and the step needs no mask at all. Sparse families replace the causal predicate with a position-based pattern (§23.4).

23.3.5 Absolute Positional Encodings

Additive absolute encodings (sinusoidal, learned) are applied once to the embeddings. In cached decoding the embedding lookup for the new token receives the same offset, so token p is augmented with row p of the table:

$$\mathbf{x}_i \;\leftarrow\; \mathbf{x}_i + \mathrm{PE}\big[\pi_0 + i\big].$$

23.3.6 Interlocks with Mixture-of-Depths, mHC, Hyperloop and AttnRes

Mixture-of-Depths routes a fraction of tokens through each layer. Routing a prompt would produce ragged, token-dependent caches, which v1 deliberately excludes; when the cache is active, prefill is forced dense while decode degenerates to dense automatically (one token is always within capacity). The divergence is one-directional: forcing dense can only add keys, never attend to keys that do not exist. mHC and Hyperloop residual streams are per-token state and flow through a cached forward without cross-step carry; the loop-level hyper-connections fire per iteration and need no slot of their own. Attention-residual variants keep cross-layer buffers that must accumulate across steps, a mode not yet implemented; the combination disables caching for that forward with a warning and falls back to full recompute. PaPE augmentations are content-dependent, so caching post-augmentation keys is not provably exact; such layers fall back per layer [119].

23.4 Incremental Recipes by Family

23.4.1 Quadratic Softmax Family

For a dense attention layer with H query heads, H_{kv} key/value heads and head dimension d, the cached branch appends the new keys and values and attends over the whole retained prefix:

$$\mathbf{K} \leftarrow \big[\mathbf{K}_{\text{cached}};\ \mathbf{k}_{\text{new}}\big], \quad \mathbf{V} \leftarrow \big[\mathbf{V}_{\text{cached}};\ \mathbf{v}_{\text{new}}\big], \quad \mathbf{S} = \frac{\mathbf{q}\mathbf{K}^{\top}}{\sqrt{d}} + \mathbf{B}, \quad \mathbf{o} = \mathrm{softmax}(\mathbf{S})\,\mathbf{V},$$

with $\mathbf{B}$ the positional bias of §23.3. Grouped-query attention caches only the H_{kv} compact heads and expands head groups after the read, which is where its cache advantage originates [2]. The family's variations are all captured by substituting the normalizer or the bias:

- **Sigmoid attention.** The normalizer is replaced by an ℓ_1 row normalization, $\hat{\mathbf{S}}_{ij} = \sigma(\mathbf{S}_{ij})/\big(\sum_{j'} \sigma(\mathbf{S}_{ij'}) + \varepsilon\big)$. Because sigmoid is not shift-invariant, the denominator must be re-summed over the full cached prefix at every step; no online-softmax maximum trick exists [134].

- **Continuum attention.** The normalizer is replaced by the exact segment integration: the effective weights are a closed-form function (exponential moments of the endpoint-score differences) of the full score row, so decode recomputes them over the whole cached prefix each step — row-local, hence exact, in the sigmoid re-normalization pattern [79].

- **Gated softmax.** The gate $\sigma(\mathbf{x}\mathbf{W}_g)$ multiplies the attention output and depends only on the current token, so nothing extra is cached [129].

- **Forgetting attention.** The logit bias is $D_{ij} = c_i - c_j$ with $c_i = \sum_{l \le i} \log f_l$ the cumulative log-forget sum. The cache stores one scalar per token and head as an extra; decode extends the running sum from the last cached token and evaluates a single row of D against the cached column [99].

- **First passage attention.** The logit bias is $\beta \min(0, \min_{s<u\le t} \phi_u - \psi_s)$, a range minimum of a per-token field ϕ below the key's level ψ. The cache stores *two* scalars per token and head as extras (ϕ_u and ψ_s); decode updates $\bar{\phi}_{t,s} = \min(\bar{\phi}_{t-1,s}, \phi_t)$ in $O(t)$ and evaluates one bias row against the cached levels, which is exact (Proposition 2.4(b)). Dead keys (bias below a threshold) never revive, so they are evictable [104].

- **Titans.** The implemented layer is the attention window; the memory module is outside the cached path. Standard append/read applies [11].

- **Memory Attention.** Class (i): prefill caches the post-RoPE keys and the constructed values $V = K + \mathrm{RMSNorm}(E[s])$, so decode looks up only the new token's memory row and the slot is byte-identical to standard KV [76].

- **Interleaved head attention.** Pseudo-key/value tensors are static learned head mixings computed on the fly; only the real per-token K/V is cached, and the mixing is applied after the read (the mixing is a per-token head-space map, so applying it post-read is exact) [38].

23.4.2 Latent Family

Latent mixers compress keys and values into a low-rank latent and cache only that latent. Writing $\mathbf{c}_{KV} = \mathbf{W}_{DKV}\mathbf{x}$, decode re-expands and rotates per step:

$$\mathbf{k} = \mathbf{W}_{UK}\mathbf{c}_{KV}, \qquad \mathbf{v} = \mathbf{W}_{UV}\mathbf{c}_{KV}, \qquad [\mathbf{q}, \mathbf{k}] \leftarrow \mathrm{RoPE}\left([\mathbf{q}, \mathbf{k}], \pi_0\right),$$

with attention as in the dense case [111]. DeepSeek-style MLA caches a decoupled RoPE key and can absorb the up-projections into the query, avoiding per-step expansion; this toolkit's variant applies RoPE to the decompressed query and key, so absorption is unavailable and expansion is the documented cost of halving the cache. GQLA reuses one weight set with two algebraically equivalent decode paths, and the slot layout follows the configured path [113]. MLRA partitions the latent into disjoint sub-heads, which is a tensor-parallel sharding win rather than a compression win; the total cache is MLA-sized [102]. Tucker attention caches the factorized key and value latents and expands through the core projections [86]. GTA caches one key per head *group* plus latent values; because rotation at a fixed position is head-independent, group means commute with the rotation and caching the mean is exact [151]. MTLA is the only method that compresses along time: a hyper-network merges adjacent cache vectors at stride boundaries, the schedule is dynamic, and its parity with a chunk-level training schedule is a named risk, so it currently falls back [32].

23.4.3 Sparse and Window Families

Sparse patterns are evaluated as predicates over absolute positions, never over array indices [22]:

$$\text{attend}(q_i, \kappa_j) \iff \text{pattern}\big(\pi_0 + i,\ \kappa_j\big)\ \wedge\ \kappa_j \le \pi_0 + i.$$

- **Sliding window with sinks.** Every query attends to the first s positions plus the last $w - s$; the ring buffer's retention policy *is* the attention pattern, so a decode step attends the cached set directly and a multi-token chunk is pattern-masked [75, 166]. Removing the sinks collapses quality, which is why they are pinned both by the pattern and by the ring.
- **Longformer.** Symmetric windows plus task-specified global tokens; because a global token may sit anywhere, the slot is unbounded and the pattern is evaluated over positions [12].
- **BigBird.** Window, globals and random links. Random links must be reproducible across incremental steps, so they are drawn from a generator seeded by (ℓ, q_i) over $[0, q_i]$: links beyond the query are causal-masked anyway, so the effective mask is unchanged while becoming deterministic [175].
- **Sparse Transformers.** Strided and fixed factorized patterns, where a token's summary-column role changes as blocks complete; position-based evaluation handles this, and an automatic stride is resolved from the accumulated length so the full and cached paths agree [22].
- **FASA.** Per-head importance is computed from dominant RoPE-frequency chunks and re-ranked per query; it is row-local, so a decode step re-ranks the cached keys exactly [164]. SpargeAttn, NSA, MiniMax and SparDA select blocks from chunk-level statistics and therefore fall back in v1 [180, 173, 91, 45].

23.4.4 Recurrent and Linear State Family

Recurrent mixers carry a fixed-size state; caching means persisting that state, so memory is constant and no eviction is meaningful. With the paper convention $\mathbf{S}_t \in \mathbb{R}^{d_v \times d_k}$ and read $\mathbf{o}_t = \mathbf{S}_t \mathbf{q}_t$:

$$\text{RetNet:}\quad \mathbf{S}_t = \gamma_h\, \mathbf{S}_{t-1} + \mathbf{v}_t \mathbf{k}_t^\top, \qquad \text{per-head multi-scale } \gamma_h, \tag{242}$$

$$\text{GLA:}\quad \mathbf{S}_t = \mathbf{G}_t \odot \mathbf{S}_{t-1} + \mathbf{v}_t \mathbf{k}_t^\top, \qquad \mathbf{G}_t = \text{diag}\left(\exp(\text{logsigmoid}(\mathbf{W}_{gk}\mathbf{x}_t)/c)\right), \tag{243}$$

$$\text{DeltaNet:}\quad \mathbf{S}_t = \mathbf{S}_{t-1}\big(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top\big) + \beta_t \mathbf{v}_t \mathbf{k}_t^\top, \tag{244}$$

$$\text{Gated DeltaNet:}\quad \mathbf{S}_t = \alpha_t\, \mathbf{S}_{t-1}\big(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top\big) + \beta_t \mathbf{v}_t \mathbf{k}_t^\top, \tag{245}$$

$$\text{GDN-2:}\quad \mathbf{S}_t = \big(\mathbf{I} - \mathbf{k}_t(\mathbf{b}_t \odot \mathbf{k}_t)^\top\big)\mathbf{D}_t \mathbf{S}_{t-1} + \mathbf{k}_t(\mathbf{w}_t \odot \mathbf{v}_t)^\top, \tag{246}$$

$$\text{HGRN2:}\quad \mathbf{S}_t = \text{diag}(\mathbf{g}_t)\mathbf{S}_{t-1} + \mathbf{v}_t \mathbf{k}_t^\top, \qquad g_{t,j} = b_\ell + (1 - b_\ell)\sigma(\cdot), \tag{247}$$

$$\text{KDA:}\quad \mathbf{S}_t = \big(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top\big)\mathbf{D}_t \mathbf{S}_{t-1} + \beta_t \mathbf{k}_t \mathbf{v}_t^\top, \tag{248}$$

$$\text{GMA:}\quad \tilde{\mathbf{V}}_t = \tilde{\mathbf{V}}_{t-1} + \boldsymbol{\gamma}_t^K \otimes \mathbf{v}_t, \quad \mathbf{Z}_t = \mathbf{Z}_{t-1} + \boldsymbol{\gamma}_t^K, \quad \mathbf{o}_t = \frac{\sum_k \gamma_{t,k}^Q \tilde{\mathbf{V}}_{t,k}}{\sum_k \gamma_{t,k}^Q Z_{t,k} + \varepsilon}, \tag{249}$$

referring to Sun et al. [152], Yang et al. [169], Yang et al. [170], Yang et al. [171], Hatamizadeh et al. [60], Qin et al. [128], Kimi Team et al. [83] and Huang and Raza [72].

Two subtleties deserve emphasis. First, RetNet's parallel retention sum and the recurrence above produce the same row for the same query — algebraically the parallel form factors through the running state — so the cached path may use the recurrent form while the uncached path uses the parallel one, and the parity test compares the shared row. Second, GDN-2 and KDA

decay along the *key* axis ($\mathbf{D}_t$ multiplies the state from the left), whereas GLA and HGRN2 gate the key axis element-wise; both are diagonal in the key dimension, which is what makes a simple in-place recurrence exact. The ODE-attention block is excluded: its solver re-derives every token's intermediate state at each integration step, so a converged-state cache cannot reproduce the full path.

23.4.5 Auxiliary State and Documented Fallbacks

Some mixers need a small FIFO beside their activation cache. Falcon's chunk-parallel variants bound a rehearsal window, so exact continuation requires the last $B-1$ prefix-aligned pairs; CCA/CCGQA stack two causal convolutions and need the last $k-1$ pre-convolution latents plus a shifted value; Engram needs the last $n_{\max}-1$ token ids only. These are the `conv_tail` and `token_tail` kinds.

Mixer	Why it runs uncached in v1
`mamba`	linear placeholder; stateless per token, inherently incremental
`ssog`	bidirectional vision grid; no causal formulation exists
`ode`	solver re-derives all tokens' intermediate states each step
`falcon1..3a`	chunk-parallel rehearsal FIFO (`conv_tail` design pending)
`cca`, `ccgqa`	two causal convolutions + shifted value
`engram`	token-id tail (`token_tail` design pending)
`nsa`, `msa`, `sparda`, `sparge`	block selection uses chunk-level statistics, not row-local
`mtla`	dynamic hyper-network merge schedule (risk R1)
`sparsek`	eviction-policy design (SparseK is a policy, not a slot)
PaPE layers	content-dependent q/k augmentation; exactness unproven

Table 27: Documented uncached fallbacks. Each entry carries a paper-based rationale in the capability registry and a loader warning when the cache is enabled; none of them can make inference fail.

23.5 KV-Cache Quantization

Quantization is a write-time transform: the slot encodes on append and dequantizes on read, so the stored footprint shrinks while the mixer consumes ordinary floating-point tensors. This placement follows the literature and keeps codecs orthogonal to mixers: the same codec serves dense, latent and ring slots, and latent slots may declare a different granularity per stream. Recurrent states are *not* quantized in v1: the evidence base there is thinnest, and their constant size makes the payoff small.

23.5.1 Symmetric int8 with Asymmetric Axes

Keys exhibit persistent per-channel outliers (a RoPE-frequency structure), while values exhibit per-token outliers. The `int8` codec therefore uses different reduction axes:

$$\text{keys (channel axis):}\quad s_{h,c} = \frac{\max_{t\in g}|K_{t,h,c}|}{127}, \qquad \hat{K} = \operatorname{round}\big(K/s\big) \in [-127, 127];$$

$$\text{values (token axis):}\quad s_{t,h} = \frac{\max_{c}|V_{t,h,c}|}{127}, \qquad \hat{V} = \operatorname{round}\big(V/s\big) \in [-127, 127].$$

Each stored block carries its own scales, so a token-by-token decode block gets exact per-token scales and the relative rounding error stays bounded by $1/127 \approx 0.8\%$ independently of block size. Dequantization is $\hat{K} \odot s$; merging two payloads is dequantize, concatenate along time, and requantize.

23.5.2 Group-Wise int4 with a Protected Window

The `int4_kivi` codec combines KIVI's group-wise low-bit quantization with the two-region protection pattern of streaming sinks and SKVQ. Signed levels are $[-7, 7]$; g tokens form a group along time; granularity follows the int8 codec. Two regions are kept in full precision:

- the most recent w_p tokens (the *protected window*), never quantized;
- the first s tokens (*sinks*), never quantized and never evicted.

Append semantics mirror KIVI's residual window: a decode step pushes rows into the full-precision window, and as soon as the oldest chunk completes a group of g tokens that group is quantized and moved into the history; prefill quantizes whole groups immediately. Reading concatenates [quantized history | protected window]. With signed 4-bit levels, the worst-case absolute error of a group quantizer is half a step,

$$|K - \hat{K}| \;\le\; \frac{s}{2} \;=\; \frac{\max_g |K|}{2 \cdot 7} \;\approx\; 7.1\% \text{ of the group maximum},$$

which is the bound asserted by the codec tests. Because sinks and the protected window stay exact, the tokens that dominate attention mass are never perturbed [105, 68, 35, 166].

23.5.3 Interface and Future Codecs

Both codecs implement the same three-method contract: `encode_write`, `decode_read` and `update`. Codecs must never raise on unknown tensors: on any internal error they fall back to identity and record a one-time warning, preserving D5 at the storage layer as well. The v1 registry contains `none`, `int8` and `int4_kivi`; further methods are interface-ready but deliberately unimplemented, and requesting one is a configuration error at generation start: `int2_kivi` (2-bit with dual-axis variance normalization), `palu_lowrank` (rank-r projection with reconstruction), `commvq` (RoPE-commutative codebooks, attention in codebook space), `qjl_sketch` (1-bit Johnson–Lindenstrauss signs with an unbiased inner-product estimator) and `block_gtq` (per-RoPE-frequency bit allocation). Slot metadata already records the positional-encoding type and the scaling version, which is what a codebook or pre-RoPE codec needs to declare its compatibility. Sensitivity-aware per-layer bit allocation (`k_bits` $\neq$ `v_bits`) is exposed by configuration, since attention mass is pyramidal in depth.

23.6 Compression and Eviction

Eviction is a read-time transform; keeping it orthogonal to quantization is what allows a quantized cache to be evicted and an evicted cache to be quantized. The v1 policy is StreamingLLM's window-with-sinks: pin the first s positions, retain the last w positions, drop everything in between. The retained budget is

$$\mathcal{B} \;=\; s + w,$$

optionally overridden per layer by a budget vector $\mathcal{B}_\ell$ (attention mass is pyramidal in depth, so deeper layers tolerate smaller budgets). Policies act only on position-indexed slots; recurrent states are constant-size and MTLA manages its own merging schedule, so external policies are excluded there by construction. Future enum-ready policies are H2O (accumulate per-token attention mass and evict the lightest), SnapKV (one-shot post-prefill compression from a look-ahead window), PyramidKV (static depth-dependent budgets) and SparseK (learned scorer with permanent within-generation eviction, which is why it is modeled as a policy rather than a slot).

23.7 Configuration and Deployment Interface

The runtime block is validated by a dedicated schema module and flattened into a runtime dataclass that is deliberately *not* part of the architectural configuration, mirroring the `training.*` separation.

Key	Default	Description
`inference.generation.max_new_tokens`	64	tokens generated after the prompt
`inference.generation.temperature`	1.0	softmax temperature
`inference.generation.top_k`	50	top-k filter; 0 disables
`inference.generation.top_p`	1.0	nucleus mass; 1.0 disables
`inference.generation.greedy`	`false`	argmax decoding
`kv_cache.enabled`	`false`	opt-in; unset keeps the legacy loop bit-identical
`quantization.method`	`none`	`none`, `int8`, `int4_kivi`
`quantization.k_bits` / `v_bits`	4 / 4	int4 bit widths (int8 fixes 8)
`quantization.group_size`	64	tokens per group
`quantization.protected_window`	32	most recent tokens kept in full precision
`compression.method`	`none`	`none`, `window_sinks`
`compression.num_sinks`	4	pinned leading positions
`compression.window`	512	sliding window size
`compression.budget`	`null`	scalar budget per layer; `null` = unlimited
`compression.per_layer_budgets`	[]	per-layer budget vector, overrides the scalar
`per_mixer[].enabled`	`true`	per-mixer opt-out

Table 28: Inference runtime configuration. Cross-key rules: compression requires `enabled`, and `window` $\geq$ `protected_window` so the protected region can never be evicted.

Resolution order is **CLI flags** > **YAML** `inference:` > **schema defaults**. Every CLI flag defaults to "unset", so a YAML preset is patched leaf-by-leaf rather than replaced:

```
infer -task causal_lm -config <preset> -prompt "..." -kv-cache -kv-quant int4_kivi -kv-compress
                                window_sinks -kv-window 512
```

The deployer serializes the resolved block into the artifact's `config.json` under `"inference"`, and the inference engine reads it back, so cached generation is reproducible without the original YAML. The HuggingFace export's causal-LM wrapper accepts `past_key_values`, `use_cache` and `cache_position` and forwards them to the decoder instead of discarding extras, and implements `prepare_inputs_for_generation` so that a standard transformers loop feeds only the last token once a cache exists.

23.8 Validation

The central test is a parity protocol: for a mixer and a positional encoding, compare a full forward over S tokens against prefill over $S - k$ tokens followed by k decode steps, requiring agreement on the shared row. Greedy generation is compared token-for-token first; if a numerical discrepancy appears, last-position logits are compared with rtol $= 10^{-4}$, atol $= 10^{-5}$ in fp32 with quantizer `none` and BitNet disabled.

Suite	Assertions
slots	append/read/truncate/device moves; ring wrap with position metadata; in-place state; conv-tail FIFO; token tail
codecs	int8 error $\leq 10^{-2}$; int4 error $\leq 7.5 \times 10^{-2}$; protected window exactly fp; group boundaries; `update` = dequant→concat→requant
mixers	every implemented mixer × {rope, none, ALiBi}: full forward vs prefill+decode
fallback	all 45 enum values run with the cache on; unimplemented mixers bit-identical on/off; PaPE per-layer fallback
mechanisms	looped depth (distinct slots per logical layer), Hyperloop, mHC, MoD dense prefill, AttnRes fallback
quantization	int8/int4 drift bounds with BitNet on and off; int4 engages on short prompts; protected window exact
compression	sinks pinned after eviction; retained region correct; composition with int8
generation	greedy parity exact; seeded sampling determinism; `truncate` rollback
config	schema-default parity, flattening, precedence, cross-key rules, fallback warnings, serialization

Table 29: Validation matrix. The capability registry and the schema's layer-pattern enum are derived from the same component registry, so a new mixer cannot be added without a cache decision.

23.9 Relation to Prior Work

- **KV-cache quantization** (KIVI, KVQuant, SKVQ): adopted as write-time codecs with asymmetric granularity and protected outlier regions; the group-window append semantics follow KIVI's residual window.
- **Eviction and sparsity** (StreamingLLM, H2O, SnapKV, PyramidKV, Quest): adopted as read-time policies; Quest-style page metadata and query-aware reading are future work inside the slot extras.
- **Paged and prefix sharing** (PagedAttention/vLLM, RadixAttention): structurally compatible — slots are content-agnostic and serializable — but block tables and prefix sharing are out of scope for v1.
- **Hybrid schedulers** (Jamba, Griffin, Zamba): the same slot dictionary holds attention K/V and recurrent states, which is the pattern these systems established.
- **Speculative decoding and early exit** (LayerSkip): served by the same `truncate` primitive that the tests exercise.

When to enable the cache. Any decoder-only generation workload benefits: the regime where decode dominates (long outputs, short prompts) gains the most, while a single full forward (classification, embedding, masked prediction) gains nothing and should leave `enabled` false. Quantization is worth enabling when memory, not latency, is the binding constraint, and `int4_kivi` should keep a protected window of at least the attention-sink span; window-with-sinks eviction is the right default for streaming or unbounded contexts. For recurrent layers the state is already constant-size, so eviction is meaningless and quantization is disabled by design.

24 The Experiment Harness

This annex documents the machinery behind Section 9. Its purpose is reproducibility across rounds: an experiment is simply a new folder under `experimentos/`, and the same consolidation, reporting, and publishing scripts operate on it unchanged. The harness is deliberately decoupled

from the model code in both directions—it only reads what training and deployment write—so an experiment never depends on the library version that produced it, and the library never learns about the campaign that used it. The practical consequence is that rounds remain interpretable and rebuildable long after the library has evolved.

24.1 Stages and Records

Each configuration is one YAML file and one slug. The slug is a short identifier that encodes the family and the mixers: `attn_nsa__engram` is a hybrid pair (first mixer `nsa`, second mixer `engram`), `attn_single_mamba` is a single-mixer baseline, and the ablation families carry their own prefixes (`opt_`, `norm_`, `pe_`, `trans_`, `vit_`, `task_`, `tokenizer_`, `batch_`, `realdata_`, `audio_`, `bitnet_`). Every run executes the same five stages: train, export the checkpoint, deploy in a standard and a quantized format, evaluate four held-out masked prompts per format, and export the Hugging Face package.

The failure policy deserves a note. A failure in one stage is recorded, does not abort the sweep, and surfaces later as a missing row rather than as a silent zero. The distinction matters for downstream statistics: a model that failed to train must not be mistaken for a model that trained and scored zero, and a missing cell in the atlas must be traceable to a recorded failure rather than to an unnoticed gap.

Three recorders write independently and are joined only at reporting time:

1. **The per-step training log.** One CSV per run with one row per optimisation step: loss, accuracy, learning rate, gradient norms per parameter group, clipping ratio, step time and throughput. Round 2 logged 2,656,372 rows, which are streamed and reduced to per-epoch means before any chart is drawn—no one plots 2.6 million points, and the reduction is deterministic, so the published curve is an exact function of the raw log.

2. **The parameter census.** For every configuration, the number of parameters in each module group (attention, embeddings, FFN, head, normalization, positional encoding, mHC, router, residual). This is what separates "the model is bigger" from "the attention block is bigger", and it allows cost to be attributed per component rather than per model.

3. **The pipeline ledger.** One row per stage execution with category, status, duration and return code. It is the only record of the failures, and therefore the only source for the reliability analysis.

24.2 The Stage Ledger of Round 2

Table 30 reproduces the ledger. Training dominates both the record count and the cost: 1,039 attempts, of which 868 succeeded and 171 failed, with a median of 68 minutes and a P90 of 84 minutes per attempt—906 CPU-hours in total, of which 905 belong to the 868 analysed runs and the remainder to the discarded attempts. Every other stage is measured in seconds. Deployment is the only stage besides training that fails (104 times out of 1,734), and the failures are symmetric: whenever a configuration's standard export fails, its quantized export fails too. The symmetry points to a shared root cause upstream of quantization—checkpoint or configuration serialisation—rather than to the quantization step itself. Evaluation and Hugging Face export never failed in the analysed population.

Table 30: Pipeline stage ledger for Round 2. Durations are seconds; totals are hours of wall-clock across the whole campaign.

Stage	Records	OK	Failed	Median (s)	P90 (s)	Total (h)
Training	1,039	868	171	4,101.45	5,037.3	906.0
HF export	867	867	0	17.41	44.7	5.4
Inference evaluation	1,630	1,630	0	10.37	13.1	4.9
Deployment	1,734	1,630	104	8.64	38.0	8.8
BitNet GGUF	1	1	0	3.32	3.3	0.0

The 275 failures are informative rather than merely inconvenient. All 171 training failures sit outside the 868 analysed configurations, which were selected precisely because they trained; the 104 deployment failures occur in symmetric pairs, as explained above. Read as a reliability measurement, the ledger says that the pipeline fails loudly and early—in training or in export—and never silently during evaluation. The failure distribution by family, and the identical-fingerprint analysis of the four mixers that collapse at 32 tokens, are charted in Appendix 25.

24.3 Consolidated Datasets

The reporting chain starts from the raw per-run folders and produces a small set of joined tables (Table 31). The join key is the configuration slug: `master.csv` carries one row per configuration with the parameter census, the training summary, the stage statuses, and the inference scores attached side by side. The tables are published with the report so that any figure can be rebuilt from them without the 702 MB of per-step logs, and the raw per-run CSVs are published separately, gzip-compressed, for anyone who needs step-level data.

Table 31: Files published with a round. `master.csv` is the join key: one row per configuration with parameters, training, pipeline and inference metrics.

File	Contents
`master.csv`	One row per configuration; parameters by group, final and averaged training metrics, stage durations and statuses, inference scores per format.
`training_per_epoch.csv`	Per-run, per-epoch means of loss, accuracy, learning rate, gradient norms and throughput (2.66 M steps reduced to 86,274 rows).
`inference_samples.csv`	Every inference record: four samples, two formats, one row per sample.
`pipeline_stages.csv`	The full stage ledger (5,271 rows).
`param_metrics.csv`	The raw parameter census by module group.
`training_metrics_avg.csv`	Per-run means of the 34 logged metrics.
`training_metrics_final.csv`	Last-step snapshot of every run.
`training_metrics_std.csv`	Per-run sample standard deviations.
`results.md.csv`	Human-readable digest of the pipeline results.
`data/raw/*.csv.gz`	The 868 per-run step logs, one file per configuration.

24.4 Rebuilding a Round

The pipeline is three scripts and a publisher, all standard-library or pandas/numpy/matplotlib only, and every step runs on CPU:

1. `python experimentos/to_csv.py N` reads `experimentos/N/` without modifying it and writes consolidated CSVs to `experimentos/N/csvs/`: the inference table with the `notes` column

expanded into one column per `key=value` pair, the parameter census as one row per configuration, the results ledger as a flat table, and the per-run training CSVs concatenated together with their per-run mean, standard deviation and final row.

2. `python experimentos/N/report/build_data.py` streams the raw logs, joins the three recorders by slug and writes `master.csv` and the derived tables.

3. `python experimentos/N/report/make_charts.py` renders every figure twice (dark and light themes) and records the captions in `registry.json`; `make_report.py` consumes both to emit the bilingual static page.

4. `bash experimentos/N/report/publish.sh` copies the page, the figures and the datasets to the GitHub Pages mirror. It never commits or pushes by itself.

The paper figures are a curated subset of the same chart builders, exported as vector PDFs by `make_paper_figures.py`. The division of labour is deliberate: one set of chart builders feeds both the web report and the paper, so a number in the paper and the corresponding pixel on the website cannot drift apart. The whole chain is CPU-only and reproduces every number in this annex from the published datasets.

24.5 Round 1: The Pilot

Round 1 exercised the same harness end-to-end at smoke scale: 5,129 stage records, training runs of roughly half a minute, and the same five-stage sequence. Its purpose was to validate the mechanics—slug generation, ledger bookkeeping, deployment symmetry and the reporting joins—before committing CPU time to the real campaign. It is not analysed further here because the runs were too short to separate architectures; its value is procedural: Round 2 inherited a harness whose bookkeeping had already been exercised and whose failure modes were already known.

25 Round 2: The Mixer Atlas

This annex reports Round 2 in full. It follows the order of the published report: the parameter landscape, the training outcomes, the learning dynamics, the A×B atlas, the ablation sweeps, the deployment and inference measurements, the reliability of the pipeline, and the correlations between the measured features. All numbers come from `master.csv` of the published dataset; all figures are regenerated by `make_paper_figures.py` from the same chart builders as the web report, so the annex and the website cannot disagree.

25.1 Setup

The population is 868 configurations: 703 hybrid-pair records (667 unique pairs after collapsing the `retnet_attn` alias duplicate), 38 single-mixer baselines, 48 vision transformers and 79 ablations. Every run trains for 100 epochs (about 3,000 steps) on a 512-token Spanish vocabulary with 32-token sequences, one seed, the same warm-up-plus-cosine schedule and BitNet enabled by default. Holding the schedule, the data and the budget fixed is what makes the cells comparable: the only moving parts are the architectural choices under study. Hybrid pairs are the two-element combinations of the 42-entry mixer list, with layer 0 running mixer A and layer 1 running mixer B; no reversed pair exists, so the first/second asymmetry is descriptive. A cell of the grid is covered only if both of its mixers train in every configuration that contains them: the grid has $42 \times 41/2 = 861$ cells and 667 were measured.

25.2 The Parameter Landscape

The 868 configurations span 127,328 to 415,904 parameters, with a median of 220,160. Hybrid pairs sit in the middle of the range because they stack two attention modules; the smallest runs are single-mixer baselines such as `attn_single_mamba` (127,328 parameters), and the largest is `engram__conformer` at 415,904, driven by the `engram` memory bank, which attaches a large key–value table to the attention block.

The composition chart (Figure 109) is the most useful diagnostic of the section. The token embeddings and the output head are fixed costs of roughly 49k parameters each, so almost every difference between configurations is paid in the attention block and the FFN—the two components the experiments actually vary. Vision transformers are the most parameter-hungry family because they add patch embeddings and a larger classification head, while the optimizer, normalization and positional-encoding ablations keep the same 210k backbone by design, so that only the ablated component differs. Parameter cost has a clear per-mixer reading: `engram` adds 100k–160k parameters wherever it appears, and `mamba` and `retnet` are among the cheapest second mixers even though they are the worst performers—cheap and bad, the worst combination on both axes.

25.3 Training Outcomes

Across the hybrid pairs, the final training loss ranges from 8.116 to 10.450 nats (median 9.595). The best pair is `nsa__engram` at 8.116, followed by `gated_deltanet2__sparda` (8.567) and `falcon3__conformer` (8.620); the worst, `gqa__tucker`, needs 10.450. The spread between the best and worst pair is wider than the spread between almost all optimizer ablations—the strongest argument in the round for choosing the attention pair deliberately: the mixer stack matters more than the optimizer.

Loss and held-out accuracy, however, disagree. Their rank correlation over the pairs is weak; the pair with the absolute best loss scores 0.0% top-1, while the best-accuracy pair (`fox__falcon1a`) reaches 12.55% with a worse training loss (9.292). The reason is resolution: with four evaluation sentences, a single lucky or unlucky token rewrites the ranking. Both signals are therefore reported throughout, and the inference tables should be read as a catastrophic-failure check rather than as a leaderboard. Table 32 lists the ten best pairs by final loss and Table 33 the ten best by held-out accuracy; the two tables have no pair in common.

Table 32: The ten best hybrid pairs by final training loss.

Pair (A + B)	Params	Final loss	Top-1	Top-5	Train (min)
`nsa + engram`	383,999	8.116	0.00%	25.0%	154.5
`gated_deltanet2 + sparda`	254,336	8.567	0.00%	25.0%	78.2
`falcon3 + conformer`	254,640	8.620	4.67%	25.0%	32.6
`falcon1a + conformer`	254,640	8.633	2.38%	25.0%	23.0
`falcon1 + conformer`	254,640	8.639	4.67%	25.0%	21.3
`fox + falcon2a`	222,768	8.666	2.40%	25.0%	84.2
`gqa + conformer`	242,720	8.670	5.87%	25.0%	88.3
`falcon3a + conformer`	254,640	8.681	3.97%	25.0%	23.2
`fox + falcon2`	222,768	8.705	2.40%	25.0%	85.8
`gla + engram`	387,296	8.736	5.17%	25.0%	149.2

Table 33: The ten best hybrid pairs by held-out top-1 accuracy.

Pair (A + B)	Params	Final loss	Top-1	Top-5	Train (min)
`fox + falcon1a`	214,232	9.292	12.55%	0.0%	79.3
`fox + falcon3`	214,232	9.618	12.55%	0.0%	106.1
`fox + falcon3a`	214,232	9.235	12.55%	0.0%	77.3
`gated_deltanet2 + falcon2`	250,216	9.105	12.41%	0.0%	80.7
`gated_deltanet2 + falcon2a`	250,216	8.968	12.41%	0.0%	82.1
`hgrn2 + gated_softmax`	238,688	9.386	12.30%	25.0%	71.7
`hgrn2 + tucker`	238,688	9.221	12.30%	25.0%	81.6
`gqla + mlra`	194,624	9.461	11.85%	0.0%	56.0
`mla + mlra`	194,624	9.464	11.85%	0.0%	86.6
`deltanet + kda`	231,304	9.024	11.37%	0.0%	78.6

25.4 Training Dynamics

Every run uses the same warm-up-plus-cosine schedule, so the learning curves are directly comparable without re-normalisation. The pair population is still descending at epoch 100: the median loss falls from about 10.2 nats at epoch 0 to 9.49 at the end. The sweep therefore measures speed of learning, not converged quality, and every ranking in this annex should be read as "which pair learns fastest in 3,000 steps" rather than "which pair is best in the limit".

The optimisation statistics are calm. Gradient norms are stable across families (1.87 for the attention block, 0.92 for the FFN, 3.47 for the head), and the median clip ratio sits at 0.90—below the clipping threshold of 1.0—so most steps pass unclipped and clipping only fires on spikes. The throughput median is 357 tokens/s for the pair population; the 3k–31k tail visible in the charts belongs to the tokenizer and audio ablations, whose step timer covers only part of the work, which is why the Pareto analysis excludes it.

25.5 The A×B Atlas

The atlas (Figure 117) is the centrepiece of the experiment: a matrix in which each cell is one hybrid pair, rows index the first mixer, columns index the second, and colour encodes the final loss. It is read in two directions. Column by column—fix the second mixer, vary the first—the second mixer dominates: a good B (`sparsek`, `engram`, `mlra`) keeps almost every A in the 8.9–9.6 nat band, while a bad B (`mamba`, `retnet`, `bigbird`) drags even the best A above 10 nats. Row by row—fix the first mixer, vary the second—no A rescues a bad B. The asymmetry is the visual form of the first finding of Section 9.

Coverage is almost complete: 667 of the 861 possible pairs were trained. The 158 pairs that never trained are exactly those containing `cca`, `ccgqa`, `msa` or `ssog`—four mixers that fail in every configuration—which accounts for the count: $4 \times 41 - \binom{4}{2} = 158$.

Which second mixer to pick. Table 34 ranks second mixers by mean final loss across every pair that contains them. The *Pairs* column is the sample size behind each mean and should be read alongside it: a single-pair mean (for example `mamba`, with one pair) is anecdotal, while a 30–40 pair mean is a solid estimate. The bottom half of the table is as informative as the top: `mamba`, `bigbird`, `longformer` and `retnet` are cheap but consistently drag their pairs down—exactly the pattern the heatmaps show, and a warning against equating parameter economy with quality.

Table 34: Second mixers ranked by mean final loss over the pairs that contain them (best eight and worst eight).

Second mixer	Pairs	Mean loss	Mean top-1	Mean params
`mlra`	22	9.368	5.74%	209,581
`conformer`	36	9.369	3.39%	259,861
`engram`	18	9.373	2.05%	379,284
`sparsek`	9	9.382	4.82%	208,926
`falcon2a`	34	9.387	4.33%	229,973
`fox`	16	9.409	2.23%	216,038
`mla`	20	9.457	4.66%	217,989
`mtla`	26	9.459	5.29%	217,223
`falcon1`	30	9.779	3.27%	221,856
`gta`	25	9.788	3.27%	219,828
`titan`	3	9.814	6.28%	204,768
`gla`	11	9.814	3.90%	222,555
`standard`	4	9.960	6.65%	206,264
`sigmoid`	5	10.045	4.37%	207,162
`sparse_transformer`	6	10.108	5.59%	207,760
`mamba`	1	10.279	2.04%	192,608

Interaction beats marginals. Ranking mixers individually cannot reconstruct the atlas, because pairs interact. The synergy chart quantifies this: for each pair it measures how much better (or worse) the pair is than the average of its two marginal means,

$$S_{A,B} = \frac{\mu_A + \mu_B}{2} - L_{A,B},$$

where μ_A and μ_B are the mean losses of all pairs containing A and B respectively. Positive synergy means the pair beats what its parts would predict. `nsa+engram` gains 1.304 nats, and the top of the table shares a pattern: a cheap, clean first mixer stacked under a memory-heavy second one. The negative tail is dominated by `gta`, `gqa` and `tucker` as B—mixers that look average in isolation but clash when stacked after a strong A. For synthesis purposes the message is that a ranked list of mixers is a necessary summary but an insufficient design tool: the atlas, or at least the synergy table, must be consulted before composing a stack.

Table 35: The best and worst mixer interactions by synergy. Positive values mean the pair is better than its parts suggest.

Pair	Pair loss	Expected	Synergy	Top-1
`nsa + engram`	8.116	9.420	+1.304	0.00%
`gated_deltanet2 + sparda`	8.567	9.448	+0.881	0.00%
`fox + falcon2`	8.705	9.446	+0.741	2.40%
`gqa + falcon1`	8.784	9.518	+0.734	2.86%
`fox + falcon2a`	8.666	9.390	+0.724	2.40%
`bigbird + gta`	10.445	9.725	−0.720	1.79%
`retnet + gated_softmax`	10.239	9.533	−0.706	7.70%
`mamba + engram`	10.277	9.543	−0.734	1.10%
`mamba + sparda`	10.391	9.643	−0.749	5.09%
`gqa + tucker`	10.450	9.432	−1.018	7.76%

25.6 Ablations

Optimizers. The optimizer sweep is the most sobering chart of the round: 23 optimizers, the same 210k backbone, and a final-loss range of only 10.10–10.77 nats (Table 36). `q_apollo` wins at 10.098 and `galore_adamw` closes the table at 10.772, but a dozen optimizers land on exactly 10.404—the run is too short for the choice of optimizer to separate from noise, and the honest summary at this budget is that most modern optimizers are interchangeable. The speed differences are real, however: mean step time varies by roughly 30%, and the second-order and orthogonalised methods (`shampoo`, `muon`) pay a visible overhead. When runs are short, the optimizer is best chosen for speed rather than for final loss.

Table 36: Optimizer ablation: final loss, held-out top-1 and throughput on the same backbone.

Optimizer	Final loss	Top-1	Tok/s
`q_apollo`	10.098	3.63%	436
`sophia`	10.104	1.79%	378
`lion`	10.245	4.64%	279
`shampoo`	10.401	7.76%	382
`adopt`	10.404	7.76%	391
`mars_adamw`	10.404	7.76%	343
`adan`	10.404	7.76%	359
`adamw`	10.404	7.76%	384
`schedulefree_adamw`	10.404	7.76%	487
`soap`	10.404	7.76%	412
`ademamix`	10.404	7.76%	366
`cautious_adamw`	10.404	7.76%	365
`prodigy`	10.439	0.00%	349
`sgd_momentum`	10.450	0.00%	347
`turbo_muon`	10.454	0.00%	536
`anon`	10.458	0.00%	387
`muon`	10.461	0.00%	507
`apollo`	10.519	0.00%	401
`adafactor`	10.530	0.00%	395
`ada_muon`	10.620	1.79%	497
`lamb`	10.653	7.76%	464
`radam`	10.738	7.76%	502
`galore_adamw`	10.772	9.54%	451

Normalization and positional encoding. Among the normalizations (Table 37), `flash_norm` wins clearly at 9.749 nats—half a nat ahead of the runner-up, a large margin by the standards of this round—while `prms_norm` and `rms_norm` lag. In the positional encodings (Table 38), the PAPE family (`pape_ri`, `pape`, `pape_efficient`) sweeps the podium, and classical RoPE is mid-table. Both results are useful calibration: "modern" is not automatically better on a 100-epoch run at this scale, and component choices that look like refinements in the literature can separate by half a nat here.

Table 37: Normalization ablation on the shared backbone.

Variant	Final loss	Top-1	Params
`flash_norm`	9.749	1.43%	209,600
`dynamic_tanh`	10.246	0.00%	210,752
`derf`	10.295	1.79%	210,762
`layer_norm`	10.401	7.76%	210,752
`rms_norm`	10.436	7.76%	210,272
`prms_norm`	10.573	0.00%	210,272

Table 38: Positional-encoding ablation.

Encoding	Final loss	Top-1	Params
`pari`	9.235	0.00%	211,520
`pape`	9.455	4.17%	216,928
`pape_efficient`	9.455	4.17%	216,928
`learned_absolute`	9.525	7.02%	216,896
`hope`	10.348	—	210,752
`sinusoidal_absolute`	10.364	6.68%	210,752
`bam`	10.397	9.54%	210,768
`nope`	10.397	9.54%	210,752
`none`	10.397	9.54%	210,752
`rope`	10.401	7.76%	210,752
`sinusoidal_rotary`	10.401	7.76%	210,752
`alibi`	10.417	7.76%	210,752

Structural switches. The structural toggles (Table 39) separate more than the optimizers do—evidence that, at this scale, architecture is a stronger lever than the update rule. The three wins are factorised embeddings (9.623), a SwiGLU FFN (9.901) and two Hyperloop loops (9.908). The BitNet toggles land at 10.016, mHC at 10.186, and the residual variants between 10.43 and 10.49. Mixture-of-Depths is the worst switch at 10.854, consistent with its documented incompatibility with mHC at this depth.

Table 39: Structural switches, ordered by final loss.

Switch	Final loss	Top-1	Params
factorized_emb	9.623	1.43%	181,056
ffn_swiglu	9.901	2.49%	228,800
num_loops_2	9.908	6.94%	154,880
fixed_softmax_ssmax_bitnet	9.936	2.60%	209,864
bitnet_conv	10.016	2.60%	209,856
bitnet_on	10.016	2.60%	209,856
mhc	10.186	1.79%	254,092
bitnet_routers	10.327	—	247,104
ssmax	10.359	9.54%	210,760
ffn_relu	10.393	3.83%	210,752
fixed_both	10.401	7.76%	210,752
embedding_conv	10.401	7.76%	210,752
fixed_ce	10.401	7.76%	210,752
pos_rope	10.401	7.76%	210,752
fixed_softmax	10.401	7.76%	210,752
residuals_full_attn	10.433	9.54%	210,944
residuals_none	10.476	5.71%	210,752
moe	10.477	—	248,384
residuals_block_attn	10.487	1.79%	210,944
mod	10.854	—	210,944

Vision and tasks. The vision block works. Segmentation reaches 90.7% mean held-out accuracy, classification 75.8% and patch prediction 69.9%. These scores are an order of magnitude above the MLM pairs (4.0% median) because the task is easier, not because the mixer matters more; the report therefore keeps the families separate and draws no cross-family conclusion. The tokenizer and task ablations behave as expected and mainly serve as scale anchors: the tokenizer moves the loss by several nats because it changes the vocabulary and the average sequence length, and the task changes it because the objective itself changes. Neither is comparable with the mixer sweep, and both are excluded from it.

25.7 Deployment and Inference

Post-training quantization is the good news of the round. Across the 810 configurations with both formats, mean top-1 moves from 5.62% to 5.43% and mean top-5 from 13.36% to 13.24%—deltas of -0.19 and -0.12 percentage points. Eight hundred of the 810 configurations are exactly unchanged, and the few that move do so by a single sample out of four. For a smoke-scale model this is the expected result, but it is worth measuring rather than assuming.

The speed story is less flattering, and the reason is structural. The quantized deployment is faster than the standard one in only 32.3% of runs (median ratio 0.64×) because the deploy stage is dominated by serialisation and format conversion, not by inference; the evaluation stage, which does measure inference, is faster 51.2% of the time with a median 1.03× speed-up. Inference itself takes about 10.5 s for four sentences—the number that matters at serving time.

The best inference configuration is not the best-loss configuration. `mamba__engram` hits three of four tokens in the top-5 (75.0%) with a training loss of 10.277, above the pair median. With four samples per configuration, top-5 is a blunt instrument: use it to catch catastrophic failures, not to crown a winner.

Table 40: Configurations with the best standard-format top-5 hit rate.

Configuration	Top-5 std	Top-1 std	Top-5 quant	Final loss	Params
mamba__engram	75%	50%	75%	10.277	347,168
mamba__nsa	75%	0%	75%	9.262	192,479
retnet__fox	75%	0%	75%	9.873	220,936
bigbird__kda	50%	0%	50%	10.196	221,128
fox__engram	50%	25%	50%	9.535	375,496
fox__gated_softmax	50%	25%	50%	9.892	220,744
fox__tucker	50%	25%	50%	9.890	220,744
gated_deltanet__tucker	50%	25%	50%	9.796	230,928
gla__falcon1	50%	0%	50%	9.511	226,032
gla__mlra	50%	25%	50%	9.538	211,808

25.8 Reliability of the Pipeline

The ledger records 5,271 stage executions. 868 training runs succeeded and 171 failed with a non-zero exit code—all of them outside the 868 analysed configurations, which were selected precisely because they trained. Deployment failed 104 times out of 1,734, always in symmetric pairs: a configuration that fails the standard export also fails the quantized one, which points to a shared root cause (checkpoint or configuration serialisation) rather than to quantization itself.

Two data-quality findings shape the interpretation of the atlas. First, four mixers never train: `cca`, `ccgqa`, `msa` and `ssog` fail in their four singles and in all 158 pairs that contain them—exactly the 158 missing cells. Second, four mixers are indistinguishable at a 32-token context: `standard`, `bigbird`, `longformer` and `titan` share bit-identical losses and parameter counts because their windows (at least 64 tokens) cover the whole sequence and the position-augmented variant reduces to plain multi-head attention; they agree on all 33 shared second mixers. Separately, 37 pairs appear as `attn_retnet__X` and `attn_retnet__X` with identical metrics—two names for one implementation—which the atlas collapses to one row. Each of these facts is a caveat carried forward rather than a defect hidden: the atlas has 667 measured cells, 158 structurally missing ones, and four mixers that are, at this context length, the same model.

25.9 What Predicts Quality

Over the hybrid pairs, parameter count correlates with final loss at $r = -0.20$ and with held-out accuracy at $r = -0.07$: size barely buys quality here. What it does predict reliably is cost ($r = 0.68$ with step time, $r = 0.65$ with wall-clock). Training loss carries only a weak signal about inference quality ($r = 0.14$ with top-5)—the quantitative form of the loss/accuracy disagreement noted above. Gradient norm and clip ratio are exactly redundant with each other ($r = 1.00$), which is expected because the clip ratio is computed from the gradient norm, and neither correlates with quality.

The correlation matrix of Figure 136 is computed over all 868 rows. Read that way, parameter count correlates with loss at -0.43 and with accuracy at $+0.42$, apparently contradicting the pair-only numbers. Both readings are correct, and the reconciliation is a textbook confound: the all-rows matrix is dominated by the contrast between vision transformers (larger, much lower loss, much higher accuracy) and MLM pairs, so it measures the gap between families rather than the effect of size within a family. Within the population the atlas is about—the hybrid pairs—size is close to irrelevant, and that is the number to quote.

25.10 Practical Reading and Next Steps

If a pair must be chosen blind, the round supports a simple heuristic: pick a memory-heavy second mixer (`engram`, `sparsek`, `mlra`) and a cheap, stable first mixer (`fox`, `gqa`, `gated_deltanet2`),

and avoid `mamba`, `retnet` and `bigbird` as B. Expect about ±0.3 nats of interaction noise—larger than the difference between most optimizers, and therefore the dominant uncertainty when composing a stack.

The round also defines its own successor, and its limitations are the specification. The atlas should be re-run at 256–512 tokens, where the four window-collapsing mixers become distinguishable and the sparse patterns actually sparsify; the evaluation should use a 200-sentence held-out set scored by per-token log-likelihood, because four sentences cannot rank 800 configurations; the top pairs should train to 300 epochs, because every curve is still descending; the `retnet` alias and the four failing mixers should be fixed or removed from the list; and the ablation families should run at least three seeds, because their current spread is within seed noise. Finally, quantization should become the default deployment format: it cost 0.12 points of top-5 and sped up evaluation in half the runs.

26 Configuration Schema Structure

The Frankenstein Transformer toolkit enforces a strict YAML configuration schema with `additionalProperties: false` at all levels. The schema is defined in `src/schema.yaml`, which references modular sub-schemas in `src/schema/`. Every configurable field must be declared in the schema; unrecognized keys are rejected at validation time. This appendix documents the complete schema structure, including all fields, their types, constraints, and cross-component validation rules.

26.1 Top-Level Structure

The root schema object has the following top-level properties. All are optional except `training`, which is required:

- `base_model` (string) — HuggingFace model identifier or local path for continual pretraining or SBERT fine-tuning. When set, `model_class` and `model` are not required (the architecture is inferred from the checkpoint).
- `tokenizer` (object) — Tokenizer configuration, available to both the `base_model` path and custom from-scratch models. Two modes are selected via `source`: `hf_repo` (default) loads a pretrained tokenizer from `name_or_path`; `train_from_dataset` trains a new tokenizer from the dataset text (see §26.4). Required when using `base_model` with the `mlm` task.
- `model_class` (enum) — Model architecture variant: `frankenstein_text_encoder` (mixed-architecture encoder), `frankenstein_text_decoder` (autoregressive causal decoder), or `frankenstein_vit` (vision transformer). Required when `base_model` is not set.
- `model` (object) — Model architecture parameters (see §26.2). Required when `base_model` is not set.
- `training` (object, **required**) — Training configuration (see §26.3).

The schema enforces a mutual-exclusivity rule: either `base_model` must be present, or both `model_class` and `model` must be present.

26.2 Model Configuration

The `model` object defines all architectural hyperparameters. Required fields are `dims.vocab_size`, `dims.hidden_size`, `dims.num_layers`, `dims.num_heads`, and `dims.layer_pattern`. Table 41 lists all model fields with their types and descriptions.

Field	Type	Description

continued on next page

Field	Type	Description
*model.dims.**		
`dims.mode`	enum (encoder, decoder)	Attention masking mode. `encoder` = bidirectional; `decoder` = causal.
`dims.vocab_size`	int ≥ 1 (required)	Number of unique tokens in the vocabulary. Must match the tokenizer's vocab size.
`dims.hidden_size`	int ≥ 1 (required)	Hidden dimension. Must be divisible by `num_heads`. Typical: 512–2048.
`dims.num_layers`	int ≥ 1 (required)	Number of physical transformer layers. Typical: 6–24.
`dims.num_loops`	int ≥ 1	Logical loop count. Each layer in `layer_pattern` runs `num_loops` times. Default: 1.
`dims.num_heads`	int ≥ 1 (required)	Number of parallel attention heads. Must divide `hidden_size` evenly.
`dims.dropout`	float [0,1]	Global dropout rate. Typical: 0.1 (BERT standard).
`dims.layer_pattern`	array of enum (required)	Ordered sequence of layer types. Length must equal `num_layers`. 48 mixer codes available (see Table 26.2.1).
*model.norm.**		
`norm.type`	enum	Normalization strategy: `layer_norm`, `dynamic_tanh`, `derf`, `rms_norm`, `prms_norm`, `flash_norm`.
`norm.partial_ratio`	float (0,1]	Fraction of hidden dims for pRMSNorm estimation. Default: 0.0625.
*model.embedding.factorized.**		
`embedding.factorized.enabled`	bool	Enable embedding matrix factorization to reduce parameters.
`embedding.factorized.dim`	int ≥ 1	Intermediate dimension for factorized embeddings. Typical: 64–256.
*model.embedding.conv.**		
`embedding.conv.enabled`	bool	Apply Conv1d over token embeddings to capture local n-gram patterns.
`embedding.conv.kernel`	int ≥ 1	Kernel size for embedding Conv1d. Typical: 3.
*model.attention.ode.**		
`attention.ode.solver`	enum (rk4, euler)	Numerical integration method for ODE layers. Default: rk4.
`attention.ode.steps`	int ≥ 1	Integration steps between consecutive transformer layers. Default: 2.
*model.attention.retnet.**		

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Field	Type	Description
attention.retnet.heads	int ≥ 1	Retention heads for retnet layers. Must divide hidden_size. Default: 8.
*model.attention.mla.**		
attention.mla.latent_rank	int ≥ 1	Rank of joint KV latent for MLA. Default: hidden_size // 2.
*model.attention.gqla.**		
attention.gqla.latent_rank	int ≥ 1	Rank of joint KV latent for GQLA. Default: hidden_size // 2.
attention.gqla.num_groups	int ≥ 1	Number of GQA groups for GQLA decode path. Must divide num_heads.
attention.gqla.decode_path	enum (mqa_absorb, gqa)	Decoding path for GQLA.
*model.attention.mlra.**		
attention.mlra.latent_rank	int ≥ 1	Total latent rank for MLRA. Default: hidden_size // 2.
attention.mlra.num_latent_heads	int ≥ 1	Number of disjoint latent sub-spaces for MLRA. Default: 4.
*model.attention.tucker.**		
attention.tucker.query_rank	int ≥ 1	Tucker rank for query weight tensor. Default: hidden_size.
attention.tucker.key_rank	int ≥ 1	Tucker rank for key weight tensor. Default: hidden_size // 2.
attention.tucker.value_rank	int ≥ 1	Tucker rank for value weight tensor. Default: hidden_size // 2.
*model.attention.iha.**		
attention.iha.num_pseudo_heads	int ≥ 1	Number of pseudo-heads per head for IHA. Default: num_heads.
*model.attention.gta.**		
attention.gta.num_shared_groups	int ≥ 1	Number of head groups sharing an attention map for GTA. Must divide num_heads.
attention.gta.value_latent_rank	int ≥ 1	Rank of value-cache latent for GTA. Default: hidden_size // 2.
*model.attention.mtla.**		
attention.mtla.latent_rank	int ≥ 1	Latent rank for MTLA. Default: hidden_size // 2.
attention.mtla.merge_factor	int ≥ 1	Number of consecutive KV entries merged per temporal slot. Default: 2.
attention.mtla.stride	int ≥ 1	Stride between merged temporal slots. Default: mtla_merge_factor.
*model.attention.cca.**		

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Field	Type	Description
attention.cca.latent_rank	int ≥ 1	Latent width for CCA. Default: hidden_size // 4. Must be divisible by num_heads.
attention.cca.num_conv_layers	enum (0, 1, 2)	Number of convolution layers in CCA. Default: 2.
attention.cca.conv_kernel_seq	int ≥ 1	Kernel size of depth-wise causal sequence conv. Default: 4.
attention.cca.conv_kernel_ch	int ≥ 1	Kernel size of head-wise grouped channel conv. Default: 3.
attention.cca.qk_mean	bool	Enable q-k-mean bias in CCA. Default: true.
attention.cca.value_shift	bool	Enable value-shift in CCA. Default: true.
*model.attention.ccgqa.**		
attention.ccgqa.query_latent_rank	int ≥ 1	Query latent width for CCGQA. Default: hidden_size // 2.
attention.ccgqa.kv_latent_rank	int ≥ 1	KV latent width for CCGQA. Default: hidden_size // 8.
attention.ccgqa.num_kv_heads	int ≥ 1	Number of KV heads for CCGQA. Must divide num_heads.
attention.ccgqa.num_conv_layers	enum (0, 1, 2)	Number of convolution layers in CCGQA. Default: 2.
attention.ccgqa.conv_kernel_seq	int ≥ 1	Kernel size of depth-wise causal sequence conv. Default: 4.
attention.ccgqa.conv_kernel_ch	int ≥ 1	Kernel size of head-wise grouped channel conv. Default: 3.
attention.ccgqa.qk_mean	bool	Enable q-k-mean bias in CCGQA. Default: true.
attention.ccgqa.value_shift	bool	Enable value-shift in CCGQA. Default: true.
*model.attention.msa.**		
attention.msa.block_size	int ≥ 1	Block size for MiniMax Sparse Attention. Default: 128.
attention.msa.topk_blocks	int ≥ 1	Number of blocks selected per GQA group per query. Default: 16.
attention.msa.index_dim	int ≥ 1	Dimensionality of Index Branch heads. Default: 64.
attention.msa.kl_loss_weight	float ≥ 0	Weight of KL alignment loss for MSA index branch. Default: 0.0.
*model.attention.sparda.**		
attention.sparda.block_size	int ≥ 1	KV block size for SparDA. Default: 128.
attention.sparda.topk_blocks	int ≥ 1	Number of KV blocks selected per GQA group. Default: 16.
attention.sparda.forecast_dim	int ≥ 1	Dimensionality of Forecast projection. Default: 64.
*model.attention.lsa.**		

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Field	Type	Description
attention.lsa.num_long_range	int ≥ 1	Long-range key budget k selected per query. Default: 2048.
attention.lsa.num_buckets	int ≥ 1	Buckets scored per query p; raised so $p \cdot$ pool_size $\geq k$. Default: 512.
attention.lsa.num_sinks	int ≥ 1	Always-attended leading sink keys. Default: 4.
attention.lsa.pool_size	int ≥ 1	Keys summed per pool key before Ward clustering. Default: 4.
attention.lsa.cluster_block	int ≥ 2	Cluster block size; also the window n_w. Default: 512.
attention.lsa.keys_per_bucket	int ≥ 1	Average keys per bucket; yields cluster_block / keys_per_bucket buckets per block. Default: 8.
attention.lsa.max_keys_per_bucket	int ≥ 1	Merge cap per bucket. Default: 16.
*model.attention.fastconformer.**		
attention.fastconformer.window_size	int ≥ 2	Total local span w; per-side context $w/2$ (paper: 128 each side). Default: 256.
attention.fastconformer.global_tokens	int ≥ 0	Leading global attention tokens g (paper: 1; 0 = pure limited context). Default: 1.
attention.fastconformer.global_tokens_spacing	int ≥ 1	Stride s between global positions $\{0, s, \ldots, (g-1)s\}$. Default: 1.
attention.fastconformer.global_attn_separate	bool	Separate global Q/K/V projections ($+3H^2$; §2.2) vs shared (§3.4). Default: false.
*model.attention.engram.**		
attention.engram.max_ngram_size	int ≥ 2	Maximum N-gram order for Engram memory module.
attention.engram.n_heads_per_ngram	int ≥ 1	Number of hash heads per N-gram order.
attention.engram.embed_dim_per_head	int ≥ 1	Embedding dimension per hash head.
attention.engram.kernel_size	int ≥ 1	Causal depthwise conv kernel width for Engram.
attention.engram.seed	int	Random seed for N-gram hash multipliers.
*model.mhc.**		
mhc.enabled	bool	Enable the manifold-constrained n-stream residual (mHC, arXiv:2512.24880). Default: false.
mhc.expansion_rate	int ≥ 1	Residual-stream expansion factor n. The stream becomes $n\times$ hidden_size. 1 recovers the identity mapping. Default: 4.
mhc.sinkhorn_iters	int ≥ 1	Sinkhorn-Knopp normalisation rounds constraining H^{res} to be doubly stochastic. Default: 20.

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Field	Type	Description
mhc.gating_init	float > 0	Initial value of the mHC gating scalars α. Default: 0.01.
mhc.checkpoint	bool	Gradient checkpointing on mHC layers to mitigate the $\sim n\times$ activation-memory increase. Default: false.
mhc.full_prec_under_bitnet	bool	Keep the mHC coefficient projection full-precision under BitNet. Default: true.
*model.moe.**		
moe.enabled	bool	Enable Mixture of Experts in FFN layers. Default: true.
moe.num_experts	int ≥ 1	Total number of expert networks. Default: 8.
moe.top_k	int ≥ 1	Experts activated per token. Must be $\leq$ num_experts. Default: 2.
*model.bitnet.**		
bitnet.enabled	bool	Enable BitLinear ternary weight quantization. Default: true.
bitnet.routers	bool	Also quantize routing/scoring projections. Requires bitnet.enabled=true. Default: false.
bitnet.conv	bool	Quantize embedding Conv1d with BitConv1d. Requires bitnet.enabled=true and embedding.conv.enabled=true. Default: false.
*model.mixture_of_depths.**		
mixture_of_depths.enabled	bool	Enable Mixture-of-Depths token routing. Default: false.
mixture_of_depths.capacity_ratio	float (0,1]	Fraction of tokens updated per layer. Default: 0.5.
mixture_of_depths.router_aux_loss_weight	float ≥ 0	Weight for the router auxiliary loss. Default: 0.0.
*model.ssmax.**		
ssmax.enabled	bool	Enable Scalable Softmax (arXiv:2505.22842). Default: false.
ssmax.s_init	float > 0	Initial per-head SSMax scalar. Default: 1.0.
*model.numerics.**		
numerics.fixed_softmax	bool	Cancellation-free softmax/log-softmax (arXiv:2610.31642). Default: false.
numerics.fixed_cross_entropy	bool	Hard-target CE with zero-sum gradient by construction (arXiv:2610.31642). Default: false.

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Field	Type	Description
*model.ffn.**		
ffn.hidden_size	int ≥ 1	Intermediate FFN dimension. Default: 2× hidden_size.
ffn.activation	enum	FFN non-linearity (43 values). Default: silu.
ffn.activation_config	object	Parameters for learnable/parametric activations.
*model.classification_head.**		
classification_head.enabled	bool	Enable the NLP sequence-classification head. Default: false.
classification_head.num_labels	int or null	Number of classes.
classification_head.pooling_mode	enum (cls, gap)	Pooling for the classification head. Default: cls.

Table 41: Complete model configuration fields. All fields are optional unless marked required.

Fields are grouped under hierarchical sub-objects (dims, norm, embedding, attention, moe, bitnet, mixture_of_depths, ssmax, numerics, ffn, classification_head, mhc, residuals); each sub-header row marks the prefix in effect for the rows that follow it. There are no flat feature keys at model.*.

26.2.1 Layer Pattern Enumeration

The model.dims.layer_pattern array accepts the following 48 mixer codes. Each element defines one physical layer in the model stack. Table 42 lists all available types with their training and inference support.

Type	Description	Train	Infer
standard	Classic multi-head self-attention (BERT-style)	✓	✓
sigmoid	Sigmoid-gated attention	✓	✓
ca	Continuum Attention — exact continuous integration over piecewise-linear memory trajectories	✓	✓
memory	Memory Attention — token-ID memory values (V = K + Norm(E[s]))	✓	✓
gated_softmax	Gated softmax attention with post-SDPA sigmoid	✓	✓
titan	Titan attention with positional encoding (HoPE/RoPE)	✓	✓
retnet	Retention Network (RNN+transformer hybrid)	✓	✓

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Type	Description	Train	Infer
mamba	Selective state-space model (SSM)	✓	✓
ode	Continuous-depth ODE layer	✓	✓
gla	Gated Linear Attention with multiplicative decay	✓	✓
deltanet / gated_deltanet	DeltaNet with corrective delta rule	✓	✓
gated_deltanet2	Gated DeltaNet-2 with decoupled channel-wise erase/write gates	✓	✓
hgrn2	HGRN2 with hierarchical forget gates	✓	✓
fox	Forgetting Transformer (FoX) with forget bias in logits	✓	✓
fpa	First Passage Attention with a range-minimum forget bias	✓	✓
nsa	Native Sparse Attention (three-branch)	✓	✓
engram	N-gram conditional memory lookup	✓	✓
gqa	Grouped-Query Attention with configurable KV heads	✓	✓
longformer	Longformer: sliding windows + global tokens	✓	✓
bigbird	BigBird: local + random + global attention	✓	✓
sparse_transformer	Sparse Transformer: factorized patterns	✓	✓
sparsek	SparseK: differentiable top-k KV selection	✓	✓
sparge	SpargeAttn — two-stage block pruning; dense training, sparse inference	✓	✓
fasa	FASA — frequency-aware token selection; dense training, sparse inference	✓	✓
msa	MiniMax Sparse Attention — block-sparse on GQA	✓	✓
sparda	SparDA — decoupled sparse attention with Forecast projection	✓	✓
lsa	Latent Sparse Attention — semantic key clustering with top-k selection	✓	✓
fastconformer	Fast Conformer — limited-context window + head-pinned global token	✓	✓

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Type	Description	Train	Infer
`kda`	Kimi Delta Attention — channel-wise decay + scalar write gate	✓	✓
`falcon1 / falcon2 / falcon3`	Falcon regression family — RAW prefix-aligned NLMS delta rule	✓	✓
`falcon1a / falcon2a / falcon3a`	Falcon inner-product family — shifted additive writes with ridge decay	✓	✓
`mla`	Multi-Head Latent Attention + RoPE	✓	✓
`gqla`	Group-Query Latent Attention — two decoding paths	✓	✓
`mlra`	Multi-Head Low-Rank Attention — partitionable latent	✓	✓
`tucker`	Tucker Attention — generalised low-rank factorisation	✓	✓
`iha`	Interleaved Head Attention — cross-head pseudo-heads	✓	✓
`gta`	Grouped-head laTenT Attention — shared map + latent values	✓	✓
`mtla`	Multi-head Temporal Latent Attention — temporal KV merge	✓	✓
`cca`	Compressed Convolutional Attention — latent-space attention + convs	✓	✓
`ccgqa`	Compressed Convolutional Grouped Query Attention — CCA + GQA in latent	✓	✓
`gma`	Gaussian Mixture Attention — probabilistic latent routing, $\mathcal{O}(NK)$ linear-time	✓	✓
`ssog`	Separable Sum of Gaussians — Gaussian field over relative position, near-linear $\mathcal{O}(RN\sqrt{N})$	✓	✓

Table 42: Complete `layer_pattern` enumeration. ✓ = supported, × = not supported. `fasa` and `sparge` train as standard dense attention over their own projections and apply their sparse heuristic only at inference.

26.3 Training Configuration

The `training` object controls the entire training pipeline. The only required field is `task`. Table 43 lists all training fields.

Field	Type	Description
task	enum (mlm, sbert, causal_lm) (required)	Training objective. mlm = Masked Language Modeling; sbert = Sentence-BERT fine-tuning; causal_lm = autoregressive next-token prediction (requires model_class='frankenstein_text_decoder').
num_epochs	int ≥ 1	Complete passes through the training dataset (MLM only).
batch_size	int ≥ 1	Examples per optimizer step (before gradient accumulation).
dataloader_workers	int ≥ 0	Parallel worker processes for data loading.
max_length	int ≥ 1	Maximum token sequence length. Typical: 128–2048.
mlm_probability	float [0,1]	Fraction of tokens masked for MLM prediction. Standard: 0.15.
max_samples	int ≥ 1	Total samples before stopping, independent of epochs.
test_percentage	float [0,100]	Fraction of the dataset held out from training and evaluated at the end; the result is written to test_metrics.json. Default: 0 (disabled).
dataset_batch_size	int ≥ 1	Internal batch size for streaming dataset.
num_workers	int ≥ 0	Parallel workers for dataset streaming pipeline.
cache_dir	string	Directory for caching processed datasets.
local_parquet_dir	string	Path to local parquet files for offline streaming.
prefer_local_cache	bool	Check local cache first before remote download.
stream_local_parquet	bool	Stream from local parquet when local_parquet_dir is set.
join_temp_data_context_window	int ≥ 0	If > 0, joins cached token chunks into sequences of this length.
join_temp_data_min_remainder_tokens	int ≥ 0	Minimum content tokens in last partial joined window.
use_amp	bool	Enable FP16 mixed precision.
gradient_accumulation_steps	int ≥ 1	Accumulate gradients over N mini-batches before optimizer step.
optimizer	object	Optimizer configuration (see §26.5). Required when task=mlm or task=causal_lm.
sbert	object	SBERT training configuration (see §26.6). Required when task=sbert.
scheduler_total_steps	int ≥ 1	Total optimizer steps for LR schedule.
scheduler_warmup_ratio	float [0,1]	Fraction of scheduler_total_steps for LR warmup.

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Field	Type	Description
scheduler_type	enum (cosine, constant, linear_warmup_then_constant)	LR decay strategy.
grad_clip_max_norm	float ≥ 0	Maximum allowed gradient norm. 0 = disabled.
inf_post_clip_threshold	float ≥ 0	Threshold for flagging post-clip gradient explosions.
max_nan_retries	int ≥ 0	Maximum retries on NaN/Inf gradients before stopping.
checkpoint_every_n_steps	int ≥ 1	Save rolling checkpoint every N steps.
max_rolling_checkpoints	int ≥ 1	Maximum rolling checkpoints kept. Oldest deleted when exceeded.
num_best_checkpoints	int ≥ 1	Number of best checkpoints kept permanently (by validation loss).
nan_check_interval	int ≥ 1	Steps between NaN/Inf checks.
log_gradient_stats	bool	Log per-layer gradient norms, min/max.
gradient_log_interval	int ≥ 1	Steps between gradient stat logs.
csv_log_path	string	File path for step-level training metrics in CSV format.
csv_rotate_on_schema_change	bool	Rotate CSV file when schema changes.
gpu_metrics_backend	enum (nvml, none)	GPU telemetry backend.
nvml_device_index	int ≥ 0	GPU index to monitor (0 = first GPU).
enable_block_grad_norms	bool	Log gradient norms per block (embeddings, attention, FFN, etc.).
telemetry_log_interval	int ≥ 1	Steps between heavy telemetry logs.
gpu_temp_guard_enabled	bool	Strict GPU temperature monitoring. Requires `gpu_metrics_backend=nvml`.
switch_on_thermal	bool	Critical temp triggers GPU→CPU fallback, auto-resume on cooldown.
gpu_temp_pause_threshold_c	float	Temperature (°C) to pause training.
gpu_temp_resume_threshold_c	float	Temperature (°C) to resume training after pause.
gpu_temp_critical_threshold_c	float	Informational logging marker for severe thermal events.
gpu_temp_poll_interval_seconds	float	Seconds between temp checks during cooldown wait.
use_galore	bool	Enable Gradient Low-Rank Projection for memory-efficient training.
galore_rank	int ≥ 1	Low-rank projection dimension for GaLore.
galore_update_interval	int ≥ 1	Steps between projection matrix recomputation.
galore_scale	float ≥ 0	Scaling factor for projected gradients.
galore_max_dim	int ≥ 1	Maximum tensor dimension for GaLore projection.

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Field	Type	Description
hf_output_dir	string	Directory for saving final model via save_pretrained().

Table 43: Complete training configuration fields.

26.4 Tokenizer Configuration

The top-level tokenizer object is available to both training paths: the base_model route (continual pretraining / fine-tuning, where the tokenizer must match the pretrained checkpoint) and the custom from-scratch route (where it resolves the vocabulary before the embedding matrix is allocated). The source field selects between two modes:

- hf_repo (default) — loads a pretrained tokenizer from name_or_path (a HuggingFace repo id or local path) through AutoTokenizer.from_pretrained(). Complementary flags: use_fast (Rust-backed fast tokenizer, default true) and trust_remote_code (default false).
- train_from_dataset — trains a *new* tokenizer from the corpus text with the HuggingFace tokenizers library. The corpus is the top-level text_dataset block (HF hub id, optionally streamed, or a local parquet/json directory); when the config is driven by an embedding host such as dashAI, the host's run dataset can supply it instead. The resulting vocabulary size is injected into model.dims.vocab_size before model construction, so the embedding matrix always matches the tokenizer. In this mode name_or_path must be omitted.

Table 44 lists the tokenizer.training sub-object fields.

Field	Type	Description
algorithm	enum (bpe, wordpiece, wordlevel, unigram)	Tokenizer training algorithm. bpe (Byte Pair Encoding, GPT-2 style, default); wordpiece (original BERT); wordlevel (word lookup table); unigram (subword language model, auto-pruning).
vocab_size	int ≥ 1	Target vocabulary size. Required for bpe/wordpiece/wordlevel; optional for unigram.
min_frequency	int ≥ 1	Minimum token frequency to keep (default 2).
special_tokens	array of string	Special tokens registered in the tokenizer (default [PAD]/[UNK]/[CLS]/[SEP]/[MASK]).
save_dir	string	Directory persisted via save_pretrained().
lowercase	bool	Lowercase all text before training (default false).
strip_accents	bool	Strip diacritics before training (default false).
pre_tokenizer	enum (whitespace, bytelevel, bert)	Pre-tokenization strategy (default whitespace).

max_token_length	int ≥ 0	Per-token character cap; 0 disables the limit.

Table 44: tokenizer.training.* fields (required when source=train_from_dataset).

26.5 Optimizer Configuration

The training.optimizer object has two fields:

- optimizer_class (enum, required) — One of 23 optimizer algorithms: sgd_momentum, adamw, adafactor, galore_adamw, prodigy, lion, sophia, muon, turbo_muon, radam, adan, adopt, ademamix, mars_adamw, cautious_adamw, lamb, schedulefree_adamw, shampoo, soap, anon, apollo, apollo_mini, q_apollo.
- parameters (object) — Optimizer hyperparameters with prefixed keys in the format <optimizer>-<group>_<param>.

26.5.1 Shared Per-Group Suffixes

All optimizers support the following per-parameter-group suffixes, prefixed by the optimizer class name (e.g., adamw-lr_embeddings):

- **Learning rate groups:** lr_embeddings, lr_norms, lr_attention, lr_other
- **Weight decay groups:** wd_embeddings, wd_norms, wd_attention, wd_other
- **Beta groups:** betas_embeddings, betas_norms, betas_attention, betas_other
- **Epsilon groups:** eps_embeddings, eps_norms, eps_attention, eps_other

Note: parameters of ODE, RetNet, and Mamba mixers are routed into the attention group rather than having dedicated groups.

26.5.2 Optimizer-Specific Global Suffixes

Table 45 lists the additional global suffixes supported by each optimizer (prefixed by the optimizer name).

Optimizer	Specific Global Suffixes
sgd_momentum	momentum, nesterov
adamw	(none — shared suffixes only)
adafactor	beta2_decay, clip_threshold, eps1, eps2
galore_adamw	rank, update_proj_gap
prodigy	d_coef
lion	(none — shared suffixes only)
sophia	rho, update_k
muon	momentum, nesterov, ns_steps, ns_eps
turbo_muon	momentum, nesterov, ns_steps, ns_eps
radam	(none — shared suffixes only)
adan	(none — shared suffixes only)
adopt	(none — shared suffixes only)
ademamix	(none — shared suffixes only)
mars_adamw	(none — shared suffixes only)
cautious_adamw	cautious_clip
lamb	(none — shared suffixes only)
schedulefree_adamw	(none — shared suffixes only)

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Optimizer	Specific Global Suffixes
shampoo	(none — shared suffixes only)
soap	(none — shared suffixes only)
anon	gamma
apollo	rank, update_proj_gap, scale, scale_type, proj_type, scale_front, disable_nl
apollo_mini	update_proj_gap, scale, proj_type, scale_front, disable_nl
q_apollo	rank, update_proj_gap, scale, scale_type, proj_type, scale_front, disable_nl, quant_bits

Table 45: Optimizer-specific global parameter suffixes. All keys are prefixed with the optimizer class name (e.g., sgd_momentum-momentum).

26.6 SBERT Configuration

The training.sbert object configures Sentence-BERT fine-tuning. It is required when training.task=sbert. Key fields include:

- dataset_name (string) — HuggingFace dataset ID or local path.
- dataset_type (enum) — paired_similarity, triplets, or qa.
- columns (object) — Optional column name overrides for dataset fields.
- query_prefix / document_prefix (string) — Text prepended to query/document fields during tokenization.
- output_dir (string) — Directory for SBERT artifacts.
- batch_size (int) — Examples per batch. Typical: 16–64.
- gradient_accumulation_steps (int) — Accumulation steps for SBERT.
- grad_clip_max_norm (float) — Gradient clipping threshold. Typical: 0.5–2.0.
- num_epochs (int) — Complete passes through SBERT dataset. Typical: 1–10.
- warmup_steps (int) — Steps for LR warmup.
- evaluation_steps (int) — Evaluate every N steps.
- checkpoint_save_steps (int) — Save rolling checkpoint every N steps.
- resume_from_checkpoint (string or null) — auto resumes from the latest checkpoint; null starts fresh.
- learning_rate (float) — Initial LR. Typical: 1×10^{-5} to 1×10^{-4}.
- max_train_samples / max_eval_samples (int) — Limit training/evaluation samples.
- max_length (int) — Max token length for input texts. Typical: 128–512.
- pooling_mode (enum) — mean, cls, or max.
- trust_remote_code (bool) — Allow custom code from model repo.
- use_amp (bool) — Enable FP16 mixed precision.
- resample_balanced (bool) — Resample to balanced distribution over similarity scores.
- standardize_scores (bool) — Z-score normalize similarity scores.
- resample_std (float) — Target std dev when standardize_scores=true.
- optimizer (object) — Custom optimizer for SBERT (same classes as MLM training); constructor kwargs go under optimizer.optimizer_kwargs.
- wandb_project (string) — Weights & Biases project name for telemetry.

26.7 Validation Rules

The schema enforces the following cross-component validation rules, implemented both in the JSON Schema (_conditional_rules.yaml) and at runtime in the configuration loader:

1. `additionalProperties: false` is enforced at all nesting levels. Any unrecognized key in any object triggers a validation error.
2. `model.dims.hidden_size` must be divisible by `model.dims.num_heads`. The per-head dimension is `hidden_size/num_heads`.
3. `model.dims.num_kv_heads` must divide `model.dims.num_heads` exactly when Grouped-Query Attention is used.
4. `vocab_size` must match the tokenizer's vocabulary size. Mismatch causes token ID corruption and unrecoverable training failure.
5. When `base_model` is set, `training.task` is required. If `task=mlm`, `tokenizer.name_or_path` is also required — unless `tokenizer.source=train_from_dataset`, in which case the tokenizer is trained from the corpus and no name is needed.
6. `tokenizer.source=train_from_dataset` requires the `tokenizer.training` sub-object (with `vocab_size` for `bpe`/`wordpiece`/`wordlevel`) and is mutually exclusive with `tokenizer.name_or_path`. The trained vocabulary overrides `model.dims.vocab_size` before model construction.
7. `task: mlm` requires `training.optimizer`; `task: sbert` requires `training.sbert`; `task: causal_lm` requires `training.optimizer` and `model_class='frankenstein_text_decoder'`.
8. `bitnet.routers=true` requires `bitnet.enabled=true`. The flag has no effect otherwise.
9. `mhc.enabled=true` (mHC n-stream residual) and `mixture_of_depths.enabled=true` are mutually exclusive. Enabling both raises a `ValueError`: MoD token routing operates on a single C-dimensional stream, which conflicts with the n-stream residual.
10. `fasa` and `sparge` are inference-sparse layer types: they apply their sparse heuristic only in evaluation mode. During training the enclosing hybrid layer substitutes standard dense attention over the same **q**/**k**/**v**/**o** projections, so they can appear in `model.dims.layer_pattern` for training runs; the trained weights are the ones the sparse path reads at inference.
11. `model_class: frankenstein_text_decoder` forces `model.dims.mode: decoder` at runtime. Setting `model.dims.mode: encoder` with this class is overridden.
12. Either `base_model` must be present, or both `model_class` and `model` must be present. The schema rejects configurations that provide neither or an incomplete combination.

27 Comprehensive Summary Tables

This appendix provides a complete reference of all components implemented in the Frankenstein Transformer codebase: 49 sequence mixers across nine families, 38 optimizers, 6 normalization variants, 43 activation functions, and 4 embedding methods. Each table is self-contained and cross-references the bibliography.

27.1 Sequence Mixer Summary

Table 46 enumerates every sequence mixer supported by the `layer_pattern` schema field, organized by architectural family. Training and inference complexity are expressed in terms of sequence length n, hidden dimension d, and family-specific parameters.

Name	Family	Training	Inference	Key Characteristic	Reference
Dense and convolution-augmented (6)					
standard	Dense	$\mathcal{O}(n^2 d)$	$\mathcal{O}(n)$ cache	Full global context; KV bottleneck at long sequences	[159]

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Name	Family	Training	Inference	Key Characteristic	Reference
sigmoid	Dense	$\mathcal{O}(n^2 d)$	$\mathcal{O}(n)$ cache	Element-wise gating; hardware-efficient replacement for softmax	[134]
gqa	Dense	$\mathcal{O}(n^2 d)$	$\mathcal{O}(n)$ cache	Tunable KV compression; $G\times$ cache reduction via shared heads	[2]
ca	Dense	$\mathcal{O}(n^2 d)$	$\mathcal{O}(n)$ cache	Exact integration over piecewise-linear key/value trajectories; half-token caps; nonnegative normalized effective weights	[79]
conformer	Conv-augmented	$\mathcal{O}(n^2 d + nKd)$	$\mathcal{O}(n)$ (full recompute)	Macaron FFNs around relative-sinusoidal MHSA and a depthwise-conv module ($K = 32$); local and global context in one block	[56]
fastconformer	Conv-augmented	$\mathcal{O}(n \cdot (w+g) \cdot d)$	$\mathcal{O}(w+g)$	Limited-context window + head-pinned global token; optional separate global projections; linear-memory long-form	[136]
Recurrent (3)					
retnet	Recurrent	$\mathcal{O}(n^2 d)$ parallel	$\mathcal{O}(1)$	Multi-scale decay; triple computation mode (parallel, recurrent, chunkwise)	[152]
mamba	Recurrent	$\mathcal{O}(nd)$	$\mathcal{O}(1)$	Selective state-space model; hardware-aware parallel scan	[54]
ode	Recurrent	$\mathcal{O}(k \cdot n^2 d)$	$\mathcal{O}(kn)$	Continuous-depth RK4 integration; adaptive step solver	[178]
Sparse (11)					
sparse_transformer	Sparse	$\mathcal{O}(n\sqrt{n} \cdot d)$	$\mathcal{O}(n)$	Factorized strided + fixed masks; $\sqrt{n}$ sparsity pattern	[22]
longformer	Sparse	$\mathcal{O}(n \cdot w \cdot d)$	$\mathcal{O}(n)$	Sliding window + dilation + global tokens; linear in n	[12]
swa	Sparse	$\mathcal{O}(n \cdot w \cdot d)$	$\mathcal{O}(w)$	Sliding window + attention sinks; training-free, fixed KV cache	[75]
bigbird	Sparse	$\mathcal{O}(n)$	$\mathcal{O}(n)$	Random + local + global; Turing complete attention	[175]
sparsek	Sparse	$\mathcal{O}(n \cdot d)$	$\mathcal{O}(k)$	Differentiable top-k selection; learned sparsity mask	[108]
nsa	Sparse	$\mathcal{O}((t/d + n_l + w)d)$	$\mathcal{O}(n)$	Hardware-aligned 3-branch design; token/neighbor/window	[173]

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Name	Family	Training	Inference	Key Characteristic	Reference
sparge	Sparse	$\mathcal{O}(n^2 \cdot s \cdot d)$	$\mathcal{O}(n)$	Two-stage block filtering; dense training, sparse inference	[180]
fasa	Sparse	$\mathcal{O}(t \cdot N_{\text{tip}} + N_{\text{fac}} \cdot d)$	$\mathcal{O}(n)$	RoPE frequency-aware sparse; dense training, sparse inference	[164]
msa	Sparse	$\mathcal{O}(H_{\text{kv}} \cdot d_{\text{idx}} \cdot N^2 + H_q \cdot d_h \cdot N \cdot k \cdot B_k)$	$\mathcal{O}(n)$	Index-branch block sparse; deployed at 109B scale	[91]
sparda	Sparse	$\mathcal{O}(H_{\text{kv}} \cdot d_f \cdot N \cdot N_b + H_q \cdot d_h \cdot N \cdot k \cdot B_k)$	$\mathcal{O}(n)$	Forecast-guided lookahead prefetch; predictive sparsity	[45]
lsa	Sparse	$\mathcal{O}(N^2/\bar{n}_b + N\,(k + n_w + n_{\text{sink}})\,d_h)$	$\mathcal{O}(N/\bar{n}_b + k + n_w + n_{\text{sink}})$	Semantic Ward clustering of keys; top-k long-range selection; natively trainable, no extra parameters	[89]
Gated (9)					
gla	Gated	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Data-dependent diagonal decay; linear attention with gating	[169]
deltanet	Gated	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Delta rule; error-correcting linear attention	[170]
gated_ deltanet	Gated	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Decay + write gating synthesis; unified delta framework	[171]
gated_ deltanet2	Gated	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Decoupled channel-wise erase/write; improved gating	[60]
hgrn2	Gated	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Hierarchical bounded forget gates; multi-scale recurrence	[128]
fox	Gated	$\mathcal{O}(L^2d)$	$\mathcal{O}(L)$ KV	Logit-space recency bias; FlashAttn compatible	[99]
gated_ softmax	Gated	$\mathcal{O}(L^2d)$	$\mathcal{O}(L)$ KV	Post-SDPA sigmoid gating; attention-sink mitigation	[129]
kda	Gated	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Channel-wise decay + scalar write; Kimi Linear design	[83]
fpa	Gated	$\mathcal{O}(L^2d)$	$\mathcal{O}(L)$ KV	Path-extremum (range-min) forget bias; contains FoX; two extra scalars per cached key/head	[104]
Fast-Weight (6)					
falcon1	Fast-weight	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Prefix-aligned NLMS delta rule; best LM perplexity of the family	[181]
falcon2	Fast-weight	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Per-value-channel NLMS step sizes	[181]
falcon3	Fast-weight	$\mathcal{O}(Ld^2B)$	$\mathcal{O}(d^2) + \mathcal{O}(B)$ tail	Sliding-window minibatch rehearsal with exact window statistics	[181]

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Name	Family	Training	Inference	Key Characteristic	Reference
falcon1a	Fast-weight	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Shifted additive linear attention with ridge decay	[181]
falcon2a	Fast-weight	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2)$	Per-channel additive decays	[181]
falcon3a	Fast-weight	$\mathcal{O}(Ld^2)$	$\mathcal{O}(d^2) + \mathcal{O}(B)$ tail	Windowed additive; best length extrapolation	[181]
Latent (10)					
mla	Latent	$\mathcal{O}(n^2 d)$	$\mathcal{O}(r_{\text{kv}} \cdot n)$	KV latent of rank $r_{\text{kv}} \approx d/2$; DeepSeek-style	[111]
gqla	Latent	$\mathcal{O}(n^2 d)$	$\mathcal{O}(r_{\text{kv}} \cdot n)$	Hardware-adaptive dual decode paths; GQA + latent fusion	[113]
mlra	Latent	$\mathcal{O}(n^2 d)$	$\mathcal{O}(r_{\text{kv}} \cdot n)$	L sub-heads; 4-way tensor-parallel compatible	[102]
tucker	Latent	$\mathcal{O}(n^2 d)$	$\mathcal{O}(k_{\text{rank}} \cdot n)$	Tucker factorization; unifies MHA/GQA/MLA	[86]
iha	Latent	$\mathcal{O}(n^2 H^2)$	$\mathcal{O}(H \cdot n \cdot d_h)$	Pseudo-heads; $\Theta(\sqrt{k})$ parameter scaling	[38]
gta	Latent	$\mathcal{O}(n^2 \cdot G \cdot d_h)$	$\mathcal{O}(r_v \cdot n)$	Shared map + SiLU decoder; 62.5% FLOP reduction	[151]
mtla	Latent	$\mathcal{O}(n^2 d/m)$	$\mathcal{O}(r_{\text{kv}} \cdot n/m)$	Temporal merge; 5.3× decode speedup	[32]
cca	Latent	$\mathcal{O}(n^2 d/C)$	$\mathcal{O}(\tilde{e} \cdot n)$	Full attention in compressed latent; $C\times$ fewer FLOPs	[44]
ccgqa	Latent	$\mathcal{O}(n^2 d/C_1)$	$\mathcal{O}(\tilde{e}_{\text{kv}} \cdot n)$	CCA + decoupled GQA; best loss at 8× cache reduction	[44]
gma	Latent	$\mathcal{O}(nKd_r + nKd_v)$	$\mathcal{O}(K)$/step	Gaussian Mixture Attention; probabilistic latent routing through K learned GMM components per head; linear-time for fixed K; causal via prefix cumsums	[72]
Geometric (1)					
ssog	Geometric field	$\mathcal{O}(R \cdot n\sqrt{n} \cdot d)$	$\mathcal{O}(R \cdot n\sqrt{n} \cdot d)$/fwd	SSOG: separable Gaussian field over relative position; content steers (bounded $\mu/\sigma/\lambda$ residuals behind cold-started gates), never scores; encoder-only	[125]
Memory (3)					

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Name	Family	Training	Inference	Key Characteristic	Reference
titan	Memory	$\mathcal{O}(c^2 + nf)$	$\mathcal{O}(c^2 + 1)$	Test-time memory adaptation; surprise-driven write dynamics (attention window only in this implementation)	[11]
engram	Memory	$\mathcal{O}(n \cdot N \cdot H)$	$\mathcal{O}(1)$ lookup	Conditional N-gram memory; hash-based scalable retrieval	[21]
memory	Memory	$\mathcal{O}(n^2 d)$	$\mathcal{O}(n)$ cache	Token-ID memory values $V = K + \text{Norm}(E[s])$; removes the value projection; $+Nd_v$ parameters per layer	[76]

Table 46: Complete sequence mixer catalog. n = sequence length, d = hidden size, w = window size, r = low-rank dimension, k = sparsity parameter, C = compression ratio. Family-specific parameters are defined in the corresponding annex.

27.2 Optimizer Summary

Table 47 catalogs all 38 supported optimizers with their memory footprint, per-step computational cost, and key hyperparameters. State buffers are expressed as the number of optimizer-owned tensors per parameter; per-step cost focuses on the dominant term beyond gradient computation.

Optimizer	Family	State Buffers	Per-Step Cost	Key Hyperparams	Reference
Classical & Adaptive Baselines					
SGD	Classical	0	$\mathcal{O}(n)$	lr, wd	[67]
SGD+ Momentum	Classical	1 (m)	$\mathcal{O}(n)$	lr, momentum, wd	[126]
RMSProp	Adaptive	1 (v)	$\mathcal{O}(n)$	lr, decay α (0.9), eps, wd	[157]
AdamW	Adaptive	2 (m, v)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd	[107]
RAdam	Adaptive	2 (m, v)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd	[101]
Variance Reduction & Advanced Momentum					
Adan	Variance reduction	3 (m, v, s)	$\mathcal{O}(n)$	lr, β_1, β_2, β_3, eps, wd	[167]
ADOPT	Adam variant	2 (m, v)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd	[156]
AdEMAMix	Multi-EMA	3 (m_1, m_2, v)	$\mathcal{O}(n)$	lr, β_1, β_2, β_3, eps, wd	[120]
MARS	Variance-reduced	3 (m, v, z)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd, γ	[174]
Cautious AdamW	Masked momentum	2 (m, v)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd	[97]
Large-Batch, Memory-Efficient & Parameter-Free					
LAMB	Layer-wise adaptive	2 (m, v)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd	[172]
Schedule-Free	Scheduler-free	3 (z, x, n)	$\mathcal{O}(n)$	lr, β_1, β_2, wd	[31]
Adafactor	Memory-efficient	1–2 (row/col)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd	[144]
AdamPM	Pooled second moment	2 (m, $\bar{v}$ at $1/s$)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd, group size s, row/col	[73]
GaLore	Low-rank projection	2 (m, v) + SVD	$\mathcal{O}(nr)$	lr, rank r, β_1, β_2, eps, wd	[182]

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Optimizer	Family	State Buffers	Per-Step Cost	Key Hyperparams	Reference
Prodigy	Parameter-free	3 (m, v, d)	$\mathcal{O}(n)$	β_1, β_2, eps, wd, d_0	[115]
Lion	Sign momentum	1 (m)	$\mathcal{O}(n)$	lr, β_1, β_2, wd	[20]
Second-Order, Geometric & Orthogonality					
Shampoo	Second-order	$2d$ (L_i, R_i)	$\mathcal{O}(n^{1+2/d})$	lr, eps, wd	[57]
SOAP	Second-order	$2d+2$ (m, v)	$\mathcal{O}(n^{1+1/d})$	lr, β_1, β_2, eps, wd	[160]
Sophia	Curvature-aware	3 (m, v, h)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd, k	[100]
Muon	Orthogonality (hybrid)	1 (m); + v on the AdamW fallback	$\mathcal{O}(n \cdot k)$	lr, momentum, wd, NS steps k; β, eps for embeddings/pos/heads/vectors	[146]
Turbo-Muon	Orthogonality (hybrid)	1 (m); + v on the AdamW fallback	$\mathcal{O}(n \cdot k)$	lr, momentum, wd, NS steps k; β, eps for embeddings/pos/heads/vectors	[14]
AF-Muon	AdamW-free tied table	1 (m)	$\mathcal{O}(n \cdot k + V d I)$	lr, momentum, wd, cap c, table scale s, ρ_{out}/ρ_{hid}, NS steps k	[90]
AdaMuon	Orthogonality + adaptive	2 (m, v)	$\mathcal{O}(n \cdot k)$	lr, momentum, eps, wd, NS steps k	[147]
Dion3	Orthogonality (fractional)	1 (m); + v rows if NorMuon	$\mathcal{O}(r^2(m+r))$, $r = \lceil fn \rceil$	lr, momentum, wd, fraction f, NS steps k, β_2 (NorMuon)	[4]
ROOT	Orthogonality (robust)	1 (m); + v on the AdamW fallback	$\mathcal{O}(n \cdot k)$ + quantile sort	lr, momentum, wd, NS steps k, quantile p; β, eps for embeddings/pos/heads/vectors	[64]
Muon²	Orthogonality + adaptive	2 (m, v); factored: 1 (m) + row/col; + v on the AdamW fallback	$\mathcal{O}(n \cdot k)$	lr, β_1, β_2, eps, wd, NS steps k, factored	[106]
NorMuon	Orthogonality + neuron-wise adaptive	2 (m, row v); + v on the AdamW fallback	$\mathcal{O}(n \cdot k)$	lr, momentum, β_2, eps, wd, NS steps k	[96]
Conda	Orthogonality + column-wise adaptive	3 (m, v_{proj}, basis P); + v on the AdamW fallback	$\mathcal{O}(mn\,r)$, $r = \min(m,n)$; SVD every T steps	lr, β_1, β_2, eps, wd, gap T, scale α	[162]
TUON	Orthogonality (tensor, cross-layer)	1 (m per slice); + $K \times K$ basis per stack (refresh T); + v on the AdamW fallback	$\mathcal{O}(Kmn \cdot \min(m,n) + K^2 mn)$; eigh every T steps	lr, momentum, wd, NS steps k, refresh T, transform Q, stack K	[155]
NAMO	Orthogonality + norm-based adaptive	2 (m, scalar v); + v on the AdamW fallback	$\mathcal{O}(n \cdot k)$	lr, μ_1, μ_2, eps, wd, NS steps k	[179]
NAMO-D	Orthogonality + diagonal adaptive	2 (m, column v); + v on the AdamW fallback	$\mathcal{O}(n \cdot k + mn)$	lr, μ_1, μ_2, eps, wd, NS steps k, clamp c	[179]
Low-Rank Adaptive & Tunable Adaptivity					
APOLLO	Low-rank adaptive	2 low-rank + proj	$\mathcal{O}(nr)$	lr, rank r, update gap, scale, betas, eps, wd	[184]

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Optimizer	Family	State Buffers	Per-Step Cost	Key Hyperparams	Reference
APOLLO-Mini	Low-rank adaptive	2 rank-1 + proj	$\mathcal{O}(n)$	lr, update gap, scale, betas, eps, wd	[184]
Q-APOLLO	Quantized low-rank	2 quantized + proj	$\mathcal{O}(nr)$	lr, rank r, update gap, scale, quant bits, betas, eps, wd	[184]
Anon	Adaptivity-tunable	3 (m, v, fixed_lr)	$\mathcal{O}(n)$	lr, β_1, β_2, eps, wd, γ	[6]
Population-Based & Evolutionary					
EvE	Evolutionary hybrid (DE + Adam rescue)	12 ($4z$, $4m$, $4v$)	$\mathcal{O}(n)$ + up to n_{adam} fwd/bwd	lr (η_{lr}), F, σ, n_{adam}, jitter γ, wd (L2)	[132]
EA4LLM	Gradient-free evolution strategies	0 (RNG + counters only)	N fwd/bwd evals, no state	lr (α), population N, σ, shaping, antithetic, $1/\sigma$	[103]

Table 47: Complete optimizer catalog. n = number of parameters, d = tensor dimensionality, V = vocabulary size, r = low-rank dimension, k = Newton–Schulz iteration count, I = per-row bisection iteration count. State buffers lists the number of optimizer state tensors maintained per parameter group.

27.3 Normalization Variants

Table 48 summarizes the six normalization methods available through the `norm_type` schema field, ranging from classical statistics-based approaches to normalization-free bounded transforms and weightless RMSNorm with fused norm-then-project execution.

Method	Formula	Stats Needed	Reference
LayerNorm	$\gamma_i(x_i - \mu)/\sqrt{\sigma^2 + \epsilon} + \beta_i$	mean + variance	[7]
RMSNorm	$\gamma_i x_i / \sqrt{\frac{1}{d}\sum_j x_j^2 + \epsilon}$	RMS only	[177]
pRMSNorm	$\gamma_i x_i / \sqrt{\frac{1}{k}\sum_{j \le k} x_j^2 + \epsilon}$	Partial RMS ($p\,\%$)	[177]
Dynamic Tanh (DyT)	$\tanh(\alpha x)$	none	[185]
Dynamic Erf (Derf)	$\mathrm{erf}(\alpha x + s)$	none	[19]
FlashNorm	$x_i / \sqrt{\frac{1}{d}\sum_j x_j^2 + \epsilon}$	RMS only (weightless)	[51]

Table 48: Normalization variants. d = hidden dimension, $k = \lceil d \cdot p \rceil$ for partial ratio p, α and s are learnable parameters. FlashNorm is weightless (no learnable parameters); the per-dim scale is folded into the subsequent linear layer per Prop. 1 of [51].

27.4 Residual Connections

Table 49 compares the residual-connection schemes available in the codebase: the standard identity residual, the four Attention Residuals (AttnRes) strategies [153], and the manifold-constrained n-stream residual (mHC) [168]. L is the logical depth, C the hidden dimension, n the stream expansion factor, $t_{\max}$ the Sinkhorn–Knopp iteration count, α the mHC gating scalars, and $\mathbf{w}_l$ the per-layer AttnRes pseudo-query (initialized to zero). HC (unconstrained Hyper-Connections) is listed for reference but is *not* exposed in this codebase.

Scheme	Stream Update	Constraint	Reference
Standard residual	$x_{l+1} = x_l + F(x_l, W_l)$	Identity mapping (by construction)	[62]
No residual (experimental)	$x_{l+1} = F(x_l, W_l)$	No skip path	—
Full AttnRes	$x_{l+1} = \sum_{i=0}^{l} \mathrm{softmax}_i(\mathbf{w}_l^\top \mathrm{RMSNorm}(v_i))\, v_i$	$\mathbf{w}_l$ learned, zero-init; RMSNorm on keys; $L \cdot C$ parameters	[153]
Block AttnRes	$x_{l+1} = \sum_{m=0}^{n} \mathrm{softmax}_m(\mathbf{w}_l^\top \mathrm{RMSNorm}(v_m))\, v_m$	$\mathbf{w}_l$ learned, zero-init; RMSNorm on keys; N block reps + partial sum; $L \cdot C$ parameters	[153]
Hyper-Connections (HC)	$x_{l+1} = H_l^{\mathrm{res}} x_l + (H_l^{\mathrm{post}})^\top F(H_l^{\mathrm{pre}} x_l, W_l)$	Unconstrained	[168]
mHC (this codebase)	same update as HC	H_l^{res} doubly stochastic via Sinkhorn–Knopp ($t_{\max} = 20$), $H^{\mathrm{pre}} = \sigma(\cdot)$, $H^{\mathrm{post}} = 2\sigma(\cdot)$, α-init 0.01	[168]
Hyperloop (this codebase)	$y_t^{(l+1)} = H_l^{\mathrm{res}} y_t^{(l)} + H_l^{\mathrm{post}} \otimes (F(H_l^{\mathrm{pre}} y_t^{(l)}) + e_l)$	Loop-level HC on a middle-cycle looped middle block; per-loop $\{W_l, b_l, \alpha_l, e_l\}$; copy/avg expansion; H^{res} diagonal (default), Sinkhorn or identity	[176]

Table 49: Residual-connection schemes. $x_l \in \mathbb{R}^C$ (standard / AttnRes / no residual) or $\mathbb{R}^{n \times C}$ (mHC) is the stream at layer l, F the layer function, $H^{\mathrm{pre}}, H^{\mathrm{post}}, H^{\mathrm{res}}$ the learned mHC mappings. AttnRes strategies are controlled by `model.residuals.type`; mHC by `model.mhc`. Full details in the mHC and AttnRes appendices.

27.5 Activation Functions

Table 50 catalogs the 43 feed-forward activation functions available through the `ffn_activation` schema field (40 elementwise plus 3 gated-FFN variants), grouped by family. "Learn." marks activations with trainable parameters; the full formulas are in Appendix 14.

Name	Family	Formula	Learn.	Reference
Classical / Sigmoid–Tanh (12)				
silu / swish	Classical	$x\,\sigma(\beta x)$	no / β	[133]
gelu	Classical	$x\,\Phi(x)$	no	[65]
gelu_tanh	Classical	$\frac{x}{2}\left(1 + \tanh\sqrt{\frac{2}{\pi}}(x + 0.044715x^3)\right)$	no	[65]
relu	Classical	$\max(0, x)$	no	[118]
sigmoid	Classical	$1/(1 + e^{-x})$	no	[92]
tanh	Classical	$(e^x - e^{-x})/(e^x + e^{-x})$	no	[92]
arctan	Classical	$\arctan(x)$	no	[92]
softsign / elliott	Classical	$x/(1 + \lvert x\rvert)$	no	[92]
identity	Classical	x	no	(internal)
softplus	Classical	$\log(1 + e^x)$	no	[48]
mish	Classical	$x \tanh(\log(1 + e^x))$	no	[116]
Rectified family (12)				
leaky_relu	Rectified	$\max(0, x) + a \min(0, x)$, a=0.01	no	[110]
relu6	Rectified	$\min(\max(0, x), 6)$	no	[69]
hardswish	Rectified	$x\,\mathrm{ReLU6}(x+3)/6$	no	[69]
prelu	Rectified	$\max(0, x) + \mathbf{p} \odot \min(0, x)$	**yes**	[61]
abs_relu	Rectified	$\max(0, x) - \max(0, -x)$	no	[36]
nl_relu	Rectified	$\beta \log(1 + \max(0, x))$	no	[36]
brelu	Rectified	$\min(\max(0, x), t)$	no	[36]
vrelu	Rectified	$\lvert x\rvert$	no	[36]

continued on next page

Name	Family	Formula	Learn.	Reference
hexpo	Rectified	$a\max(0,x) - c\max(0,-x)$	no	[36]
ptanh	Rectified	$\max(0, \tanh(x))$	no	[36]
dis_relu	Rectified	$\max(0, x-\delta)$	no	[36]
lisht	Rectified	$x\tanh(x)$	no	[139]
Exponential / ELU family (12)				
elu	Exponential	x if $x \geq 0$ else $\alpha(e^x-1)$	no	[23]
selu	Exponential	$\lambda\,\mathrm{ELU}_{\alpha'}(x)$	no	[85]
celu	Exponential	x if $x \geq 0$ else $\alpha(e^{x/\alpha}-1)$	no	[8]
pelu	Exponential	$\frac{\alpha}{\beta}x$ / $\alpha(e^{x/\beta}-1)$	**yes**	[36]
mpelu	Exponential	ELU + learnable α, β	**yes**	[36]
felu / eelu / pdelu / preu	Exponential	ELU variants w/ learnable params	**yes**	[36]
softexp	Exponential	exp/linear/log interpolation	**yes**	[36]
elish / hardelish	Exponential	Swish $\cup$ ELU gated branches	no	[10]
Learnable / Adaptive (4)				
swish_trainable	Learnable	$x\,\sigma(\beta x)$, β trainable	**yes**	[133]
maxout	Learnable	$\max_i(W_i x + b_i)$	**yes**	[50]
raf	Learnable	$P(x)/Q(x)$ Padé$(5,4)$, version A	**yes**	[42]
Gated FFN variants (3)				
swiglu	Gated	$\mathrm{SiLU}(xW_g) \odot (xW_u)\,W_d$	**yes**	[143]
geglu	Gated	$\mathrm{GELU}(xW_g) \odot (xW_u)\,W_d$	**yes**	[143]
reglu	Gated	$\mathrm{ReLU}(xW_g) \odot (xW_u)\,W_d$	**yes**	[143]

Table 50: Activation function catalog. "Learn." indicates the presence of trainable parameters. RAF = Rational Activation Function (learnable Padé ratio). σ is the logistic sigmoid.

27.6 Embedding Variants

Table 51 catalogs the positional encodings and structural embedding optimizations supported by the codebase. The 12 positional encodings are configured via the model-wide `positional_encoding` field (see Appendix 19 for full formulations); the two structural optimizations are controlled by their own flags.

Method	Description	Reference
Positional Encodings (12)		
RoPE	Rotary Position Embedding; frequency-based rotation of Q/K, encoding relative position via inner-product geometry	[150]
HoPE	Hyperbolic Rotary Positional Encoding; Lorentz-inspired rotation using hyperbolic functions; monotonic attention decay with distance; RoPE is a special case	[25]
NoPE	No Positional Encoding; model learns position implicitly from causal mask; outperforms explicit PE on length generalization	[77]
ALiBi	Attention with Linear Biases; static distance-proportional penalty on attention scores; enables length extrapolation	[127]
BAM	Bayesian Attention Mechanism; Generalized-Gaussian relative-position bias with learnable per-head shape (β) and scale (α); recovers ALiBi at $\beta = 1$; SSMax rescale via `use_ssmax`	[13]
PaPE	Parabolic Position Encoding; parabolic basis functions concatenated to Q/K; vision-centric, n-D, extrapolatable	[119]
PaPE-Efficient	Pure-PyTorch PaPE; no custom Triton kernels, same math	[119]

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Method	Description	Reference
PaPE-RI	Rotation-invariant PaPE (PaPE-RI)	[119]
Sinusoidal Absolute	Fixed sinusoidal added to embeddings (original Transformer)	[159]
Sinusoidal Rotary	Sinusoidal frequencies as a rotation matrix on Q/K	[159]
Learned Absolute	Learnable parameter matrix added to embeddings	[159]
None	Null module; no positional information injected	—
Structural Embedding Optimizations (4)		
Factorized Embeddings	Decomposes $V \times H$ embedding matrix into $V \times R$ and $R \times H$ factors; reduces parameters from $\mathcal{O}(VH)$ to $\mathcal{O}(VR + RH)$; controlled by `factorized_embedding_dim`	(internal)
Embedding Convolution	Depthwise Conv1d applied over the embedding stream before the first transformer block; provides lightweight local-context filtering; controlled by `embedding_conv_kernel`	(internal)
Dynamic Embeddings (EELBERT)	Replaces the $V \times H$ table with an on-the-fly n-gram pooling hash of the token strings; frozen, no trainable embedding parameters; controlled by the `embedding.dynamic.*` block (Appendix 20)	[24]
Audio frontend	Log-mel/spectrogram/raw frames through two stride-2 Conv2d stages and a linear projection (4× frame reduction); input side of `frankenstein_audio`; controlled by the `audio.*` block (Appendix 22)	[56]

Table 51: Embedding variants. V = vocabulary size, H = hidden size, R = factorized intermediate dimension. The first 12 rows are positional encodings (Appendix 19); the last four are structural embedding optimizations, including the audio input frontend.

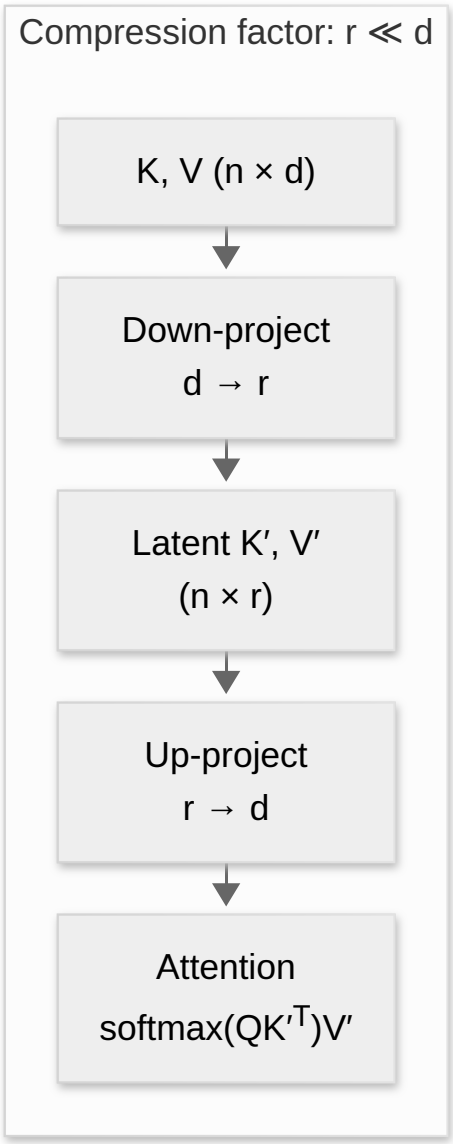


Figure 6: Latent and low-rank attention. Key–value pairs are projected into a low-rank latent space, where attention is computed at reduced cost, and projected back. Variants differ in factorization, head interleaving, temporal compression, cross-covariance formulation, and probabilistic routing.

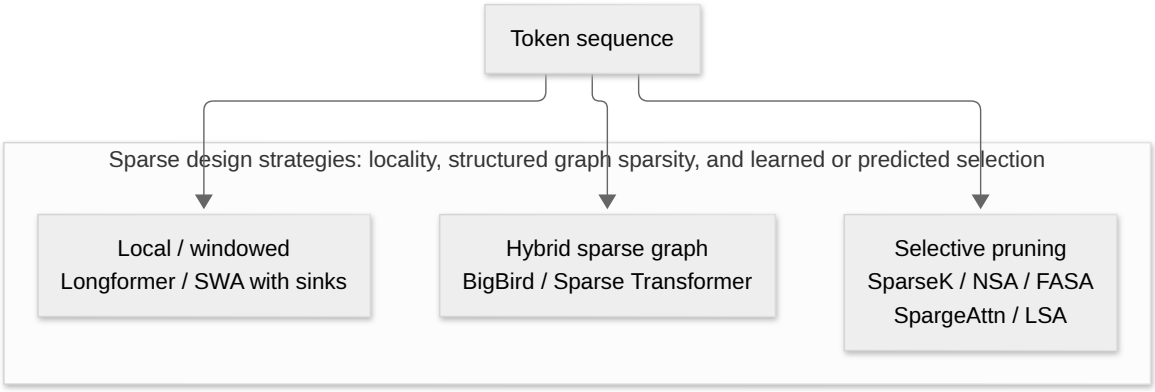


Figure 7: Sparse attention design choices. Methods reduce cost by restricting neighborhoods, constructing sparse graphs, or selecting only high-value tokens or blocks.

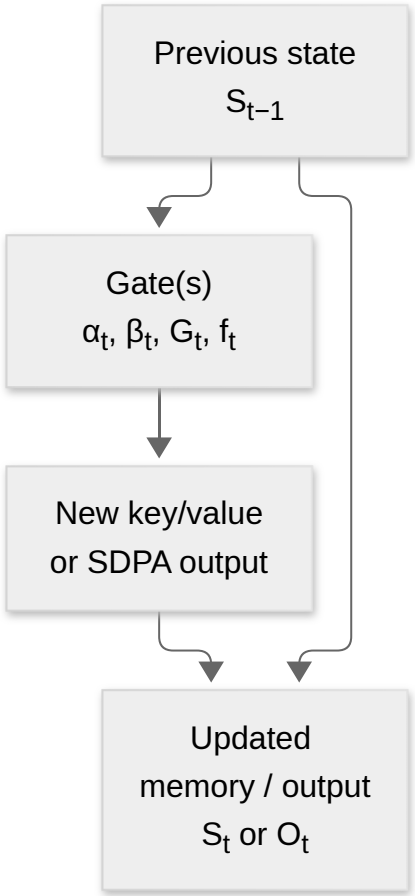


Figure 8: Generic gating template. A gate can decay existing memory, regulate write strength, or modulate dense attention outputs, depending on the block family.

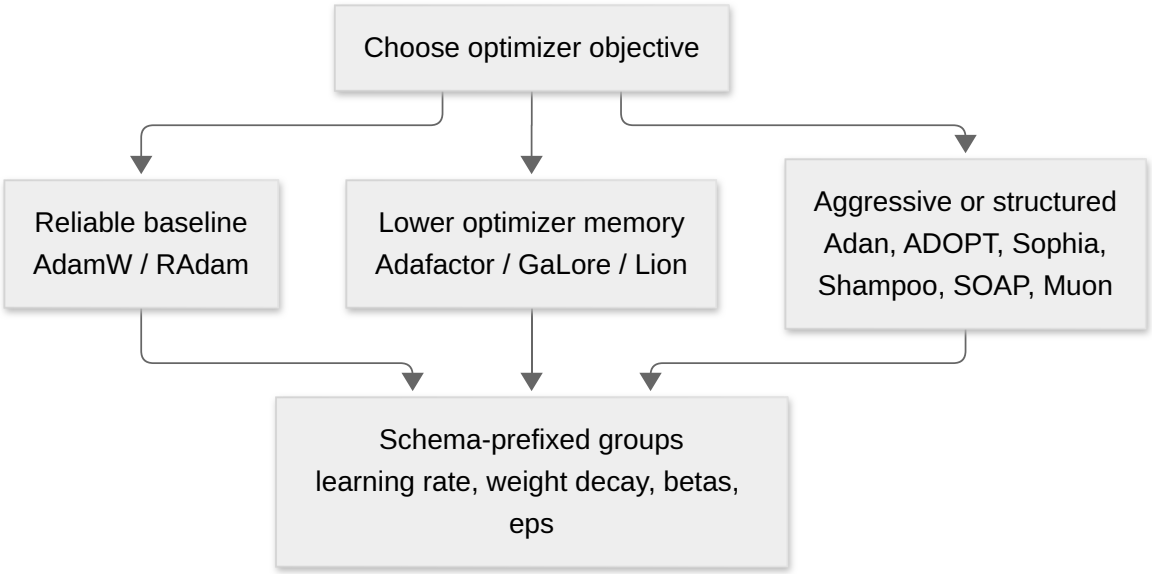


Figure 9: Optimizer selection in practice: the class determines the update rule, while the schema controls how hyperparameters apply across the ten canonical parameter blocks (embeddings, positional encodings, attention, feed-forward, normalization, heads, mHC, routers, residuals, and the remaining parameters).

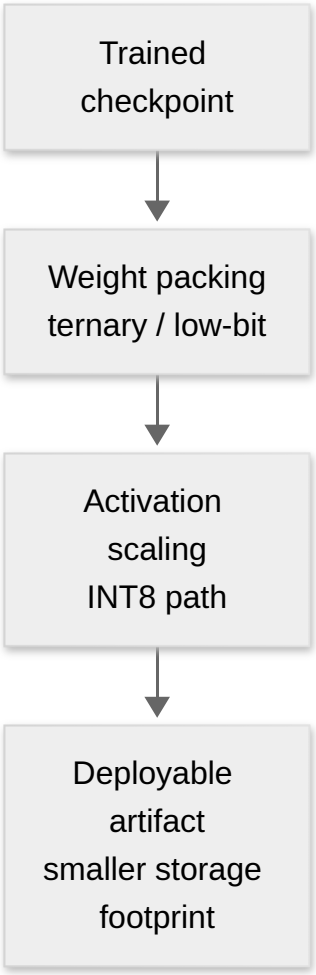


Figure 10: Deployment path from a trained checkpoint to a compact artifact. Quantization is a deploy-stage transformation rather than a separate model family.

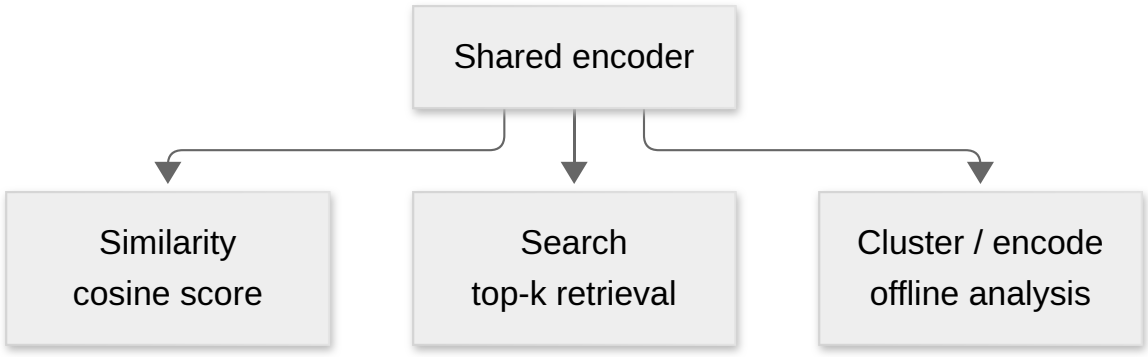


Figure 11: SBERT workflow reuse. A single encoder supports online pair scoring, corpus retrieval, clustering, and persistent embedding export.

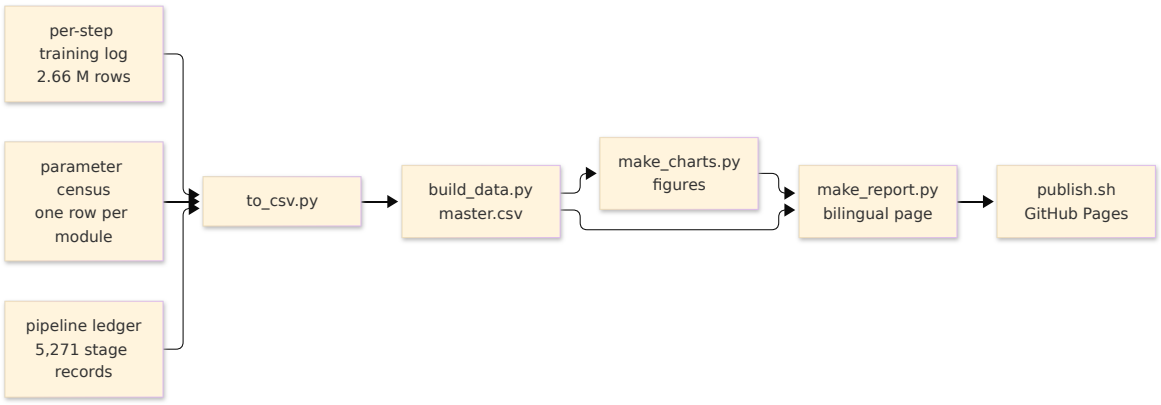


Figure 108: The offline reporting toolchain. The three recorders are consolidated into CSV files, joined by configuration into `master.csv`, charted in both themes, and rendered into a bilingual static page with the joined datasets published next to it. Nothing in this chain needs a GPU.

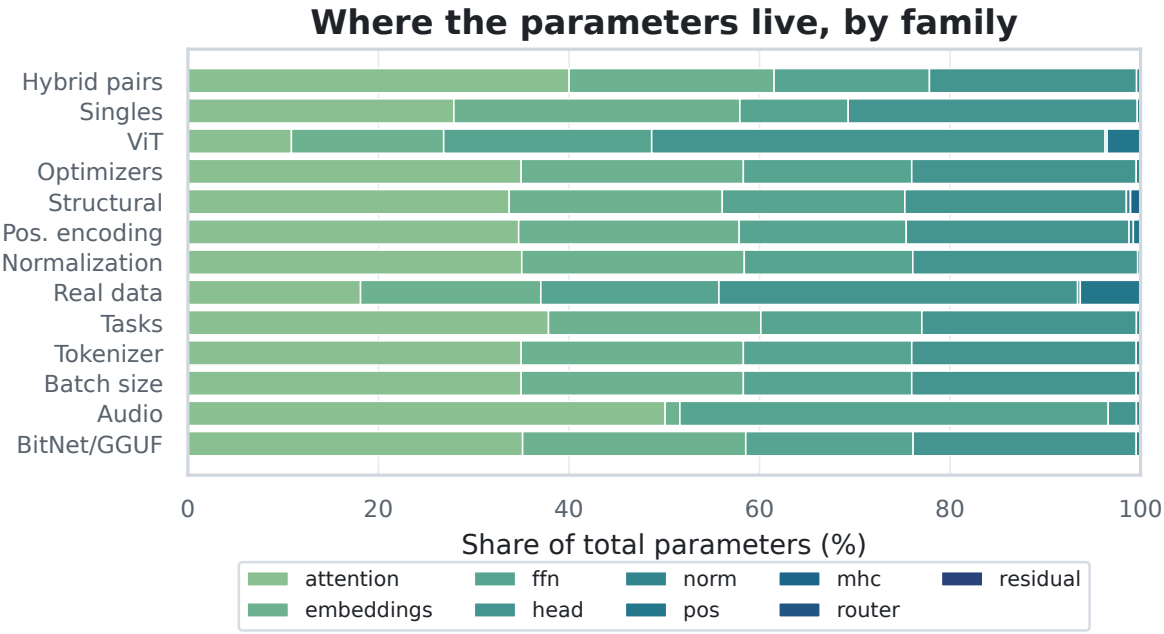


Figure 109: Where the parameters live, by family. Embeddings and the head are fixed costs; the attention block and the FFN carry the architectural differences.

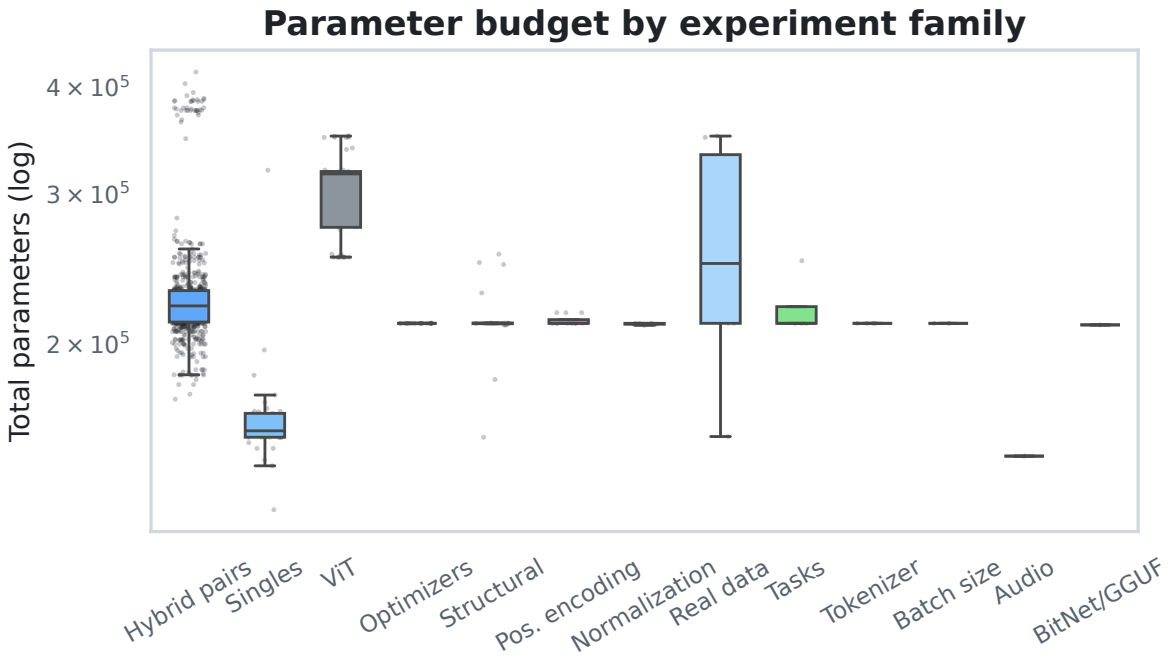


Figure 110: Parameter budget by experiment family (log scale). The ViT family is the most expensive; the ablation families share one backbone.

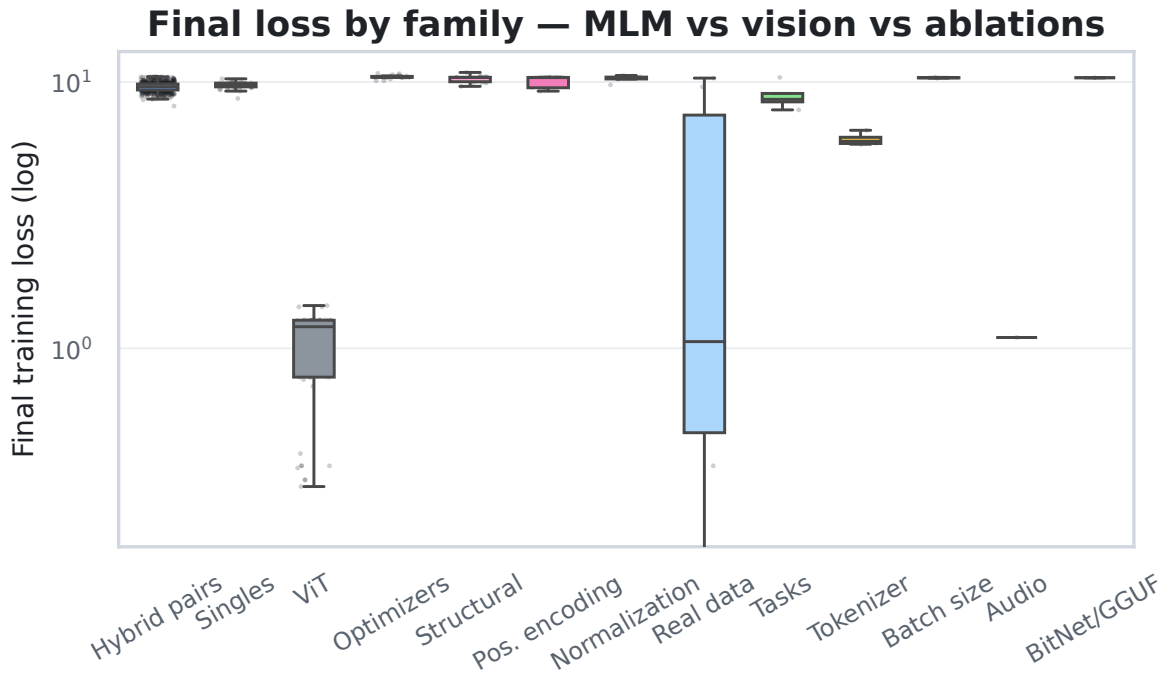


Figure 111: Final training loss by family (log scale). The vision and real-data families are not comparable with MLM losses; the families are kept separate throughout.

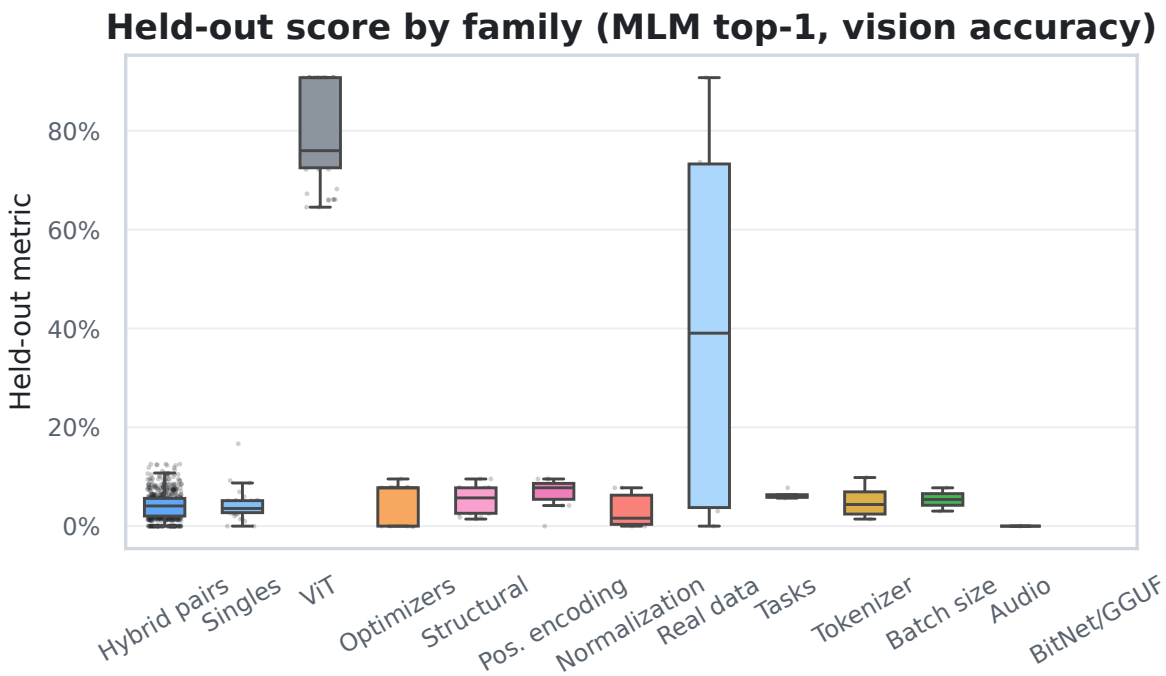


Figure 112: Held-out score by family. Vision tasks score an order of magnitude above MLM because the task is easier, not because the mixer matters more.

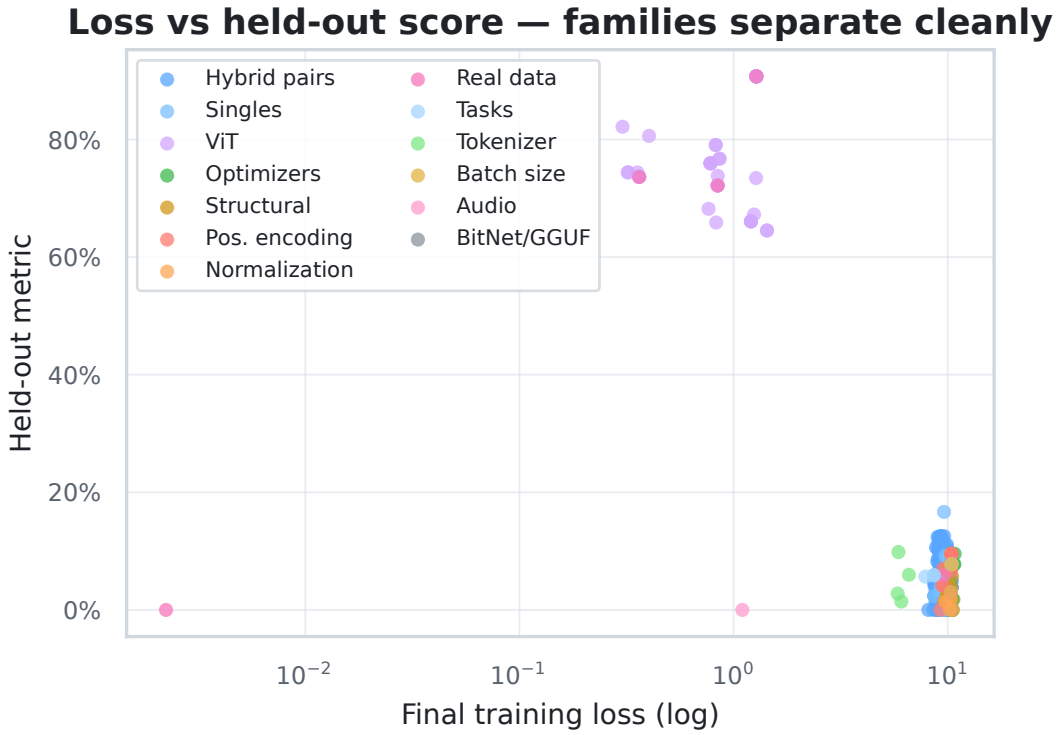


Figure 113: Final loss against held-out score. The relation is weak within the pairs: the best-loss pair scores 0.0% top-1, and the best-accuracy pair has a worse loss.

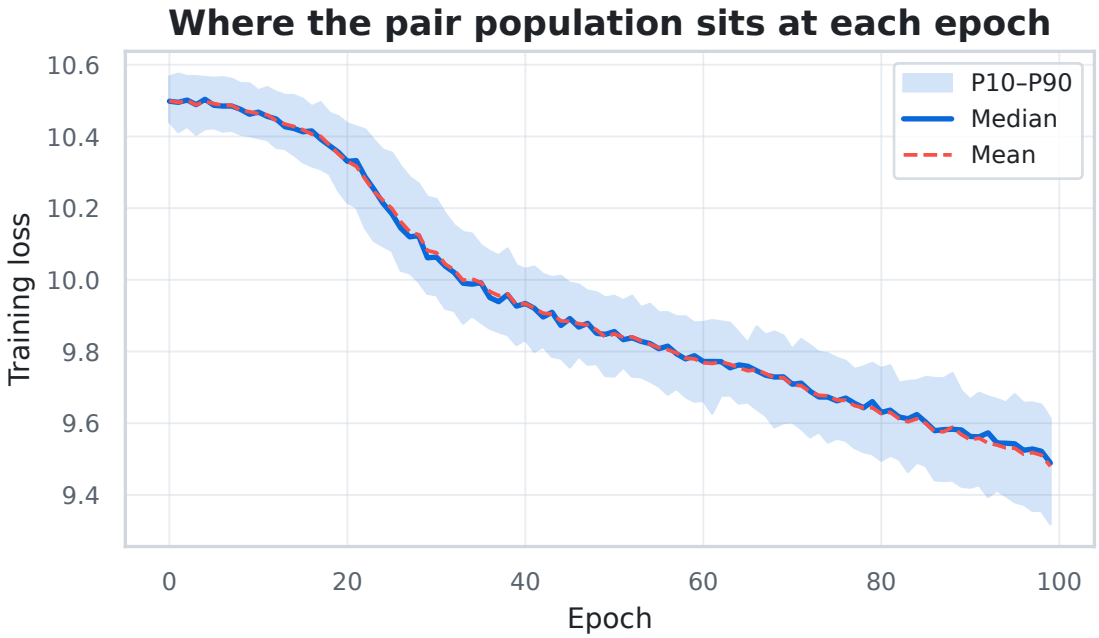


Figure 114: Population learning curves with the P10–P90 band. The curves are still descending at epoch 100, so the sweep measures learning speed rather than converged quality.

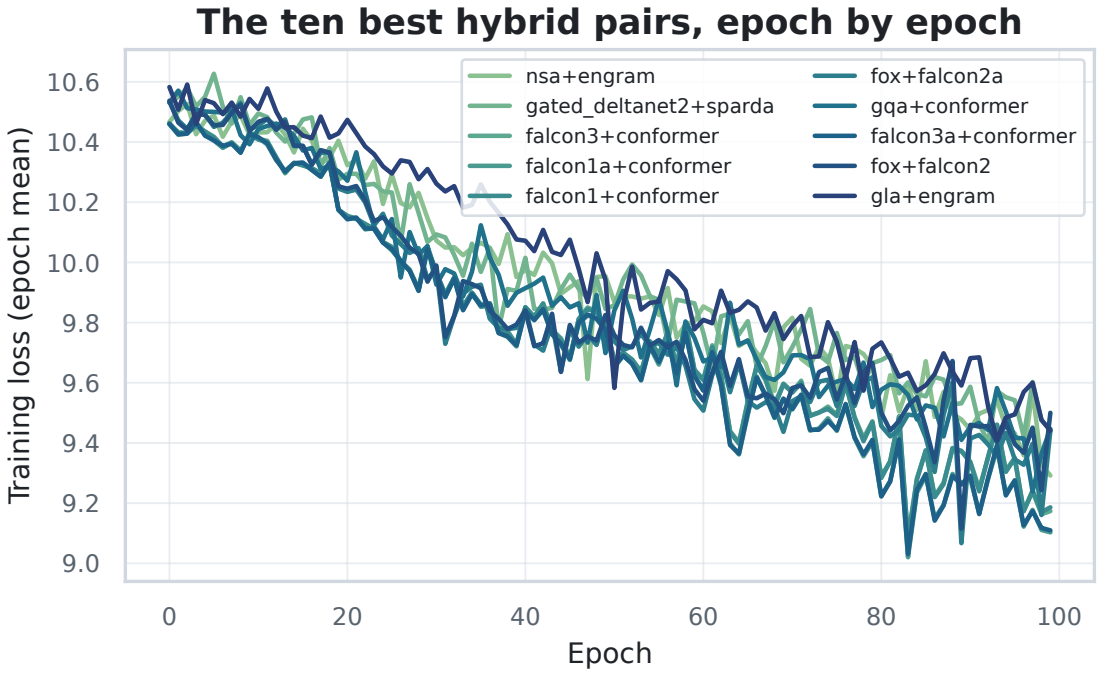


Figure 115: The ten best pairs over time. The ordering is established early and rarely changes after epoch 50.

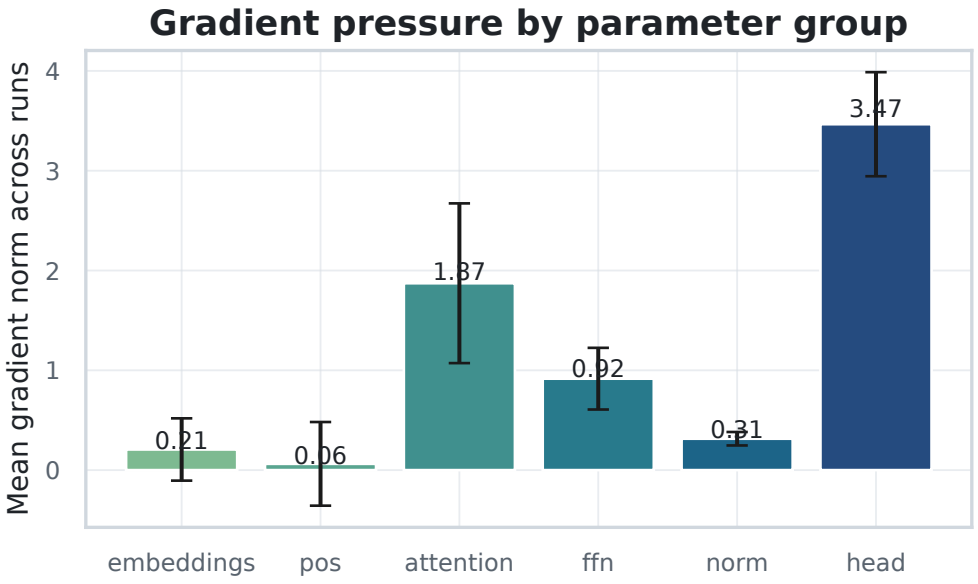


Figure 116: Gradient pressure by parameter group. The head carries the largest gradient, the FFN the smallest; clipping fires rarely because the median clip ratio is 0.90.

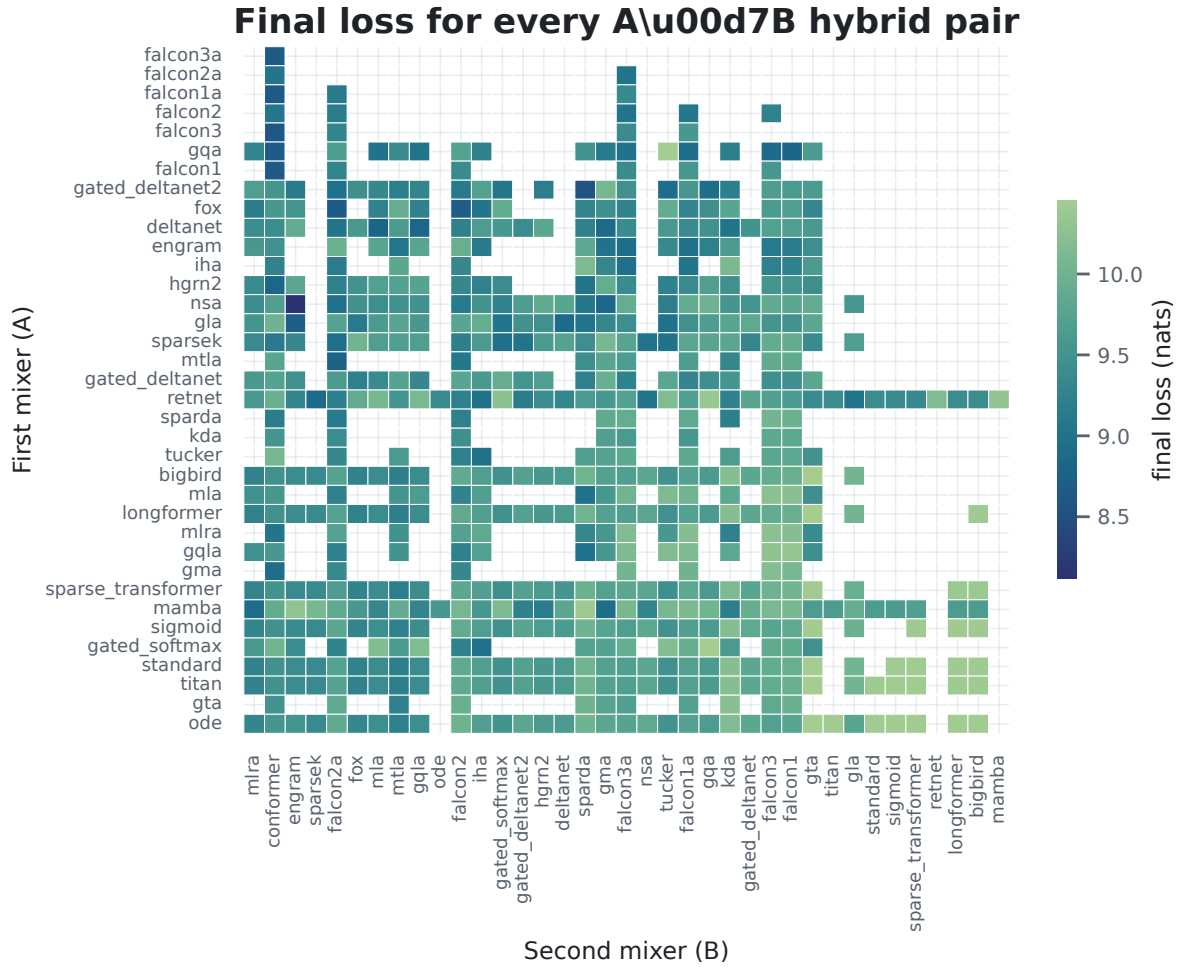


Figure 117: The A×B atlas of final loss. Rows are the first mixer, columns the second; empty cells are pairs that never trained.

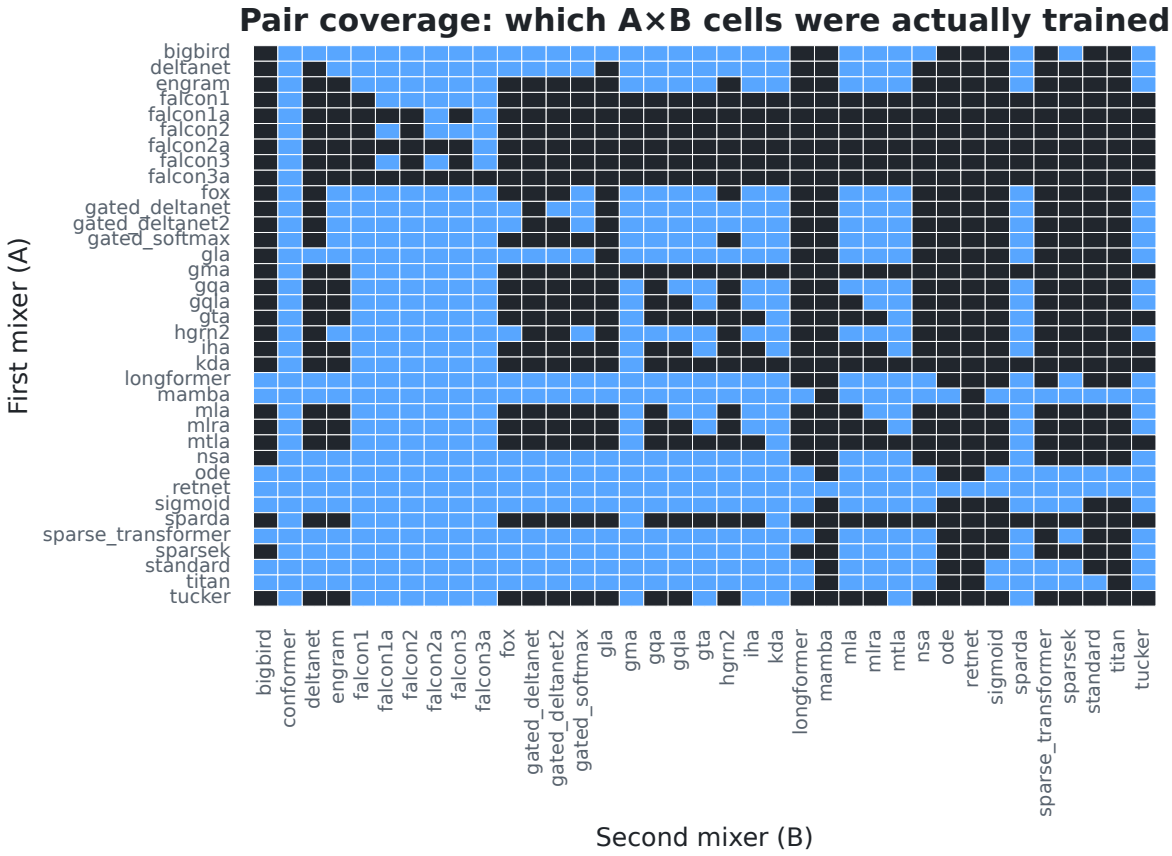


Figure 118: Coverage of the A×B grid. The missing cells are the 158 pairs that contain one of the four mixers that never train.

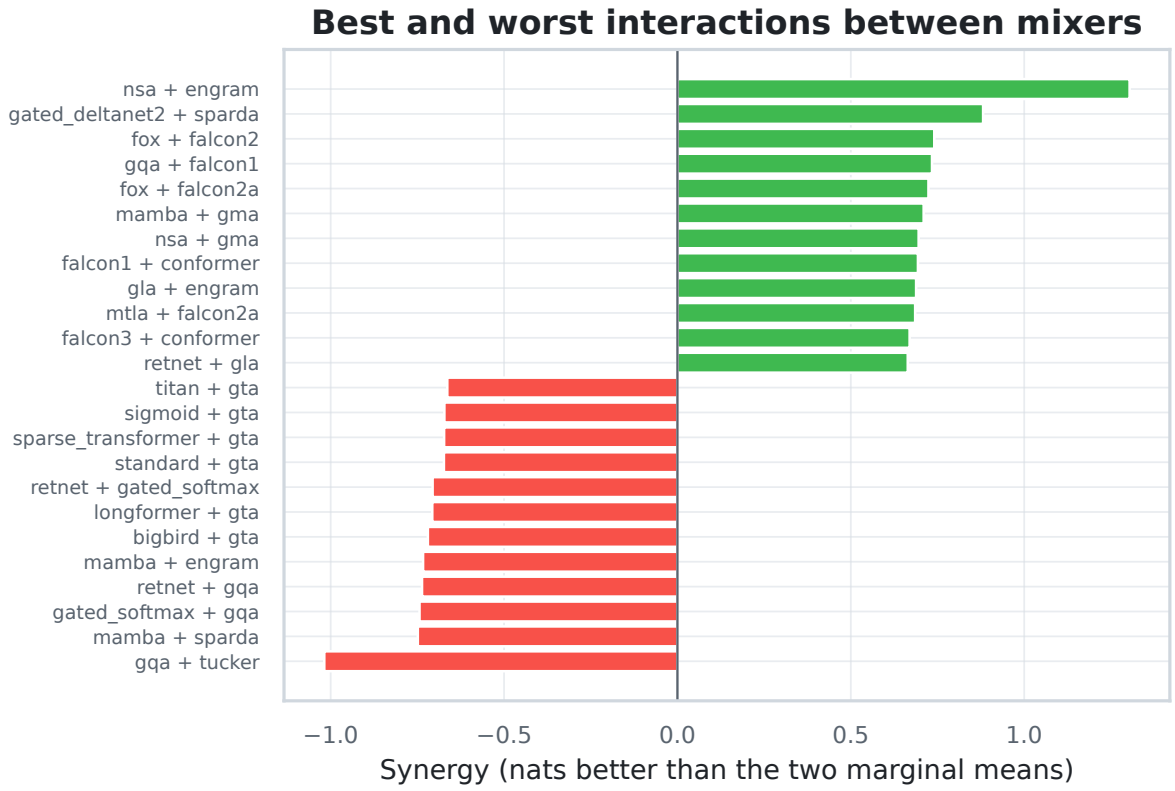


Figure 119: Best and worst mixer interactions. The positive tail pairs a cheap, stable first mixer with a memory-heavy second one; the negative tail is dominated by second mixers that clash after a strong A.

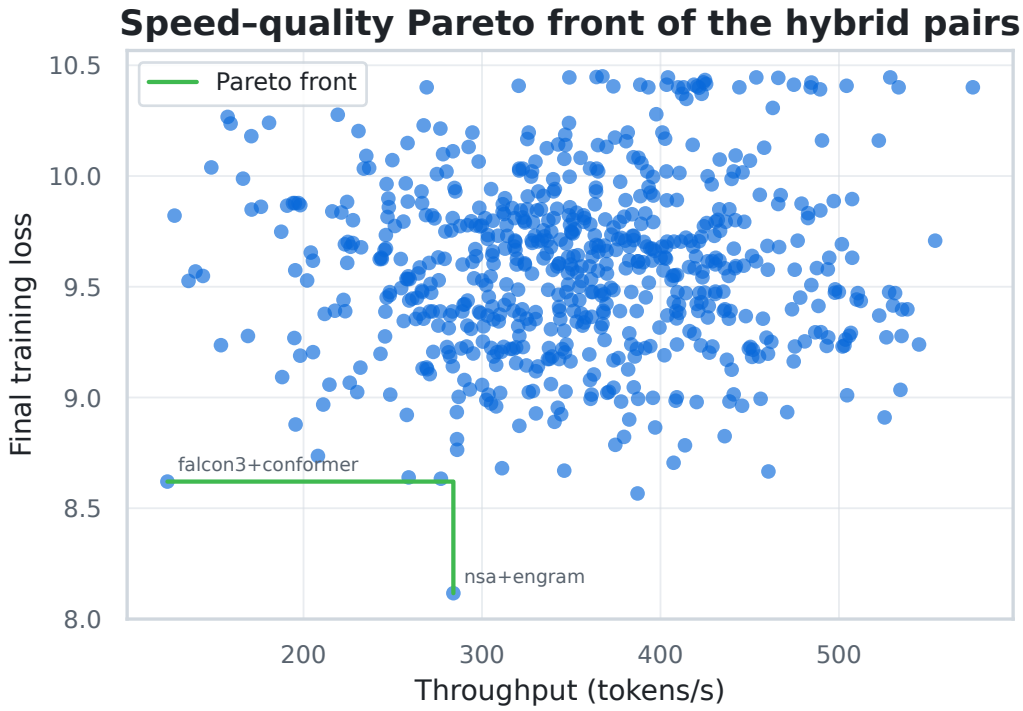


Figure 120: Speed–quality Pareto front of the hybrid pairs. A pair sits on the front when no other pair is both faster and better; the front is populated by pairs whose second mixer is memory-heavy but whose first mixer is cheap.

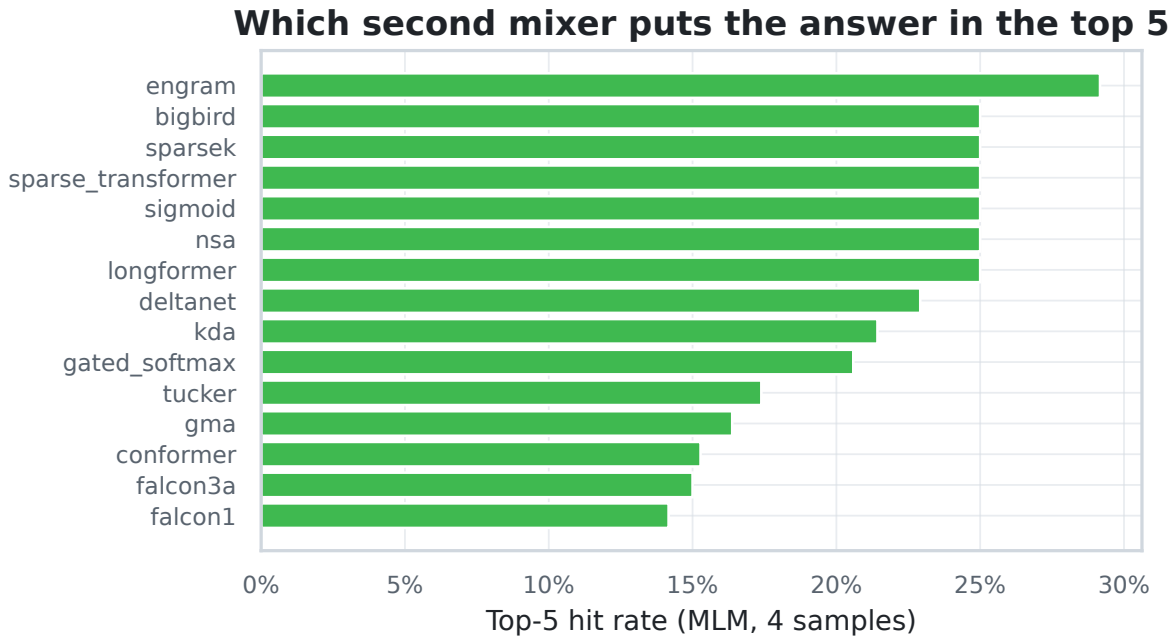


Figure 121: Top-5 hit rate by second mixer. With four samples per configuration the interval between adjacent bars is one sample, so the ordering is indicative.



Figure 122: Inference quality against training loss. The correlation is weak ($r = 0.14$): training loss is not a substitute for held-out evaluation.

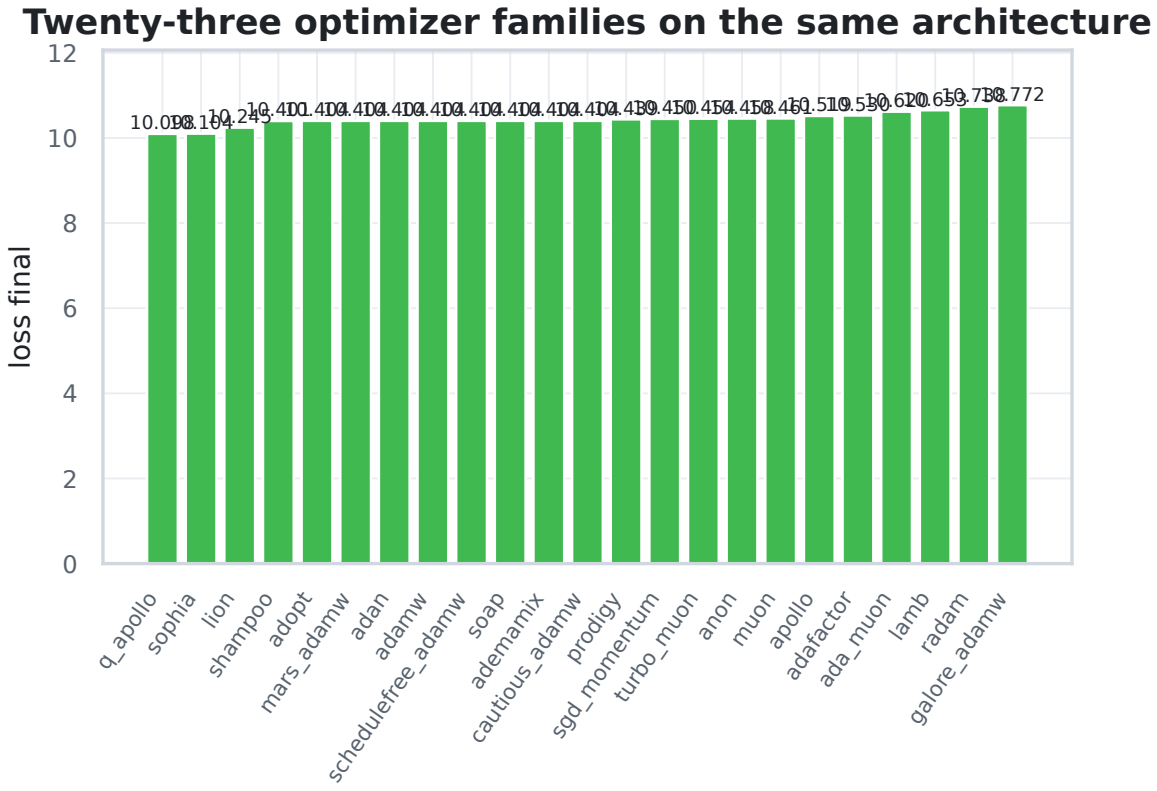


Figure 123: Optimizer ablation: final loss. A dozen optimizers converge to the same value, which measures the shortness of the run more than the quality of the optimizer.

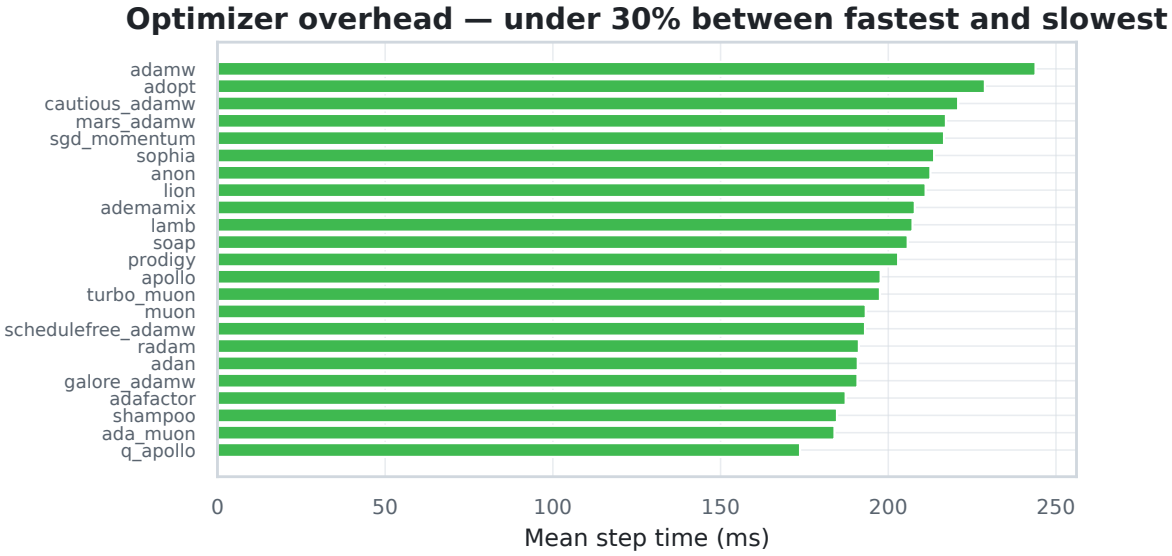


Figure 124: Optimizer ablation: mean step time. The spread is about 30%; second-order and orthogonalised methods pay the visible overhead.

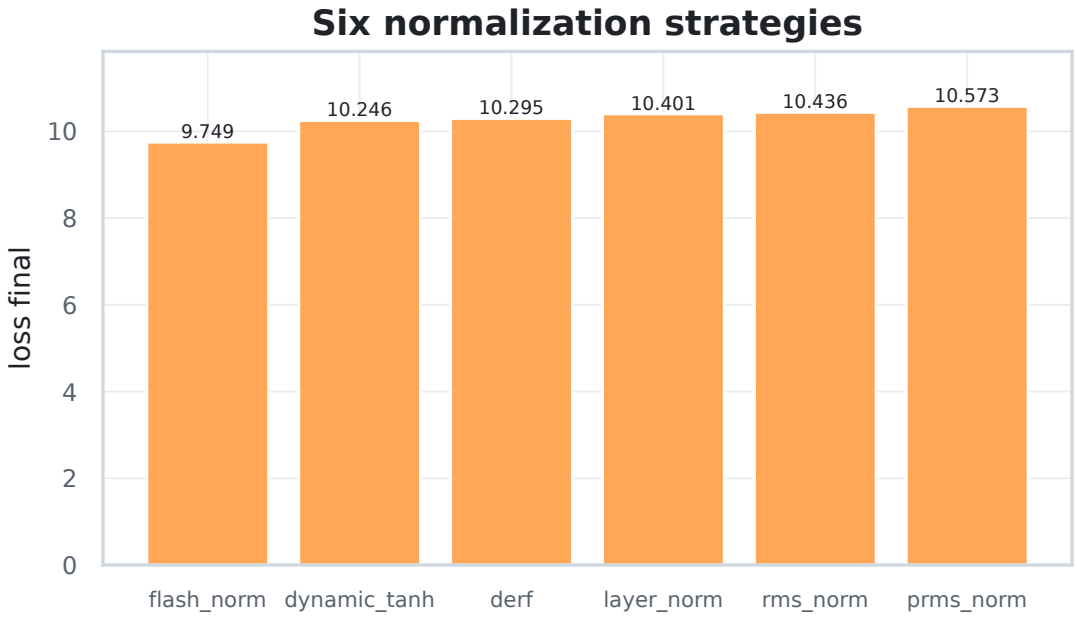


Figure 125: Normalization ablation: `flash_norm` separates from the rest by about half a nat.

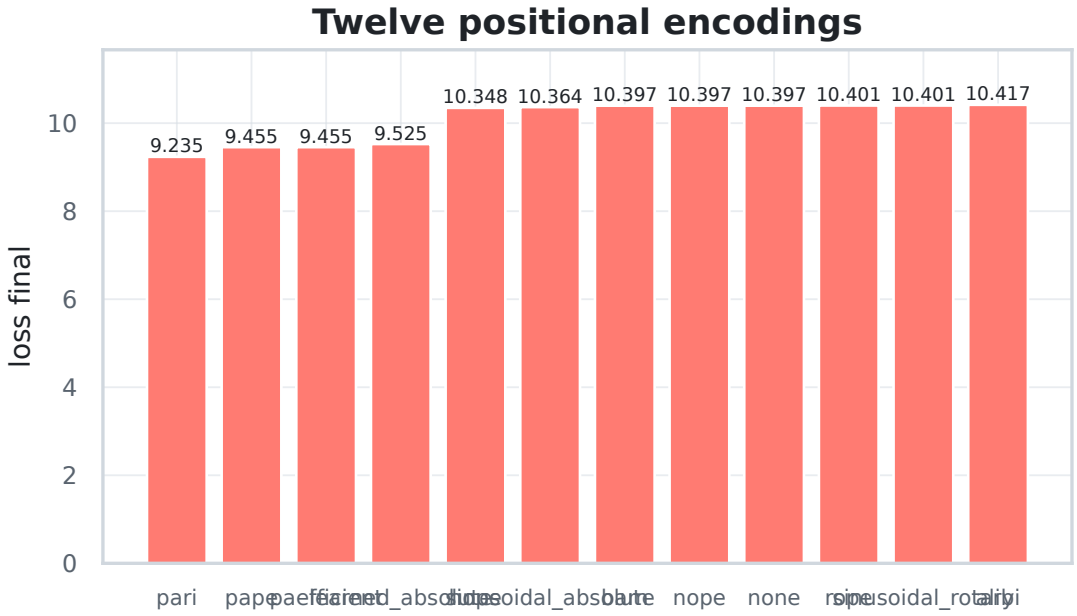


Figure 126: Positional-encoding ablation: the PAPE family leads, and classical RoPE is mid-table at this scale.

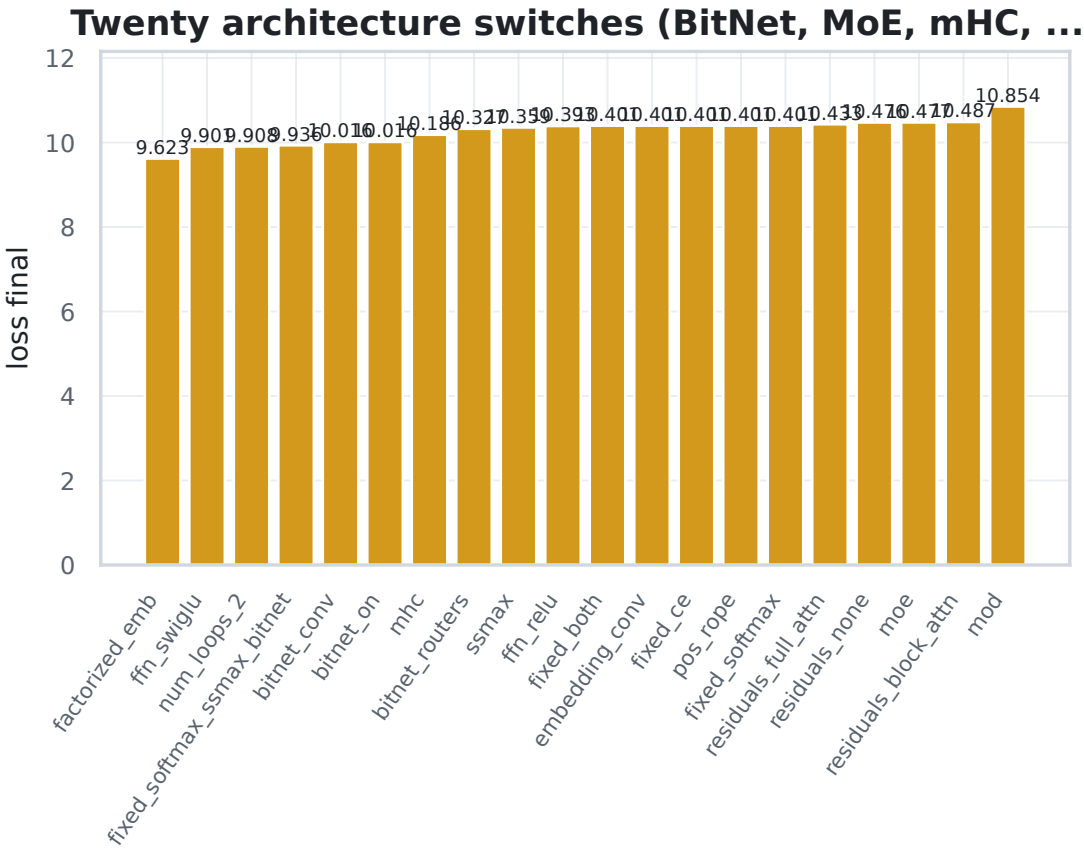


Figure 127: Structural toggles. Factorised embeddings, SwiGLU and Hyperloop are the three wins; Mixture-of-Depths is the worst switch.

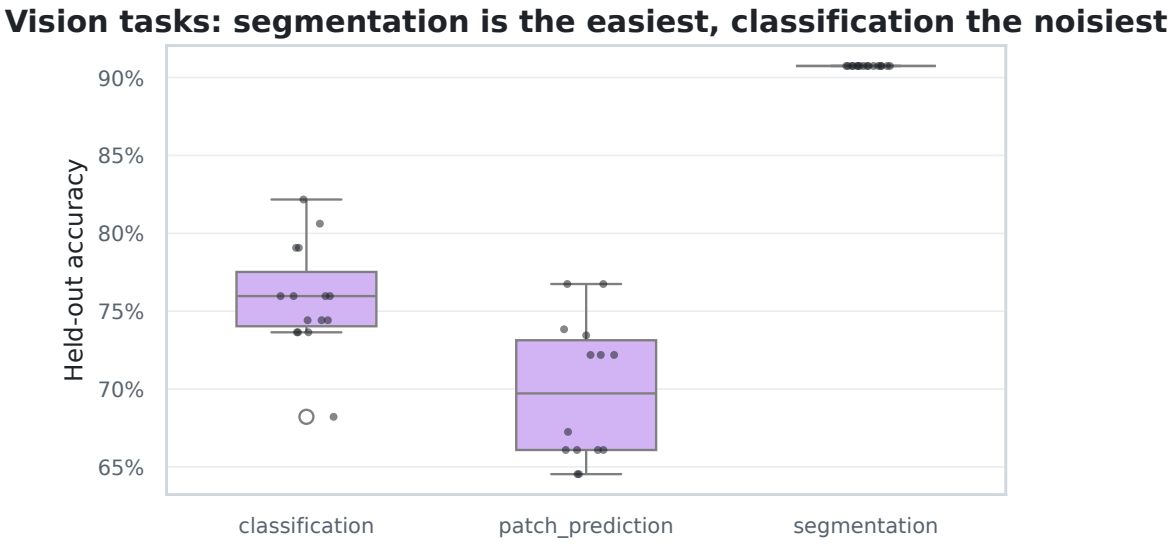


Figure 128: Vision tasks by objective. Segmentation is the easiest, patch prediction the hardest.

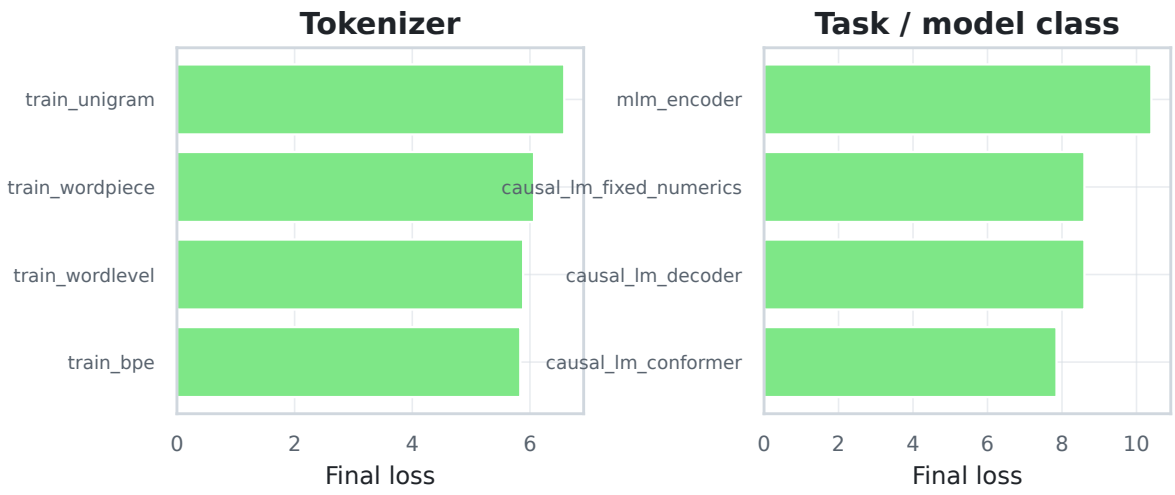


Figure 129: Tokenizer and task ablations. Both change the loss scale, so they are not comparable with the mixer sweep.

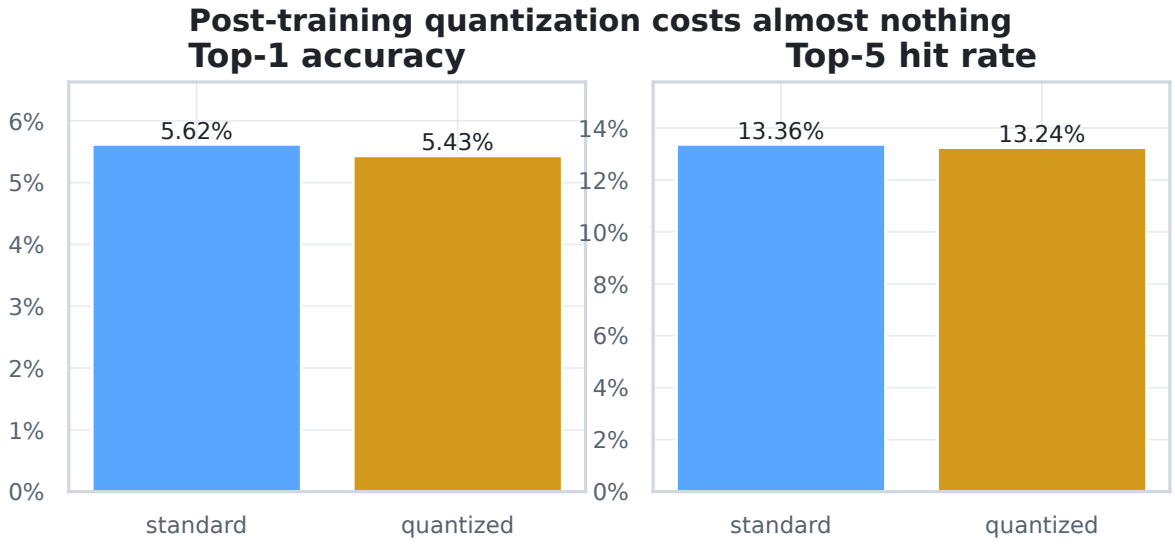


Figure 130: Standard versus quantized inference. The distributions overlap almost completely.

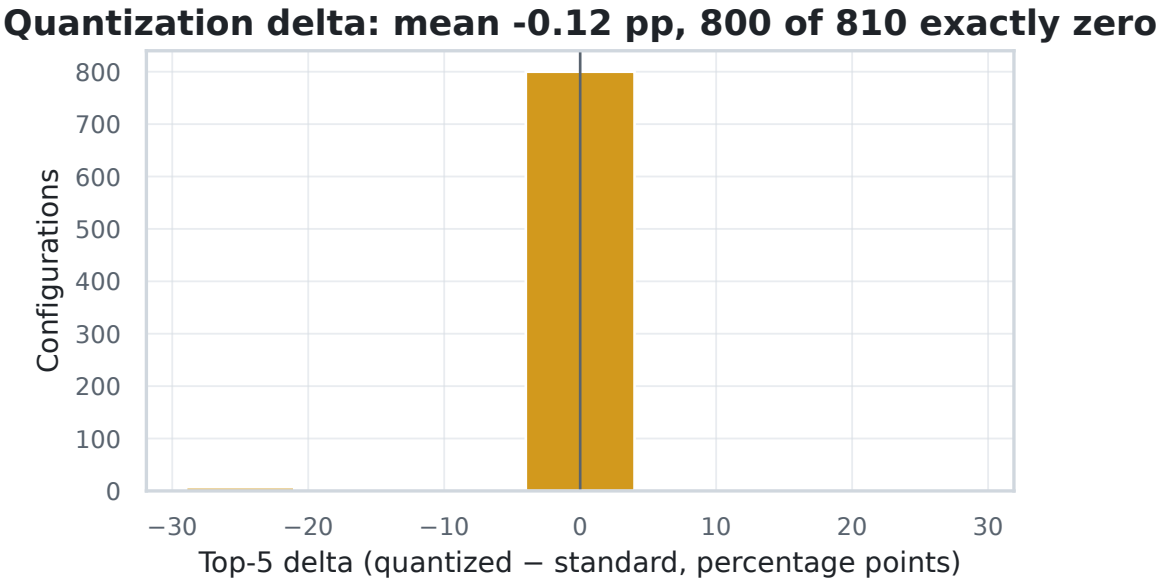


Figure 131: Quantization delta distribution. Eight hundred of 810 configurations have a delta of exactly zero.

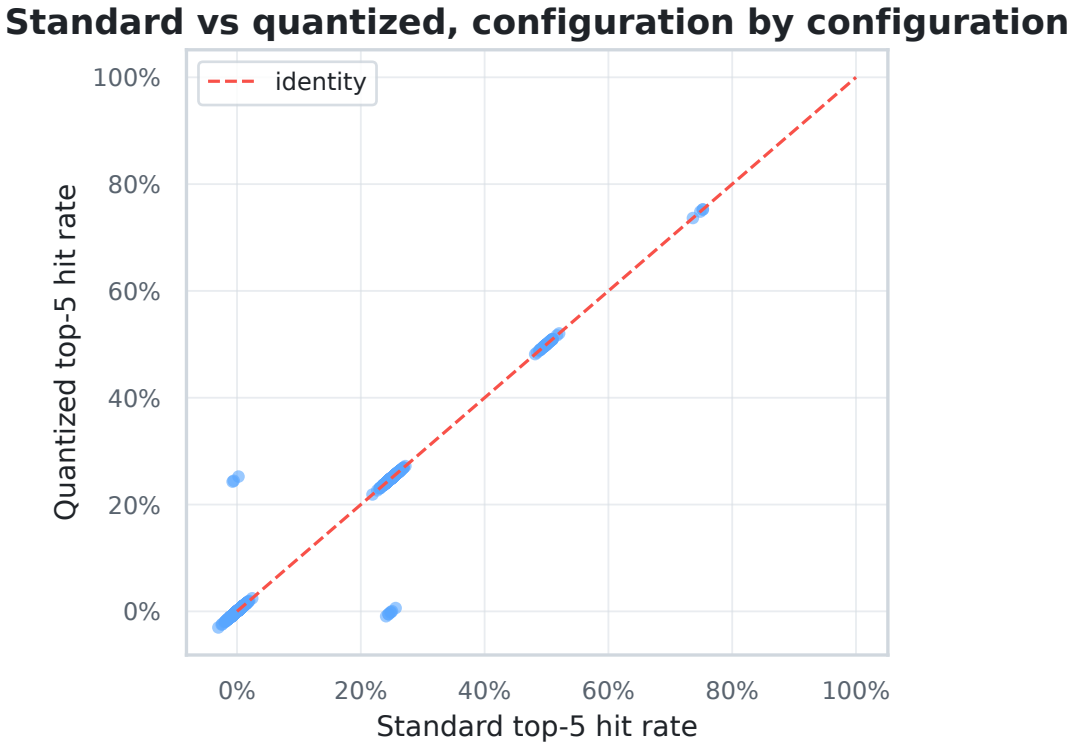


Figure 132: Standard versus quantized top-5. Every point on the diagonal is a configuration that quantization did not change.

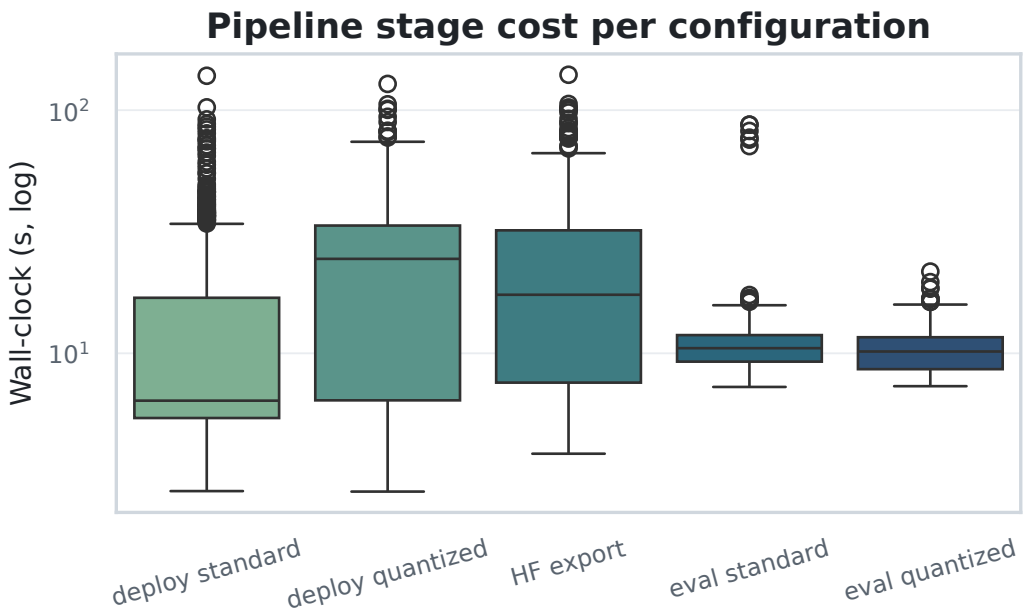


Figure 133: Pipeline stage cost. Deployment and evaluation cost seconds; training costs hours.

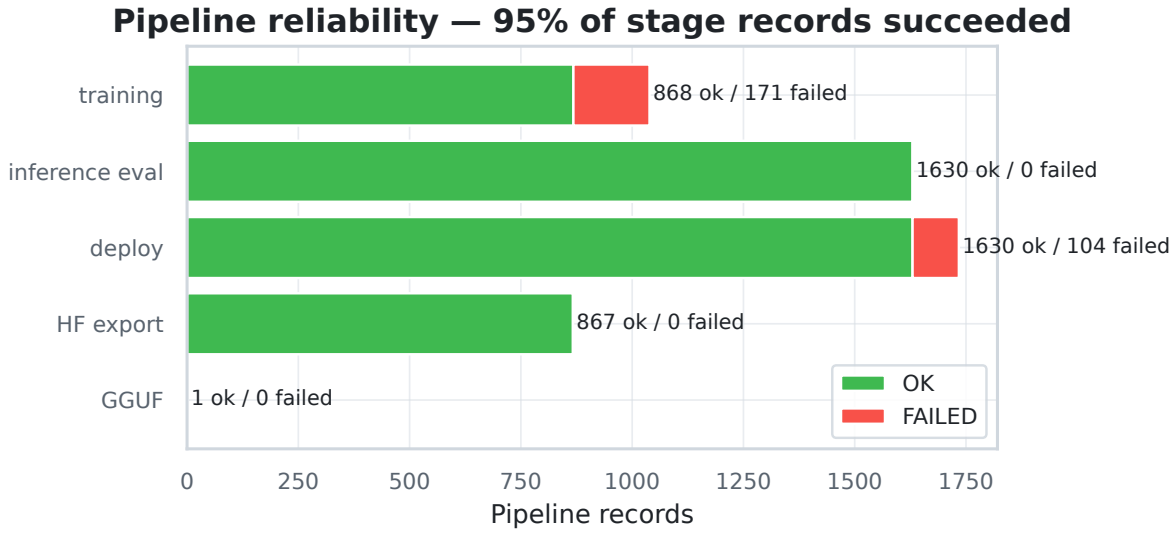


Figure 134: Pipeline reliability. 95% of stage records succeeded; training and deployment account for all failures.

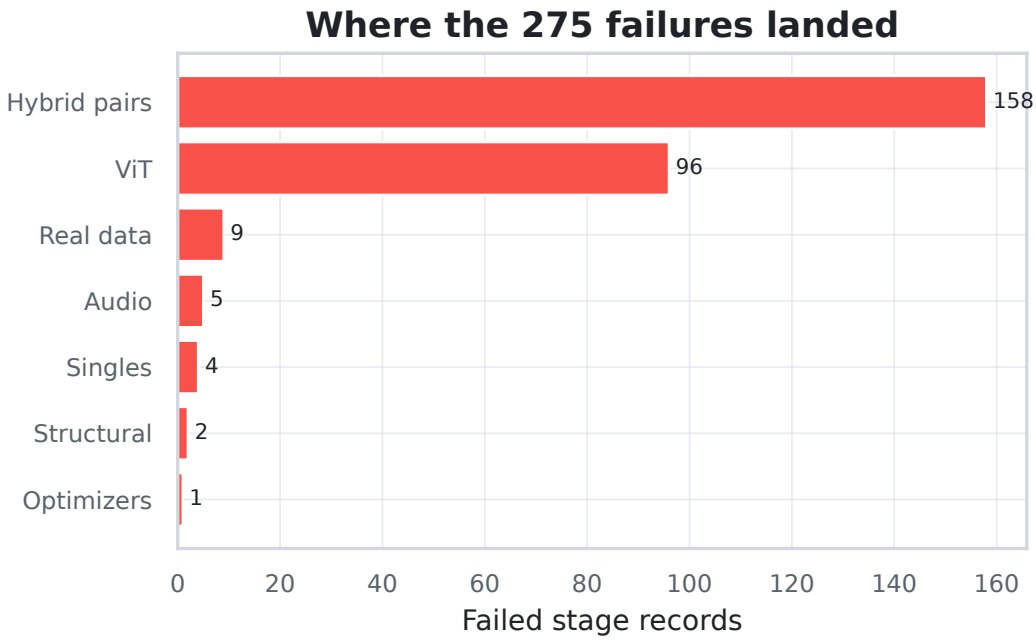


Figure 135: Where the 275 failures landed. Training failures cluster in the families whose mixers never train; deployment failures are symmetric across formats.

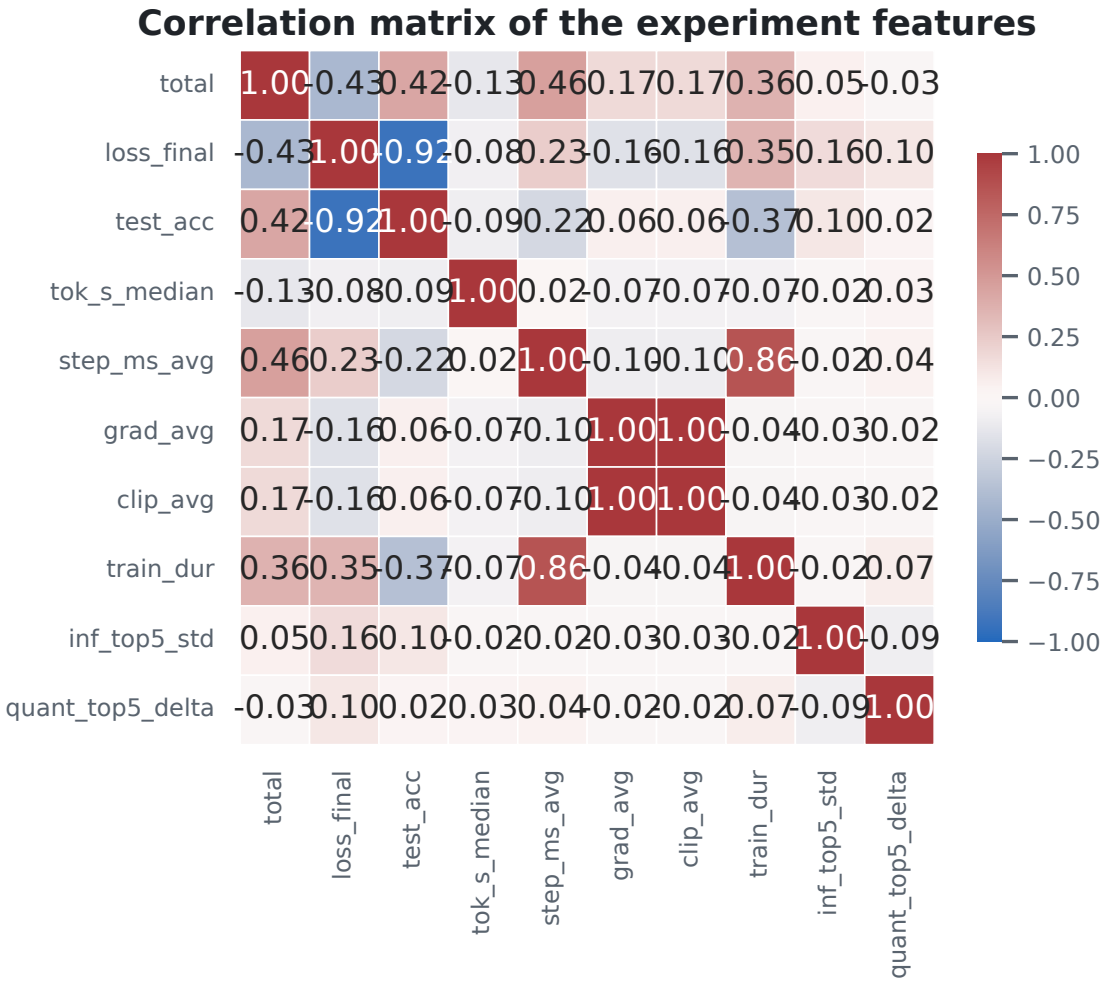


Figure 136: Correlation matrix over all 868 configurations. The strong parameter–quality correlations are a family confound; within the hybrid pairs they fall to −0.20 and −0.07.